GE Medical Systems

Technical Publications

Direction 2223752

Revision 16

Signa® MR/*i* & CV/*i* System Installation

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **“damage in shipment”** written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366. Select “Install Support Services for FOA and MIS”.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
A	Nov 10, 1998	Preliminary release
B	Nov 30, 1998	Preliminary release. Tab 1 – Addded Wide-Open System Power Cable information. Tab 2 – Added Revision B of Wide-Open Magnet Enclosure Installation. Tab 3 – Added Water Chiller cabinet installtion. Tab 11 – Added Wide-Open Pneumatic Patient Alert Installation.
0	Feb 1, 1999	Initial Release. Overview Tab, Section 1 – Added PDU E–Stop to installation Flowchart. Tab 2 – Revised Wide-Open installation instructions. Tab 3 – Revised Water Chiller installation procedures. Tab 13 – Added note to move CERD Spike Noise Test Cable to TPS RF Service Interface Kit.
1	Mar 31, 1999	Added CV/*i* information where applicable.. Overview Tab, Section 1 – Added CV/*i* info and steps related to T/R and Mobile installations. Section 2 – Added cable maps for T/R and Mobile. Tab 1 – Added Mobile and T/R references where needed. Tab 2 – Revised Fixed & T/R Wide-Open to include new rear pedestal installation, new rear End Bell installation, added photo–plethysmograph cable installation and revised Remote PAC Interface installation. Added Mobile Wide–Open Installation, Direction 2223858. Other minor revisions included in other Tabs.
2	Jul 27, 1999	Overview Tab, Section 1 – Updated flowcharts to include new references to CD–ROM procedure locations. Added Water Chiller temperature set point check. Tab 2 – Updated fixed site enclosure installation with routing of ECG cable in foam tube, adding anti–seize to four screws holding MR/*i* bridge to bridge support bracket, and other minor changes. Tab 3 – Updated Water Chiller instructions to include control set point of 20°C. Tab 4 – Added tightening instructions for penetration panel mounting hardware. Tab 13 – Added new DASM part numbers. Installation instructions should be included with DASM packaging.
3	Oct 12, 1999	Overview Tab, Section 1 – Updated catalogs and flowchart for 8.3 release. Removed references to PDU and RF/PEN 2 installation and calibration procedures. Inserted 1.5T RF/PDU installation and calibration procedures. Added camera calibration reference to flowchart. Section 2 – Added Caution referring to removal of sharp edges on ty–wraps. Removed cable runs 790 and 816 and replaced with Run 836. Added 1.5T RF/PDU to cable maps. Tab 1 – Updated fiber optic installation instructions for Run 836. Updated Penetration Panel on SGD Output cable installation. Tab 2 – Added copper tape installation to SRI module. Tab 3 – Added installation of teflon tape around hose barbs. Tab 4 – Added part numbers for copper tape and updated pen panel drawing. Tab 6 – Removed Direction 2202891., PDU installation. Tab 8 – Revised System cabinet installation for connection of Run 836. Tab 13 – Removed Run 816 to Bit 3 module and added Run 836 installation to Octane computer with bracket. Tab 14 – Added 1.5T RF/PDU installation, Direction 2244712. Corrected 1.0T Extension cables part #s.
4	Nov 1, 2000	Overview Tab, Section 1 – Added ACGD catalog information and updated installation flowchart with same information. Section 2 – Updated cabel tables to include new ACGD cables. Added new cable maps for ACGD and SRF cabinets. Section 3 – Added instruction for attaching correct language MSDS labels to phantoms. Tab 1 – Added ACGD Gradietn Cable, Fiber Optic Cable, and System Power Cables installatino instructions. Tab 13 – Updated Operator Workspace instalaltion instructions. Added two new tabs to manual: Tab 15 – ACGD/PDU Cabinet Installation and Tab 16 – SRF Cabinet Installation.
5	Apr 16, 2001	Overview Tab, Section 1 – Updated ACGD catalog information. Removed references to SGD installation. Updated installation flowchart with same information. Section 2 – Updated cable tables and cable map with existing cables that were just assigned cable run numbers. Removed obsolete cables. Added instructions to Sorting and Routing Cables section. Tab 1 – Removed instructions for all SGD installation. Updated 1.0T and 1.5T heliax cable installation. Tab 8 – Revised fiber optic cable connection to Bit3 module. Removed Tab 9. Tab 13 – Removed black & white monitor installation. Removed installation of Runs 792 and 809. Removed Tab 14. Tab 15 – Updated circuit breaker DIP switch settings instructions for PDU. Tab 16 – Changed Run 229 connection from J3 to J1.
6	Jul 17, 2001	Overview Tab, Section 1 – Added ACGD/PDU Cabinet Fan Module Rotation check to installation flow chart. Section 2 – Added to all cable tables the following information: Cable Type, Voltage Rating, Actual Voltage, Temp Rating, and Flame Rating. Tab 15 – Removed note referring to the installation of 50 Hz Inverter. Tab 16 – Revised wire color connection for Run 967 to reflect cable design change.
7	Jan 23, 2002	Overview Tab, Section 1 – Updated catalog information. Section 2 – Added cable map Note 12 for customer supplied ethernet cable for InSite connection. Section 3 – Added multilingual phantom attention label installation. Tab 2 – Updated magnet enclosure installation for installing either original long bridge or new split bridge. Tab 8 – Minor revisions to systemcabinet installation. Tab 13 – Updated operator workspace for installation of new SCIM/keyboard, ethernet connection info, and other minor revisions. Tab 15 – Updated ACGD/PDU with installation of incoming power ferrite filter and added WARNING for connecting designated power cords only to the PDU.

Continued on next page

REVISION HISTORY (continued)

REV	DATE	PRIMARY REASON FOR CHANGE
8	Jul 8, 2002	Overview Tab, Section 1 – Updated catalog with the addition of Release 10.x (MGD), EXCITE Technology information. Section 2 – Added new MGD cable information to tables and added two new cable maps for MGD. Tab 1 – Added Direction 2338872 for MGD fiber optics. Tab 2 – Updated existing enclosure installation instructions for MGD. Tab 4 – Updated Penetration Panel installation for MGD. Tab 8 – Changed title to existing System Cabinet instructions by adding 8.x/9.x. Added Direction 2338502, Signa 10.x (MGD) system Cabinet Installation Instructions. Tab 13 – Updated Octane Workspace installation with MGD parts and information. Tab 15 – Updated ACGD/PDU circuit breaker information by adding “I” DIP switch settings. Tab 16 – Added Dir 2346942, SRFD2 Installation.
9	Jun 16, 2003	Tab 4 – Added more information on attaching right angle heliax connectors to Pen Panel showing correct installation of O–rings (CQA 1029849). Tab 15 – Added neutral connection for SSM in PDU power cable connections table (SPR# MRIge82869). Revised wording on DIP switch settings. Tab 16 – Revised 1.0T and 1.5T designations for identifying Runs 236/748 and 235/745 (SPR # MRIge83424). Added note for location of UFI Label kit.
10	Oct 31, 2003	Overview Tab, Section 1 – Revised Product Locator to emphasize preferred method of submitting FE Site Verification information. Updated catalog offerings for EXCITE II product. Added new Common Chiller option. Updated fow chart to include EXCITE II equipment, Times MIcrowave cables, Linux OW, and System Calibration. Section 2 – Added EXCITE II cables to information tables. Added EXCITE II cable maps. Tab 1 – Added Direction 238919, 1.5T Times Microwave RF cable installation. Added Direction 2398342, Signa EXCITE II Fiber Optic cable installation. Added Driection 2380612, ACGD Lite Gradient Cable Installation. Tab 2 – Added EXCITE Magnet Enclosure Installation, Direction 2398604. Tab 4 – Revised Penetration Panel installation for alternate Times Microwave RF cable installation. Tab 8 – Added Direction 2398344, EXCITE II System Cabinet installation. Tab 13 – Added Direction 2386129, Linux Operator Workspace installation instructions. Updated Octane installation with minor corrections. Tab 15 – Revised attachment procedure of fan to ACGD/PDU cabinet. Added Direction 2379890, ACGD Lite/PDU Cabinet Installation. Tab 16 – Added connection point for Run 038 ground wire for SRFD2.
11	Jan 15, 2004	Updated the flowchart for EXCITE II installation. Should have been part of Revsion 10. Corrected gradient cabinet humidity warning for turning on circuit breakers (SPR# MRIge90203).
12	Jan 14, 2005	Section 1 – Updated flowchart LVSHIM value from 500 Hz to 200Hz and added Generate IIS Key instruction. Section 2 – Corrected Run 1042 cable description (SPR# MRIhc01910). Added “N” designation to Run 966 power cord connection at ACGD Cabinet (SPR# MRIge82868). Tab 2 – Direction 2398604: Corrected connections for Runs 743 & 744. Added 2mm–4mm adjustment height for bridge. Added Laser Alignment Light Warning Label Installation instructions
13	May 15, 2007	Section 1 – Added OW EXCITE Direction 5234966 to Install flow chart. Section 13 – Added Direction 5234966, *Signa EXCITE Operator Workspace Installation Instructions.*
14	Aug 16, 2007	Tab 8 – Direction 2398344: Added installation step for attaching the Canadian Registration Label. (PSR 13098263).
15	Jun 16, 2009	Tab 2 – Directions 2398604, 2223857, and 2223858: Removed instructions for installation of rating plates on end of bridge. (PQR 13246367). Direction 2398604: Added instructions for applying laser alignment light warning labels. (PQR 13246299).
16	Sep 10, 2010	Overview Tab, Section 1 – Revised Ferrous Material Handling warning, Section 3: Added step in Section 3–5 for confirming blower box anchoring. Tab 1 – Directions 2286976, 2380612, 2287106, 2338872, & 2398342: Revised Ferrous Material Handling warnings. Tab 2 – Directions 2398604, 2223857, and 2223858: Revised Ferrous Material Handling warnings and instructions for installing blower box. (CAPA 1859293)

LIST OF EFFECTIVE PAGES

PAGE	REV
Title Page	16
Damage in Trans.	–
Direction 2128126	1*
A	8
B to C	16
i to ii	10
iii	13
OVERVIEW	
MR367	–
1–1	16
1–2 to 1–16	10
1–17 to 1–18	11
1–19	13
1–20 to 1–22	11
1–23 to 1–25	12
2–1 to 2–4	10
2–5	12
2–6 to 2–14	10
2–15 to 2–24	12
3–1	7
3–2	16
TAB 1 SYSTEM	
MR231	–
1–1	10
Direction 2389819	0*
Direction 2147593	2*
Direction 2223755	2*
Direction 2286976	1*
Direction 2380612	1*
Direction 2287106	2*
Direction 2338872	2*
Direction 2398342	1*
Direction 2287109	1*
TAB 2 MAGNET ENCLOSURE	
MR232	–
1–1	10
Direction 2398604	3*
Direction 2223857	7*
Direction 2223858	5*
TAB 3 BODY COIL	
2132237–100	–
Direction 2231327	4*
TAB 4 PENETRATION PANEL	
2132238	–
Direction 2147670	5*
TAB 8 SYSTEM CABINET	
MR238	–
1–1	10
Direction 2170004	6*
Direction 2338502	1*
Direction 2398344	1*
TAB 11 ACCESSORIES AND OPTIONS	
MR 241	–
1–1	1
Direction 2123158	2*
Direction 2231762	1*
TAB 13 OPERATOR WORKSPACE	
2163917	–
1–1	10
Direction 2210982	12*
Direction 2386129	0*
Direction 5234966	1*
TAB 15 ACGD/PDU CABINET	
2285569	–
1–1	10
Direction 2286811	8*
Direction 2379890	0*
TAB 16 SRF CABINET	
2285570	–
1–1	8
Direction 2286812	4*
Direction 2346942	3*

* This revision number/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

OVERVIEW TAB

TABLE OF CONTENTS (continued)

TABLE OF CONTENTS (continued)

SECTION 1 – GETTING STARTED

1–1 SIGNA MR/*i* & CV/*i* SYSTEM CONFIGURATION

This manual provides instructions to install a new 1.5T Signa MR/*i* with EXCITE Technology, 1.5T Signa® MR/*i*, 1.0T Signa® MR/*i,* and 1.5T Signa® CV/*i* System. This publication also covers Signa Select™ which is a Signa MR/*i* system with Release ASP2 software.

Additional options are installed according to the installation manual delivered with the option.

Note

There are some naming differences between the sales literature documents and the technical documentation. Listed below are the corresponding names. The Installation document uses the technical document names.

SALES LITERATURE NAME	**= TECHNICAL DOCUMENT NAME**
Signa Infinity 1.5T with EXCITE	= 1.5T Signa EXCITE MR/*i*
Signa Infinity 1.5T	= 1.5T Signa MR/*i*
Signa Infinity 1.0T	= 1.0T Signa MR/*i*
CX 100 Magnet with K4 Technology	= 1.0T LCC (Active Shield) Magnet
CX 150 Magnet with K4 Technology	= 1.5T LCC (Active Shield) Magnet

1–2 SITE READY CHECK

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2223170, Signa MR/i & CV/i Pre–installation*, Preinstallation Checklist tab.

Tools and test equipment to install and calibrate the Signa MR/*i* system are listed in *Direction 2223170, Signa MR/i & CV/i Pre–installation*, Section 10, TOOLS AND TEST EQUIPMENT.

WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM MAGNET AS POSSIBLE.

Service installation tools used in this manual that may not be included in above references lists include:

- 46–301450G1 Fiber Optic Cable Connector Repair kit
- 46–320362G1 Heliax Termination kit
- 2134776 Gradient Cable Termination kit

1–3 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. Review and confirm that order is delivered complete. Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. The labels are color coded for different installation areas as follows:

- Green – Equipment Room
- Purple – Operator’s Room
- Orange – Magnet Room
- Yellow – Post–installation (Phantoms, Coils).

“Product Delivery Instructions” (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M1000PM is for the Fixed Site Installation Kit). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summery of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1–4 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. PREFERRED METHOD – Submitting information is the *FE Site Verification Web Site* at:
   http://gein2.med.ge.com/gib
   The FE Site Verification consists of three components that are available on the web from the main menu. They are:
   - Install/deinstall product locator model and serial numbers
   - Add/modify ship to address information
   - Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – http://supportcentral.ge.com/15563

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1–5 SIGNA MR/*i* SYSTEM

Note

Throughout this document references to Signa MR/*i* apply to both Signa 1.5T MR/*i* with EXCITE Technology and Signa 1.5T MR/*i* except where explicitly documented as different.

1–5–1 Basic System

Tables 1–1 through 1–12 list the major equipment which comprise the Signa MR/*i* systems. Illustration 1–1 shows the major equipment of the Signa MR/*i* system.

TABLE 1–1

SYSTEM ELECTRONICS MAJOR EQUIPMENT (one of the following)

CATALOG	DESCRIPTION
FOR 1.5T SIGNA EXCITE II MR/*i*	
M3000DB SmartSpeed Phased Array with EXCITE II	System Cabinet consisting of MGD chassis, Remote RF chassis, and Driver Module. The equipment listed in Note 1 1.5T Quad Head Coil
M3000DC HiSpeed Plus Phased Array with EXCITE Technology	System Cabinet consisting of MGD chassis, Remote RF chassis, and Driver Module. The equipment listed in Note 1 1.5T Quad Head Coil
M3000DD EchoSpeed Plus Phased Array with EXCITE Technology	System Cabinet consisting of MGD chassis, Remote RF chassis, and Driver Module. The equipment listed in Note 1 1.5T Quad Head Coil
FOR 1.5T SIGNA MR/*i*	
M3000AB SmartSpeed, Phased Array	System Cabinet consisting of Integrated Systems Electronics subsystem, Universal Combined Exciter/Receiver (UCERD) and Phase Array Option hardware integrated 1.5T SRFD II Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. The equipment listed in Note 2 1.5T Quad Head Coil
M3000AC HiSpeed Plus, Phased Array	System Cabinet consisting of Integrated Systems Electronics subsystem, Universal Combined Exciter/Receiver (UCERD) and Phase Array Option hardware integrated 1.5T SRFD II Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. The equipment listed in Note 2 1.5T Quad Head Coil
M3000AD EchoSpeed Plus, Phased Array	System Cabinet consisting of Integrated Systems Electronics subsystem, Universal Combined Exciter/Receiver (UCERD) and with Fast Receiver 1.5T SRFD II Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. The equipment listed in Note 2 1.5T Quad Head Coil

Note 1 The following major equipment is included in above EXCITE II catalogs:

- ACGD/PDU Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480Volt, 50/60 Hz with power filter.
- Pneumatic Patient Alert System.
- Operator Workspace equipment: GOC Computer Cabinet, SCSI Tower, Mouse and Mouse Pad, LCD panel, and chair. Note, refer to Tables 1–6, 1–7, and 1–8 for catalog choices required to complete Operator Workspace equipment.
- Patient Transport Table and cradle.
- Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps.
- Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows.

Note 2 The following major equipment is included in above catalogs:

- ACGD/PDU Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480Volt, 50/60 Hz with power filter.
- Operator Workspace equipment: Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD panel, and chair. Note, refer to Tables 1–6, 1–7, and 1–8 for catalog choices required to complete Operator Workspace equipment.
- Pneumatic Patient Alert System.
- Patient Transport Table and cradle.
- Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps.
- Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows.

(Continued)

1–5–1 Basic System (Continued)

TABLE 1–1
SYSTEM ELECTRONICS MAJOR EQUIPMENT

CATALOG	DESCRIPTION
FOR 1.0T SIGNA MR/*i*	
M3800AA SmartSpeed	System Cabinet consisting of Integrated Systems Electronics subsystem, Universal Combined Exciter/Receiver (UCERD) 1.0T SRF Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. The equipment listed in Note 2 1.0T Quad Head Coil
M3800AB SmartSpeed, Phased Array	System Cabinet consisting of Integrated Systems Electronics subsystem, Universal Combined Exciter/Receiver (UCERD) and Phase Array Option hardware integrated 1.0T SRF Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. The equipment listed in Note 2 1.0T Quad Head Coil
M3800AD HiSpeed Plus, Phased Array	System Cabinet consisting of Integrated Systems Electronics subsystem, Universal Combined Exciter/Receiver (UCERD) and Phase Array Option hardware integrated 1.0T SRF Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. The equipment listed in Note 2 1.0T Quad Head Coil

Note 2 The following major equipment is included in above catalogs:

- ACGD/PDU Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480Volt, 50/60 Hz with power filter.
- Operator Workspace equipment: Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD panel, and chair. Note, refer to Tables 1–6, 1–7, and 1–8 for catalog choices required to complete Operator Workspace equipment.
- Pneumatic Patient Alert System.
- Patient Transport Table and cradle.
- Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps.
- Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows.

TABLE 1–2
EXCITE II SRFD CABINET (SELECT ONE)

CATALOG	DESCRIPTION
M3090LK EXCITE II SRFD Cabinet	1.5T SRFD II Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components.

1–5–1 Basic System (Continued)

TABLE 1–3
INTEGRATED BODY MODULE (one of the following)

CATALOG		DESCRIPTION
1.5T	M1085BG	1.5T Advanced 60 cm Integrated Body Module (BRM–F)
1.0T	M1885BG	1.0T Advanced 60 cm Integrated Body Module (BRM–F)

TABLE 1–4
MAGNET SUBSYSTEM (one of the following)

CATALOG		DESCRIPTION
1.5T	M1060LA	15 kilogauss (1.5T) LCC (Active Shield) Magnet for Fixed Site or Transportable/Relocatable Mobile (See Note 1). Includes partial Enclosure installed on magnet, Magnet Rundown Unit and Magnet Monitor (performs cryogen meter function and magnet pressure control). (**See Notes 2 & 3**)
	M1061LA	15 kilogauss (1.5T) LCC (Active Shield) Magnet for True Mobile (See Note 1). Includes partial Enclosure installed on magnet, Magnet Rundown Unit, Magnet Monitor (performs cryogen meter function and magnet pressure control), and UPS for Magnet Monitor equipment. (**See Note 2**).
1.0T	M1860LA	10 kilogauss (1.0T) LCC (Active Shield) Magnet for Fixed Site or Transportable/Relocatable Mobile (See Note 1). Includes partial Enclosure installed on magnet, Magnet Rundown Unit and Magnet Monitor (performs cryogen meter function and magnet pressure control). (**See Notes 2 & 3**)
	M1861LA	10 kilogauss (1.0T) LCC (Active Shield) Magnet for True Mobile (See Note 1). Includes partial Enclosure installed on magnet, Magnet Rundown Unit, Magnet Monitor (performs cryogen meter function and magnet pressure control), and UPS for Magnet Monitor equipment. (**See Note 2**)
Note		1 For Mobiles (Transportable, Relocatable, and True Mobile) contact your selected Van Manufacturer for site pre–installation, refer to Section 1–14, VAN MANUFACTURERS CONTACT INFORMATION. 2 **M1060JW Shield/Cryo Cooler Compressor (water cooled) must be ordered for LCC Magnet. The Shield/Cryo Cooler Compressor is not part of the LCC Magnet Catalogs but is required at magnet delivery to minimize cryogen consumption.** 3 If M1060LA or M1860LA Magnet is to be shipped by air then must order M1060SC CX Magnet Air Transport Crate (not required within Continental USA).

TABLE 1–5
LCC MAGNET 2.5 METER CEILING HEIGHT KIT (OPTIONAL)

CATALOG	DESCRIPTION
M1060SR	LCC Magnet Low Ceiling Height (2.5 Meter) Siting Option **Note:** This option reduces the minimum required ceiling height for the Magnet Room to 2.5 meters from 2.67 meters. Check with GE Installation Specialist to determine if this is a required option.

1–5–1 Basic System (Continued)

TABLE 1–6
RECON PROCESSOR WITH BAM MEMORY (one of the following)

CATALOG	DESCRIPTION
FOR 1.5T SIGNA EXCITE II MR/*i*	
M3000RH	EXCITE ReFlex200
M3000RJ	EXCITE ReFlex400
FOR 1.5T SIGNA EXCITE I MR/*i*	
M3000RD	EXCITE ReFlex100
M3000RE	EXCITE ReFlex200
M3000RF	EXCITE ReFlex400
FOR 1.5T SIGNA MR/*i* & 1.0T SIGNA MR/*i*	
M3000RA	ReFlex100 – Hex Power PC Processor with four 128 MB Bulk Acquisition Memory (BAM) boards **See Note 1**
M3000RB	ReFlex50 – Dual Power PC Processor with four 128 MB Bulk Acquisition Memory (BAM) boards **See Note 1**
M3000RC	ReFlex50 – Dual Power PC Processor with two 128 MB Bulk Acquisition Memory (BAM) boards **See Note 1**
Note 1 BAM memory is now included with the ReFlex Array Processor and is not ordered separately.	

TABLE 1–7
FILMING INTERFACE (SEE NOTE 1 IN TABLE)

CATALOG	DESCRIPTION
FOR 1.5T SIGNA EXCITE II MR/*i*	
M1000MK	Digital DASM
FOR SIGNA MR/i (1.5T & 1.0T)	
M1000MJ	Analog DASM
M1000MK	Digital DASM
Note 1 Must choose one Filming DASM Board unless DICOM Print will be used exclusively for software filming to DICOM print peripheral devices.	

1–5–1 Basic System (Continued)

TABLE 1–8
OPERATOR WORKSPACE MONITOR

CATALOG	DESCRIPTION
M1000NT	LCD Color Monitor

TABLE 1–9
OPERATOR WORKSPACE TABLE (one required)

CATALOG	DESCRIPTION
M1000MW	Table for Operator Workspace *See Note 2*
Note 2	OW Table is an integral part of the regulatory approved system. OW Table provides mounting for several assemblies (e.g. OW Interface Module, OW Power Distribution Box, Modem, DASM) and cable routing for OW interconnects.

TABLE 1–10
COUNTRY KITS (SELECT ONE)

CATALOG	DESCRIPTION
M1000MN	English Keyboard
M1000MP	French Keyboard
M1000MR	German Keyboard
M1000MS	Scandinavian Keyboard
M1000NH	Italian Keyboard
M1000NJ	Portuguese Keyboard
M1000NK	Spanish Keyboard

TABLE 1–11
PHYSICIAN CHAIR

CATALOG	DESCRIPTION
E8803BD	Physician chair for True Mobile
E8803BE	Physician chair for Fixed Site or Transportable/Relocatable Mobile

TABLE 1–12
IIP GLOBAL MODEM (ONE REQUIRED)

CATALOG	DESCRIPTION
M1000NW	MR InSite Interactive Platform Global Modem; An external InSite modem is required to be available during system installation and warranty time.

1–5–1 Basic System (Continued)

TABLE 1–13
SITE COLLECTOR KIT (one of the following)

CATALOG	DESCRIPTION
SITE COLLECTOR WITHOUT CABLES FOR 1.5T SIGNA EXCITE II MR/*i*	
M3043PD	LCC (Active Shield) Magnet Fixed Site Wide Open Enclosure Penetration Panel with Phase Array hardware integrated and Penetration Panel Covers SPT Phantom Set with Shipping/Storage Cart
M3053PD	LCC (Active Shield) Magnet True Mobile Site Wide Open Technology Enclosure Penetration Panel with Phase Array hardware integrated True Mobile Site system interconnect cables including Phased Array Option True Mobile Site Power Distribution Unit cables SPT Phantom Set
M3063PD	LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Wide Open Enclosure Penetration Panel with Phase Array hardware integrated Transportable/Relocatable Mobile Site system interconnect cables including Phased Array Option Transportable/Relocatable Mobile Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
FOR 1.5T SIGNA MR/*i*	
M3043PA Refer to Note 1 & 2	LCC (Active Shield) Magnet Fixed Site Wide Open Enclosure Penetration Panel with Phase Array hardware integrated and Penetration Panel Covers Fixed Site system interconnect cables including Phased Array Option Fixed Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
M3053PA	LCC (Active Shield) Magnet True Mobile Site Wide Open Enclosure Penetration Panel with Phase Array Option hardware integrated True Mobile Site system interconnect cables including Phased Array Option True Mobile Site Power Distribution Unit cables SPT Phantom Set
M3063PA	LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Wide Open Enclosure Penetration Panel with Phase Array hardware integrated Transportable/Relocatable Mobile Site system interconnect cables including Phased Array Option Transportable/Relocatable Mobile Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
Note 1 Operator Workspace Long Cables (M1090ZN) is available to order only if the GE Installation Specialist determines the standard length cables (supplied in listed catalogs) for interconnects from OW to other system components will not work at the site. M1090ZN provides long length interconnects for OW to PDU, OW to System Cabinet, and OW to Penetration Panel ONLY. 2 Site Collector Kit with Phased Array Hardware with Long Cables (M3044PA) is available to order only if the GE Installation Specialist determines the standard length cables in M3043PA will not work at the site. M3044PA includes OW Long Cables with the long cables for the Equipment Room.	
(Continued)	

1–5–1 Basic System (Continued)

TABLE 1–13 (Continued)
SITE COLLECTOR KIT (SELECT ONE)

CATALOG	DESCRIPTION
	FOR 1.0T SIGNA MR/*i*
M3842PA	LCC (Active Shield) Magnet Fixed Site Wide Open Enclosure Penetration Panel and Penetration Panel Covers Fixed Site system interconnect cables Fixed Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
M3843PA Refer to Notes 1 & 3	LCC (Active Shield) Magnet Fixed Site Wide Open Enclosure with Phase Array Option hardware integrated Penetration Panel with Phase Array Option hardware integrated and Penetration Panel Covers Fixed Site system interconnect cables including Phased Array Option Fixed Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
M3852PA	LCC (Active Shield) Magnet True Mobile Site Wide Open Enclosure Penetration Panel True Mobile Site system interconnect cables True Mobile Site Power Distribution Unit cables SPT Phantom Set
M3853PA	LCC (Active Shield) Magnet True Mobile Site Wide Open Enclosure with Phase Array Option hardware integrated Penetration Panel with Phase Array Option hardware integrated True Mobile Site system interconnect cables including Phased Array Option True Mobile Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
M3862PA	LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Wide Open Enclosure Penetration Panel Transportable/Relocatable Mobile Site system interconnect cables Transportable/Relocatable Mobile Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
M3863PA	LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Wide Open Enclosure with Phase Array Option hardware integrated Penetration Panel with Phase Array Option hardware integrated Transportable/Relocatable Mobile Site system interconnect cables including Phased Array Option Transportable/Relocatable Mobile Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
Note	1 Operator Workspace Long Cables (M1090ZN) is available to order only if the GE Installation Specialist determines the standard length cables (supplied in listed catalogs) for interconnects from OW to other system components will not work at the site. M1090ZN provides long length interconnects for OW to PDU, OW to System Cabinet, and OW to Penetration Panel ONLY. 3 Site Collector Kit with Phased Array Hardware with Long Cables (M3844PA) is available to order only if the GE Installation Specialist determines the standard length cables in M3843PA will not work at the site. M3844PA includes OW Long Cables with the long cables for the Equipment Room.

1–5–1 Basic System (Continued)

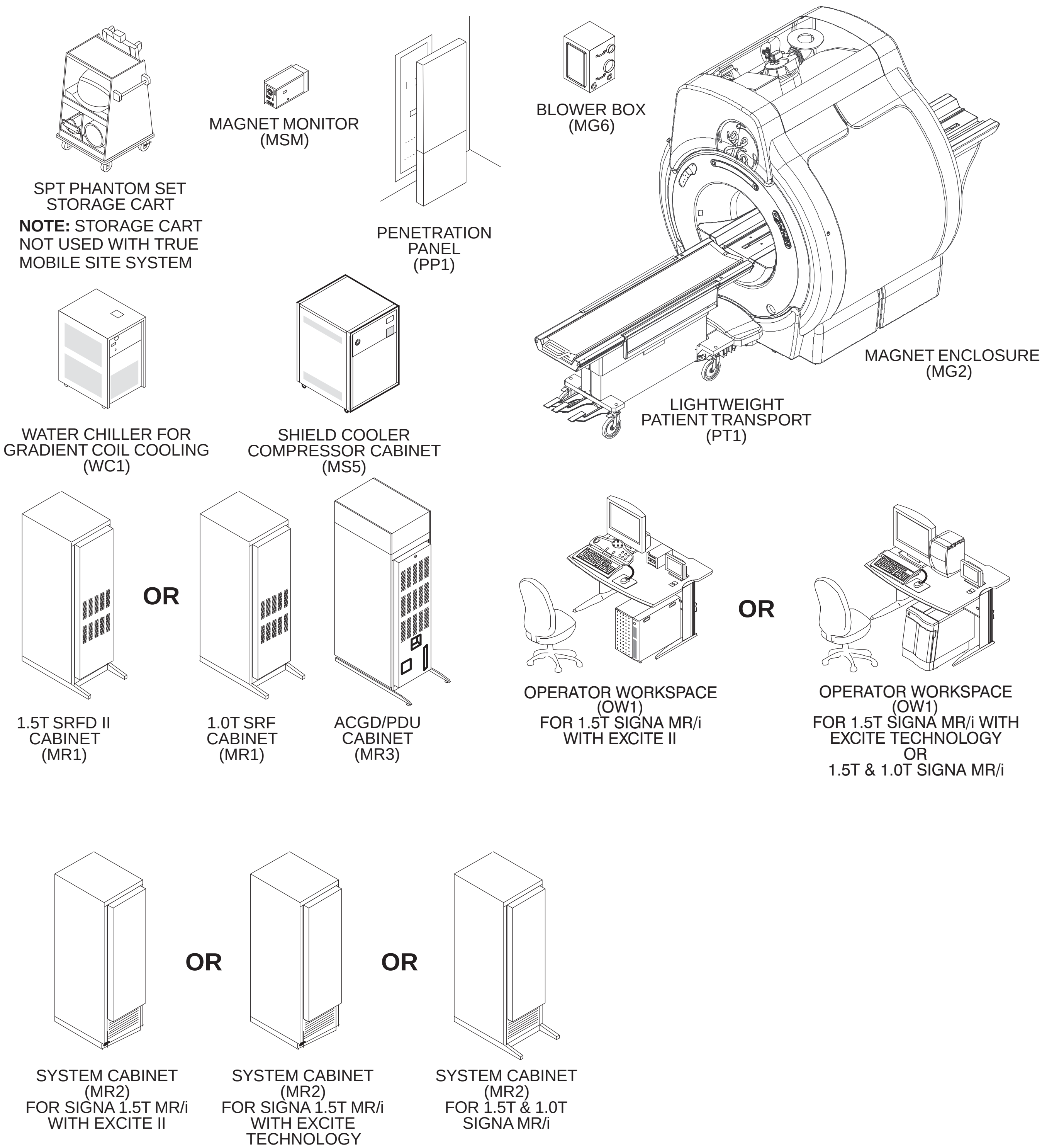
TABLE 1–14
CABLE COLLECTOR FOR 1.5T SIGNA MR/*i* WITH EXCITE II (SELECT ONE)

CATALOG	DESCRIPTION
M3143PD	Fixed Site system interconnect cables for configuration A lengths
M3144PD	Fixed Site system interconnect cables for configuration B lengths
M3145PD	Fixed Site system interconnect cables for configuration C lengths
M3153PD	True Mobile system interconnect cables
M3163PD	Transportable/Relocatable Mobile system interconnect cables

TABLE 1–15
WATER CHILLER FOR GRADIENT COIL COOLING (SELECT ONE)

CATALOG	DESCRIPTION
M1085KC	60 Hz Water–to–Air Chiller for Gradient Coil Cooling *See Note 1*
M1085KB	50 Hz Water–to–Air Chiller for Gradient Coil Cooling *See Note 1*
M1085KF	50/60 Hz Water–to–Water Heat Exchanger for Mobile systems Gradient Coil Cooling *See Note 2*
Note	1 Fixed Site must order Water Chiller for their power frequency (60 or 50 Hz). Mobile (True, Transportable, Relocatable) systems do not require this chiller, Gradient Coil chilled water will be provided by Van chilled water system. 2 True Mobile, Transportable and Relocatable systems require this heat exchanger. The Mobile van will provide chilled water for the heat exchanger operation.

1–5–1 Basic System (Continued)



SIGNA MR/*i* SYSTEM
ILLUSTRATION 1–1

1–5–2 System Options

Purchased options may also be installed concurrently with the system upgrade. Flow charts provide steering to the Option Installation Direction that is delivered with the system.

Refer to manual shipped with the option for installation of Surface Coil, Advantage Window, and Camera Options.

Refer to installation manual shipped with the option for the following:

- M1060KM – Oxygen Monitor: *Direction 15336, Oxygen Monitor Subsystem*
- M1090TL – Second Patient Transport: *Direction 15477, Second Patient Transport Option Installation*

PERFORMANCE UPGRADE OPTIONS

- M1090TZ – VCR Interface Kit
- M3000JB 18Gb Disk Drive for Octane
- M3033CB EXCITE 8 Channel (for Signa EXCITE II 1.5T MR/*i* Systems ONLY)

HARDWARE & SOFTWARE OPTIONS

- M1033BL BrainWaveHW Lite (1.5T only)
- M1033MD Multi–Nuclear Spectroscopy (1.5T only)

SOFTWARE OPTIONS

- M1033JA – *i*Drive Pro
- M1033JB – SmartPrep 2000 (Includes SmartStep)
- M1033JD – Probe2000
- M1033JF – ScanTools 2000
- M1033JH – SmartPrep 2000 Update
- M1033JJ – Probe2000 Update (for customers that have Probe software)
- M1033JK – LX FuncToll 2000
- M1033JL – LX Tools
- M1033JZ – FuncTool 2000 for AW
- M1033MG – SAGE 7
- M1033PA Performed Procedure Step
- M1090CR – Cardiac
- M1090LP – EchoPlus Diffusion Weighted Imaging Software
- M1090LR – Single Shot – FSE
- M1090LT – SmartPrep IA
- M1090MW – 8.x Software Non–Z with Multi–slice
- M1090MZ – Clariview
- M1090NN – Functool
- M1090PM – Bar Code Reader
- M1090PR – Flow Analysis
- M3033JA – Probe 3D Brain
- M3033JB – Functool CSI
- M3033JC – Ultra–Short TR for 3D Fast GRE
- M3033JD – BodyPak
- M3033JE – ACGD Plus

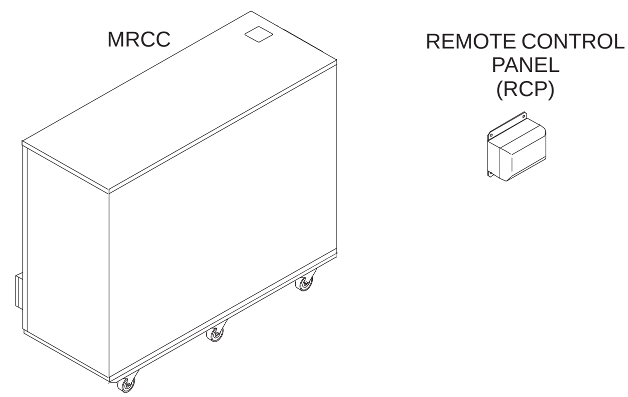
1–5–2 System Options (Continued)

EXCITE II SOFTWARE OPTIONS

- M3033KS – ASSET
- M3033LW – 3D FRFSE
- M3033LZ – Flouro Trigger MRA
- M3033MB – Fiesta C
- M3033MD – 3D Delayed Enhancement
- M3033ME – Navigator 3D
- M3033MF – 3D Fiesta
- M3033MH – Tricks
- M3033MJ – Vibrant
- M3033MM – Spiral Hi–Res
- M3033MP – FGRE–ET
- M3033MR – Real–Time FGRE–ET
- M3033MZ – Real–Time Spiral
- M3033MT – iDrivePro
- M3033MW – iDrivePro Plus
- M3033MY – Cardiac Tagging
- M3033PJ – Performed Proc Step
- M3090KJ – 2D Fiesta
- M3090KK – 3D fiesta
- M3090KR – 2D Delayed Enhancement
- M3090LF – HiSpeed Option
- M3090LG – EchoSpeed Option
- M3090LP – EchoPlus
- M3090LR – Diffusion Tensor
- M3090MA – Probe 2000
- M3090MB – Probe 2000 Upgrade
- M3090MC – Sage 7
- M3090MD – Probe 3D Brain
- M3090PL – ConnectPro

1–6 FACILITY OPTIONS

- MR Signa Main Disconnect Panel with shield cooler compressor auto restart accessory (E4503AT) for 480/277 VAC wye 5 wire or grounded delta 4 wire, surface/semi–flush mount.
- Digital Energy SG Series UPS 100KVA (E4502FB) provides reliable, clean, constant voltage power for the complete MR system. The use of a full system UPS enables the system imaging to be completed after loss of supply power and allows for saving of valuable data and orderly system shutdown. Also recommend installing a 100KVA UPS Bypass Panel (E4504CG) which feeds power to the UPS in the normal mode and enables the imaging system to operate when the UPS is in manual bypass mode for routine servicing of the UPS.
- MR Common Chiller (MRCC) (M3088TL)
  The MRCC is a single–loop 10KW 50/60 Hz water chiller that can be used to cool either Shield/Cryo Cooler Compressor. The MRCC can be installed either indoors or outdoors. There is a Remote Panel to be installed in the MR system Equipment Room that can stop or restart the chiller and can display supply water temperature, chiller setpoints and alarms. The MRCC power is supplied by the MR system Main Disconnect Panel (MDP). This chiller will be serviced by a third party vendor. See Illustration



MR COMMON CHILLER (M3088TL)
ILLUSTRATION 1–2

- Signa System Seismic Anchorage Service (R4390JA).

Note

Magnet Seismic anchoring is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems.

- Oxygen Monitor Kit (M1060KM) which includes Oxygen Monitor and Remote Oxygen Monitor Module.

Note

The Oxygen Monitor does not bear a CE monogram and therefore may not be acceptable in all European countries.

1–7 INSTALLATION PROCEDURE

1–7–1 Installation Flow

Note

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

The flow chart in Illustration 1–2 should be followed for an orderly and efficient installation of the Signa MR/*i* system. Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

Note

The flow chart references the Tab number and the PIP (Pictorial Installation Procedure) Direction number for the procedure. PIP Directions are similar to cable maps in that they are to be removed from this binder and used at the location where hardware installation is being performed. The PIP Direction may be returned to the binder for future reference upon completion of the installation. Other Directions and installation drawings delivered with kits are also referenced by the flowchart.

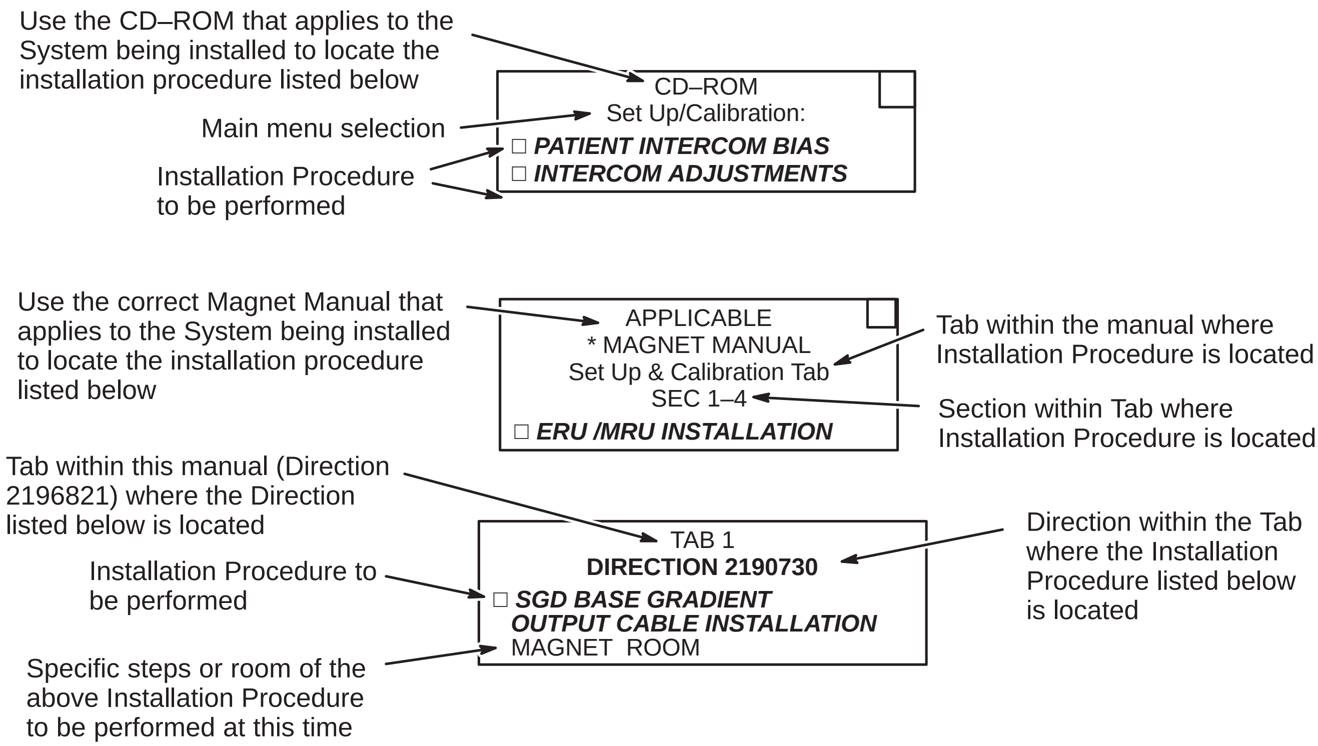
This installation flowchart has been developed assuming that all Signa MR/*i* equipment has been delivered together. Ensure that every part required is available before starting.

The general installation flow is as follows:

- Hardware installation per applicable Pictorial Installation Procedure (PIP) Direction
- Power–up and other pre–software load checks (this can be done while Magnet is being ramped)
- Software reconfiguration (this can be done while Magnet is being ramped)
- Function checks, Set–up and Calibration, and other activity requiring software running that can be done while magnet field is stabilizing
- Mechanical shim of Magnet
- System calibrations
- System performance checks
- Cover replacement
- Final sign–off and records
- Customer turn over

1–7–2 Installation Flowchart Explanation

Shown below are three examples of the flow chart Installation Procedures and an explanation of the areas within the procedures.



LEGEND FOR FOLLOWING INSTALLATION FLOWCHART

MX INDICATES MILESTONE COMPLETION GOAL IN DAYS

OR INDICATES AND OPTIONAL PATH DEPENDING ON SYSTEM CONFIGURATION

INDICATES GE PROPRIETARY PROCEDURE

1 INDICATES AN ADDITIONAL PROPRIETARY TEST OR ALTERNATIVE PROPRIETARY CALIBRATION PROCEDURE AVAILABLE FOR GE USE OR TO CUSTOMERS WITH AN ADVANCED SERVICE PACKAGE LIMITED LICENSE.

2 AFTER GRADIENT SHIM (OR LV SHIM) IT MAY BE NECESSARY TO REPARK THE MAGNET.

NOTE 1: STANDARD CORRELATED NOISE CHECK NOT REQUIRED IF PROPRIETARY SPT FULL TEST MODE PROCEDURE PERFORMED EARLIER DURING INSTALLATION (HOWEVER, MULTICOIL AND FAST RECEIVER OPTION CORRELATED NOISE CHECKS IN PROCEDURE STILL REQUIRED).

NOTE 2: IF TROUBLESHOOTING NECESSARY.

NOTE 3:

NOTE 4: A “saveINFO” SHOULD BE DONE BEFORE SPT IS RUN. IF YOU HAVE THE EPI OPTION, IT IS NECESSARY TO “saveINFO” AGAIN. IN THIS WAY, ANY DATA COLLECTED FOR CALIBRATION WILL NOT BE LOST IF A PROBLEM OCCURS.

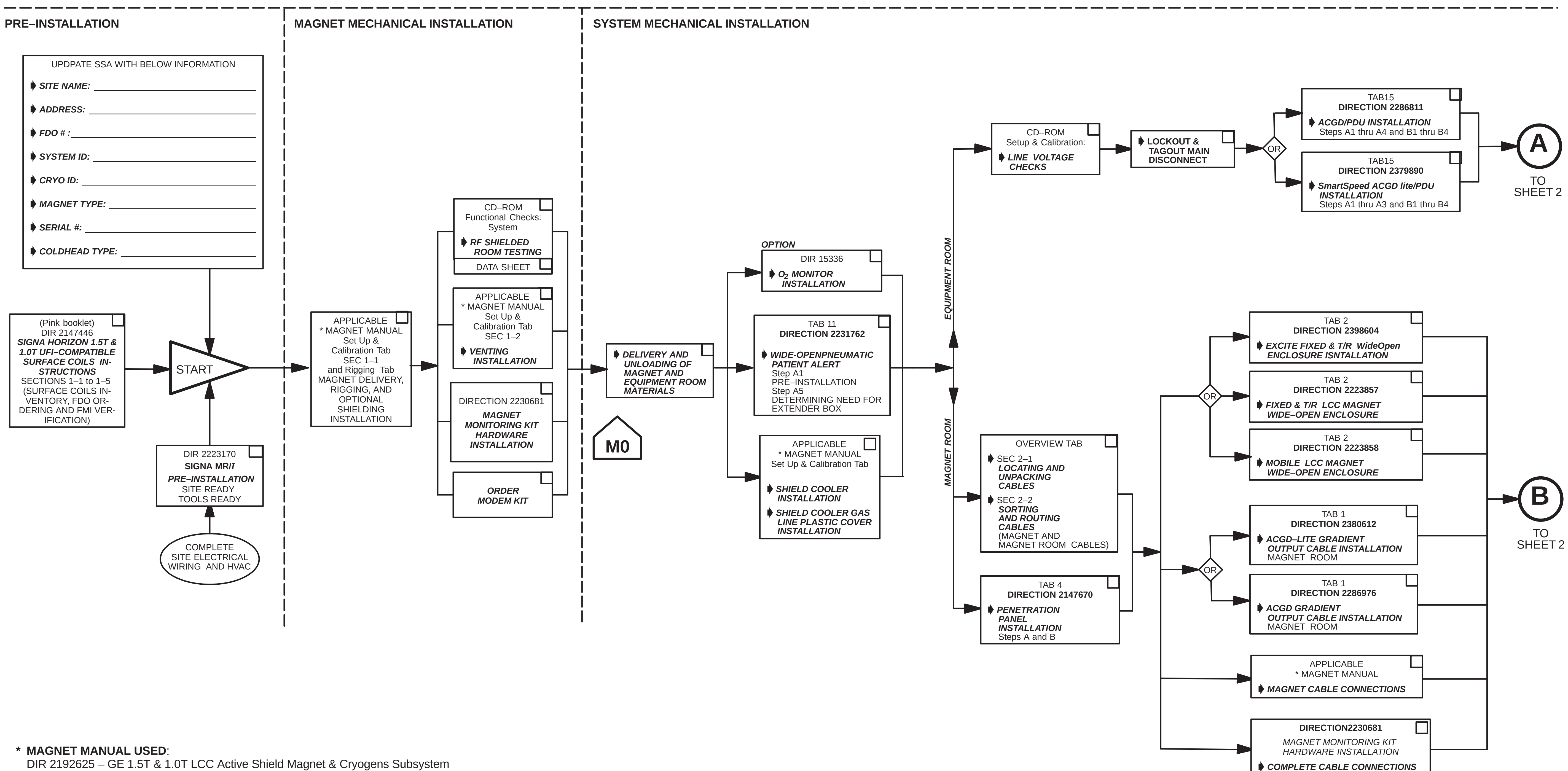
NOTE 5: VALUES OF LVSHIM ARE DEPENDENT UPON EDDY CURRENT COMPENSATION.

NOTE 6: VALUES OF EDDY CURRENT COMPENSATION ARE INDEPENDENT OF MAGNET SHIM.

NOTE 7: OVERNIGHT DIAGS MAY BE RUN AT END OF ANY DAY TO DETECT POSSIBLE ERRORS.

MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS



* **MAGNET MANUAL USED**:
DIR 2192625 – GE 1.5T & 1.0T LCC Active Shield Magnet & Cryogens Subsystem
DIR 2226318 – GE 1.5T & 1.0T LCC Magnet Signa MR/*i* & CV/i Delivery & Installation

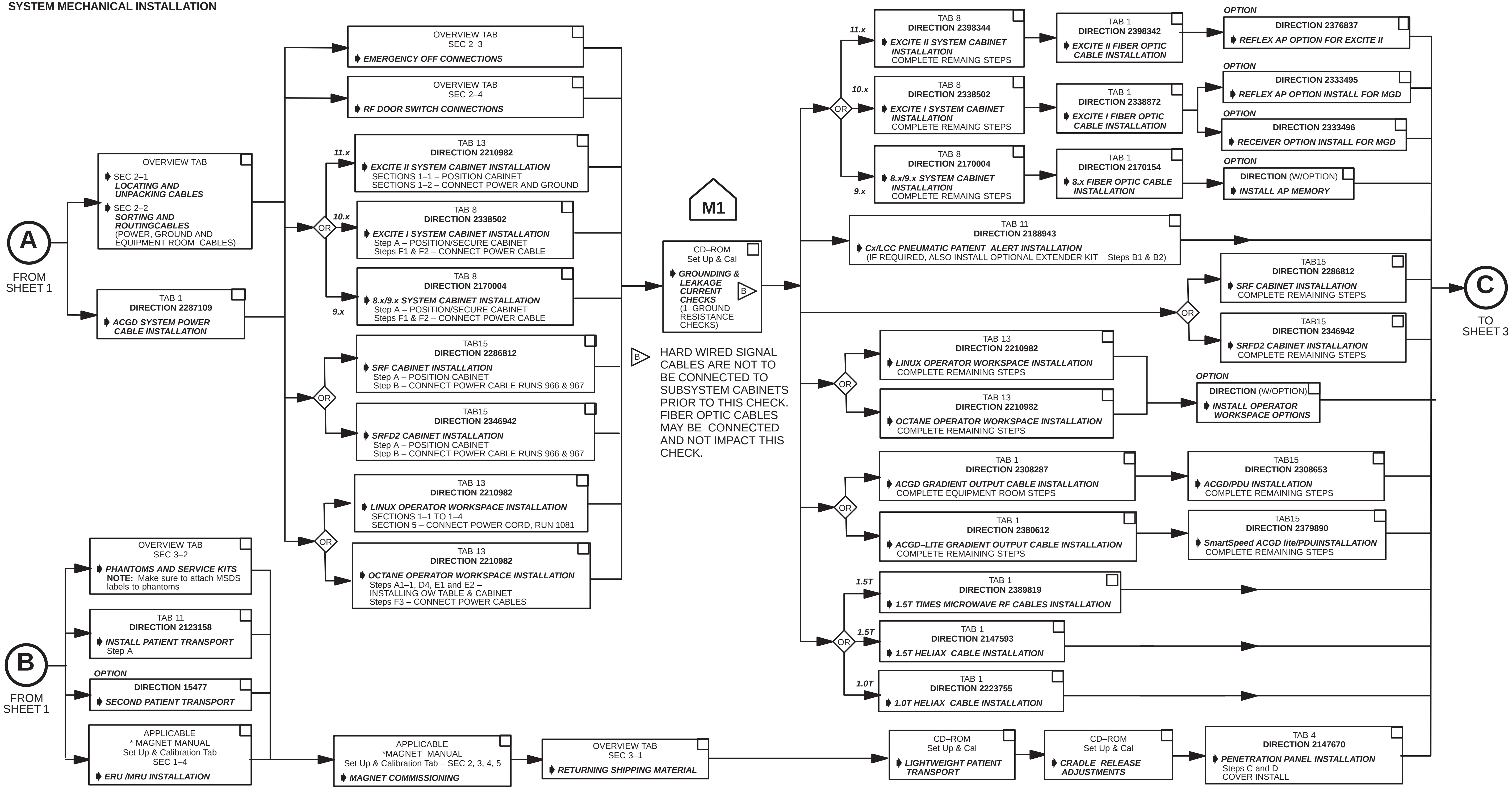
SIGNA MR/*i* & CV/*i* INSTALLATION FLOW CHART
ILLUSTRATION 1–3 (SHEET 1 OF 5)

THIS PAGE LEFT BLANK INTENTIONALLY

MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM MECHANICAL INSTALLATION

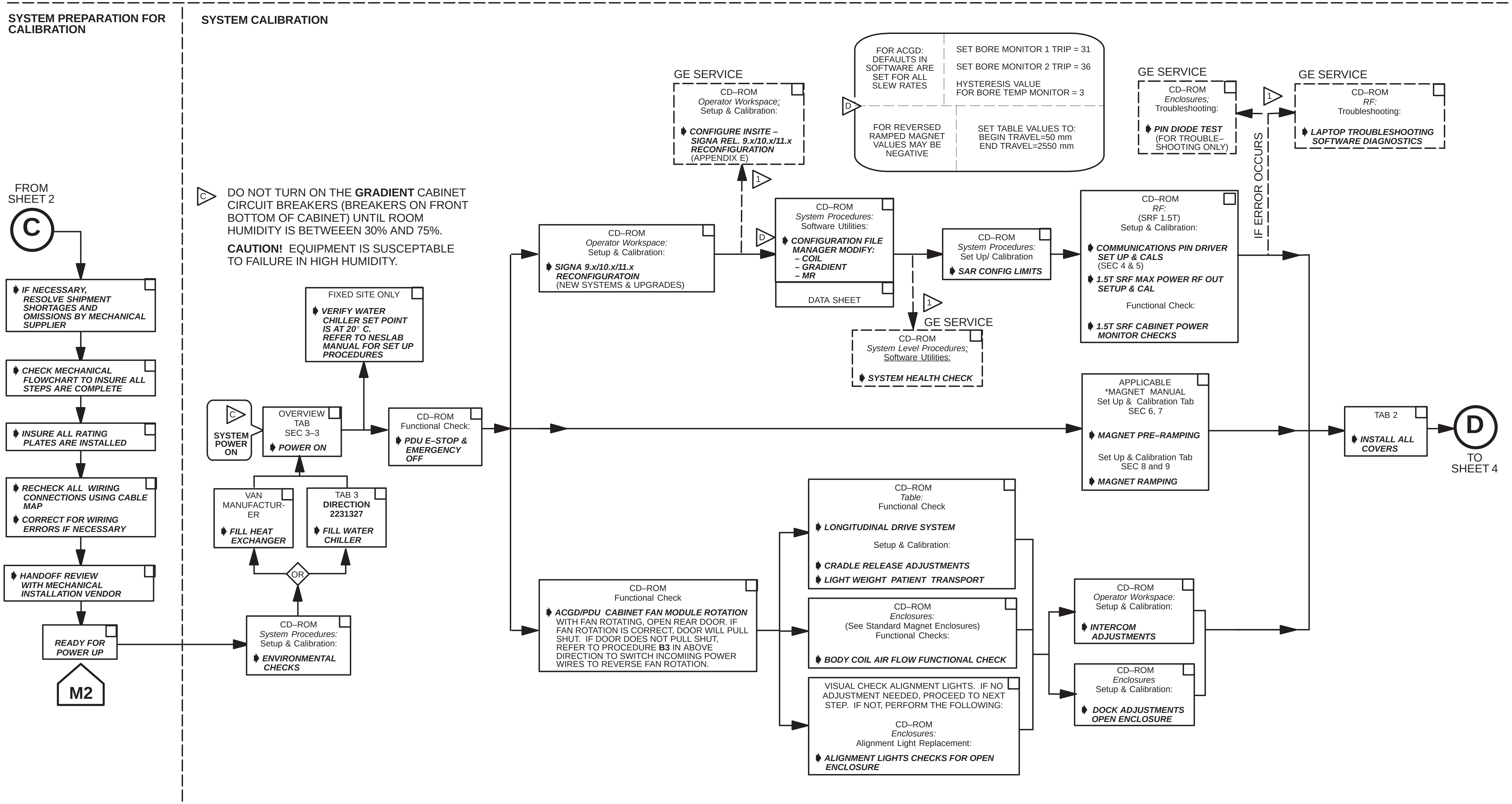


SIGNA MR/*i* & CV/*i* INSTALLATION FLOW CHART
ILLUSTRATION 1–2 (SHEET 2 OF 5)

THIS PAGE LEFT BLANK INTENTIONALLY

MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS



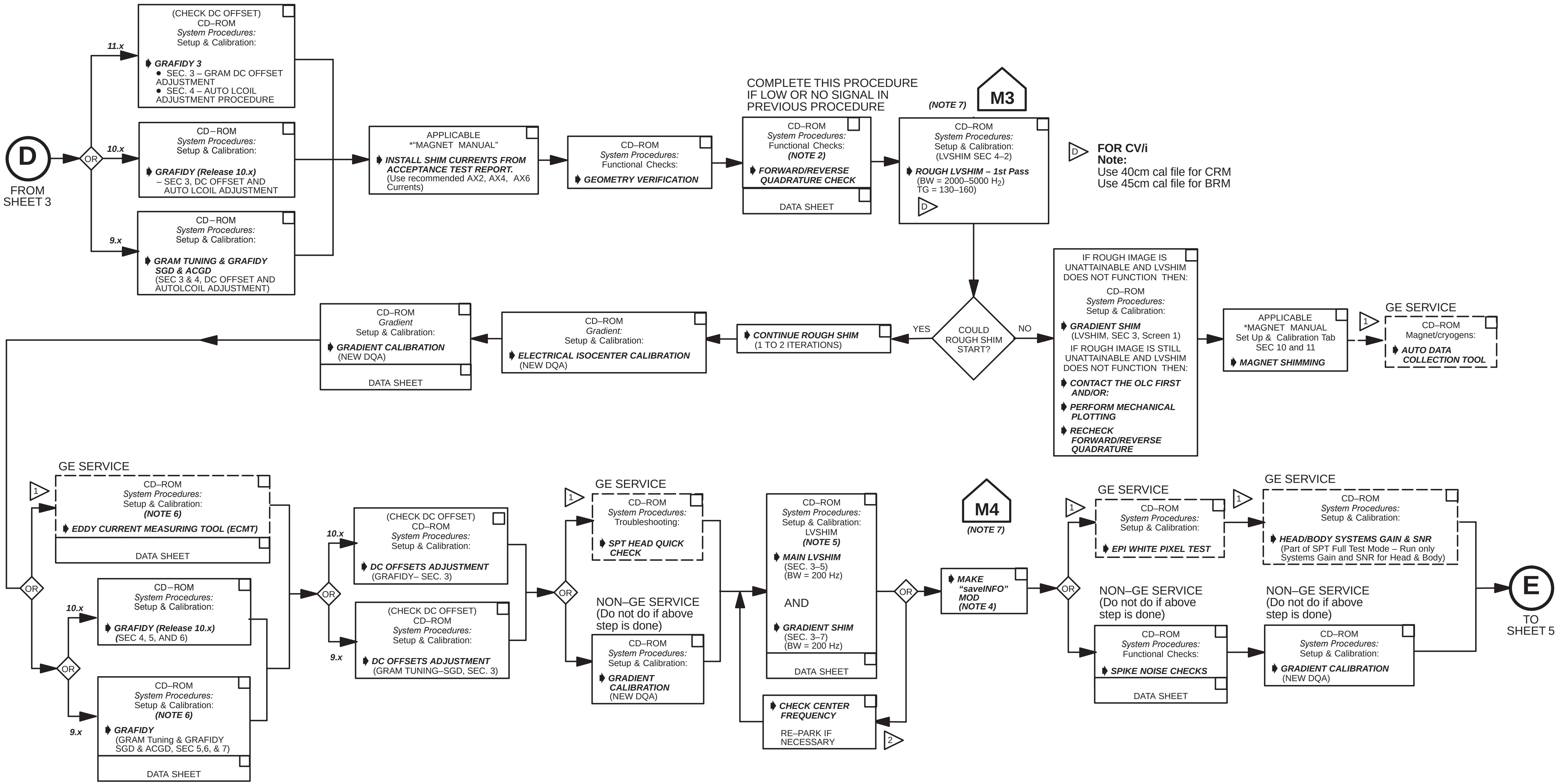
SIGNA MR/*i* & CV/*i* INSTALLATION FLOW CHART
ILLUSTRATION 1–2 (SHEET 3 OF 5)

THIS PAGE LEFT BLANK INTENTIONALLY

MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM CALIBRATION

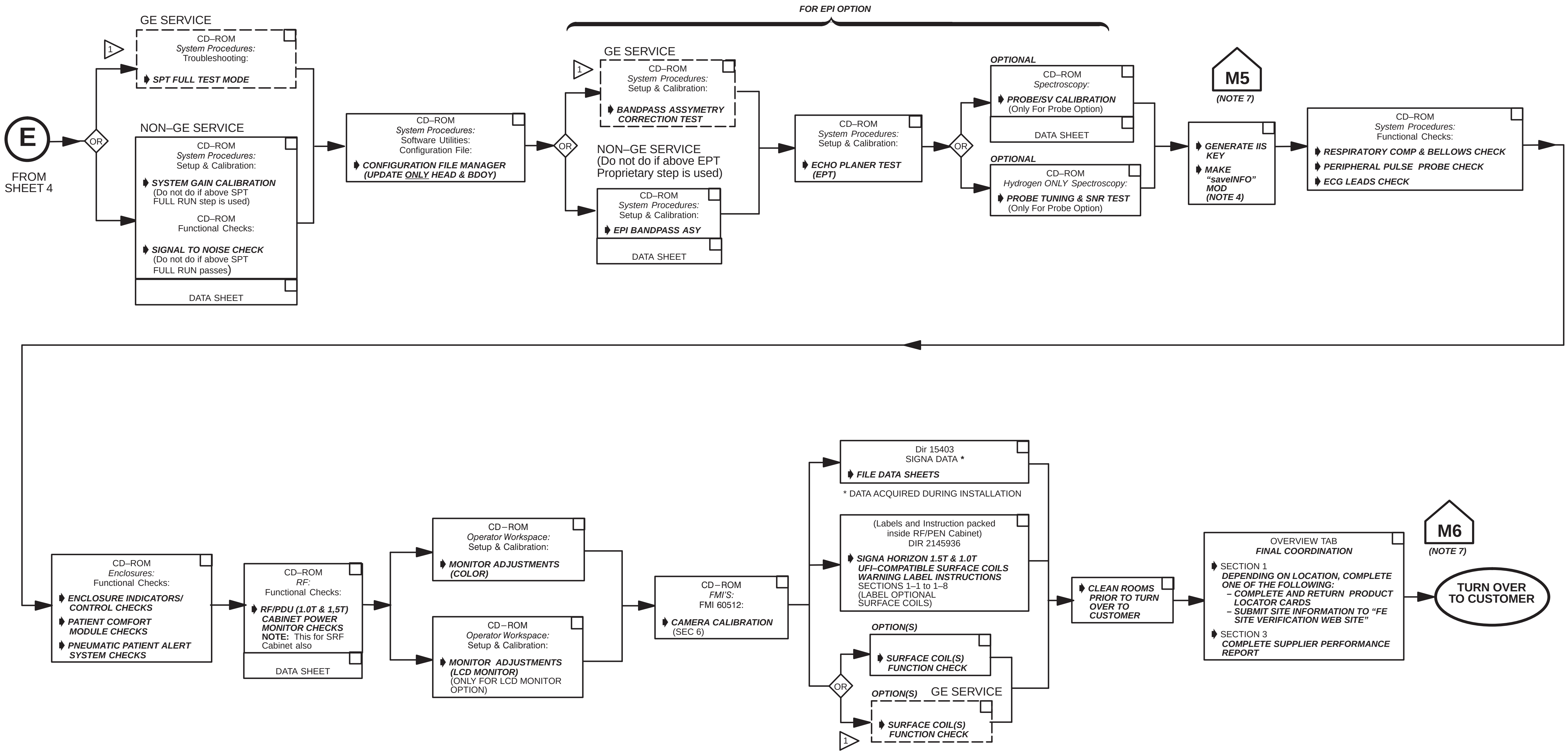


SIGNA MR/*i* & CV/*i* INSTALLATION FLOW CHART
ILLUSTRATION 1–2 (SHEET 4 OF 5)

THIS PAGE LEFT BLANK INTENTIONALLY

MARK OFF PERFORMED ITEMS **IMPORTANT : PER ISO 9001 REQUIREMENTS**

SYSTEM CALIBRATION



SIGNA MR/*i* & CV/*i* INSTALLATION FLOW CHART
ILLUSTRATION 1–2 (SHEET 5 OF 5)

SECTION 2 – SYSTEM CABLES

2–1 LOCATING AND UNPACKING CABLES

This section assumes Signa MR/*i* System and options are delivered complete with all cables present.

Many of the cables are common to both the 9.x , 10.x (EXCITE I) and 11.x (EXCITE II) System. The cables that are specific to 9.x, 10.x, or 11.x installation are designated in the applicable cable maps and in the first column of the following tables as:

* Cable used on 9.x installation only

** Cable used on 10.x installation only

*** Cable used on 11.x (EXCITE II) installation only

Cables shipment containers have color coded labels which identify the cable routing areas:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room
- Yellow – Post–installation (Phatoms, Coils)

CAUTION

Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
"FROM" & "TO"	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material my be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material's ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable's use in the MR system	

2–1 LOCATING AND UNPACKING CABLES (continued)

1. Unpack magnet room cables from the GE Magnet accessory box(s) and move them into the magnet or equipment room for routing. The shield cooler runs should have been installed previously when magnet was delivered.
   a. For magnet and equipment room standard runs delivered with GE Magnet, refer to Table 2–3.
2. Unpack and move power and ground cable boxes to the PDU area where they will be routed. Set Run 040, 044, and 045 aside. Power and ground cables are listed in Table 2–1.
   a. Move Run 040 and 045 Ground Cables into magnet room. Refer to Table 2–1.

TABLE 2–1
PDU POWER AND GROUND CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW–1	MR2 "permanent" (UL) ground
038	PD1 A2 A2	MR1 A7 GND	31 ft Ground Wire (#10)	SO	600	0	105	VW–1	SRF Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW–1	Magnet Ground Cable
044	PP1 GND	MR1 A7 GND	31 ft Ground Wire (#10)	SO	600	0	105	VW–1	RF Cabinet Separate ground
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW–1	Blower Cabinet Separate ground
048 * **	PD1 GND	OW1 A2 A5 GND	51 ft Ground Wire (#10/1)	SO	600	0	105	VW–1	Operator Workspace Cabinet ground
049 * **	OW1 A1 A1	OW1 A2 A5 J2	Power Cord	STO	600	120	90	VW–1	Operator Workspace Table to OW Cabinet
966	MR1 A7 J1	MR3 PD1	35 ft Power (5–#8)	SO	600	208Y	75	FT–4	ACGD SSM Module Power
967	MR1 CB1	MR3 PD1	35.5 ft Power (4–#8)	SO	600	208	90	FT–4	ACGD SRF Cabinet Power
968	MR1 A12 J1	MR3 PD1	35 ft Power (5–#8)	SO	600	208Y	75	FT–4	ACGD System Cabinet Power
969 * **	OW1 A2 A5 J1	MR3 PD1	65.5 ft Power (4–#10)	SO	600	120	75	FT–4	ACGD Operator Workspace Cabinet Power
970	WC1	MR3 PD1	82 ft Power (3–#10)	STO	600	208	90	FT–4	ACGD Water Chiller Power
973	HE1	MR3 PD1	82 ft Power (3–#12)	SO	600	208	75	FT–4	ACGD Heat Exchanger Power
1081 ***	OW1 A19 J1	MR3 PD1	65.5 ft Power (4–#10)	SO	600	120	75	FT–4	Operator Workspace Power
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUP-PLY	50 ft Power (5–#8) With 30A Plug and Recep-tacle						Cable for Compact PDU to GE Magnet Power Supply or SC Shim Supply Power

2–1 LOCATING AND UNPACKING CABLES (continued)

3. Unpack and move fixed site magnet room interconnect cables to magnet room. Refer to Table 2–2.
   a. Locate Body Coil RF input Cables (ship with Body Coil) and move cables to magnet room.
   b. Runs 404, 414, 421, 422, 485, and 749 are shipped with the Magnet Enclosure.
   c. Move gradient cables into magnet room. Gradient installation instructions are located in Tab 1.

TABLE 2–2
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
262	PP1 J8	MG3 A3 A5 OUT	45 ft. RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	Body Coil receive
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Table Elevation motor power
404	MG3 A17 J1	MG2 A16 A5 J6	13.67 ft. RG58/U Coax	CI2	1900 VRMS	<30	80	VW–1	Head transmit cable take up
414	MG3 A35	MG3 A2 J9	30 ft. 9–pin Sub–D	CI3	300	<30	80	FT–4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	Long Drive Encoder/Limit Switch
458	PP1 J17	OM3	9 pin Sub–D	CI3	300	<30	80	FT–4	Optional Remote O2 Sensor
485*	MG3 A17 J2	MG2 A16 A3 OUT	13.67 ft. RG58/C Coax	CI2	1900 VRMS	<30	80	VW–1	Head receive cable take up
491	PP1–J80	MG3–A36–J7	80 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
492	PP1–J81	MG3–A36–J8	80 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC3 Receive
493	PP1–J82	MG3–A36–J9	80 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
494*	PP1–A14–J87	MG3–A36–J6	80 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC1 Receive
495*	PP1–A15–J78	MG3–A36–J10	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	Multi–coil Switching
700*	PP1 A14 J86	MG2/MG3 A17 J2	45 ft. RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	Head or Surface Coil receive
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	PAC Power
727*	PP1–A16–J77	MG3–A36–J5	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	Multi–coil Bias
728	MG3 A36 J11	MG2 A16 OUT J1	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
729	MG3 A36 J12	MG2 A16 OUT J2	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
730	MG3 A36 J13	MG2 A16 OUT J3	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
731	MG3 A36 J14	MG2 A16 OUT J4	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
732	MG3 A36 J15	MG2 A16 OUT J5	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable

2–1 LOCATING AND UNPACKING CABLES (continued)

TABLE 2–2 (CONTINUED)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
733	MG3 A36 J16	MG2 A16 OUT J6	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
734	MG3 A36 J17	MG2 A16 OUT J7	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
735	MG3 A36 J18	MG2 A16 OUT J8	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
736*	MG3 A36 J19	MG2 A16 A8 J9	12 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	RF Cable
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F “N”	STO	2500 VDC	950	85	VW–1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F “N”	STO	2500 VDC	950	85	VW–1	Body Coil Drive
749	MG2 A33 J10	MG2 A36	35 ft 3 cond w/ 9–Sub D	CI3	300	<30	105	FT–4	Bore Temperature Sensor
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power cable	SO	600	120	80	FT–4	Patient cooling blower power
786	PP1 F7, F8	MG6 J2	40 ft. 2 cond power cable	SO	600	120	80	FT–4	Gradient cooling blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub–D/rt. angl	CI3	300	<30	80	FT–4	Operator Display to PAC II
861 * **	MG2 A4 J4	MG2 A33J5,6,7,8,9	20 ft 5/37 pin Sub–D/80 pin	CI2	300	<30	80	VW–1	Operator Display to SRI
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW–1	Operator Display to Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW–1	Operator Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW–1	Spliiter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 37 pin Sub–D/26 pin	CI3	300	<30	80	FT–4	Operator Display to SRI
1011 ***	MG2 A4 J4	MG2 A33 J5	20 ft. Data Cable (80–#28)	CI2	300	<30	80	VW–1	SRI Splitter to Oper Display
1035 ** / ***	MG2 A16 A3/A5 J4	MG3 A36 J21	40 ft. RG58/C Coax	CI2	1900 VRMS	<30	80	VW–1	Head or Surface Coil Receive
1036 *x / ***	PP1 J84	MG3 A36 J6	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1037 ** / ***	PP1 J6	MG3 A36 J1	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1038 ** / ***	PP1 J7	MG3 A36 J2	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1039 ** / ***	PP1 J58	MG3 A36 J3	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1040 ** / ***	PP1 J59	MG3 A36 J4	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive

2–1 LOCATING AND UNPACKING CABLES (continued)

TABLE 2–2 (CONTINUED)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1041 ** / ***	PP1 A15 J78	MG3 A36 J10	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	SRI Power
1042 ** / ***	PP1 A16 J77	MG3 A36 J5	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	CrossPoint/8 Channel SW Control Power Cable
1043 ** / ***	MG3 A8 J11	MG3 A5 J1	0.5ft Discrete Wires (13–#22)	CI3	300	<30	60	VW–1	Head T/R Power/Coil ID LED
1049 ** / ***	MG3 A36 J19	MG2 A16 A8 J9	12 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Multi–Coil Select Cable
1054 ***	MG3 A36 J20	MG2 A33 J15	Data Cable (9–#22)	CI3	300	<30	80	vw–1	MXA Signal Cable
1057 ***	PP1 J85	MG3 A8 J1	32 ft. R223/U Coax	CMX	1900 VRMS	<30	80	VW–1	UNTS Antenna

TABLE 2–3
GE MAGNET CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp and Shim Cable Assembly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12–#22)	CI3	150	60 VDC	80	FT–4	Switch Heaters
605	MS1	MR2 A8	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Instrumentation Cable, 80 feet
606	MS4	MS1 A3 A1	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Emergency Run Down Unit
608	MS1	MR2 A8	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Instrumentation Cable, 15 feet
609	MS7 OR MS8	PDU	Power Cable (5–#16)	SO	300	208Y	105	FT–4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW–1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW–1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4–#18)	SO	600	200	105	VW–1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4–#18)	SO	600	200	105	VW–1	Cold Head Control Cable

2–1 LOCATING AND UNPACKING CABLES (continued)

4. Unpack and move fiber optic cables into routing location. Refer to Table 2–4.

TABLE 2–4
FIBER OPTIC CABLES DELIVERED WITH FIXED–SITE KIT

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
708*	MR2 A11 J16	MR1 A7 TX	1/2 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
709	MR3 A11 J2	MR1 A7 RX	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
710*	MR2 A11 J17	MR3 A11 J1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic and one plastic fiber optic cables
711/ 712*	MR2 A11 J13	PP1–J91	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
711/ 712	PP1–J91	MG2 A33 and MG2 A11 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC–II
836*	MR2 A11 J30 BIT3	OW1 A15 BIT3	3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables wiith two spares
1033 ** / ***	MR2 A11 J16	MR1 A7 TX	20 ft. 1/2 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
1034 ** / ***	MR2 A11 J17	MR3 A11 J1	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic and one plastic fiber optic cables
1044**	MR2 A11 J22	OW1 A15 SLOT2	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1045 ** / ***	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
1078 ***	MG2 A33 J30/J29	MG3 A36 J18/J19	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables, SRI to Cross Point Switch
1083 ***	MR2 A11 J22	OW1 A15 XMT1	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares

2–1 LOCATING AND UNPACKING CABLES (continued)

5. Unpack and move interconnect cables which are cut to length to the applicable room. Refer to Table 2–5.

TABLE 2–5
EQUIPMENT ROOM AND MAGNET ROOM CABLES CUT TO LENGTH

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
536	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
537	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
235	MR1 A7 J13	PP1 J44	7/8 in. HELIAX® Coax or	STO	6000 VDC	<30	120	VW–1	1.5T Body Coil Run created from kit
			Times Microwave LMR–900FR	CATVR	5000 VDC		85	FT–4	
236	MR1 A7 J12	PP1 J4	1/2 in. HELIAX® Coax or	STO	4000 VDC	<30	120	VW–1	1.5T Head Coil Run created from kit
			Times Microwave LMR–600FR	CATVR	4000 VDC		85	FT–4	
257	PP1 J44	MG3 A3 A1 J1	7/8 in. HELIAX® Coax or	STO	6000 VDC	<30	120	VW–1	1.5T Body Coil Transmit
			Times Microwave LMR–900FR	CATVR	5000 VDC		85	FT–4	
258	PP1 J4	MG3 A17 J1	1/2 in. HELIAX® Coax or	STO	4000 VDC	<30	120	VW–1	1.5T Head Coil Transmit
			Times Microwave LMR–600FR	CATVR	4000 VDC		85	FT–4	
745	MG3 A7 J13	PP1 J44	50 ft 1/2" Heliax (HN)	STO	4000 VDC	950	120	VW–1	1.0T Body RF Transmit
746	PP1 J44	MG3 A3 J1	55 ft 1/2" Heliax (HN)	STO	4000 VDC	950	120	VW–1	1.0T Body Coil Transmit
747	PP1 J4	MG3 A17 J1	50 ft 3/8" Heliax (N)	STO	2500 VDC	950	85	VW–1	1.0T Head Coil Transmit
748	MG3 A7 J12	PP1 J4	50 ft 3/8" Heliax (N)	STO	2500 VDC	950	85	VW–1	1.0T Head RF Transmit
762	MR3 CABI/F J3	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT–4	X Gradient output
762	MG3 A32 1/2	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT–4	X Gradient output
763	MR3 CABI/F J4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT–4	Y Gradient output
763	MG3 A32 3/4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT–4	Y Gradient output
764	MR3 CABI/F J5	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT–4	Z Gradient output
764	MG3 A32 5/6	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT–4	Z Gradient output

2–1 LOCATING AND UNPACKING CABLES (continued)

6. Verify magnet monitoring cables for GE Magnet are delivered (or if previously installed, remain in place). Refer to Table 2–6.

TABLE 2–6
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15–#22)	CI3	300	<30	80	FT–4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9–#22)	CI3	300	<30	80	FT–4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25–#22)	CI3	300	<30	80	FT–4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15–#22)	CI3	300	<30	80	FT–4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8–#22) & Data Cable (4–#22)	CI3	300	<30	80	FT–4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9–#22)	CI3	300	<30	80	FT–4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25–#22)	CI3	300	<30	80	FT–4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT–4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8–#24)	CI3	300	<30	80	FT–4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8–#22)	CM	300	<30	80	FT–4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6–#24)	CM	300	<30	80	FT–4	Compressor to Compressor I/F
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9–#22)	CI3	300	<30	80	FT–4	Monitor to Modem
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

2–1 LOCATING AND UNPACKING CABLES (continued)

7. Unpack and move equipment room system cables into routing location (cabinets, Operator Console, Operator Workstation, PDU). Refer to Table 2–8.

TABLE 2–7
EQUIPMENT ROOM RUNS

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
229	MR2 A11 J1	MR1 A7 J3	20 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	Exciter RF output
231	MR2 A11 J2	PP1 J8	40 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	Body Coil receive
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub–D	CI3	300	<30	80	FT–4	Optional O2 Monitor cable
486*	PP1–A14–J85	MR1–A7–J22	30 ft. 9 Sub–D	CI3	300	<30	80	FT–4	Head/MC1 Receive Select Switch
487*	PP1–A15–J78	MR1–A7–J20	40 ft.15 Sub–D	CI3	300	<30	80	FT–4	Multi–coil Switching
488	PP1–J80	MR2–A11–J8	40 ft. BNC	CMX	300	<30	80	VW–1	MC2 Receive
489	PP1–J81	MR2–A11–J9	40 ft. BNC	CMX	300	<30	80	VW–1	MC3 Receive
490	PP1–J82	MR2–A11–J10	40 ft. BNC	CMX	300	<30	80	VW–1	MC4 Receive
499/ 233	MR2 A11 J5	PP1 J7	40 ft RG223/U Coax	CMX	300	<30	80	VW–1	Head and Surface Coil receive
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	RF Door switch
702	MR2 A11 J18	MR1 A7 J5	20 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Unblank/EFB
703	MR2 A11 J23	PD1 A14 A1 J2/ MR1 PD1 J2	40 ft 15 pin Sub–D	CI3	300	<30	80	FT–4	PDU Status
706*	MR2 A11 J14	PD1 A14 A1 J3/ MR1 PD1 J3	40 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Remote PDU
726*	PP1–A16–J77	MR1–A7–J18	30 ft. 15 Sub–D	CI3	300	<30	80	FT–4	Multi–coil Bias
768	MR1 A7 J31	PP1 J20	30 ft 37 pin Sub–D	CI3	300	<30	80	FT–4	Patient Alignment Light power
769	MR1 A7 J34	PP1 J15	30 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	Bore Light power
770	MR1 A7 J41	PP1 J11	30 ft 9 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Long Drive Motor power
771	MR1 A7 J36	PP1 J47	30 ft 25 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Dock Power
772	MR1 A7 J40	PP1 J14	30 ft 37 pin Sub–D	CI3	300	<30	80	FT–4	SRI Power
773	MR1 A7 J35	PP1 A17 J1	30 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	PAC & Fiber Optic I/F Power
774	MR2 A11 J25	MR1 A7 J32	20 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	Hydraulic Manifold Interface
775	MR1 A7 J42	PP1 A11 J92	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	100V Dynamic Disable
776	MR1 A7 J44	PP1 A11 J93	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	100V Dynamic Disable
777	MR1 A7 J43	PP1 A11 J94	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	100V Dynamic Disable
783	MR1 A7 J37	PP1 F5/F6	30 ft 2–#8 AWG	SO	600	120	80	FT–4	Blower Cabinet Power
785	MR1 A7 J18	PP1 F7/F8	30 ft 2–#8 AWG	SO	600	120	80	FT–4	Blower Cabinet Power
788	OW1 A1 A4 J7	PP1 J12	40 ft Data (25–#22)	CI3	300	<30	80	FT–4	Intercom
789	OW1 A1 A4 J18	MR2 A11 J26	51 ft Data (9–#22)	CI3	300	<30	80	FT–4	E–stop
791	OW1 A16 PORT1	MR2 A11 J30	51 ft Data (25–#22)	CI3	300	<30	80	FT–4	TNF monitoring (Octane)

2–1 LOCATING AND UNPACKING CABLES (continued)

TABLE 2–7 (CONT'D)
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
797	OW1 A2 A1 PC	OW1 A3 LCD	2 line; 3.5 mm jacks	CI3	300	<30	80	FT–4	LCD Display
798	OW1 A2 A3 HOST	OW1 A6	10.5 ft. Data (3–#24)	CI3	300	<30	80	FT–4	Color Monitor cable
805	OW1 A1 A4 J6	OW1 A7	3.3 ft. Data	CI3	300	<30	80	FT–4	Keyboard cable
807	OW1 A2 A3 HOST	OW1 A5	9.8 ft. Data	CI3	300	<30	80	FT–4	DASM SCSI cable
810	OW1 A1 A2 J8	OW1 A15 KBD,MS	9 ft. 25 pin Sub–D	CI3	300	<30	80	UL–1581	Keyboard/Mouse cable (Octane)
811	OW1 A2 A4	OW1 A15	12 ft. Ethernet	CI3	300	<30	80	FT–4	Communication
812	OW1 A15	OW1 A13	12 ft. 25/9 pn Sub–D	CM	300	<30	80	UL–1581	Modem
813	OW1 A2 A1 COM2	OW1 A415 PORT7	12 ft. 9 pin Sub–D	CI3	600	<30	105	VW–1	Communication
814	OW1 A2 A1 KBD,MS	OW1 A1 A4 J9	9 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Keyboard/Mouse cable (PC)
815	OW1 A16	OW1 A15 SCSI	6 ft. 68/50 pin	CI3	300	<30	80	FT–4	SCSI Cable (Octane)
817	OW1 A2 A5 J3	OW1 A1 A2 J17	8 ft. 9 pin Sub –D	CI3	300	<30	80	FT–4	Workspace I/F Module
818	OW1 A16 J6,7,8	MR2 A11 J31	70 ft. 25/3–25 pin	CI3	300	<30	80	FT–4	TNF monitoring
819	OW1 A2 A7 SCSI	OW1 A15	10 ft. 68/50 pin	CI3	300	<30	80	FT–4	SCSI Cable
821	OW1 A15 MON-ITOR	OW1 A6 R,B,G	10.5 ft. 3 BNC/SubD	CI3	300	<30	80	FT–4	Color Monitor cable
971	MR2 A11 J23	MR3 PD1 J2	40 ft 15 pin Sub–D	CI3	300	<30	80	FT–4	PDU Status (ACGD)
972*	MR2 A11 J14	MR3 PD1 J3	40 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Remote PDU (ACGD)
1025 **/***	MR2 A11 J40	PP1 J6	40 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
1026 **/***	MR2 A11 J41	PP1 J7	40 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
1027 **/***	MR2 A11 J42	PP1 J58	40 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
1028 **/***	MR2 A11 J43	PP1 J59	40 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
1029 **/***	MR2 A11 J62	PP1 A16 J77	60 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	Monitor to Compresssor I/F
1030 **/***	MR2 A11 J61	PP1 A16 J78	30 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	SRI Power
1031**	MR2 A11 J63	OW1 A15 SLOT1	65 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Communication
1032	MR2–A11–J31	OW1–A16–FJ100	65 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Serial Communication
1046	FJ100	OW1–A16–PORT 1,6,7,8	4.5 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Serial Communication
1053 ***	PP1 J85	MR2 A11 J54	34.5 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	UNTS Antenna

2–1 LOCATING AND UNPACKING CABLES (continued)

TABLE 2–7 (CONT'D)
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1056 ***	OW1 A17 PRT6	OW1–A12 ETHRNET	3.3 ft. Ethernet Cat 5	CI3	30	<30	80	FT–4	Laptop Access cable
1059 ***	OW1 A17 PRT2	OW1–A15 ETHRNET	5.9 ft. Ethernet Cat 5	CI3	30	<30	80	FT–4	Lan Hub cable
1060 ***	OW1 A19 J3	OW1 A21 J17	3.9 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	WIM Power cord
1061 ***	OW1 A15 KYBD/ MS	OW1 A21 J8	2.9 ft. 25 pin Sub–D 2 PS/2s	CI3	300	<30	90	CL–2	Keyboard/Mouse cable
1062 ***	OW1 A20	OW1 A21 J6	13.2 ft. 50 pin SCSI, 9 pin Sub–D	CI3	300	<30	80	VW–1	SCIM cable
1063 ***	OW1 A15 Video 1	OW1 A6 HOST LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW–1	Host LCD Display Video cable
1064 ***	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	9.9 ft. SCSI cable 50	CI3	300	<30	80	VW–1	SCSI Tower Data cable
1065 ***	OW1 A11 J2	OW1 A6 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	Host LCD Display Power cable
1066 ***	OW1 A15	OW1 A13	3.9 ft. 25/9 pn Sub–D	CM	300	<30	80	FT–4	Modem I/O cable
1067 ***	OW1 A15 SCSI 2	OW1 A5 DASM	5.9 ft. SCSI cable 50	CI3	30	<30	80	VW–1	DASM I/O cable
1068 ***	OW1 A15 Video2	OW1 A3 LCD	9.9 ft. Data (15–#28)	CI3	300	<30	80	VW–1	LCD Display Video cable
1069 ***	OW1 A11 J8	OW1 A3 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	LCD Display Power cable
1070 ***	OW1 A11 J11	OW1 A16 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	SCSI Tower Power cable
1071 ***	OW1 A11 J1	OW1 A15 Pwr	9.9 ft. Power (3–#18)	SVT	300	120	80	VW–1	Host Computer Power cable
1074 ***	OW1 A19 J2	OW1 A11	3.3 ft. Power (3–#14)	SVT	600	120	105	VW–1	Power Strip cord
1075 ***	OW1 A19 J4	OW1 A11	3.3 ft. Power (3–#14)	SVT	600	120	105	VW–1	Power Strip cord
1082 ***	OW1 A15 ETH SLOT 2	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT–4	EXCITE Ethernet C'cable
1084 ***	OW1 A21 J6 via Run 1062	MR2 A11 J26	60 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	E–stop
1085 ***	OW1 A21 J7	PP1 J12	65 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	Intercom

8. Unpack additional equipment room cables for Signa MR/*i* Options:
   a. Refer to the service/installation manual shipped with options.
   b. Refer to *Direction 2151387, Signa Horizon Spectroscopy Subsystem,* for 1.5T Spectroscopy (M1040JB) equipment room cables.

2–2 SORTING AND ROUTING CABLES

CAUTION

Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

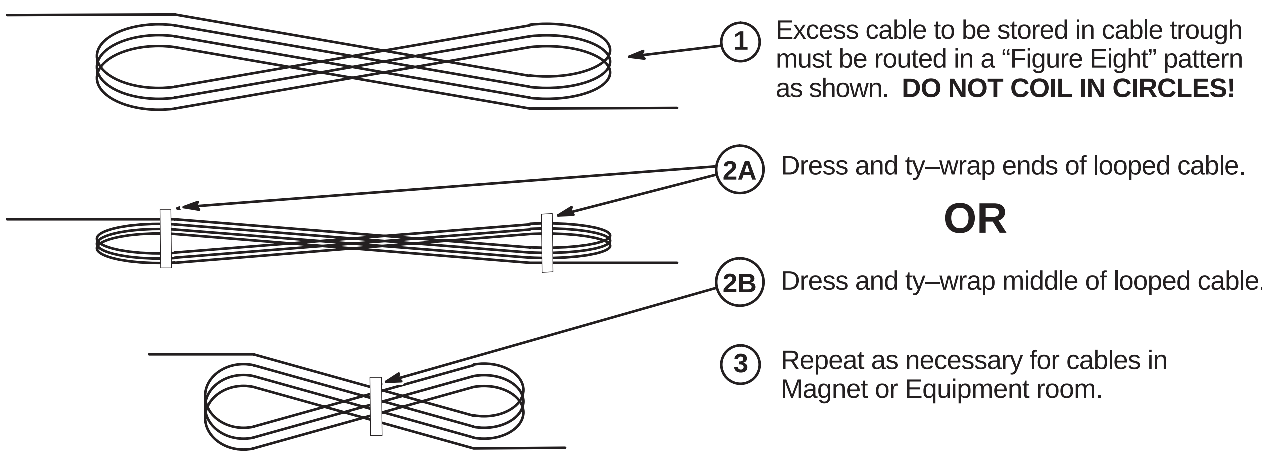
System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

   Note

   Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 2–1. Some cables are usually cut to length such as gradient cables, power cables, Heliax, and Fiber–optic cables.



PROPER STORAGE OF EXCESS CABLES
ILLUSTRATION 2–1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

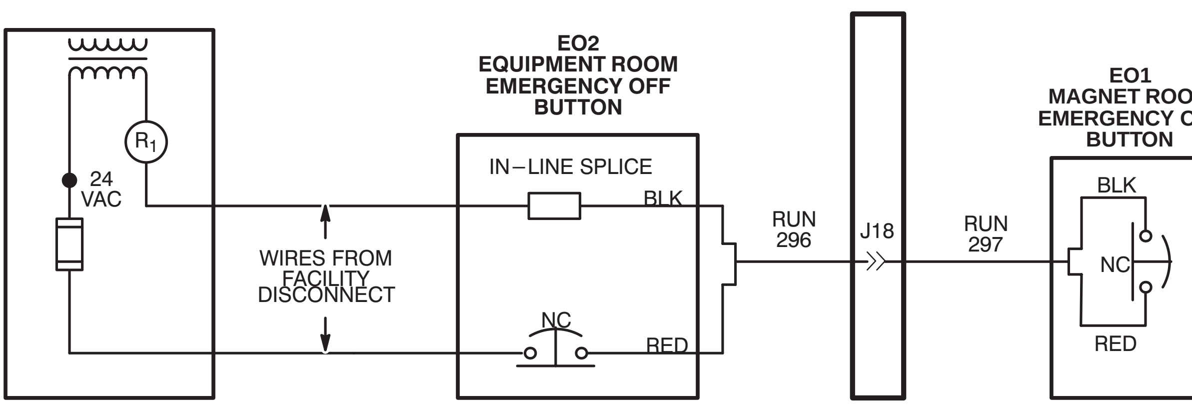
2–3 EMERGENCY OFF CONNECTIONS

Run 296 is routed to EO2 (Equipment Room Emergency Off) and connected to EO2 and wire from Facility Disconnect. Run 297 is routed from Penetration Panel to EO1 (Magnet Room Emergency Off Button). Refer to *Direction 2223170, Signa MR/i Pre–Installation*, Section 5–3–1, Protective Disconnect Device, for detailed wiring diagram.

1. Complete routing of Run 296 to EO2 (Equipment Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.

   Aside: In Illustration 2–3, black and red wires are used for connections in Runs 295 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. Runs 296 and 297 are actually nine wire cables.

2. At equipment room room “Emergency Off” (EO2) location, terminate red wire at end of Run 296 with local supplied terminal and connect to customer supplied Equipment Room Emergency Off Button (EO2). See Illustration 2–3.
3. Terminate customer supplied wire (from fuse in Protective Disconnect Device) with local supplied applicable terminal and connect to customer supplied Emergency Off Button (EO2). See Illustration 2–3.
4. Terminate customer supplied wire from R1 in Protective Disconnect Device with local supplied push–on terminal.
5. Terminate black wire at end of Run 296 with local supplied “push–on” terminal.
6. Connect black wire from end of Run 296 to customer supplied wire from protective disconnect device with local supplied in–line splice. See Illustration 2–3.
7. Complete routing of Run 297 to EO1 (Magnet Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.
8. Terminate red and black wires at end of Run 297 with local supplied terminals and connect to customer supplied Magnet Room Emergency Off Button (EO1). See Illustration 2–3.



EMERGENCY OFF WIRING
ILLUSTRATION 2–2

2–4 RUN 701 (RF DOOR SWITCH) CONNECTIONS

1. Route Run #701 (from System Cabinet) to RF Door Switch. This 100 ft cable has a 9–pin subminature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminature D plug.
2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.

2–5 CABLE INTERCONNECTION DOCUMENTATION

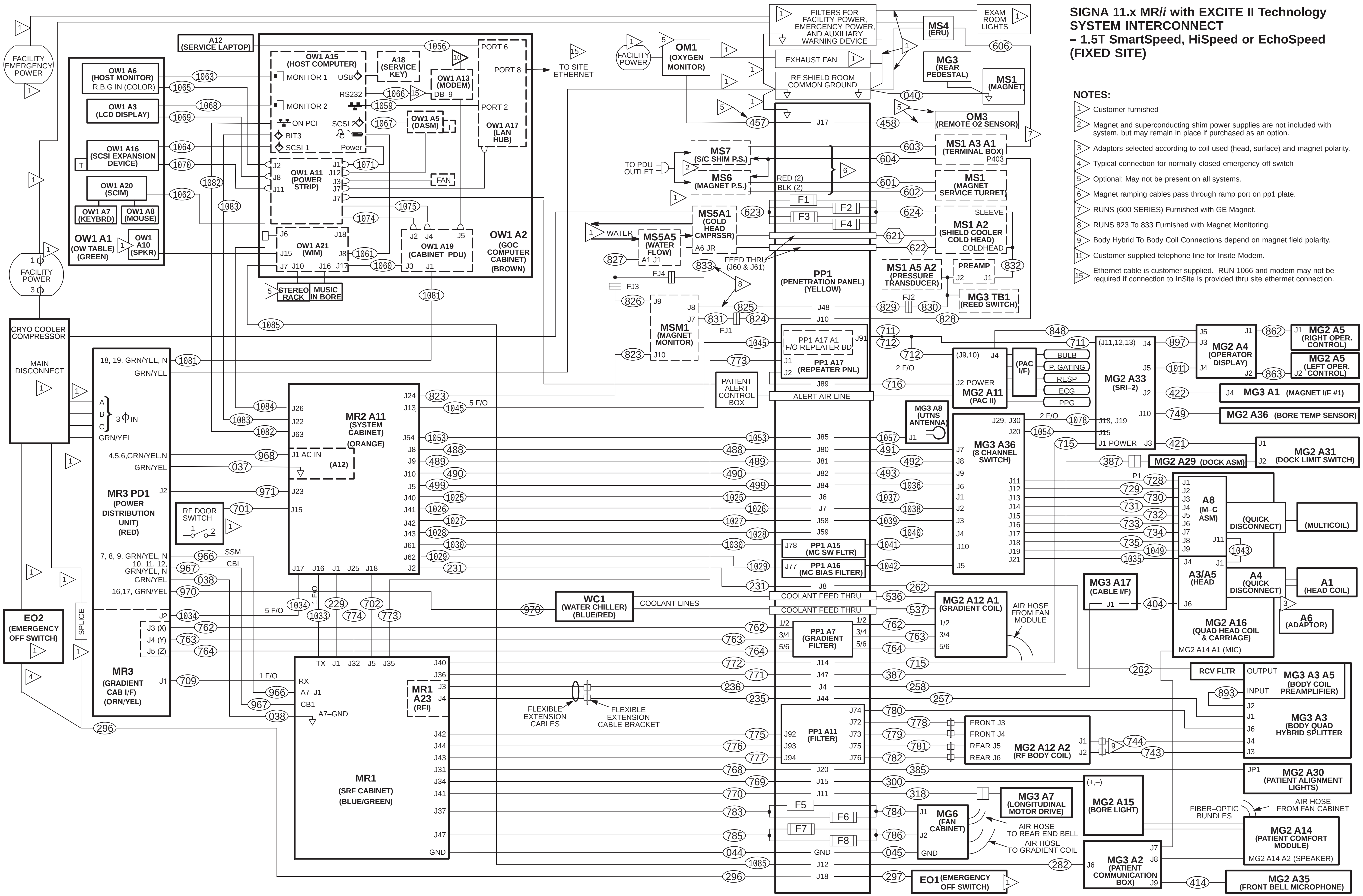
The cable maps illustrate the Signa MR/*i* (or CV/*i*) System interconnect for all supplied cables and interface with customer supplied wiring.

1. Interconnects for additional cables supplied with other options (Spectroscopy, etc.) are illustrated in the installation manual shipped with the option.
   - Connections for 1.5T Spectroscopy are in *Direction 2151389, Signa Horizon Spectroscopy Subsystem*.
   - Instructions for connections for optional laser cameras are supplied with option.
   - Connection procedures for other options are covered in the installation manual shipped with the option.
2. Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.
3. Interconnect details for magnet subsystem is covered in the applicable magnet subsystem manual.

Fold–out MR/*i* (or System Interconnect Cable Maps are provided at the end of this section. They are:

Select the map that applies to the system being installed. Tape map in a convenient location to check off routed cable. Upon completion of connecting cables, return fold–out map to the binder for future reference.

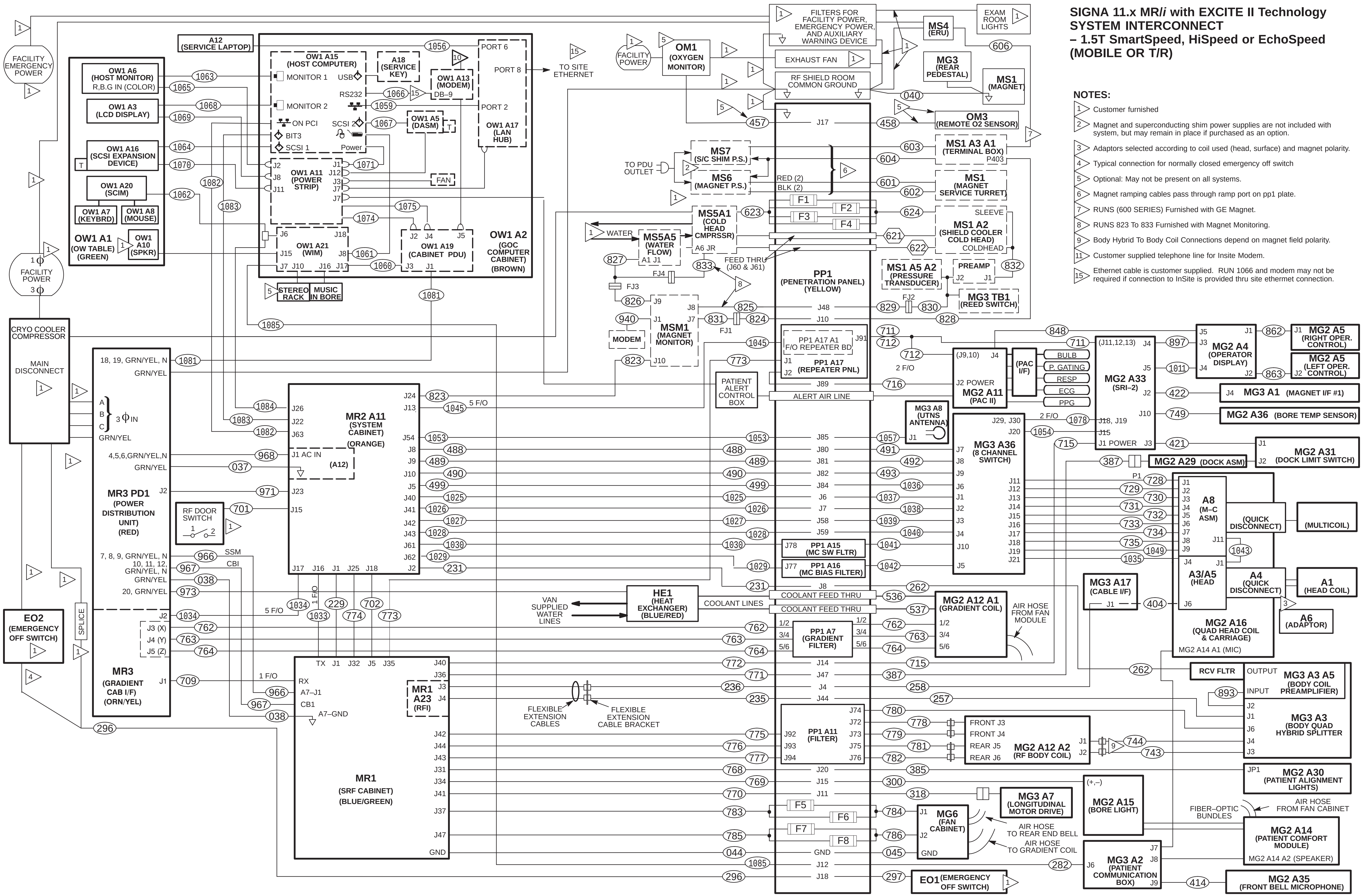
SIGNA 11.x MR/*i* with EXCITE II Technology SYSTEM INTERCONNECT – 1.5T SmartSpeed, HiSpeed or EchoSpeed (FIXED SITE)



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

SIGNA 11.x MR/*i* with EXCITE II Technology SYSTEM INTERCONNECT – 1.5T SmartSpeed, HiSpeed or EchoSpeed (MOBILE OR T/R)



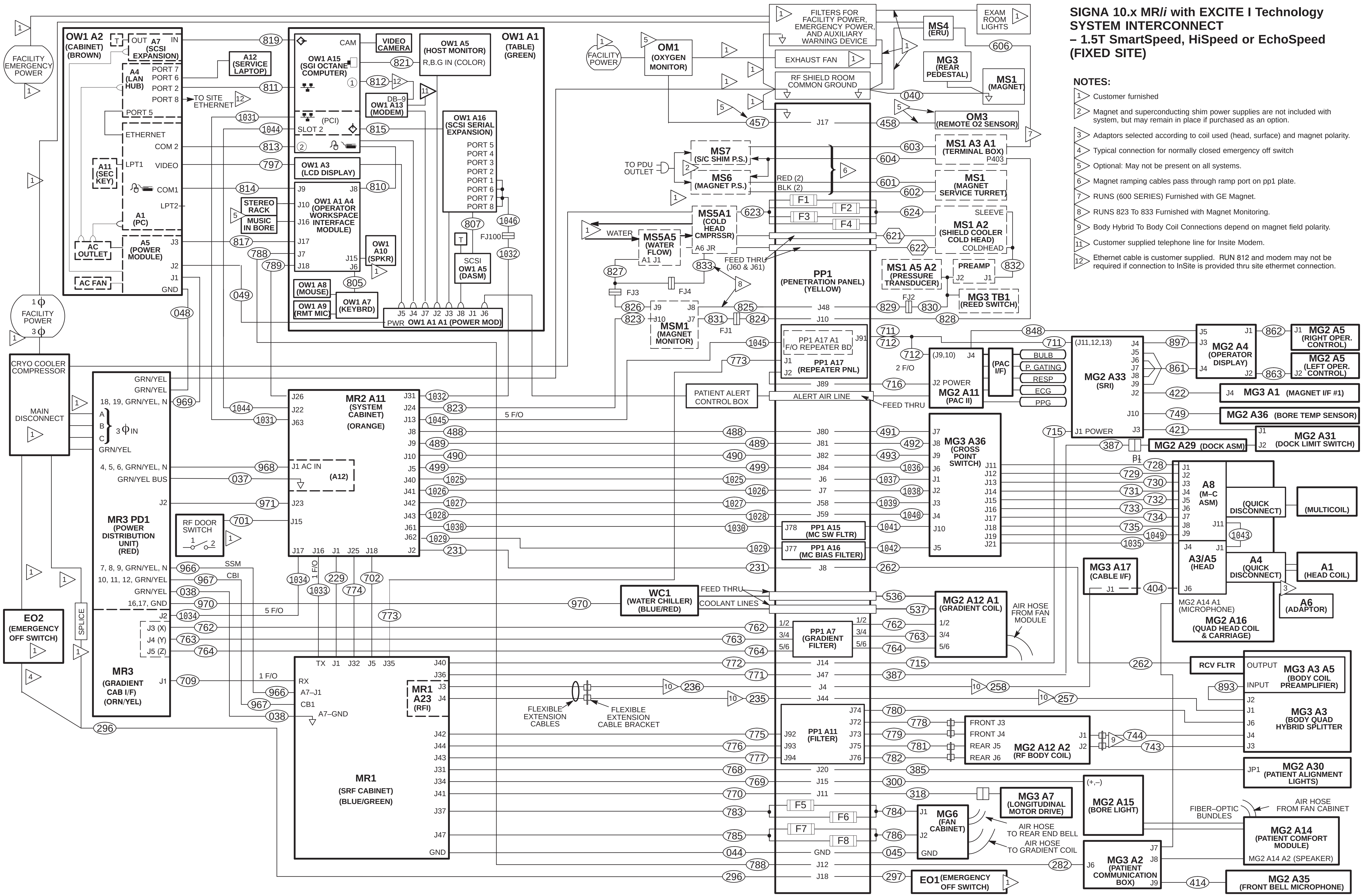
NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
15. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.

SIGNA 10.x MR/*i* with EXCITE I Technology SYSTEM INTERCONNECT – 1.5T SmartSpeed, HiSpeed or EchoSpeed (FIXED SITE)

NOTES:

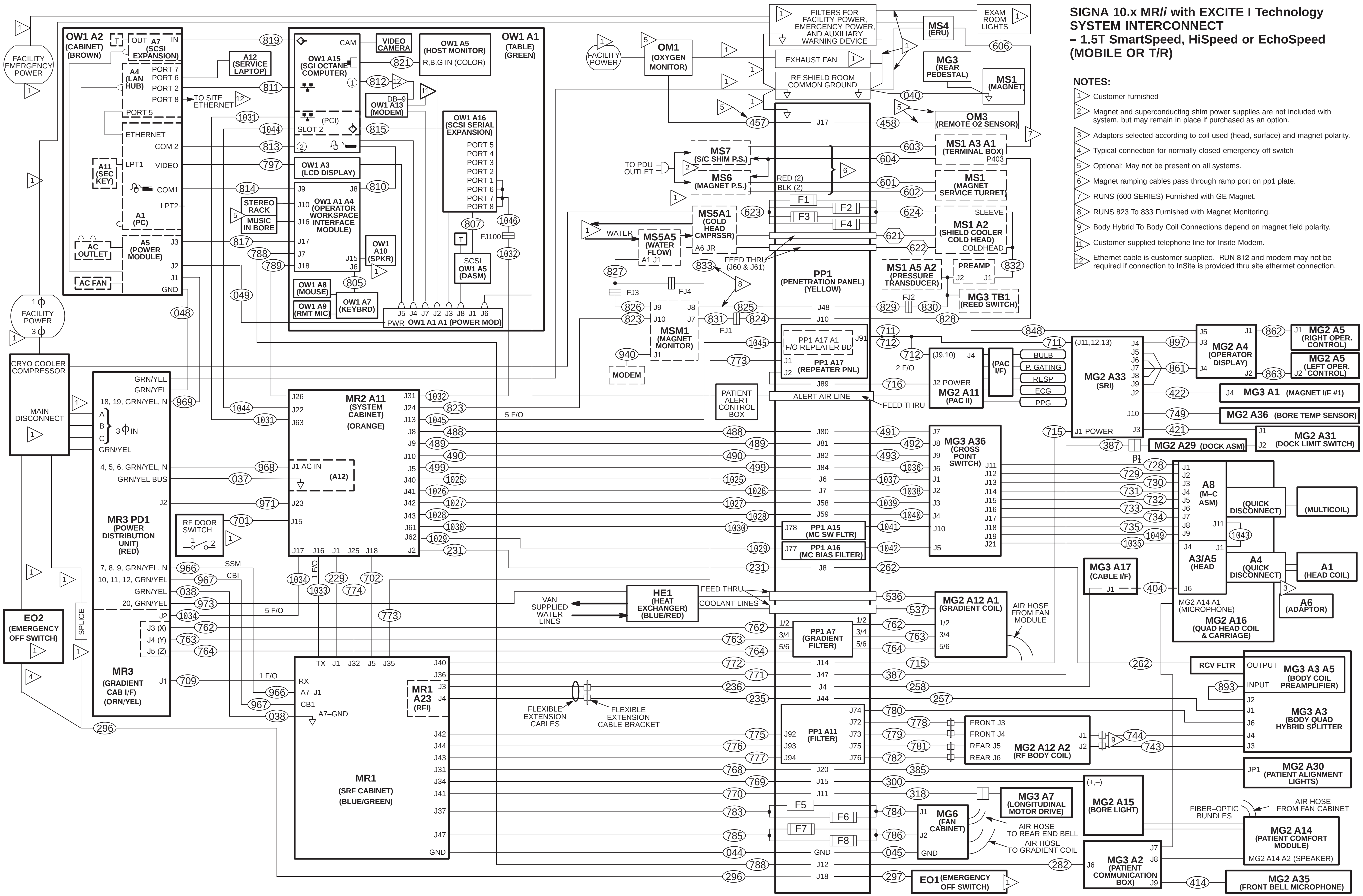
1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



SIGNA 10.x MR/*i* with EXCITE I Technology SYSTEM INTERCONNECT – 1.5T SmartSpeed, HiSpeed or EchoSpeed (MOBILE OR T/R)

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



SIGNA 9.x MR/*i* SYSTEM INTERCONNECT – 1.5T SmartSpeed, HiSpeed or EchoSpeed

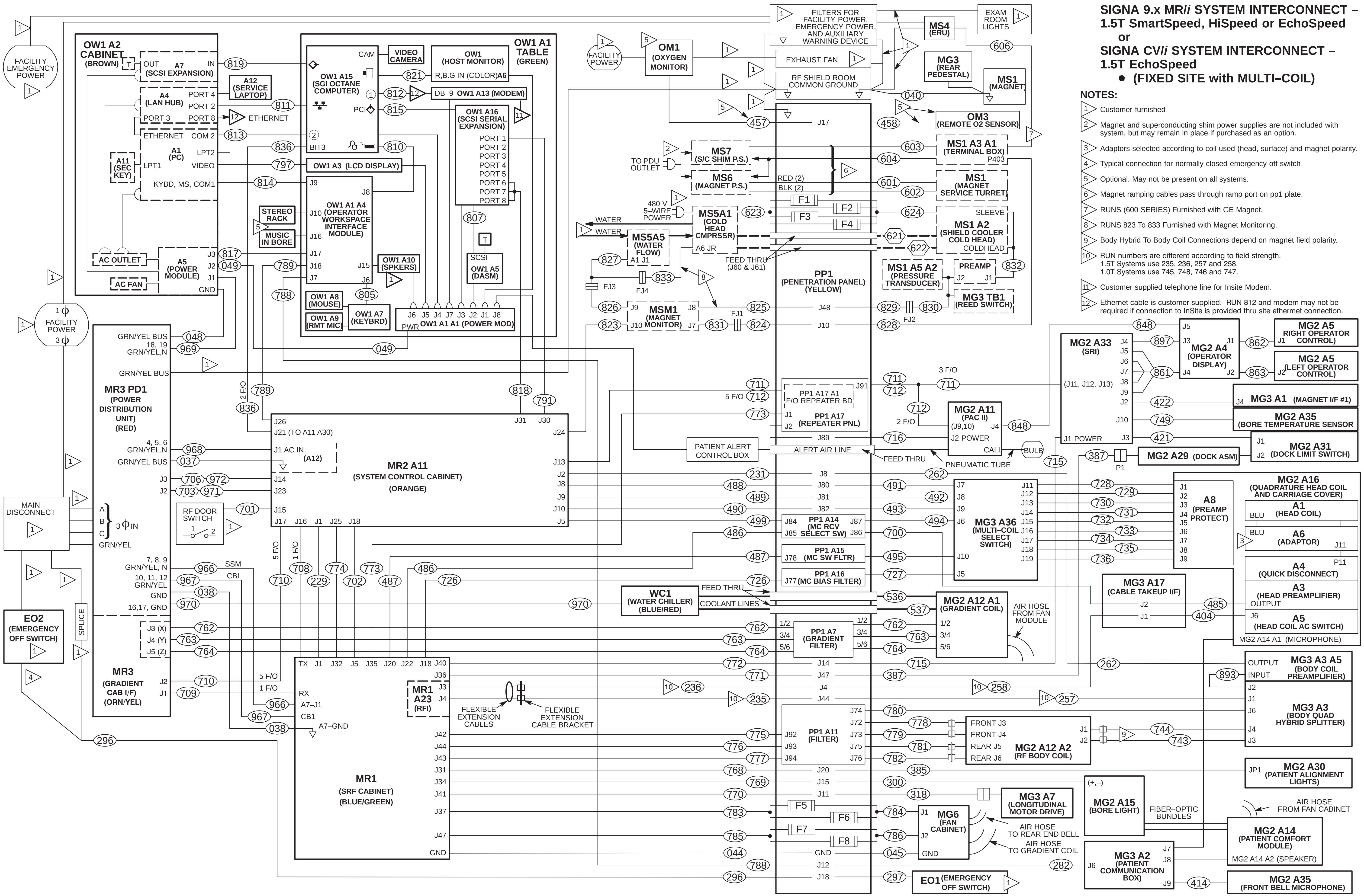
or

SIGNA CV/*i* SYSTEM INTERCONNECT – 1.5T EchoSpeed

• (FIXED SITE with MULTI–COIL)

NOTES:

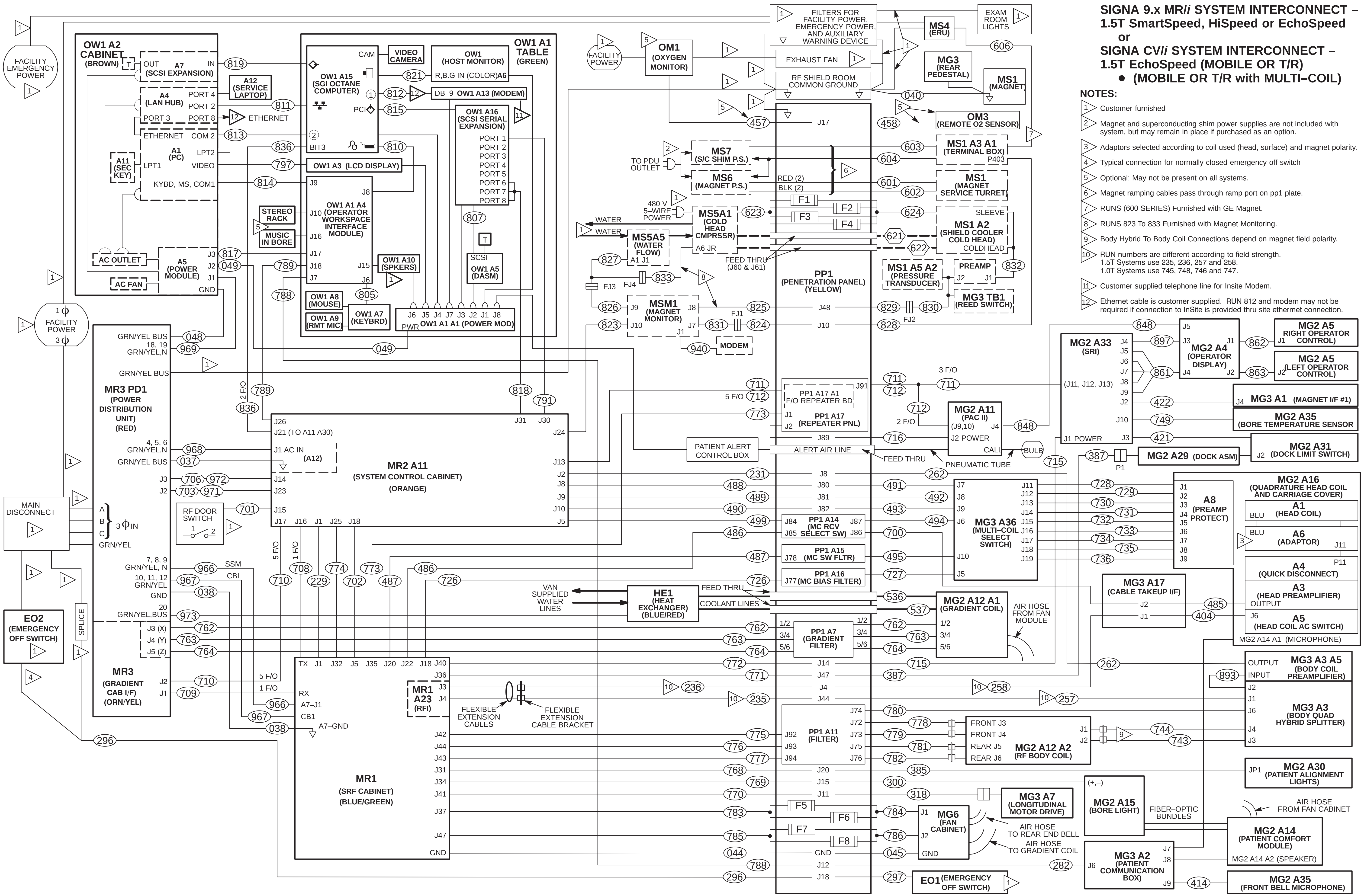
1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



SIGNA 9.x MR/*i* SYSTEM INTERCONNECT – 1.5T SmartSpeed, HiSpeed or EchoSpeed
or
SIGNA CV/*i* SYSTEM INTERCONNECT – 1.5T EchoSpeed (MOBILE OR T/R)
• (MOBILE OR T/R with MULTI–COIL)

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

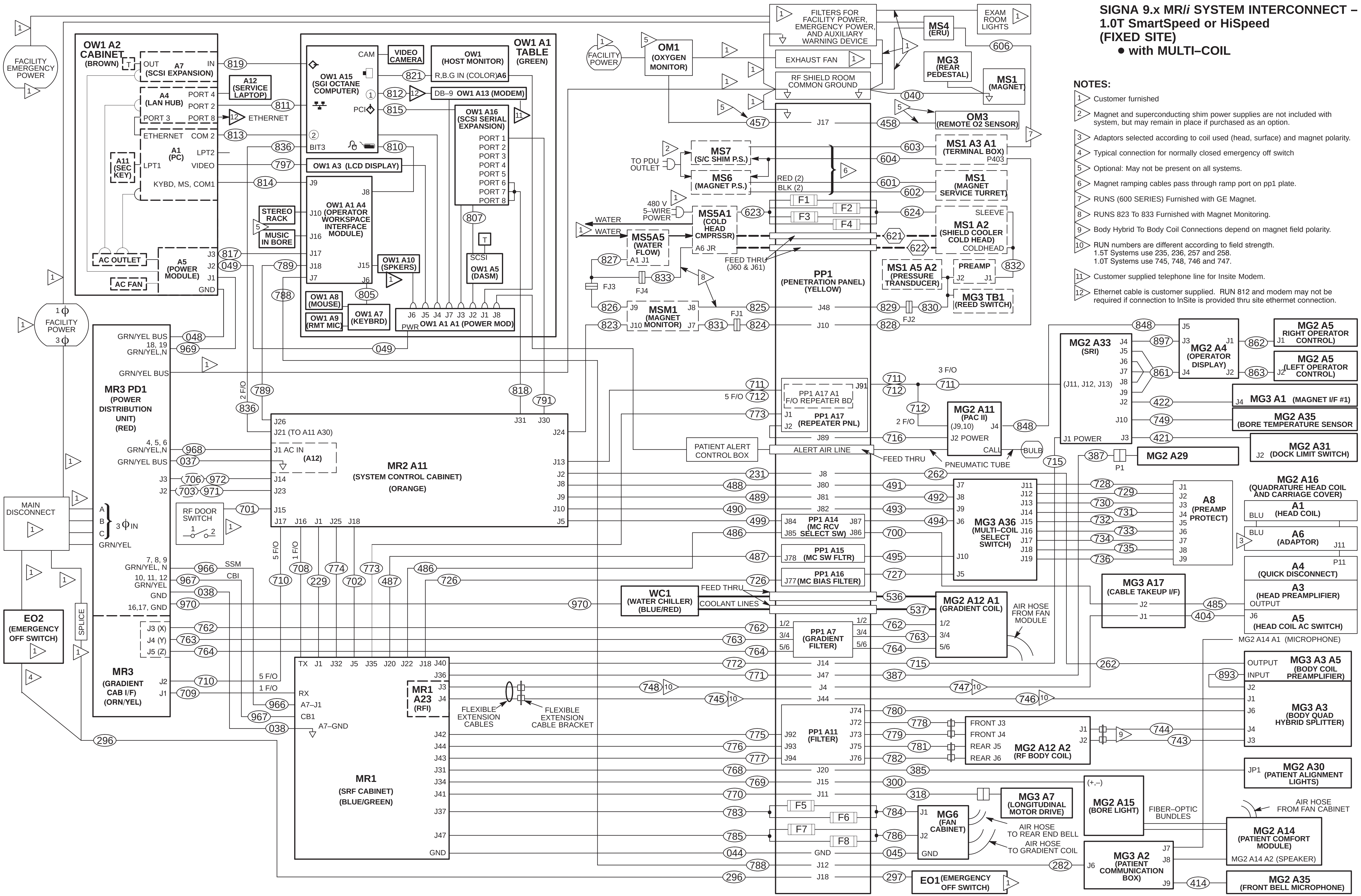


SIGNA 9.x MR/*i* SYSTEM INTERCONNECT – 1.0T SmartSpeed or HiSpeed (FIXED SITE)

- with MULTI–COIL

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

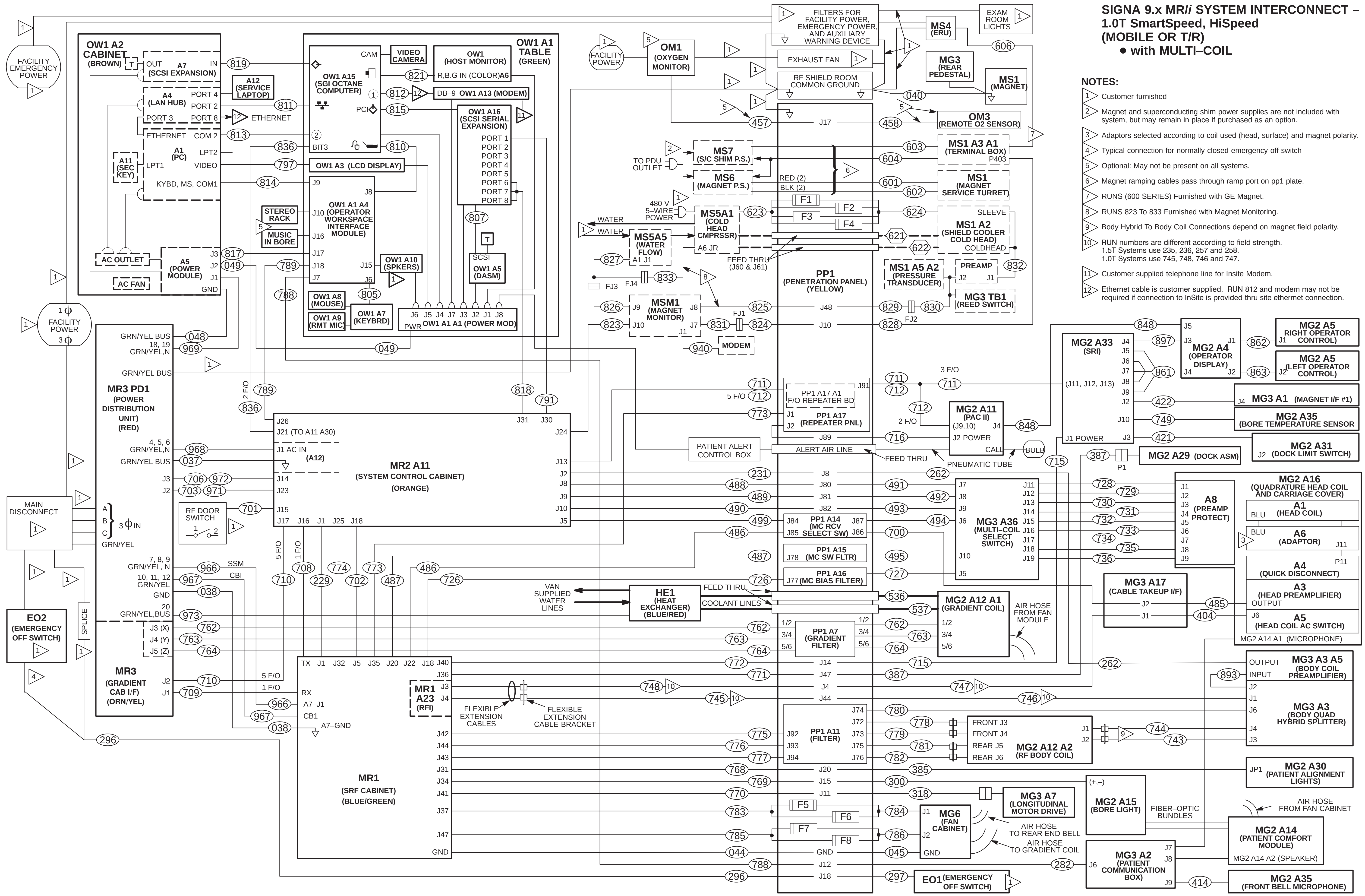


SIGNA 9.x MR/*i* SYSTEM INTERCONNECT – 1.0T SmartSpeed, HiSpeed (MOBILE OR T/R)

● with MULTI–COIL

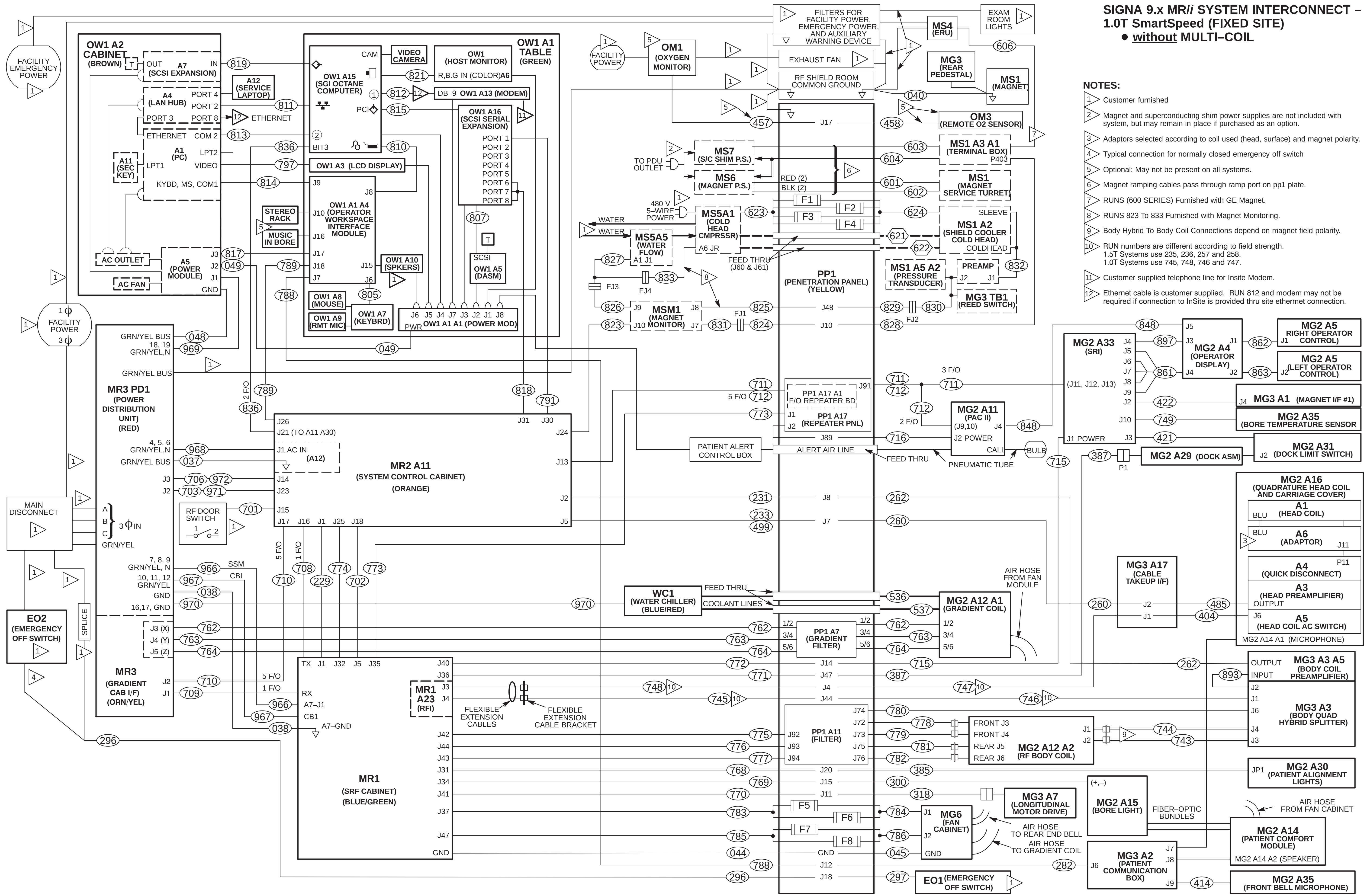
NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



SIGNA 9.x MR/*i* SYSTEM INTERCONNECT – 1.0T SmartSpeed (FIXED SITE)

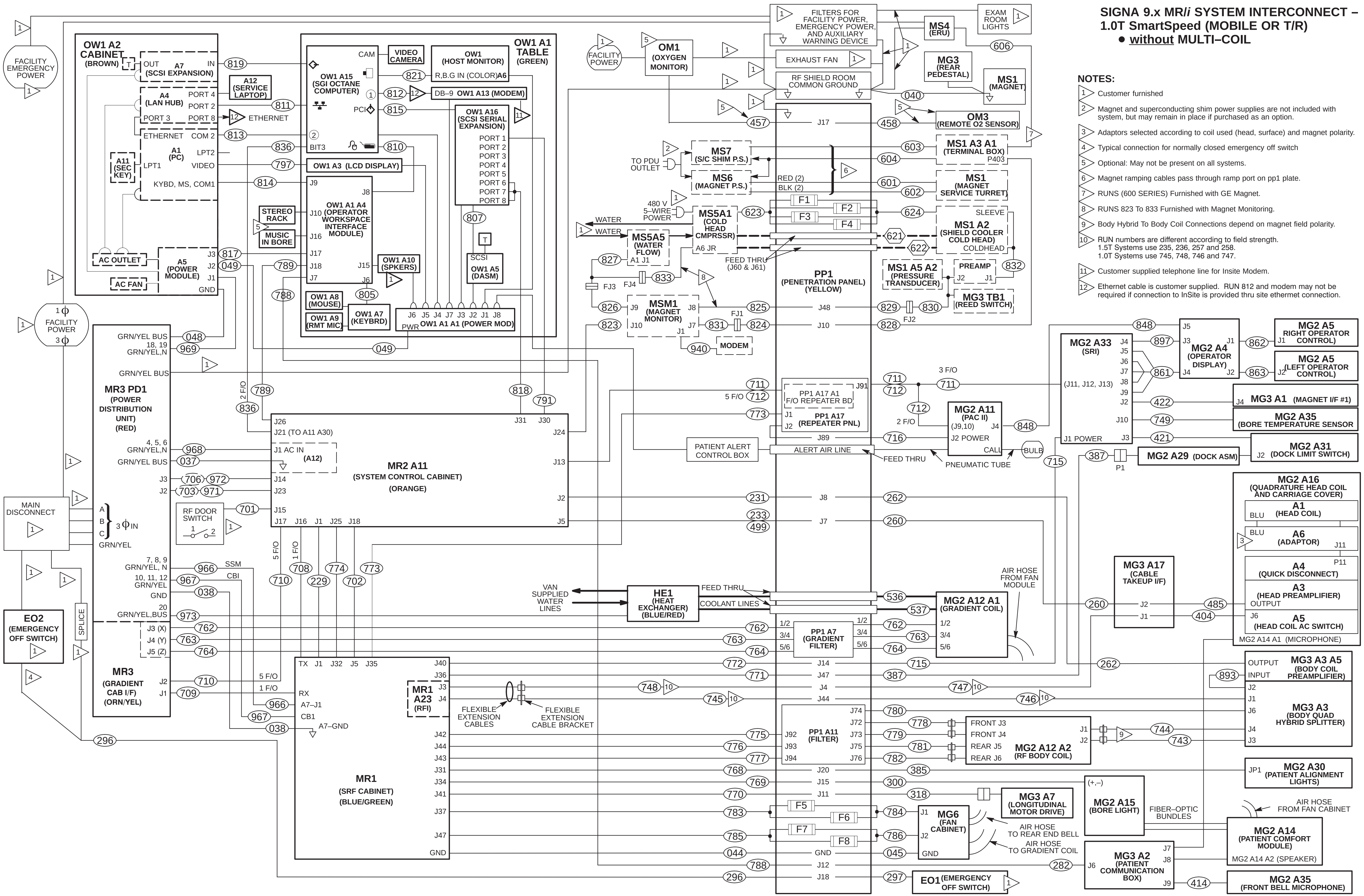
● **without** MULTI–COIL



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

SIGNA 9.x MR/*i* SYSTEM INTERCONNECT – 1.0T SmartSpeed (MOBILE OR T/R)
● __without__ MULTI–COIL



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

SECTION 3 – SYSTEM START UP

3–1 RETURNING SHIPPING MATERIAL

The shipping material must be returned **as soon as possible.** Prompt return is required so that production shipments can continue with these labor saving dollies and carts. These can be included with the "Return pile", Recycling Operation will route them to appropriate location.

3–2 PHANTOMS AND SERVICE KITS

1. Locate Phantom Kit (Loader with Head TLT sphere, 4.38 inch Surface Coil) and SPT Phantom Cart Kit.
2. Unpack the phantoms carefully to reduce risk of damage, breakage, or chemical spill.
3. Locate the multilingual (provided in eight languages) phantom attention labels shipped with each of the phantoms.

CAUTION

Three types of multilingual phantom attention labels are provided and each type must be applied to the correct phantom. The directions at the top of each label sheet describe which phantoms that can receive that label.

4. Read the directions provided at the top of each of the multilingual attention label sheets and apply the appropriate language attention label for the site to each of the phantoms that should receive that type of label.
   a. Each phantom will receive the correct type of attention label written in the appropriate language for that site.
   b. Note that the LVshim phantom assembly will receive 3 labels, one for each phantom component.
5. Place service phantoms into SPT Phantom Cart or service storage cabinet to be organized and ready to be used during applicable checks and calibrations. The DQA Phantom is to be stored in a customer designated location.
6. Locate TPS RF I/F Kit (if applicable) and Signa Spares Kit. Place kits into service storage cabinet for ready use.

3–3 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD–ROM, Set–Up & Calibration, and perform the following:
   - PDU Voltage Checks Compact
   - Startup Checks Compact
   - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the ACGD Gradient power modules, switch all cabinet circuit breakers to ON. The ACGD power module circuit breakers should remain OFF until room humidity stabilizes under 60%.

3–4 CALIBRATION AND SYSTEM CHECKS

Refer to Installation Flow Chart to perform:

- Software load and Configuration file edit.
- Functional checks
- Set–up and calibration
- System performance checks

3–5 FINAL COORDINATION

1. Confirm that the blower box has been anchored properly and installed outside the minimum service area.
2. Complete assembly of PPG cable and probe head.
3. Verify that Operator Alert squeeze bulb will operate alarm.
4. Replace cabinet covers that have not been previously replaced.
5. Make sure all Magnet Enclosure and Rear Pedestal covers that have not been installed.
6. Dock Patient Transport in position at front of bore.
7. Record and enter applicable data into applicable site configuration files and records.
8. Complete Product Locator information for all serialized components, new or updated, that were installed via either:
   - (**U.S. Only)** *FE Site Verification Web Site*
     or
   - Process and return product locater installation cards for all serialized components to:
     Product Locator File, P.O. Box 414, W–523, Milwaukee, WI 53201–0414

Refer to Section 1 for details on submitting Product Locator information.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

9. Process the **Supplier Performance Report** located in Common Forms. A reference copy of the form is provided at the end of this section.
10. Store the delivered site's set of service tools and spares kit in service cabinet at site.
11. Locate surface coil ("quick disconnect") adapters and any optional surface coils and positioning accessories and place on Patient Transport. Be sure only the correct polarity (normal or reverse **but not both**) head coil adapter is left at site.
12. Locate Box "To be opened by Applications Specialist" and set aside for later visit by Applications Specialist.
13. Leave the site's set of Manuals and CD–ROM at site. Organize and set up reference cabinet for the Manuals.
14. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
15. Verify that coordination of application support for instruction of site users on Software Release 5.5 or 8.0 has been made before returning system to the users. Turn site over to applications who will instruct users.

GE Medical Systems Field Service

SUPPLIER PERFORMANCE REPORT

Send completed forms to the 4 addresses at the end of this form.

Supplier Name: ______________________________ Date: ______________

Work Performed / Evaluated: ______________________________

Commodity / Item Purchased: ______________________________

GE Customer Name: ______________________________

FDO Number: ______________________________ Test Equipment Bar–code No; _

GEMS Evaluator Name: ______________________________

Please rate suppliers as 1 to 5 on each of the 15 questions below:
1 = Unacceptable 2 = Poor 3 = Average 4 = Excellent 5 = Outstanding

There are 5 groups with three related questions in each group. Groups can be rated either as a group or 3 separate questions.

Any rating below 3 triggers a discrepancy review with the supplier.

Corrective actions will be faster and more effective when you provide examples and/or detail information.

	Supplier schedules well and:					
1	starts on time?	1	2	3	4	5
2	finishes on time?	1	2	3	4	5
3	avoids disruptions?	1	2	3	4	5
	Supplier is flexible and:					
4	responds to GEMS/customer requests?	1	2	3	4	5
5	cooperates with changes in schedule?	1	2	3	4	5
6	deals with unexpected problems?	1	2	3	4	5
	Supplier tools and equipment are:					
7	appropriate for contracted work?	1	2	3	4	5
8	adequate for contracted work?	1	2	3	4	5
9	calibrated and in good working order?	1	2	3	4	5
	Supplier meets GEMS quality standards with employees who:					
10	are trained, skilled and technically competent?	1	2	3	4	5
11	always keep job sites clean and free of debris?	1	2	3	4	5
12	ensure both function and appearance are as expected?	1	2	3	4	5
	Supplier employees enhance GEMS image by their:					
13	appearance?	1	2	3	4	5
14	behavior?	1	2	3	4	5
15	attitude?	1	2	3	4	5

Examples / Details: ______________________________

____

____

Any rating below 3 triggers a discrepancy review with the supplier.

SEND COMPLETED FORMS TO:

1. Senior Operations Specialist for your LCT
2. The contractor who performed the work
3. ISS (already in default distribution)
4. Sourcing Supplier Quality (already in default distribution)

SYSTEM INSTALLATION INSTRUCTIONS

1–1 Tab 1, System, contains installation instructions for the following:

Use only the applicable System Installation Instruction Directions

- *DIRECTION 2389819* – 1.5T Times Microwave Installation Instructions
- *DIRECTION 2147593* – 1.5T Heliax Cable Installation Instructions
- *DIRECTION 2223755* – 1.0T Heliax Cable Installation Instructions
- *DIRECTION 2286976* – Signa ACGD Gradient Cable Installation Instructions
- *DIRECTION 2380612* – Signa ACGD Lite Gradient Cable Installation Instructions
- *DIRECTION 2287106* – Signa ACGD Fiber Optic Cable Installation Instructions
- *DIRECTION 2338872* – Signa EXCITE I Fiber Optic Cable Installation Instructions
- *DIRECTION 2398342* – Signa EXCITE II Fiber Optic Cable Installation Instructions
- *DIRECTION 2287109* – Signa ACGD System Power Cable Installation Instructions

TABLE OF CONTENTS

SECTION **PAGE**

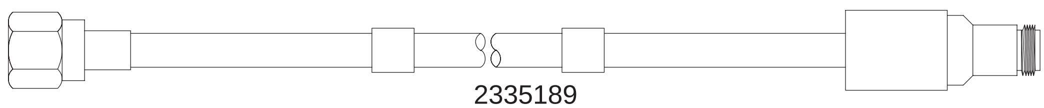
1–1 TIMES MICROWAVE RF CABLES INFORMATION

Four cables and other associated parts to be installed are provided for this procedure.

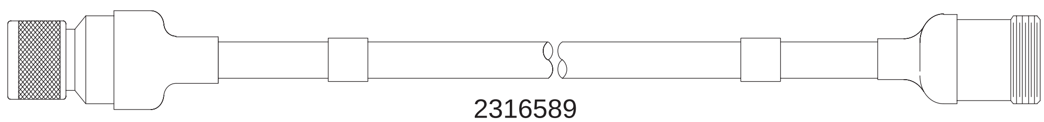
CAUTION

Do not subject LMR–600FR cable (Head) to a bend radius less than 1.5 inches (138mm). Do not subject LMR–900FR cable (Body) to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

The two short cables, 2335189 (Head) and 2316589 (Body) are connected to the SRFD2 cabinet.

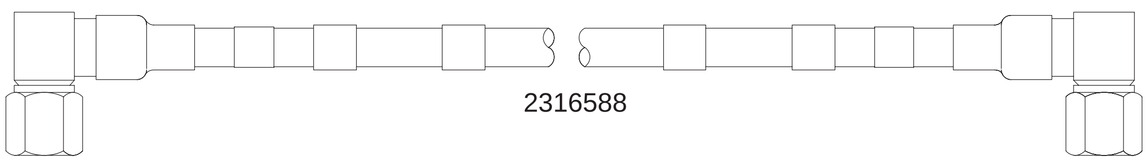


2335189

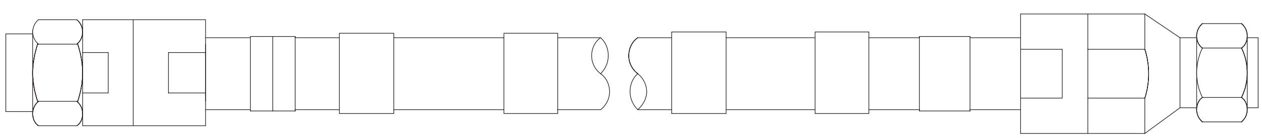


2316589

The two long cables, 2316588 (Head, Runs 236 & 258) and 2316584 (Body, Runs 235 & 257), will be cut to length and terminated for routing in the equipment and scan rooms.



2316588



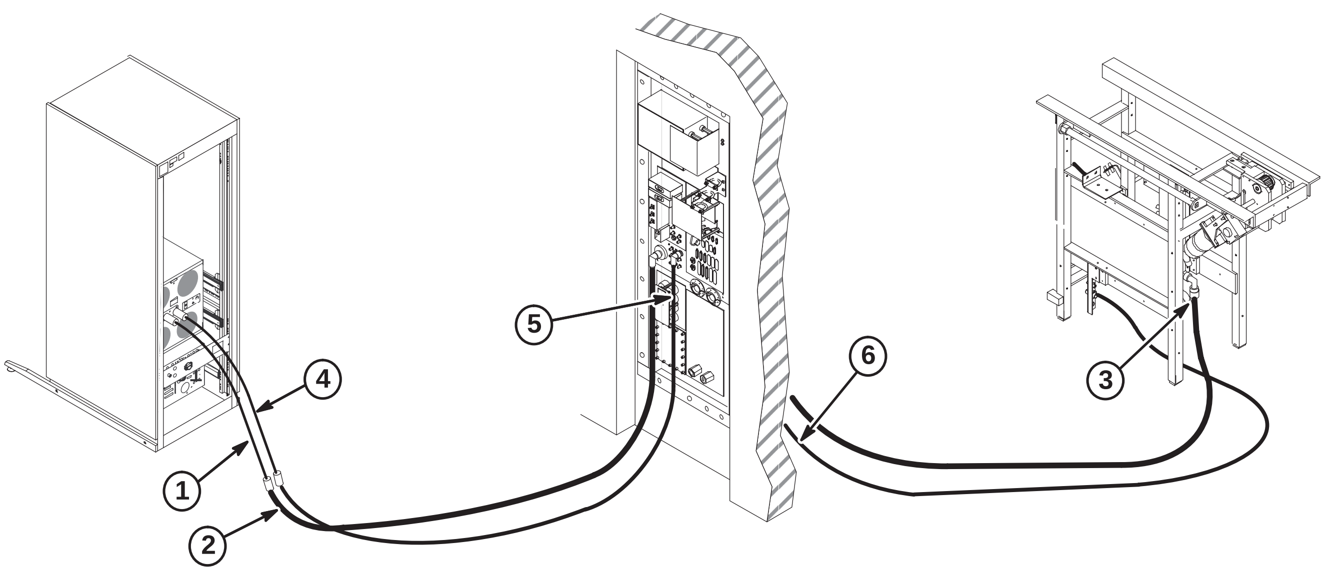
2316584

The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

1–2 LAYOUT OF CABLES

RF cables will be routed and connected as indicated in the following steps.

1. Connect extension cable 2316589 to J4 Body Output at rear of RF module in RF cabinet.
2. Connect end of Times RF Body cable (2316584) marked "(RUN 235) FJ5" to Body extension cable. Route to J44 at Penetration Panel. Determine if any take–up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1–3 for termination procedure of cut end.
3. Move remainder of RF Body cable to Scan Room and connect other end of cable (2316584) marked "(RUN 257) MG3A3–J1" to Rear Pedestal as shown below. Route to J44 at Penetration Panel. Determine if any take–up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1–3 for termination procedure of cut end.
4. Connect extension cable 2335189 to J3 Head Output at rear of RF module in RF cabinet.
5. Connect end of Times RF Head cable (2316588) marked "(RUN 236) PP1–J4" to J4 at Penetration Panel. Route to RF Head extension cable at rear of RF cabinet. Determine if any take–up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1–4 for termination procedure of cut end.
6. Move remainder of RF Head cable to Scan Room and connect other end of cable marked (RUN 258) PP1–J4" to Penetration Panel at J4. Route to Rear Pedestal Takeup bracket J1. Determine if any take–up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1–4 for termination procedure of cut end.



1–3 TERMINATE AND CONNECT RF BODY CABLES AT PENETRATION PANEL

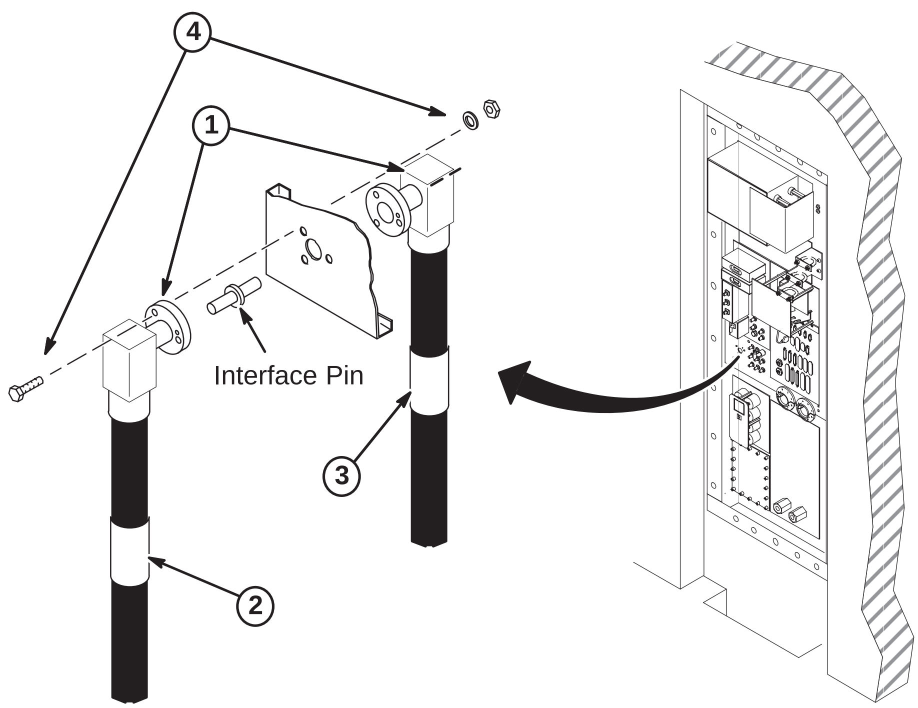
Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an EIA Right Angle Connector (2319986) on each end of the RF cables. Use provided LMR900 Stripping Tool (2335190).

Note

Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

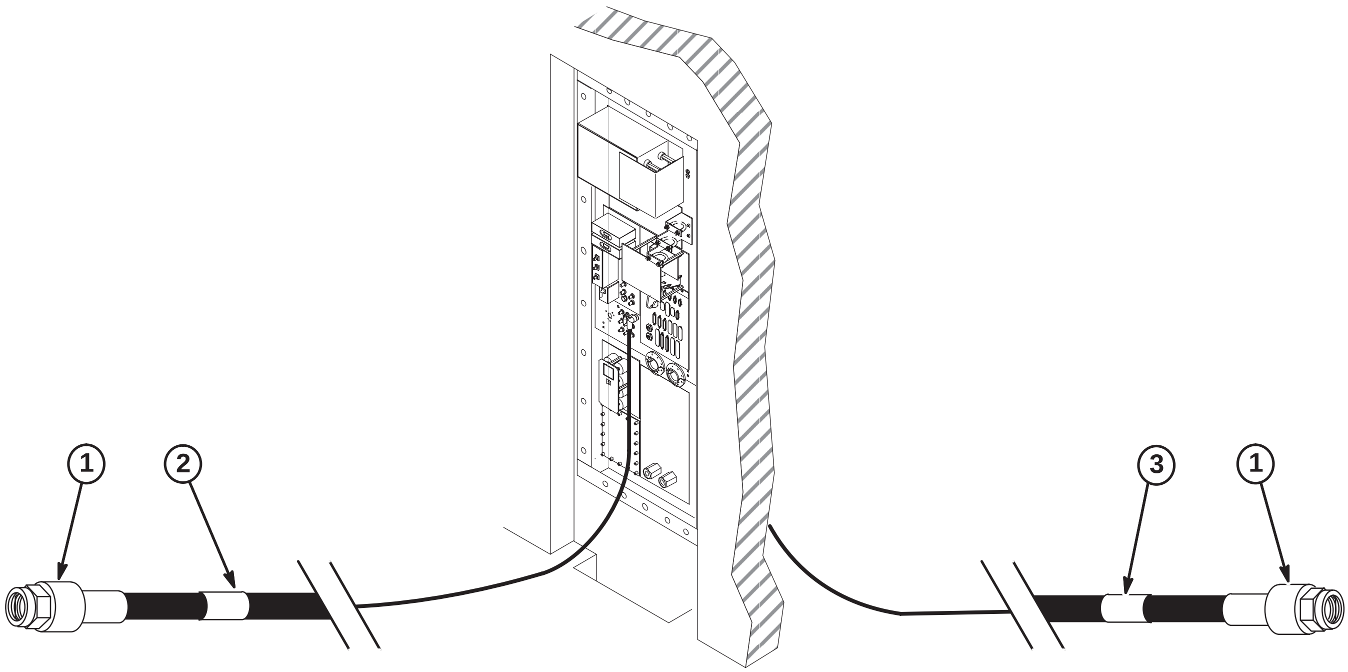
2. Attach label "RUN 235, PP1–J44" (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label "RUN 257, PP1–J44" (2357230–2) to cable as shown on Scan room side of Penetration Panel.
4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
   - 1/4–20 x 1.25 Hex Head Bolt (brass)
   - 1/4 Screw Size Washer (phos bronze)
   - 1/4–20, 7/16" Hex x 3/16" thk Nut (brass)



1–4 TERMINATE AND CONNECT RF HEAD COIL CABLES

Perform the following steps for terminating and connecting the RF Head cables in the Equipment and Scan rooms.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an Straight Male N Connector (2352195) on each end of the RF cable. Use provided LMR600 Stripping Tool (2352193).
   - Connect Equipment room end to Extension cable at FJ6.
   - Connect Scan room end to MG3A17–J1 at Interface bracket on Rear Pedestal.
2. Attach label “RUN 236 FJ6” (2357230–3) to cable as shown in Equipment room.
3. Attach label “RUN 258 MG3A17–J1” (2357230–4) to cable as shown in Scan room.



1.5T HELIAX CABLE INSTALLATION

INSTALLATION STEPS SUMMARY

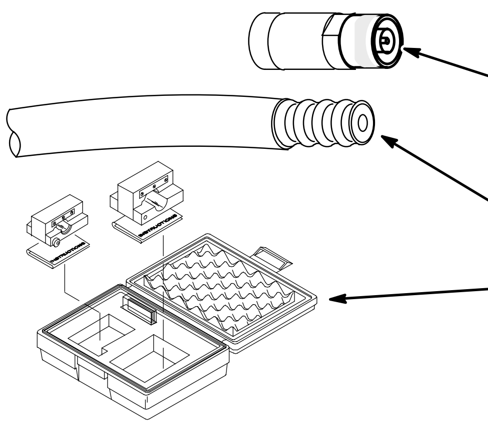
- ❑ **A – Routing, Cutting, Terminating Heliax Cables**
- ❑ **B – Connecting Runs 235, 236, 257, and 258**

CAUTION

Do not subject 1/2 inch diameter heliax cable to a bend radius less than 5 inches (127mm). Do not subject 7/8 inch diameter heliax cable to a bend radius less than 10 inches (254mm). This cable will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

❑ A – ROUTING, CUTTING, TERMINATING HELIAX CABLES

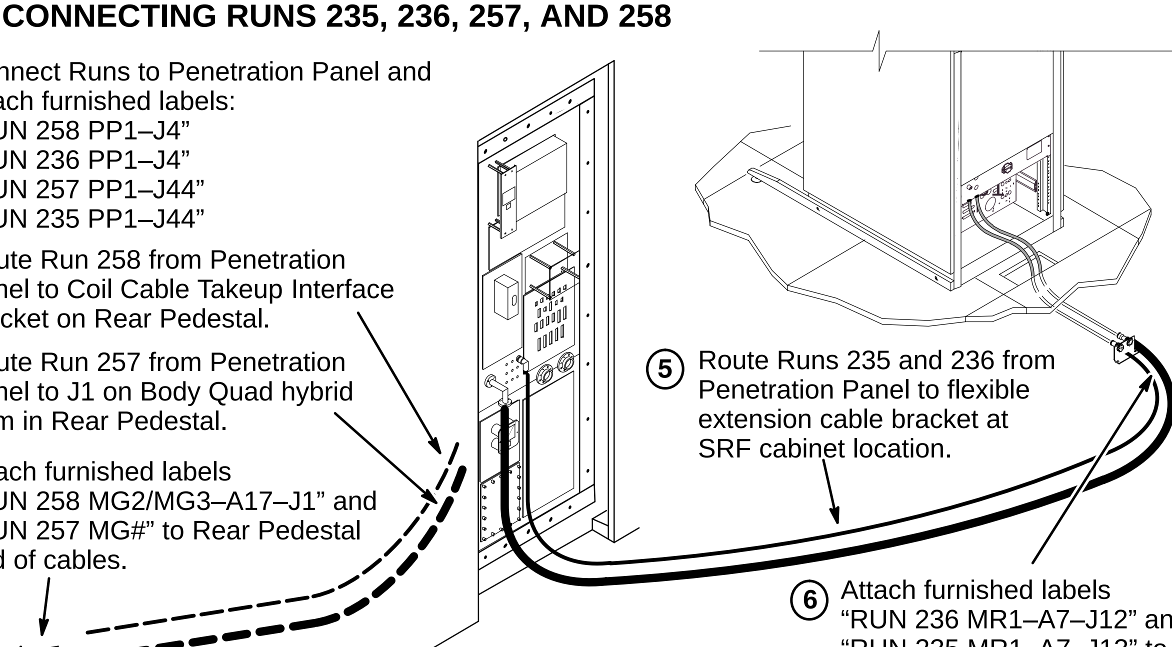
Runs 236 and 258 are created during installation from 50 foot and/or 55 foot cables along with four pre–marked labels. The installer may use both lengths of cables as provided, or if storage of excess length is a problem, cut one length (If runs are short, it may be feasible to cut one cable into two lengths) or both lengths and re–terminate each cut end. Either way, labels are attached after length is decided and cables are installed. Runs 235 and 257 are created during installation from a 100 foot cable along with four pre–marked labels.



1. Decide if cables are to be cut and re–terminated. If yes, continue to Step 2. If no, proceed to Step B.
2. Procure a new connector (46–230011P1) for each cable that needs to be shortened and re–terminated.
3. Route cables and mark at length for cut–off.
4. Cut cables to length with a fine–tooth hack saw. Remove about one inch of jacket to expose outer conductor.
5. Obtain the Heliax Termination Kit (46–320362G1) field service tool and follow instructions supplied with the kit. This will enable precise cut–off of cable.
6. Attach new connector(s) to cable(s) according to instruction sheets supplied with connector.

❑ B – CONNECTING RUNS 235, 236, 257, AND 258

1. Connect Runs to Penetration Panel and attach furnished labels:
   "RUN 258 PP1–J4"
   "RUN 236 PP1–J4"
   "RUN 257 PP1–J44"
   "RUN 235 PP1–J44"
2. Route Run 258 from Penetration Panel to Coil Cable Takeup Interface bracket on Rear Pedestal.
3. Route Run 257 from Penetration Panel to J1 on Body Quad hybrid Asm in Rear Pedestal.
4. Attach furnished labels "RUN 258 MG2/MG3–A17–J1" and "RUN 257 MG#" to Rear Pedestal end of cables.



5. Route Runs 235 and 236 from Penetration Panel to flexible extension cable bracket at SRF cabinet location.
6. Attach furnished labels "RUN 236 MR1–A7–J12" and "RUN 235 MR1–A7–J13" to RF Cabinet end of cables.

1.0T HELIAX CABLE INSTALLATION

INSTALLATION STEPS SUMMARY

- ❑ **A – Routing, Cutting, Terminating Heliax Cables**
- ❑ **B – Connecting Runs 745, 746, 747, and 748**

CAUTION

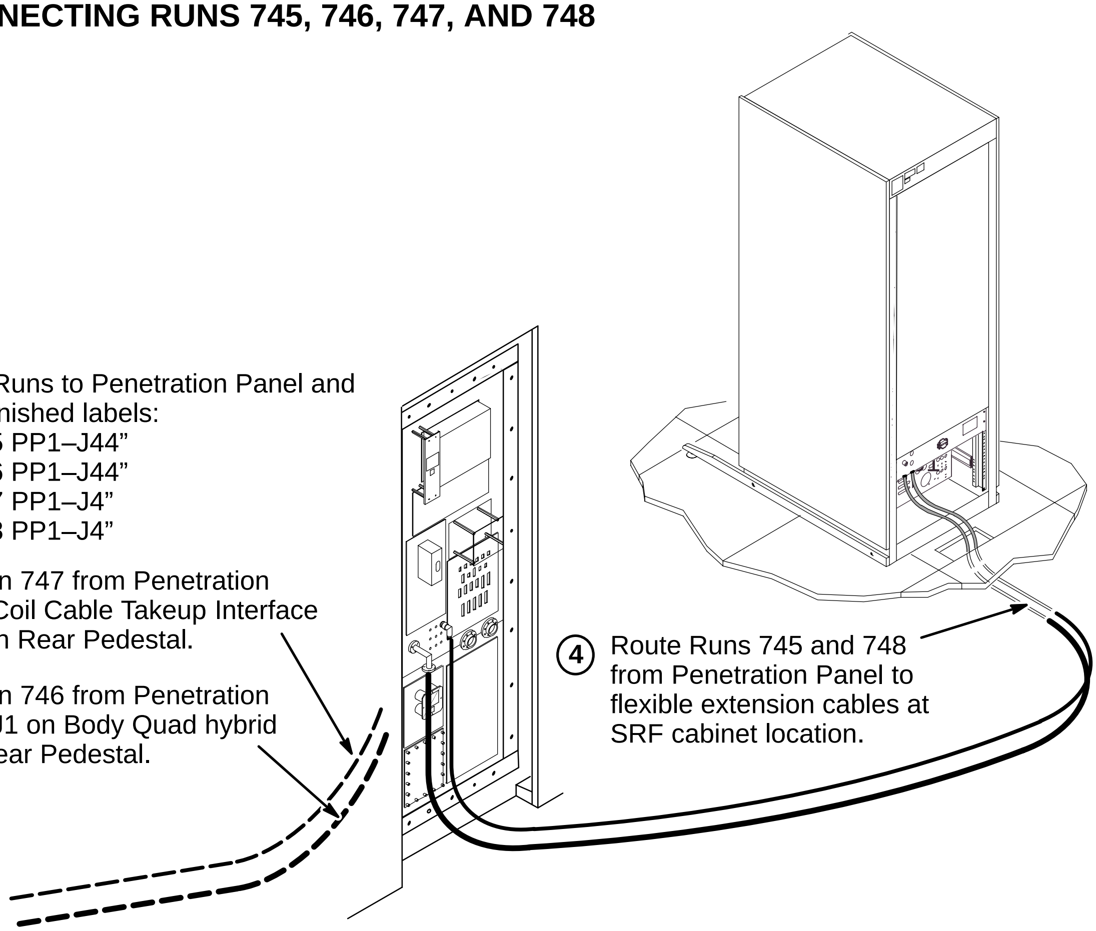
Do not subject 0.6 inch diameter heliax cable to a bend radius less than 1.5 inches (138mm). Do not subject 7/8 inch diameter heliax cable to a bend radius less than 3 inches (75mm). This cable will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

❑ A – ROUTING, CUTTING, TERMINATING HELIAX CABLES

Runs 747 and 748 are created during installation from 100 foot cables along with two labels. Labels are attached after length is decided and cables are installed. Runs 745 and 746 are created during installation from a 100 foot cable along with two labels. Kit (2207145) includes cable with terminal ends and 4 connectors.

1. Route cables and mark at length for cut–off.
2. Cut cables to length with a fine–tooth hack saw.
3. Follow instructions supplied with the kit. This will enable precise cut–off of cable.
4. Attach new connector(s) to cable(s) according to instruction sheets supplied with connector.

❑ B – CONNECTING RUNS 745, 746, 747, AND 748



1. Connect Runs to Penetration Panel and attach furnished labels:
   "RUN 745 PP1–J44"
   "RUN 746 PP1–J44"
   "RUN 747 PP1–J4"
   "RUN 748 PP1–J4"
2. Route Run 747 from Penetration Panel to Coil Cable Takeup Interface bracket on Rear Pedestal.
3. Route Run 746 from Penetration Panel to J1 on Body Quad hybrid Asm in Rear Pedestal.
4. Route Runs 745 and 748 from Penetration Panel to flexible extension cables at SRF cabinet location.

SIGNA ACGD GRADIENT OUTPUT CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

- ❑ **A – Cut cables to length and route in Equipment and Magnet Rooms**
- ❑ **B – Terminate Runs 762, 763, and 764 for ACGD**
- ❑ **C – Terminate Runs 762, 763, and 764 in Magnet Room**
- ❑ **D – Connect Cables at Penetration Panel**
- ❑ **E – Connection of Body Coil Gradient Cables to Runs 762, 763 & 764**

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

❑ A – CUT CABLES TO LENGTH AND ROUTE IN EQUIPMENT AND MAGNET ROOMS

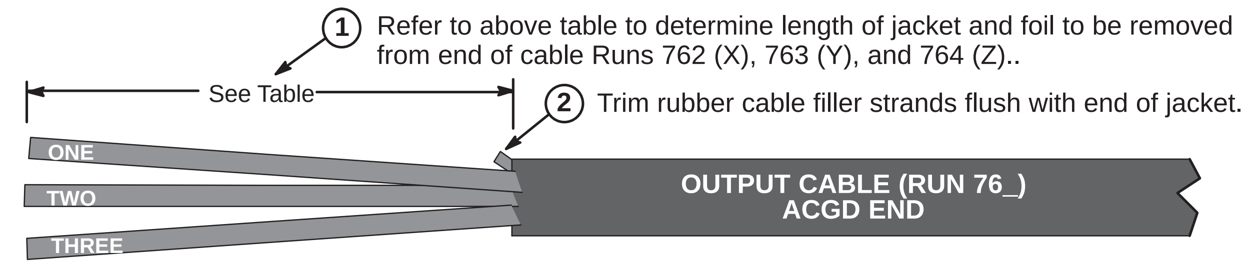
1. Verify that 100 ft. (30.5 M) length of supplied Run 762, 763, and 764 cable is sufficient to route from ACGD/PDU Cabinet to Penetration Panel and Penetration Panel to Body Coil Gradient Cables at rear of magnet.
   Each cable has identical terminations at each end for connection to <u>both sides of the Penetration Panel Gradient Filter</u> after completing Steps 2 and 3 below.
2. Unroll each cable. Route cables from equipment room side of Penetration Panel to ACGD/PDU Cabinet. The end of cable with terminals will be connected to Penetration Panel. Make sure enough slack is allowed for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
3. Move cable to magnet room. Route cables from Gradient Filter of Penetration Panel to Body Coil gradient cables at rear of magnet.
4. Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. **Do not cut Body Coil gradient cables.**
5. Locate furnished bag of parts and attach terminal and labels to each cable which was cut to length. See Procedures **B** and **C.**
6. Route Run 762, 763, and 764 from equipment room side of Penetration Panel to ACGD/PDU cabinet.
7. See Procedures **D** for connection to Penetration Panel.
8. Route Run 762, 763, and 764 from magnet room side of Penetration Panel to Body Coil gradient cables.
9. See Procedure **D** for connection to Penetration Panel.
10. See Procedure **E** for connection of Runs 762, 763, and 764 to Body Coil gradient cables.
11. See ACGD/PDU Cabinet installation instructions **(DIRECTION 2286811 IN TAB 15)** for connection of Runs 762, 763, and 764 to ACGD/PDU cabinet.

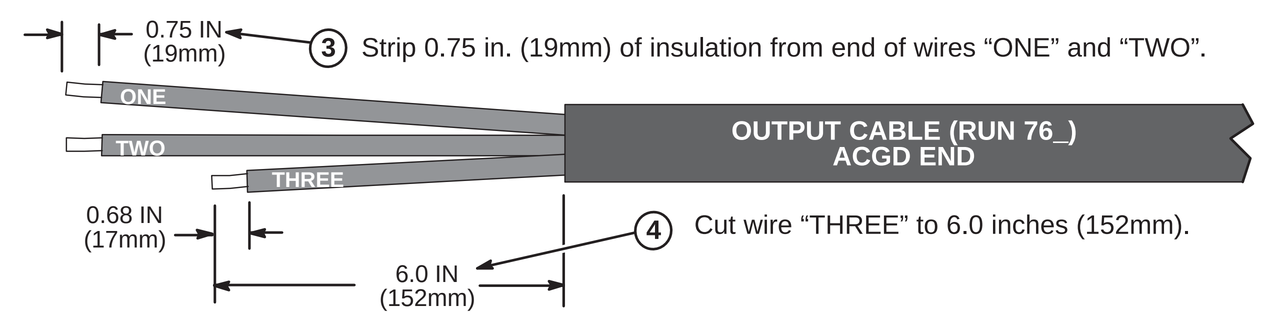
TOOLS REQUIRED: 2135776 – Cable Crimper/Stripper Kit

❑ B – TERMINATE RUNS 762, 763, AND 764 FOR ACGD

Note: The following steps are required to terminate the end of the cable that was cut in Step **A3**. The other end of the cable is connected to the Penetration Panel.

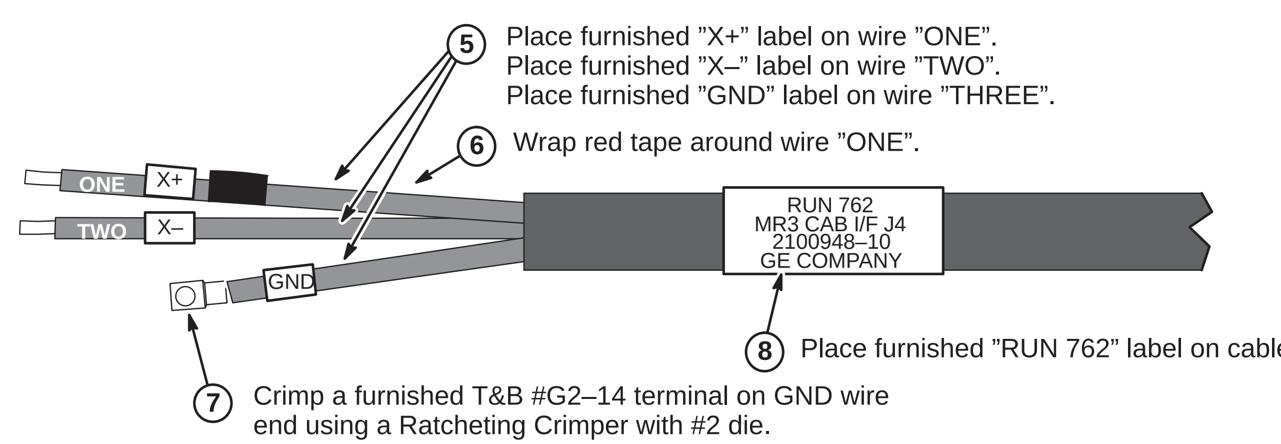
	Run 762	Run 763	Run 764
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)





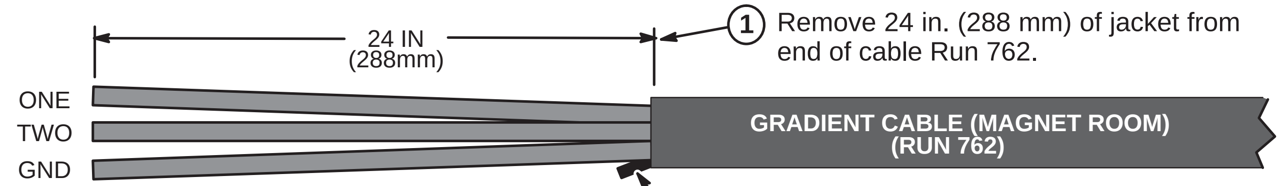
Note:
Steps in illustration below show completion of Run 762. Repeat same process for Runs 763 (Y+, Y–) and 764 (Z+, Z–).

Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire ”ONE”.**

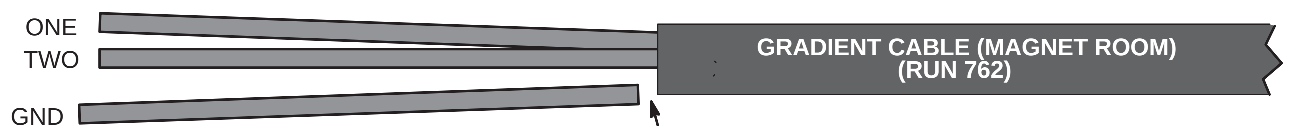


❑ C – TERMINATE RUNS 762, 763, AND 764 FOR MAGNET ROOM

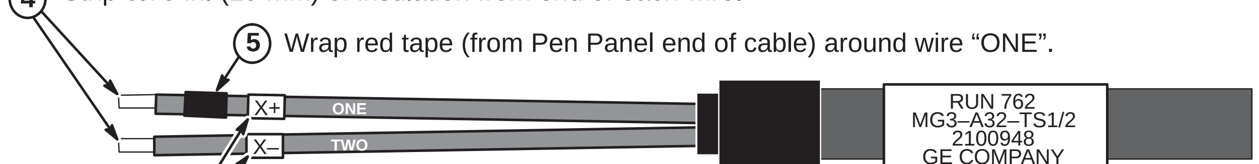
Note: The following steps are required to terminate the end of the cable that was cut in Step **A3**. The other end of the cable is connected to the Penetration Panel.



1. Remove 24 in. (288 mm) of jacket from end of cable Run 762.
2. Trim rubber cable filler strands flush with end of jacket.

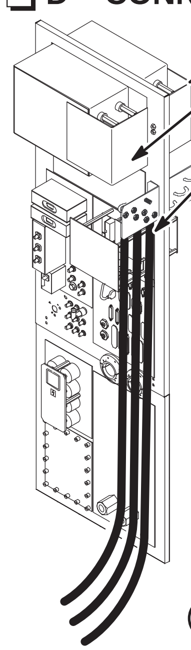


3. Cut and remove GROUND wire.
4. Strip 0.75 in. (19 mm) of insulation from end of each wire.
5. Wrap red tape (from Pen Panel end of cable) around wire "ONE".

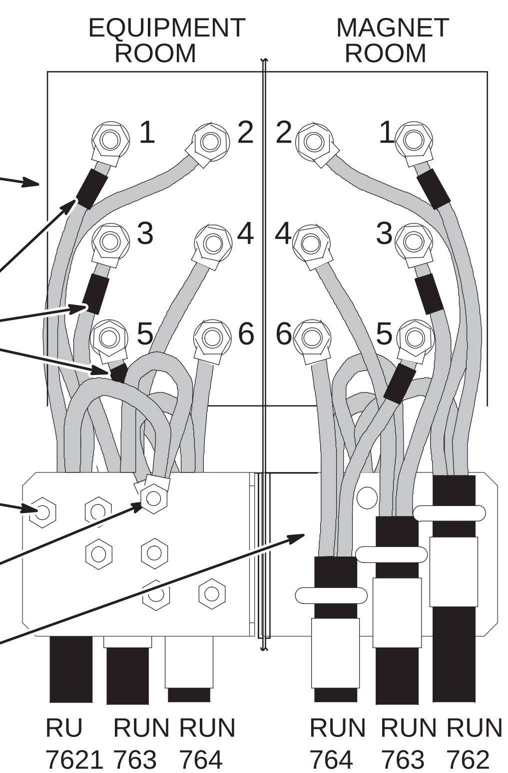


6. Place furnished "X+" label on wire "ONE". Place furnished "X–" label on wire "TWO".
7. Place furnished "Run 762" label on cable.
8. Repeat above process for Runs 763 and 764. **Note: Red tape, Y+** (RUN 763) **and Z+** (RUN 764) **labels are attached to wire "ONE".**

❑ D – CONNECT CABLES AT PENETRATION PANEL



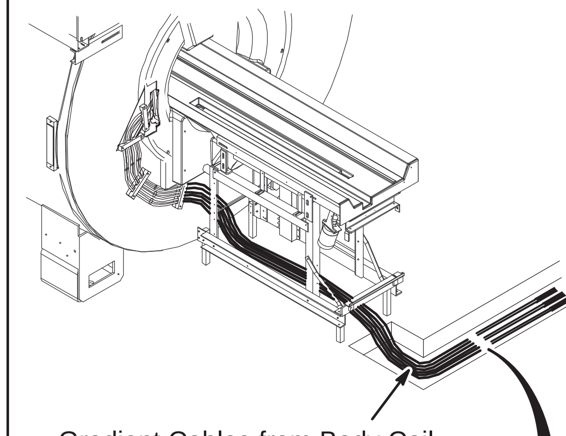
1. Remove gradient filter terminal covers.
2. Route Run 762, 763, and 764 to gradient filter on Penetration Panel.
4. Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs. **Note:** Red tape indicates positive (+) wires #1, #3, and #5.
3. Secure each cable Run to strain–relief bracket with U–bolts.
5. Secure GND lead wire on Ground Stud for each cable Run.
6. Repeat procedure and connect cables to Gradient Filter on other side of Penetration Panel.
7. Replace gradient filter terminal covers.



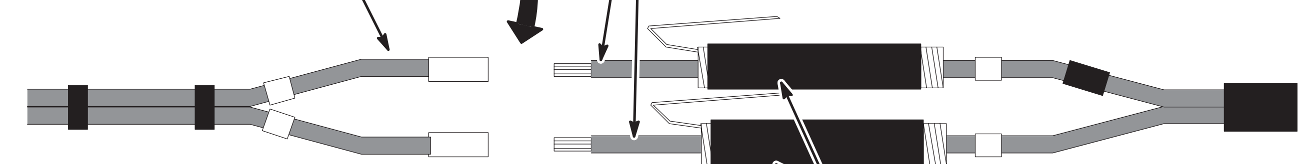
❑ E – CONNECTION OF BODY COIL GRADIENT CABLES TO RUNS 762, 763 & 764

Note:
Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

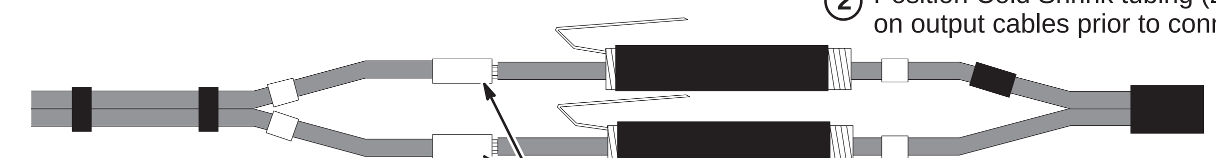
Note:
Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet



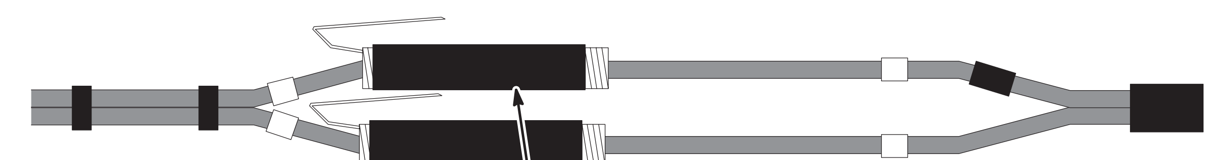
1. Cable connection process shown below is for Run 762. Repeat process for Runs 763 and 764. **Note: Red tape, X+** (RUN 762), **Y+** (RUN 763) **and Z+** (RUN 764) **labels are attached to wire "ONE".**



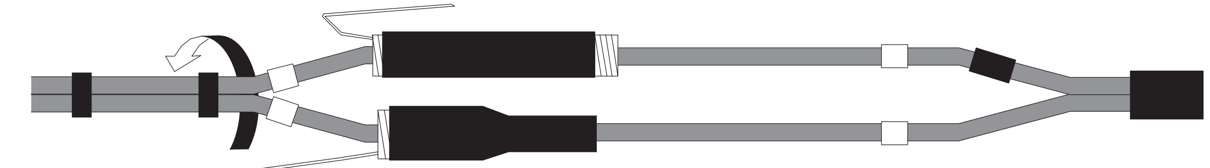
2. Position Cold Shrink tubing (2218148) on output cables prior to connecting.



3. Insert ends of Run 762 into correct butt splice (2218149) on Body Coil gradient cable.
4. Crimp butt splices using a T&B WT 3185 Gradient Cable Crimper (2144091) with #2 die.



5. Locate Cold Shrink tubing at desired starting point. Position tubing 1–1/2 to 2 inches (38 to 50mm) beyond end of connector.



6. Pull loose tab gently while unwinding in a counter–clockwise direction. Repeat process for other cable.

7. Connection of cables and installation of Cold Shrink tubing is now complete.

SIGNA ACGD LITE GRADIENT OUTPUT CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

- ❑ **A – Cut cables to length and route in Equipment and Magnet Rooms**
- ❑ **B – Terminate Runs 762, 763, and 764 for ACGD Lite Cabinet Gradients**
- ❑ **C – Terminate Runs 762, 763, and 764 in Magnet Room**
- ❑ **D – Connect Cables at Penetration Panel**
- ❑ **E – Connection of Body Coil Gradient Cables to Runs 762, 763 & 764**

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

❑ A – CUT CABLES TO LENGTH AND ROUTE IN EQUIPMENT AND MAGNET ROOMS

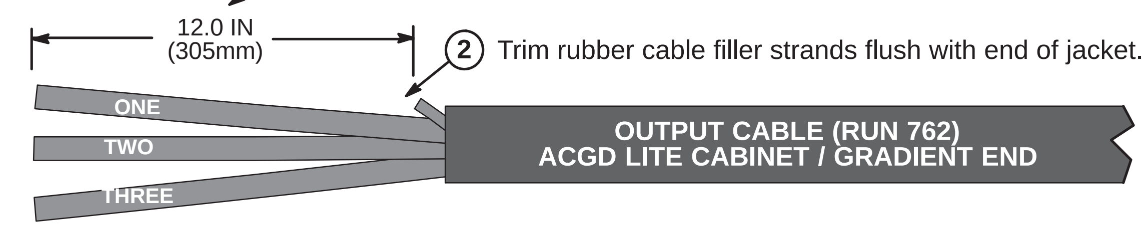
1. Verify that 100 ft. (30.5 M) length of supplied Run 762, 763, and 764 cable is sufficient to route from ACGD/PDU Cabinet to Penetration Panel and Penetration Panel to Body Coil Gradient Cables at rear of magnet.
   Each cable has identical terminations at each end for connection to <u>both sides of the Penetration Panel Gradient Filter</u> after completing Steps 2 and 3 below.
2. Unroll each cable. Route cables from equipment room side of Penetration Panel to ACGD Lite/PDU Cabinet. The end of cable with terminals will be connected to Penetration Panel. Make sure enough slack is allowed for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
3. Move cable to magnet room. Route cables from Gradient Filter of Penetration Panel to Body Coil gradient cables at rear of magnet.
4. Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. **Do not cut Body Coil gradient cables.**
5. Locate furnished bag of parts and attach terminal and labels to each cable which was cut to length. See Procedures **B** and **C.**
6. Route Run 762, 763, and 764 from equipment room side of Penetration Panel to ACGD Lite/PDU cabinet.
7. See Procedures **D** for connection to Penetration Panel.
8. Route Run 762, 763, and 764 from magnet room side of Penetration Panel to Body Coil gradient cables.
9. See Procedure **D** for connection to Penetration Panel.
10. See Procedure **E** for connection of Runs 762, 763, and 764 to Body Coil gradient cables.
11. See ACGD Lite/PDU Cabinet installation instructions **(TAB 15)** for connection of Runs 762, 763, and 764 to ACGD Lite/PDU cabinet.

TOOLS REQUIRED: 2135776 – Cable Crimper/Stripper Kit

❑ C – TERMINATE RUNS 762, 763, AND 764 FOR ACGD LITE CABINET GRADIENTS

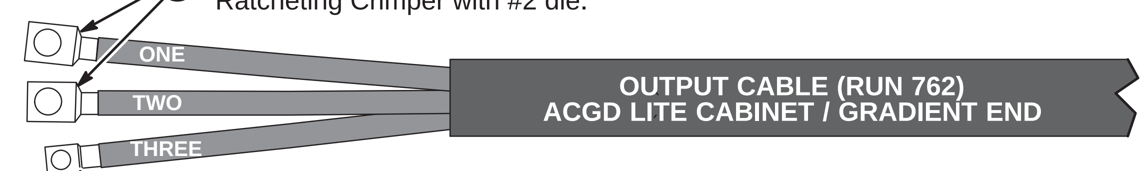
Note: The following steps are required to terminate the end of the cable that was cut in Step **A3**. The other end of the cable is connected to the Penetration Panel.

1. Remove 8.0 in. (203 mm) of jacket and foil from end of cable Run 762.
2. Trim rubber cable filler strands flush with end of jacket.

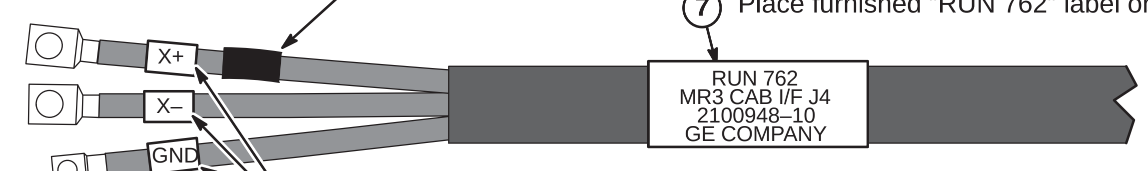


3. Strip 0.68 in. (17mm) of insulation from end of all wires.

4. Crimp a furnished T&B #G2–12 terminal on ONE and TWO wire ends using a Ratcheting Crimper with #2 die.



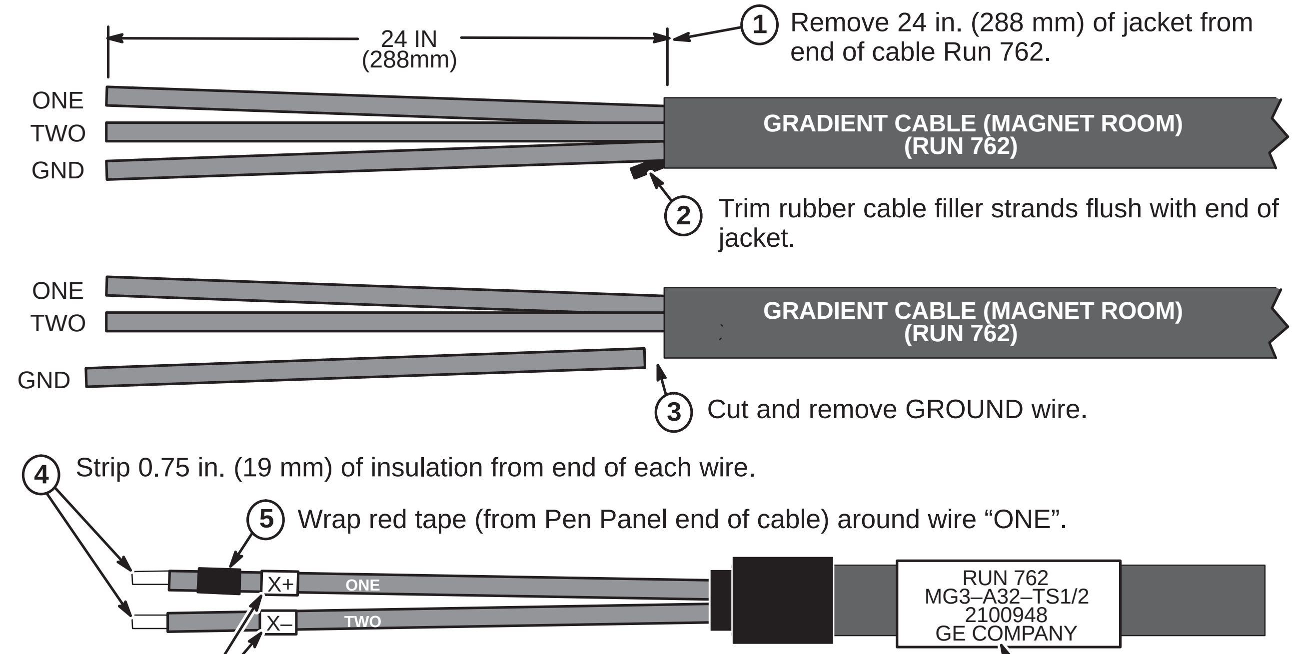
5. Crimp a furnished T&B #G2–14 terminal on GND wire end using a Ratcheting Crimper with #2 die.
6. Wrap red tape around wire "ONE".
7. Place furnished "RUN 762" label on cable.



8. Place furnished "X+" label on wire "ONE".
   Place furnished "X–" label on wire "TWO".
   Place furnished "GND" label on wire "THREE".
9. Repeat above process for Runs 763 (Y+, Y–) and 764 (Z+, Z–).
   Note: Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire "ONE".**

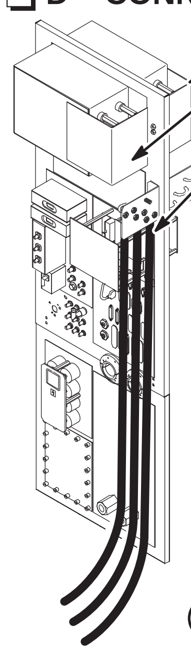
❑ C – TERMINATE RUNS 762, 763, AND 764 FOR MAGNET ROOM

Note: The following steps are required to terminate the end of the cable that was cut in Step **A3**. The other end of the cable is connected to the Penetration Panel.

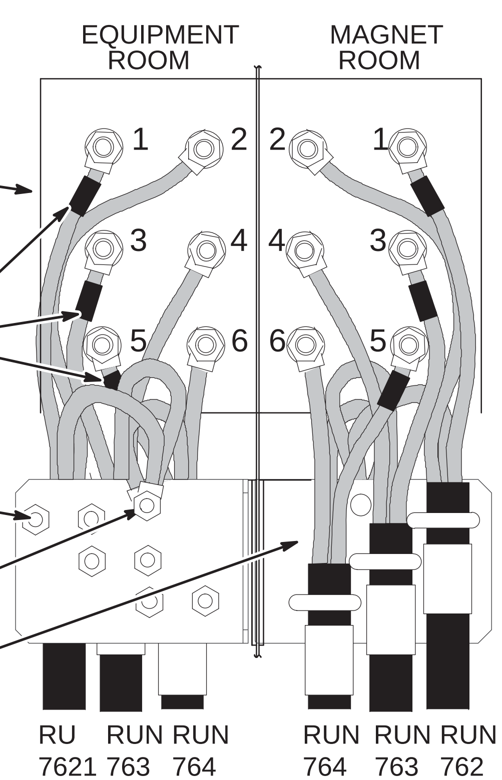


1. Remove 24 in. (288 mm) of jacket from end of cable Run 762.
2. Trim rubber cable filler strands flush with end of jacket.
3. Cut and remove GROUND wire.
4. Strip 0.75 in. (19 mm) of insulation from end of each wire.
5. Wrap red tape (from Pen Panel end of cable) around wire “ONE”.
6. Place furnished ”X+” label on wire “ONE”.
   Place furnished ”X–” label on wire ”TWO”.
7. Place furnished ”Run 762” label on cable.
8. Repeat above process for Runs 763 and 764.
   Note: Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire ”ONE”.**

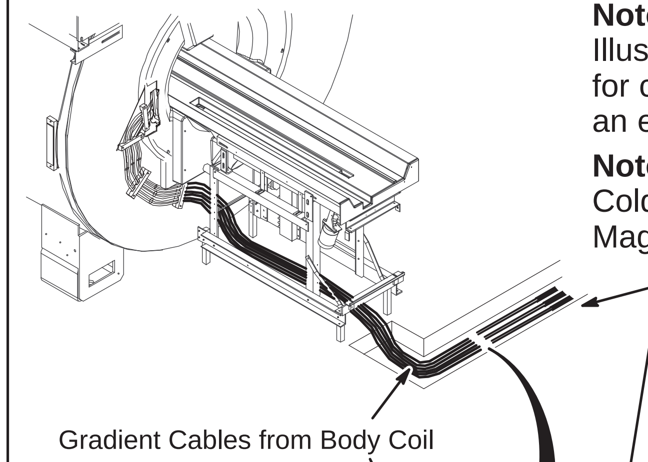
❑ D – CONNECT CABLES AT PENETRATION PANEL



1. Remove gradient filter terminal covers.
2. Route Run 762, 763, and 764 to gradient filter on Penetration Panel.
4. Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs.
   Note: Red tape indicates positive (+) wires #1, #3, and #5.
3. Secure each cable Run to strain–relief bracket with U–bolts.
5. Secure GND lead wire on Ground Stud for each cable Run.
6. Repeat procedure and connect cables to Gradient Filter on other side of Penetration Panel.
7. Replace gradient filter terminal covers.

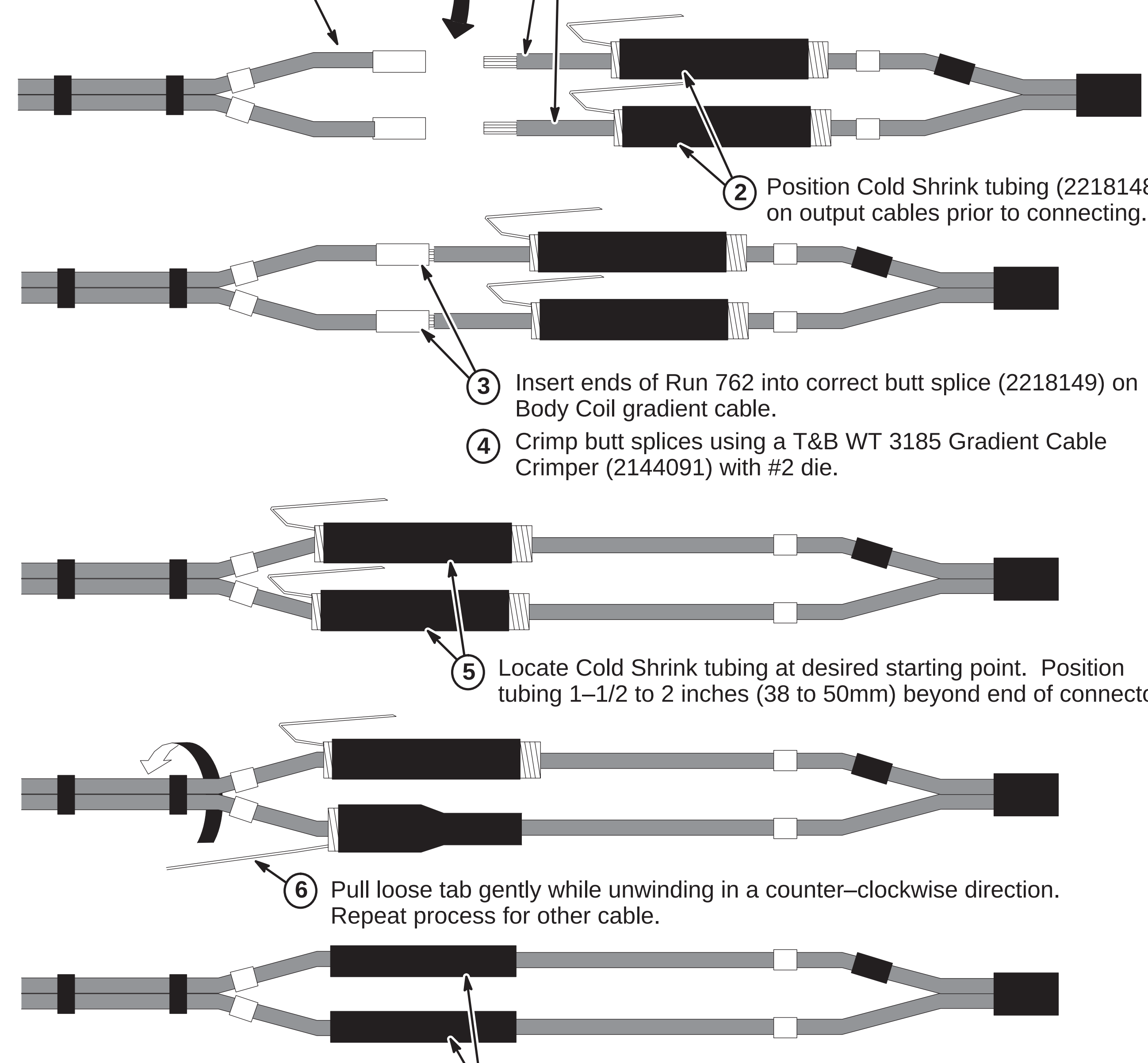


❑ E – CONNECTION OF BODY COIL GRADIENT CABLES TO RUNS 762, 763 & 764



Note:
Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

Note:
Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet



1. Cable connection process shown below is for Run 762. Repeat process for Runs 763 and 764.
   Note: Red tape, X+ (RUN 762), **Y+** (RUN 763) **and Z+** (RUN 764) **labels are attached to wire ”ONE”.**
2. Position Cold Shrink tubing (2218148) on output cables prior to connecting.
3. Insert ends of Run 762 into correct butt splice (2218149) on Body Coil gradient cable.
4. Crimp butt splices using a T&B WT 3185 Gradient Cable Crimper (2144091) with #2 die.
5. Locate Cold Shrink tubing at desired starting point. Position tubing 1–1/2 to 2 inches (38 to 50mm) beyond end of connector.
6. Pull loose tab gently while unwinding in a counter–clockwise direction. Repeat process for other cable.
7. Connection of cables and installation of Cold Shrink tubing is now complete.

SIGNA ACGD FIBER OPTIC CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

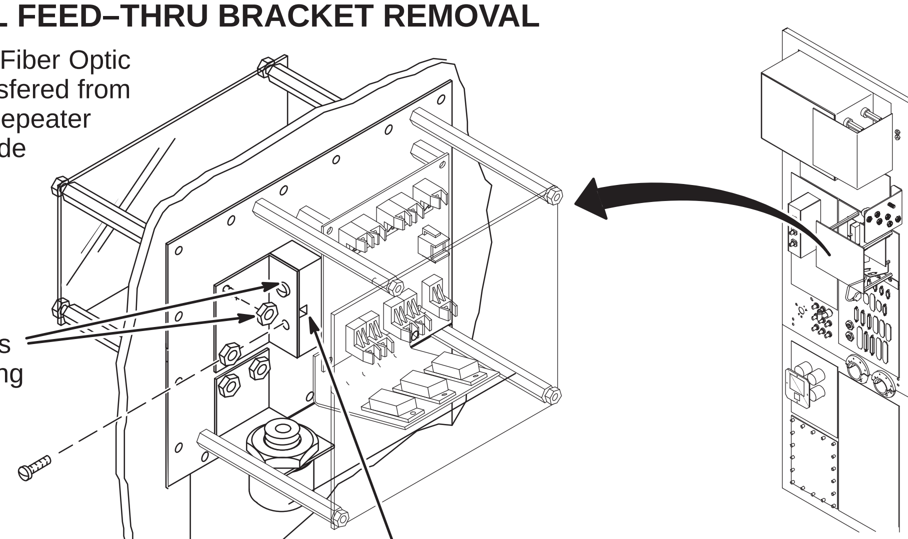
- ❑ **A1 – Repeater Panel Feed–thru Bracket Removal**
- ❑ **A2 – Routing Run 711/712 in Magnet Room**
- ❑ **A3 – Connecting Magnet Room Run 711/712 to Repeater Panel**
- ❑ **A4 – Route/Connect Equipment Room Run 711/712 to Repeater Panel**
- ❑ **A5 – Route/Connect Fiber Optic Equipment Room Runs 708, 709, 710, 711/712, and 836**

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

❑ A1 – REPEATER PANEL FEED–THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transfered from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.

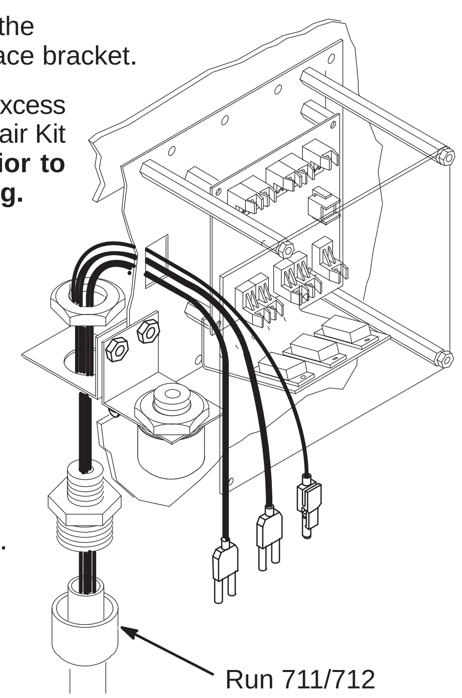


1. Unfasten two #10–32 screws and two #10–32 nuts securing feed–thru access bracket to J91 feed–thru block.
2. Remove feed–thru access bracket and fastening hardware from panel.

❑ A2 – ROUTING RUN 711/712 IN MAGNET ROOM

1. Move Run 711/712 (46–328079G9) to the Magnet Room side of the Penetration Panel. The "PP1–J91" end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

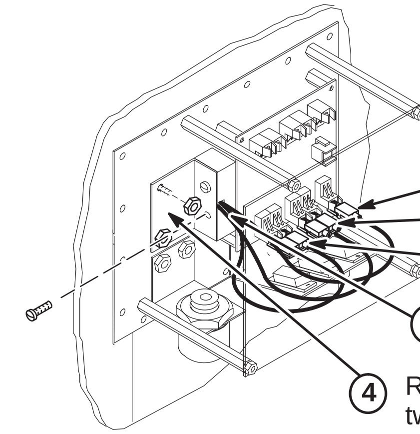


2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
3. Route ends of individual fiber–optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
4. Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC–II.

5. Route Run 711/712 to the front right side of the magnet (SRI location). .
6. Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC–II.

❑ A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

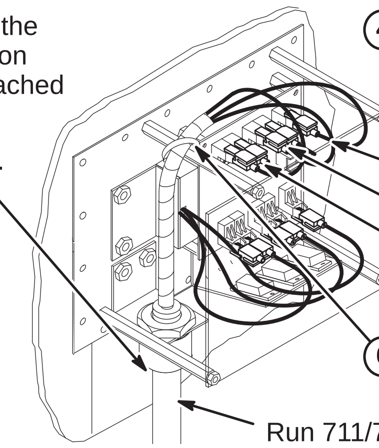


Note: Take care to maintain the minimum 2 inch bend radius.

1. Adjust route of individual fiber optic runs to maintain large radius.
2. Connect Runs 712–1, 711–1, and 711–2 as shown.
   Run 711–2 (J5)
   Run 711–1 (J3, J4)
   Run 712–1 (J1, J2)
3. Gather fiber optic runs and route through wave guide notch in feed–thru block.
4. Replace feed–thru access bracket and fasten with two #10–32 screws and two #10–32 nuts removed in step A1.

❑ A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 711/712 TO REPEATER PANEL

1. Move Run 711/712 (46–328078G5) to the Equipment Room side of the Penetration Panel. The "PP1–A17" end will be attached to interface bracket.
2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
3. Route other end of Run 711/712 to System Cabinet. Refer to Direction 2170004 (Tab 8) for connecting fiber optics in System Cabinet.

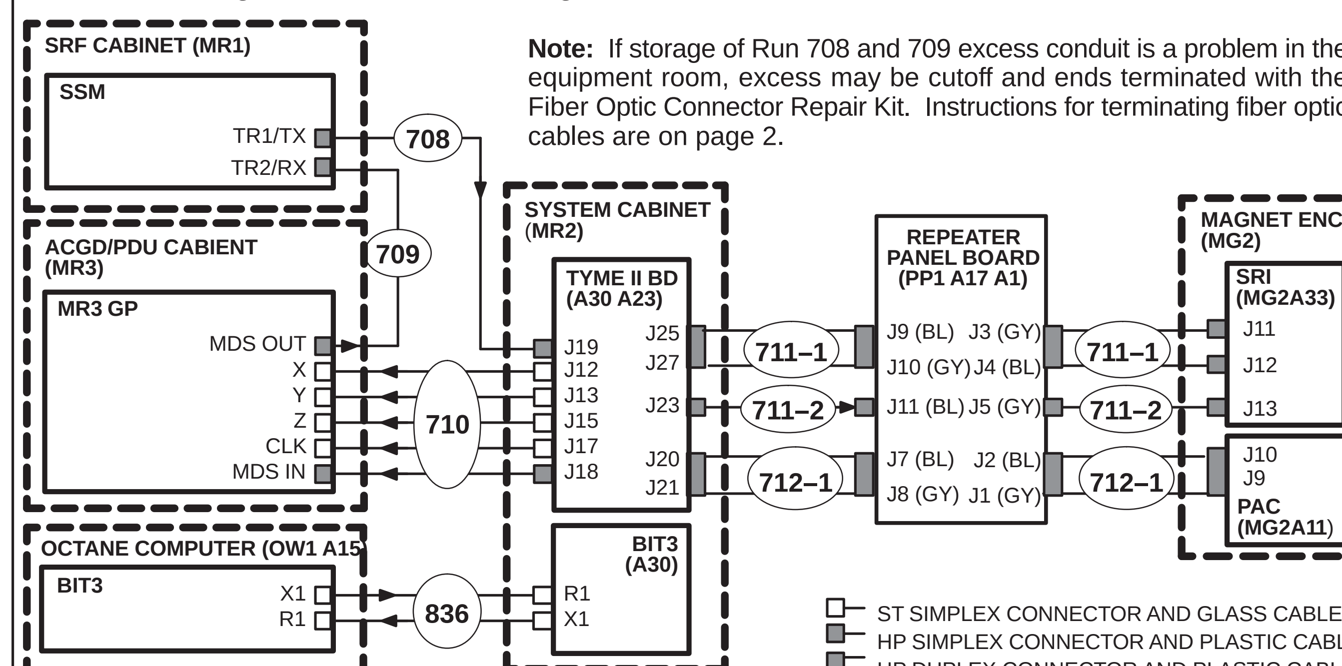


4. Adjust route of individual fiber optic runs to maintain large radius.
5. Connect Runs 712–1, 711–1, and 711–2 as shown.
   Run 711–2 (J11)
   Run 711–1 (J9, J10)
   Run 712–1 (J7, J8)
6. Adjust position of protective spiral wrap and secure with tie–wrap.

❑ A5 – ROUTE/CONNECT F/O EQUIPMENT ROOM RUNS 708, 709, 710, 711/712 & 836

1. Unroll coiled fiber optic conduits for Runs 708, 709, and 710 along intended route. Connection of Runs is shown in diagram below. Plan for storage of excess Run 710 conduit of as it cannot be reterminated.

Note: If storage of Run 708 and 709 excess conduit is a problem in the equipment room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit. Instructions for terminating fiber optic cables are on page 2.



INSTALLATION STEPS SUMMARY

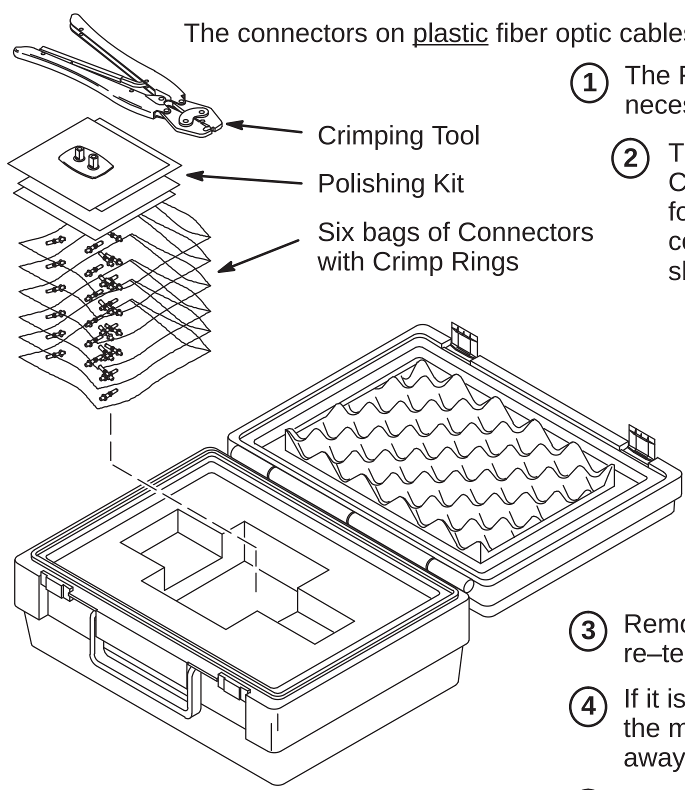
- ❑ **B1 – Fiber Optic Cable Termination Kit**
- ❑ **B2 – Stripping Fiber Optic Outer Jacket**
- ❑ **B3 – Crimping Connector to Fiber Optic Cable**
- ❑ **B4 – Trimming Excess Fiber Optic from Connector**
- ❑ **B5 – Polishing Fiber Optic End**

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

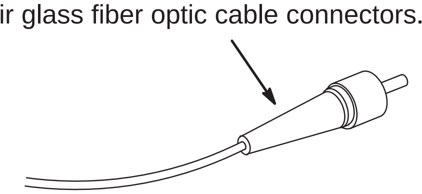
❑ B1 – FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.



1. The Fiber Optic Repair Kit, 46–301450G1, contains all the necessary materials to replace a damaged connector.
2. The "Spares Kit", delivered with the Shipping Collector, contains a 46–320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



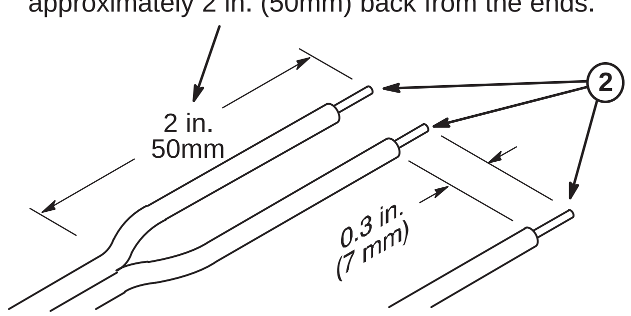
3. Remove the end of fiber optic cable to be re–terminated from magnet room if possible.
4. If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

1. For Duplex Cables, seperate the two fibers approximately 2 in. (50mm) back from the ends.

Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

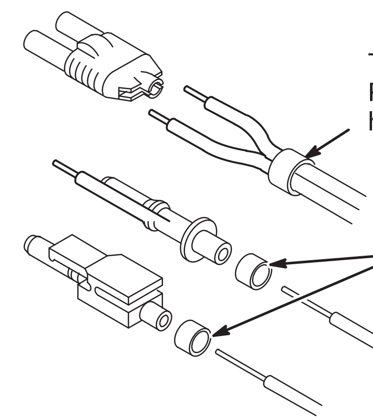
2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.



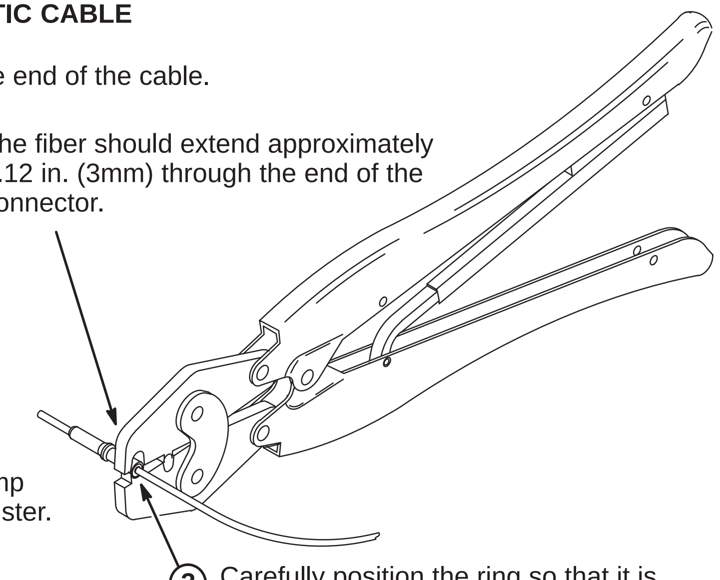
❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

1. Place the Crimp Ring and Connector over the end of the cable.

Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

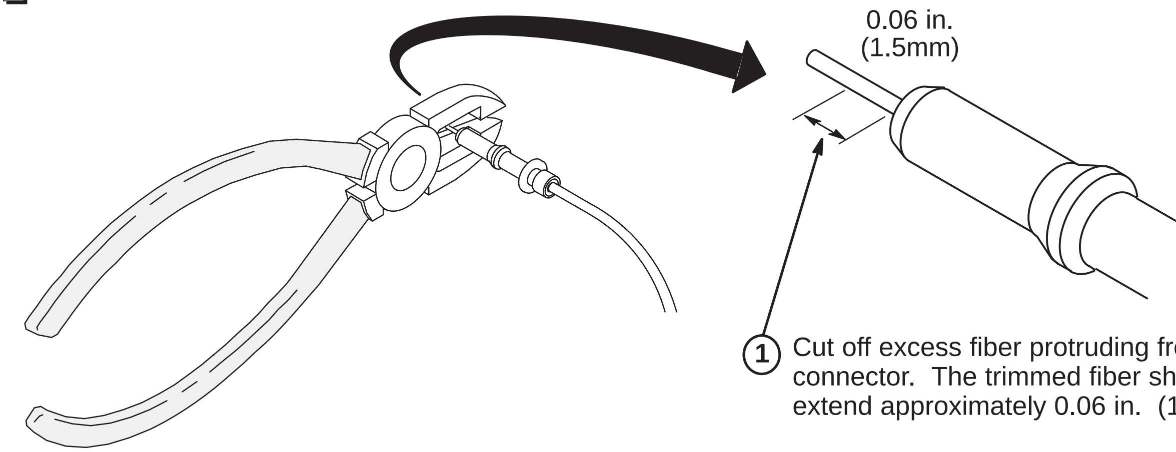


2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



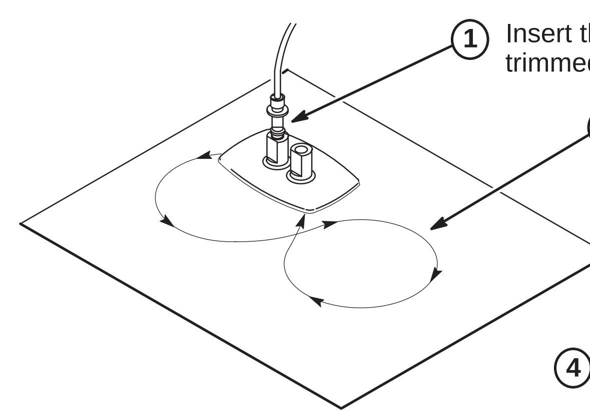
3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR



1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1–1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

SIGNA EXCITE I FIBER OPTIC CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

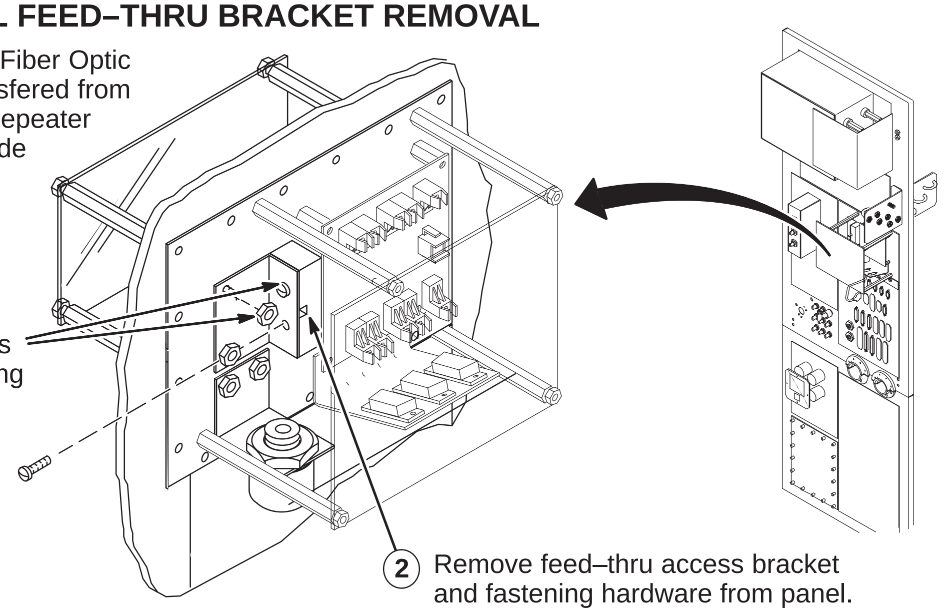
- ❑ **A1 – Repeater Panel Feed–thru Bracket Removal**
- ❑ **A2 – Routing Run 711/712 in Magnet Room**
- ❑ **A3 – Connecting Magnet Room Run 711/712 to Repeater Panel**
- ❑ **A4 – Route/Connect Equipment Room Run 1045 to Repeater Panel**
- ❑ **A5 – Cable Map of Fiber Optic Runs 709, 711/712, 1032, 1033, 1034, 1044, and 1045**

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

❑ A1 – REPEATER PANEL FEED–THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transfered from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.

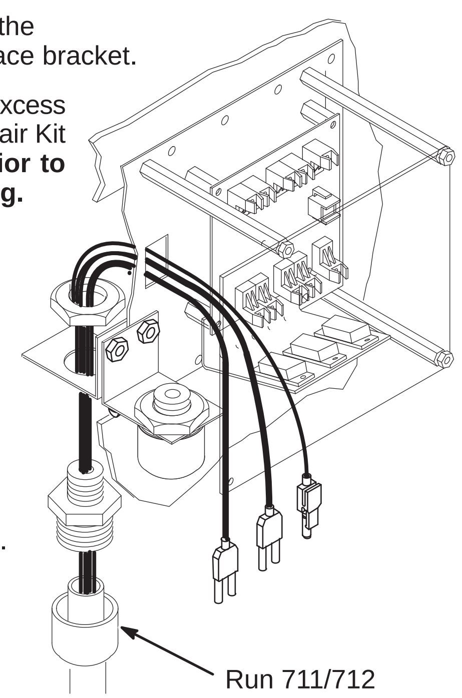


1. Unfasten two #10–32 screws and two #10–32 nuts securing feed–thru access bracket to J91 feed–thru block.
2. Remove feed–thru access bracket and fastening hardware from panel.

❑ A2 – ROUTING RUN 711/712 IN MAGNET ROOM

1. Move Run 711/712 (46–328079G9) to the Magnet Room side of the Penetration Panel. The “PP1–J91” end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**



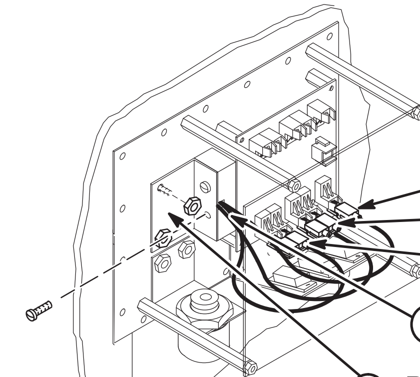
Run 711/712

2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
3. Route ends of individual fiber–optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
4. Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC–II.

5. Route Run 711/712 to the front right side of the magnet (SRI location). .
6. Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC–II.

❑ A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

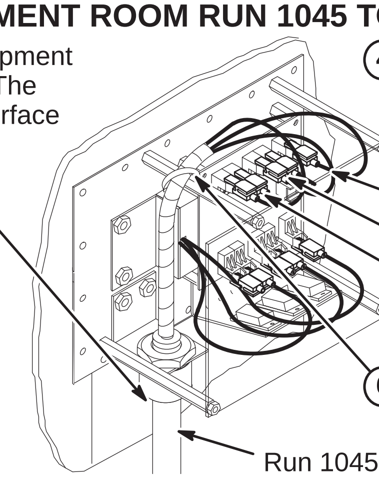


Note: Take care to maintain the minimum 2 inch bend radius.

1. Adjust route of individual fiber optic runs to maintain large radius.
2. Connect Runs 712–1, 711–1, and 711–2 as shown.
   - Run 711–2 (J5)
   - Run 711–1 (J3, J4)
   - Run 712–1 (J1, J2)
3. Gather fiber optic runs and route through wave guide notch in feed–thru block.
4. Replace feed–thru access bracket and fasten with two #10–32 screws and two #10–32 nuts removed in step A1.

❑ A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 1045 TO REPEATER PANEL

1. Move Run 1045 (2331846) to the Equipment Room side of the Penetration Panel. The “PP1–A17” end will be attached to interface bracket.
2. Attach Run 1045 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
3. Route other end of Run 1045 to System Cabinet. Refer to Direction 2338502 (Tab 8) for connecting fiber optics in System Cabinet.



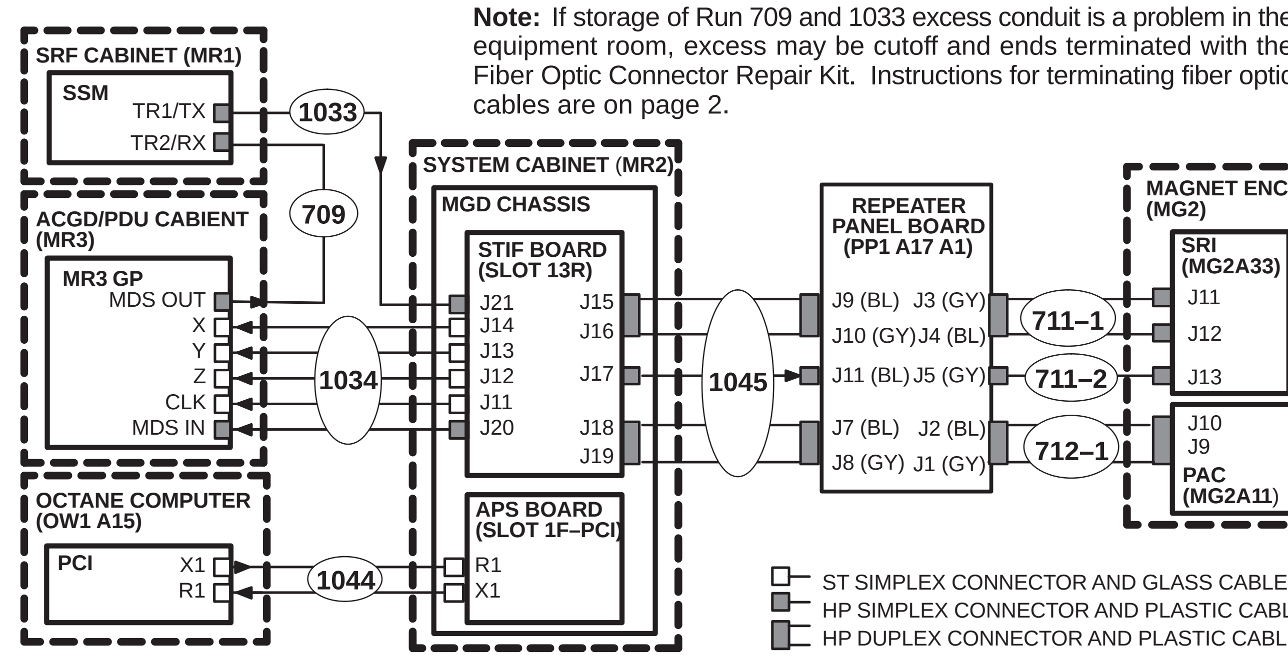
Run 1045

4. Adjust route of individual fiber optic runs to maintain large radius.
5. Connect Run 1045 as shown.
   - 2338146 (J11)
   - 2338146–3 (J9, J10)
   - 2338146–2 (J7, J8)
6. Adjust position of protective spiral wrap and secure with tie–wrap.

❑ A5 – CABLE MAP OF F/O RUNS 709, 711/712, 1032, 1033, 1034, 1044, AND 1045

1. Unroll coiled fiber optic conduits for Runs 709, 1033, and 1034 along intended route. Connection of Runs is shown in diagram below. Plan for storage of excess Run 1034 conduit of as it cannot be reterminated.

Note: If storage of Run 709 and 1033 excess conduit is a problem in the equipment room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit. Instructions for terminating fiber optic cables are on page 2.



INSTALLATION STEPS SUMMARY

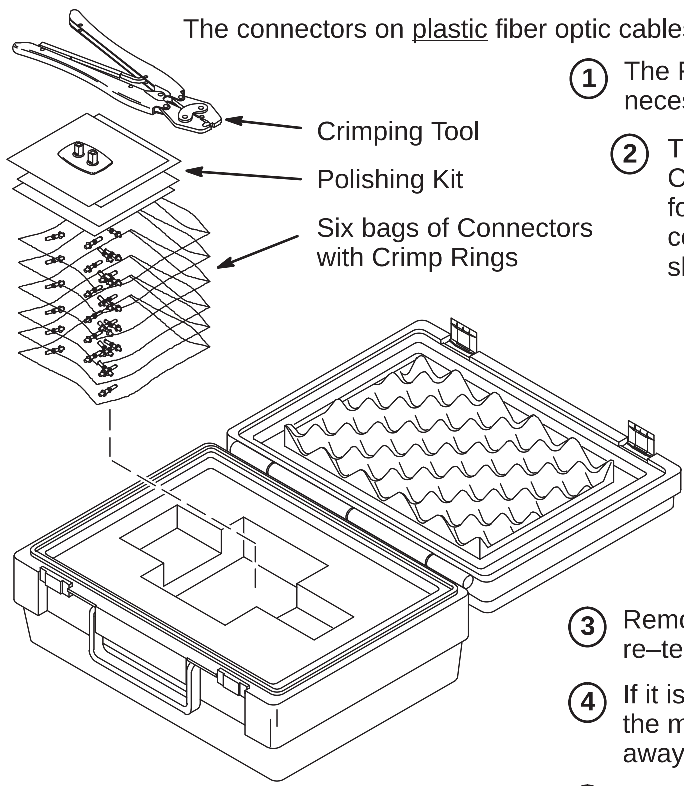
- ❑ **B1 – Fiber Optic Cable Termination Kit**
- ❑ **B2 – Stripping Fiber Optic Outer Jacket**
- ❑ **B3 – Crimping Connector to Fiber Optic Cable**
- ❑ **B4 – Trimming Excess Fiber Optic from Connector**
- ❑ **B5 – Polishing Fiber Optic End**

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

❑ B1 – FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.



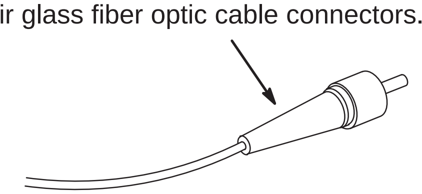
Crimping Tool

Polishing Kit

Six bags of Connectors with Crimp Rings

1. The Fiber Optic Repair Kit, 46–301450G1, contains all the necessary materials to replace a damaged connector.
2. The "Spares Kit", delivered with the Shipping Collector, contains a 46–320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



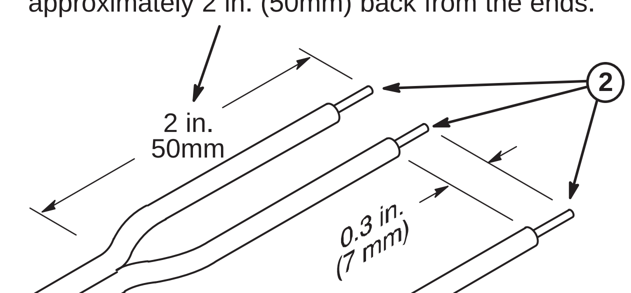
3. Remove the end of fiber optic cable to be re–terminated from magnet room if possible.
4. If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

1. For Duplex Cables, seperate the two fibers approximately 2 in. (50mm) back from the ends.

Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

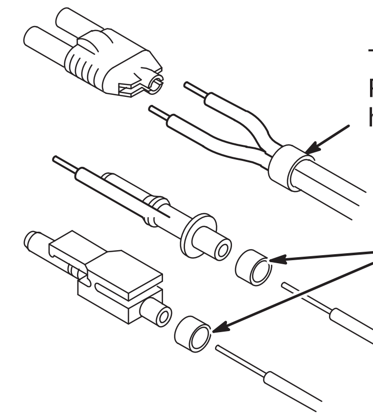
2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.



❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

1. Place the Crimp Ring and Connector over the end of the cable.

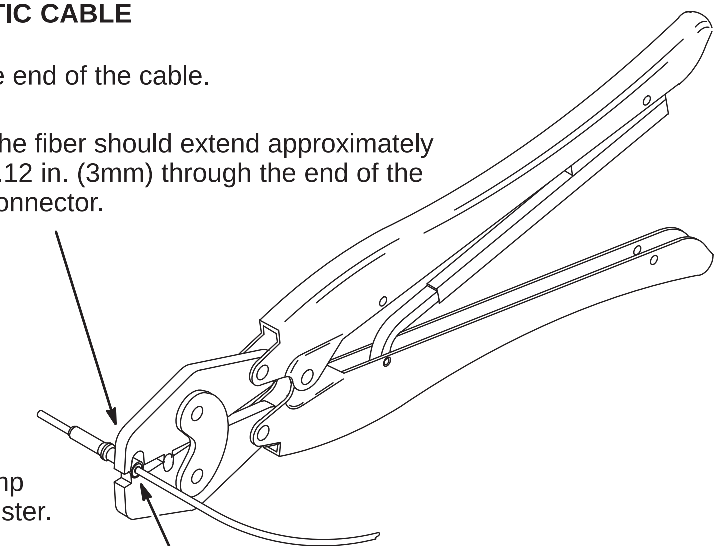
Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.



The Duplex Crimp Ring is shiny and has a thinner wall

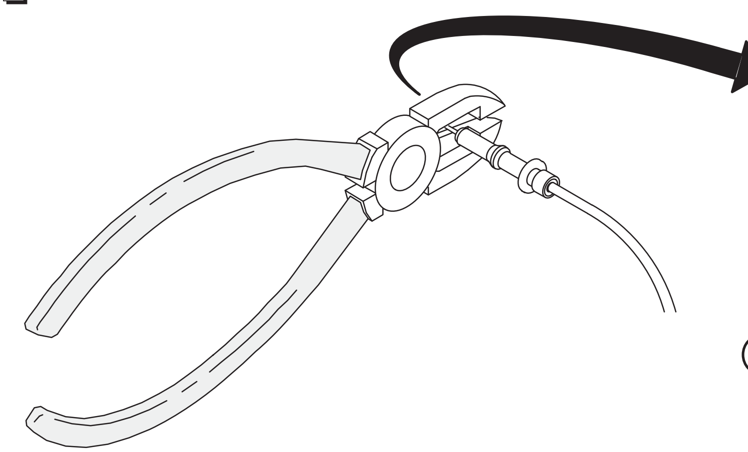
The Simplex Crimp Ring has a dull luster.

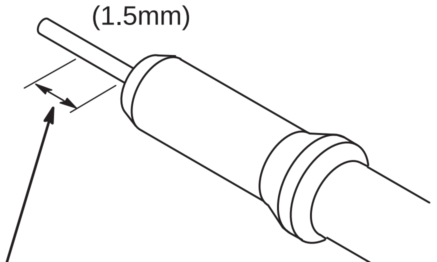
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

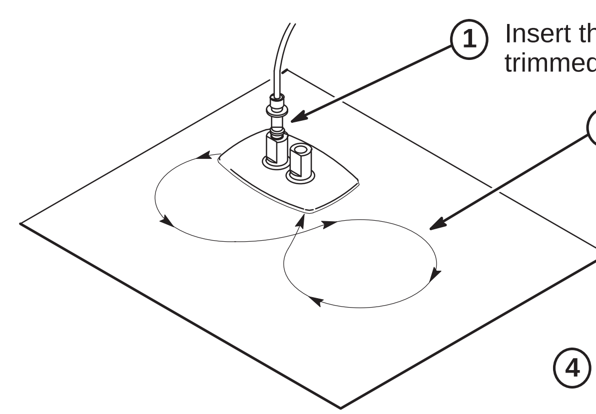
❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR





1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1–1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

SIGNA EXCITE II FIBER OPTIC CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

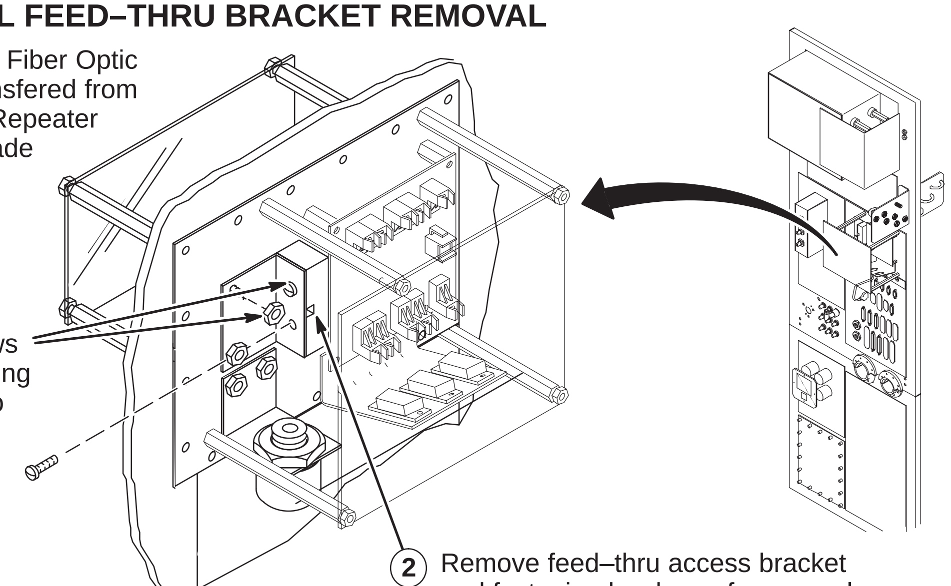
- ❑ **A1 – Repeater Panel Feed–thru Bracket Removal**
- ❑ **A2 – Routing Run 711/712 in Magnet Room**
- ❑ **A3 – Connecting Magnet Room Run 711/712 to Repeater Panel**
- ❑ **A4 – Route/Connect Equipment Room Run 1045 to Repeater Panel**
- ❑ **A5 – Cable Map of Fiber Optic Runs 709, 711/712, 1032, 1033, 1034, 1044, and 1045**

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

❑ A1 – REPEATER PANEL FEED–THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transfered from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



1. Unfasten two #10–32 screws and two #10–32 nuts securing feed–thru access bracket to J91 feed–thru block.
2. Remove feed–thru access bracket and fastening hardware from panel.

❑ A2 – ROUTING RUN 711/712 IN MAGNET ROOM

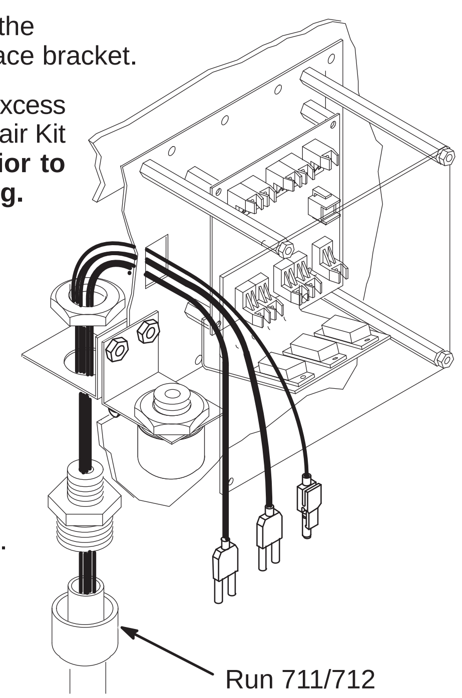
1. Move Run 711/712 (46–328079G9) to the Magnet Room side of the Penetration Panel. The "PP1–J91" end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
3. Route ends of individual fiber–optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
4. Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

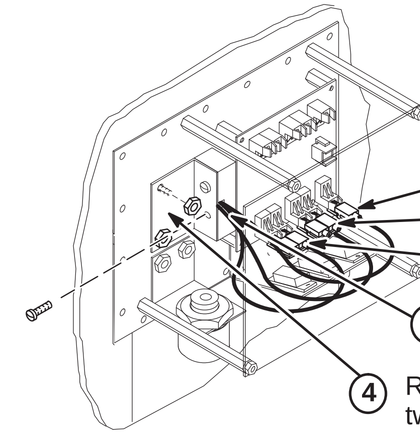
Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC–II.

5. Route Run 711/712 to the front right side of the magnet (SRI location). .
6. Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC–II.



❑ A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

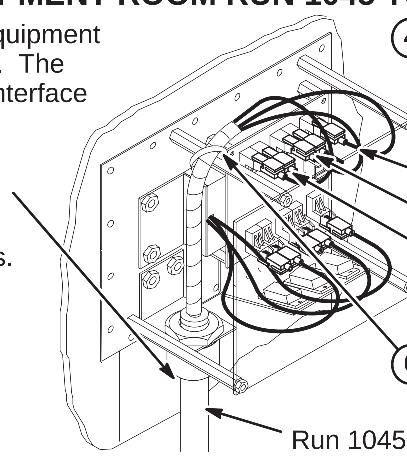
Note: Take care to maintain the minimum 2 inch bend radius.



1. Adjust route of individual fiber optic runs to maintain large radius.
2. Connect Runs 712–1, 711–1, and 711–2 as shown.
3. Gather fiber optic runs and route through wave guide notch in feed–thru block.
4. Replace feed–thru access bracket and fasten with two #10–32 screws and two #10–32 nuts removed in step A1.

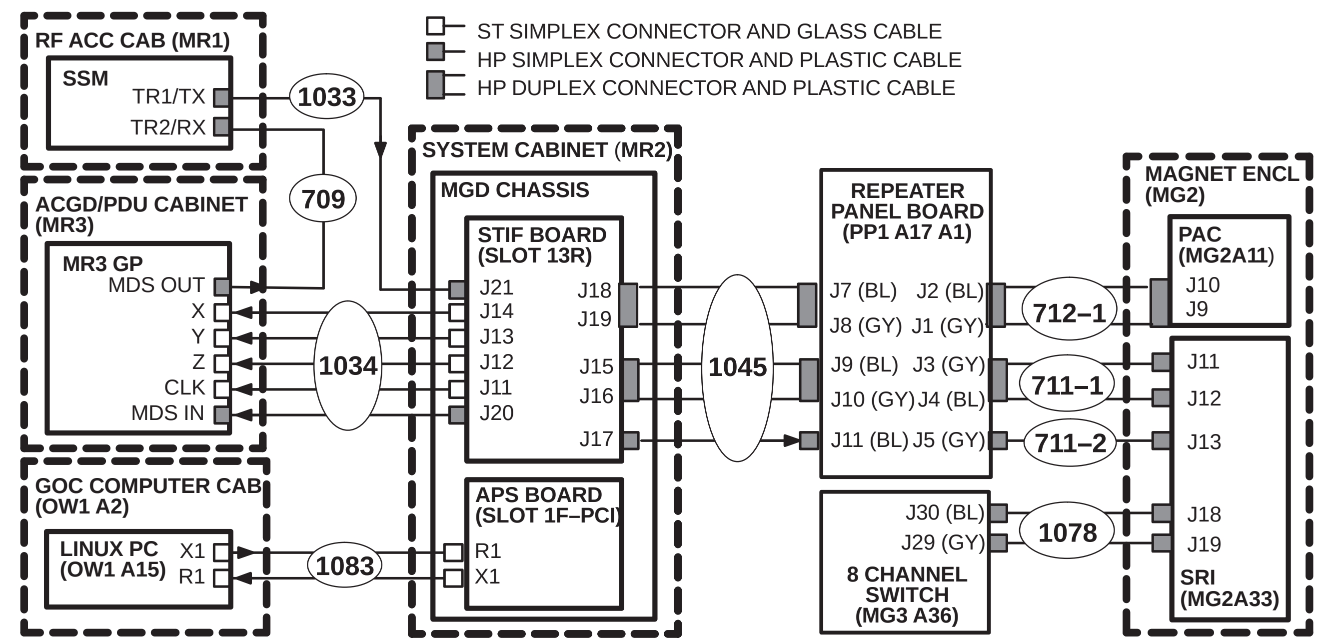
❑ A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 1045 TO REPEATER PANEL

1. Move Run 1045 (2331846) to the Equipment Room side of the Penetration Panel. The "PP1–A17" end will be attached to interface bracket.
2. Attach Run 1045 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
3. Route other end of Run 1045 to System Cabinet. Refer to Direction 2338502 (Tab 8) for connecting fiber optics in System Cabinet.
4. Adjust route of individual fiber optic runs to maintain large radius.
5. Connect Run 1045 as shown.
6. Adjust position of protective spiral wrap and secure with tie–wrap.



❑ A5 – CABLE MAP OF F/O RUNS 709, 1033, 1034, 1045, 1078, AND 1083

1. Unroll coiled fiber optic conduits for Runs 709, 1033, 1034, 1045, 1078, and 1083 along intended routes. Connection of Runs is shown in diagram below. Plan for storage of excess Run 1034 conduit of as it cannot be reterminated.



INSTALLATION STEPS SUMMARY

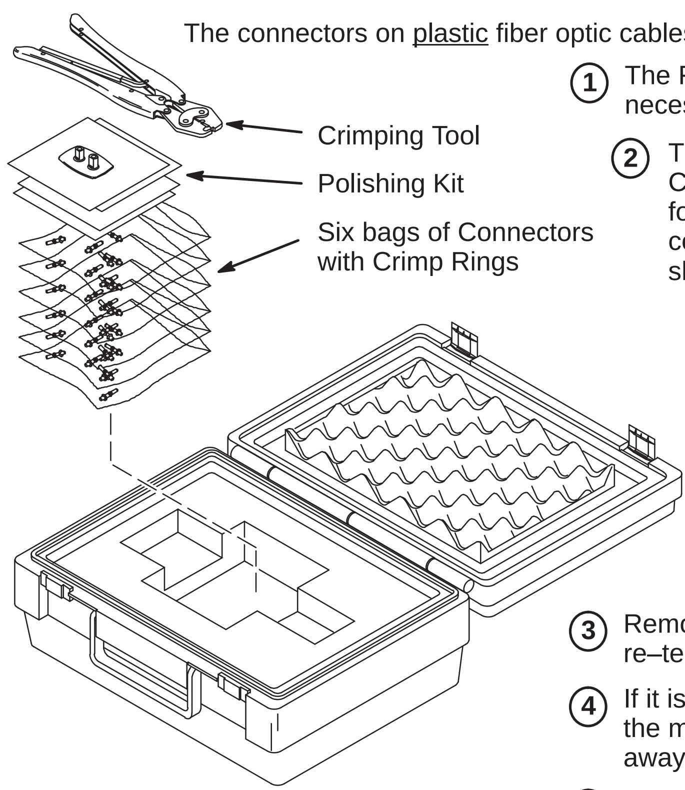
- ❑ **B1 – Fiber Optic Cable Termination Kit**
- ❑ **B2 – Stripping Fiber Optic Outer Jacket**
- ❑ **B3 – Crimping Connector to Fiber Optic Cable**
- ❑ **B4 – Trimming Excess Fiber Optic from Connector**
- ❑ **B5 – Polishing Fiber Optic End**

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

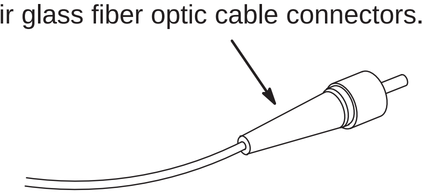
❑ B1 – FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.



1. The Fiber Optic Repair Kit, 46–301450G1, contains all the necessary materials to replace a damaged connector.
2. The "Spares Kit", delivered with the Shipping Collector, contains a 46–320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

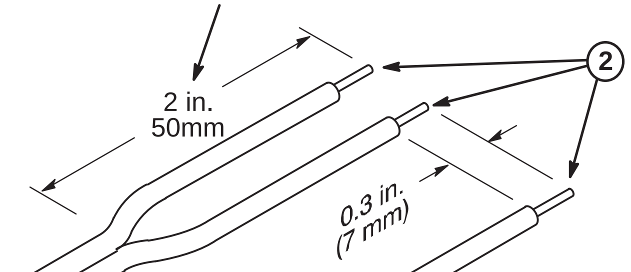
Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



3. Remove the end of fiber optic cable to be re–terminated from magnet room if possible.
4. If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

1. For Duplex Cables, seperate the two fibers approximately 2 in. (50mm) back from the ends.



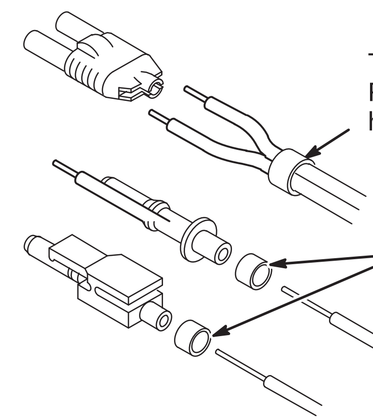
Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.

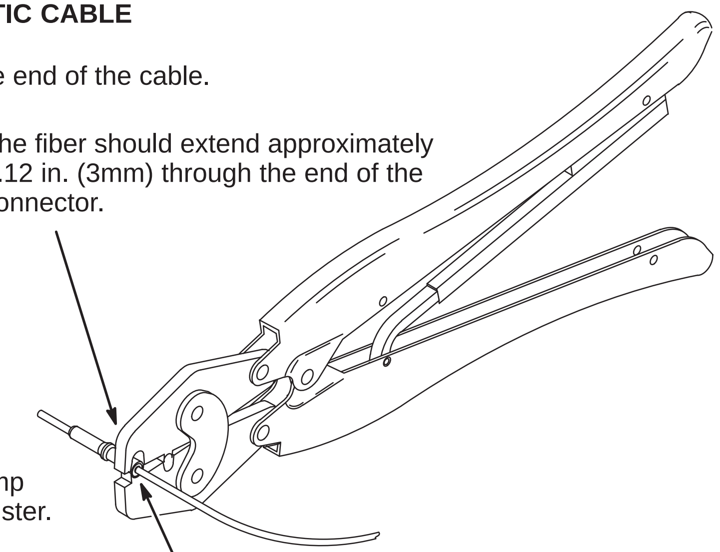
❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

1. Place the Crimp Ring and Connector over the end of the cable.

Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

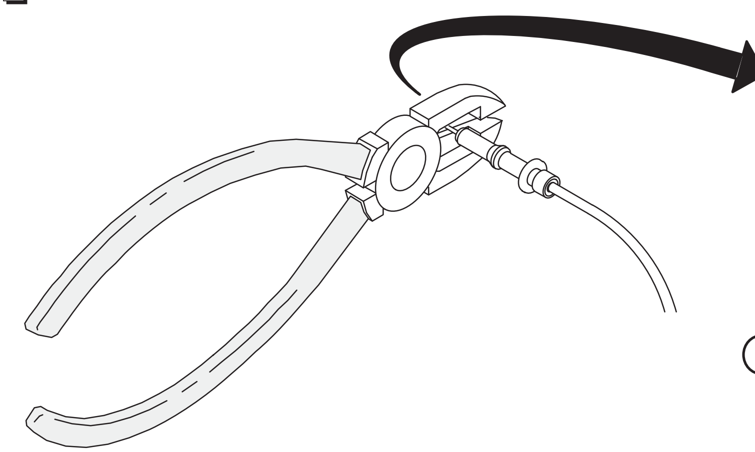


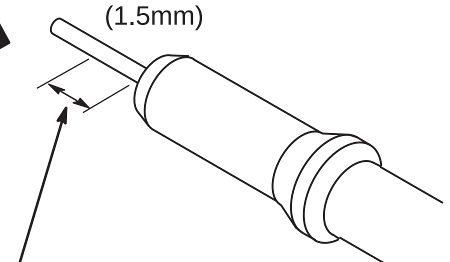
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

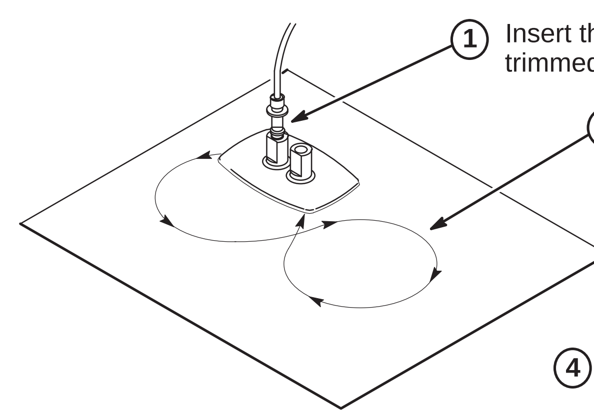
❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR





1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1–1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

INSTALLATION STEPS SUMMARY

- ❑ **A – System Power Cable Installation**
- ❑ **B – Power Cable Re–termination**
- ❑ **C – Power Cable Pre–connect checks**
- ❑ **D – Power Cable Connections Details**

❑ A – SYSTEM POWER CABLE INSTALLATION

1. Verify that all power and ground cables for the Signa System with ACGD are present. Refer to Section 2–1, LOCATING AND UNPACKING CABLES.
2. **Lengths of PDU Power cables for:**

Lengths of PDU Power cables for:	**FIXED SITE**	**MOBILE**	**T/R**
Run 966, SSM	35.0 ft. (10.7M)	35.0 ft. (10.7M)	48.0 ft. (14.6M)
Run 967, SRF Cabinet	35.5 ft. (11.1M)	29.5 ft. (9.0M)	49.0 ft. (15.0M)
Run 968, System Cabinet	35.0 ft. (10.7M)	25.0 ft. (7.6M)	45.0 ft. (13.7M)
Run 969, Operator Workspace Computer	65.5 ft. (20.0M)	33.0 ft. (10.0M)	33.0 ft. (10.0M)
Run 970, Water Chiller (Fixed–Site) **(Note 1)**	82.0 ft. (25.0M)		
Run 973, Heat Exchanger (Mob & T/R) **(Note 1)**		82.0 ft. (25.0M)	82.0 ft. (25.0M)
Run 989, Twin Accessory Cabinet **(Note 1)**	25.0 ft. (7.62M)	25.0 ft. (7.62M)	25.0 ft. (7.62M)

Note 1: Runs 970 and 973 are for Non–Twin*Speed* systems, Run 989 is only used on Twin*Speed.*

If insufficient room exists to store excess cables:
Plan for storage location if cables are not being cut to length. Proceed to Step **C**.

If insufficient room exists to store excess cables:
They can be cut shorter and re–terminated at the PDU. Complete Step **B**. Then proceed to Step **C**.

3. Route all power cables per applicable System Interconnect Diagram. Unroll cables to lay without twists or kinks from cabinet to PDU through provided troughs, conduit, or ducts.

❑ B – POWER CABLE RE–TERMINATION

1. Cut power and ground cables to length as necessary. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans.
2. Cut and trim insulation for connection of power and ground cable conductors to connectors at PDU.
3. Transfer color code tape, run–number labels, and applicable wire marking labels as each cable is cut to size.

❑ C – POWER CABLE PRE–CONNECT CHECKS

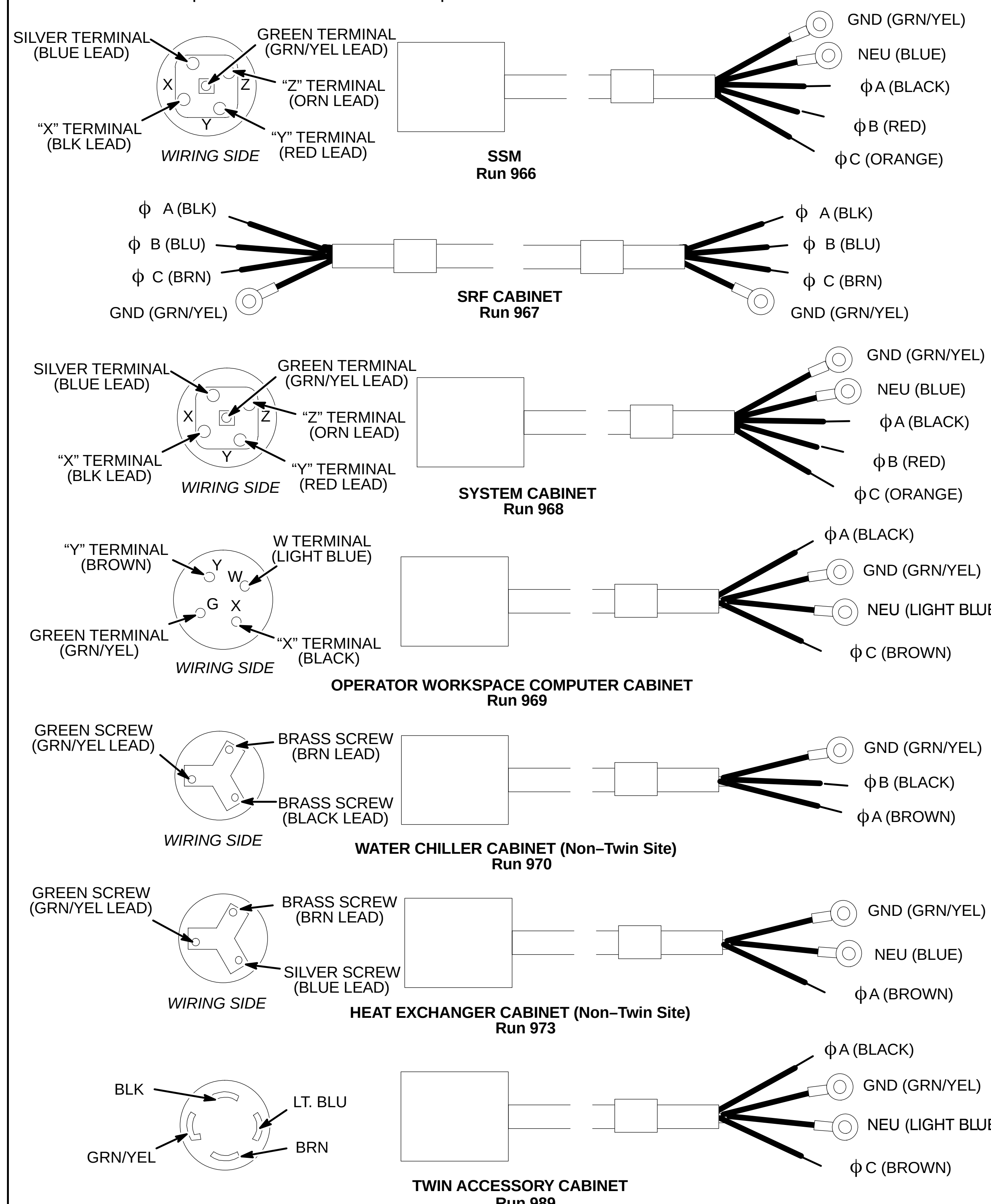
Checks for opens, shorts, and crossed connections are made for field re–terminated power cables prior to connection to PDU and cabinets. This does not provide ground resistance checks for cabinets. Ground resistance checks are in MR CD–ROM, Set Up and Calibration: System, GROUNDING AND LEAKAGE CURRENT CHECKS.

1. Use the ohmmeter and the cable plug connector Illustration to check each power cable for the following:
   - Open circuit between cable neutral and cable ground.
   - Open circuit between cable neutral and each line on the cable.
   - Open circuit between cable ground and each line on the cable.
   - Less than 1 ohm between cable neutral and the light blue (or white) conductor.
   - Less than 1 ohm between cable ground and the green/yellow conductor.
2. Any out–of–spec conditions must be immediately corrected. See Procedure **D** for plug connection details. After all power cables have been satisfactorily checked out, continue installation.

❑ D – POWER CABLE CONNECTIONS DETAILS

Note

After performing cable pre–connect checks, **avoid potential problems by taping up exposed ACGD Gradient Cabinet cables** or immediately install ACGD/PDU Gradient Cabinet and connect terminals to power modules before PDU power is turned on.



MAGNET ENCLOSURE INSTALLATION INSTRUCTIONS

1–1 Tab 2, Magnet Enclosure, contains installation instructions for the following:

Use only the applicable Magnet Enclosure Installation Instruction Directions

- *DIRECTION 2398604* – EXCITE Fixed & T/R WideOpen Enclosure Installation Instructions
- *DIRECTION 2223857* – Fixed & T/R LCC Magnet WideOpen Enclosure Installation Instructions
- *DIRECTION 2223858* – Mobile LCC Magnet WideOpen Enclosure Installation Instructions

GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.
- A 1–ounce tube of Permatex "Anti–Seize" lubricant (2119594) is provided. Follow instructions on back of tube. "Anti–Seize" lubricant is to be applied to threads of stainless–steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

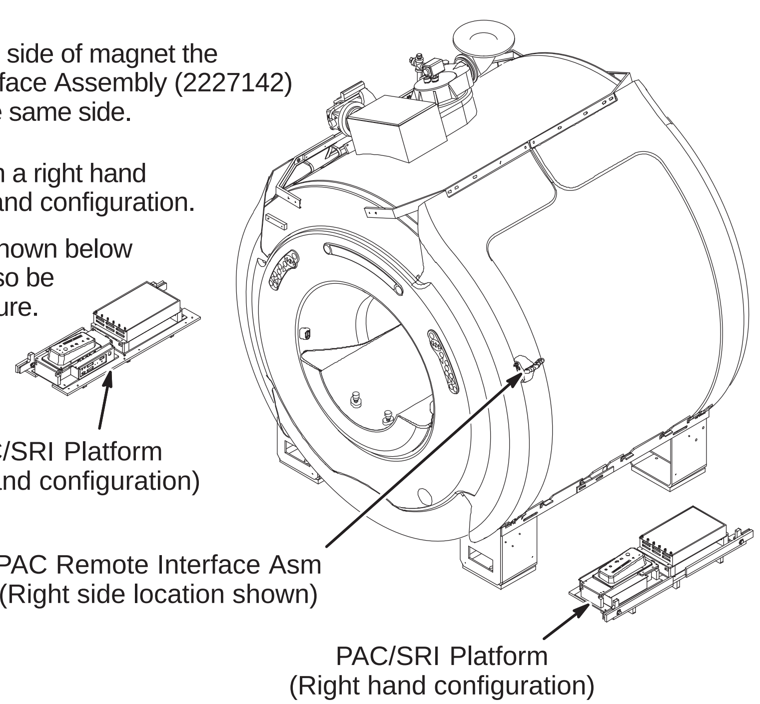
LIST OF EFFECTIVE PAGES

INSTALLATION PROCEDURES SUMMARY

- ❑ **A1 – Determine PAC/SRI Platform and PAC Remote Interface Location**
- ❑ **A2 – PAC/SRI Platform Left Hand Configuration (If Required)**
- ❑ **A3 – Location of Magnet Ground Cable**

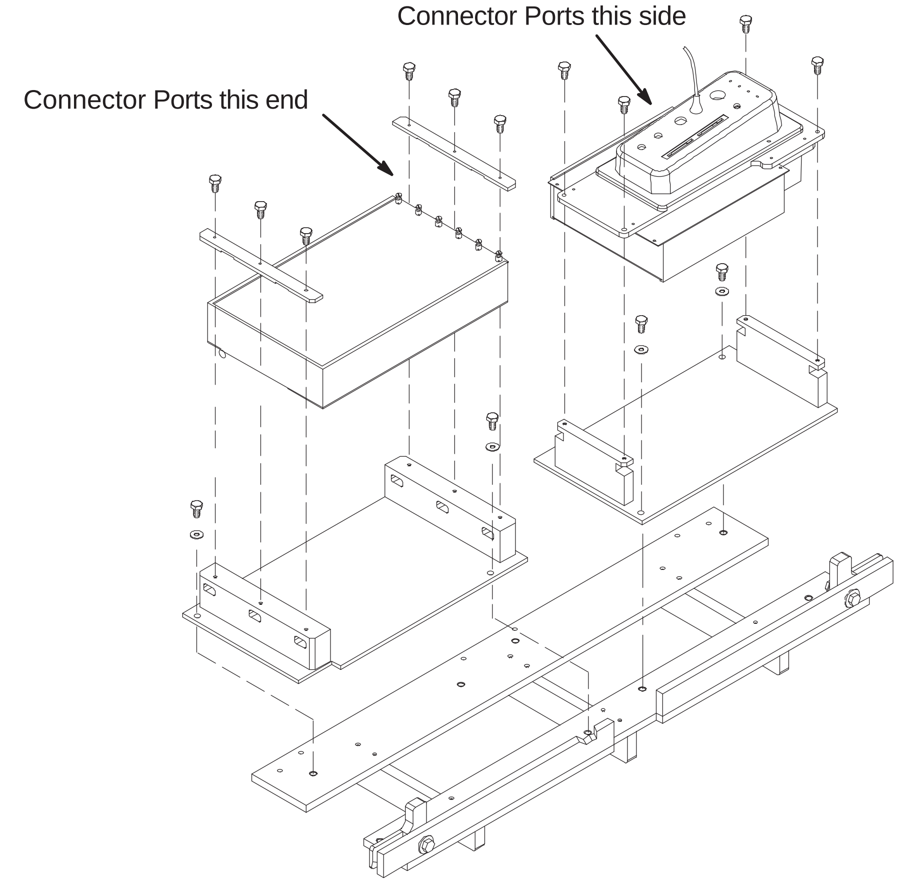
❑ A1 – DETERMINE PAC/SRI PLATFORM AND PAC REMOTE INTERFACE LOCATION

1. Consult with customer to determine which side of magnet the PAC/SRI Platform and PAC Remote Interface Assembly (2227142) should be located. They have to be on the same side.
2. The PAC/SRI Platform arrives at the site in a right hand configuration.See Procedure **A2** for left hand configuration.
3. The PAC Remote Interface Assembly is shown below installed in the right side location. It can also be mounted on the left side of magnet enclosure.

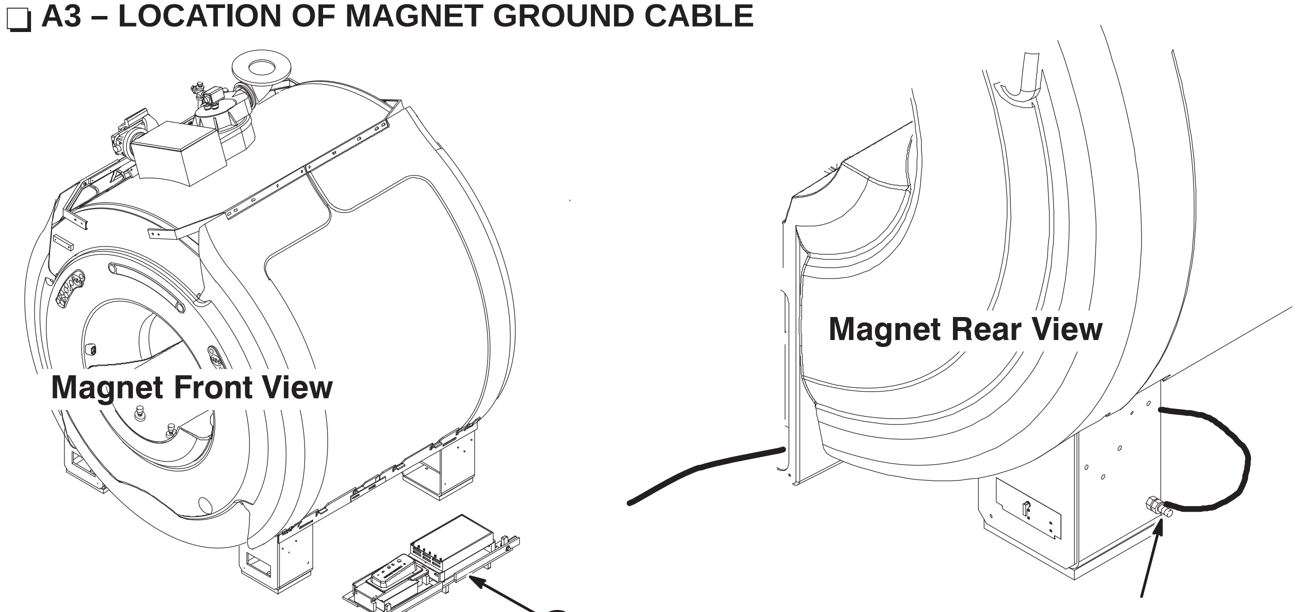


❑ A2 – PAC/SRI PLATFORM LEFT HAND CONFIGURATION (IF REQUIRED)

1. For left hand configuration, reassemble PAC/SRI as shown.



❑ A3 – LOCATION OF MAGNET GROUND CABLE

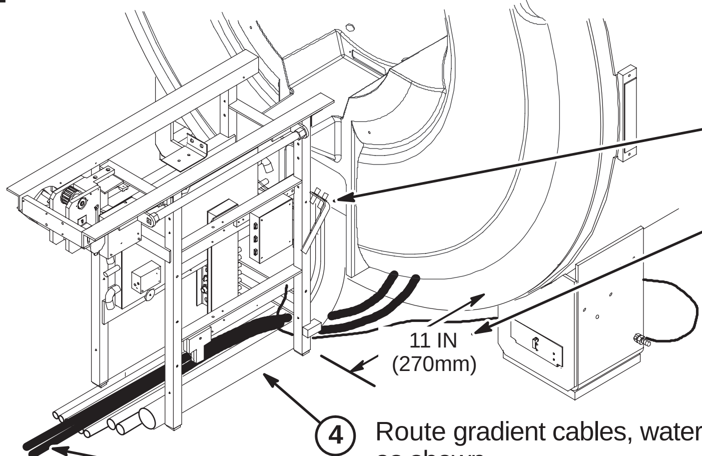


1. If PAC/SRI Platform is located on right side of magnet, then the Magnet ground cable , RUN 040, has to be connected to the opposite side rear left leg as shown above. The other end of Run 040 is connected to the RF Screen Room common ground.
2. If PAC/SRI Platform is located on left side of magnet, then the Magnet ground cable has to be connected to the opposite side rear right leg.

INSTALLATION PROCEDURES SUMMARY

- ❑ **B1 – Location of Rear Pedestal**
- ❑ **B2 – Routing and Connecting Gradient Output Cables to Runs 762, 763, and 764**
- ❑ **B3 – Route/Connect Runs 257/746, 262, 743, 744, and 780**
- ❑ **B4 – Route/Connect Runs 258/747, 318, and Drive Cables**

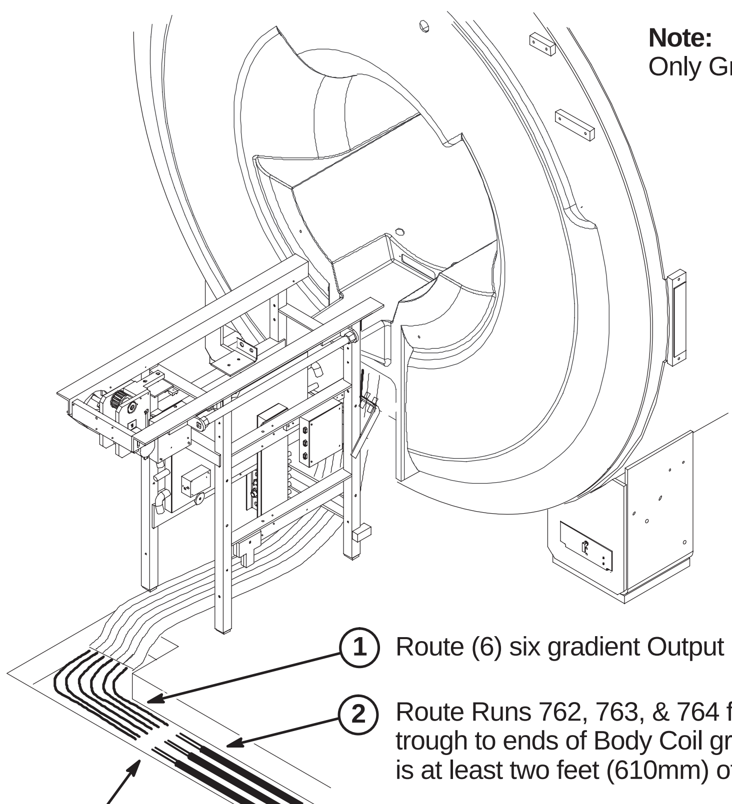
❑ B1 – LOCATION OF REAR PEDESTAL



Note:
Cables and hoses can be routed to exit out of rear or side of Rear Pedestal.

1. For **CV/*i*** install bracket on other side.
2. Center Rear Pedestal on Rear Endbell and locate approximately 11 inches (270mm) from face of magnet.
3. Route Ground cable from Rear Pedestal under magnet and connect to same Ground Stud as Magnet ground.
4. Route gradient cables, water lines, gas lines, and 2" air hose under Rear Pedestal as shown.
5. Route and connect coolant hosees from Water Chiller to 10 ft. (300cm) coolant hoses attached to Body Coil.
6. Route all other cables on top of the above cables and hoses.

❑ B2 – ROUTING AND CONNECTING GRADIENT OUTPUT CABLES



Note:
Only Gradient Cables are shown for clarity.

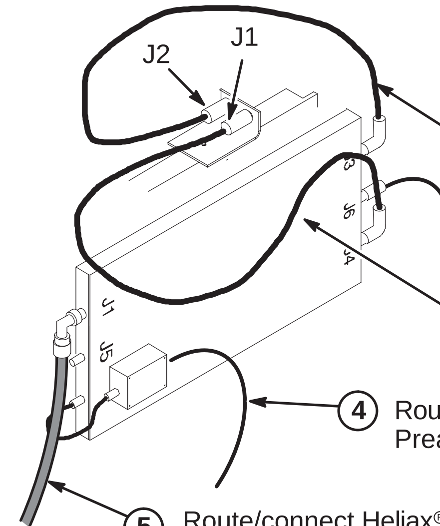
1. Route (6) six gradient Output Cables from Body Coil to cable trough.
2. Route Runs 762, 763, & 764 from Penetration Panel through cable trough to ends of Body Coil gradient cables. Make sure that there is at least two feet (610mm) of slack before cutting to length.
3. Refer to Direction 2286976 in Tab 1 for connecting Output cables from Body Coil to Runs 762, 763, & 764.

❑ B3 – ROUTE/CONNECT RUNS 257/746, 262, 743, 744, AND 780

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

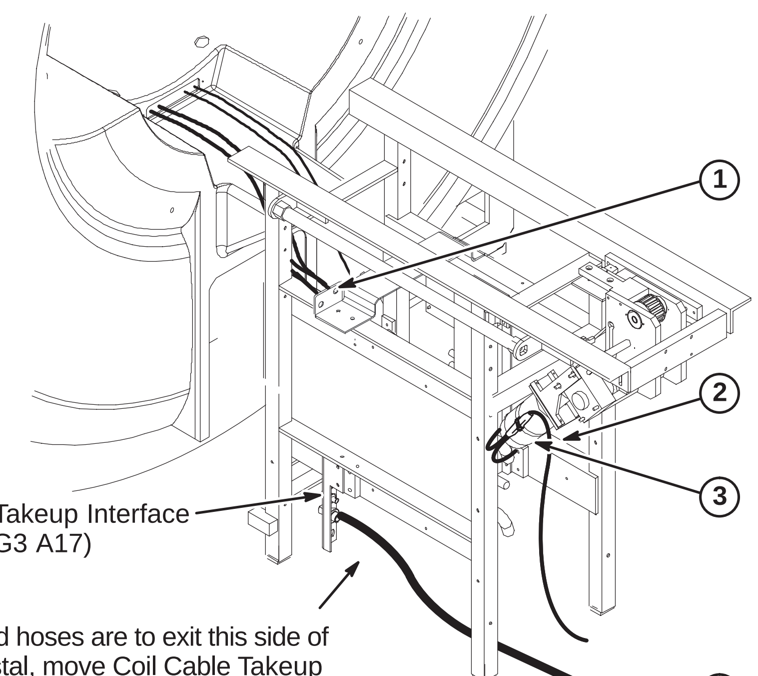
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

1. Route and connect Run 744 from J2 on bracket to J3 on the RF Splitter.
2. Route/connect Run 780 from J74 on Pen Panel to J6 on RF Splitter.
3. Route and connect Run 743 from J1 on the bracket to J4 on the RF Splitter.
4. Route/connect Run 262 from J8 on Pen Panel to Body Coil Preamplifier. (For MGD, connect to Receive Filter on Preamp)
5. Route/connect Heliax® to J1 on RF Splitter:
   Run 257 (1.5T, 7/8") or
   Run 746 (1.0T, 1/2")

❑ B4 – ROUTE/CONNECT RUNS 258/747, 318, AND DRIVE CABLES



Coil Cable Takeup Interface
(MG3 A17)

Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.

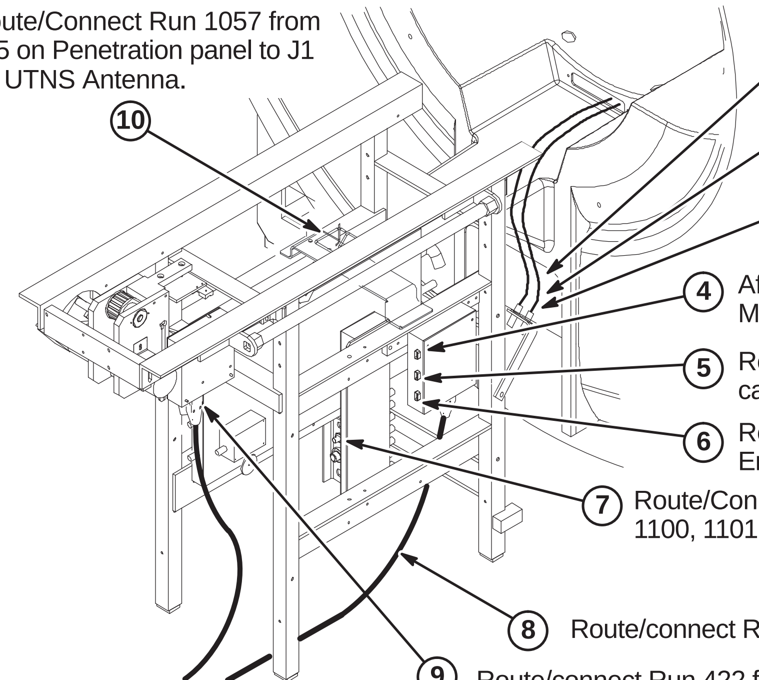
If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

1. Route I & Q RF Drive Input cables from RF Coil and connect as marked to J1 and J2 on bracket.
2. Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).
3. Ty–wrap cable to motor to prevent movement.
4. Route/connect to J1 on Coil Cable Takeup Interface:
   Run 258 (1.5T, 1/2") or
   Run 747 (1.0T, 3/8")

INSTALLATION PROCEDURES SUMMARY

- ❑ **C1 – Route/Connect Cables at Rear Pedestal**
- ❑ **C2 – Install Bore Lamp Assembly**
- ❑ **C3 – Route/Connect Runs 257/746, 262, 743, 744, and 780**
- ❑ **C4 – Install Blower Box and Route/Connect Air Hoses**

❑ C1 – ROUTE/CONNECT CABLES AT REAR PEDESTAL

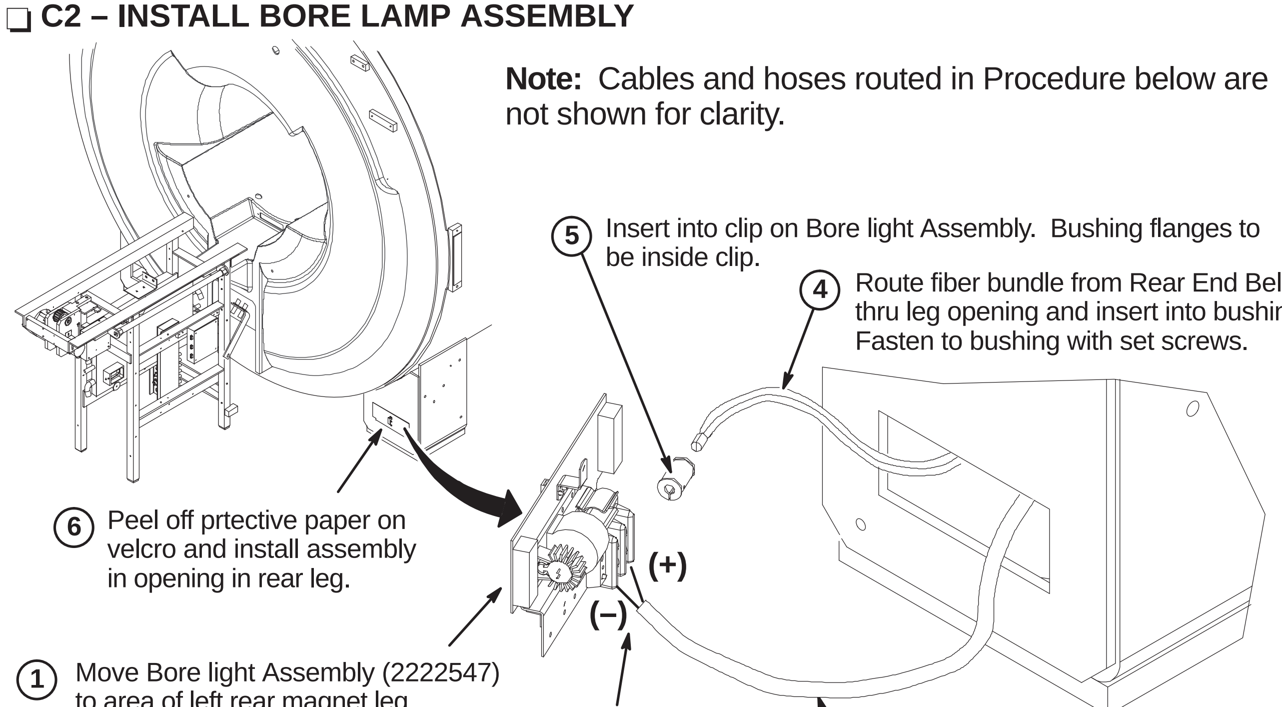


10. Route/Connect Run 1057 from J85 on Penetration panel to J1 on UTNS Antenna.

1. Route and connect RF Bias cables from RF Coil to bracket as marked.
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. After Carriage cover is installed, connect Microphone cable from Carriage Cover to J7.
5. Route/connect Front End Bell microphone cable Run 414 to J9.
6. Route/connect speaker cable from Front End Bell to J8.
7. Route/Connect Runs 1095, 1096, 1097, 1098, 1099, 1100, 1101, and 1102 to J1 thru J9 on 8 Channel Switch.
8. Route/connect Run 282 from J12 on Pen Panel to J6.
9. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.

❑ C2 – INSTALL BORE LAMP ASSEMBLY

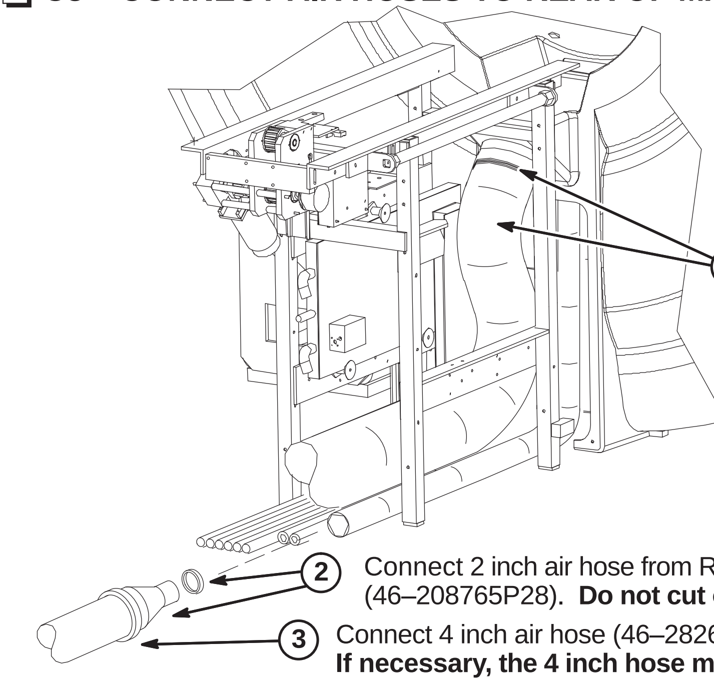
Note: Cables and hoses routed in Procedure below are not shown for clarity.



1. Move Bore light Assembly (2222547) to area of left rear magnet leg.
2. Connect the (+) and (–) wires of Run 300 to the terminal strip as shown.
3. Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
4. Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
5. Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
6. Peel off prtective paper on velcro and install assembly in opening in rear leg.

❑ C3 – CONNECT AIR HOSES TO REAR OF MAGNET

Note:
The following items are part of Blower Hose Kit, 2129412–3 or 2129412–4.

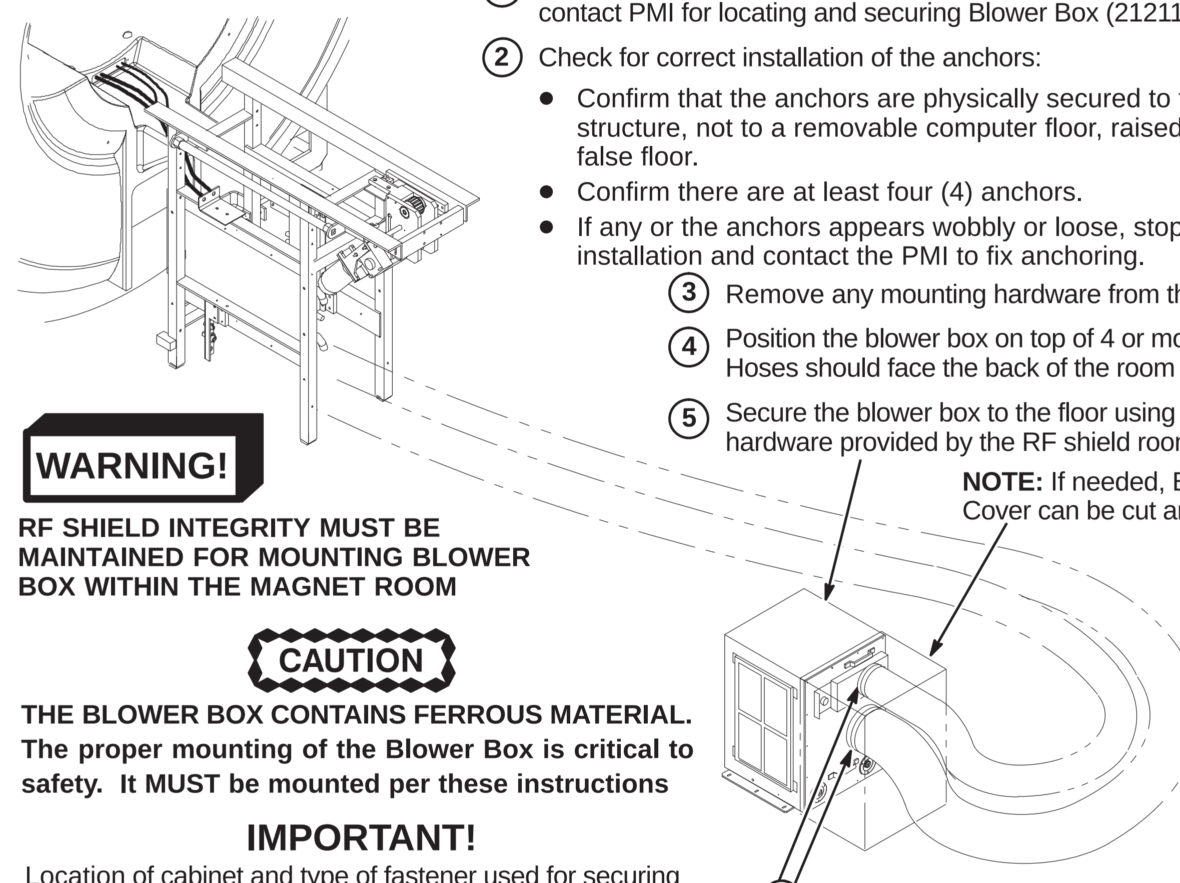


1. Connect 6 inch air hose (46–282666P12) to Rear End Bell with 6 inch clamp (46–208675P29).
2. Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). **Do not cut or reduce the length of the 2 inch hose.**
3. Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). **If necessary, the 4 inch hose may need to be cut to reduce length.**

❑ C4 – INSTALL BLOWER BOX AND ROUTE/CONNECT AIR HOSES

1. Refer to site design (architectural)/customer construction drawings or contact PMI for locating and securing Blower Box (2121199).
2. Check for correct installation of the anchors:
   - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
   - Confirm there are at least four (4) anchors.
   - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.
3. Remove any mounting hardware from the anchors.
4. Position the blower box on top of 4 or more anchors. Hoses should face the back of the room
5. Secure the blower box to the floor using the fastening hardware provided by the RF shield room vendor.

NOTE: If needed, Blower Hose Cover can be cut and modified.



6. Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:
   (1) 46–208765P34 4 inch Hose Clamp
   (1) 46–208765P29 6 inch Hose Clamp

WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

INSTALLATION PROCEDURES SUMMARY

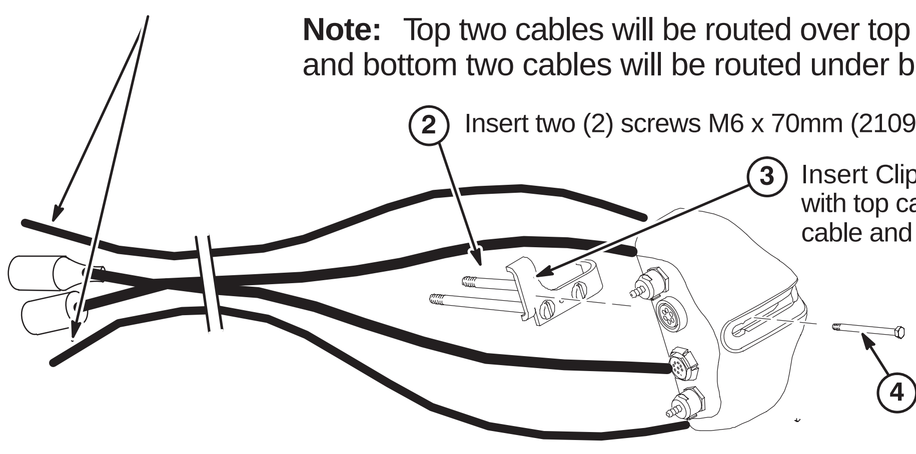
- ❑ **D1 – Prepare Cables and Tubes for Routing**
- ❑ **D2 – Position Remote PAC Interface Mounting Clip**
- ❑ **D3 – Route Remote PAC Interface Cables and Tubes**
- ❑ **D4 – Installing Remote PAC Interface to Arc Cover**
- ❑ **D5 – Install PAC Hanger Bar**

❑ D1 – PREPARE CABLES AND TUBES FOR ROUTING

Note:
Consult with customer for correct location of Remote PAC Interface assembly. It must be on the same side as the PAC/SRI Platform.

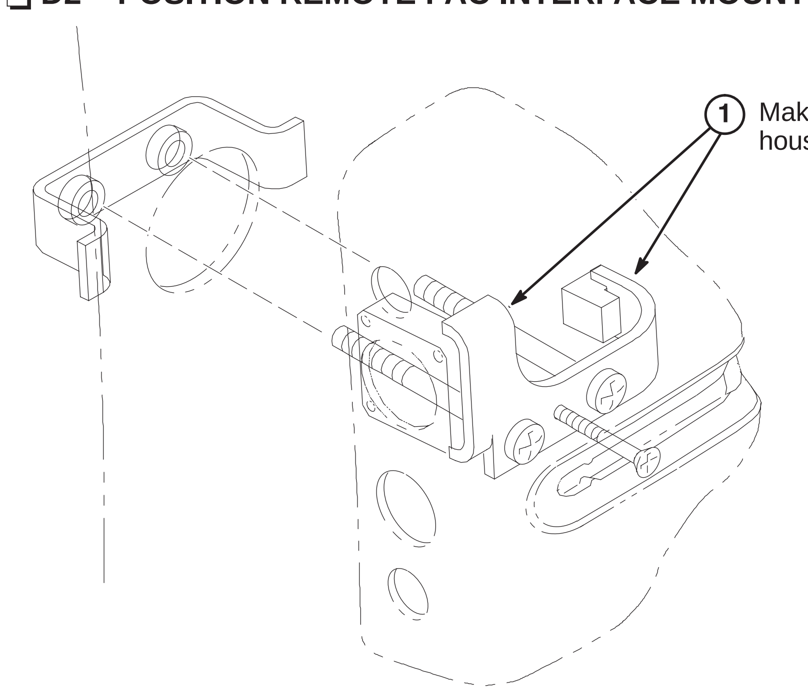
1. Apply tape around end of pneumatic tubes and mark "TOP" to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and "BOTTOM" to the tube connected to the lower connector.

Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.



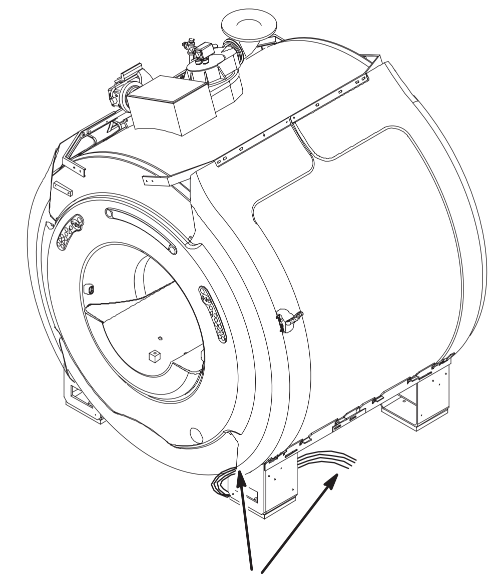
2. Insert two (2) screws M6 x 70mm (2109873–33) screws into Clip (2221023).
3. Insert Clip with screws into shell as shown with top cable and tube above Clip and bottom cable and tube below.
4. Hold in place by inserting provided M4 screw thru slot and turning about 3 or 4 turns into Clip.

❑ D2 – POSITION REMOTE PAC INTERFACE MOUNTING CLIP



1. Make sure each side of clip is mounted behind housing blocks.

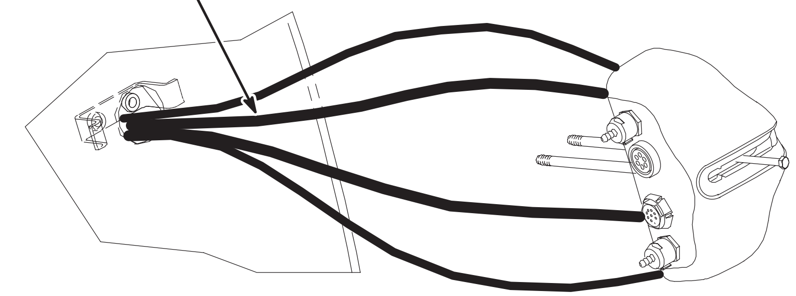
❑ D3 – ROUTE REMOTE PAC INTERFACE CABLES AND TUBES



Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

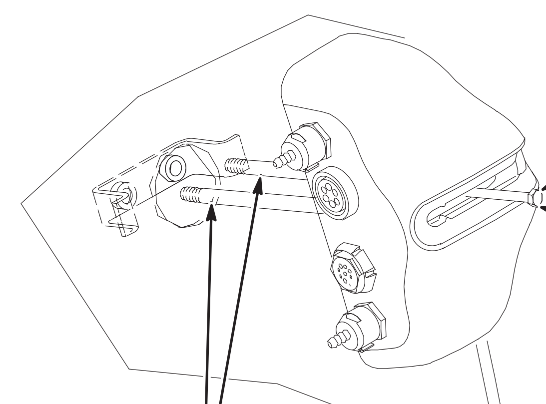
Note: Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.

1. **Insert ECG Cable first.** Route this thru pre–mounted foam tube (if available) located on inside of Arc cover.
2. Route Peripheral Pulse cable and tubes next.



3. Route around foot of magnet to PAC/SRI Platform location.

❑ D4 – INSTALLING REMOTE PAC INTERFACE TO ARC COVER

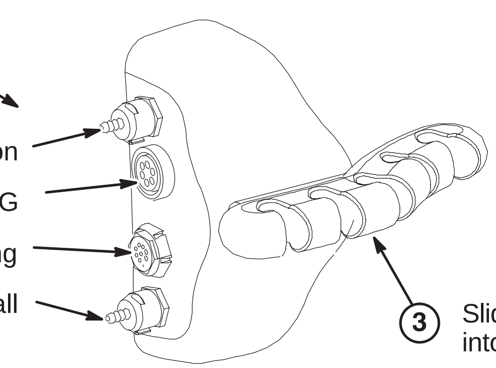


Note:
Routed cables and tubes not shown for clarity.

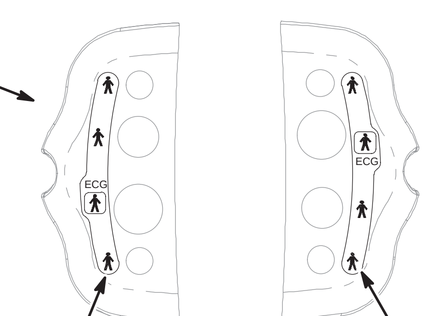
1. Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.
3. Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.
OR
Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in **T5** into place.
2. Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place

❑ D5 – INSTALL PAC HANGER BAR

1. Depending on left or right side installation of the PAC Remote Interface Asm on the magnet enclosure, attach the appropriate label that is provided with the kit.
2. Attach cables as indicated below.



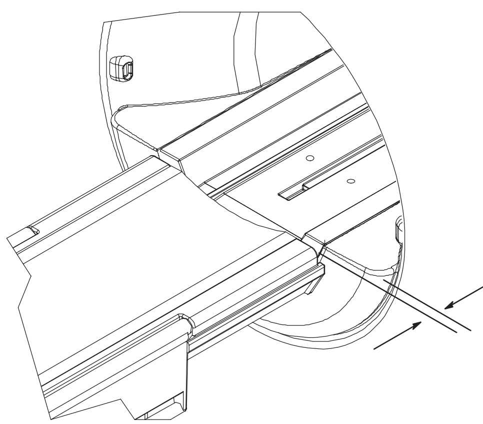
3. Slide PAC Hanger Bar (2221023) into position.



INSTALLATION PROCEDURES SUMMARY

- ❑ E1 – Install Dock Mounting Bracket and Dock
- ❑ E2 – Route and Connect Magnet Enclosure Front Area Cables
- ❑ E3 – Cable Connection to PAC Module

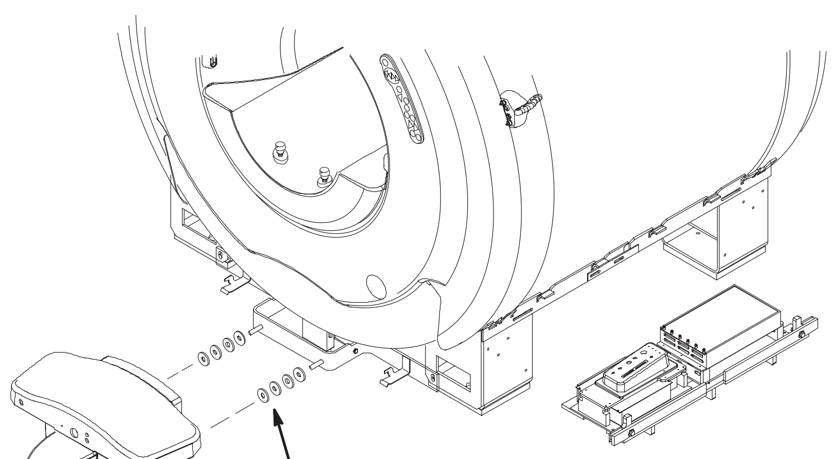
❑ E1 –INSTALL DOCK MOUNTING BRACKET AND DOCK



Note:
Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches ±0.25 (19.0±6.4mm). See Step 2 below.

WARNING!

DOCK CONTAINS FERROUS MATERIAL.
HANDLE WITH CARE DURING INSTALLATION.

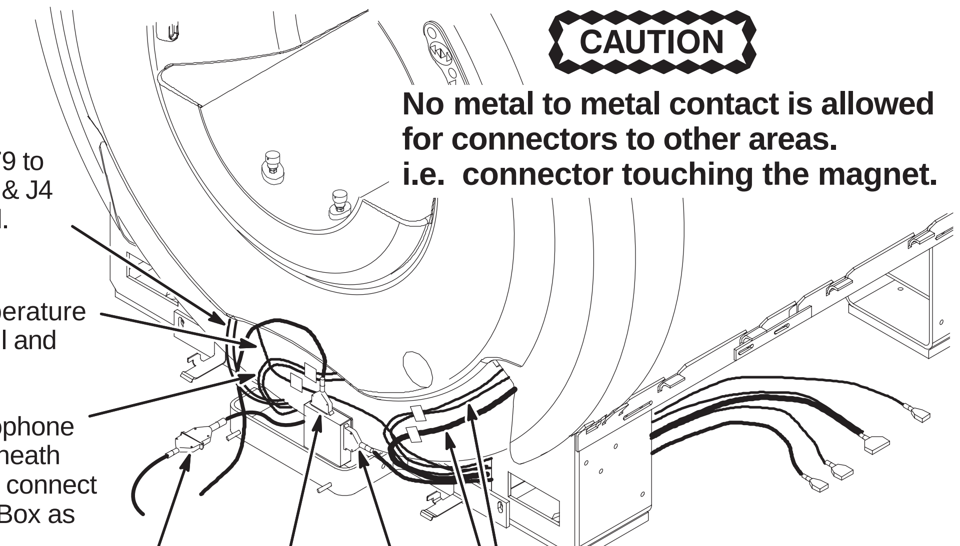


1. Start with two (2) 11mm washers on each side between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is acheived.

Note:
Floor stud installed by RF room vendor/mechanical contractor.

❑ E2 – ROUTE AND CONNECT MAGNET ENCLOSURE FRONT AREA CABLES

Note:
Dock Assembly not shown for clarity.



CAUTION

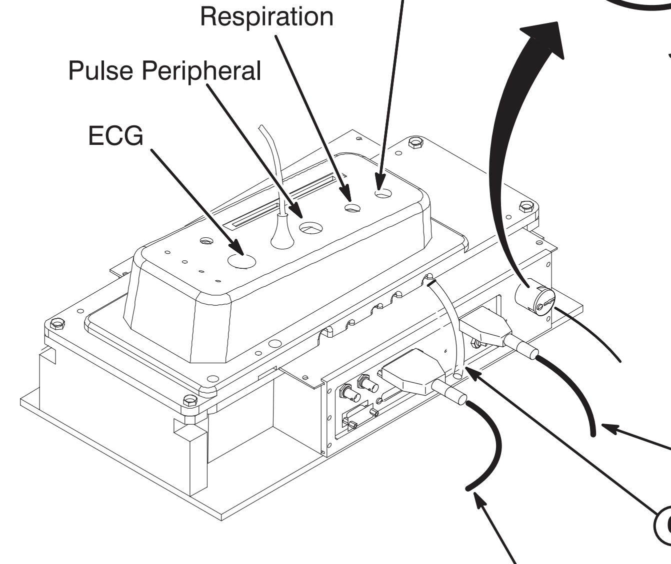
No metal to metal contact is allowed for connectors to other areas. i.e. connector touching the magnet.

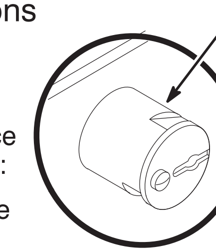
1. Route Runs 778 and 779 to front RF Bias cables J3 & J4 and connect as marked.
2. Route Run 749, Bore Temperature Sensor, from Front End Bell and connect ot J10 on SRI.
3. Route Front End Bell Microphone and Speaker cables underneath manet to ReaPedestal and connect to Patient Communication Box as marked.
4. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.
5. Connect Dock Cable to J2 on Dock Limit Box
6. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.
7. Route cables from Operator Display to area of PAC/SRI Platform and connect as marked to SRI and PAC.

❑ E3 – CABLE CONNECTION TO PAC MODULE

Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

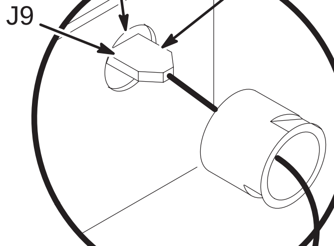
1. Connectfrom the Remote PAC Interface to the PAC Module the following below:



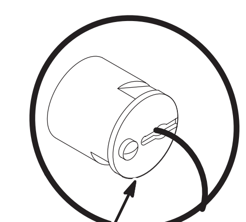


2. Remove shield cover by removing one screw and unscrew shield from module.

Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.



3. Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.



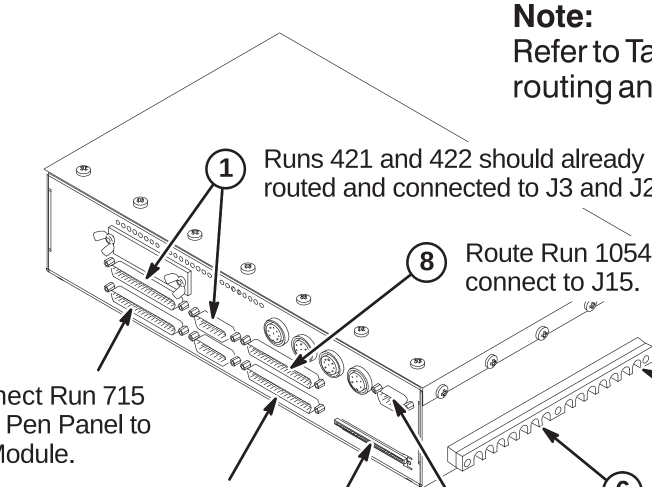
4. Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.
5. Connect Run 716
6. Connect Pneumatic Tubing from Pen Panel to tubing extension on PAC Module.
7. Connect cable from J5 on Operator Display to J4 on PAC Module.

INSTALLATION PROCEDURES SUMMARY

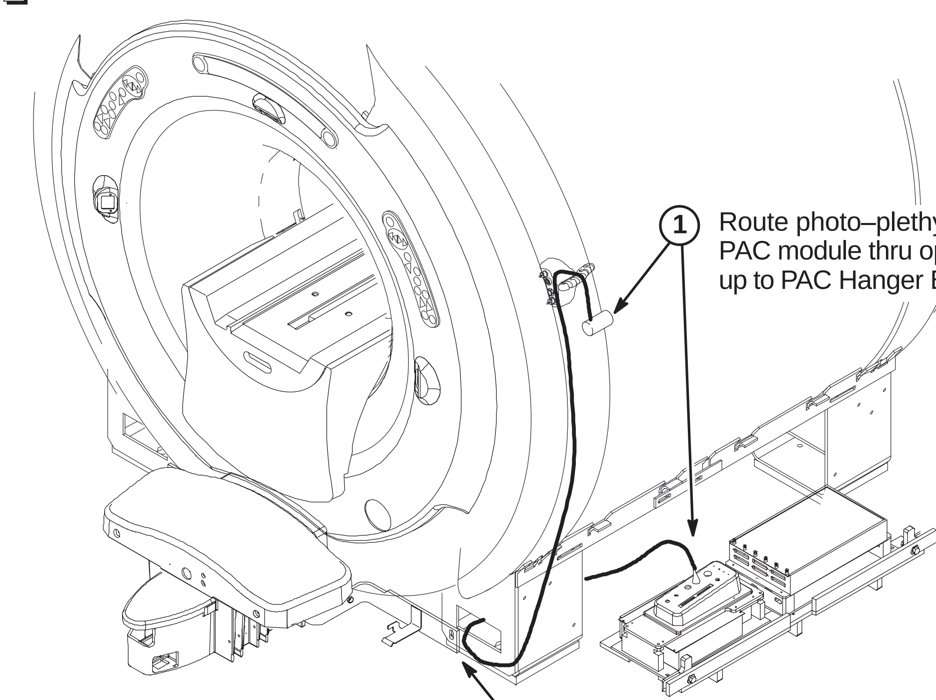
- ❑ **F1 – Connect Cables to SRI Module**
- ❑ **F2 – Routing of Photo–Plethysmograph Cable**
- ❑ **F3 – Route and Tie–wrap Cables to PAC/SRI Platform**
- ❑ **F4 – Install PAC/SRI Platform under Magnet**

❑ E1 – CONNECT CABLES TO SRI 2 MODULE

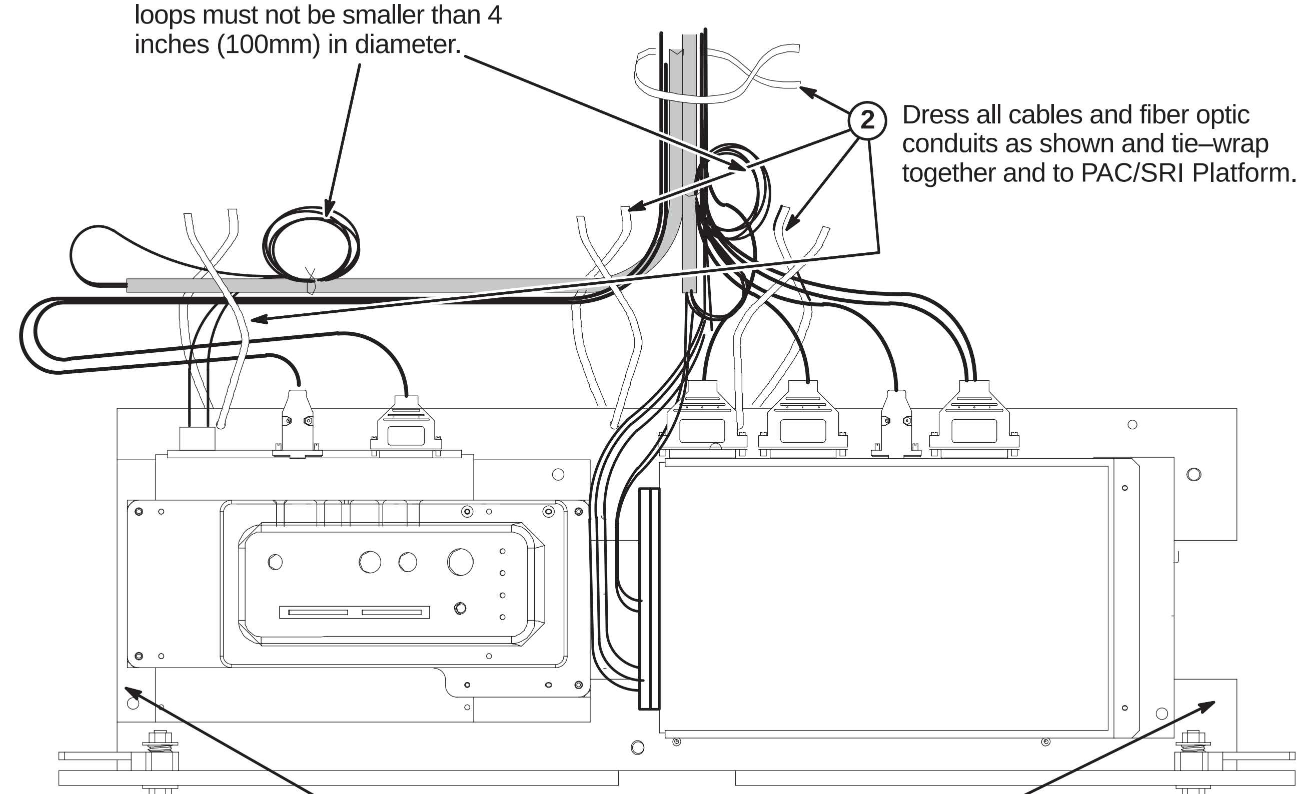
Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.



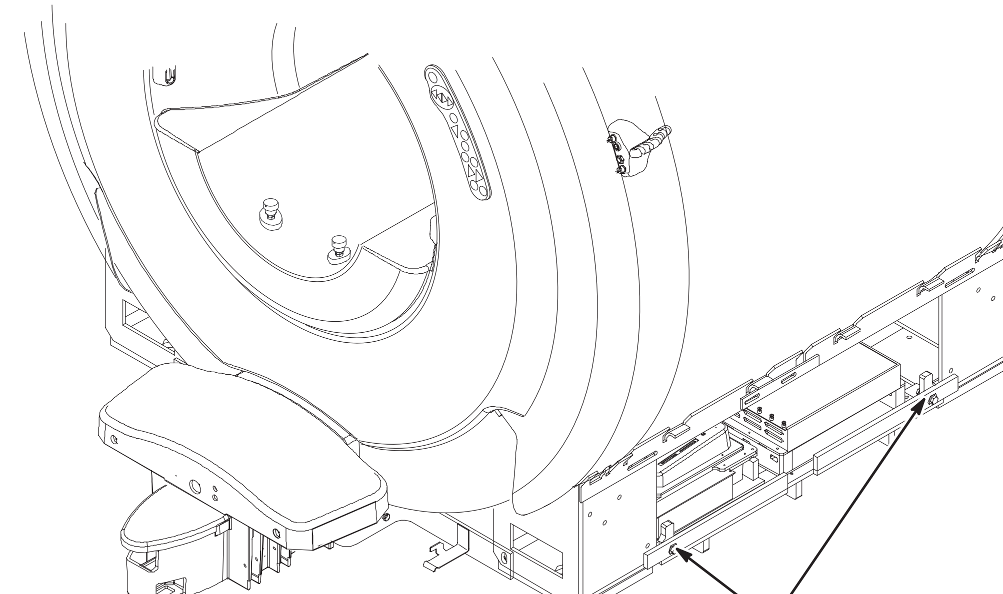
❑ E2 – ROUTING OF PHOTO–PLETHYSMOGRAPH CABLE



❑ F3 – ROUTE AND TIE–WRAP CABLES TO PAC/SRI PLATFORM



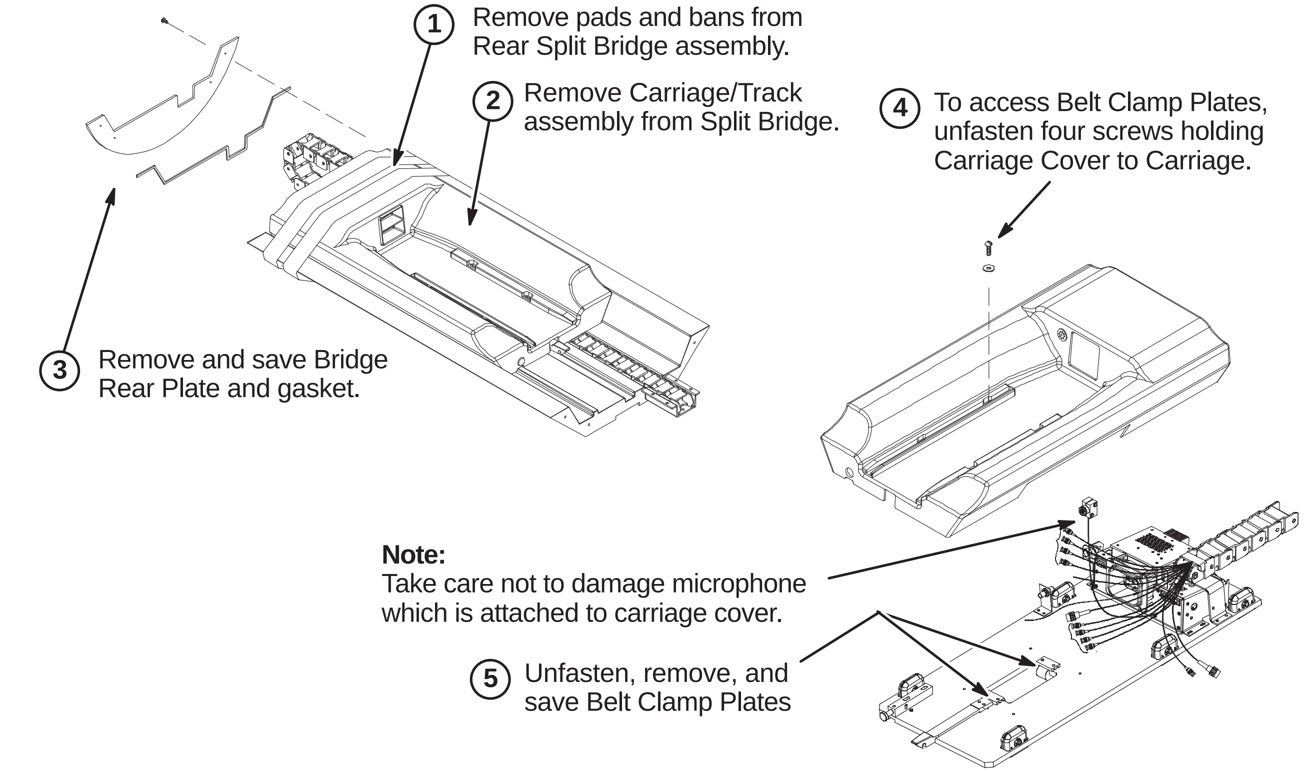
❑ F4 – INSTALL PAC/SRI PLATFORM UNDER MAGNET



INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ❑ K1 – Preparing Rear Split Bridge for Installation
- ❑ K2 – Attaching Rear Split Bridge to Rear Pedestal
- ❑ K3 – Attaching Rear Split Bridge to Front Split Bridge
- ❑ K4 – Aligning Bridge on Rear Pedestal
- ❑ K5 – Installing Bridge Locators

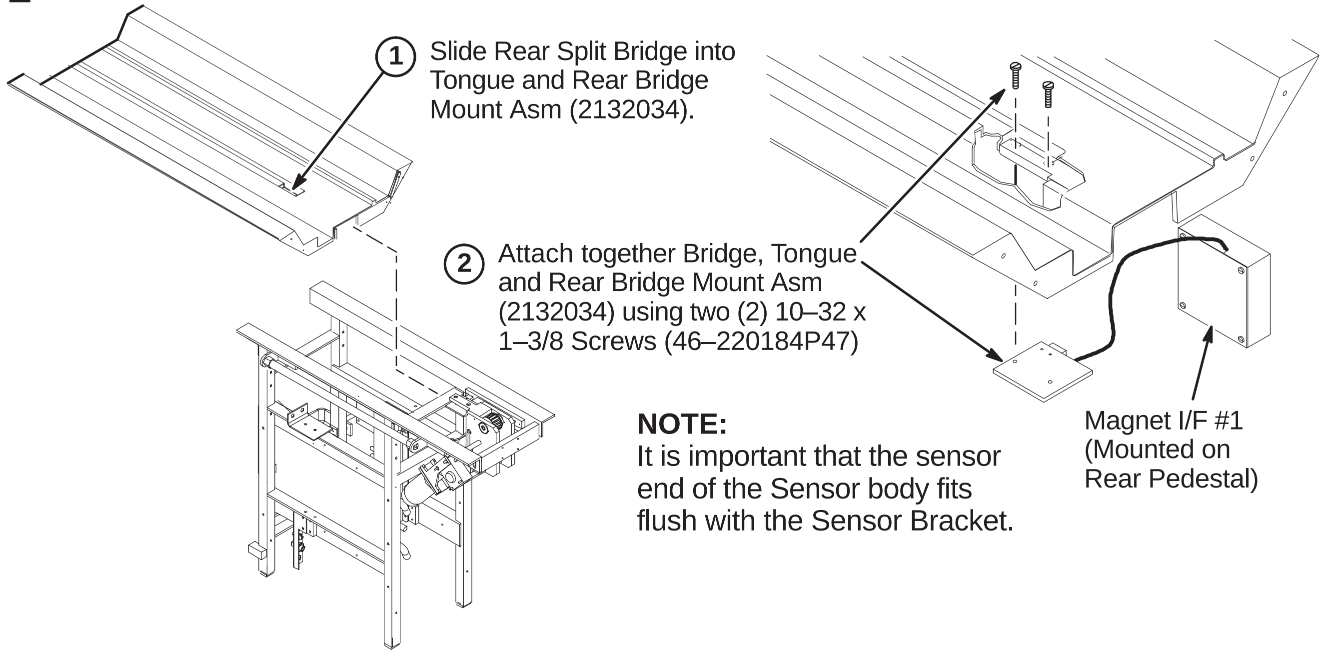
❑ K1 – PREPARING REAR SPLIT BRIDGE FOR INSTALLATION



1. Remove pads and bans from Rear Split Bridge assembly.
2. Remove Carriage/Track assembly from Split Bridge.
3. Remove and save Bridge Rear Plate and gasket.
4. To access Belt Clamp Plates, unfasten four screws holding Carriage Cover to Carriage.
5. Unfasten, remove, and save Belt Clamp Plates

Note:
Take care not to damage microphone which is attached to carriage cover.

❑ K2 – ATTACHING REAR SPLIT BRIDGE TO REAR PEDESTAL

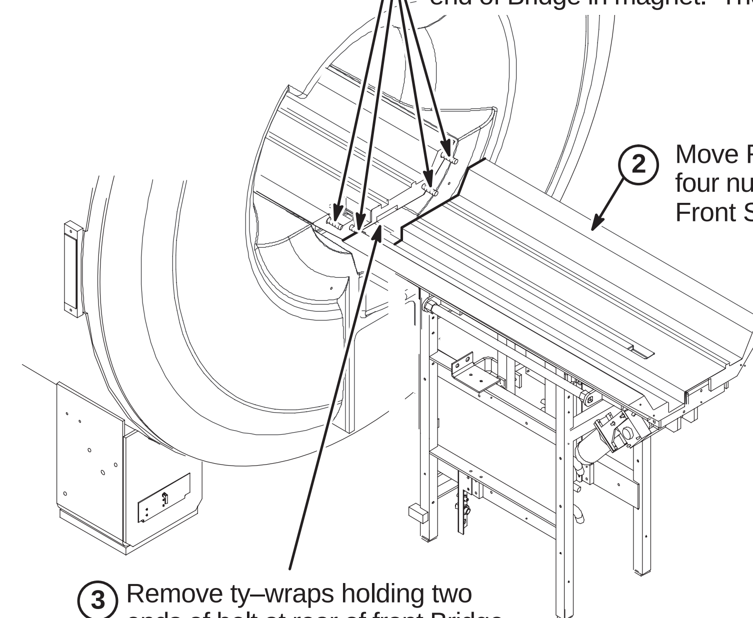


1. Slide Rear Split Bridge into Tongue and Rear Bridge Mount Asm (2132034).
2. Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) 10–32 x 1–3/8 Screws (46–220184P47)

Magnet I/F #1 (Mounted on Rear Pedestal)

NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.

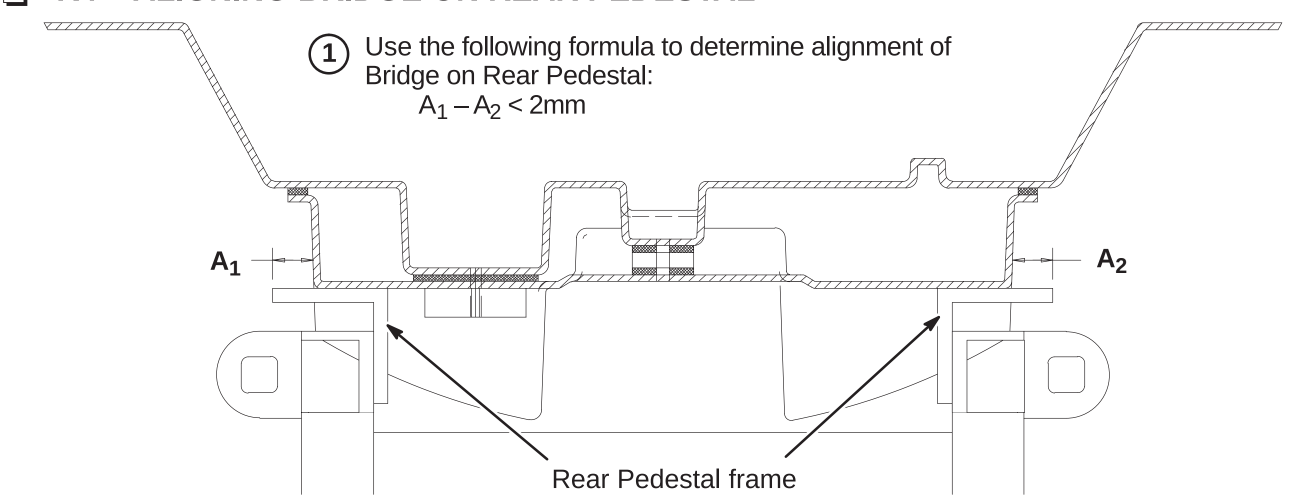
❑ K3 – ATTACHING REAR SPLIT BRIDGE TO FRONT SPLIT BRIDGE



1. Remove four 10M hex nuts (2109877–3) and washers (2228717–2) from end of Bridge in magnet. These will be used for attaching the Rear Pedestal.
2. Move Rear Pedestal/Bridge into position and fasten with four nuts and washers that were removed form end of Front Split Bridge.
3. Remove ty–wraps holding two ends of belt at rear of front Bridge.

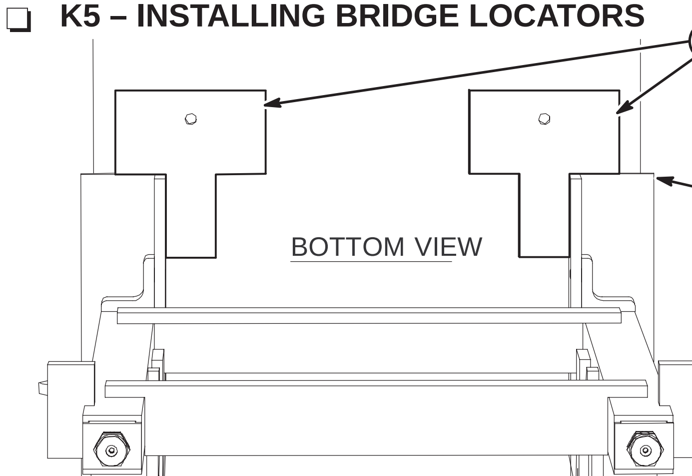
NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

❑ K4 – ALIGNING BRIDGE ON REAR PEDESTAL



1. Use the following formula to determine alignment of Bridge on Rear Pedestal:
   $A_1 - A_2 < 2\text{mm}$

❑ K5 – INSTALLING BRIDGE LOCATORS



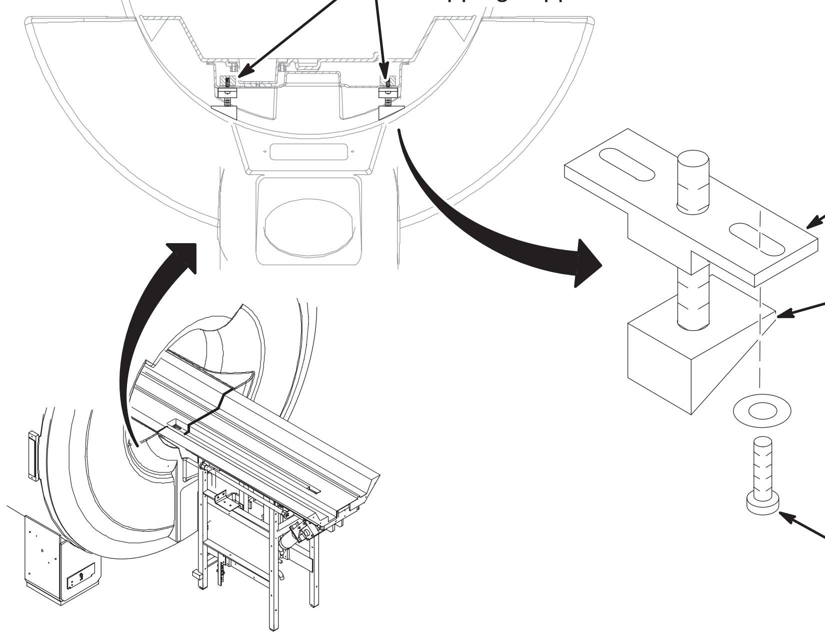
1. After Bridge and Rear Pedestal are aligned, peel off protective paper on each Bridge Locator and attach to bottom side of Bridge as shown.

End of Rear Pedestal frame closest to Magnet.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ❑ L1 – Removing Front Split Bridge Shipping Support Brackets
- ❑ L2 – Leveling Bridge
- ❑ L3 – Height Adjustment at Front of Bridge (if needed)
- ❑ L4 – Installing Carriage on Bridge
- ❑ L5 – Routing of Drive Belt

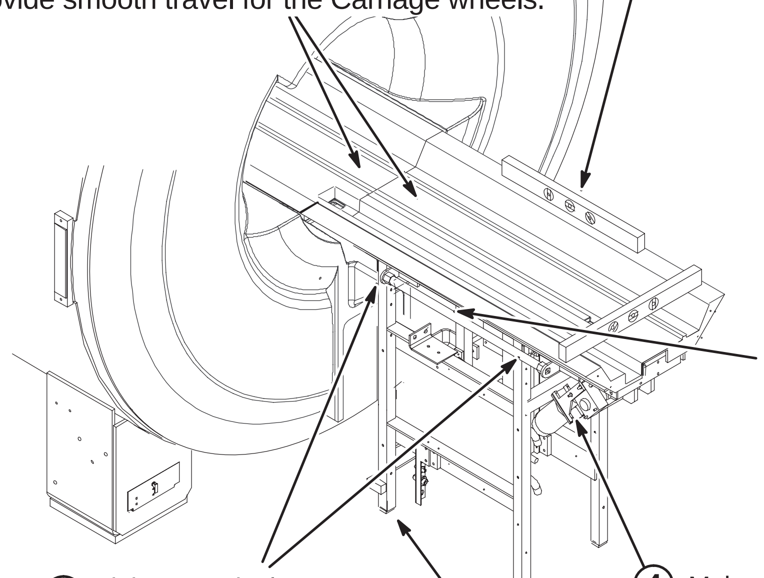
❑ L1 – REMOVING FRONT SPLIT BRIDGE SHIPPING SUPPORT BRACKETS

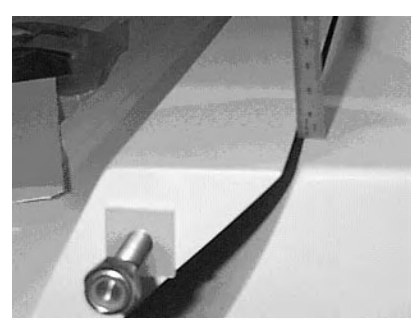


1. After Front and Rear Split Bridge parts are fastened together, the two shipping supports at the rear of the Front Split Bridge need to be removed.
2. Remove the M6 x 16mm screw and washer that attaches each bracket to Bridge
3. Unscrew downword the Bridge Mounting Flange.
4. Support Base is attached to End Bell with adhesive tape. To remove, carefully pry base away from end bell. Remove base, stud, and mounting flange

❑ L2 – LEVELING BRIDGE

NOTE: Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

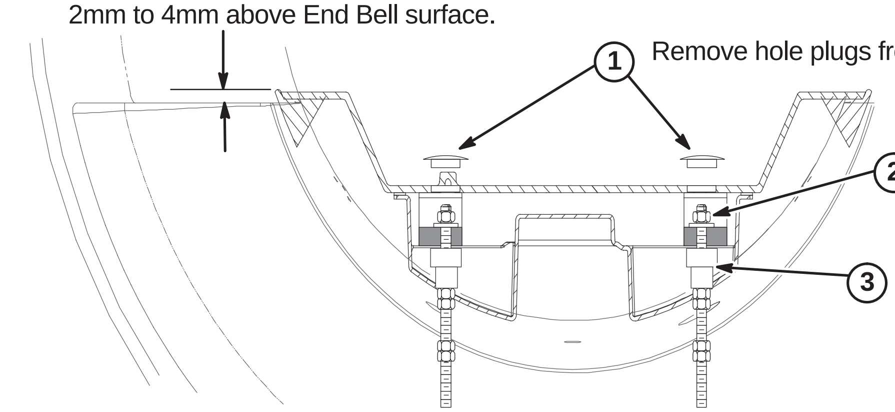




1. Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of End Bells, as indicated in photos at below.
2. Tighten Vertical Adjustment Fasteners.
3. Adjust leg screws on both sides as needed to level Rear Pedetal.
4. Make sure that Longitudinal Drive Asm is level.

NOTE:
Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

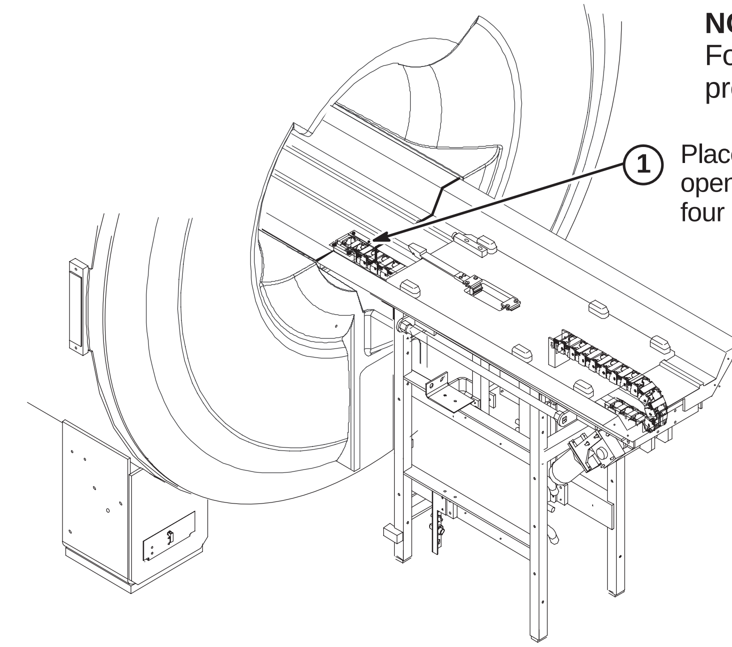
❑ L3 – HEIGHT ADJUSTMENT AT FRONT OF BRIDGE (IF NEEDED)



1. Remove hole plugs from each side of Bridge.
2. Remove M10 nut and washer from each stud.
3. Adjust Support Nut (2293381) as needed to achieve correct height.

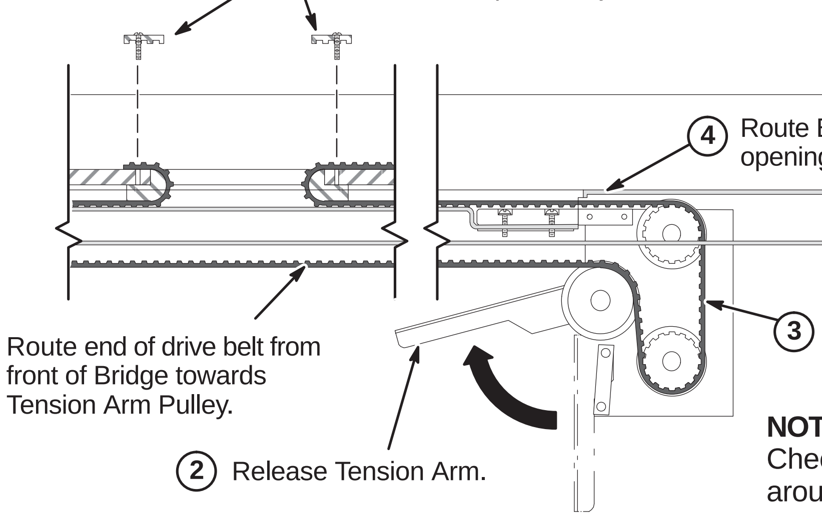
❑ L4 – INSTALLING CARRIAGE ON BRIDGE

NOTE:
For clarity, the carriage is shown without preamps, preamp protect module, and cables.



1. Place Carriage on Bridge and route all cables thru opening in Bridge and fasten plate to Front Bridge with four M6 x 12mm nylon screws (2296958) provided.

❑ L5 – ROUTING OF DRIVE BELT



1. Route end of drive belt from front of Bridge towards Tension Arm Pulley.
2. Release Tension Arm.
3. Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.
4. Route Belt under and through opening to top of Carriage.
5. Position Belt under Clamp Plates and re–attach with previously removed screws.

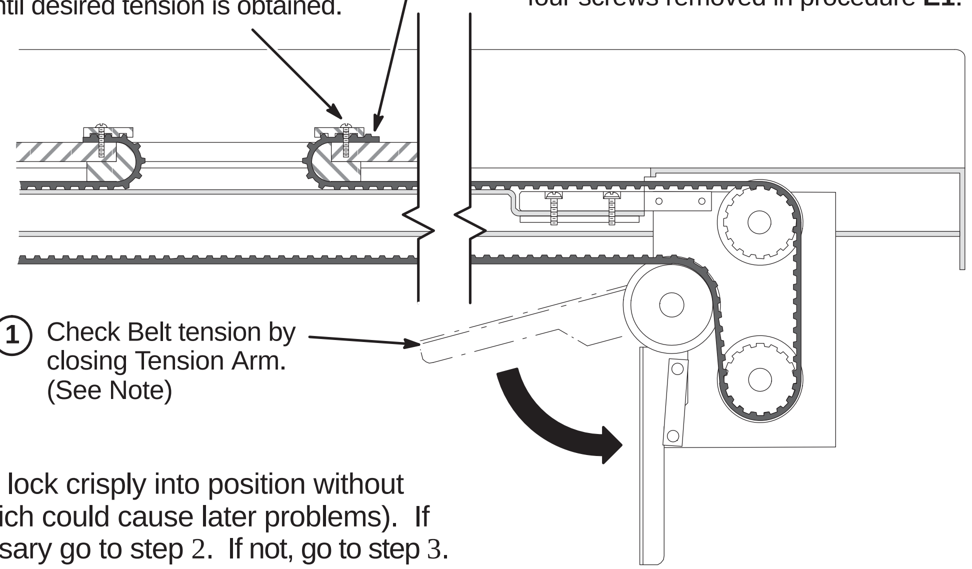
NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ❑ **M1 – Final Adjustment of Drive Belt**
- ❑ **M2 – Installation of Sensor Flag**
- ❑ **M3 – Route and Connect Runs from LPCA and SRI to Rear Pedestal**
- ❑ **M4 – Installing Carriage Cover to Trolley and Bridge Cover**

❑ M1 – FINAL ADJUSTMENT OF DRIVE BELT

1. Check Belt tension by closing Tension Arm. (See Note)
2. **(IF NECESSARY):** Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.
3. Tighten Clamps and trim excess Belt length.
4. Fasten Carriage Cover to Carriage with four screws removed in procedure **E1**.



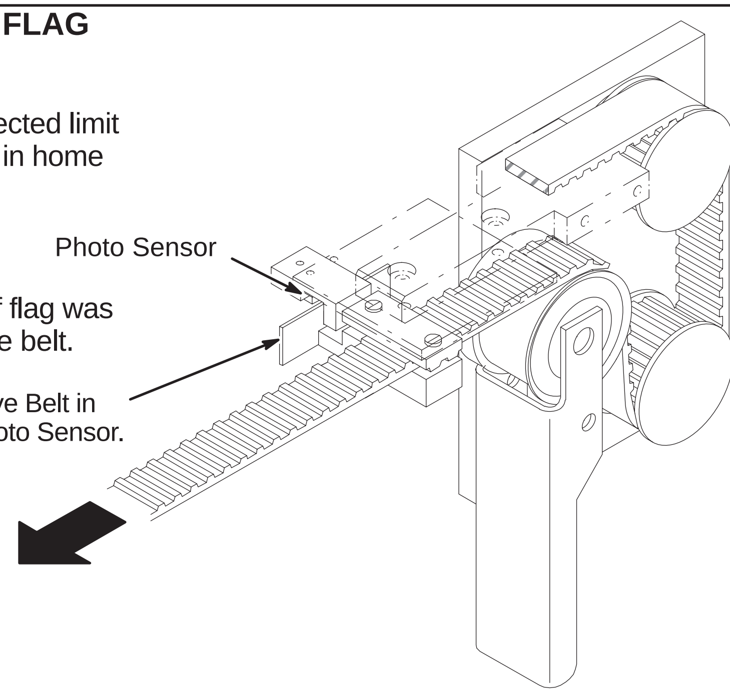
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

❑ M2 – INSTALLATION OF SENSOR FLAG

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

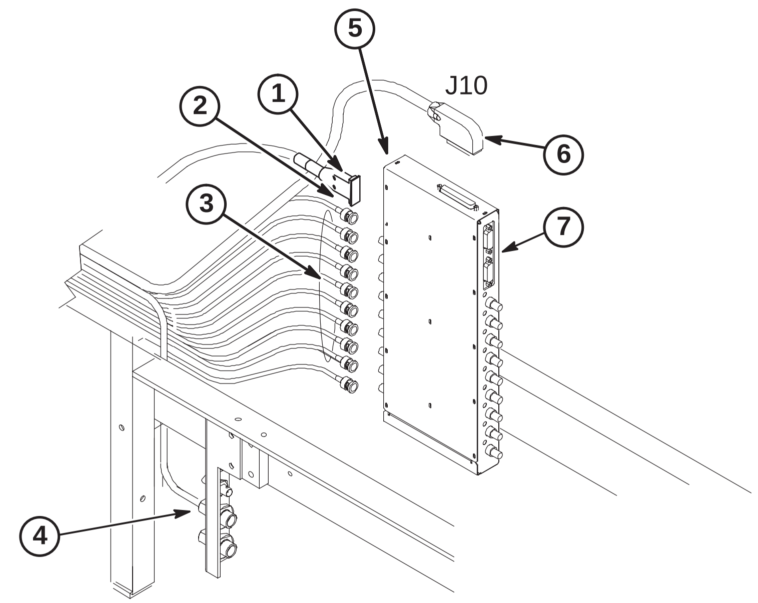
NOTE:
The following Step is needed if flag was removed during routing of drive belt.

1. Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



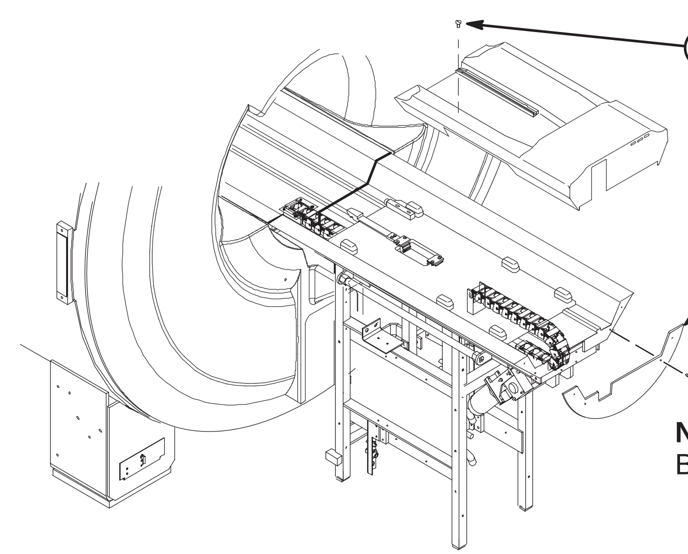
❑ M3 – ROUTE AND CONNECT RUNS FROM LPCA AND SRI TO REAR PEDESTAL

1. Route/Connect Run 1054 to J20.
2. Route/Connect Run 1035 to J21.
3. Route/Connect Runs 728 thru 735 to J11 to J18 and Run 1049 to J19.
4. Route Run 404 and 485 from Carriage and connect to J1 and J3 respectively.
5. Route/Connect Run 1078 from SRI to J29 and J30.
6. Route/Connect Run 1041 from J78 on Penetration Panel to J19.
7. Route/Connect Run 1042 from J77 on Penetration Panel to J5.



❑ M4 – INSTALLING CARRIAGE COVER TO TROLLEY AND BRIDGE COVER

1. Attach Carriage Cover to Trolley using four previously removed fasteners.
2. Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.

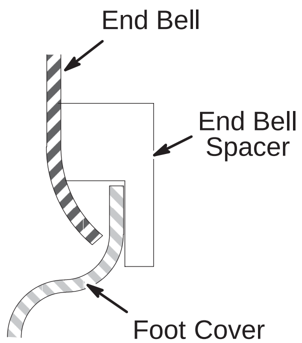


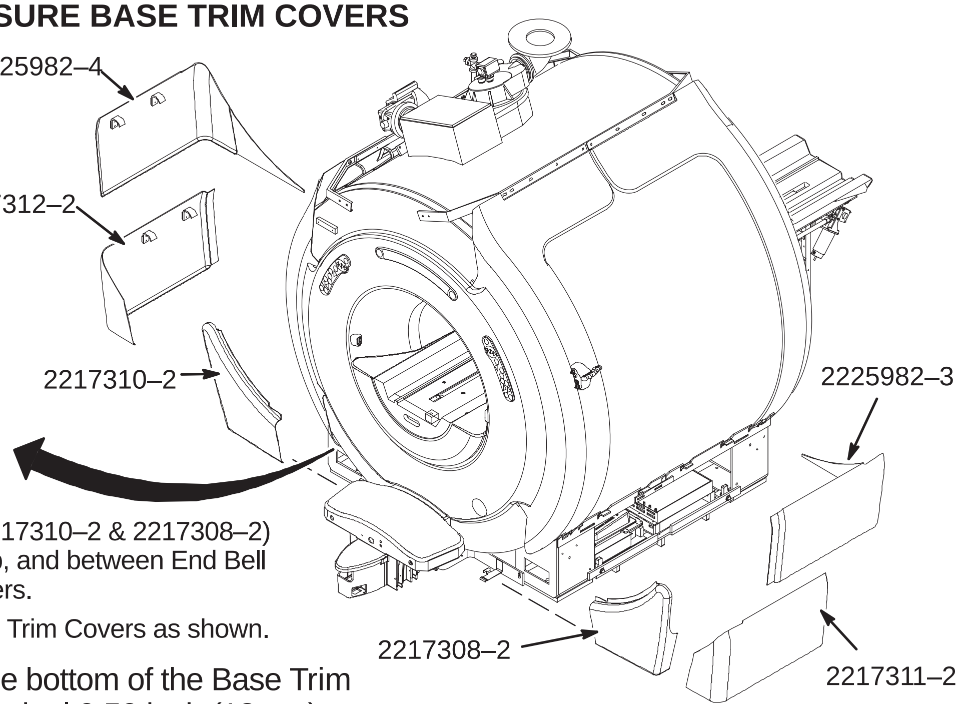
Note:
Bridge Cover has enclosure rating plates on it.

INSTALLATION PROCEDURES SUMMARY

- ❑ N1 – Install Enclosure Base Trim Covers
- ❑ N2 – Install Enclosure Top Covers
- ❑ N3 – Install Rear Pedestal Covers

❑ N1 – INSTALL ENCLOSURE BASE TRIM COVERS





1. Slide front Foot Covers (2217310–2 & 2217308–2) from side, above spring clip, and between End Bell Flange and End Bell Spacers.
2. Install remaining side Base Trim Covers as shown.

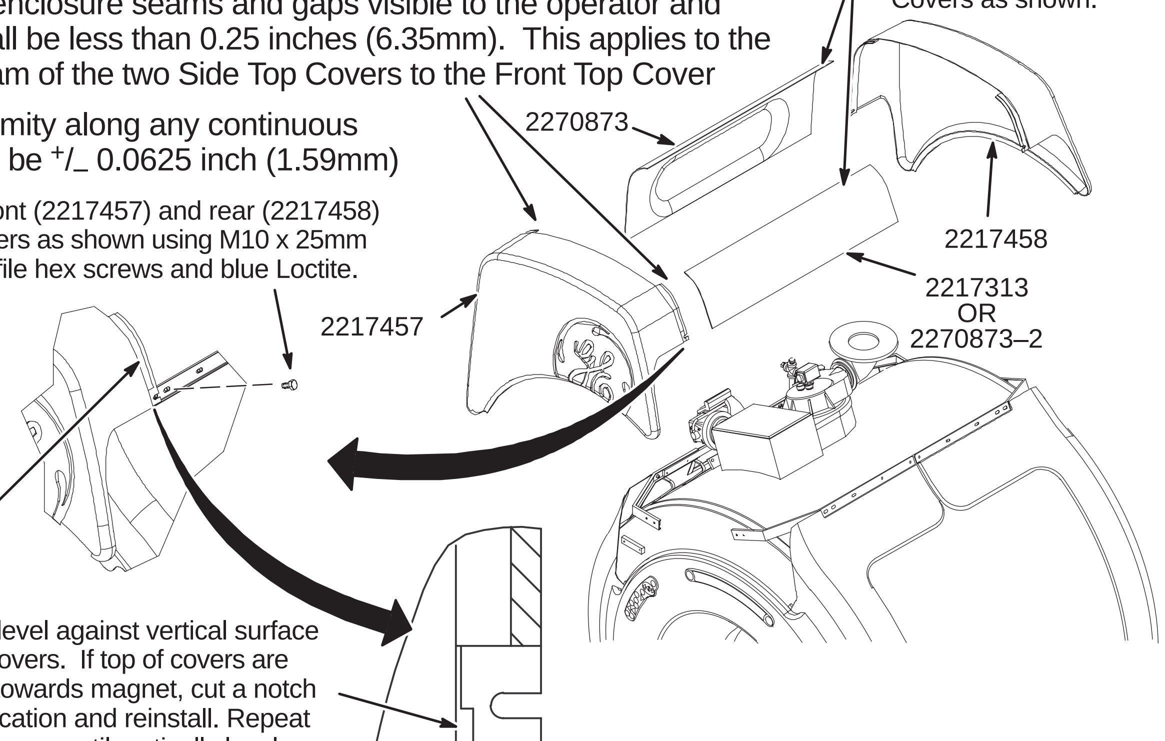
Note: The gap from the bottom of the Base Trim to floor should be a nominal 0.50 inch (13mm).

❑ N2 – INSTALL ENCLOSURE TOP COVERS

Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

Gap uniformity along any continuous seam shall be $^{+}/_{-}$ 0.0625 inch (1.59mm)

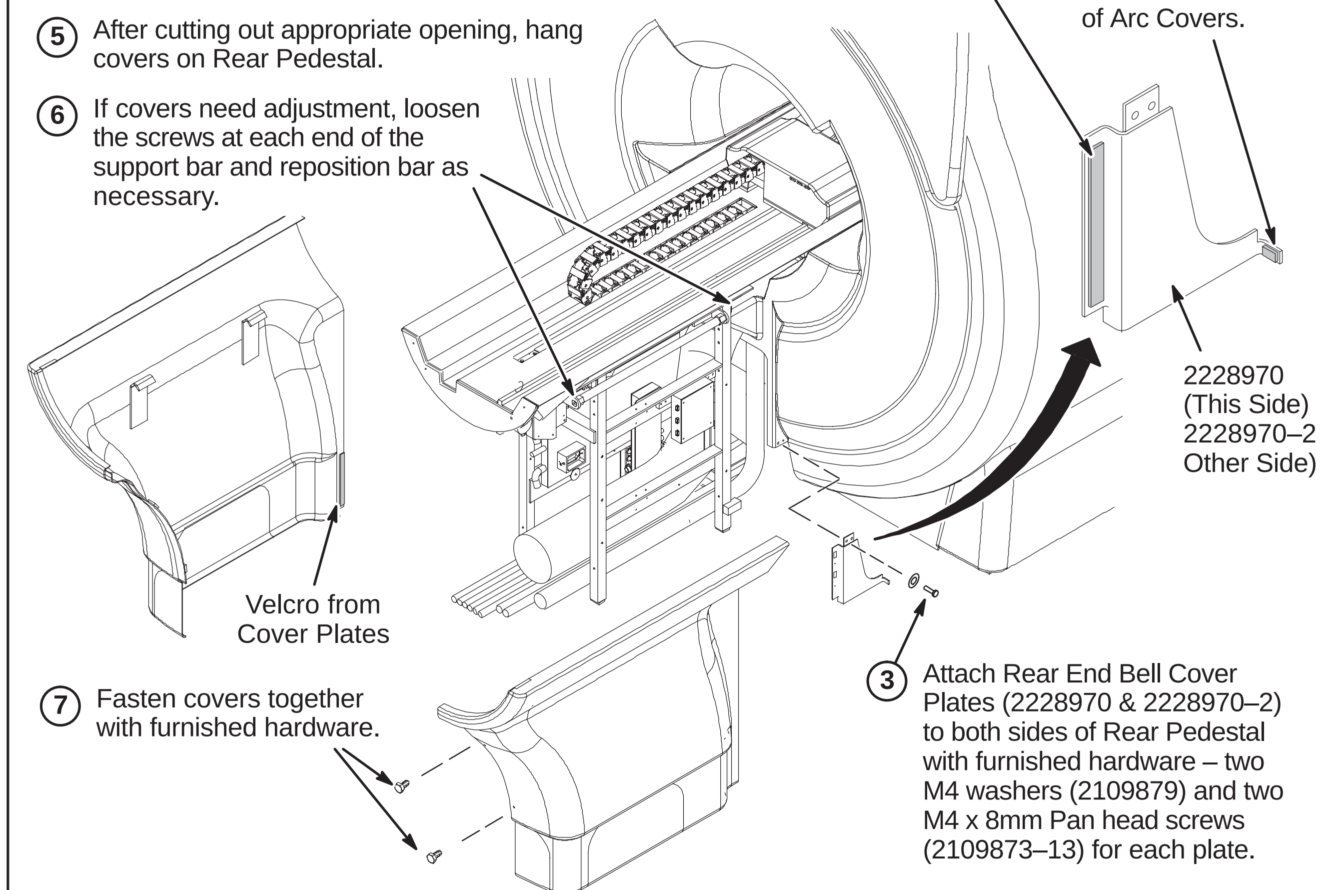
1. Install front (2217457) and rear (2217458) Top Covers as shown using M10 x 25mm Low Profile hex screws and blue Loctite.
2. Install side Top Covers as shown.



2. Place a level against vertical surface of Top Covers. If top of covers are leaning towards magnet, cut a notch at this location and reinstall. Repeat as necessary until vertically level.

❑ N3 – INSTALL REAR PEDESTAL COVERS

1. Remove mating piece of velcro from Rear End Bell Cover Plates (2228970 & 2228970–2) and attach to Rear Pedestal Covers
2. Remove mating velcro and attach to back side of Arc Covers.
3. Attach Rear End Bell Cover Plates (2228970 & 2228970–2) to both sides of Rear Pedestal with furnished hardware – two M4 washers (2109879) and two M4 x 8mm Pan head screws (2109873–13) for each plate.
4. Pedestal Covers (2217704 & 2217705) require an opening to be cut out depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting.
5. After cutting out appropriate opening, hang covers on Rear Pedestal.
6. If covers need adjustment, loosen the screws at each end of the support bar and reposition bar as necessary.
7. Fasten covers together with furnished hardware.



INSTALLATION PROCEDURES SUMMARY

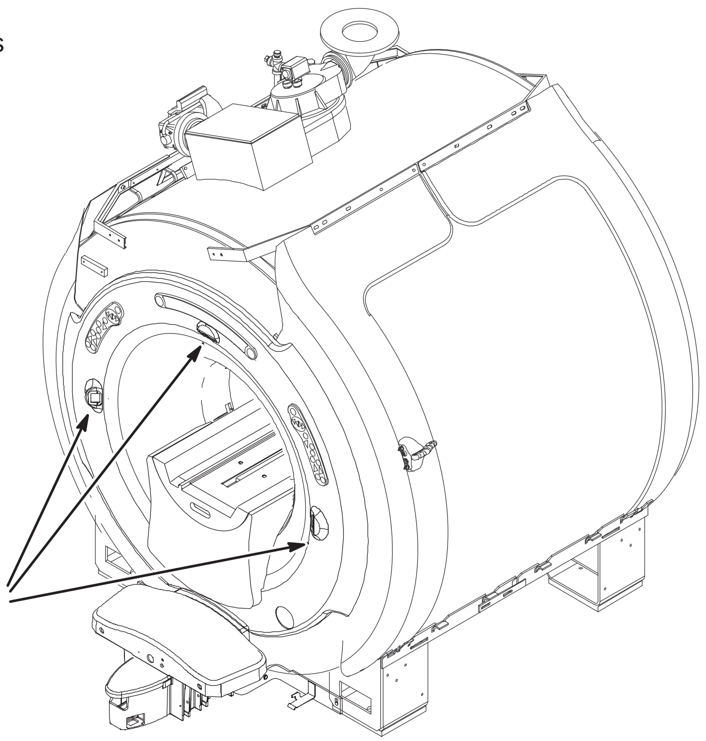
❑ **M1 – Install Laser Alignment Light Warning Labels**

❑ M1 – INSTALL LASER ALIGNMENT LIGHT WARINING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

English



① For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.

French

German

Portuguese

Spanish

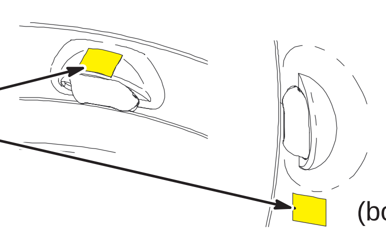
Italian

Swedish

Japanese

Chinese

② If enclosure does not have labels attached, attach new labels as indicated.



(both sides)

GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.
- A 1–ounce tube of Permatex "Anti–Seize" lubricant (2119594) is provided. Follow instructions on back of tube. "Anti–Seize" lubricant is to be applied to threads of stainless–steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

LIST OF EFFECTIVE PAGES

BRIDGE INSTALLATION INFORMATION

TWO TYPES OF BRIDGES ARE AVAILABLE FOR INSTALLATION: The original style Long Bridge and the new style Split Bridge. See items 1 and 2 below for explanation of which installation procedures to use.

1. If original style Long Bridge is delivered to site, complete procedures on pages 7, 8, and 9. This bridge would be used for pre–Split Bridge MR/i and all CV/i installations.
2. If new style Split Bridge is delivered to site, complete procedures on pages 10, 11, and 12. This bridge would be used for MR/i installations only, not CV/i.

WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

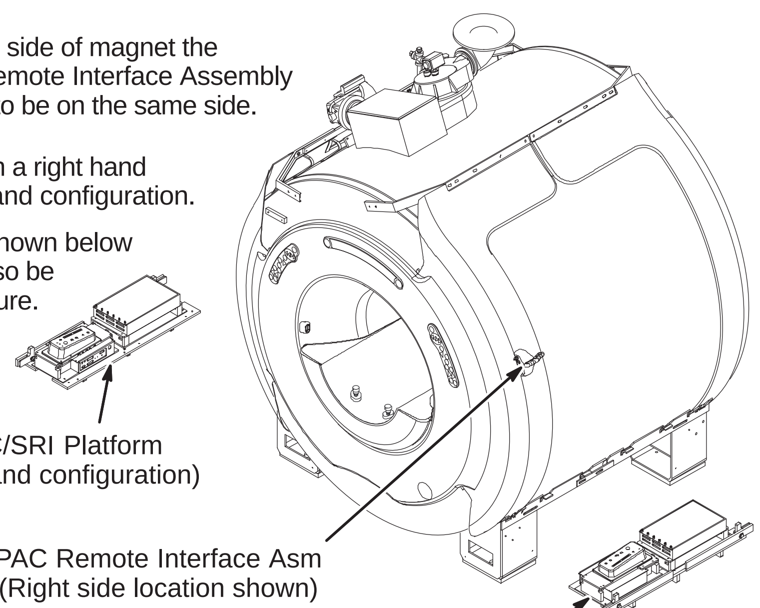
IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

INSTALLATION PROCEDURES SUMMARY

- ❑ **A1 – Determine PAC/SRI Platform and PAC Remote Interface Location**
- ❑ **A2 – PAC/SRI Platform Left Hand Configuration (If Required)**
- ❑ **A3 – Location of Magnet Ground Cable**

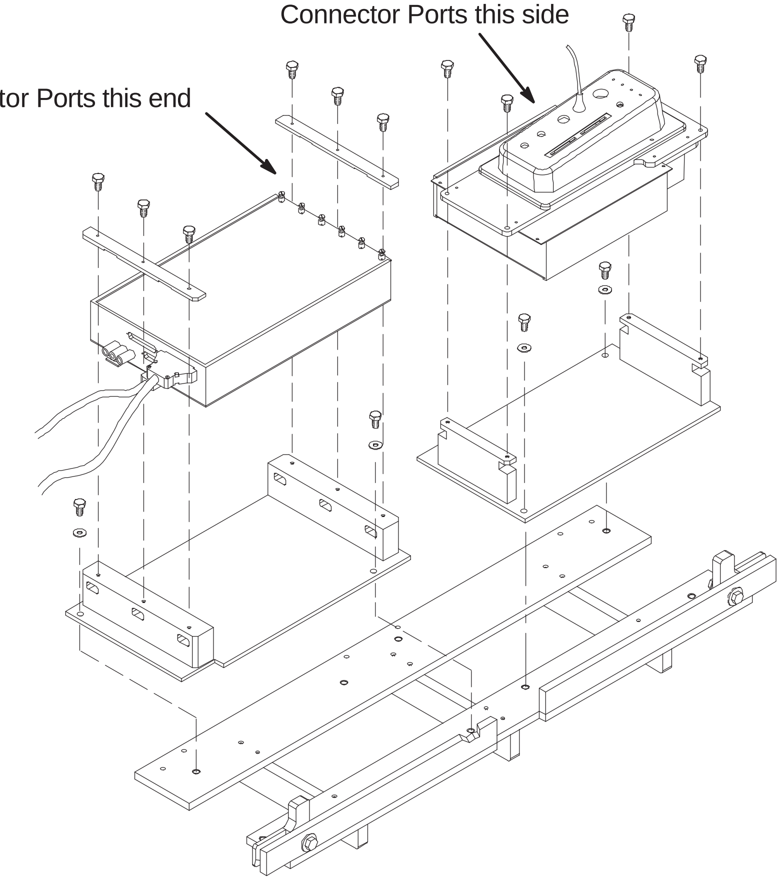
❑ A1 – DETERMINE PAC/SRI PLATFORM AND PAC REMOTE INTERFACE LOCATION

1. Consult with customer to determine which side of magnet the PAC/SRI Platform (2223440) and PAC Remote Interface Assembly (2227142) should be located. They have to be on the same side.
2. The PAC/SRI Platform arrives at the site in a right hand configuration.See Procedure **A2** for left hand configuration.
3. The PAC Remote Interface Assembly is shown below installed in the right side location. It can also be mounted on the left side of magnet enclosure.

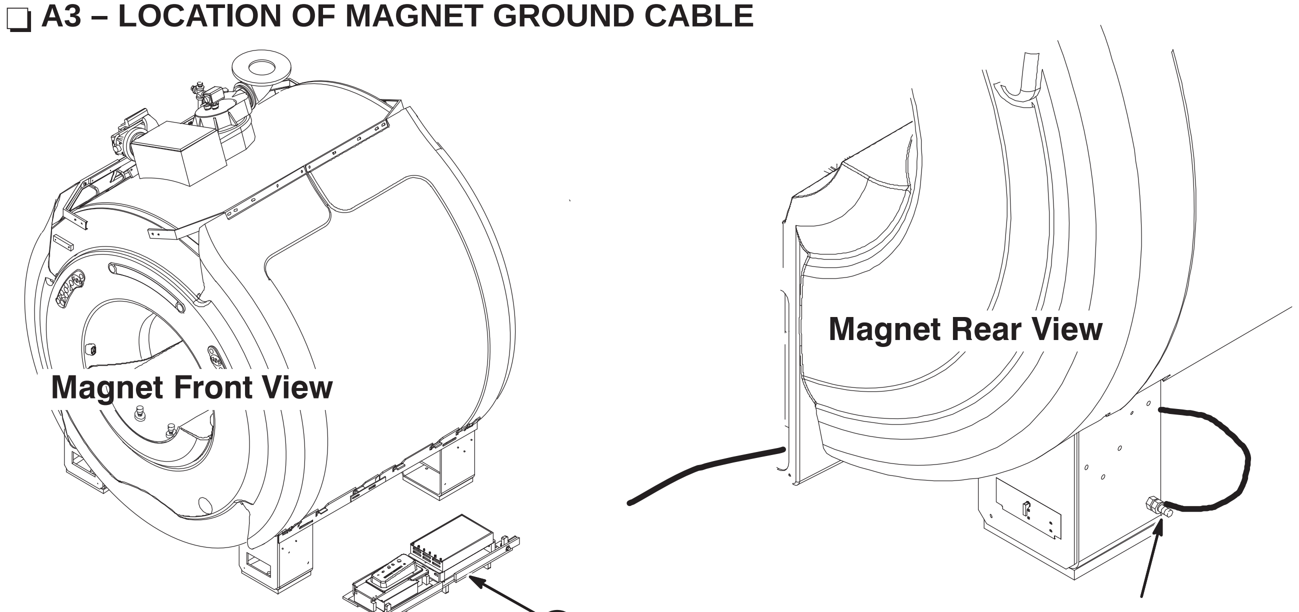


❑ A2 – PAC/SRI PLATFORM LEFT HAND CONFIGURATION (IF REQUIRED)

1. For left hand configuration, reassemble PAC/SRI as shown.



❑ A3 – LOCATION OF MAGNET GROUND CABLE



1. If PAC/SRI Platform is located on right side of magnet, then the Magnet ground cable , RUN 040, has to be connected to the opposite side rear left leg as shown above. The other end of Run 040 is connected to the RF Screen Room common ground.
2. If PAC/SRI Platform is located on left side of magnet, then the Magnet ground cable has to be connected to the opposite side rear right leg.

INSTALLATION PROCEDURES SUMMARY

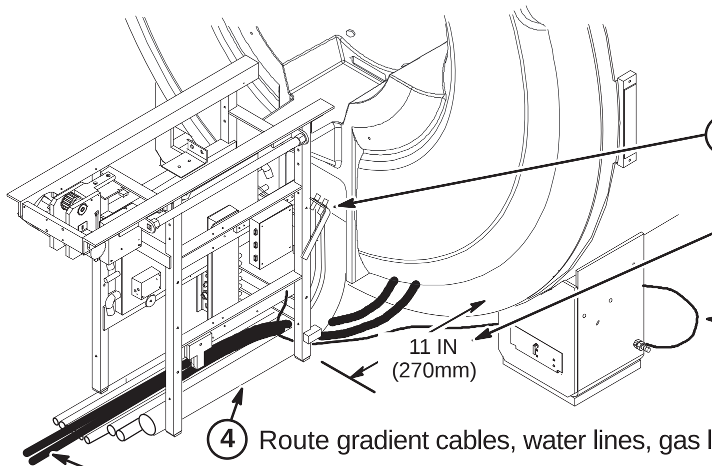
- ❑ B1 – Location of Rear Pedestal
- ❑ B2 – Routing and Connecting Gradient Output Cables to Runs 762, 763, and 764
- ❑ B3 – Route/Connect Runs 257/746, 262, 743, 744, and 780
- ❑ B4 – Route/Connect Runs 258/747, 318, 700, and Drive Cables

❑ B1 – LOCATION OF REAR PEDESTAL

WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

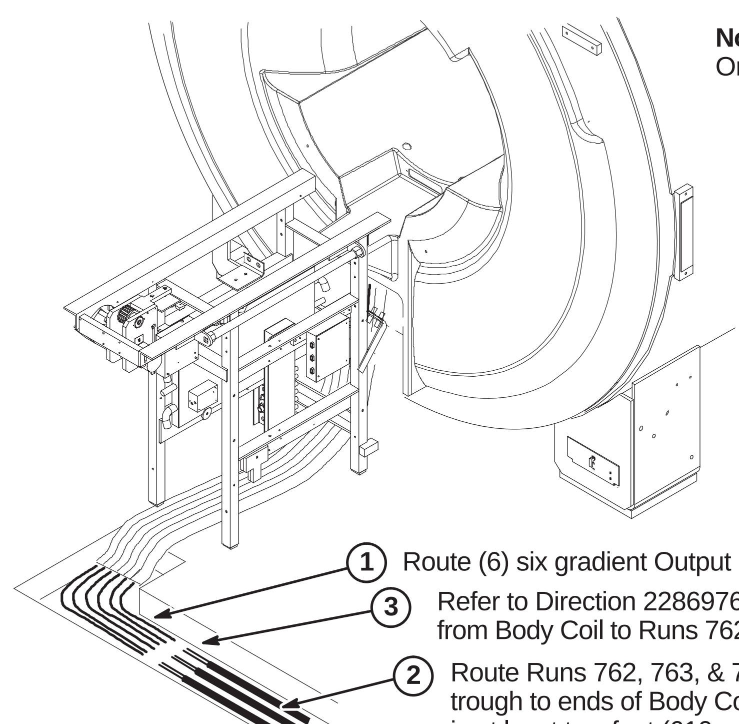
Note:
Cables and hoses can be routed to exit out of rear or side of Rear Pedestal.



1. For **CV/*i*** install bracket on other side.
2. Center Rear Pedestal on Rear Endbell and locate approximately 11 inches (270mm) from face of magnet.
3. Route Ground cable from Rear Pedestal under magnet and connect to same Ground Stud as Magnet ground.
4. Route gradient cables, water lines, gas lines, and 2” air hose under Rear Pedestal as shown.
5. Route and connect coolant hosees from Water Chiller to 10 ft. (300cm) coolant hoses attached to Body Coil.
6. Route all other cables on top of the above cables and hoses.

❑ B2 – ROUTING AND CONNECTING GRADIENT OUTPUT CABLES

Note:
Only Gradient Cables are shown for clarity.



1. Route (6) six gradient Output Cables from Body Coil to cable trough.
3. Refer to Direction 2286976 in Tab 1 for connecting Output cables from Body Coil to Runs 762, 763, & 764.
2. Route Runs 762, 763, & 764 from Penetration Panel through cable trough to ends of Body Coil gradient cables. Make sure that there is at least two feet (610mm) of slack before cutting to length.

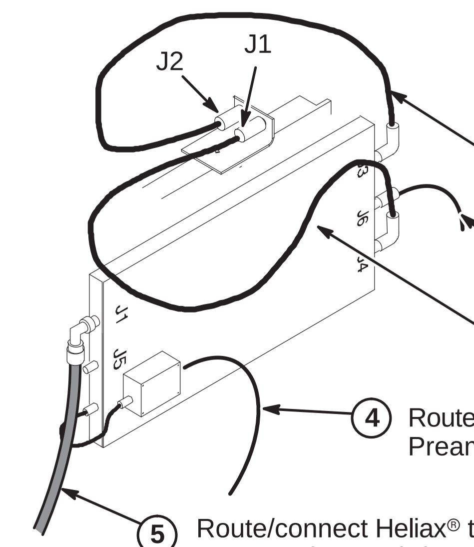
❑ B3 – ROUTE/CONNECT RUNS 257/746, 262, 743, 744, AND 780

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

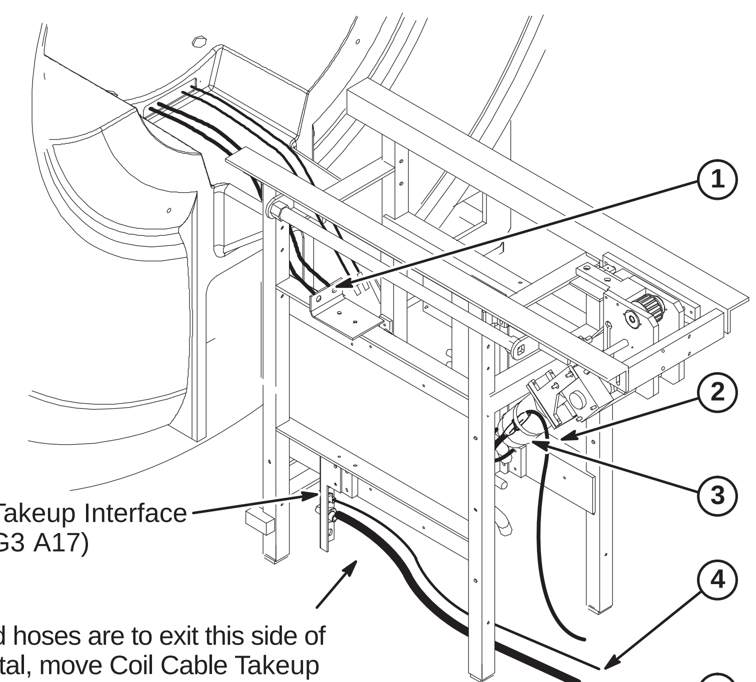
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.



1. Route and connect Run 744 from J2 on bracket to J4 on the RF Splitter.
2. Route/connect Run 780 from J74 on Pen Panel to J6 on RF Splitter.
3. Route and connect Run 743 from J1 on the bracket to J3 on the RF Splitter.
4. Route/connect Run 262 from J8 on Pen Panel to Body Coil Preamplifier. (For MGD, connect to Receive Filter on Preamp)
5. Route/connect Heliax® to J1 on RF Splitter: Run 257 (1.5T, 7/8”) or Run 746 (1.0T, 1/2”)

❑ B4 – ROUTE/CONNECT RUNS 258/747, 318, 700, AND DRIVE CABLES



Coil Cable Takeup Interface (MG3 A17)

1. Route I & Q RF Drive Input cables from RF Coil and connect as marked to J1 and J2 on bracket.
2. Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).
3. Ty–wrap cable to motor to prevent movement.
4. Route/connect Run 700 to J2 on Coil Cable Takeup Interface (NON–MGD ONLY)
5. Route/connect to J1 on Coil Cable Takeup Interface: Run 258 (1.5T, 1/2”) or Run 747 (1.0T, 3/8”)

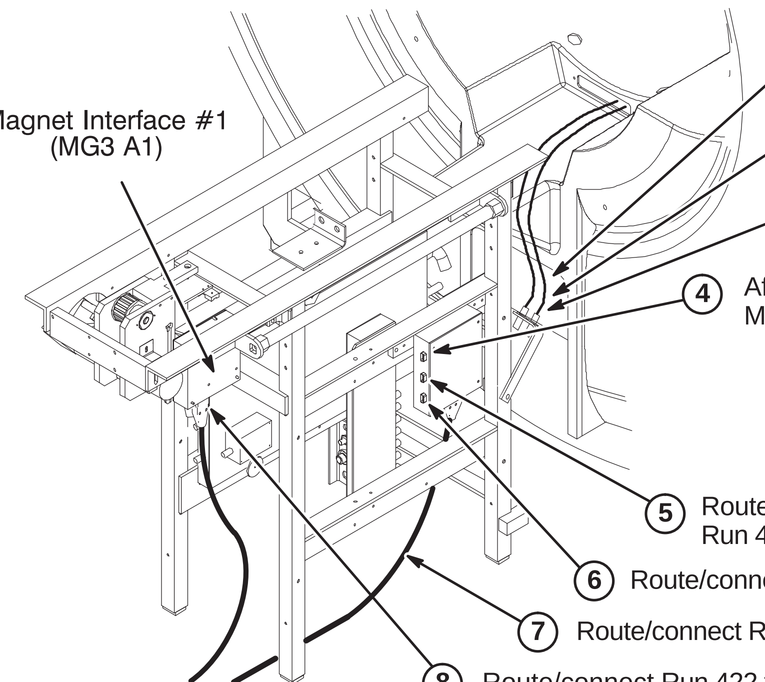
Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.

If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

INSTALLATION PROCEDURES SUMMARY

- ❑ **C1 – Route/Connect Runs 282, 414, 422, 781, 782, and Front End Bell Speaker Cable**
- ❑ **C2 – Install Bore Lamp Assembly**
- ❑ **C3 – Route/Connect Runs 257/746, 262, 743, 744, and 780**
- ❑ **C4 – Install Blower Box and Route/Connect Air Hoses**

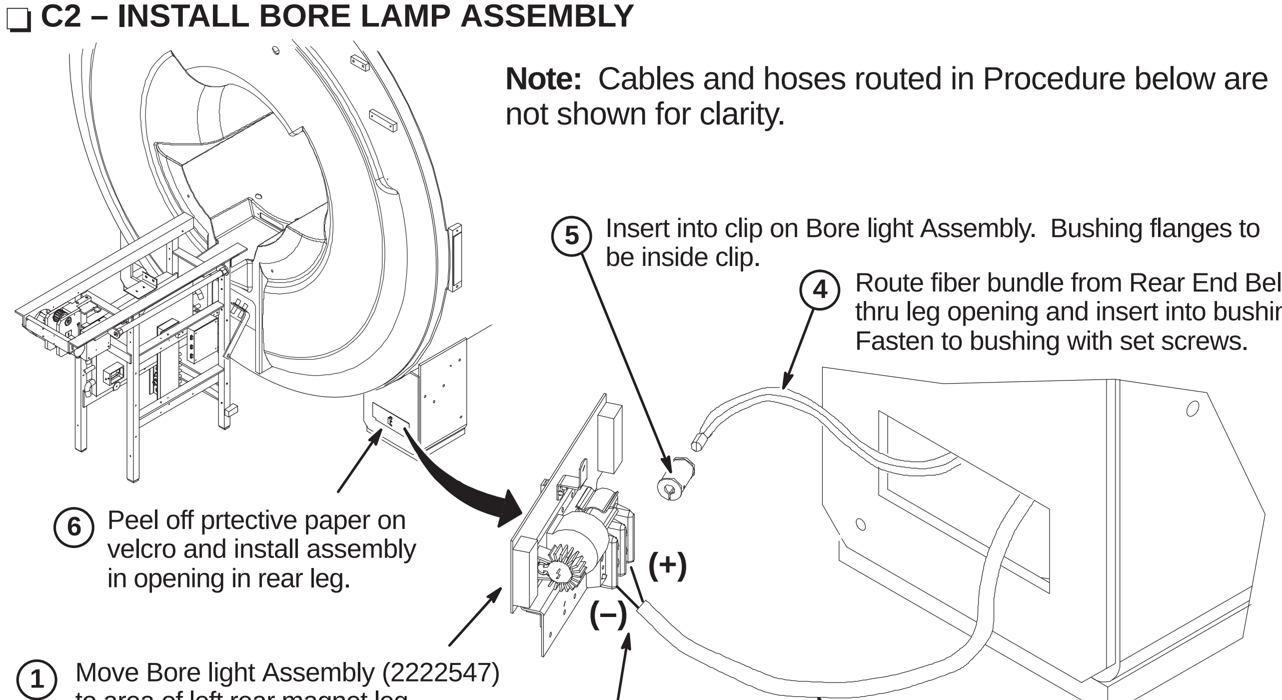
❑ C1 – ROUTE/CONNECT RUNS 282, 414, 422, 781, 782, & FRONT END BELL CABLE



1. Route and connect RF Bias cables from RF Coil to bracket as marked.
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. After Carriage cover is installed, connect Microphone cable from Carriage Cover to J7.
5. Route/connect Front End Bell microphone cable Run 414 to J9.
6. Route/connect speaker cable from Front End Bell to J8.
7. Route/connect Run 282 from J12 on Pen Panel to J6.
8. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.

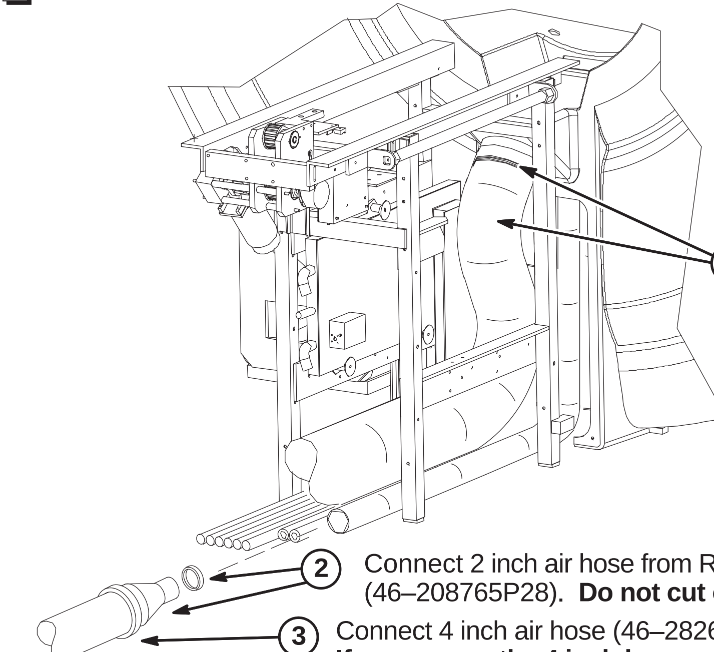
❑ C2 – INSTALL BORE LAMP ASSEMBLY

Note: Cables and hoses routed in Procedure below are not shown for clarity.



1. Move Bore light Assembly (2222547) to area of left rear magnet leg.
2. Connect the (+) and (–) wires of Run 300 to the terminal strip as shown.
3. Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
4. Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
5. Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
6. Peel off prtective paper on velcro and install assembly in opening in rear leg.

❑ C3 – CONNECT AIR HOSES TO REAR OF MAGNET



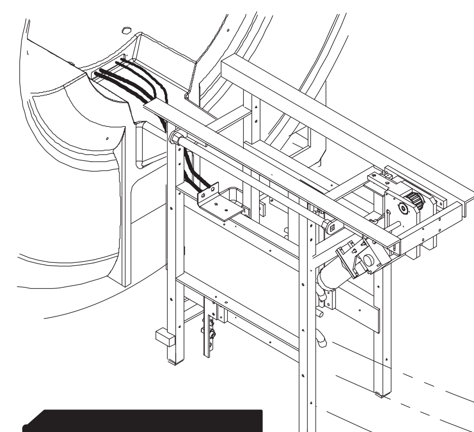
Note:
The following items are part of Blower Hose Kit, 2129412–3 or 2129412–4.

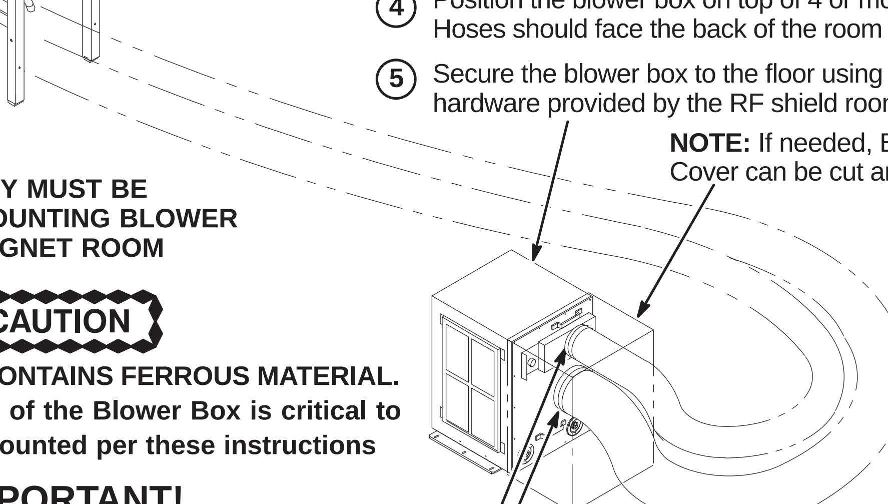
1. Connect 6 inch air hose (46–282666P12) to Rear End Bell with 6 inch clamp (46–208675P29).
2. Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). **Do not cut or reduce the length of the 2 inch hose.**
3. Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). **If necessary, the 4 inch hose may need to be cut to reduce length.**

❑ C4 – INSTALL BLOWER BOX AND ROUTE/CONNECT AIR HOSES

1. Refer to site design (architectural)/customer construction drawings or contact PMI for locating and securing Blower Box (2121199).
2. Check for correct installation of the anchors:
   - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
   - Confirm there are at least four (4) anchors.
   - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.
3. Remove any mounting hardware from the anchors.
4. Position the blower box on top of 4 or more anchors. Hoses should face the back of the room
5. Secure the blower box to the floor using the fastening hardware provided by the RF shield room vendor.

NOTE: If needed, Blower Hose Cover can be cut and modified.





WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

6. Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:
   (1) 46–208765P34 4 inch Hose Clamp
   (1) 46–208765P29 6 inch Hose Clamp

INSTALLATION PROCEDURES SUMMARY

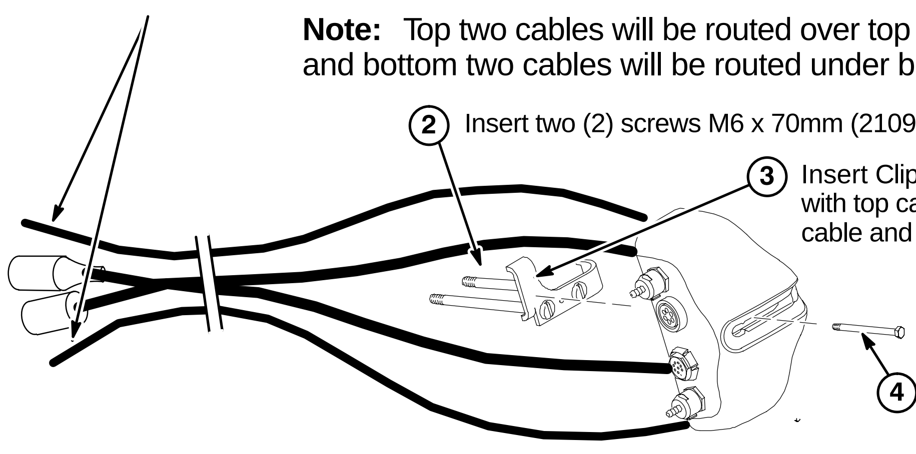
- ❑ **D1 – Prepare Cables and Tubes for Routing**
- ❑ **D2 – Position Remote PAC Interface Mounting Clip**
- ❑ **D3 – Route Remote PAC Interface Cables and Tubes**
- ❑ **D4 – Installing Remote PAC Interface to Arc Cover**
- ❑ **D5 – Install PAC Hanger Bar**

❑ D1 – PREPARE CABLES AND TUBES FOR ROUTING

Note:
Consult with customer for correct location of Remote PAC Interface assembly. It must be on the same side as the PAC/SRI Platform.

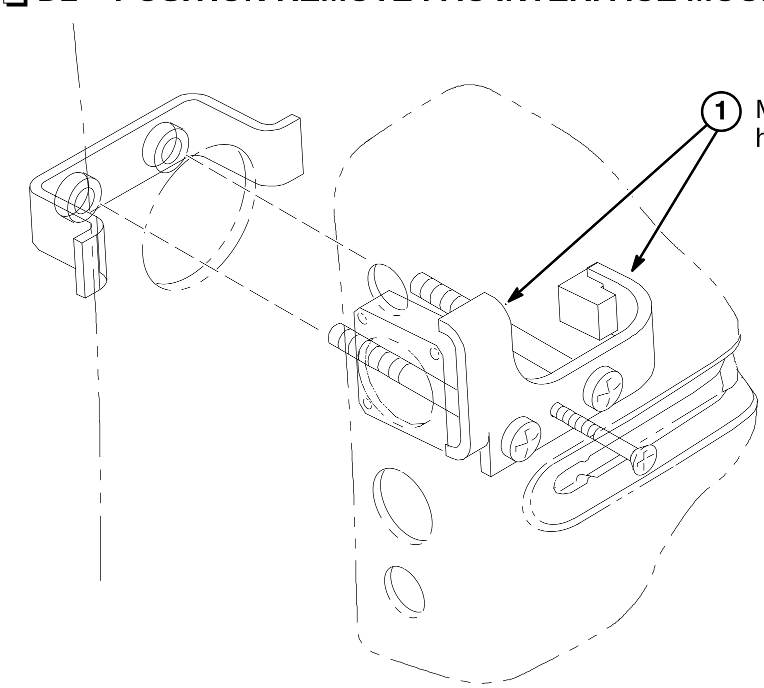
1. Apply tape around end of pneumatic tubes and mark "TOP" to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and "BOTTOM" to the tube connected to the lower connector.

Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.



2. Insert two (2) screws M6 x 70mm (2109873–33) screws into Clip (2221023).
3. Insert Clip with screws into shell as shown with top cable and tube above Clip and bottom cable and tube below.
4. Hold in place by inserting provided M4 screw thru slot and turning about 3 or 4 turns into Clip.

❑ D2 – POSITION REMOTE PAC INTERFACE MOUNTING CLIP

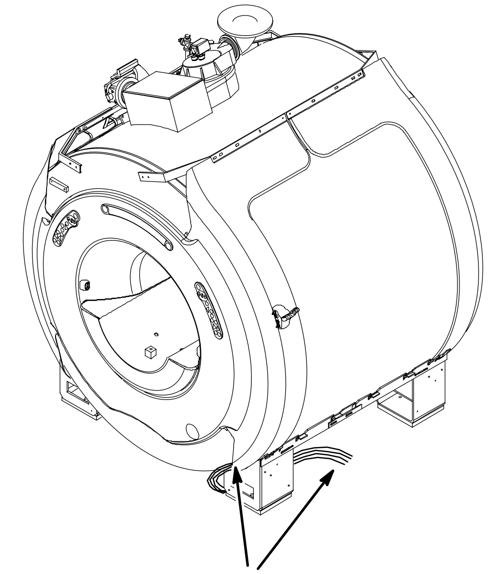


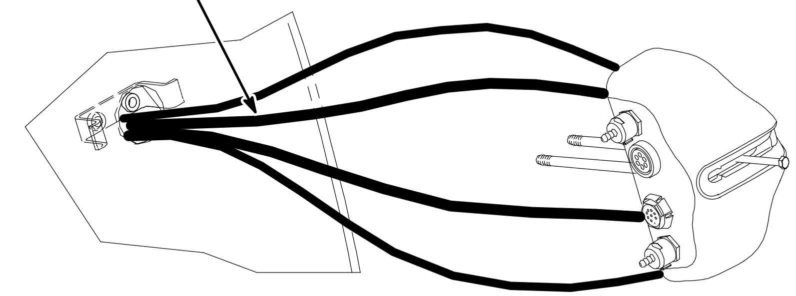
1. Make sure each side of clip is mounted behind housing blocks.

❑ D3 – ROUTE REMOTE PAC INTERFACE CABLES AND TUBES

Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

Note: Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.

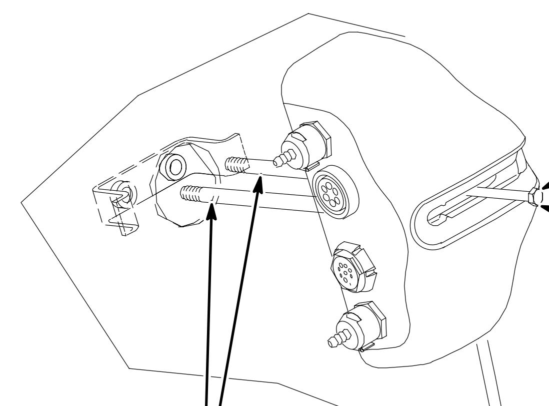




1. **Insert ECG Cable first.** Route this thru pre–mounted foam tube (if available) located on inside of Arc cover.
2. Route Peripheral Pulse cable and tubes next.
3. Route around foot of magnet to PAC/SRI Platform location.

❑ D4 – INSTALLING REMOTE PAC INTERFACE TO ARC COVER

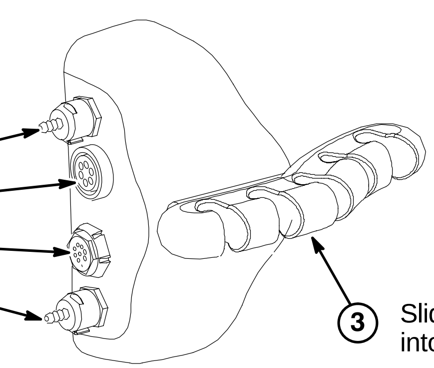
Note:
Routed cables and tubes not shown for clarity.



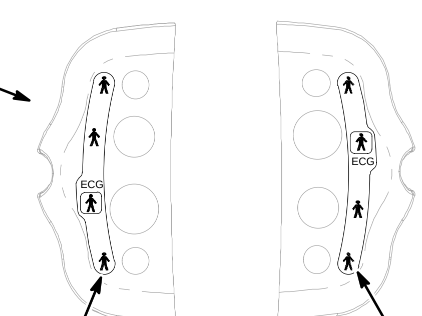
1. Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.
3. Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.
   OR
   Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in **T5** into place.
2. Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place

❑ D5 – INSTALL PAC HANGER BAR

1. Depending on left or right side installation of the PAC Remote Interface Asm on the magnet enclosure, attach the appropriate label that is provided with the kit.
2. Attach cables as indicated below.



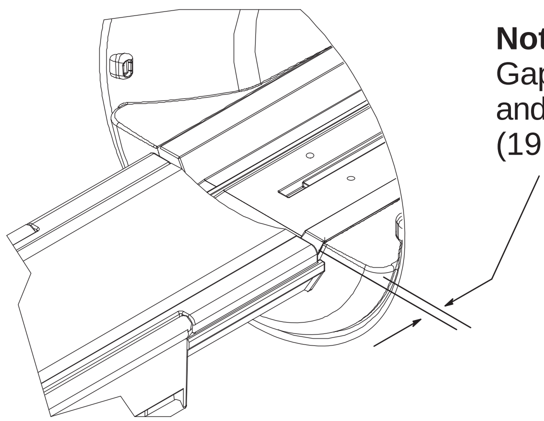
3. Slide PAC Hanger Bar (2221023) into position.



INSTALLATION PROCEDURES SUMMARY

- ❑ E1 – Install Dock Mounting Bracket and Dock
- ❑ E2 – Route and Connect Magnet Enclosure Front Area Cables
- ❑ E3 – Cable Connection to PAC Module

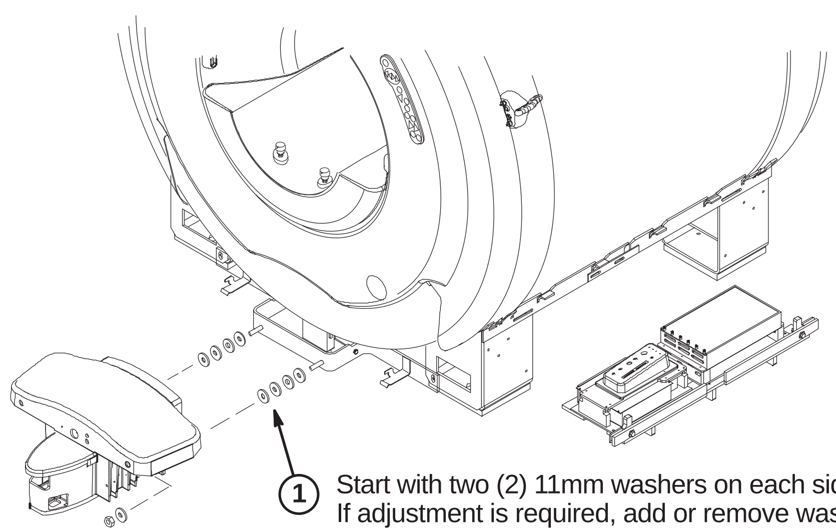
❑ E1 –INSTALL DOCK MOUNTING BRACKET AND DOCK



Note:
Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches $\pm$ 0.25 (19.0$\pm$6.4mm). See Step 2 below.

WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION.
IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY.



1. Start with two (2) 11mm washers on each side between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is acheived.

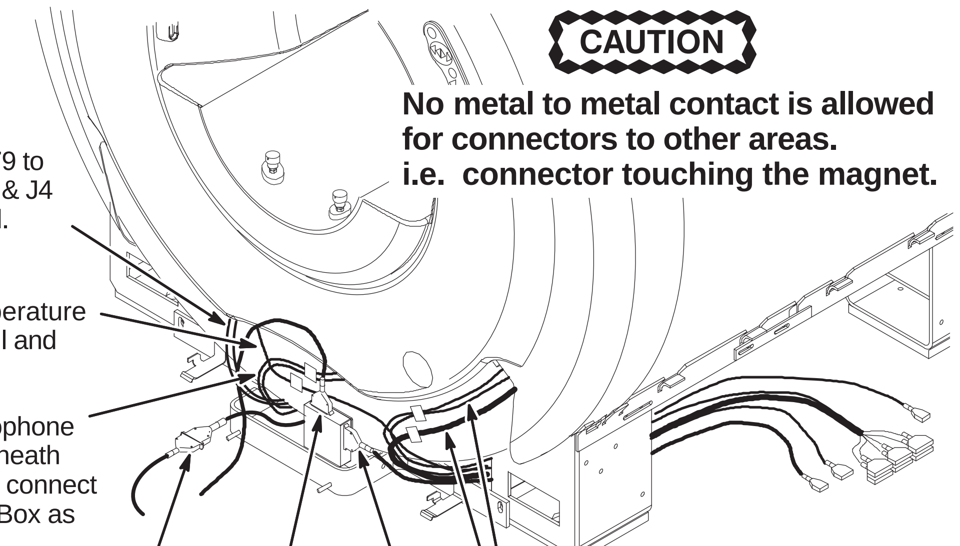
Note:
Floor stud installed by RF room vendor/mechanical contractor.

❑ E2 – ROUTE AND CONNECT MAGNET ENCLOSURE FRONT AREA CABLES

Note:
Dock Assembly not shown for clarity.

CAUTION

No metal to metal contact is allowed for connectors to other areas.
i.e. connector touching the magnet.

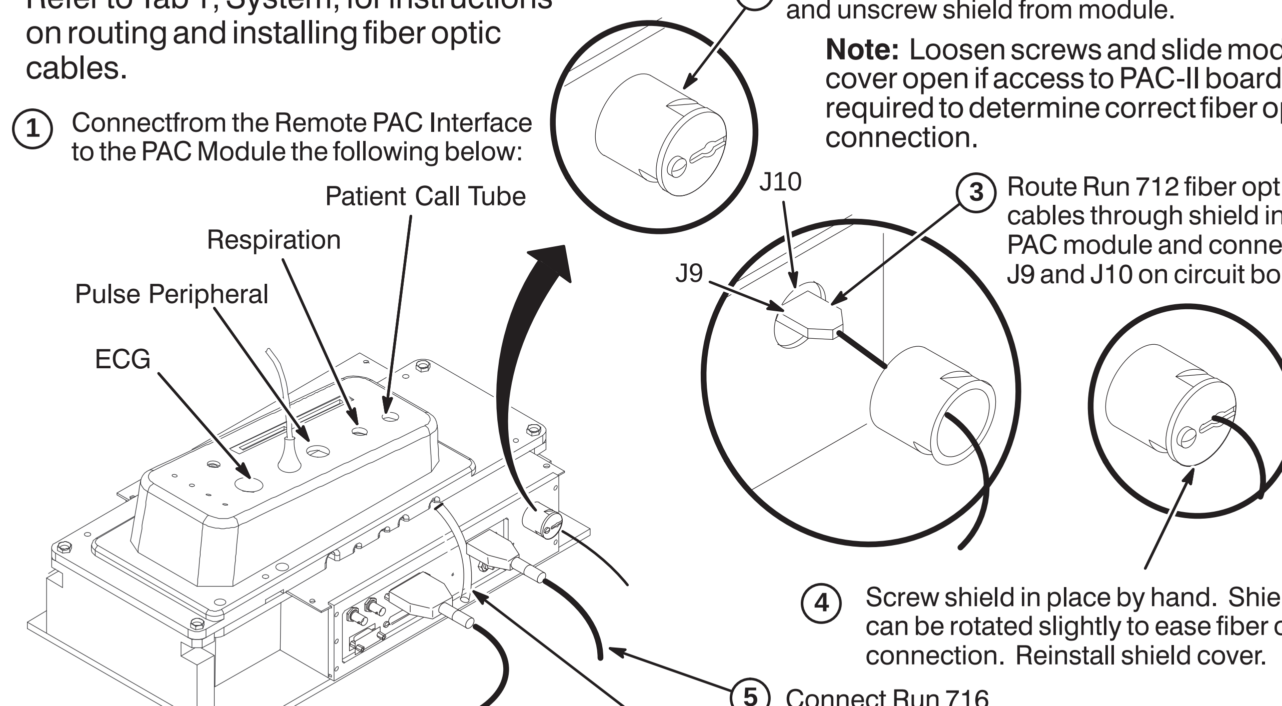


1. Route Runs 778 and 779 to front RF Bias cables J3 & J4 and connect as marked.
2. Route Run 749, Bore Temperature Sensor, from Front End Bell and connect ot J10 on SRI.
3. Route Front End Bell Microphone and Speaker cables underneath manet to ReaPedestal and connect to Patient Communication Box as marked.
4. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.
5. Connect Dock Cable to J2 on Dock Limit Box
6. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.
7. Route cables from Operator Display to area of PAC/SRI Platform and connect as marked to SRI and PAC.

❑ E3 – CABLE CONNECTION TO PAC MODULE

Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

1. Connectfrom the Remote PAC Interface to the PAC Module the following below:

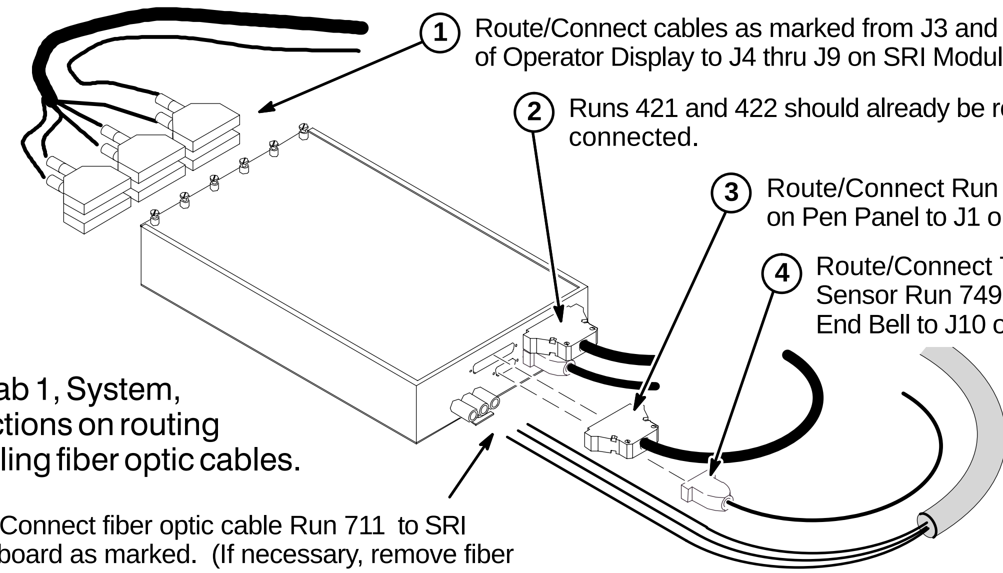


2. Remove shield cover by removing one screw and unscrew shield from module.
   Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.
3. Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.
4. Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.
5. Connect Run 716
6. Connect Pneumatic Tubing from Pen Panel to tubing extension on PAC Module.
7. Connect cable from J5 on Operator Display to J4 on PAC Module.

INSTALLATION PROCEDURES SUMMARY

- ❑ **F1 – Connect Cables to SRI Module**
- ❑ **F2 – Routing of Photo–Plethysmograph Cable**
- ❑ **F3 – Route and Tie–wrap Cables to PAC/SRI Platform**
- ❑ **F4 – Install PAC/SRI Platform under Magnet**

❑ F1 – CONNECT CABLES TO SRI MODULE

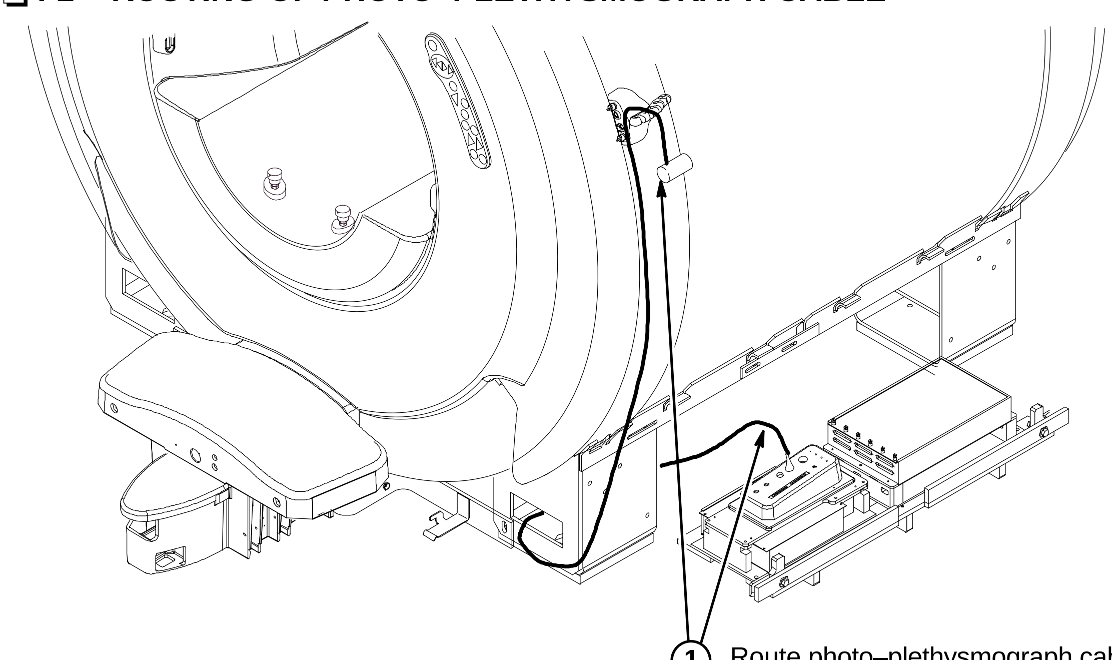


1. Route/Connect cables as marked from J3 and J4 of Operator Display to J4 thru J9 on SRI Module.
2. Runs 421 and 422 should already be routed and connected.
3. Route/Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.
4. Route/Connect Temperture Sensor Run 749 from Front End Bell to J10 on SRI Module

Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

5. Route/Connect fiber optic cable Run 711 to SRI circuit board as marked. (If necessary, remove fiber optic cable guides and pass fiber optic cables thru) Close SRI cover and tighten captive screws.

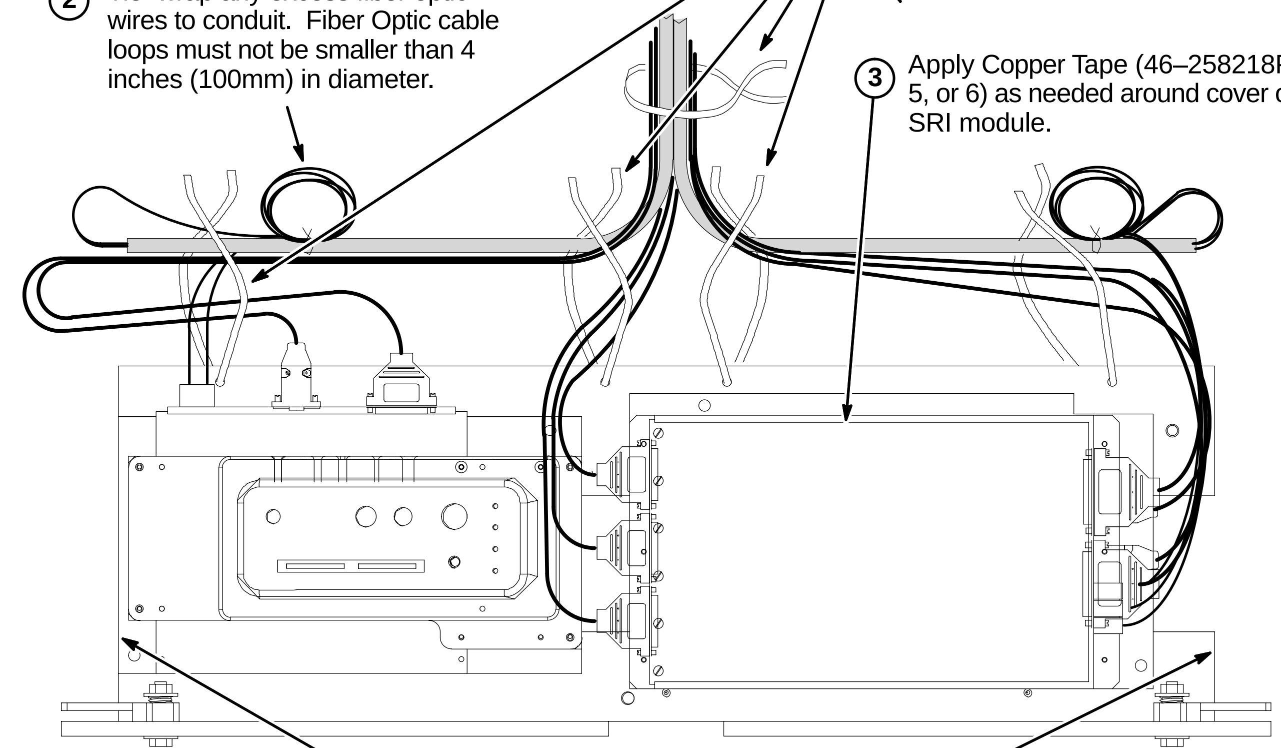
❑ F2 – ROUTING OF PHOTO–PLETHYSMOGRAPH CABLE



1. Route photo–plethysmograph cable from PAC module thru opening in front leg and up to PAC Hanger Bar.
2. Cable must fit between front Foot Cover and side Trim Cover (which are installed later).

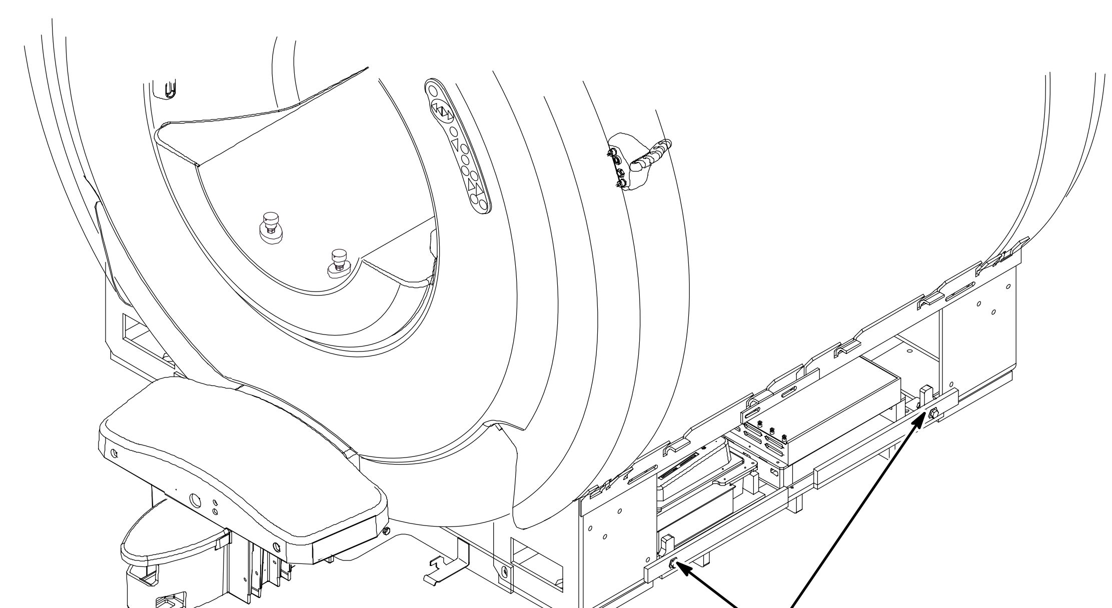
❑ F3 – ROUTE AND TIE–WRAP CABLES TO PAC/SRI PLATFORM

1. Dress all cables and fiber optic conduits as shown and tie–wrap together and to PAC/SRI Platform.
2. Tie–wrap any excess fiber optic wires to conduit. Fiber Optic cable loops must not be smaller than 4 inches (100mm) in diameter.
3. Apply Copper Tape (46–258218P4, 5, or 6) as needed around cover of SRI module.



Note: Keep cables dressed inside edge so unit slides between legs easily for service.

❑ F4 – INSTALL PAC/SRI PLATFORM UNDER MAGNET



1. Carefully slide PAC/SRI Platform between legs of magnet and lock in place at each end.

BRIDGE INSTALLATION INFORMATION

TWO TYPES OF BRIDGES ARE AVAILABLE FOR INSTALLATION: The original style Long Bridge and the new style Split Bridge. See items 1 and 2 below for explanation of which installation procedures to use.

1. **If original style Long Bridge is delivered to site, complete procedures on pages 7, 8, and 9. This bridge would be used for pre–Split Bridge MR/i and all CV/i installations.**

2. **If new style Split Bridge is delivered to site, complete procedures on pages 10, 11, and 12. This bridge would be used for MR/i installations only, not CV/i.**

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

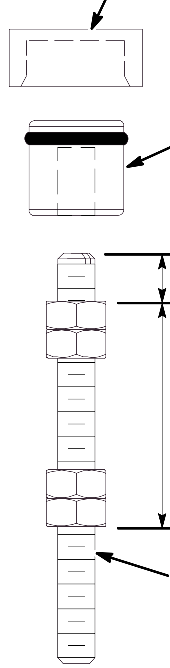
- ❑ **G1 – Front Bridge Support Adjustments**
- ❑ **G2 – Bridge Support Adjustment Tolerances**
- ❑ **G3 – Bridge Assembly Preparation Before Installing in Magnet**
- ❑ **G4 – Install Bridge**

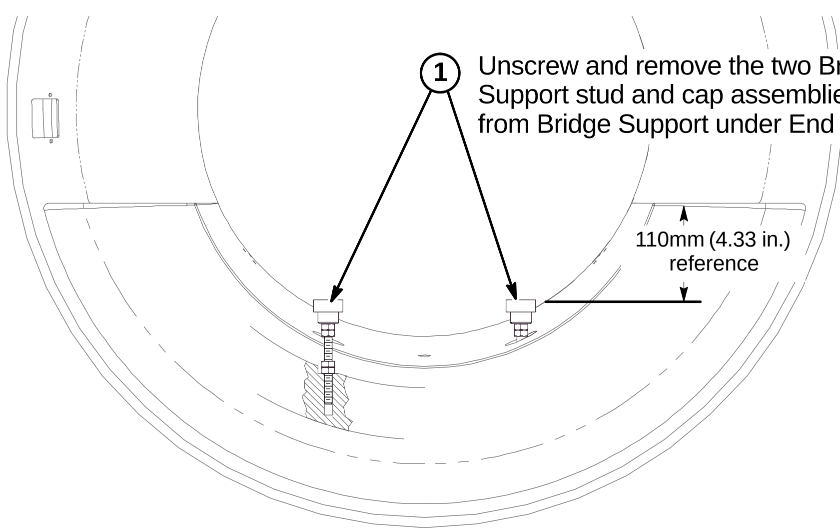
NOTE:
Measure to confirm that the distance between the top of the Support Cap Cup and the flat surface of the End Bell is110mm as shown in Procedure **G1**. Also measure to confirm that the 10mm and 62mm M10 Jam Nut distance locations on the Stud are also in place.

- If the dimensions are found to be in place for this site, proceed to Procedure **G2**.
- If these dimensions are not in place, complete Steps 1 thru 6 in Procedure **G1** and then proceed to **G2**.

❑ G1 – FRONT BRIDGE SUPPORT ADJUSTMENTS

(2) Remove Support Cap Cup and unscrew Support Cap





(1) Unscrew and remove the two Bridge Support stud and cap assemblies from Bridge Support under End Bell.

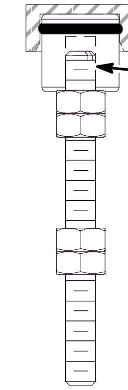
(3) Set top M10 Nut at a distance of 10mm from top of stud. Lock in position with second M10 Nut.

(4) Set bottom M10 Nut at a distance of 62mm from top of first M10 Nut. Lock in position with fourth M10 Nut.

(5) Fasten Support Cap to top of Stud and tighten to top M10 Nut. Install Support Cap Cup to Support Cap

(6) Re–install the two Bridge Support stud and cap assemblies into Bridge Support.

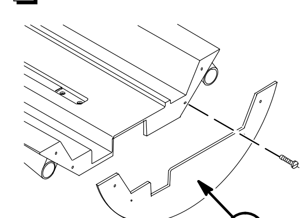
❑ G2 – BRIDGE SUPPORT ADJUSTMENT TOLERANCES

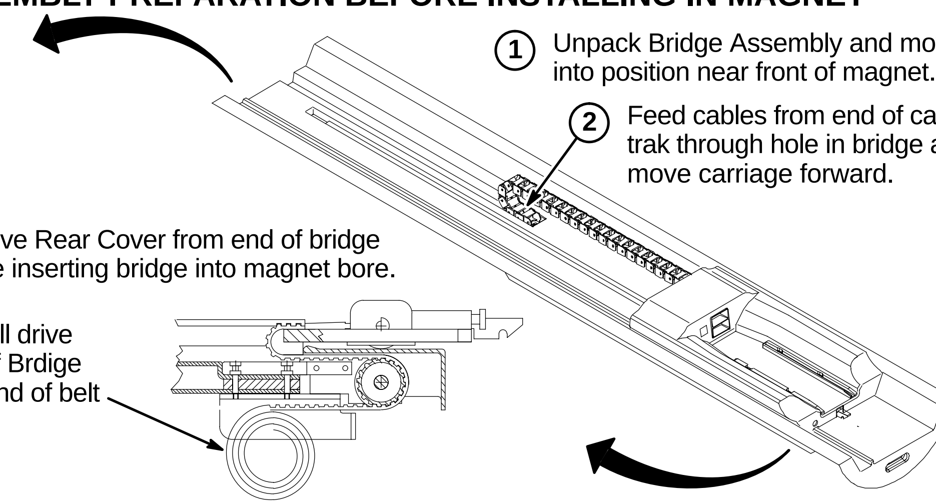


(1) The 10mm location distance noted in Procedure **G1** is a starting point for installation and leveling of Bridge.

This starting point gives the installer an adjustment of +7.5mm and –5mm. The amount of adjustment will be determined during bridge leveling.

❑ G3 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET





(1) Unpack Bridge Assembly and move into position near front of magnet.

(2) Feed cables from end of cable trak through hole in bridge and move carriage forward.

(3) If not already done, unroll drive belt along bottom side of Brdige and temporarilyy tape end of belt to rear of bridge.

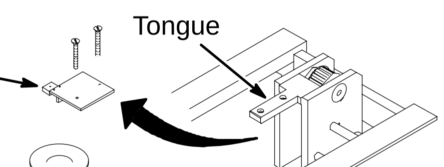
(4) Remove Rear Cover from end of bridge before inserting bridge into magnet bore.

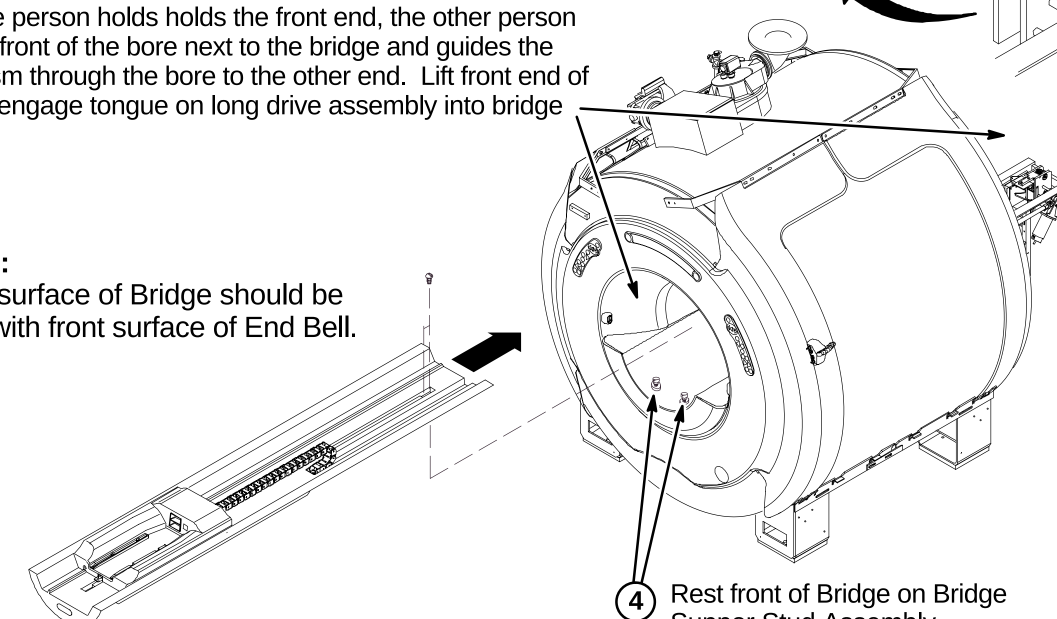
❑ G4 – INSTALL BRIDGE

(1) Unfasten Rear Bridge Mount Asm. It will be reinstalled later.

(2) Position rear end of Bridge Asm at front of Body Coil.

(3) While one person holds holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.





NOTE:
Front surface of Bridge should be flush with front surface of End Bell.

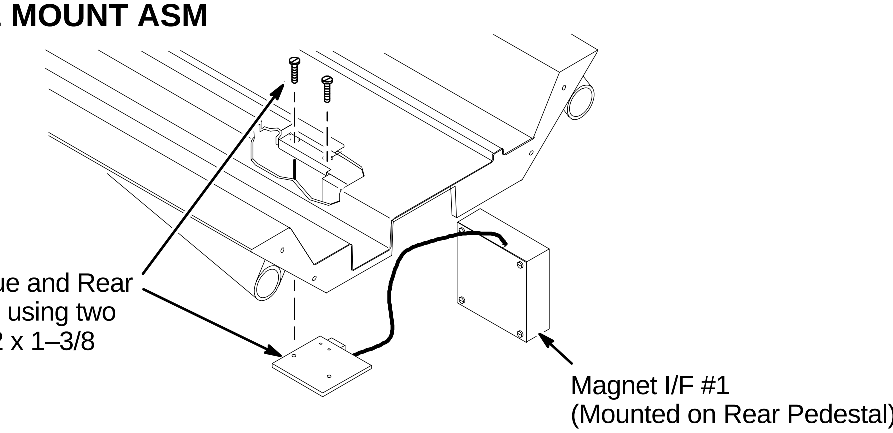
(4) Rest front of Bridge on Bridge Suppor Stud Assembly.

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

- ❑ H1 – Reattach Bridge Mount Assembly
- ❑ H2 – Align Front of Bridge to End Bell
- ❑ H3 – Applying Adhesive to Bridge Support Caps
- ❑ H4 – Leveling Bridge

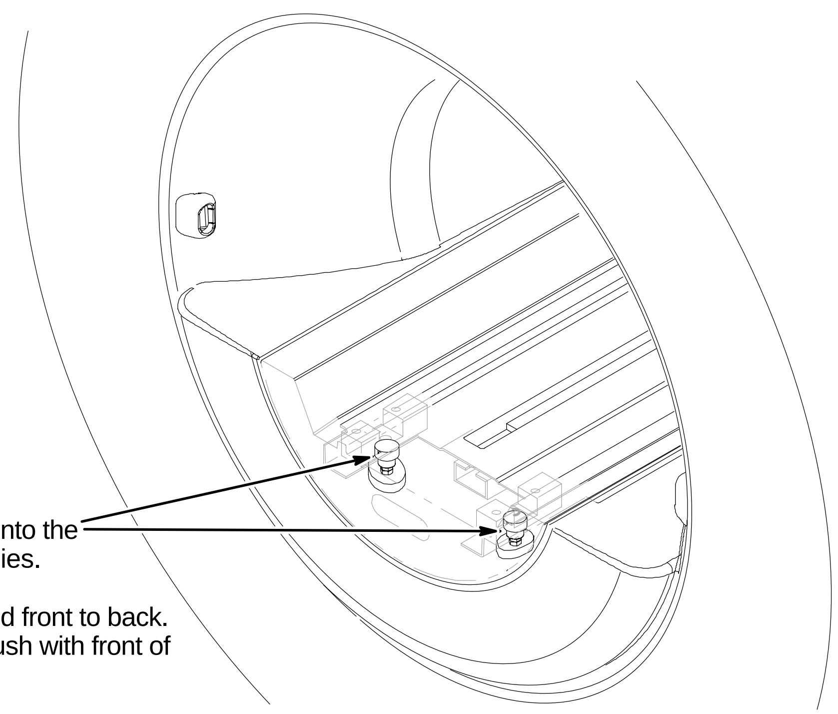
❑ H1 – REATTACH BRIDGE MOUNT ASM

NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.



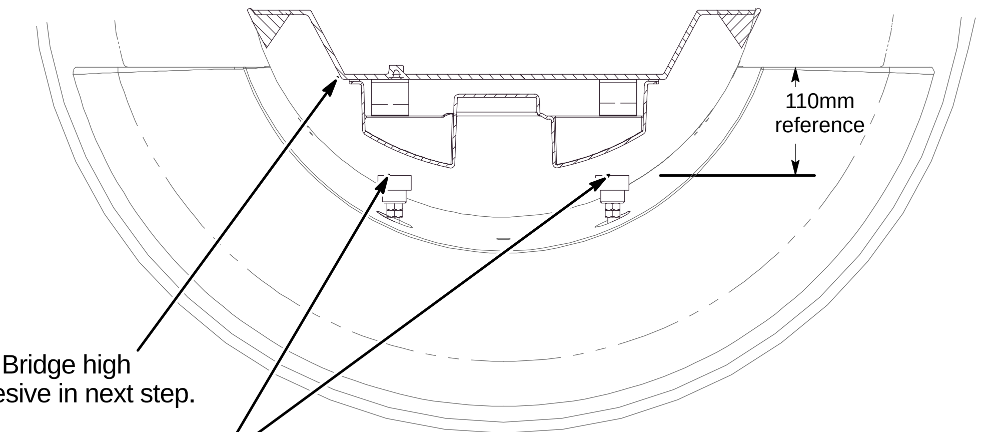
1. Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) previously removed 10–32 x 1–3/8 Screws (46–220184P47)

❑ H2 – ALIGN FRONT OF BRIDGE TO END BELL



1. Lower the front of the Bridge onto the Bridge Support Stud Assemblies.
2. Align the Bridge left to right and front to back. Make sure front of Bridge is flush with front of end bell.

❑ H3 – APPLYING ADHESIVE TO BRIDGE SUPPORT CAPS



1. Raise the front of the Bridge high enough to apply adhesive in next step.
2. Apply epoxy adhesive (46–282264P1) to top of each Bridge Support Cap.
3. Lower the Bridge to rest firmly to the Bridge Support Caps.
4. Allow adhesive to cure.
5. After adhesive has cured, adjust the Bridge to the final vertical position. Make sure bridge is level.

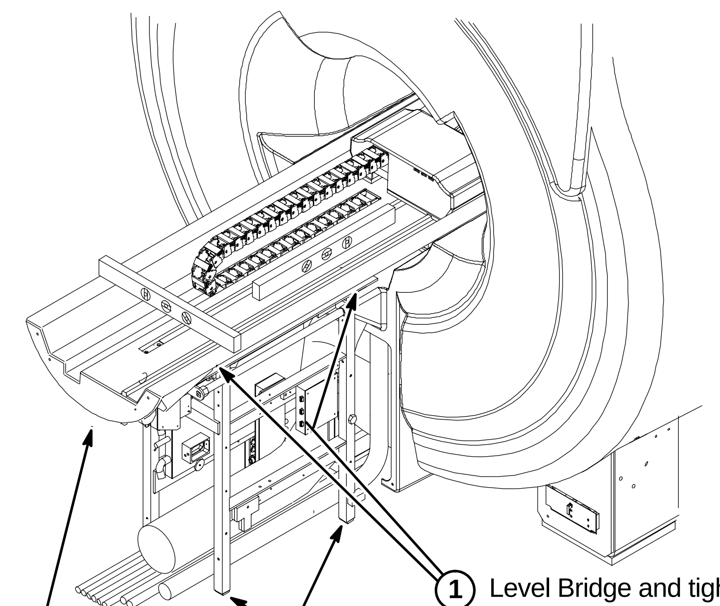
   Note: The final position will leave the top surface of the Bridge 1mm (or more) above the top surface of the end bell.
6. Repeat the vertical adjustment as needed. Make sure the two M10 jam nuts are tight against each Support Cap. This will prevent change of position.

❑ H4 – LEVELING BRIDGE

NOTE:
PVC Spacers positioned thru End Bells are to support Bridge.

NOTE:
Rear Pedestal to be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

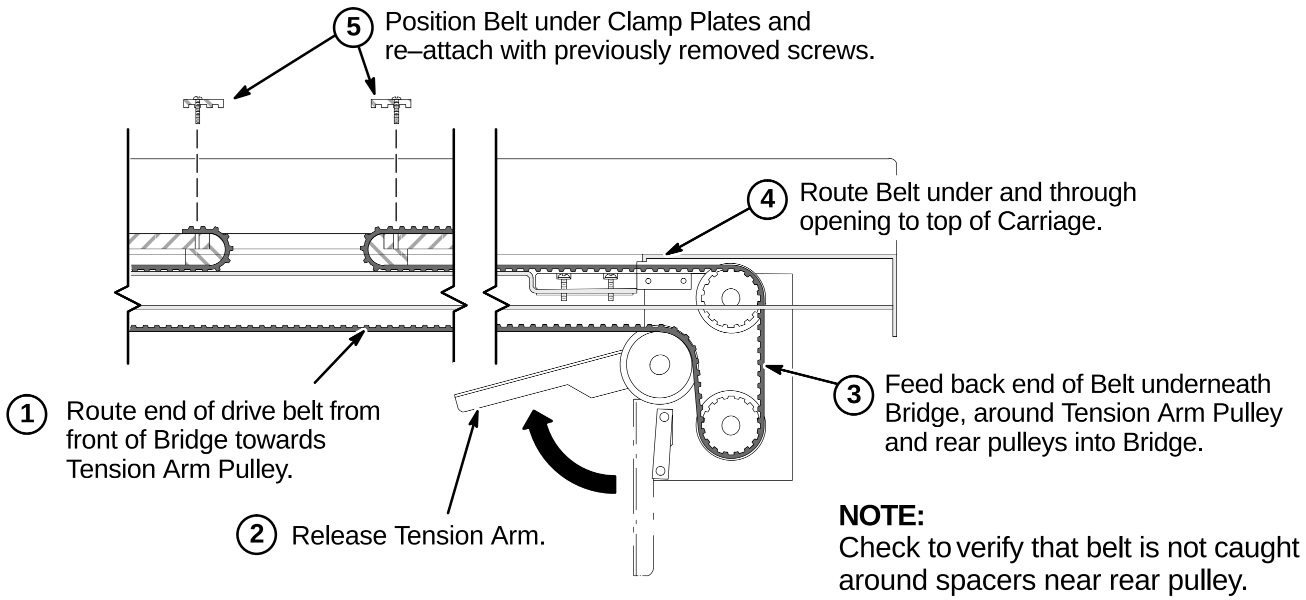


1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

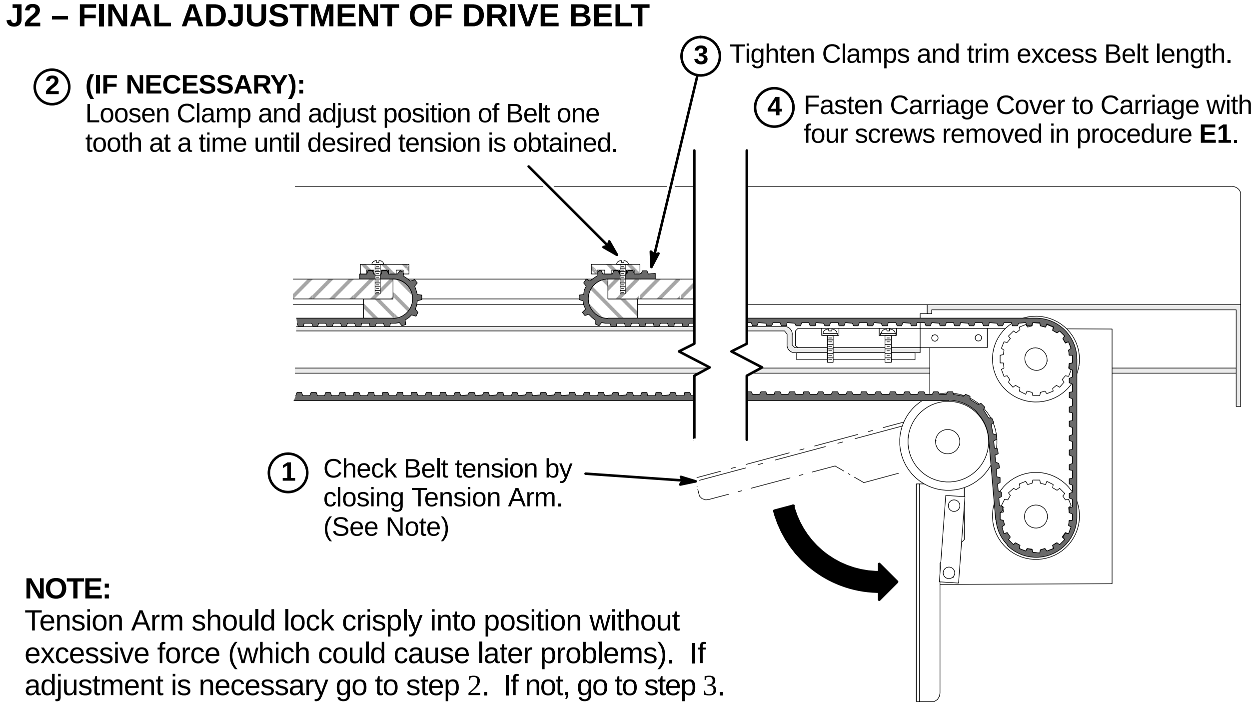
- ❑ J1 – Routing Drive Belt
- ❑ J2 – Final Adjustment of Drive Belt
- ❑ J3 – Installation of Sensor Flag
- ❑ J4 – Attach Bridge Cover
- ❑ J5 – Route/Connect Runs 404 and 485 (and Multi–coil Cables if Required)

❑ J1 – ROUTING OF DRIVE BELT



NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

❑ J2 – FINAL ADJUSTMENT OF DRIVE BELT



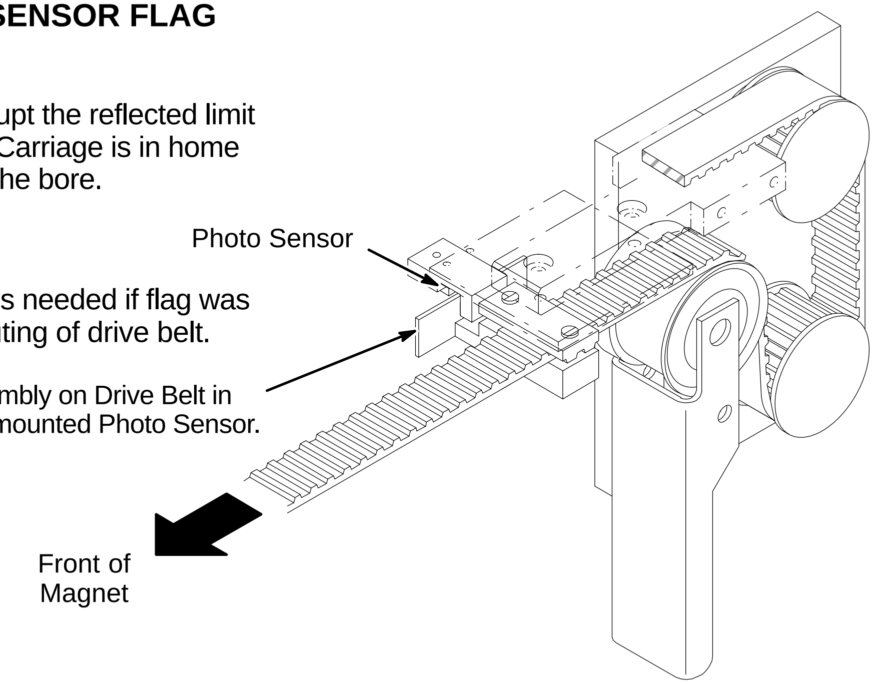
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

❑ J3 – INSTALLATION OF SENSOR FLAG

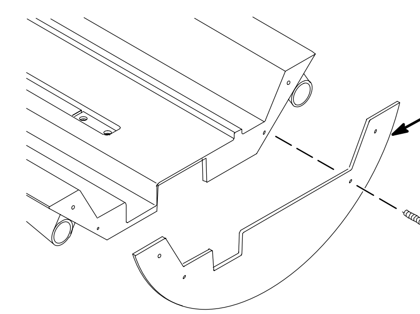
NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

1. Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



❑ J4 – ATTACH BRIDGE COVER

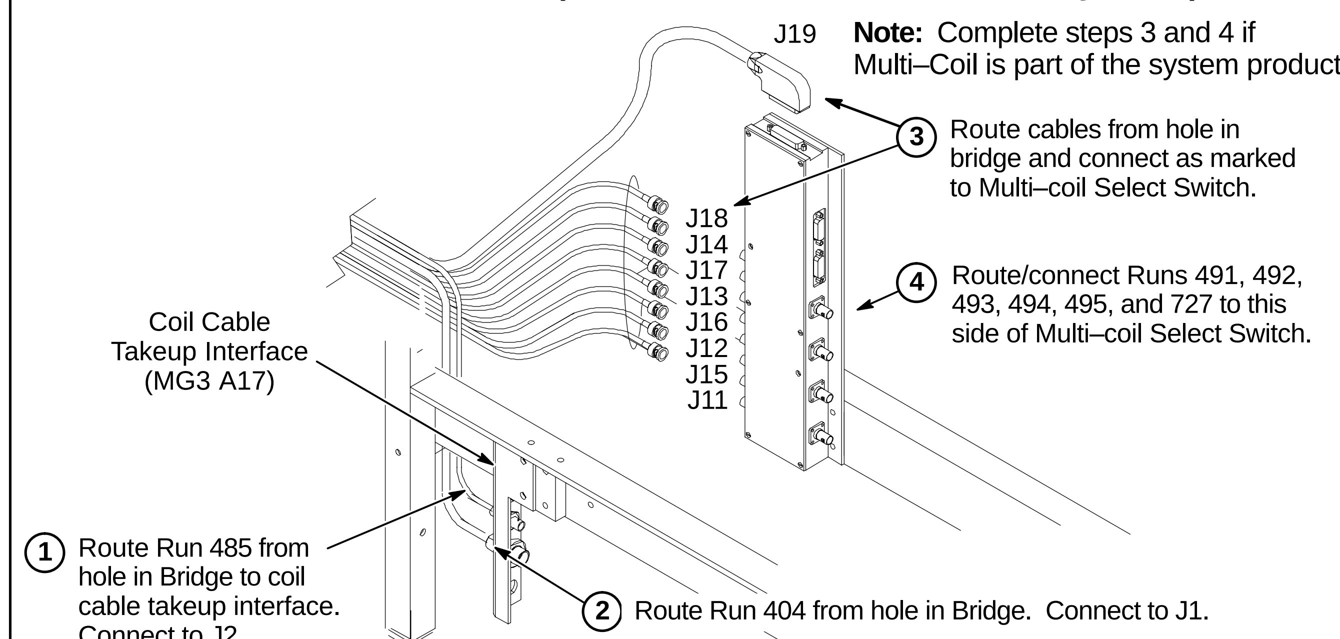


1. Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.

Note:
Bridge Cover has enclosure rating plates on it.

❑ J5 – ROUTE RUNS 404 AND 485 (AND MULTI–COIL CABLES IF REQUIRED)

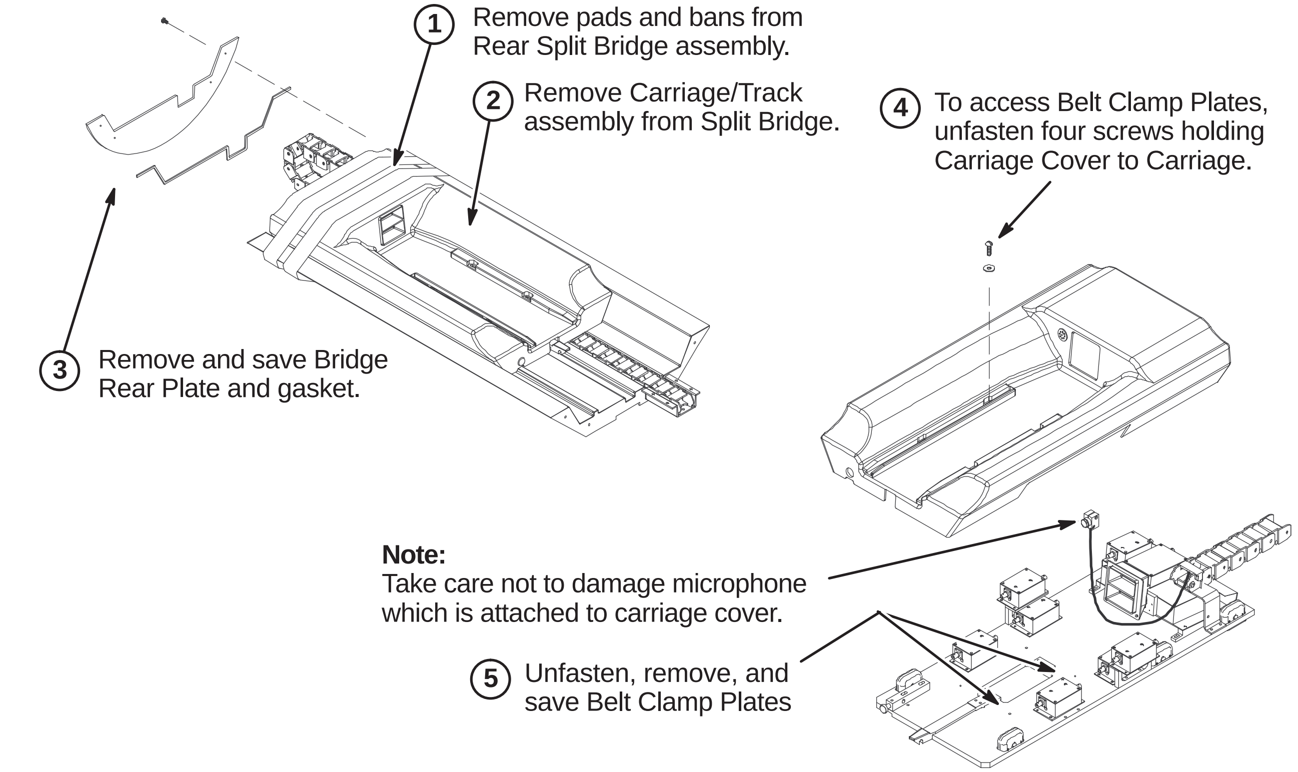
Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ❑ K1 – Preparing Rear Split Bridge for Installation
- ❑ K2 – Attaching Rear Split Bridge to Rear Pedestal
- ❑ K3 – Attaching Rear Split Bridge to Front Split Bridge
- ❑ K4 – Aligning Bridge on Rear Pedestal
- ❑ K5 – Installing Bridge Locators

❑ K1 – PREPARING REAR SPLIT BRIDGE FOR INSTALLATION



1 Remove pads and bans from Rear Split Bridge assembly.

2 Remove Carriage/Track assembly from Split Bridge.

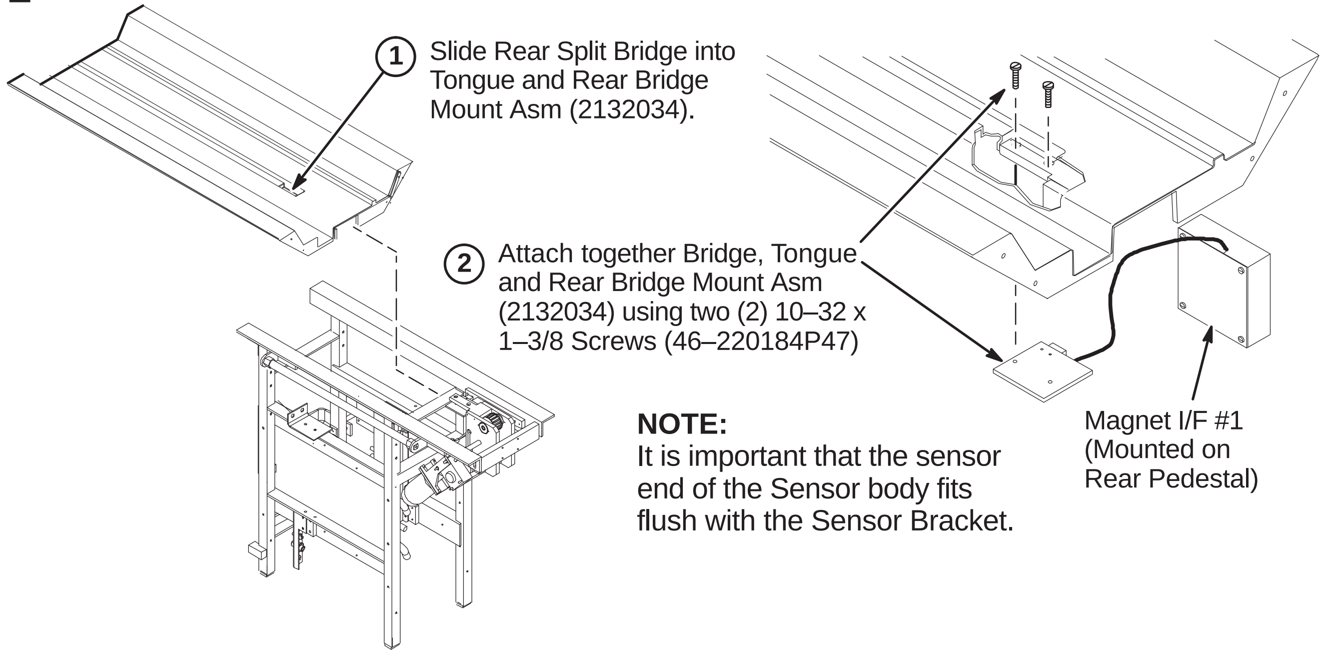
3 Remove and save Bridge Rear Plate and gasket.

4 To access Belt Clamp Plates, unfasten four screws holding Carriage Cover to Carriage.

Note:
Take care not to damage microphone which is attached to carriage cover.

5 Unfasten, remove, and save Belt Clamp Plates

❑ K2 – ATTACHING REAR SPLIT BRIDGE TO REAR PEDESTAL



1 Slide Rear Split Bridge into Tongue and Rear Bridge Mount Asm (2132034).

2 Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) 10–32 x 1–3/8 Screws (46–220184P47)

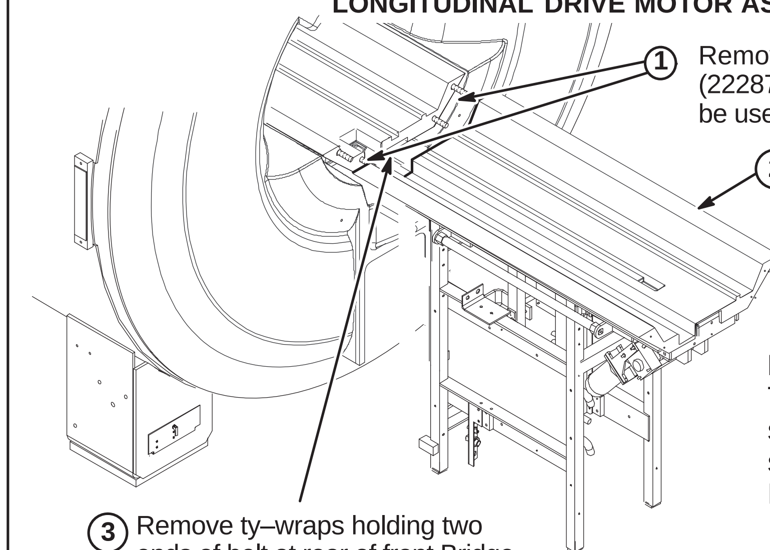
NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.

Magnet I/F #1 (Mounted on Rear Pedestal)

❑ K3 – ATTACHING REAR SPLIT BRIDGE TO FRONT SPLIT BRIDGE

WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.



1 Remove four 10M hex nuts (2109877–3) and washers (2228717–2) from end of Bridge in magnet. These will be used for attaching the Rear Pedestal.

2 Move Rear Pedestal/Bridge into position and fasten with four nuts and washers that were removed form end of Front Split Bridge.

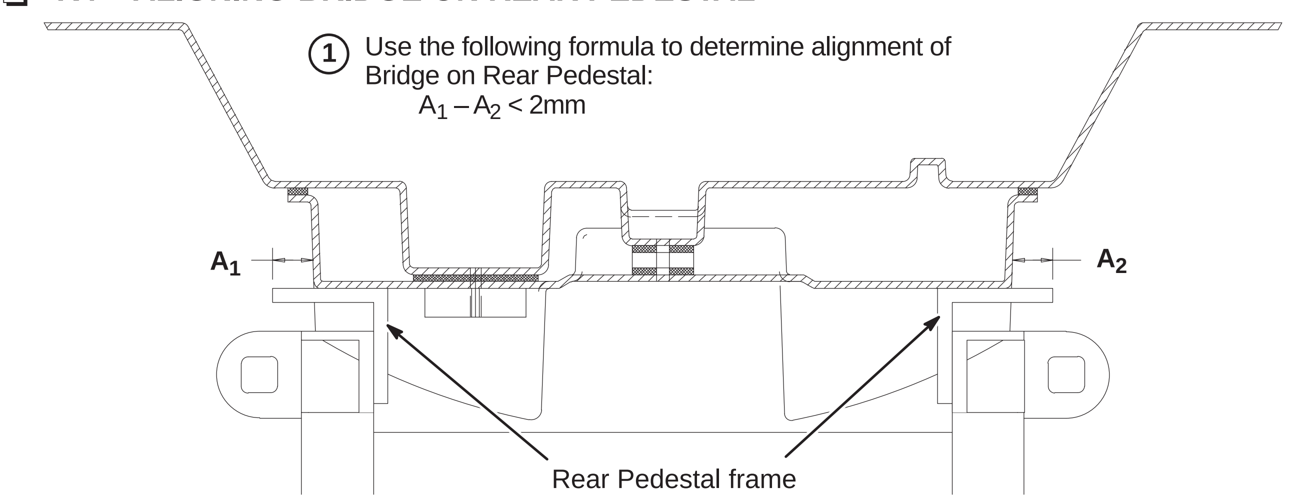
3 Remove ty–wraps holding two ends of belt at rear of front Bridge.

NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

❑ K4 – ALIGNING BRIDGE ON REAR PEDESTAL

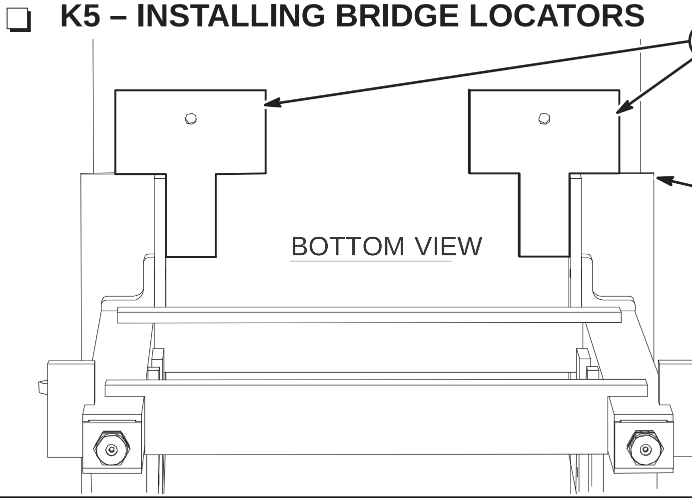
1 Use the following formula to determine alignment of Bridge on Rear Pedestal:

$A_1 – A_2 < 2mm$



❑ K5 – INSTALLING BRIDGE LOCATORS

1 After Bridge and Rear Pedestal are aligned, peel off protective paper on each Bridge Locator and attach to bottom side of Bridge as shown.

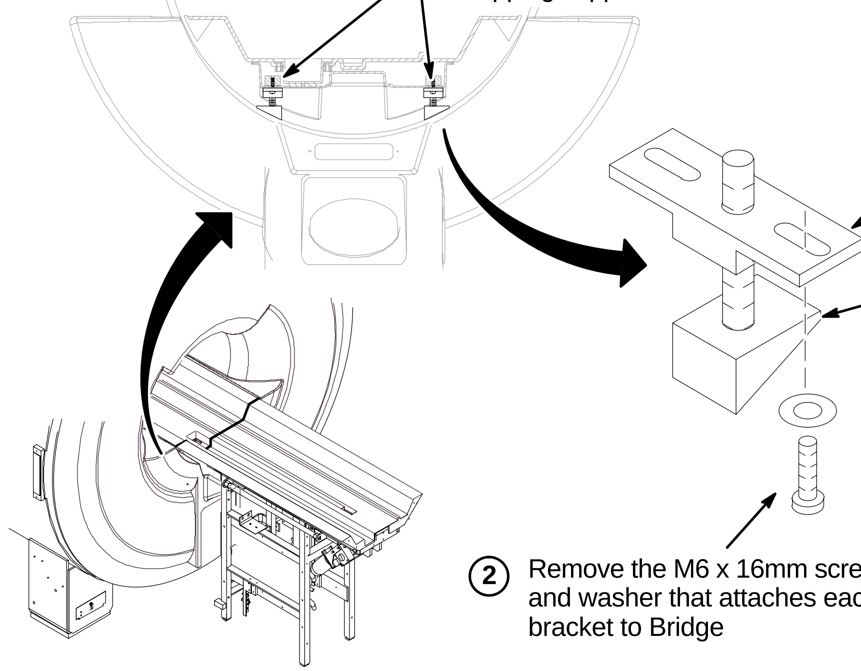


End of Rear Pedestal frame closest to Magnet.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

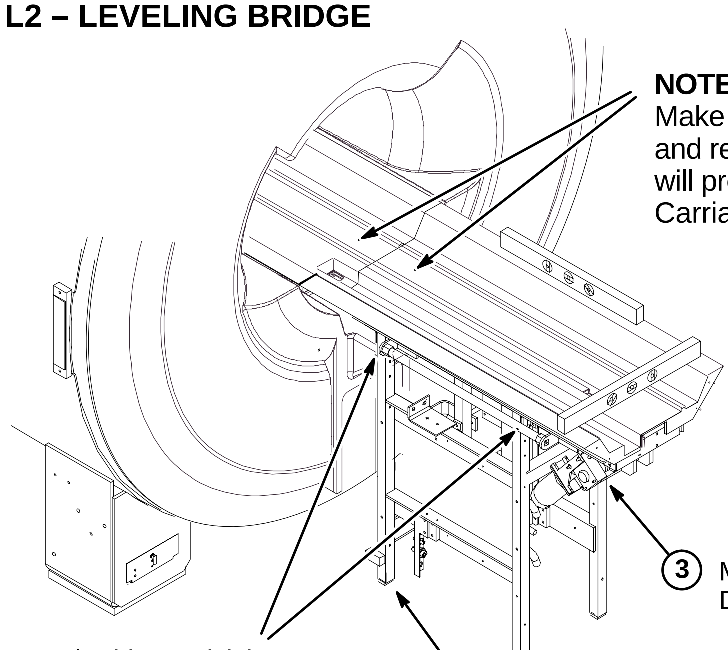
- ❑ L1 – Removing Front Split Bridge Shipping Support Brackets
- ❑ L2 – Leveling Bridge
- ❑ L3 – Height Adjustment at Front of Bridge (if needed)
- ❑ L4 – Installing Carriage on Bridge
- ❑ L5 – Routing of Drive Belt

❑ L1 – REMOVING FRONT SPLIT BRIDGE SHIPPING SUPPORT BRACKETS



1. After Front and Rear Split Bridge parts are fastened together, the two shipping supports at the rear of the Front Split Bridge need to be removed.
2. Remove the M6 x 16mm screw and washer that attaches each bracket to Bridge
3. Unscrew downword the Bridge Mounting Flange.
4. Support Base is attached to End Bell with adhesive tape. To remove, carefully pry base away from end bell. Remove base, stud, and mounting flange

❑ L2 – LEVELING BRIDGE

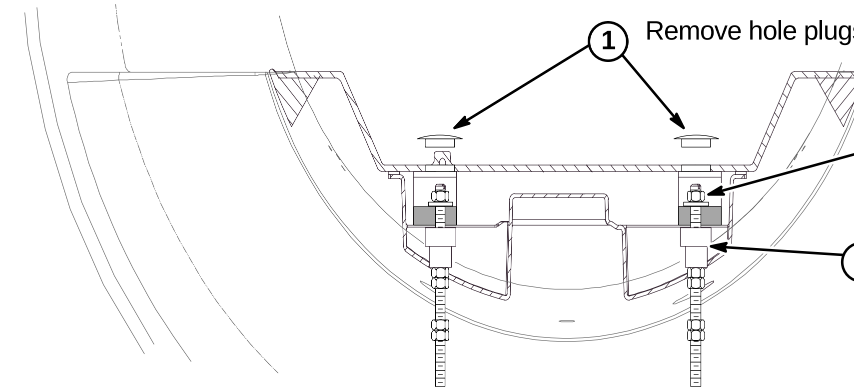


NOTE:
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:
Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

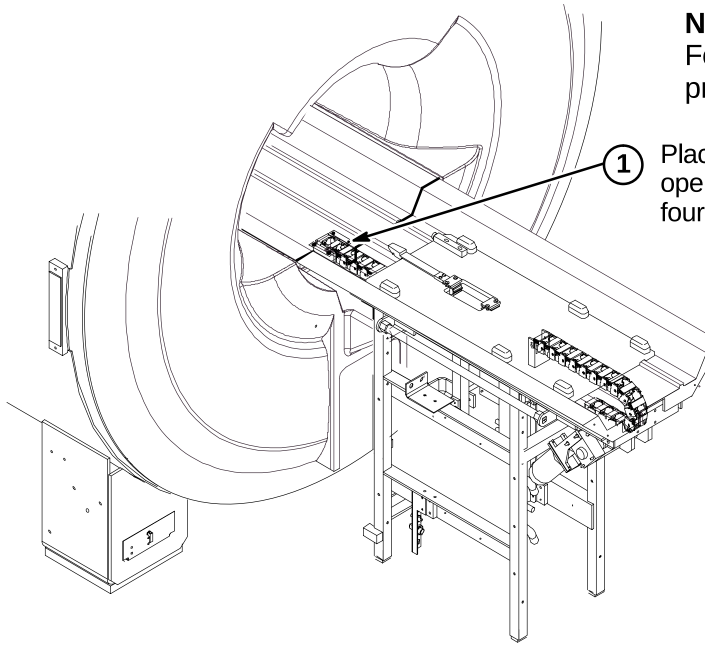
1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

❑ L3 – HEIGHT ADJUSTMENT AT FRONT OF BRIDGE (IF NEEDED)



1. Remove hole plugs from each side of Bridge.
2. Remove M10 nut and washer from each stud.
3. Adjust Support Nut (2293381) as needed to achieve correct height.

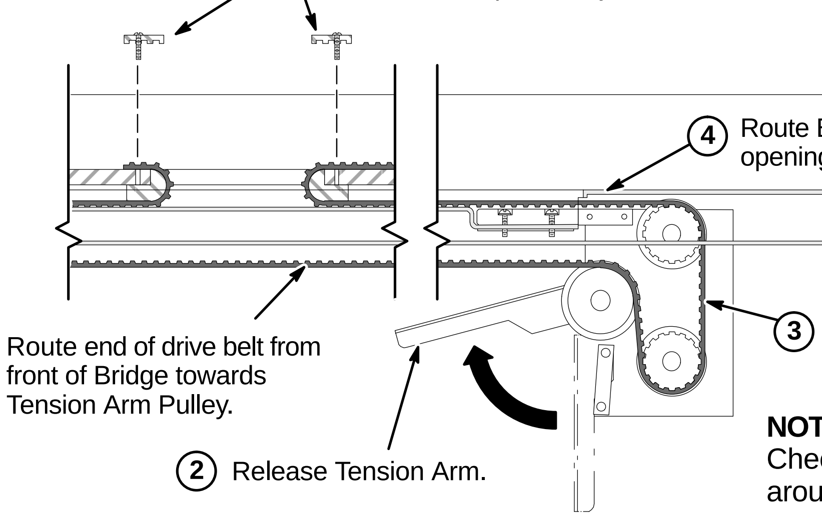
❑ L4 – INSTALLING CARRIAGE ON BRIDGE



NOTE:
For clarity, the carriage is shown without preamps, preamp protect module, and cables.

1. Place Carriage on Bridge and route all cables thru opening in Bridge and fasten plate to Front Bridge with four M6 x 12mm nylon screws (2296958) provided.

❑ L5 – ROUTING OF DRIVE BELT



1. Route end of drive belt from front of Bridge towards Tension Arm Pulley.
2. Release Tension Arm.
3. Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.
4. Route Belt under and through opening to top of Carriage.
5. Position Belt under Clamp Plates and re–attach with previously removed screws.

NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

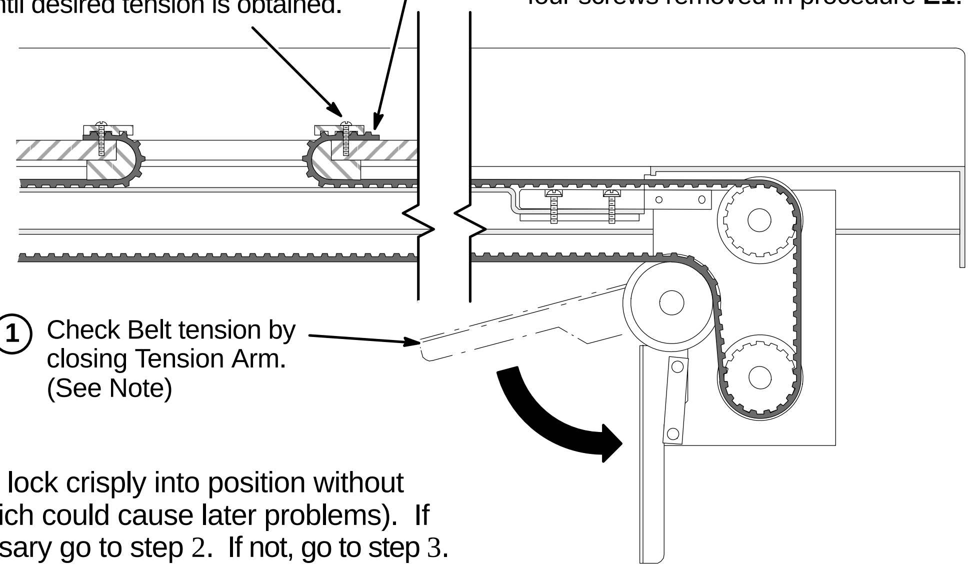
- ❑ **M1 – Final Adjustment of Drive Belt**
- ❑ **M2 – Installation of Sensor Flag**
- ❑ **M3A – Route Non–MGD Runs 404 and 485 (and Multi–coil Cables if Required)**
  OR
- ❑ **M3B – Route/Connect MGD Cables from LPCA to Rear Pedestal**
- ❑ **M4 – Installing Carriage Cover to Trolley and Bridge Cover**

❑ M1 – FINAL ADJUSTMENT OF DRIVE BELT

2 **(IF NECESSARY):**
Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.

3 Tighten Clamps and trim excess Belt length.

4 Fasten Carriage Cover to Carriage with four screws removed in procedure **E1**.



1 Check Belt tension by closing Tension Arm. (See Note)

NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

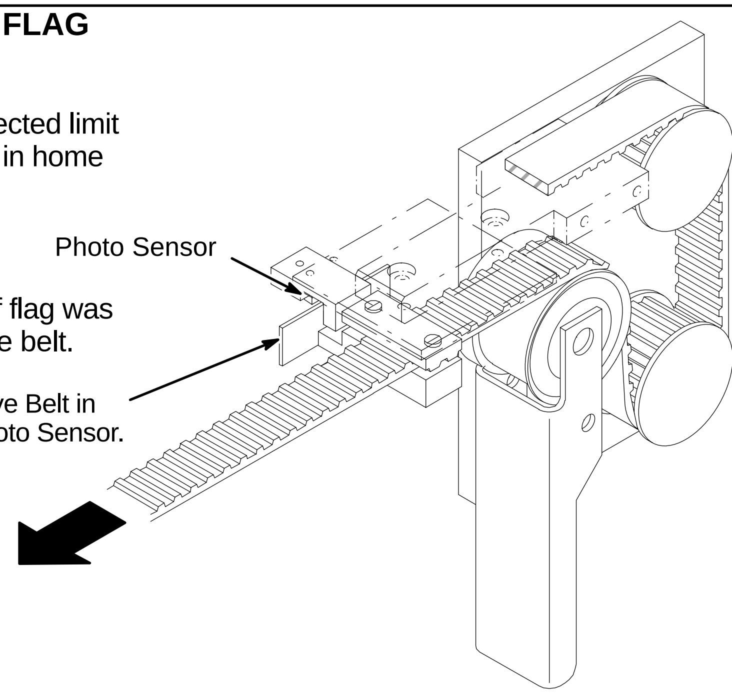
❑ M2 – INSTALLATION OF SENSOR FLAG

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

Photo Sensor

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

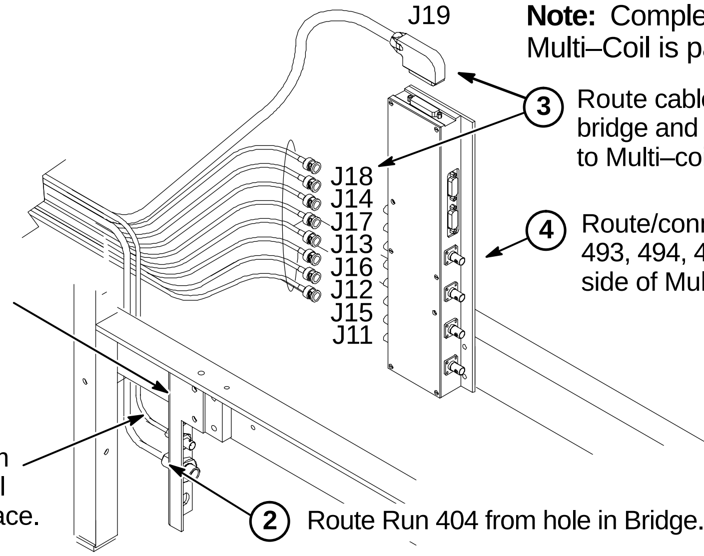
1 Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



Front of Magnet

❑ M3A – ROUTE NON–MGD RUNS 404 AND 485 (& Multi–Coil Cables if Required)

Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



3 Route cables from hole in bridge and connect as marked to Multi–coil Select Switch.

4 Route/connect Runs 491, 492, 493, 494, 495, and 727 to this side of Multi–coil Select Switch.

Coil Cable Takeup Interface (MG3 A17)

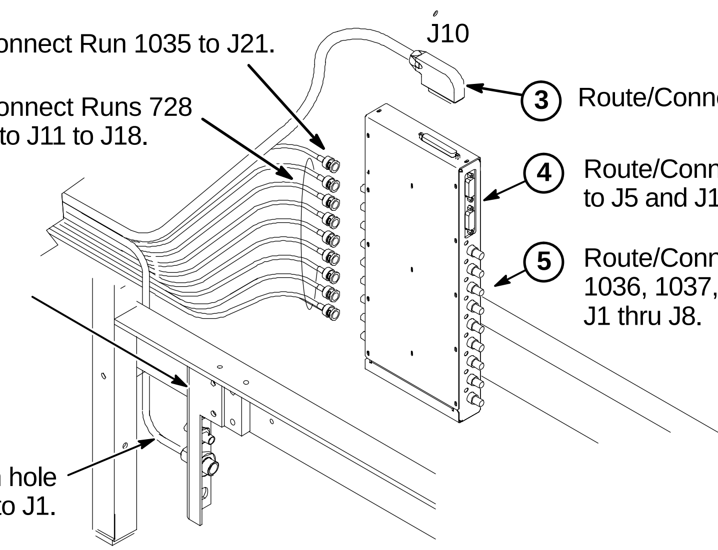
1 Route Run 485 from hole in Bridge to coil cable takeup interface. Connect to J2.

2 Route Run 404 from hole in Bridge. Connect to J1.

❑ M3B – ROUTE/CONNECT MGD CABLES FROM LPCA TO REAR PEDESTAL

1 Route/Connect Run 1035 to J21.

2 Route/Connect Runs 728 thru 735 to J11 to J18.



3 Route/Connect Run 1041 to J19.

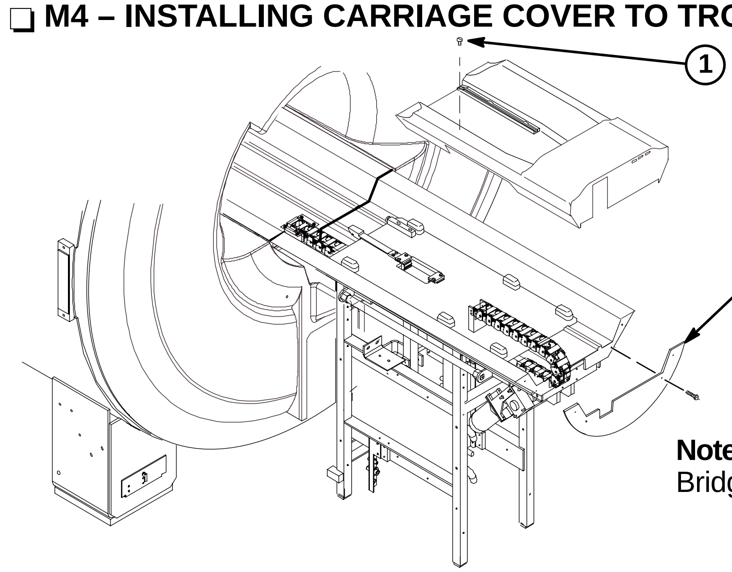
4 Route/Connect Runs 1042 and 1049 to J5 and J19.

5 Route/Connect Runs 491, 492, 493, 1036, 1037, 1038, 1039, and 1040 to J1 thru J8.

6 Route Run 404 from hole in Bridge. Connect to J1.

❑ M4 – INSTALLING CARRIAGE COVER TO TROLLEY AND BRIDGE COVER

1 Attach Carriage Cover to Trolley using four previously removed fasteners.



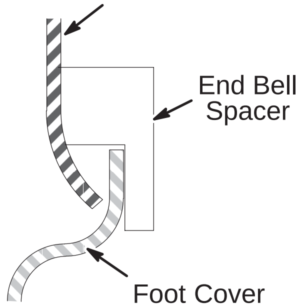
Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.

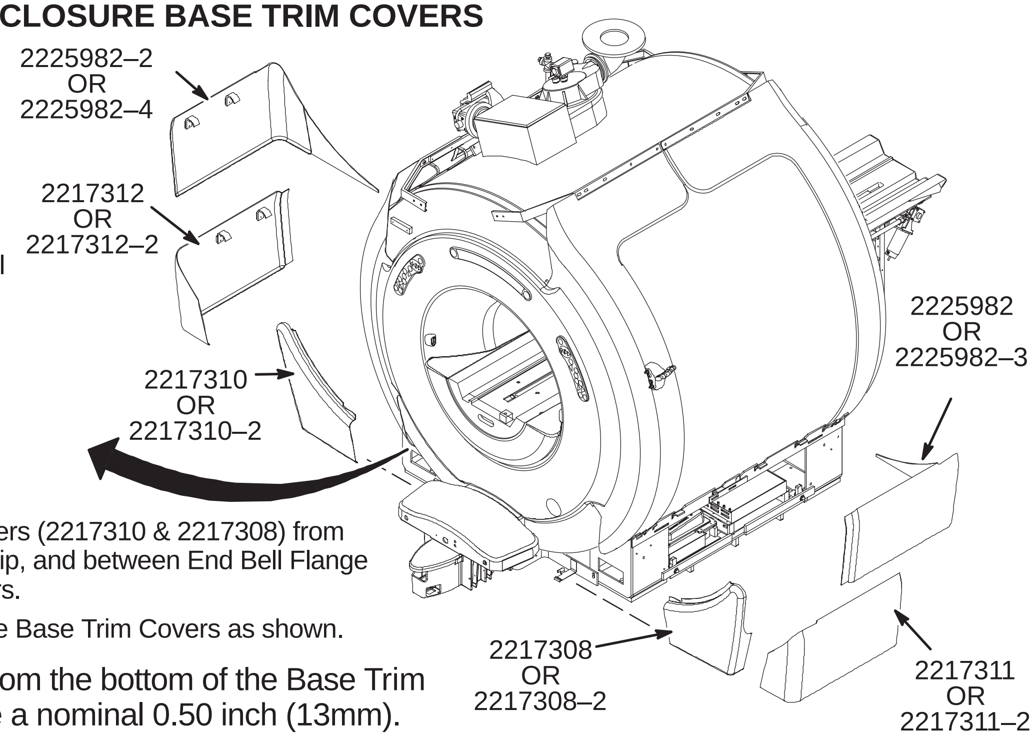
Note:
Bridge Cover has enclosure rating plates on it.

INSTALLATION PROCEDURES SUMMARY

- ❑ **N1 – Install Enclosure Base Trim Covers**
- ❑ **N2 – Install Enclosure Top Covers**
- ❑ **N3 – Install Rear Pedestal Covers**

❑ N1 – INSTALL ENCLOSURE BASE TRIM COVERS





1. Slide front Foot Covers (2217310 & 2217308) from side, above spring clip, and between End Bell Flange and End Bell Spacers.
2. Install remaining side Base Trim Covers as shown.

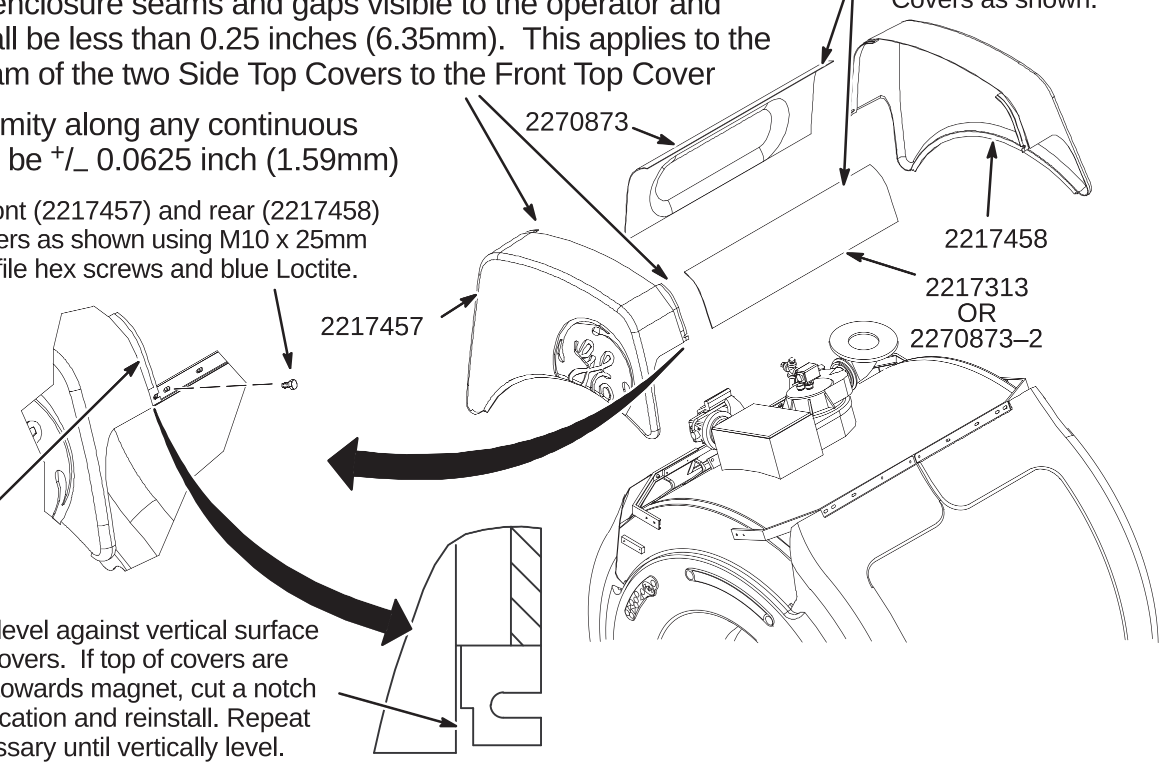
Note: The gap from the bottom of the Base Trim to floor should be a nominal 0.50 inch (13mm).

❑ N2 – INSTALL ENCLOSURE TOP COVERS

Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

Gap uniformity along any continuous seam shall be $^{+}/_{-}$ 0.0625 inch (1.59mm)

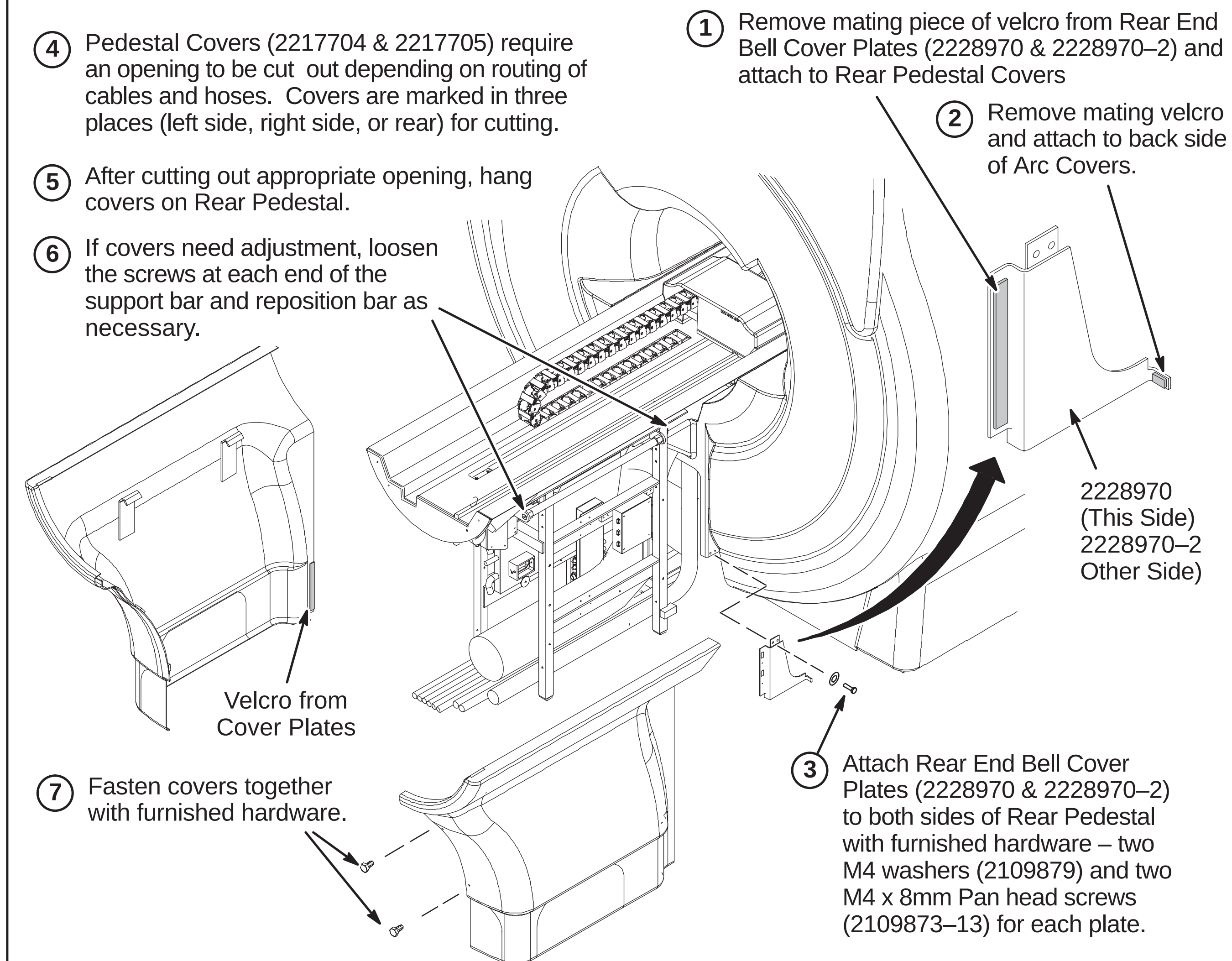
1. Install front (2217457) and rear (2217458) Top Covers as shown using M10 x 25mm Low Profile hex screws and blue Loctite.
2. Install side Top Covers as shown.



2. Place a level against vertical surface of Top Covers. If top of covers are leaning towards magnet, cut a notch at this location and reinstall. Repeat as necessary until vertically level.

❑ N3 – INSTALL REAR PEDESTAL COVERS

1. Remove mating piece of velcro from Rear End Bell Cover Plates (2228970 & 2228970–2) and attach to Rear Pedestal Covers
2. Remove mating velcro and attach to back side of Arc Covers.
3. Attach Rear End Bell Cover Plates (2228970 & 2228970–2) to both sides of Rear Pedestal with furnished hardware – two M4 washers (2109879) and two M4 x 8mm Pan head screws (2109873–13) for each plate.
4. Pedestal Covers (2217704 & 2217705) require an opening to be cut out depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting.
5. After cutting out appropriate opening, hang covers on Rear Pedestal.
6. If covers need adjustment, loosen the screws at each end of the support bar and reposition bar as necessary.
7. Fasten covers together with furnished hardware.



GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.
- A 1–ounce tube of Permatex "Anti–Seize" lubricant (2119594) is provided. Follow instructions on back of tube. "Anti–Seize" lubricant is to be applied to threads of stainless–steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV
1	5	8	2
2	5	9	2
3	5	10	5
4	2	11	2
5	5	12	3
6	2	13	4
7	2		

BRIDGE INSTALLATION INFORMATION

TWO TYPES OF BRIDGES ARE AVAILABLE FOR INSTALLATION: The original style Long Bridge and the new style Split Bridge. See items 1 and 2 below for explanation of which installation procedures to use.

1. If original style Long Bridge is delivered to site, complete procedures on pages 7, 8, and 9. This bridge would be used for pre–Split Bridge MR/i and all CV/i installations.
2. If new style Split Bridge is delivered to site, complete procedures on pages 10, 11, and 12. This bridge would be used for MR/i installations only, not CV/i.

WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

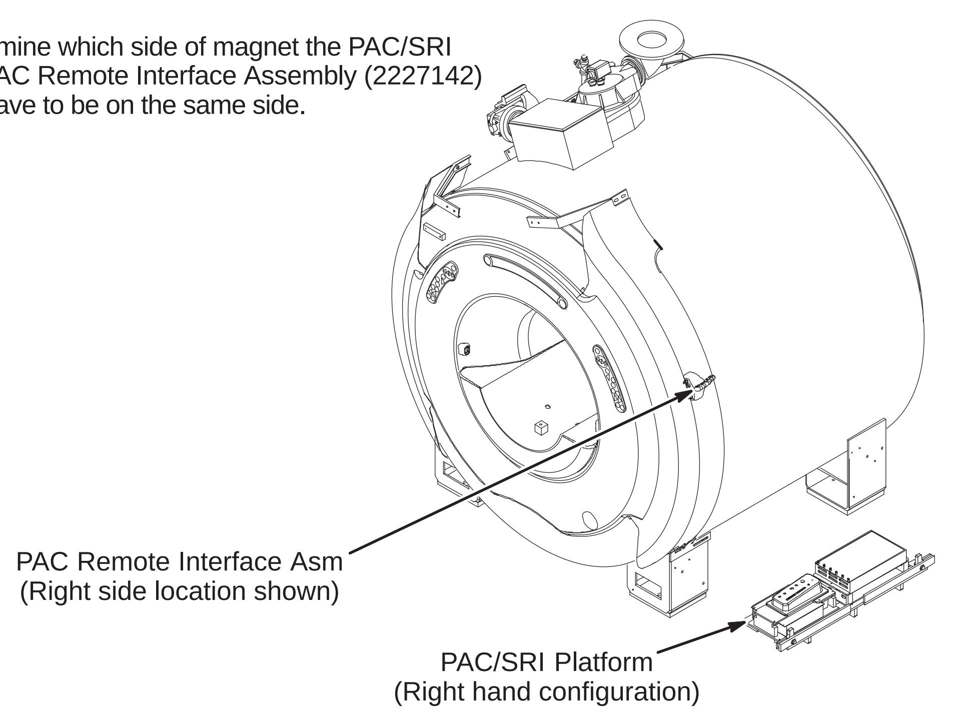
IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

INSTALLATION PROCEDURES SUMMARY

- ❑ **A1 – PAC/SRI Platform and PAC Remote Interface Location**
- ❑ **A2 – PAC/SRI Platform Left Hand Configuration (If Required)**
- ❑ **A3 – Location of Magnet Ground Cable**

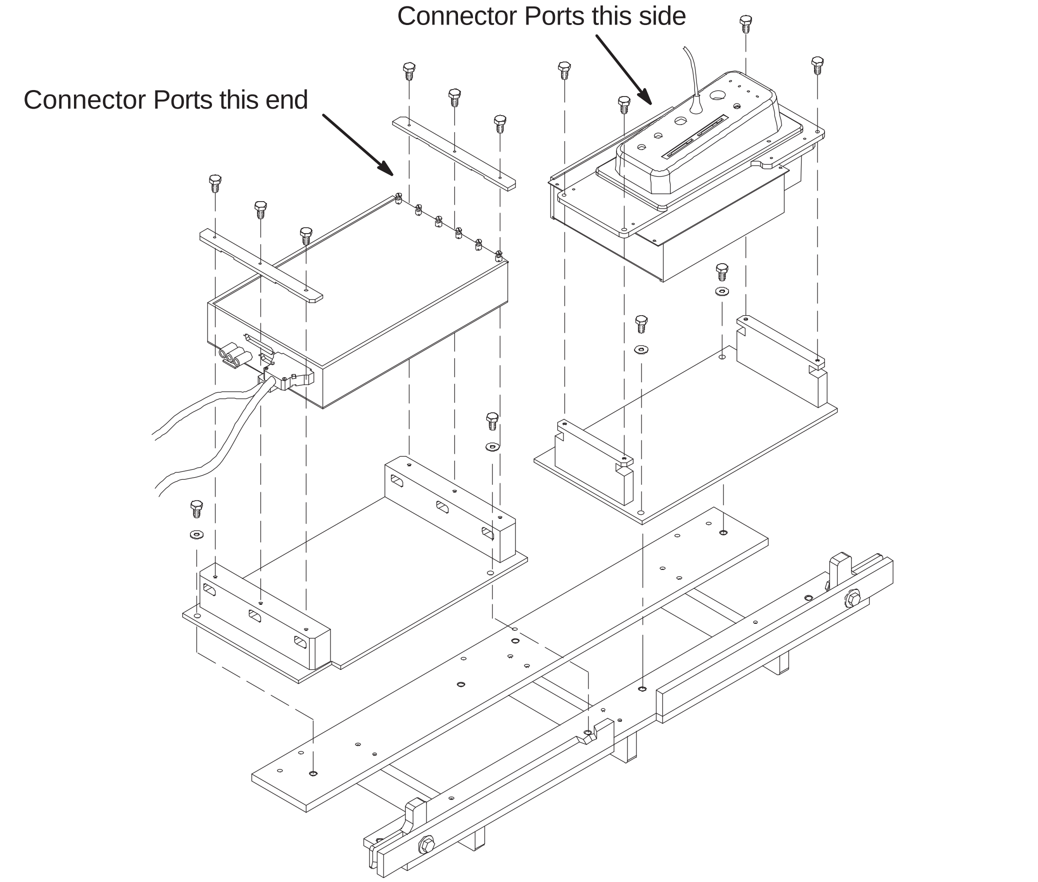
❑ A1 – PAC/SRI PLATFORM AND PAC REMOTE INTERFACE LOCATION

(1) Van manufacturer to determine which side of magnet the PAC/SRI Platform (2223440) and PAC Remote Interface Assembly (2227142) should be located. They have to be on the same side.

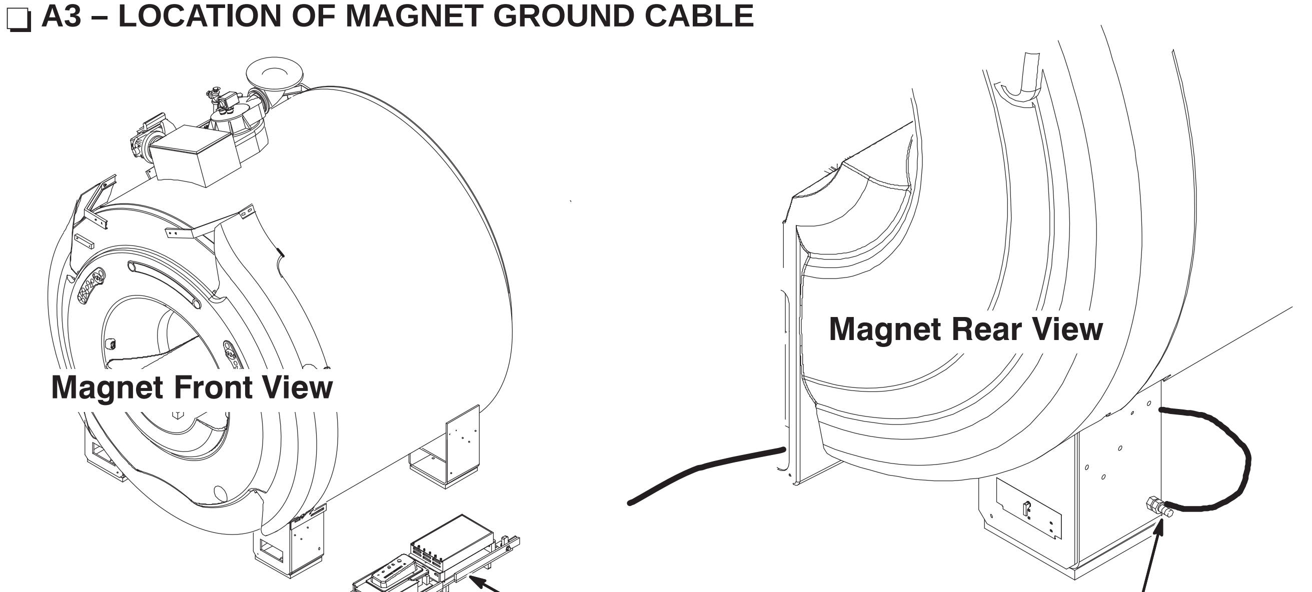


❑ A2 – PAC/SRI PLATFORM LEFT HAND CONFIGURATION (IF REQUIRED)

(1) For left hand configuration, reassemble PAC/SRI as shown.



❑ A3 – LOCATION OF MAGNET GROUND CABLE



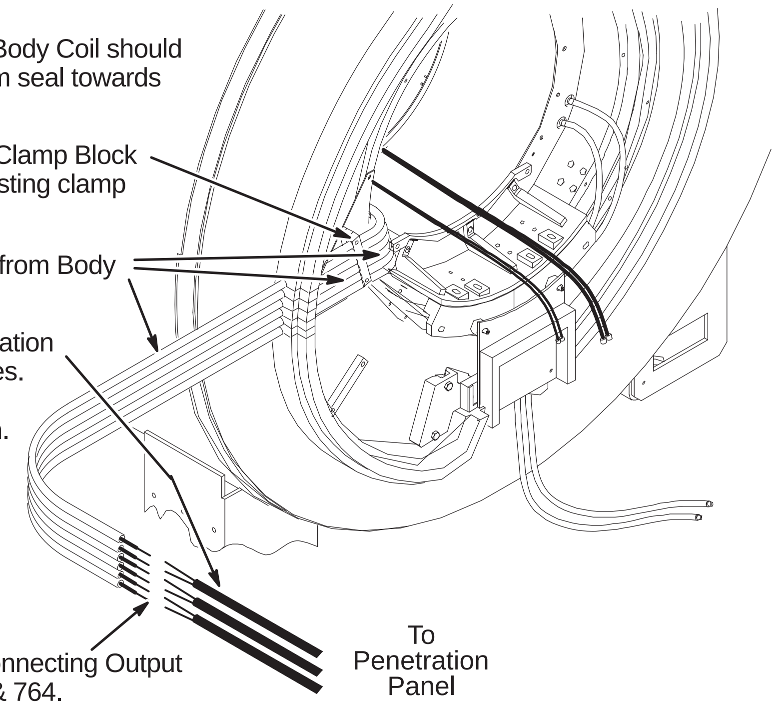
(1) If PAC/SRI Platform is located on right side of magnet, then the Magnet ground cable , RUN 040, has to be connected to the opposite side rear left leg as shown above. The other end of Run 040 is connected to the RF Screen Room common ground.

INSTALLATION PROCEDURES SUMMARY

- ❑ **B1 – Routing and Connecting Gradient Output Cables to Runs 762, 763, and 764 (If Needed)**
- ❑ **B2 – Locating Rear Pedestal and Connecting Run 318 & I and Q Drive Cables**
- ❑ **B3 – Route Cables & Hoses, Connect Ground Cable, and Install Coil Cable Takeup Interface**
- ❑ **B4 – Route/Connect Runs 257/746, 262, 743, 744, and 780**

❑ B1 – ROUTING AND CONNECTING GRADIENT OUTPUT CABLES (If Needed)

1. The (6) six gradient Output Cables from Body Coil should have been routed through slits in the foam seal towards wall and secured to wall.
2. Hold cables in place with 6 slot Gradient Clamp Block (2214162). This is mounted on top of existing clamp block using longer screws.
3. Around each cable install pipe insulation from Body Coil to Penetration Panel.
4. Route Runs 762, 763, & 764 from Penetration Panel to ends of Body Coil gradient cables. Make sure that there is at least two feet (610mm) of slack before cutting to length.



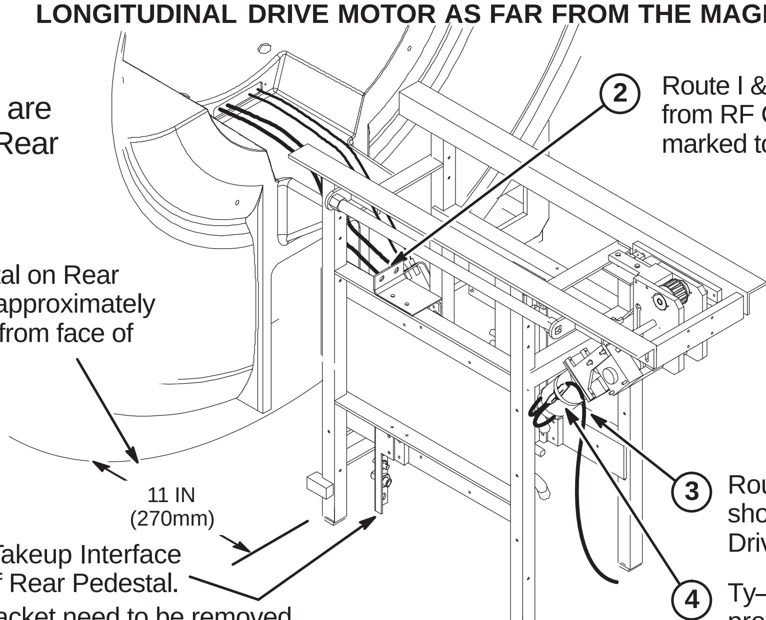
5. Refer to Direction 2286976 in Tab 1 for connecting Output cables from Body Coil to Runs 762, 763, & 764.

❑ B2 – LOCATING REAR PEDESTAL AND CONNECTING RUN 318 & DRIVE CABLES

WARNING!

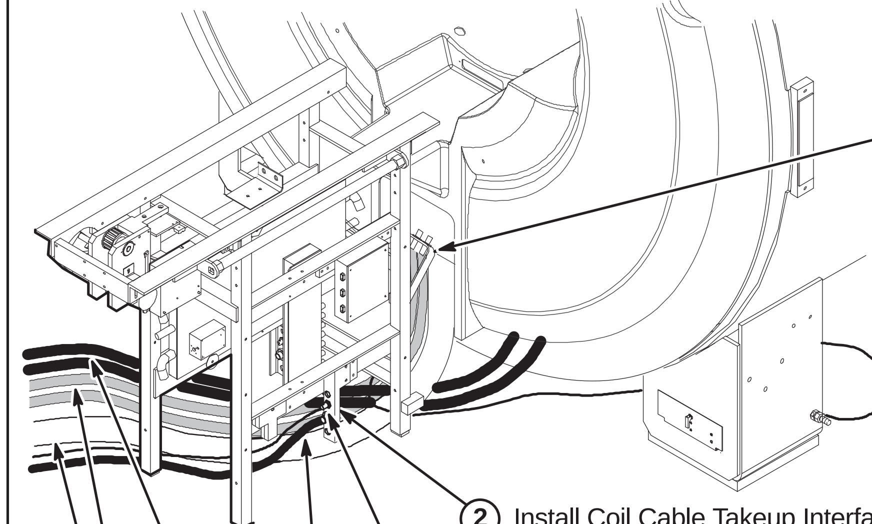
THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

Note:
Cables and hoses are to exit this side of Rear Pedestal



1. Center Rear Pedestal on Rear Endbell and locate approximately 11 inches (270mm) from face of magnet.
2. Route I & Q RF Drive Input cables from RF Coil and connect as marked to J1 and J2 on bracket.
3. Route/connect Run 318 to short cable on Longitudinal Drive Motor (MG3 A7).
4. Ty–wrap cable to motor to prevent movement.
5. Move Coil Cable Takeup Interface to opposite side of Rear Pedestal.
6. Connectors on bracket need to be removed, turned end for end, and reinstalled.

❑ B3 – ROUTE CABLES & HOSES, CONNECT GROUND CABLE, AND INSTALL TAKEUP



Note: For CV/i, install bracket on other side of Reat Pedestal.

1. Route Ground cable from Rear Pedestal under magnet and connect to same Ground Stud as Magnet ground.
2. Install Coil Cable Takeup Interface with mounting block. Make sure connectors have been turned end for end.
3. Route/connect Run 700 to J2 on Coil Cable Takeup Interface.
4. Route/connect to J1 on Coil Cable Takeup Interface: Run 258 (1.5T, 1/2") or Run 747 (1.0T, 3/8")
5. Route and connect coolant hoses from Heat Exchanger to 10 ft. (300cm) coolant hoses attached to Body Coil.
6. Route water lines, gas lines, and 2" air hose under Rear Pedestal as shown.
7. Route all other cables on top of the above cables and hoses.

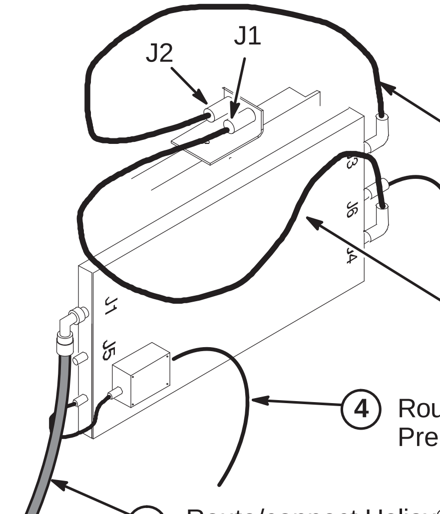
❑ B4 – ROUTE/CONNECT RUNS 257/746, 262, 743, 744, AND 780

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.



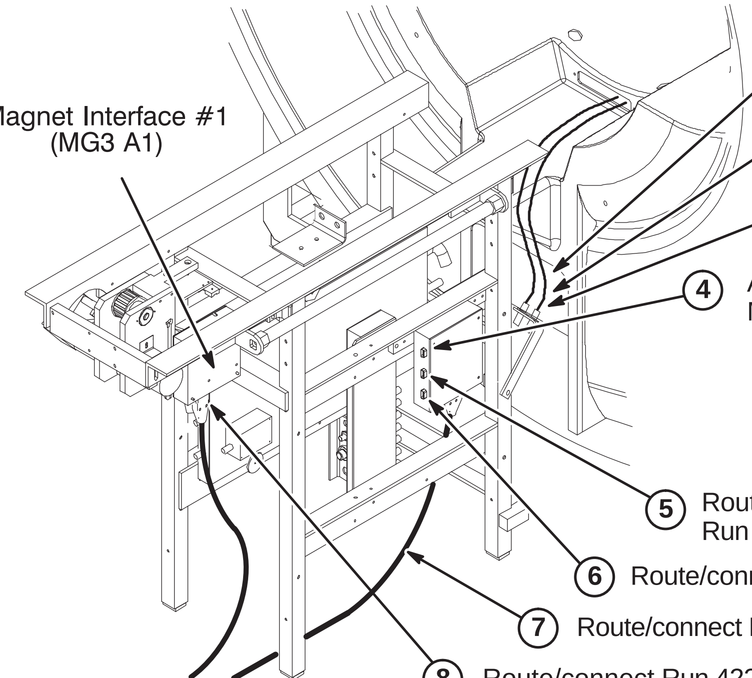
1. Route and connect Run 744 from J2 on bracket to J4 on the RF Splitter.
2. Route/connect Run 780 from J74 on Pen Panel to J6 on RF Splitter.
3. Route and connect Run 743 from J1 on the bracket to J3 on the RF Splitter.
4. Route/connect Run 262 from J8 on Pen Panel to Body Coil Preamplifier. (For MGD, connect to Receive Filter on Preamp)
5. Route/connect Helix® to J1 on RF Splitter: Run 257 (1.5T, 7/8") or Run 746 (1.0T, 1/2")

INSTALLATION PROCEDURES SUMMARY

- ❑ **C1 – Route/Connect Runs 282, 414, 422, 781, 782, and Front End Bell Speaker Cable**
- ❑ **C2 – Install Bore Lamp Assembly**
- ❑ **C3 – Route/Connect Runs 257/746, 262, 743, 744, and 780**
- ❑ **C4 – Install Blower Box and Route/Connect Air Hoses**

❑ C1 – ROUTE/CONNECT RUNS 282, 414, 422, 781, 782, & FRONT END BELL CABLE

Magnet Interface #1
(MG3 A1)

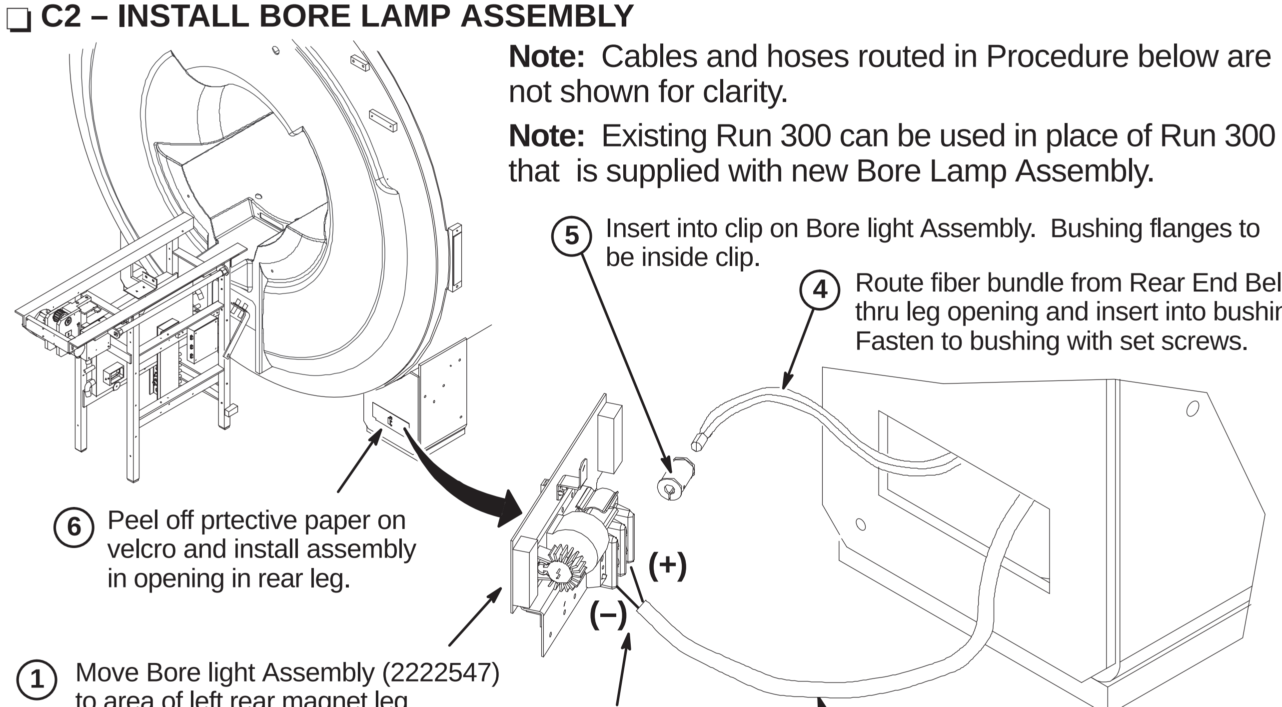


1. Route and connect RF Bias cables from RF Coil to bracket as marked.
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. After Carriage cover is installed, connect Microphone cable from Carriage Cover to J7.
5. Route/connect Front End Bell microphone cable Run 414 to J9.
6. Route/connect speaker cable from Front End Bell to J8.
7. Route/connect Run 282 from J12 on Pen Panel to J6.
8. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.

❑ C2 – INSTALL BORE LAMP ASSEMBLY

Note: Cables and hoses routed in Procedure below are not shown for clarity.

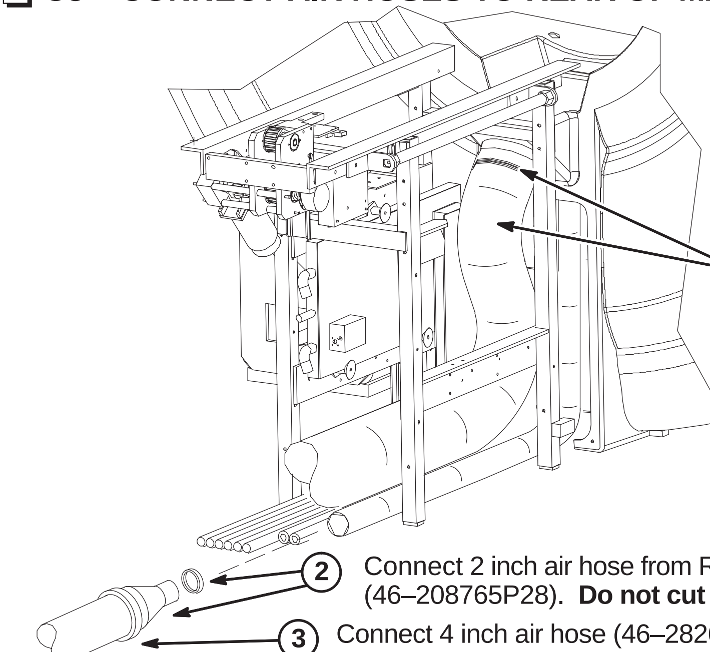
Note: Existing Run 300 can be used in place of Run 300 that is supplied with new Bore Lamp Assembly.



1. Move Bore light Assembly (2222547) to area of left rear magnet leg.
2. Connect the (+) and (–) wires of Run 300 to the terminal strip as shown.
3. Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
4. Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
5. Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
6. Peel off prtective paper on velcro and install assembly in opening in rear leg.

❑ C3 – CONNECT AIR HOSES TO REAR OF MAGNET

Note:
The following items are part of Blower Hose Kit, 2129412–3 or 2129412–4.

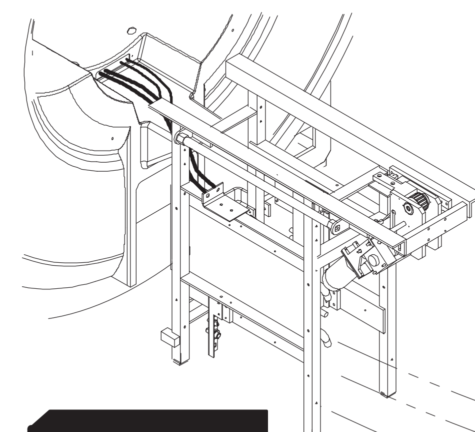


1. Connect 6 inch air hose (46–282666P12) to Rear End Bell with 6 inch clamp (46–208675P29).
2. Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). **Do not cut or reduce the length of the 2 inch hose.**
3. Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). **If necessary, the 4 inch hose may need to be cut to reduce length.**

❑ C4 – INSTALL BLOWER BOX AND ROUTE/CONNECT AIR HOSES

1. Refer to site design (architectural)/customer construction drawings or contact PMI for locating and securing Blower Box (2121199).
2. Check for correct installation of the anchors:
   - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
   - Confirm there are at least four (4) anchors.
   - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.
3. Remove any mounting hardware from the anchors.
4. Position the blower box on top of 4 or more anchors. Hoses should face the back of the room
5. Secure the blower box to the floor using the fastening hardware provided by the RF shield room vendor.

NOTE: If needed, Blower Hose Cover can be cut and modified.



WARNING!

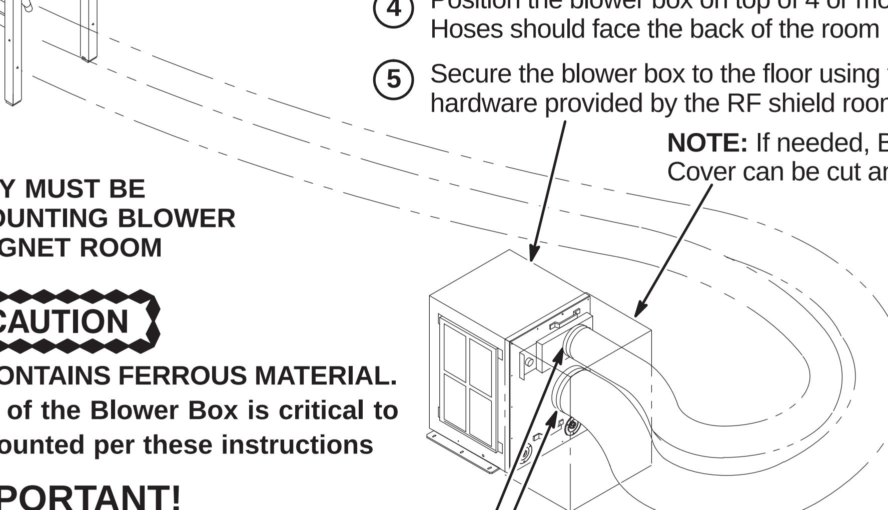
RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).



6. Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:
   (1) 46–208765P34 4 inch Hose Clamp
   (1) 46–208765P29 6 inch Hose Clamp

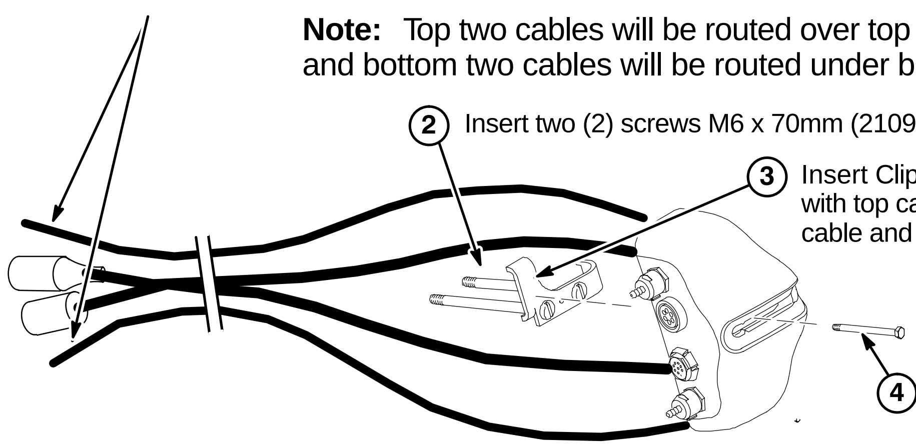
INSTALLATION PROCEDURES SUMMARY

❑ D1 – PREPARE CABLES AND TUBES FOR ROUTING

Note:
Van manufacturer to determine correct location of Remote PAC Interface assembly. It must be on the same side as the PAC/SRI Platform.

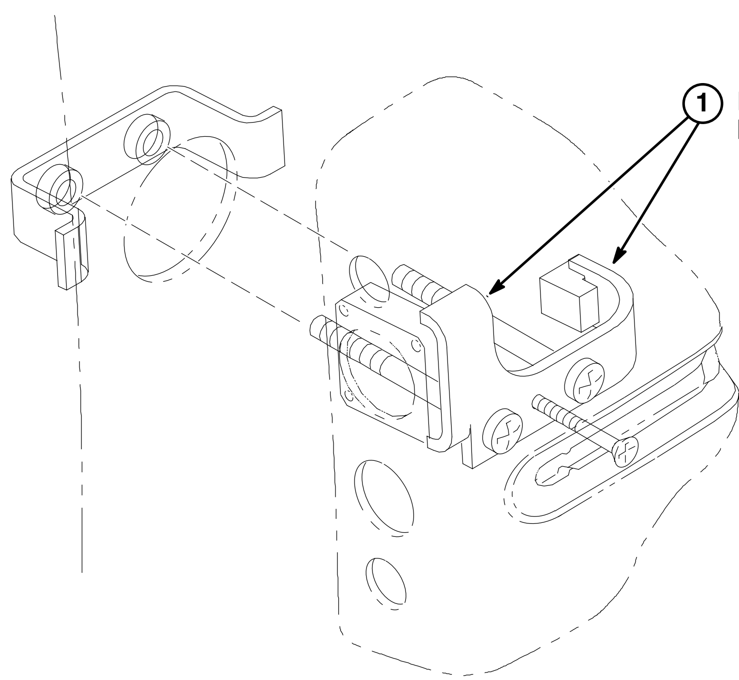
1. Apply tape around end of pneumatic tubes and mark "TOP" to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and "BOTTOM" to the tube connected to the lower connector.

Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.



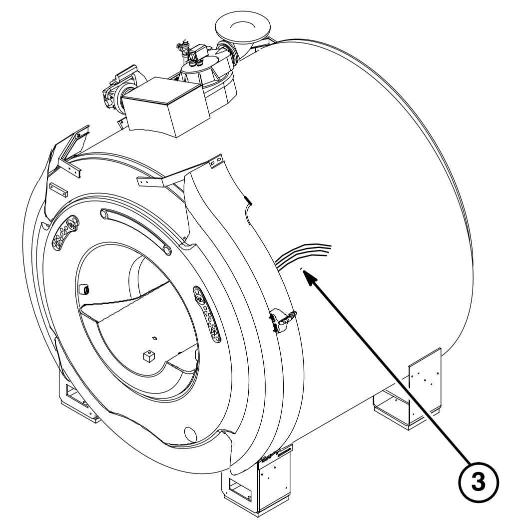
2. Insert two (2) screws M6 x 70mm (2109873–33) screws into Clip (2221023).
3. Insert Clip with screws into shell as shown with top cable and tube above Clip and bottom cable and tube below.
4. Hold in place by inserting provided M4 screw thru slot and turning about 3 or 4 turns into Clip.

❑ D2 – POSITION REMOTE PAC INTERFACE MOUNTING CLIP



1. Make sure each side of clip is mounted behind housing blocks.

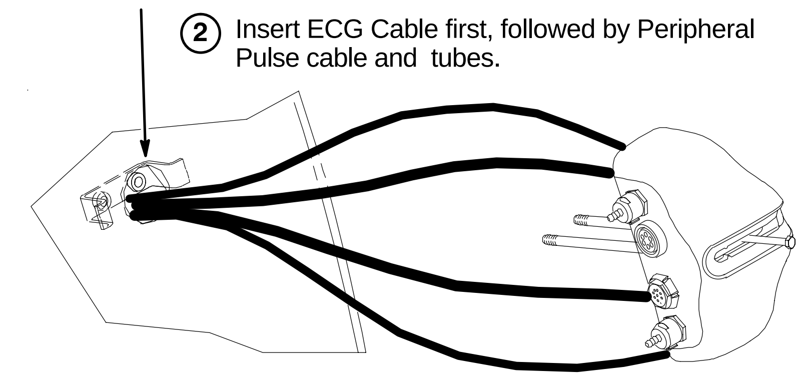
❑ D3 – ROUTE REMOTE PAC INTERFACE CABLES AND TUBES



1. Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.

Note:
Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

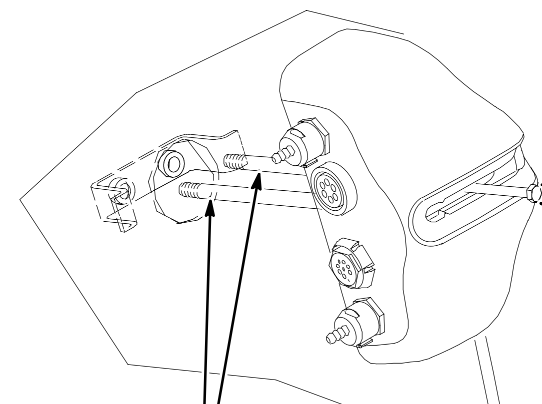
2. Insert ECG Cable first, followed by Peripheral Pulse cable and tubes.



3. Route cable to PAC/SRI Platform location.

❑ D4 – INSTALLING REMOTE PAC INTERFACE TO ARC COVER

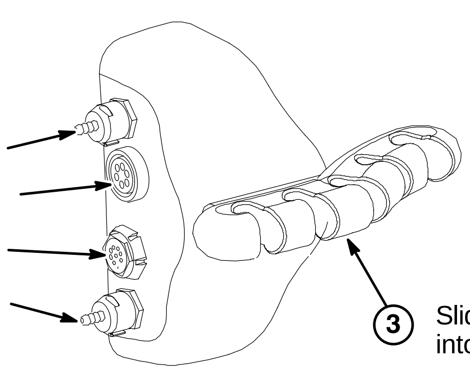
Note:
Routed cables and tubes not shown for clarity.

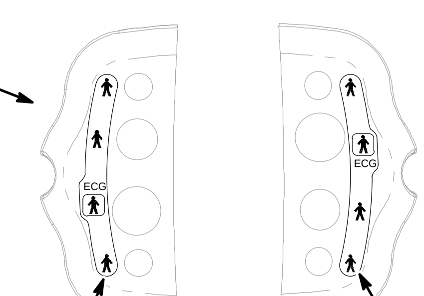


1. Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.
2. Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place
3. Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.
OR
Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in **T5** into place.

❑ D5 – INSTALL PAC HANGER BAR

1. Depending on left or right side installation of the PAC Remote Interface Asm on the magnet enclosure, attach the appropriate label that is provided with the kit.
2. Attach cables as indicated below.



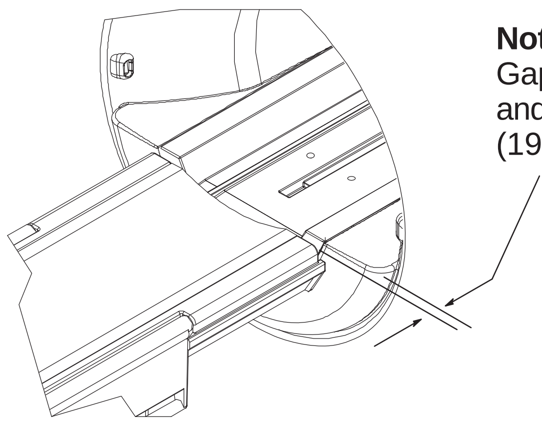


3. Slide PAC Hanger Bar (2221023) into position.

INSTALLATION PROCEDURES SUMMARY

- ❑ **E1 – Install Dock Mounting Bracket and Dock**
- ❑ **E2 – Route and Connect Magnet Enclosure Front Area Cables**

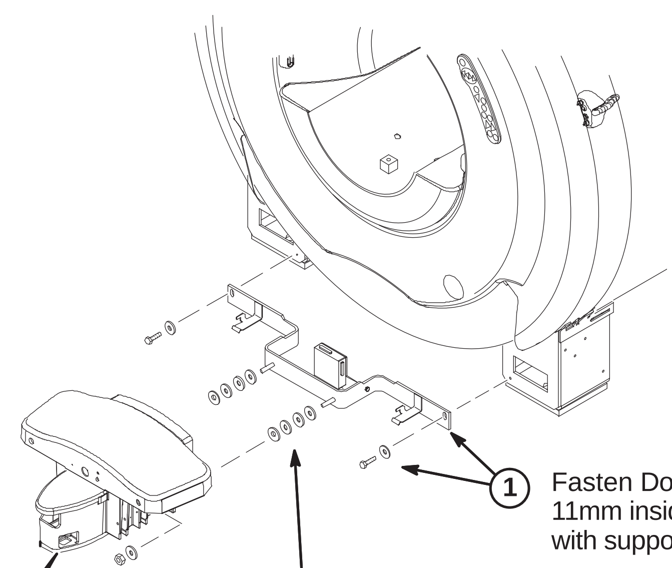
❑ E1 –INSTALL DOCK MOUNTING BRACKET AND DOCK



Note:
Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches ± 0.25 (19.0±6.4mm). See Step 2 below.

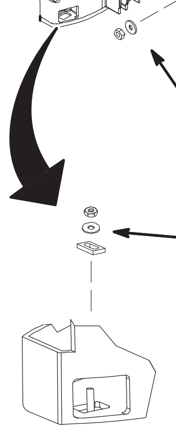
WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION. IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY.



1. Fasten Dock Mounting Support (2226264) to magnet using 11mm inside diameter washers and Hex Screws supplied with support. Apply Loctite 242 as required.
2. Start with two (2) 11mm washers on each side between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is acheived.
3. Mount Dock Asm (46–271260G3) using (2) each of the M10 Hex Nuts and 11mm washers that are provided with the Dock Mounting Support. Apply Loctite 242 as necessary.
4. Fasten Dock Asm to floor mounted stud with:
   (1) 2178874 Dock Clamp Plate
   (1) 2109878–4 10.5mm ID Washer
   (1) 2109875–4 10mm Hex Nut

   Apply Loctite 242 as required.

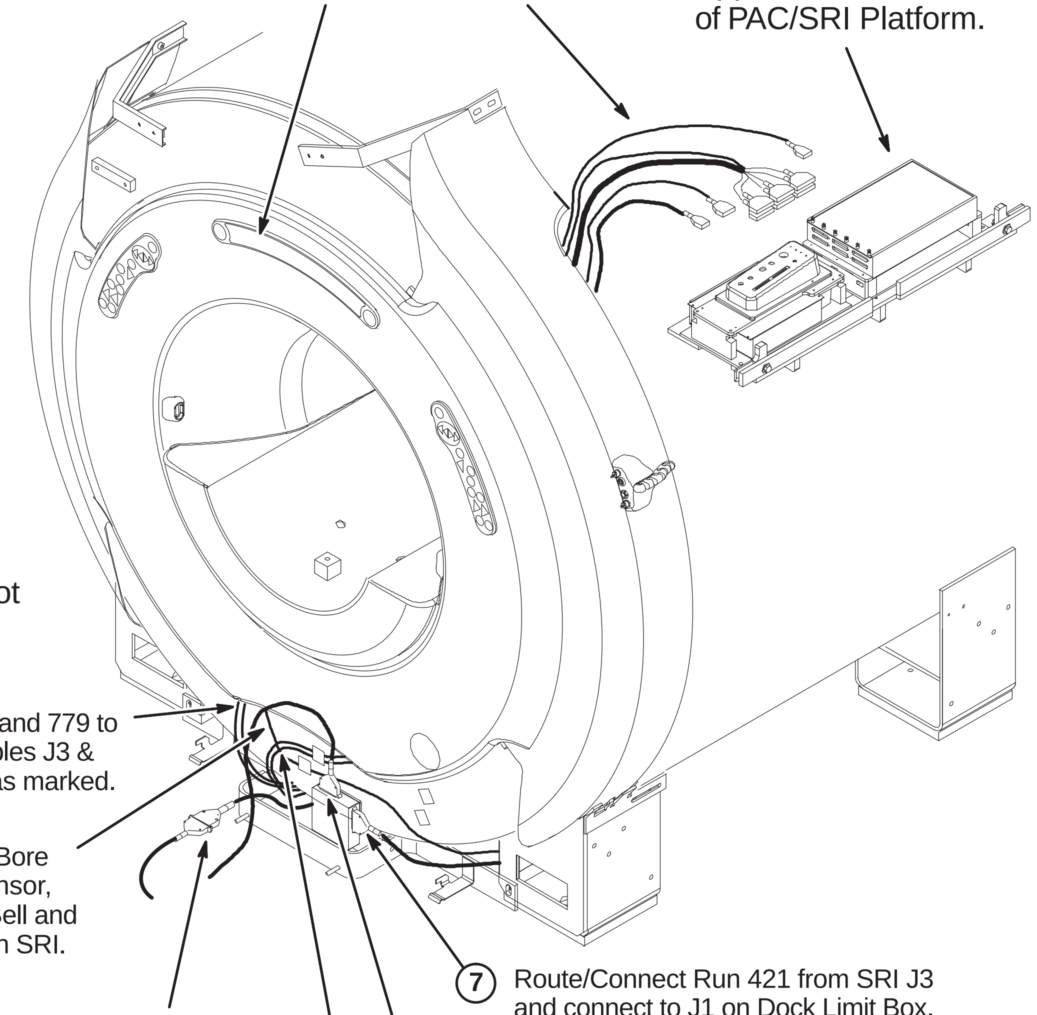


❑ E2 – ROUTE AND CONNECT MAGNET ENCLOSURE FRONT AREA CABLES

Note:
Insulating and securing of cables routed to PAC/SRI Platform assembly to be the responsibility of van manufacturer.

1. Route cables from Operator Display to area of PAC/SRI Platform and connect as marked to SRI and PAC.

Note:
Approximate location of PAC/SRI Platform.



Note:
Dock Assembly not shown for clarity.

2. Route Runs 778 and 779 to front RF Bias cables J3 & J4 and connect as marked.
3. Route Run 749, Bore Temperature Sensor, from Front End Bell and connect ot J10 on SRI.
4. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with tape or with provided velcro jacket.
5. Route Front End Bell Microphone and Speaker cables underneath magnet to Rear Pedestal and connect to Patient Communication Box as marked.
6. Connect Dock Cable to J2 on Dock Limit Box.
7. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.

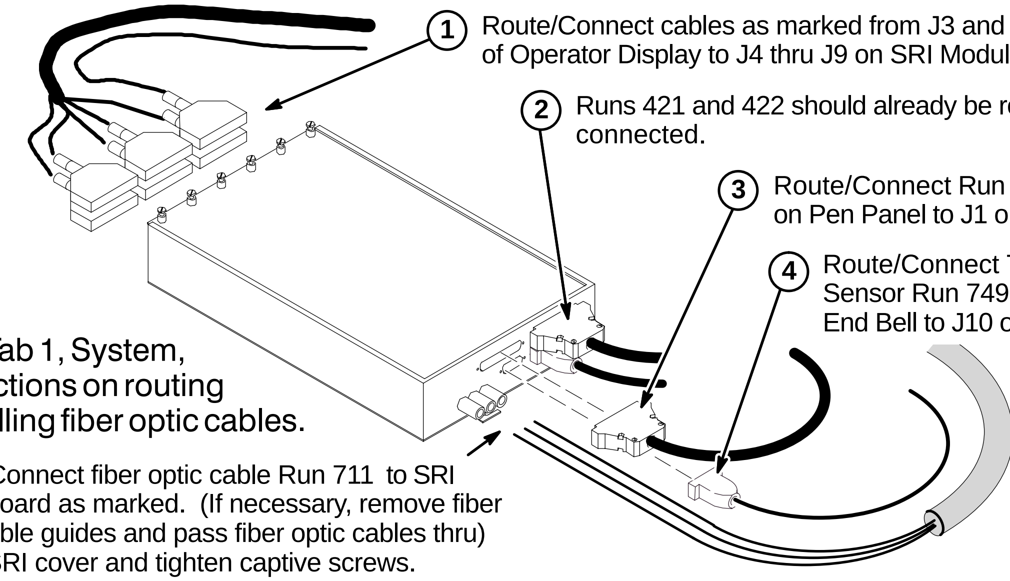
CAUTION

No metal to metal contact is allowed for connectors to other areas. i.e. connector touching the magnet.

INSTALLATION PROCEDURES SUMMARY

- ❑ F1 – Connect Cables to SRI Module
- ❑ F2 – Cable Connection to PAC Module
- ❑ F3 – Route and Tie–wrap Cables to PAC/SRI Platform
- ❑ F4 – Routing of Photo–Plethysmograph Cable

❑ F1 – CONNECT CABLES TO SRI MODULE



1. Route/Connect cables as marked from J3 and J4 of Operator Display to J4 thru J9 on SRI Module.
2. Runs 421 and 422 should already be routed and connected.
3. Route/Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.
4. Route/Connect Temperture Sensor Run 749 from Front End Bell to J10 on SRI Module

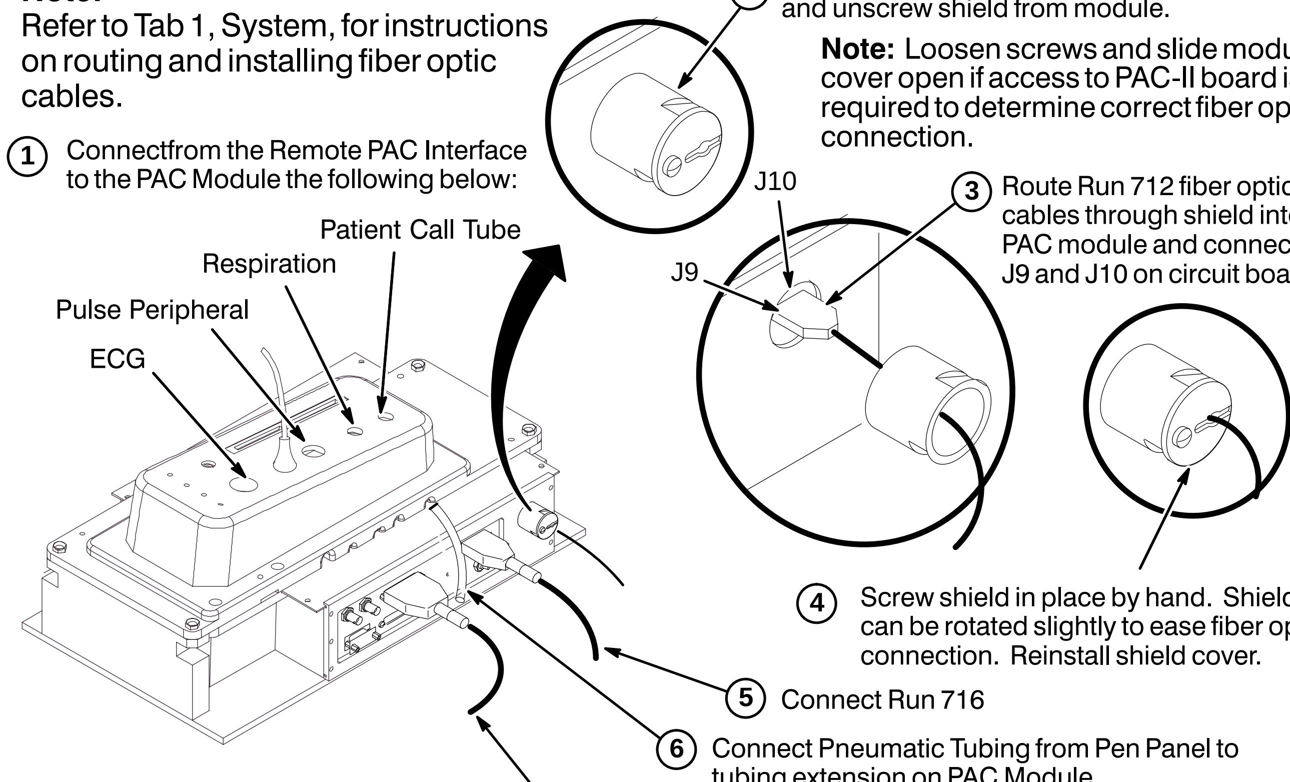
Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

5. Route/Connect fiber optic cable Run 711 to SRI circuit board as marked. (If necessary, remove fiber optic cable guides and pass fiber optic cables thru) Close SRI cover and tighten captive screws.

❑ F2 – CABLE CONNECTION TO PAC MODULE

Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

1. Connectfrom the Remote PAC Interface to the PAC Module the following below:

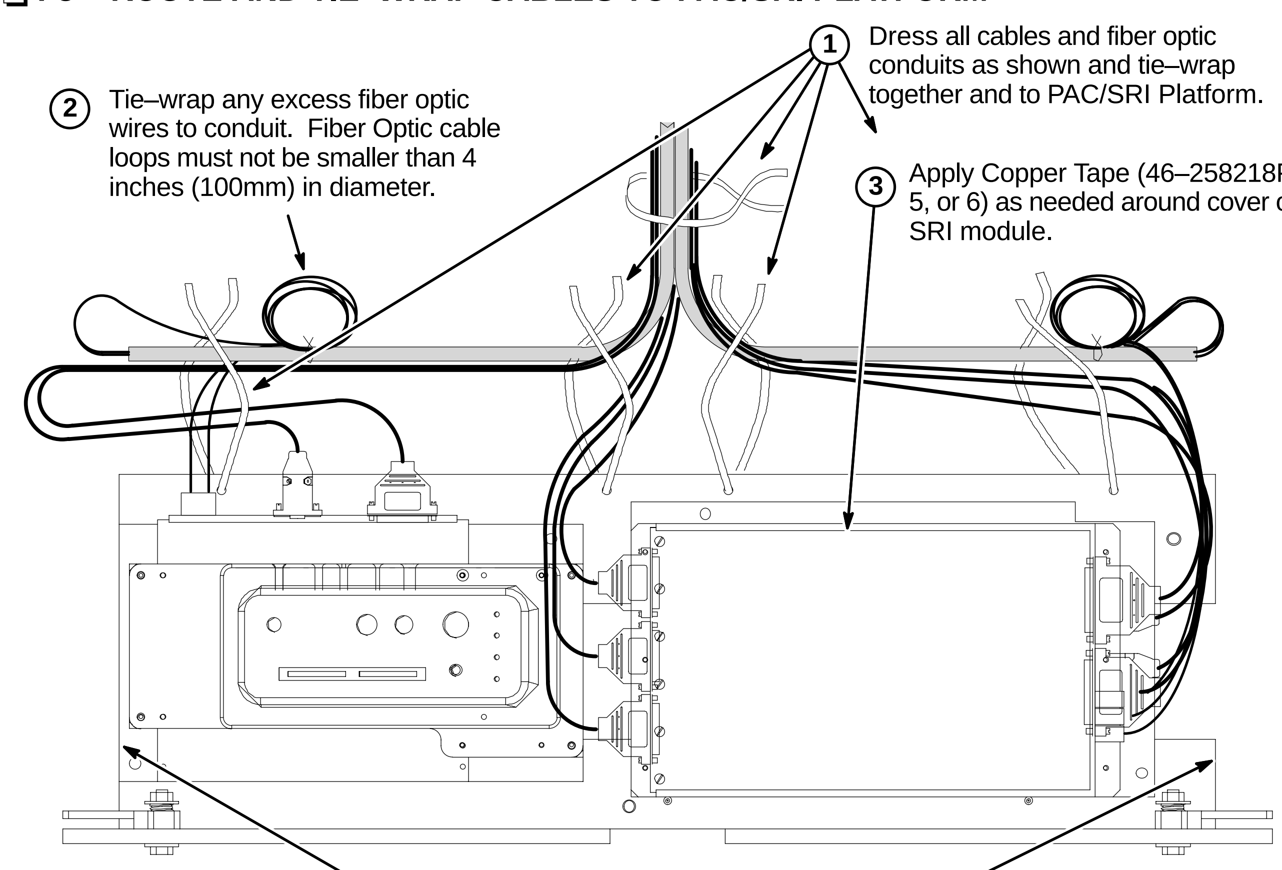


2. Remove shield cover by removing one screw and unscrew shield from module.

Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.

3. Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.
4. Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.
5. Connect Run 716
6. Connect Pneumatic Tubing from Pen Panel to tubing extension on PAC Module.
7. Connect cabler from J5 on Operator Display to J4 on PAC Module.

❑ F3 – ROUTE AND TIE–WRAP CABLES TO PAC/SRI PLATFORM

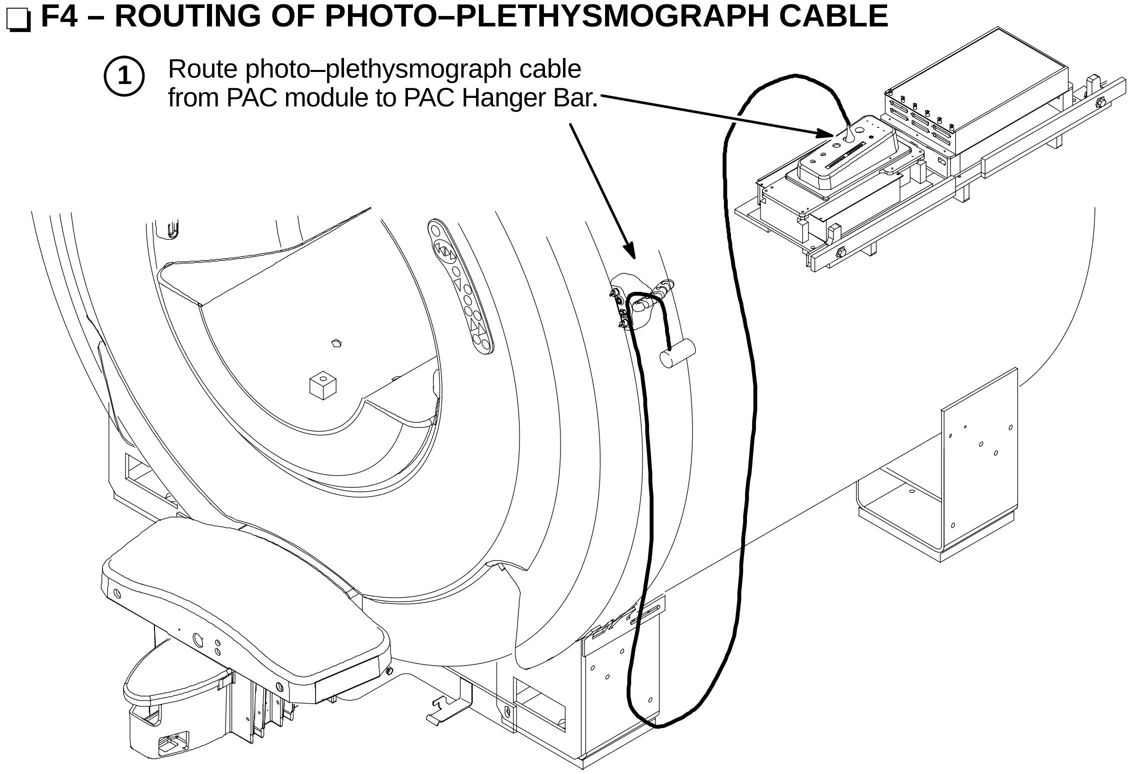


1. Dress all cables and fiber optic conduits as shown and tie–wrap together and to PAC/SRI Platform.
2. Tie–wrap any excess fiber optic wires to conduit. Fiber Optic cable loops must not be smaller than 4 inches (100mm) in diameter.
3. Apply Copper Tape (46–258218P4, 5, or 6) as needed around cover of SRI module.

Note: Keep cables dressed inside edge so unit slides between legs easily for service.

❑ F4 – ROUTING OF PHOTO–PLETHYSMOGRAPH CABLE

1. Route photo–plethysmograph cable from PAC module to PAC Hanger Bar.



BRIDGE INSTALLATION INFORMATION

TWO TYPES OF BRIDGES ARE AVAILABLE FOR INSTALLATION: The original style Long Bridge and the new style Split Bridge. See items 1 and 2 below for explanation of which installation procedures to use.

1. **If original style Long Bridge is delivered to site, complete procedures on pages 7, 8, and 9. This bridge would be used for pre–Split Bridge MR/i and all CV/i installations.**

2. **If new style Split Bridge is delivered to site, complete procedures on pages 10, 11, and 12. This bridge would be used for MR/i installations only, not CV/i.**

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

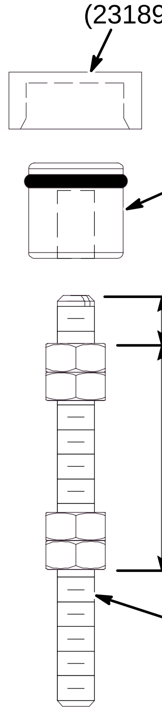
- ❑ **G1 – Front Bridge Support Adjustments**
- ❑ **G2 – Bridge Support Adjustment Tolerances**
- ❑ **G3 – Bridge Assembly Preparation Before Installing in Magnet**
- ❑ **G4 – Install Bridge**

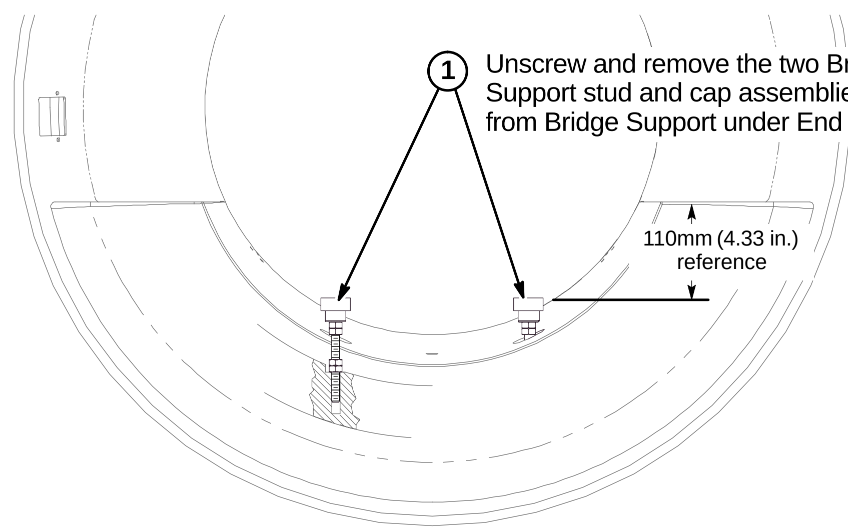
NOTE:
Measure to confirm that the distance between the top of the Support Cap Cup and the flat surface of the End Bell is110mm as shown in Procedure **G1**. Also measure to confirm that the 10mm and 62mm M10 Jam Nut distance locations on the Stud are also in place.

- If the dimensions are found to be in place for this site, proceed to Procedure **G2**.
- If these dimensions are not in place, complete Steps 1 thru 6 in Procedure **G1** and then proceed to **G2**.

❑ G1 – FRONT BRIDGE SUPPORT ADJUSTMENTS

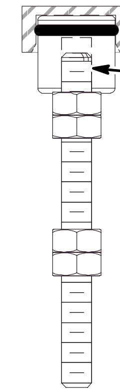
2. Remove Support Cap Cup and unscrew Support Cap





1. Unscrew and remove the two Bridge Support stud and cap assemblies from Bridge Support under End Bell.
3. Set top M10 Nut at a distance of 10mm from top of stud. Lock in position with second M10 Nut.
4. Set bottom M10 Nut at a distance of 62mm from top of first M10 Nut. Lock in position with fourth M10 Nut.
5. Fasten Support Cap to top of Stud and tighten to top M10 Nut. Install Support Cap Cup to Support Cap
6. Re–install the two Bridge Support stud and cap assemblies into Bridge Support.

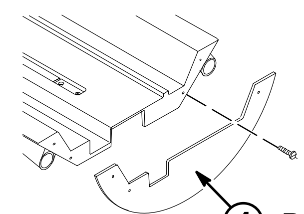
❑ G2 – BRIDGE SUPPORT ADJUSTMENT TOLERANCES

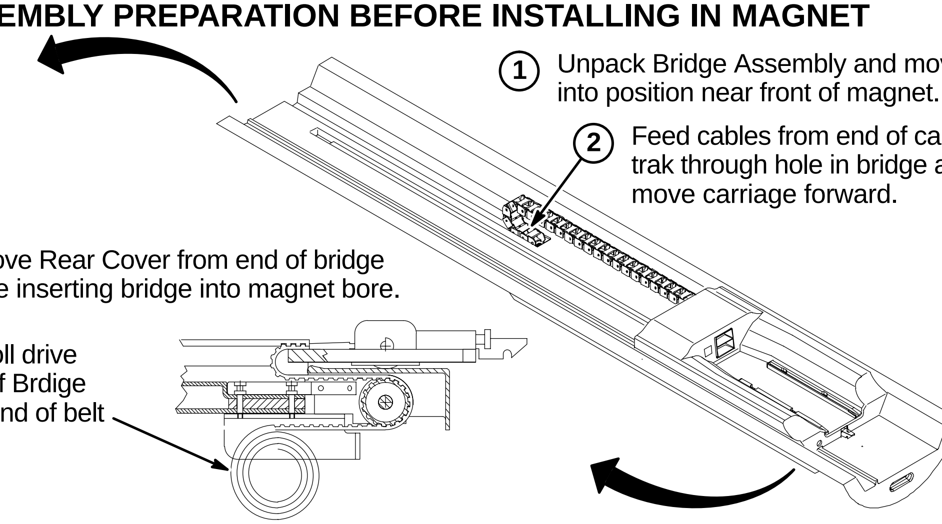


1. The 10mm location distance noted in Procedure **G1** is a starting point for installation and leveling of Bridge.

   This starting point gives the installer an adjustment of +7.5mm and –5mm. The amount of adjustment will be determined during bridge leveling.

❑ G3 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET

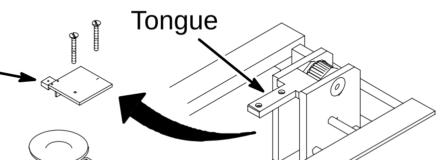


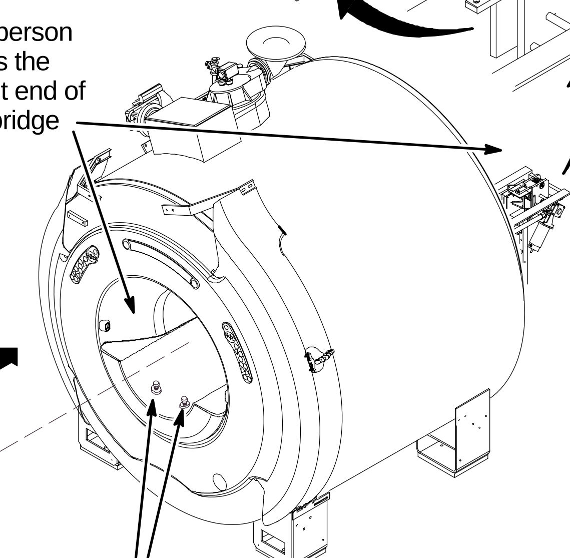


1. Unpack Bridge Assembly and move into position near front of magnet.
2. Feed cables from end of cable trak through hole in bridge and move carriage forward.
3. If not already done, unroll drive belt along bottom side of Brdige and temporarilyy tape end of belt to rear of bridge.
4. Remove Rear Cover from end of bridge before inserting bridge into magnet bore.

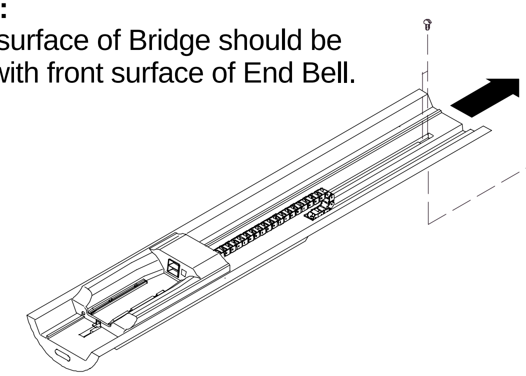
❑ G4 – INSTALL BRIDGE

1. Unfasten Rear Bridge Mount Asm. It will be reinstalled later.
2. Position rear end of Bridge Asm at front of Body Coil.
3. While one person holds holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.





NOTE:
Front surface of Bridge should be flush with front surface of End Bell.



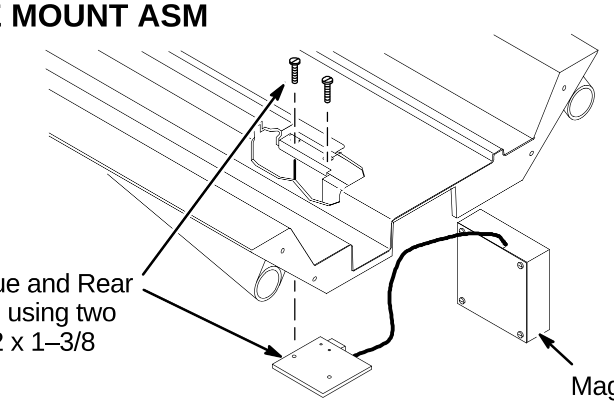
4. Rest front of Bridge on Bridge Suppor Stud Assembly.

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

- ❑ **H1 – Reattach Bridge Mount Assembly**
- ❑ **H2 – Align Front of Bridge to End Bell**
- ❑ **H3 – Applying Adhesive to Bridge Support Caps**
- ❑ **H4 – Leveling Bridge**

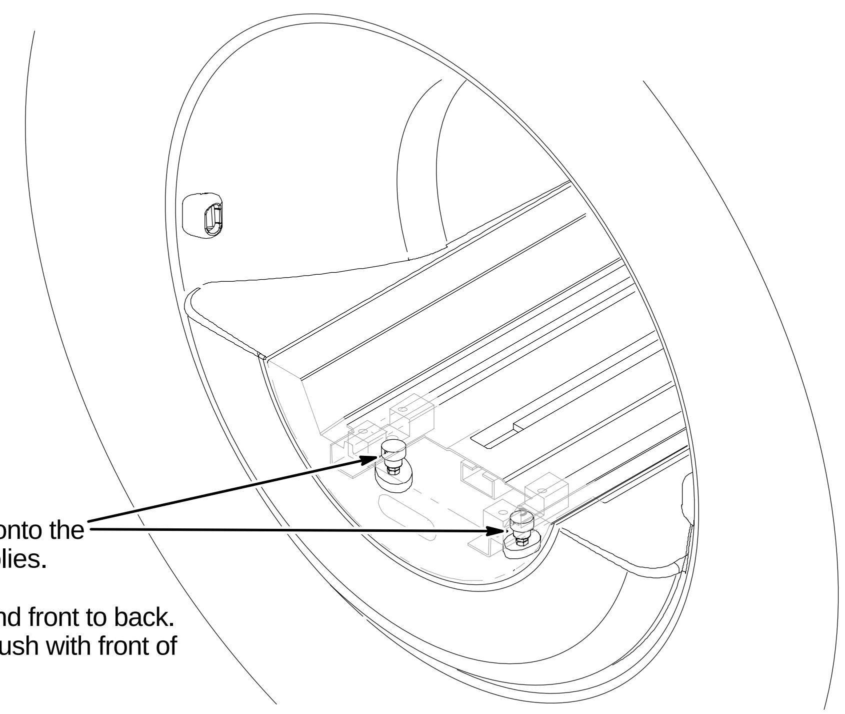
❑ H1 – REATTACH BRIDGE MOUNT ASM

NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.



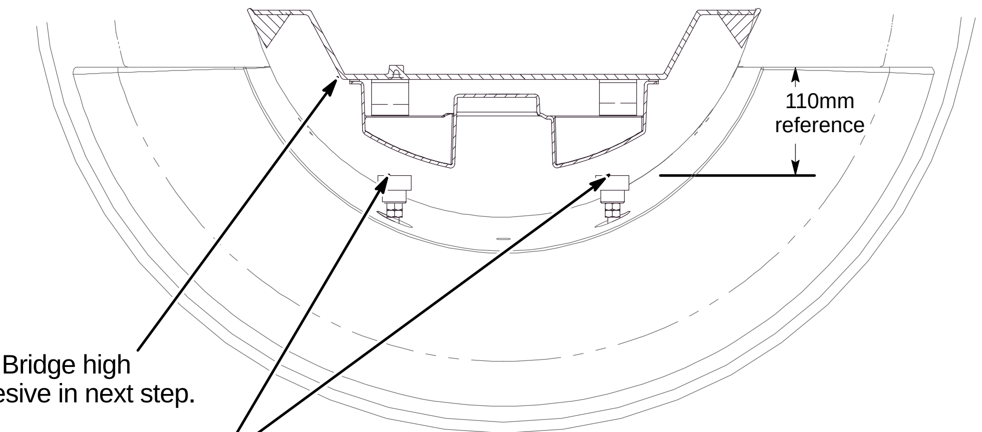
1. Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) previously removed 10–32 x 1–3/8 Screws (46–220184P47)

❑ H2 – ALIGN FRONT OF BRIDGE TO END BELL



1. Lower the front of the Bridge onto the Bridge Support Stud Assemblies.
2. Align the Bridge left to right and front to back. Make sure front of Bridge is flush with front of end bell.

❑ H3 – APPLYING ADHESIVE TO BRIDGE SUPPORT CAPS



1. Raise the front of the Bridge high enough to apply adhesive in next step.
2. Apply epoxy adhesive (46–282264P1) to top of each Bridge Support Cap.
3. Lower the Bridge to rest firmly to the Bridge Support Caps.
4. Allow adhesive to cure.
5. After adhesive has cured, adjust the Bridge to the final vertical position. Make sure bridge is level.

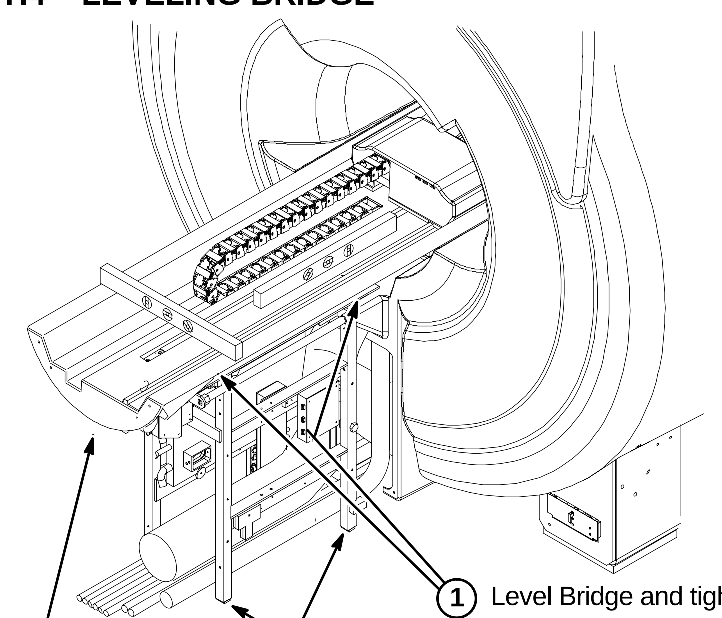
   Note: The final position will leave the top surface of the Bridge 1mm (or more) above the top surface of the end bell.
6. Repeat the vertical adjustment as needed. Make sure the two M10 jam nuts are tight against each Support Cap. This will prevent change of position.

❑ H4 – LEVELING BRIDGE

NOTE:
PVC Spacers positioned thru End Bells are to support Bridge.

NOTE:
Rear Pedestal to be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

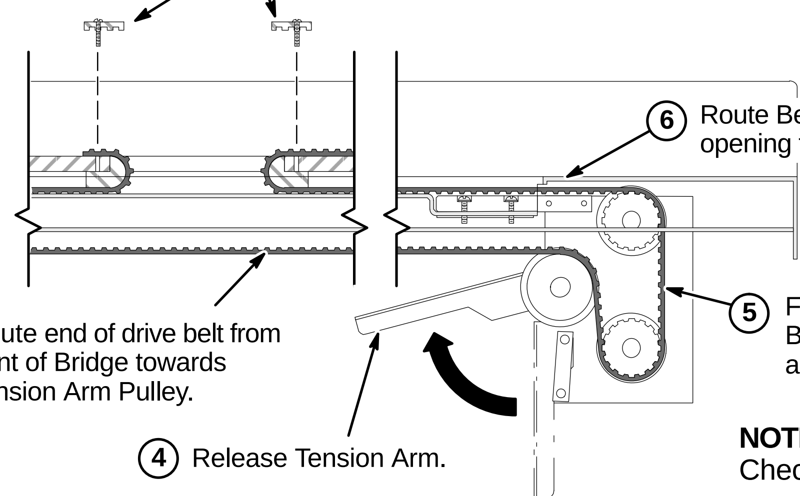


1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

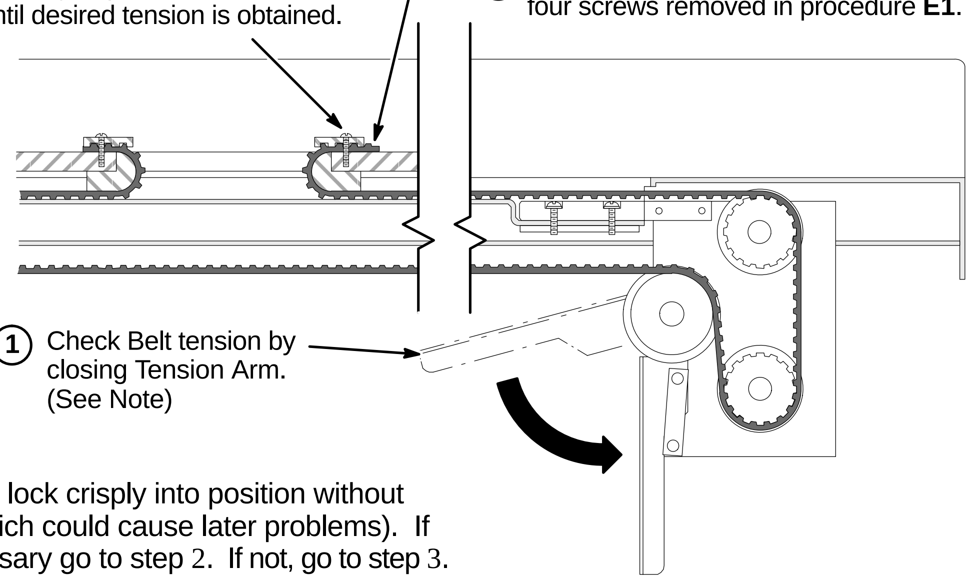
- ❑ **J1 – Routing Drive Belt**
- ❑ **J2 – Final Adjustment of Drive Belt**
- ❑ **J3 – Installation of Sensor Flag**
- ❑ **J4 – Attach Bridge Cover**
- ❑ **J5 – Route/Connect Runs 404 and 485 (and Multi–coil Cables if Required)**

❑ J1 – ROUTING OF DRIVE BELT



NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

❑ J2 – FINAL ADJUSTMENT OF DRIVE BELT

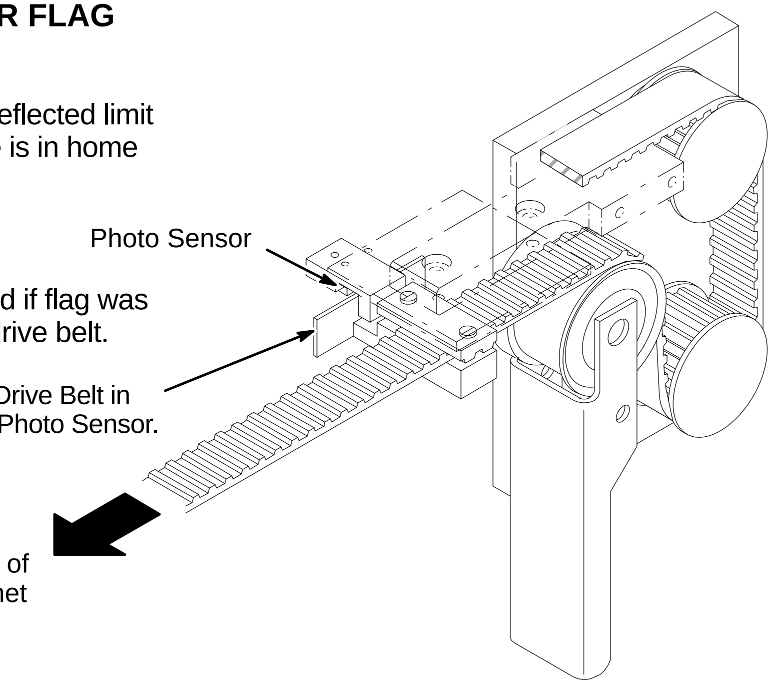


NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

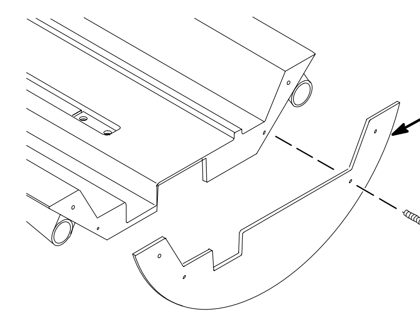
❑ J3 – INSTALLATION OF SENSOR FLAG

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.



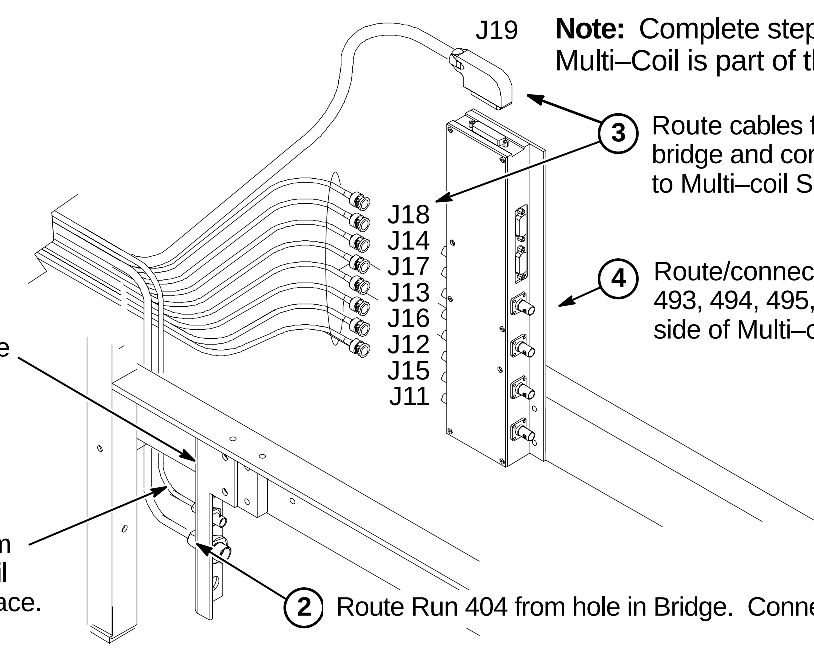
❑ J4 – ATTACH BRIDGE COVER



Note:
Bridge Cover has enclosure rating plates on it.

❑ J5 – ROUTE RUNS 404 AND 485 (AND MULTI–COIL CABLES IF REQUIRED)

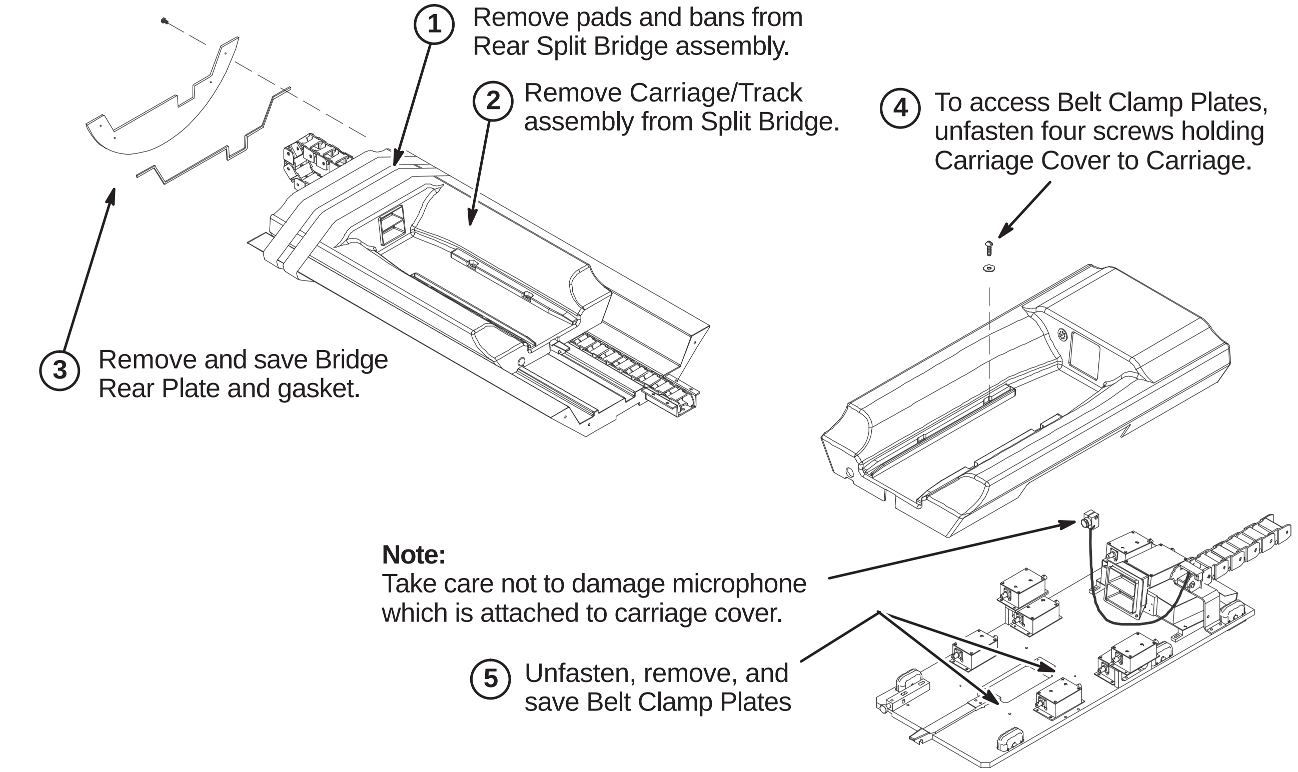
Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ❑ K1 – Preparing Rear Split Bridge for Installation
- ❑ K2 – Attaching Rear Split Bridge to Rear Pedestal
- ❑ K3 – Attaching Rear Split Bridge to Front Split Bridge
- ❑ K4 – Aligning Bridge on Rear Pedestal
- ❑ K5 – Installing Bridge Locators

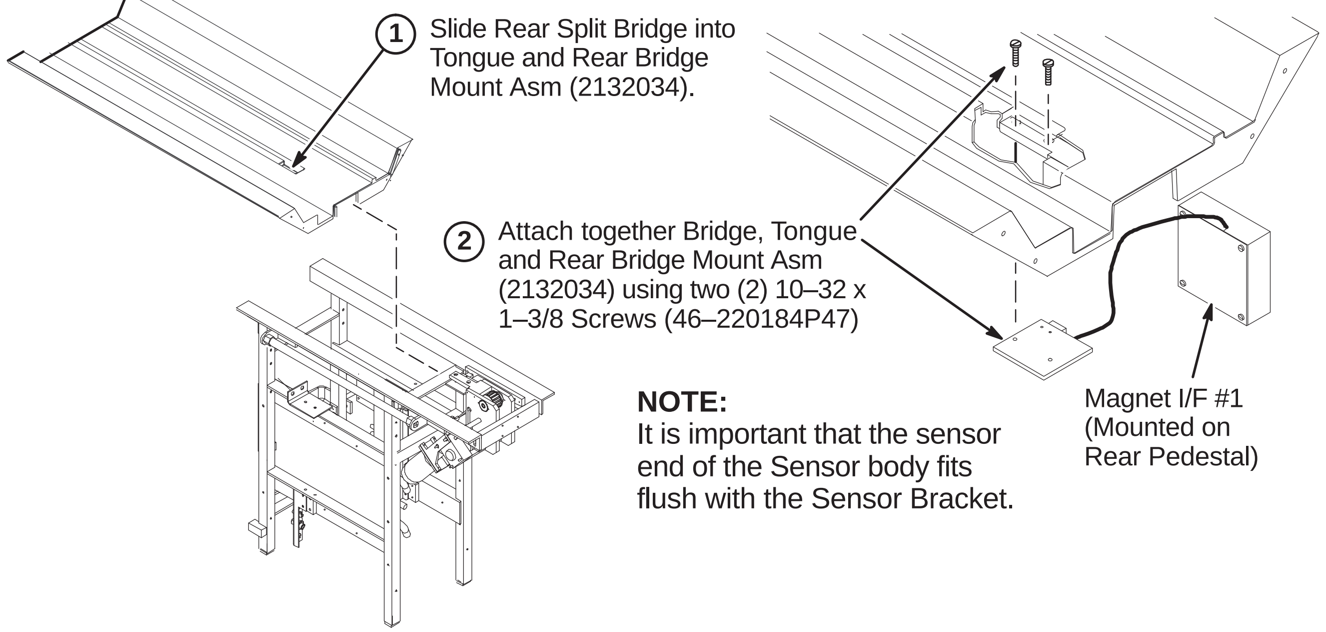
❑ K1 – PREPARING REAR SPLIT BRIDGE FOR INSTALLATION



1. Remove pads and bans from Rear Split Bridge assembly.
2. Remove Carriage/Track assembly from Split Bridge.
3. Remove and save Bridge Rear Plate and gasket.
4. To access Belt Clamp Plates, unfasten four screws holding Carriage Cover to Carriage.
5. Unfasten, remove, and save Belt Clamp Plates

Note:
Take care not to damage microphone which is attached to carriage cover.

❑ K2 – ATTACHING REAR SPLIT BRIDGE TO REAR PEDESTAL



1. Slide Rear Split Bridge into Tongue and Rear Bridge Mount Asm (2132034).
2. Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) 10–32 x 1–3/8 Screws (46–220184P47)

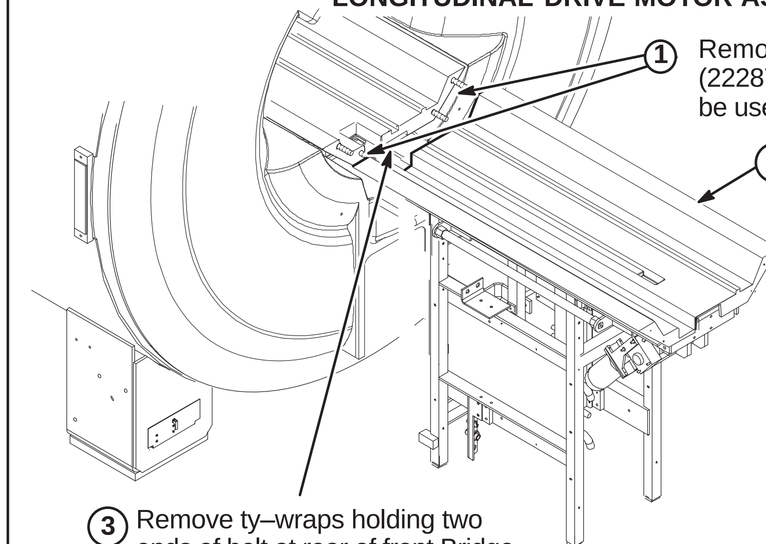
Magnet I/F #1 (Mounted on Rear Pedestal)

NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.

❑ K3 – ATTACHING REAR SPLIT BRIDGE TO FRONT SPLIT BRIDGE

WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

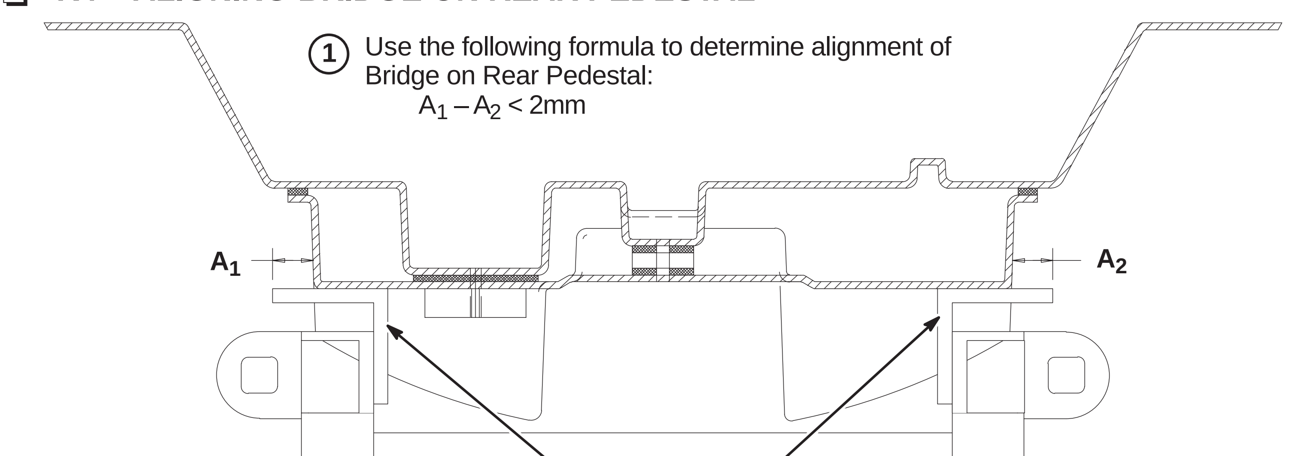


1. Remove four 10M hex nuts (2109877–3) and washers (2228717–2) from end of Bridge in magnet. These will be used for attaching the Rear Pedestal.
2. Move Rear Pedestal/Bridge into position and fasten with four nuts and washers that were removed form end of Front Split Bridge.
3. Remove ty–wraps holding two ends of belt at rear of front Bridge.

NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

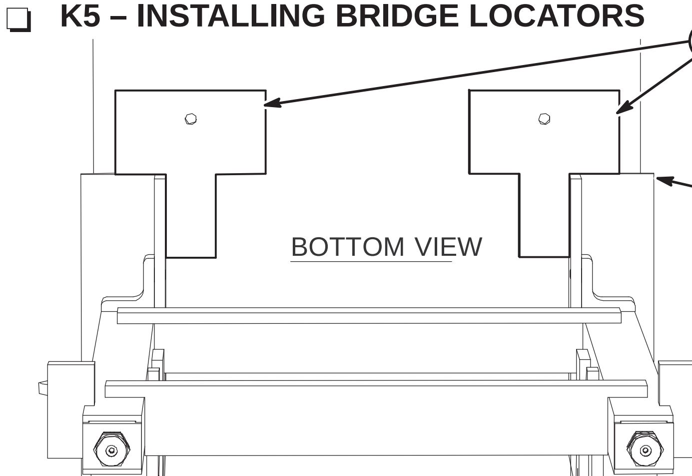
❑ K4 – ALIGNING BRIDGE ON REAR PEDESTAL

1. Use the following formula to determine alignment of Bridge on Rear Pedestal:
   $A_1 – A_2 < 2mm$



❑ K5 – INSTALLING BRIDGE LOCATORS

1. After Bridge and Rear Pedestal are aligned, peel off protective paper on each Bridge Locator and attach to bottom side of Bridge as shown.

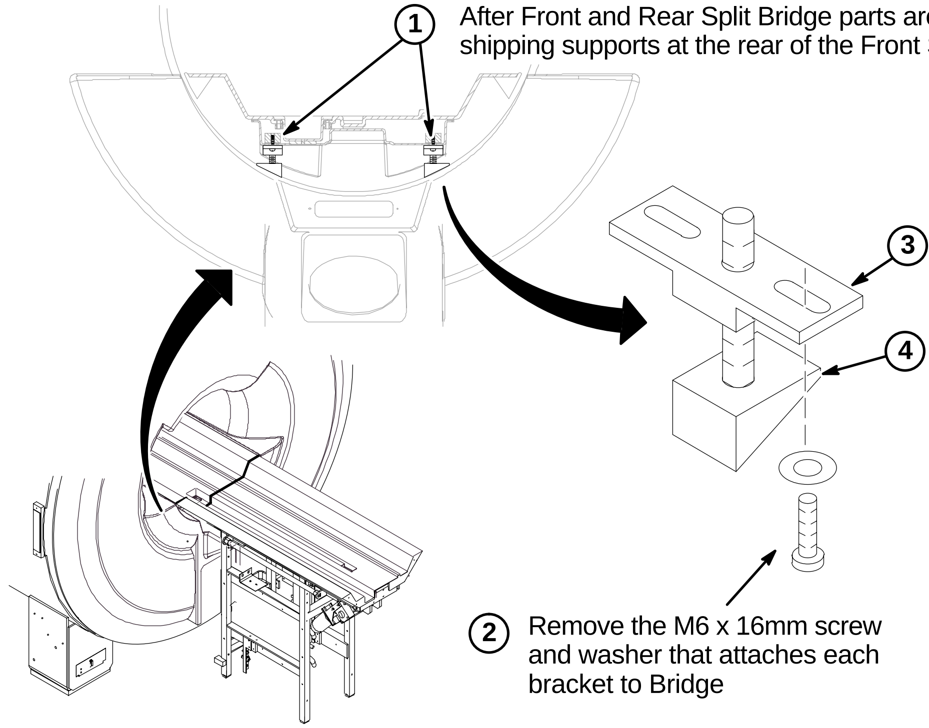


End of Rear Pedestal frame closest to Magnet.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

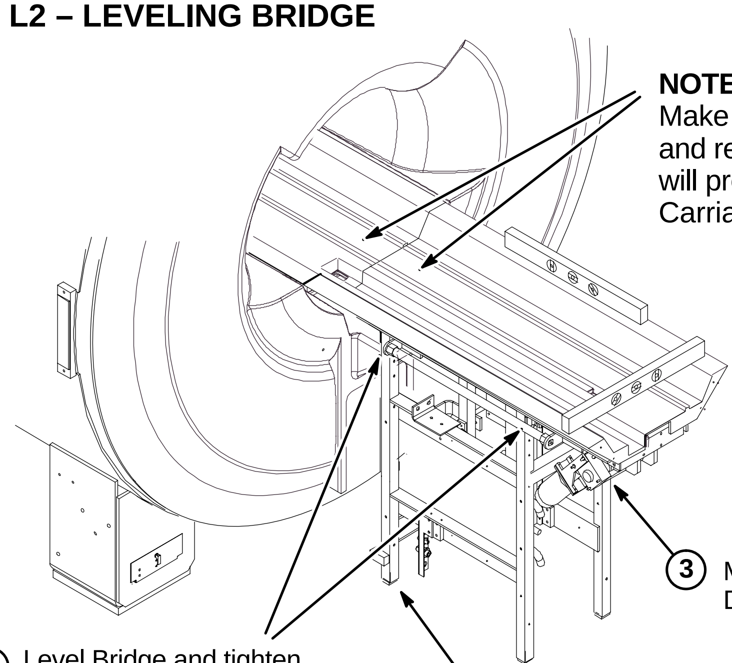
- ❏ **L1 – Removing Front Split Bridge Shipping Support Brackets**
- ❏ **L2 – Leveling Bridge**
- ❏ **L3 – Height Adjustment at Front of Bridge (if needed)**
- ❏ **L4 – Installing Carriage on Bridge**
- ❏ **L5 – Routing of Drive Belt**

❏ L1 – REMOVING FRONT SPLIT BRIDGE SHIPPING SUPPORT BRACKETS



1. After Front and Rear Split Bridge parts are fastened together, the two shipping supports at the rear of the Front Split Bridge need to be removed.
2. Remove the M6 x 16mm screw and washer that attaches each bracket to Bridge
3. Unscrew downword the Bridge Mounting Flange.
4. Support Base is attached to End Bell with adhesive tape. To remove, carefully pry base away from end bell. Remove base, stud, and mounting flange

❏ L2 – LEVELING BRIDGE

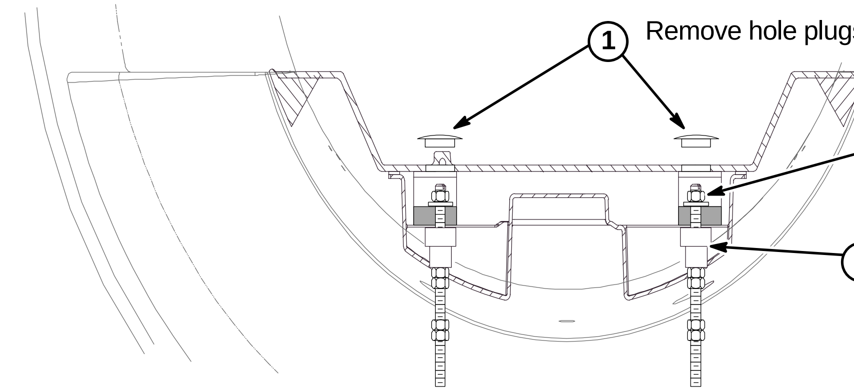


NOTE:
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:
Rear Pedestal to be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

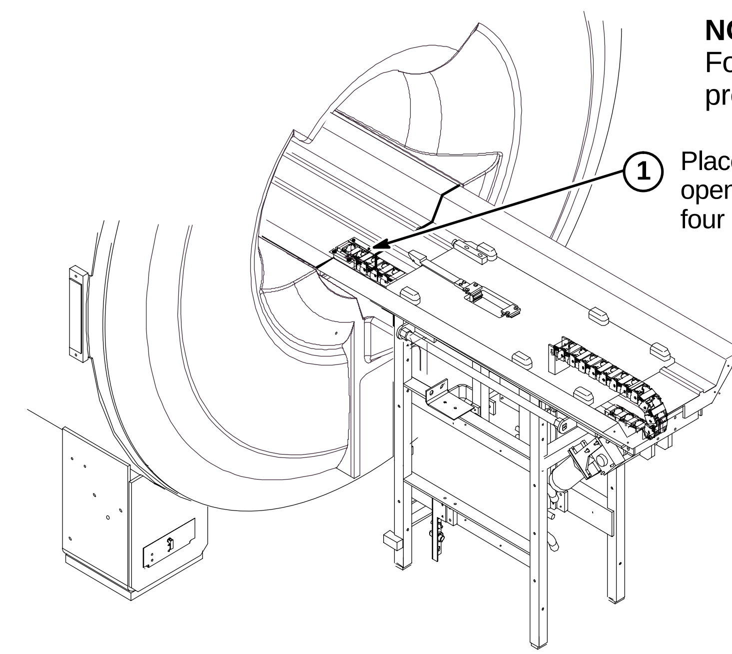
❏ L3 – HEIGHT ADJUSTMENT AT FRONT OF BRIDGE (IF NEEDED)



1. Remove hole plugs from each side of Bridge.
2. Remove M10 nut and washer from each stud.
3. Adjust Support Nut (2293381) as needed to achieve correct height.

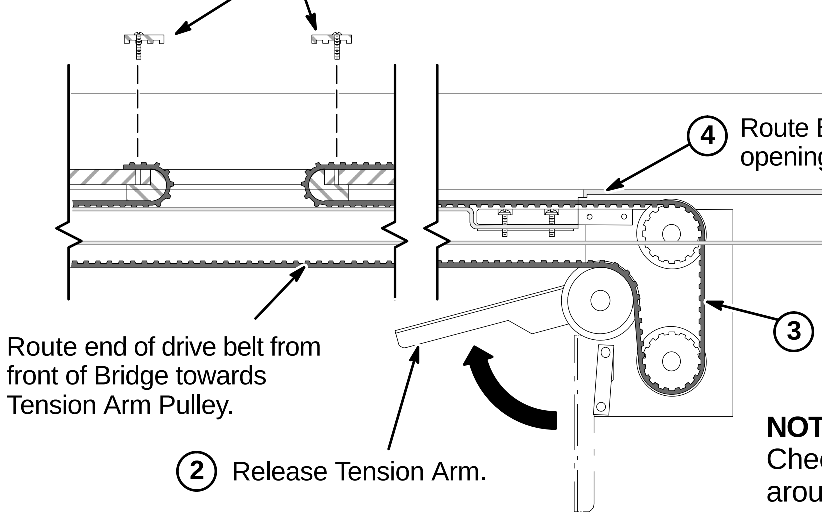
❏ L4 – INSTALLING CARRIAGE ON BRIDGE

NOTE:
For clarity, the carriage is shown without preamps, preamp protect module, and cables.



1. Place Carriage on Bridge and route all cables thru opening in Bridge and fasten plate to Front Bridge with four M6 x 12mm nylon screws (2296958) provided.

❏ L5 – ROUTING OF DRIVE BELT



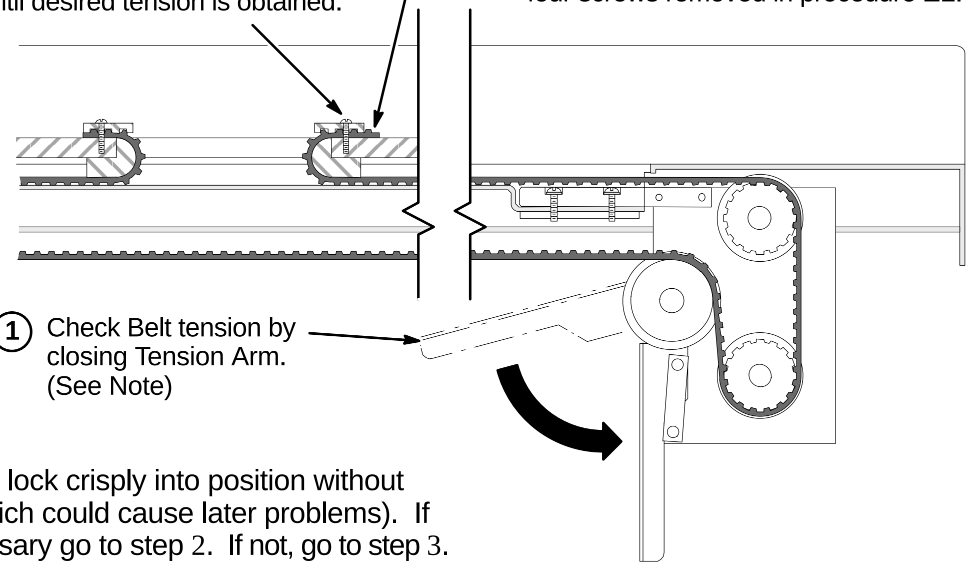
1. Route end of drive belt from front of Bridge towards Tension Arm Pulley.
2. Release Tension Arm.
3. Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.
4. Route Belt under and through opening to top of Carriage.
5. Position Belt under Clamp Plates and re–attach with previously removed screws.

NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ❑ **M1 – Final Adjustment of Drive Belt**
- ❑ **M2 – Installation of Sensor Flag**
- ❑ **M3A – Route Non–MGD Runs 404 and 485 (and Multi–coil Cables if Required)**
  OR
- ❑ **M3B – Route/Connect MGD Cables from LPCA to Rear Pedestal**
- ❑ **M4 – Installing Carriage Cover to Trolley and Bridge Cover**

❑ M1 – FINAL ADJUSTMENT OF DRIVE BELT



(1) Check Belt tension by closing Tension Arm. (See Note)

(2) **(IF NECESSARY):** Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.

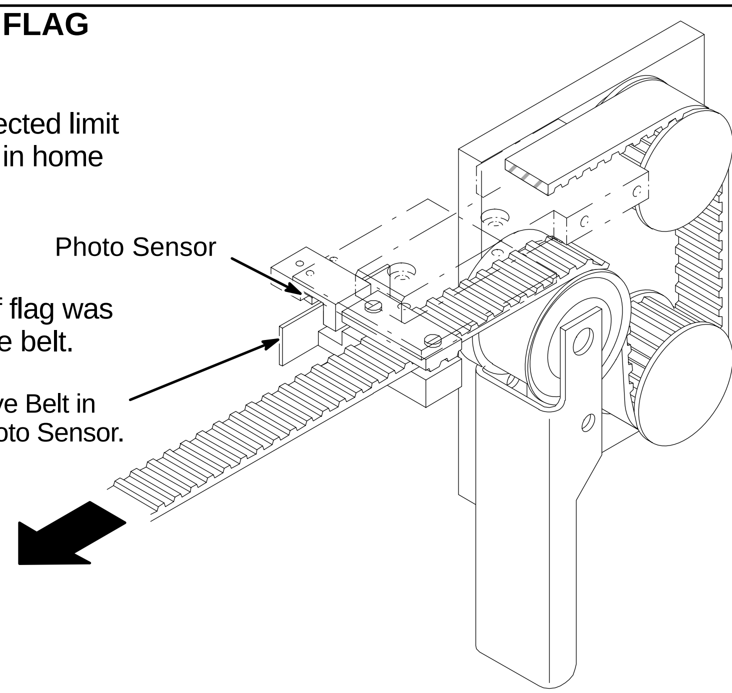
(3) Tighten Clamps and trim excess Belt length.

(4) Fasten Carriage Cover to Carriage with four screws removed in procedure **E1**.

NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

❑ M2 – INSTALLATION OF SENSOR FLAG

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

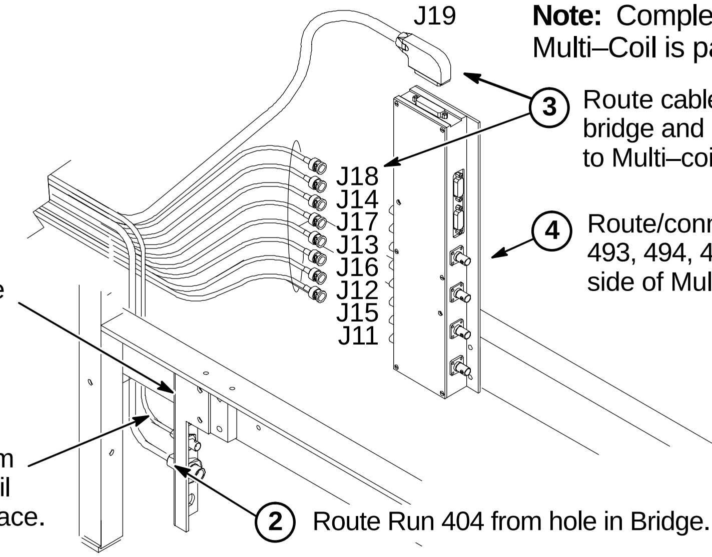


NOTE:
The following Step is needed if flag was removed during routing of drive belt.

(1) Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

❑ M3A – ROUTE NON–MGD RUNS 404 AND 485 (& Multi–Coil Cables if Required)

Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



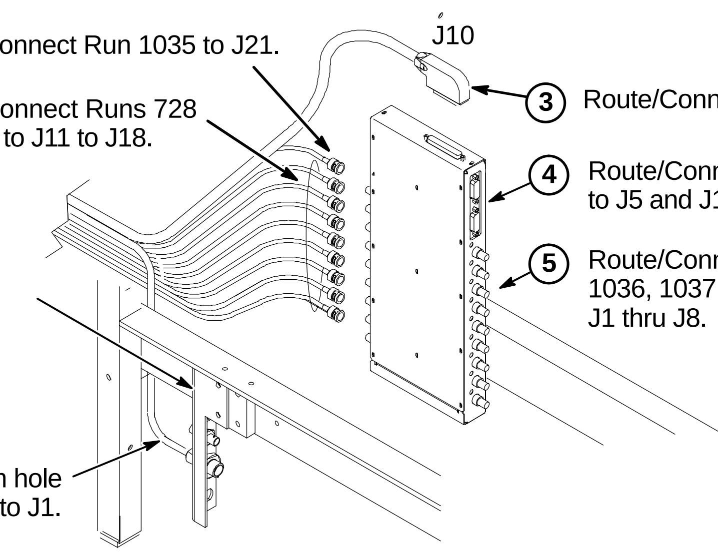
(1) Route Run 485 from hole in Bridge to coil cable takeup interface. Connect to J2.

(2) Route Run 404 from hole in Bridge. Connect to J1.

(3) Route cables from hole in bridge and connect as marked to Multi–coil Select Switch.

(4) Route/connect Runs 491, 492, 493, 494, 495, and 727 to this side of Multi–coil Select Switch.

❑ M3B – ROUTE/CONNECT MGD CABLES FROM LPCA TO REAR PEDESTAL



(1) Route/Connect Run 1035 to J21.

(2) Route/Connect Runs 728 thru 735 to J11 to J18.

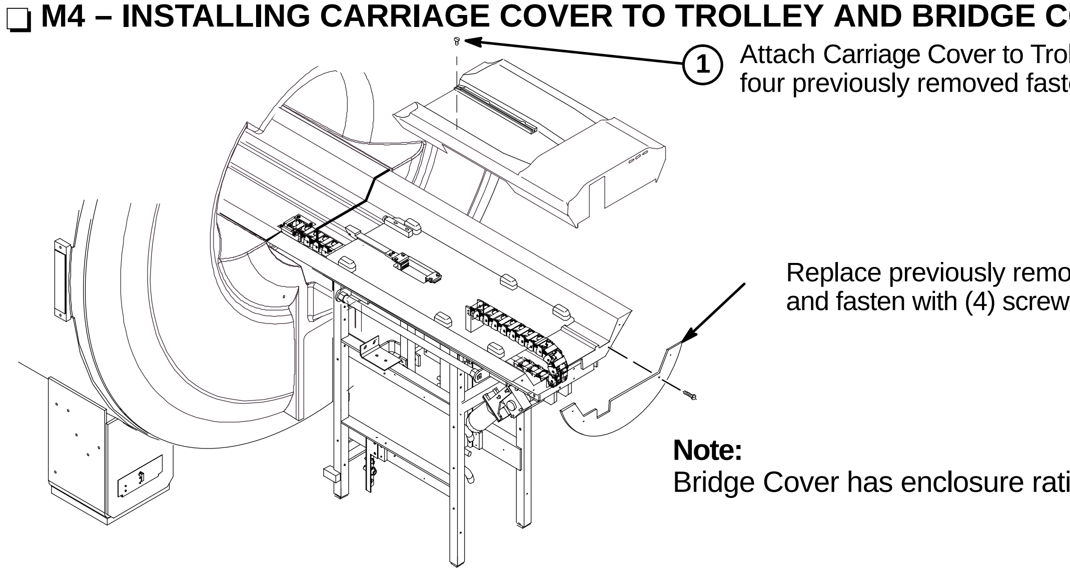
(3) Route/Connect Run 1041 to J19.

(4) Route/Connect Runs 1042 and 1049 to J5 and J19.

(5) Route/Connect Runs 491, 492, 493, 1036, 1037, 1038, 1039, and 1040 to J1 thru J8.

(6) Route Run 404 from hole in Bridge. Connect to J1.

❑ M4 – INSTALLING CARRIAGE COVER TO TROLLEY AND BRIDGE COVER



(1) Attach Carriage Cover to Trolley using four previously removed fasteners.

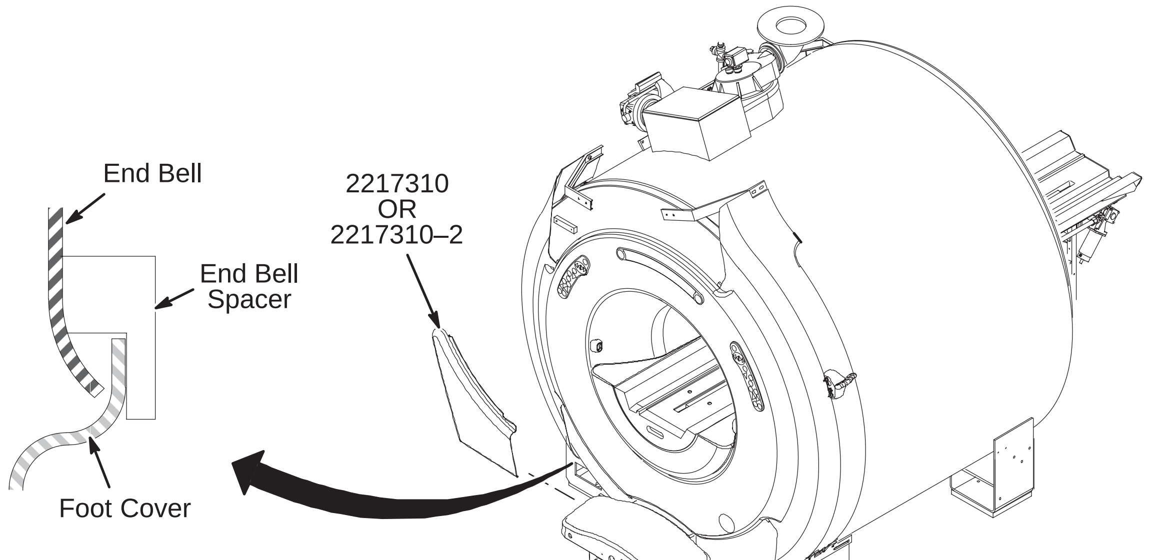
Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.

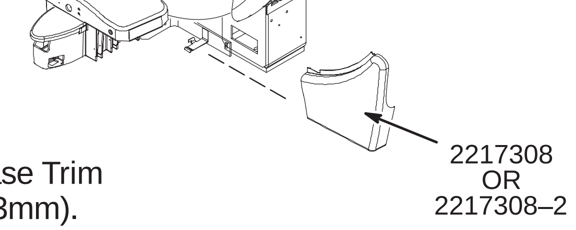
Note:
Bridge Cover has enclosure rating plates on it.

INSTALLATION PROCEDURES SUMMARY

- ❑ **N1 – Install Enclosure Base Front Foot Covers**
- ❑ **N2 – Install Enclosure Top Covers**
- ❑ **N3 – Install Rear Pedestal Covers**
- ❑ **N4 – Install Laser Alignment Light Warning Labels**

❑ N1 – INSTALL ENCLOSURE BASE FRONT FOOT COVERS





① Slide front Foot Covers from side, above spring clip, and between End Bell Flange and End Bell Spacers.

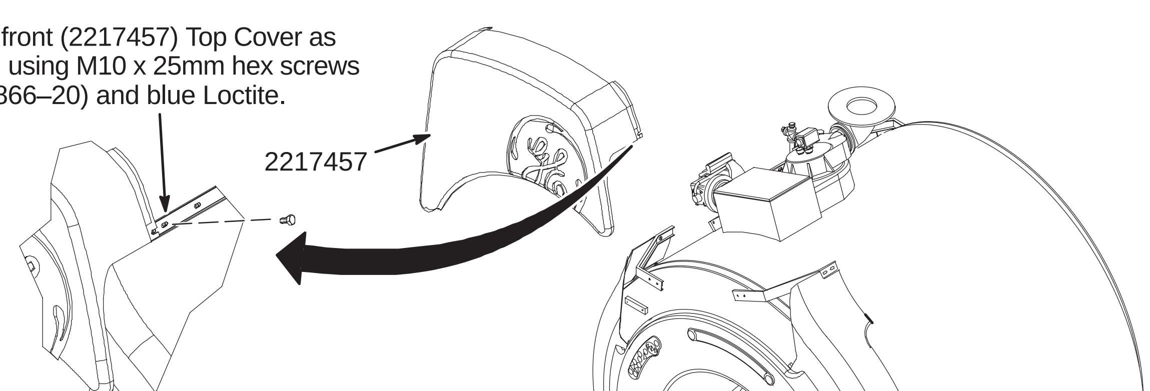
Note: The gap from the bottom of the Base Trim to floor should be a nominal 0.50 inch (13mm).

❑ N2 – INSTALL ENCLOSURE TOP COVERS

Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

Gap uniformity along any continuous seam shall be ± 0.0625 inch (1.59mm)

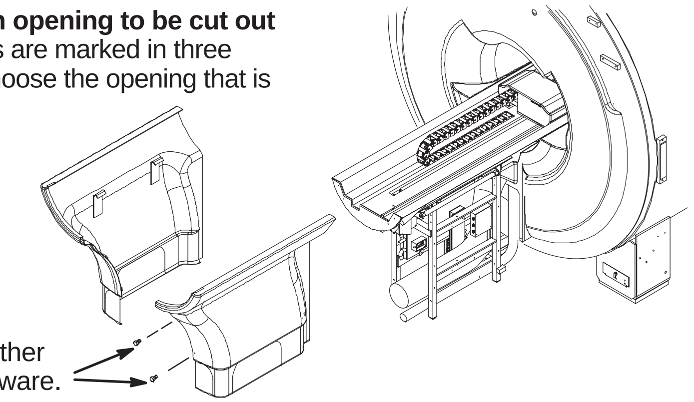
① Install front (2217457) Top Cover as shown using M10 x 25mm hex screws (2109866–20) and blue Loctite.



❑ N3 – INSTALL REAR PEDESTAL COVERS AND RATING PLATES

① Pedestal Covers (2217704 & 2217705) **require an opening to be cut out** depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting. Choose the opening that is required for the site.

② After cutting out appropriate opening, hang covers on Rear Pedestal.

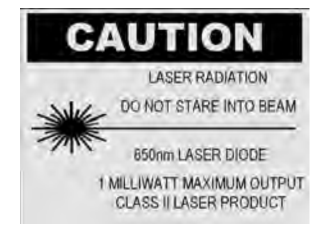


③ Fasten covers together with furnished hardware.

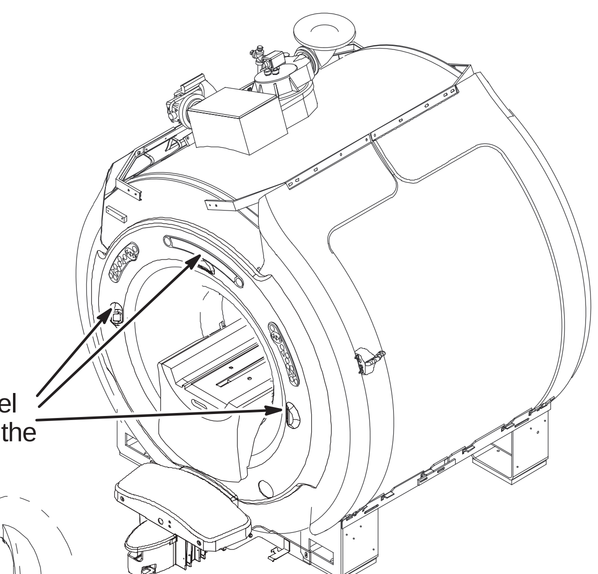
❑ N4 – INSTALL LASER ALIGNMENT LIGHT WARINING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

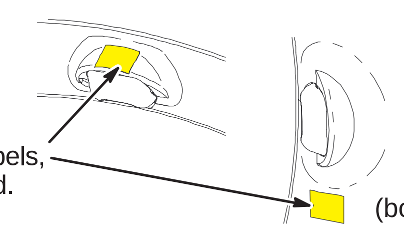
The magnet enclosure is shipped with a label in English attached below each of the three laser lights.



English



① For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



② If enclosure does not have labels, attach new labels as indicated.

(both sides)

French

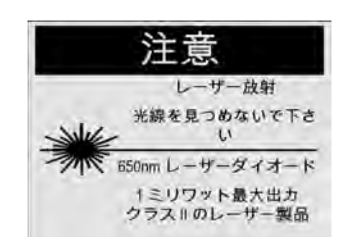
German

Portuguese

Spanish

Italian

Swedish



Japanese

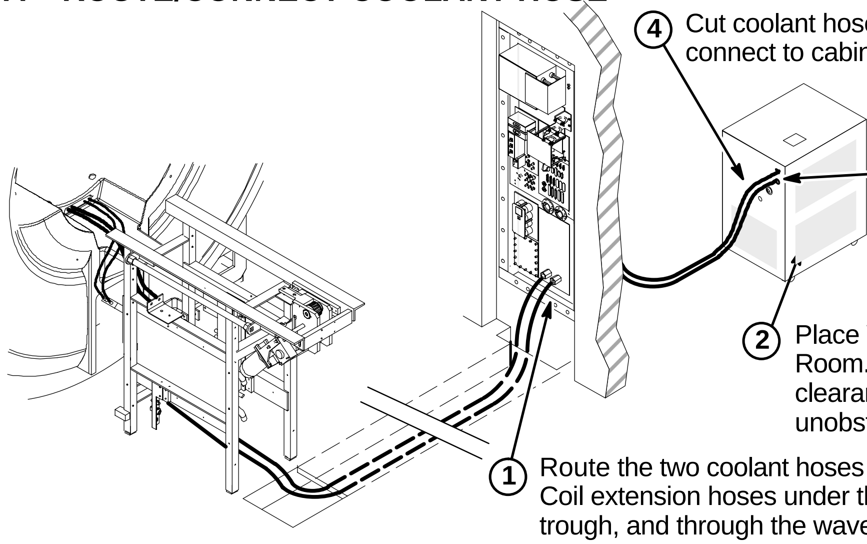
Chinese

WATER CHILLER CABINET INSTALLATION

INSTALLATION STEPS SUMMARY

- ❑ **A – Route/Connect Coolant Hose**
- ❑ **B – Install Power Cord**
- ❑ **C – Filling the System with Coolant**

❑ A – ROUTE/CONNECT COOLANT HOSE



4 Cut coolant hoses to appropriate length and connect to cabinet with Hose Clamps (2128542–2).

3 Remove two protective plugs. Wrap threads with Teflon tape and install provided $^1/_2$ inch hose barb fittings in these connections.

2 Place Water Chiller Cabinet in Equipment Room. Twenty–four inch (610mm) clearance on all sides is required for unobstructed flow of cooling air.

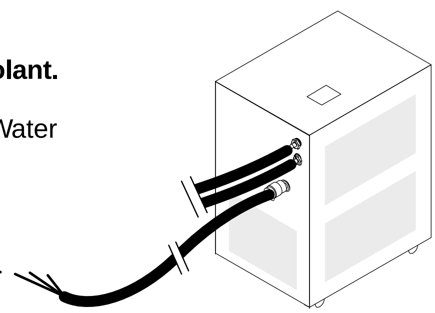
1 Route the two coolant hoses that were connected to the Body Coil extension hoses under the Rear Pedestal, through the cable trough, and through the wave guides on the Penetration Panel.

❑ B – INSTALL POWER CORD

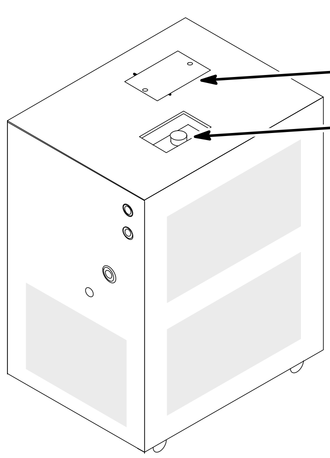
CAUTION:
Do not start Water Chiller pump before filling reservoir with coolant.
Consult vendor Water Chiller manual –
Installation, Operation, and Service Manual that is shipped with the Water Chiller.

1 **FOR RF/PDU:** Connect power cable Run 065 (2223355) to PDU as marked (TS1–26, 27, GND).

FOR ACGD/PDU: Connect power cable Run 970 (2280289) to PDU as markec (TS1–16,17,GND).



❑ C– FILLING THE SYSTEM WITH COOLANT AND START UP



1 Locate the twelve (12) gallons of Body Coil Coolant (2138791) that was shipped with System.

2 Remove cover.

3 Consult vendor Water Chiller manual – *Installation: Filling Requirements*. Initial Filling of Reservoir, and the following steps for the proper procedure on priming and filling the system with coolant.

4 Consult vendor Water Chiller manual – *Operation: Start Up.* Follow instructions in manual for starting water chiller operation.

5 **Verify unit is set to control at 20° C.**

6 If supplied, attach plastic sleeve to side of Water Chiller and insert vendor manual. This will allow easy access to any future service/ maintenance procedures. If plastic sleeve is not present, attach vendor manual to Water Chiller.

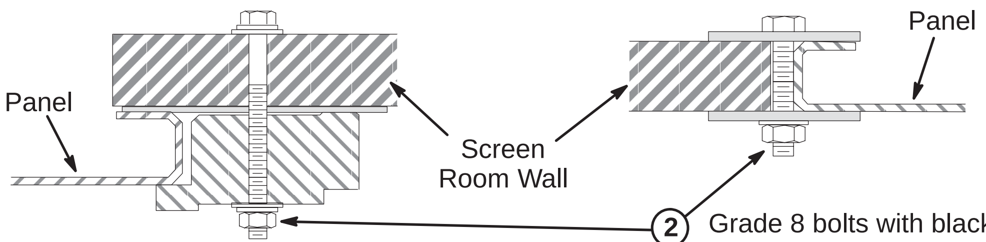
INSTALLATION STEPS SUMMARY

- ❑ **A – Remove Blank Panel/Prepare Screen Room Opening Surface**
- ❑ **B – Install Penetration Panel**
- ❑ **C – Install Penetration Panel Stand–Off Hardware**
- ❑ **D – Install Penetration Panel Covers**

❑ A – REMOVE BLANK PANEL/PREPARE SCREEN ROOM OPENING SURFACE

1. If still in place, remove vendor supplied blank panel. Save mounting hardware of either method shown below for installation of Penetration Panel.

Hold Down Method **Brass Sealing Plate Method**



2. Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb–ft (1.5 to 2 turns after lock washer flattens).

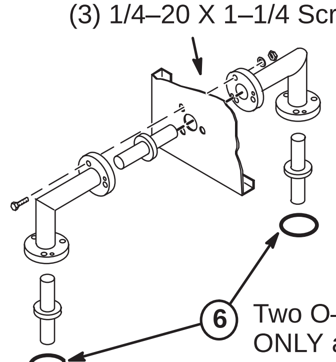
3. To insure good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**

❑ B – INSTALL PENETRATION PANEL

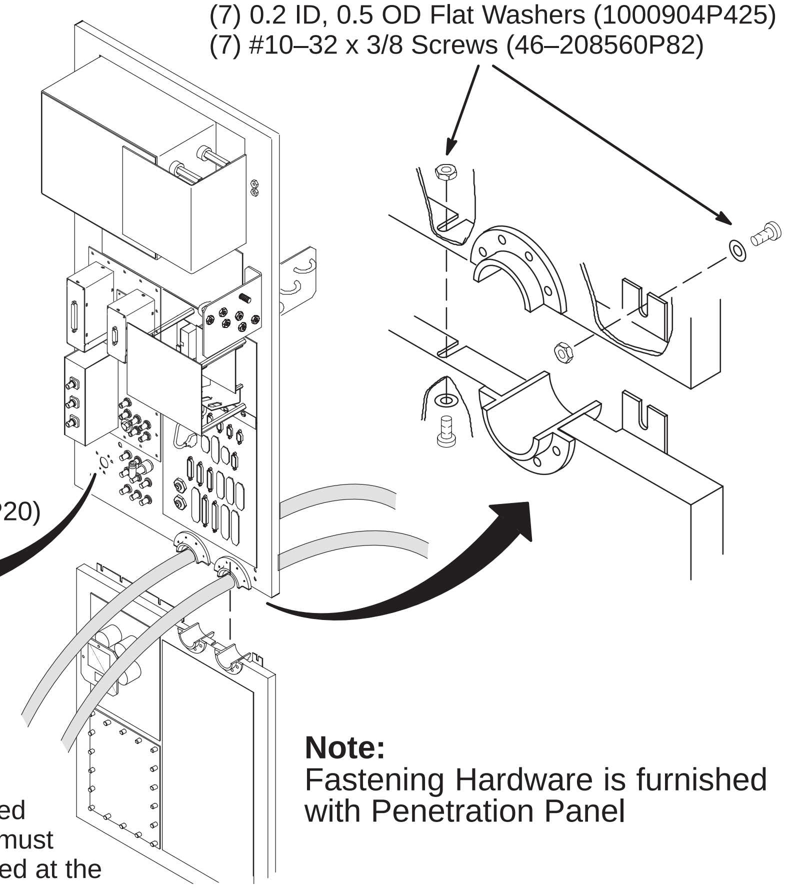
1. Assemble Penetration Panel around helium lines (Runs 621 and 622) using:
   (7) #10–32 Nylock Lock Nuts (46–208937P6)
   (7) 0.2 ID, 0.5 OD Flat Washers (1000904P425)
   (7) #10–32 x 3/8 Screws (46–208560P82)
2. Install Penetration Panel using same method and hardware for mounting of blank panel above.
3. Seal all joints with copper foil tape with a conductive adhesive (3M #1181 or 46–258218P4, 5, or 6) for RF–tight integrity.
4. Fill inside of J60 and J61 guides with Bronze Wool (46–318068P1) that was supplied with magnet.

Note: Steps 5 and 6 are not performed if Times Microwave RF cables are to be installed.

5. Attach HN Right Angle Flanged Connectors (46–230013P1) to Penetration Panel with:
   (3) 1/4–20 X 7/16 Nuts (46–208935P8)
   (3) 1/4 Flat Washers (46–221449P6)
   (3) 1/4–20 X 1–1/4 Screws (46–208939P20)



6. Two O–rings to be installed ONLY as shown. There must NOT be an O–ring installed at the Penetration Panel connection.

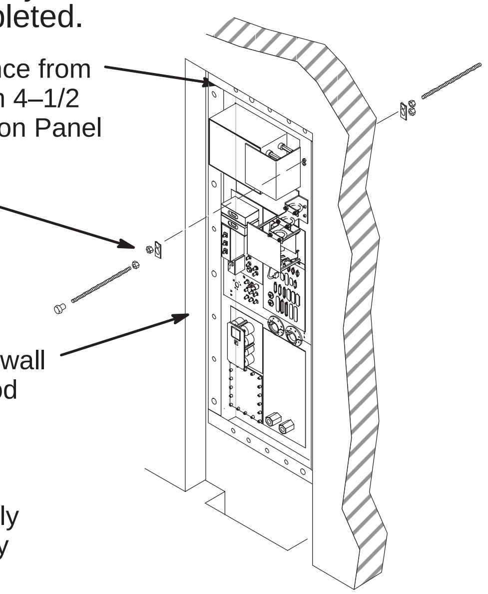


Note:
Fastening Hardware is furnished with Penetration Panel

❑ C – INSTALL PENETRATION PANEL STAND–OFF HARDWARE

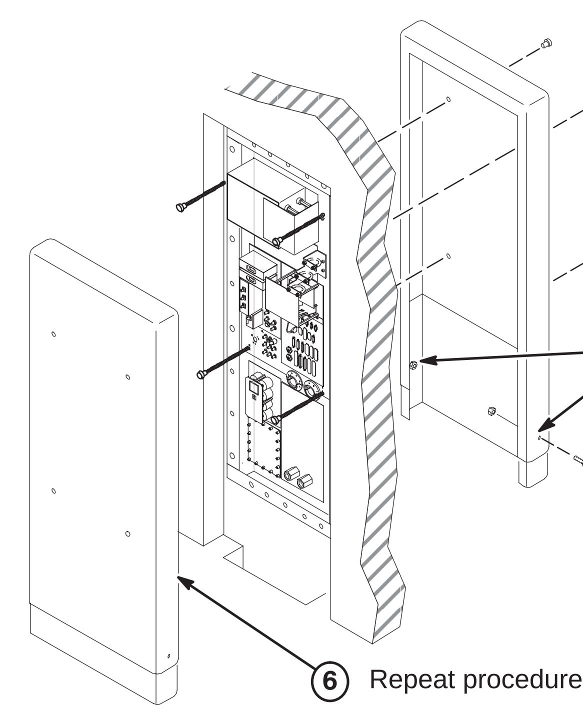
Note: The Penetration Panel Floor plate that is supplied with the cover kit is to be installed prior to routing and connecting cables to Penetration Panel. Refer to Installation Drawing furnished with the Cover Kit.

Note: The Penetration Panel Cover can be installed only after all connections and System installation has been completed.



1. Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4–1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.
2. Place (8) Reinforcing Plates (46–301211P1) over each hole pair on both sides of Penetration Panel and attach using:
   (16) 5/16–18 x 9/16 Brass Nut (46–208935P9)
   (8) 5/16–18 x 18 inch Threaded Rod (46–301213P2)
   (8) 5/16–18 x 1 inch Barrel Nut (46–301212P1)
3. Measure distance from face of Pen Panel to equipment room wall surface. Add six inches (152mm) to obtain length threaded rod should extend from panel.
4. Repeat step 3 procedure for exam room cover.
5. Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.

❑ D – INSTALL PENETRATION PANEL COVERS



1. Remove all barrel nuts.
2. In the Magnet room, temporarily install Top (46–306936P1) and Bottom (46–306964P1) Cover assembly on threaded rods and secure with four barrel nuts.
3. Adjust position of bottom cover section and mark two lower fastener holes.
4. Remove covers and drill hole in lower cover at each mark.
5. Attach top and bottom covers with furnished 10–32 nuts and screws.
6. Repeat procedure for Equipment room cover.
7. After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.

SYSTEM CABINET INSTALLATION INSTRUCTIONS

1–1 Tab 8, System, contains installation instructions for the following:

Use only the applicable System Cabinet Installation Instruction Directions

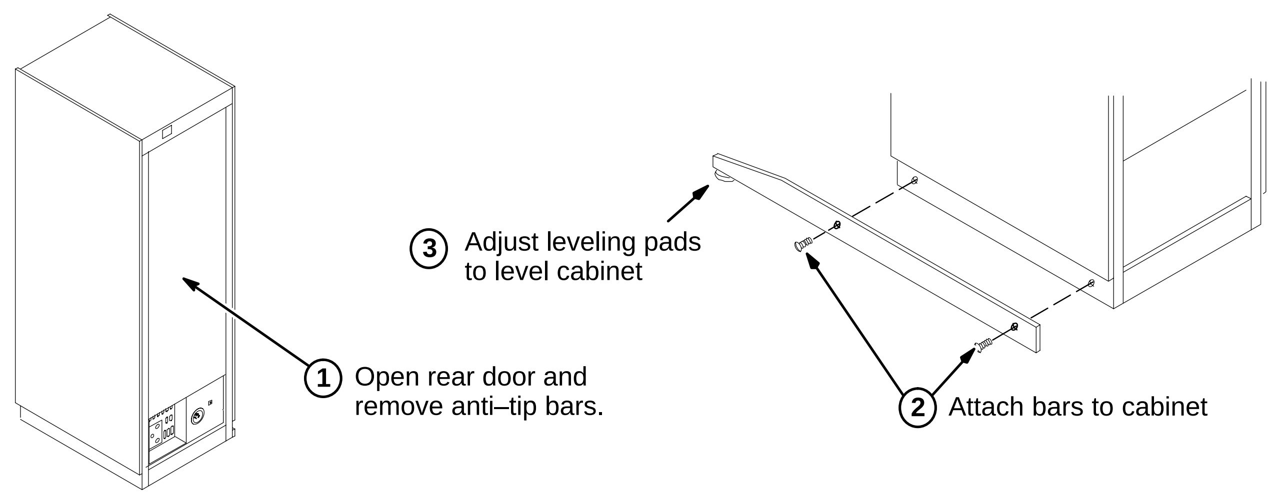
- *DIRECTION 2217004* – Signa 8.x/9.x System Cabinet Installation Instructions
- *DIRECTION 2338502* – Signa EXCITE I System Cabinet Installation Instructions
- *DIRECTION 2398344* – Signa EXCITE II System Cabinet Installation Instructions

INSTALLATION STEPS SUMMARY

- ❑ **A – Position/Secure System Cabinet**
- ❑ **B – Install Magnet Monitoring Cables**
- ❑ **C – Connect Fiber Optic Runs 708, 710, 711/712, and 836 to I/F Panel**
- ❑ **D – Route/Connect Fiber Optic Runs 836 to BIT3 Module**
- ❑ **E – Route/Connect Fiber Optic Runs 708, 710, and 711/712 to Tyme II**
- ❑ **F – Connect Power, Ground, and Data Cables**
- ❑ **G – Install InSite "SYSTEMS" Cabinet Magnet**

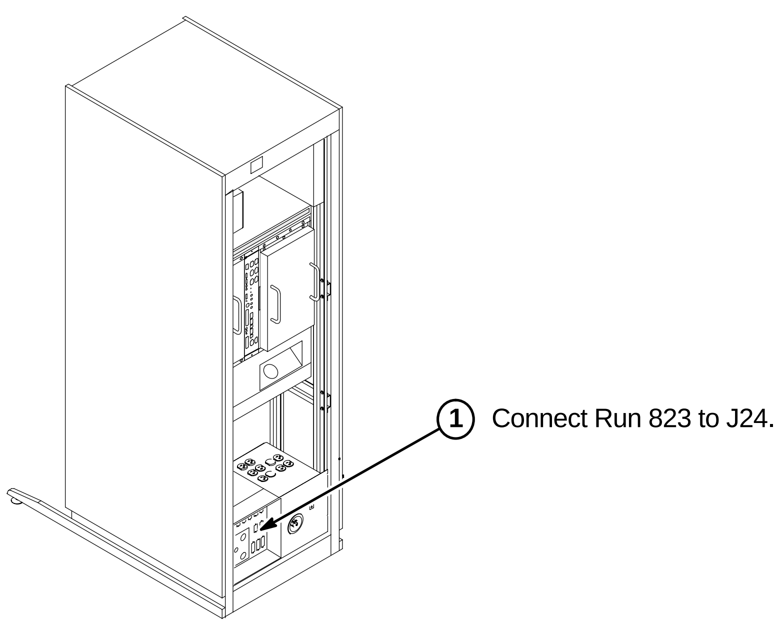
❑ A – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements by either installing anti tip legs (see illustration below) or securing the cabinet to the floor with local supplied brackets and floor anchors.



❑ B – INSTALL MAGNET MONITORING CABLES

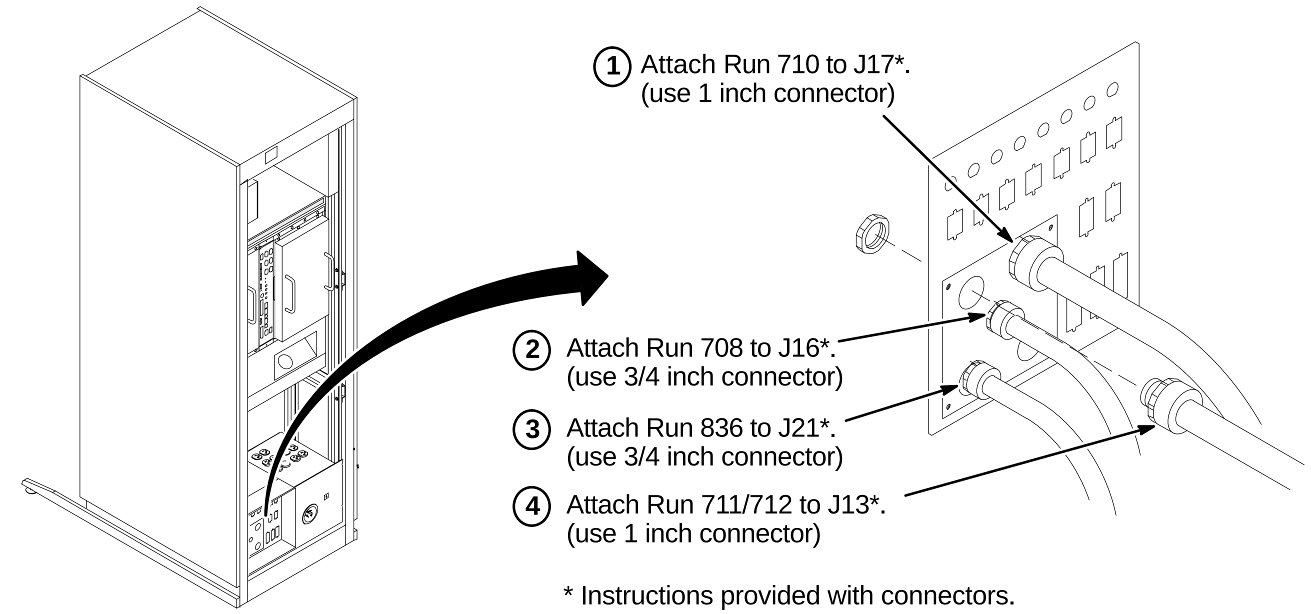
Note: Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual,* for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.



❑ C – CONNECT FIBER OPTIC RUNS 708, 710, 711/712, AND 836 TO I/F PANEL

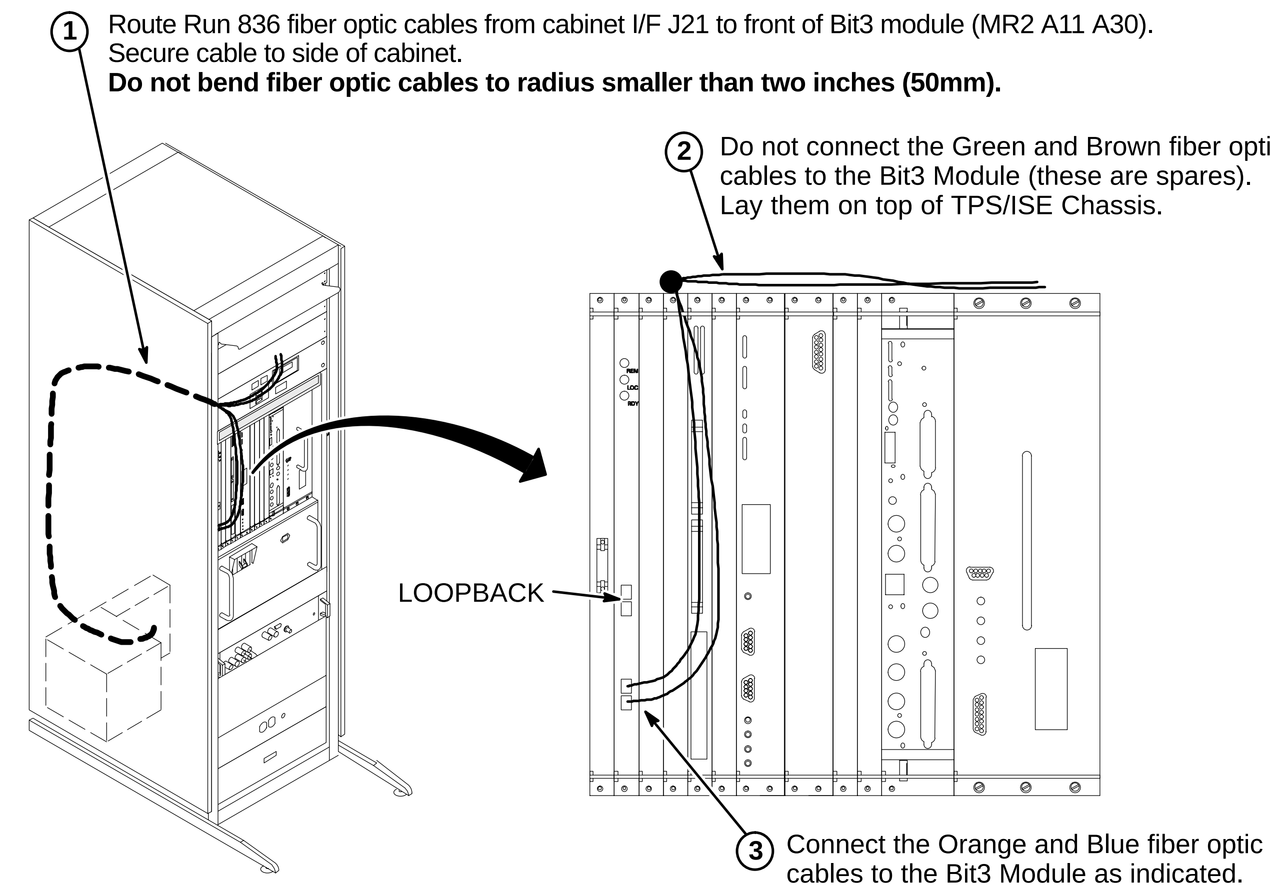
CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



* Instructions provided with connectors.

❑ D – ROUTE/CONNECT FIBER OPTIC RUN 836 TO BIT3 MODULE



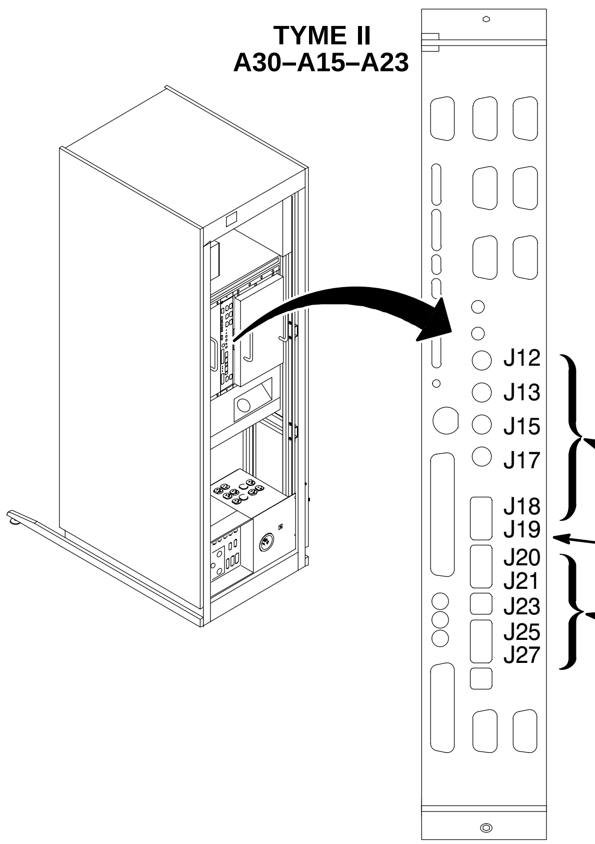
❑ E – ROUTE/CONNECT FIBER OPTIC RUNS 708, 710, AND 711/712 TO TYME II

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.

Note:
Runs 708, 710, 711, and 712 are used for both Horizon 5.x and 8.x. The fiber optic connections to the TYME board are different for 5.x and 8.x. The labels on the cables include both the 5.x and 8.x connections.

The table below indicates the connection labels as they should be marked on this end of the fiber optic cables. **Use only the 8.x connections.**



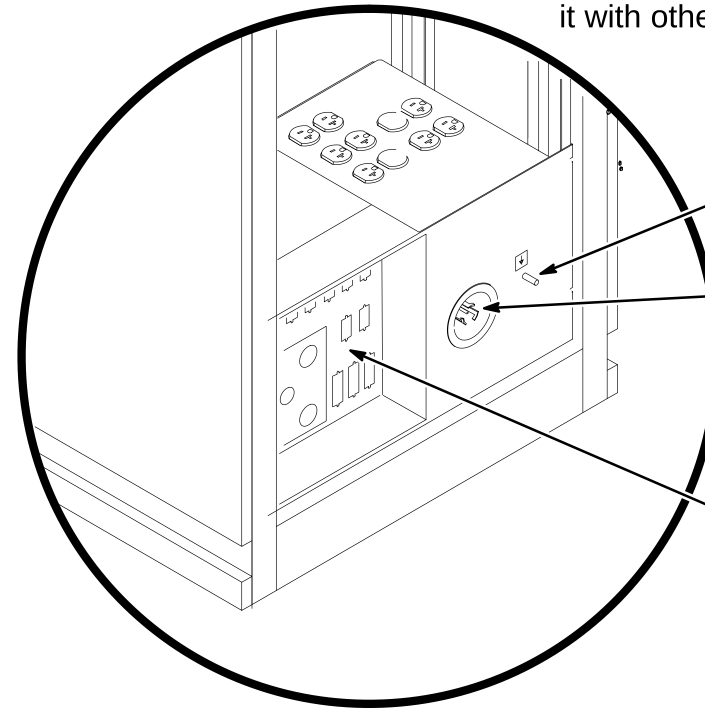
RUN #	5.x	8.x
708–1	J5	J19
710–1	J13	J12
710–2	J15	J13
710–3	J17	J15
710–4	J14	J17
710–5	J4	J18
711–1	J6,J7	J25,J27
711–2	J8	J23
712–1	J9,J10	J20,J21

1. Route/Connect as marked Run 710 fiber optic cables to J12, J13, J15, J17 and J18 on the TYME II Board.
2. Route/Connect as marked Run 708 fiber optic cable to J19 on the TYME II Board.
3. Route/Connect as marked Run 711/712 fiber optic cables to J20, J21, J23, J25, and J27 on the TYME II Board.

Note:
Use tie-wraps to secure fiber optic cables to ISE Chassis cables.

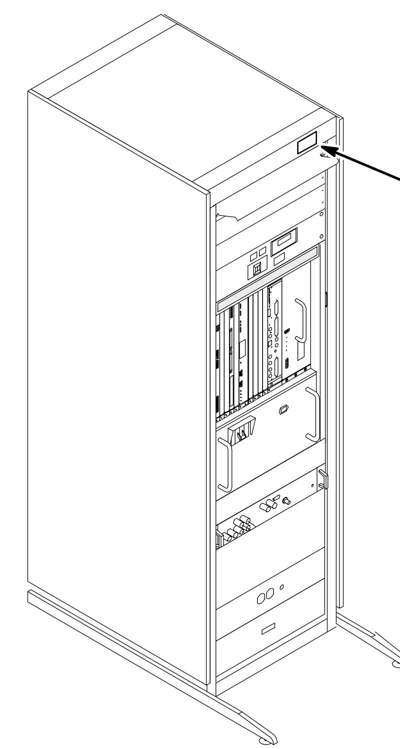
❑ F – CONNECT POWER, GROUND, AND DATA CABLES

Note:
Locate CERD Test Cable, 2168505 (it is packed with the Operator Workspace cable kit, 2154392). Store it with other test cables or in the System Cabinet.



1. Connect Run 037 GND cable to ground stud.
2. Connect Run 030 or 968 cable plug to “J1”. Be sure to push and turn plug for secure connection.
3. Perform Ground Resistance Checks before connecting remaining cabinet cables.
4. Connect remaining cabinet cables as marked.

❑ G – INSTALL INSITE “SYSTEMS” CABINET MAGNET



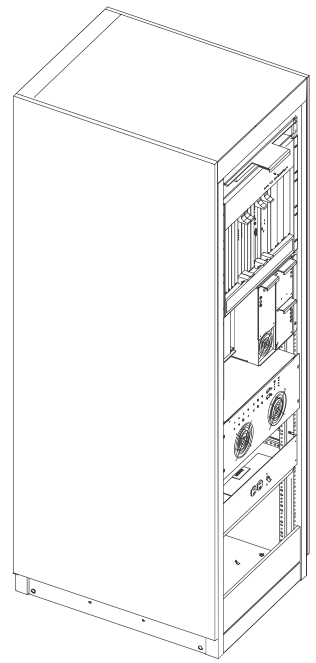
1. From the 8.0 LX Insite Kit (46–301708G14), locate the ”Systems” cabinet magnet (46–320095P5) and attach to front of cabinet as shown.

INSTALLATION STEPS SUMMARY

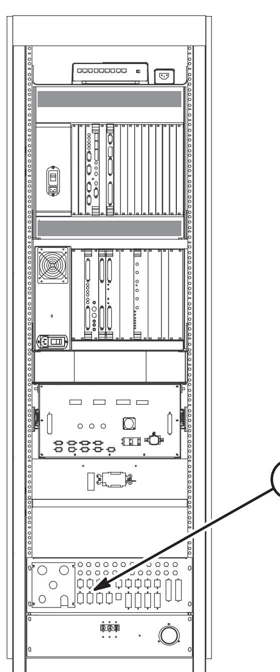
- ❑ A – Position/Secure System Cabinet
- ❑ B – Install Magnet Monitoring Cables
- ❑ C – Connect Fiber Optic Runs 1033, 1034, 1044, and 1045 to I/F Panel
- ❑ D – Route/Connect Fiber Optic Run 1044 to APS Board
- ❑ E – Route/Connect Fiber Optic Runs1033, 1034, and 1045 to STIF Board
- ❑ F – Connect Power, Ground, and Data Cables
- ❑ G – Install InSite “SYSTEMS” Cabinet Magnet

❑ A – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.



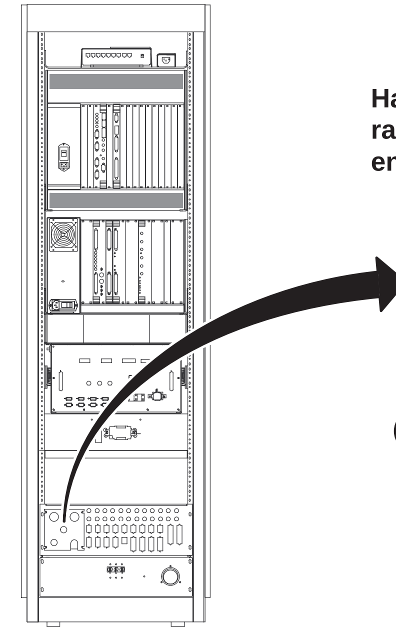
❑ B – INSTALL MAGNET MONITORING CABLES



Note: Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual,* for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.

1 Connect Run 823 to J24.

❑ C – CONNECT FIBER OPTIC RUNS 1033, 1034, 1044, AND 1045 TO I/F PANEL



CAUTION

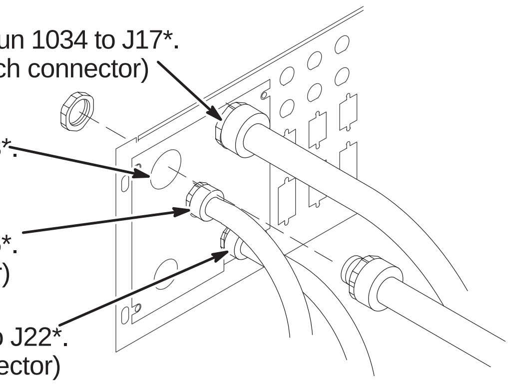
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1 Attach Run 1034 to J17*. (use 1 inch connector)

2 Attach Run 1045 to J13*. (use 1 inch connector)

3 Attach Run 1033 to J16*. (use 3/4 inch connector)

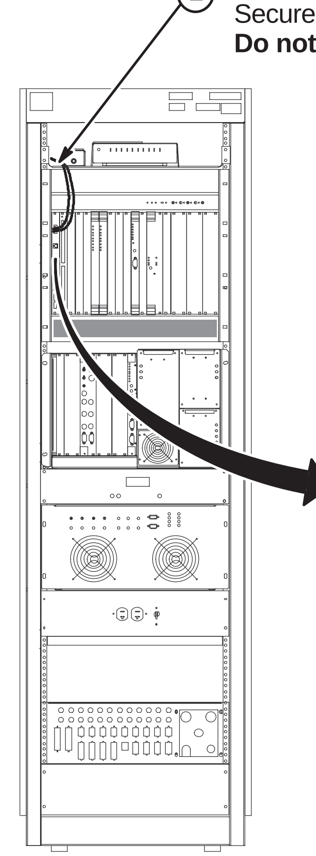
4 Attach Run 1044 to J22*. (use 3/4 inch connector)



* Instructions provided with connectors.

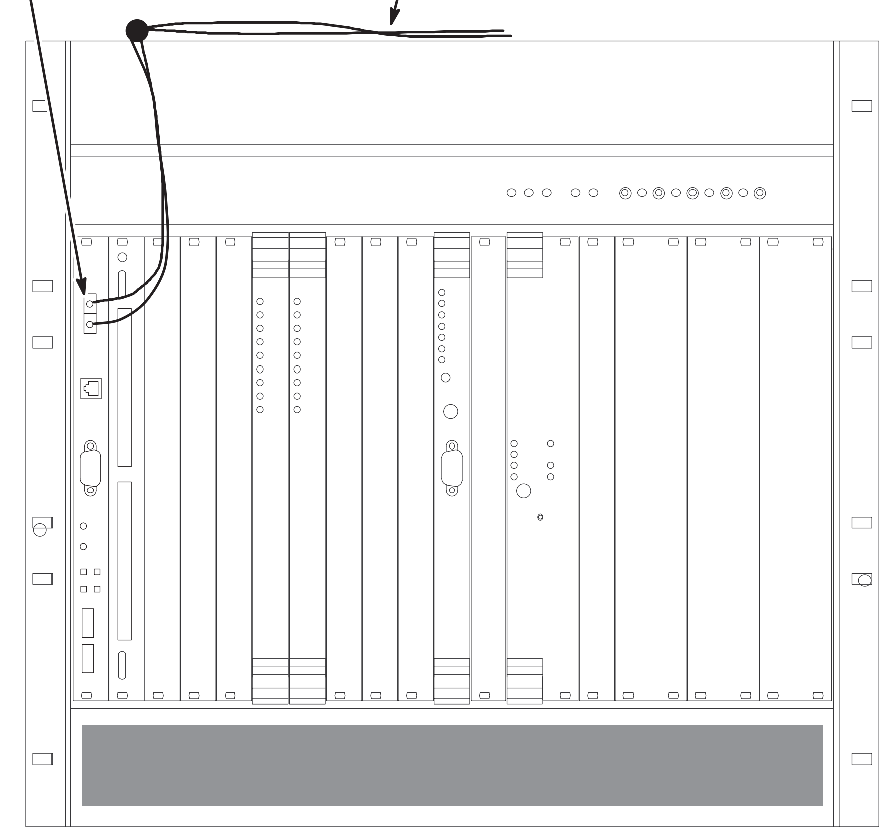
❑ D – ROUTE/CONNECT FIBER OPTIC RUN 1044 TO APS BOARD

1 Route Run 1044 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet.
Do not bend fiber optic cables to radius smaller than two inches (50mm).



2 Connect the Orange and Blue fiber optic cables to the TI–MGD Daughter Board as indicated.

3 Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.



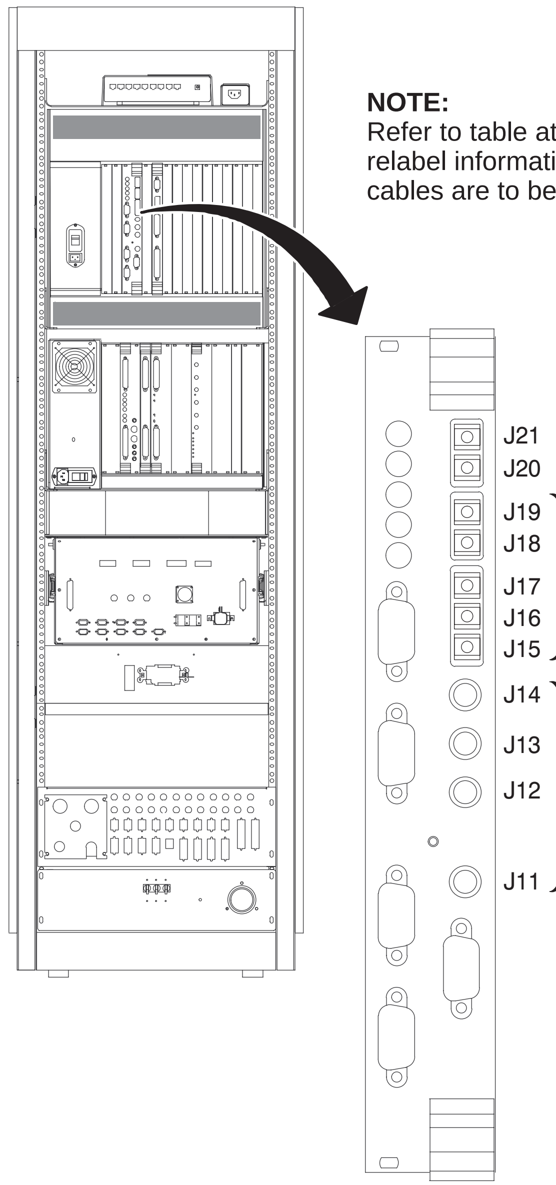
❑ E – ROUTE/CONNECT FIBER OPTIC RUNS 1033, 1034, AND 1045 TO STIF BOARD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.

NOTE:
Refer to table at right for relabel information if existing cables are to be reused.

New Cables		Existing Cables	
RUN #	10.x	RUN #	8.x/9.x
1033–1	J21	708–1	J19
1034–1	J14	710–1	J12
1034–2	J13	710–2	J13
1034–3	J12	710–3	J15
1034–4	J11	710–4	J17
1034–5	J20	710–5	J18
1045–1	J17	711–2	J23
1045–2	J18,J19	711–3	J20, J21
1045–3	J15,J16	711–1	J25, J27



1. Relabel (if old cable re–used) and Route/Connect Run 1033 fiber optic cable to J21 on the STIF Board.

Part of Run 1034 below.

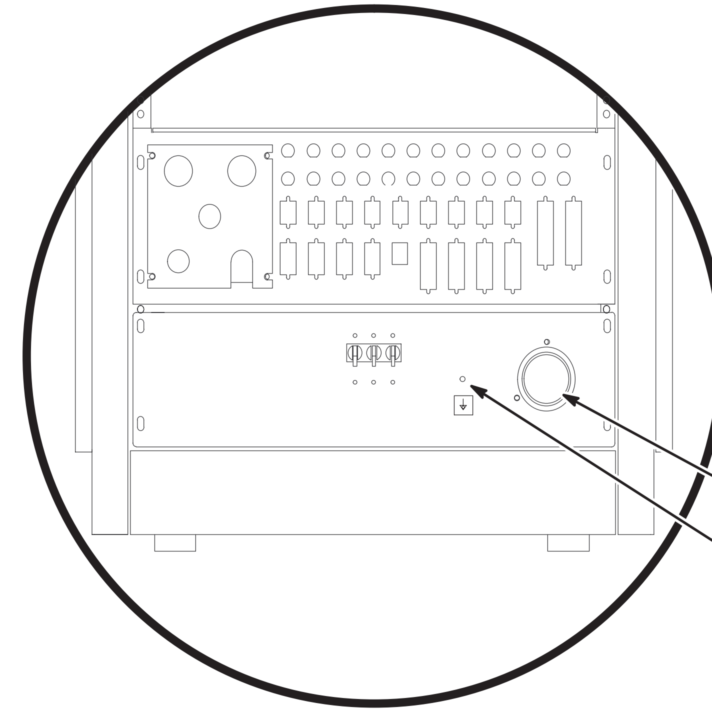
2. Relabel (if old cable re–used) and Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.

3. Relabel (if old cable re–used) and Route/Connect Run 1034 fiber optic cables to J11, J12, J13, J14, and J20 above on the STIF Board.

Note:
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

❑ F – CONNECT POWER, GROUND, AND DATA CABLES

Note:
Locate CERD Test Cable, 2168505 (it is packed with the Operator Workspace cable kit, 2154392). Store it with other test cables or in the System Cabinet.

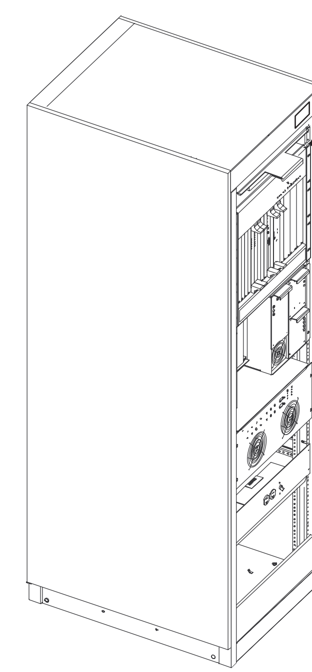


WARNING!

SYSTEM CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.

1. Connect Run 968 cable plug to “J1”. Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.
3. Perform Ground Resistance Checks before connecting remaining cabinet cables.
4. Connect remaining cabinet cables as marked.

❑ G – INSTALL INSITE “SYSTEMS” CABINET MAGNET



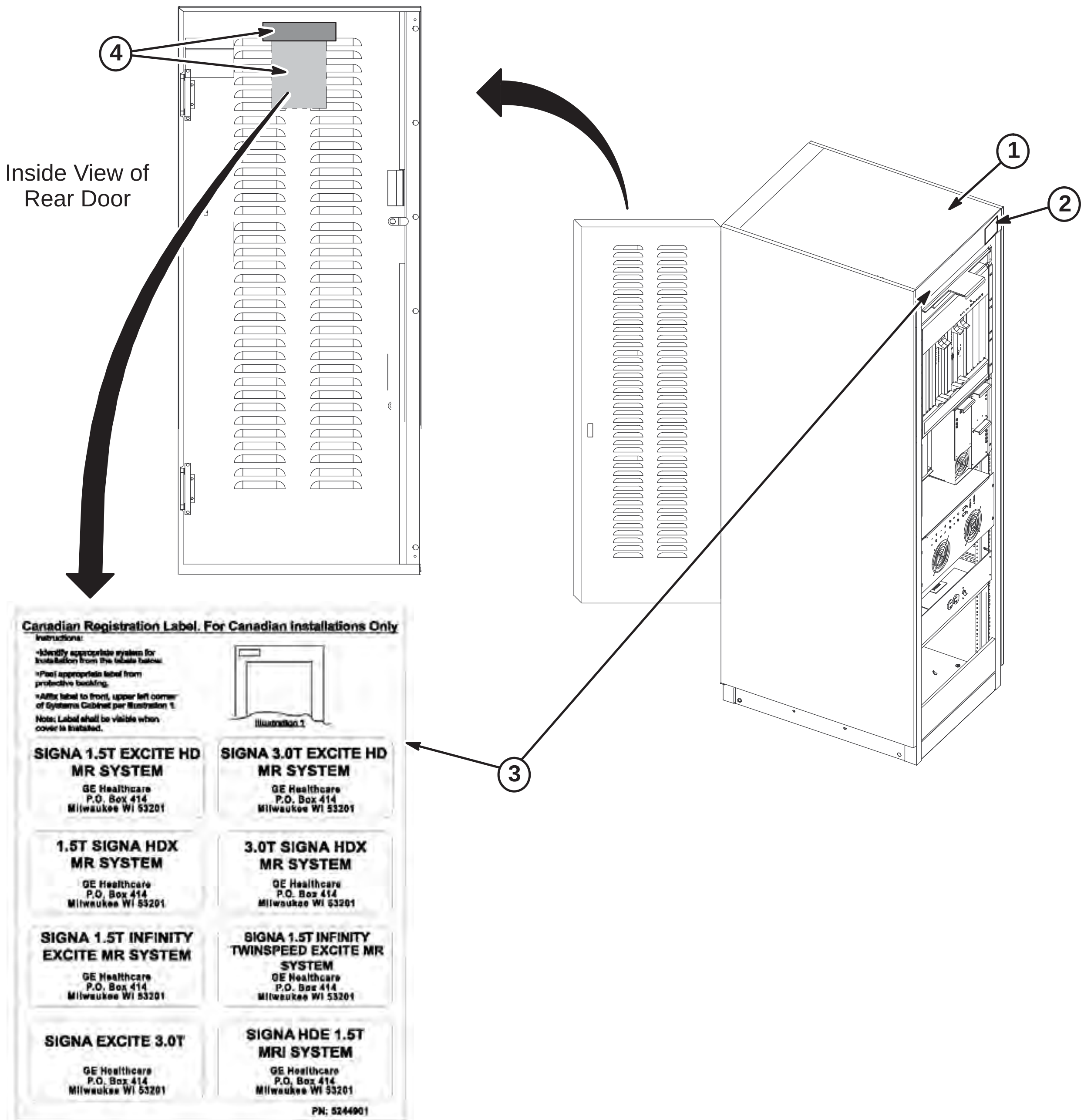
1. From the LX Insite Kit (46–301708G14), locate the ”Systems” cabinet magnet (46–320095P5) and attach to front of cabinet as shown.

TABLE OF CONTENTS

1–1 POSITION AND SECURE SYSTEM CABINET

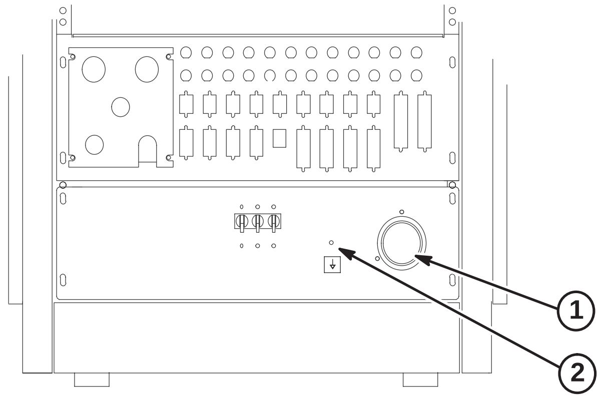
Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position System cabinet to final position as indicated by site layout drawings.
2. From the LX Insite Kit (46–301708G14), locate the ”Systems” cabinet magnet (46–320095P5) and attach to front of cabinet as shown.
3. FOR INSTALLATION IN CANADA ONLY: Remove Registration Label sheet (5244901) from plastic sleeve on inside of rear door. Follow instructions on sheet for attaching appropriate label to cabinet.
4. For all non–Canada installations, remove plastic bag with labels and dispose. Label is not required.



1–2 SYSTEM CABINET POWER AND GROUND CABLE CONNECTIONS

1. Connect Run 968 cable plug to “J1”. Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.
3. Perform Ground Resistance Checks before connecting remaining cabinet cables.

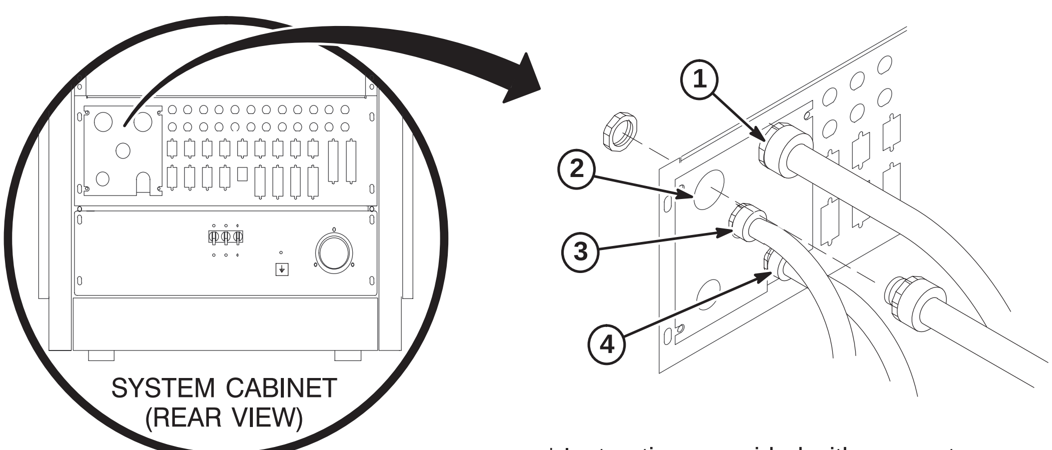


1–3 CONNECT FIBER OPTIC RUNS 1033, 1034, 1045, AND 1083 TO SYSTEM CABINET I/F PANEL

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Attach Run 1034 to J17*. (use 1 inch connector)
2. Attach Run 1045 to J13*. (use 1 inch connector)
3. Attach Run 1033 to J16*. (use 3/4 inch connector)
4. Attach Run 1083 to J22*. (use 3/4 inch connector)



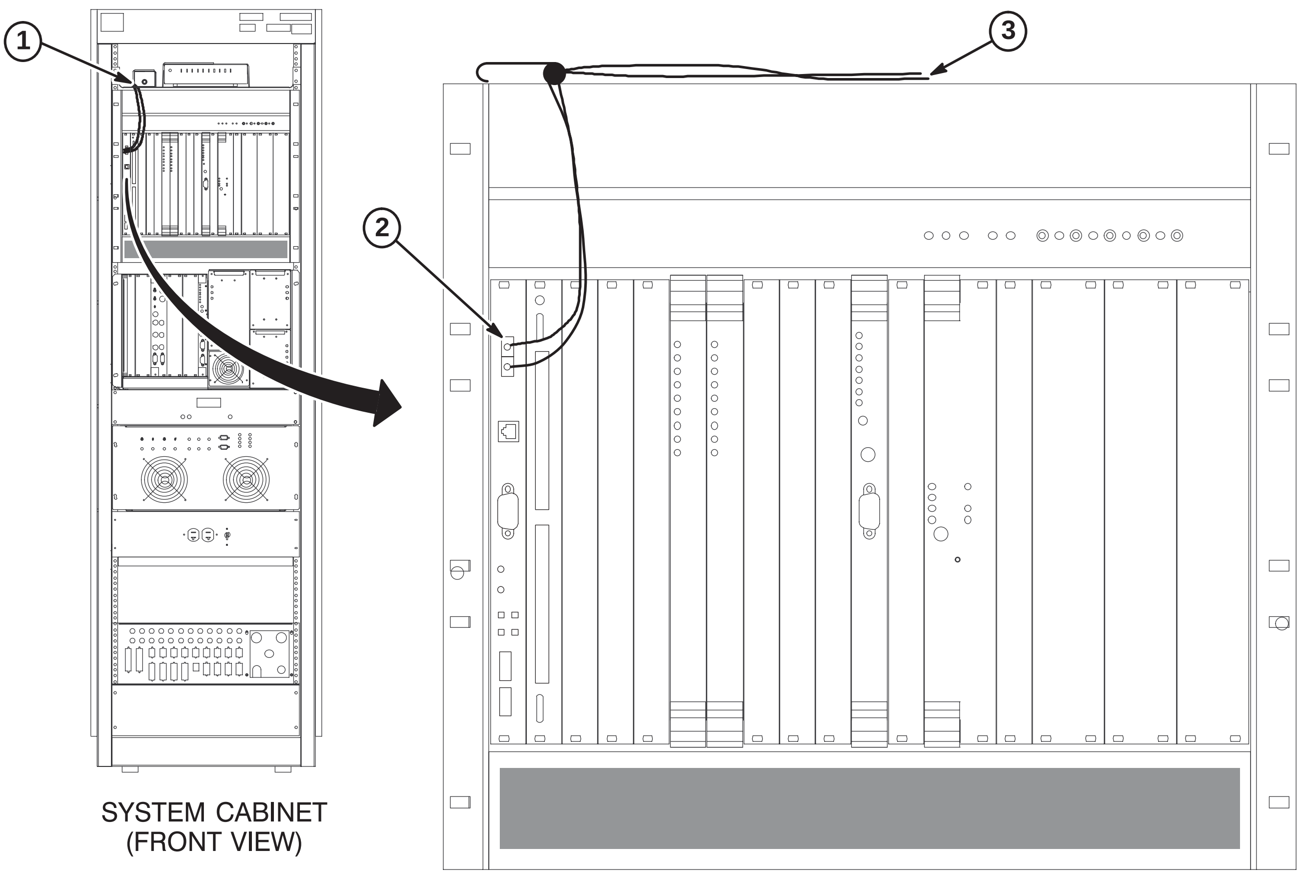
* Instructions provided with connectors.

1–4 ROUTE/CONNECT FIBER OPTIC RUN 1083 TO APS BOARD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route Run 1083 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**
2. Connect the Orange and Blue fiber optic cables to the TI–MGD Daughter Board as indicated.
3. Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.



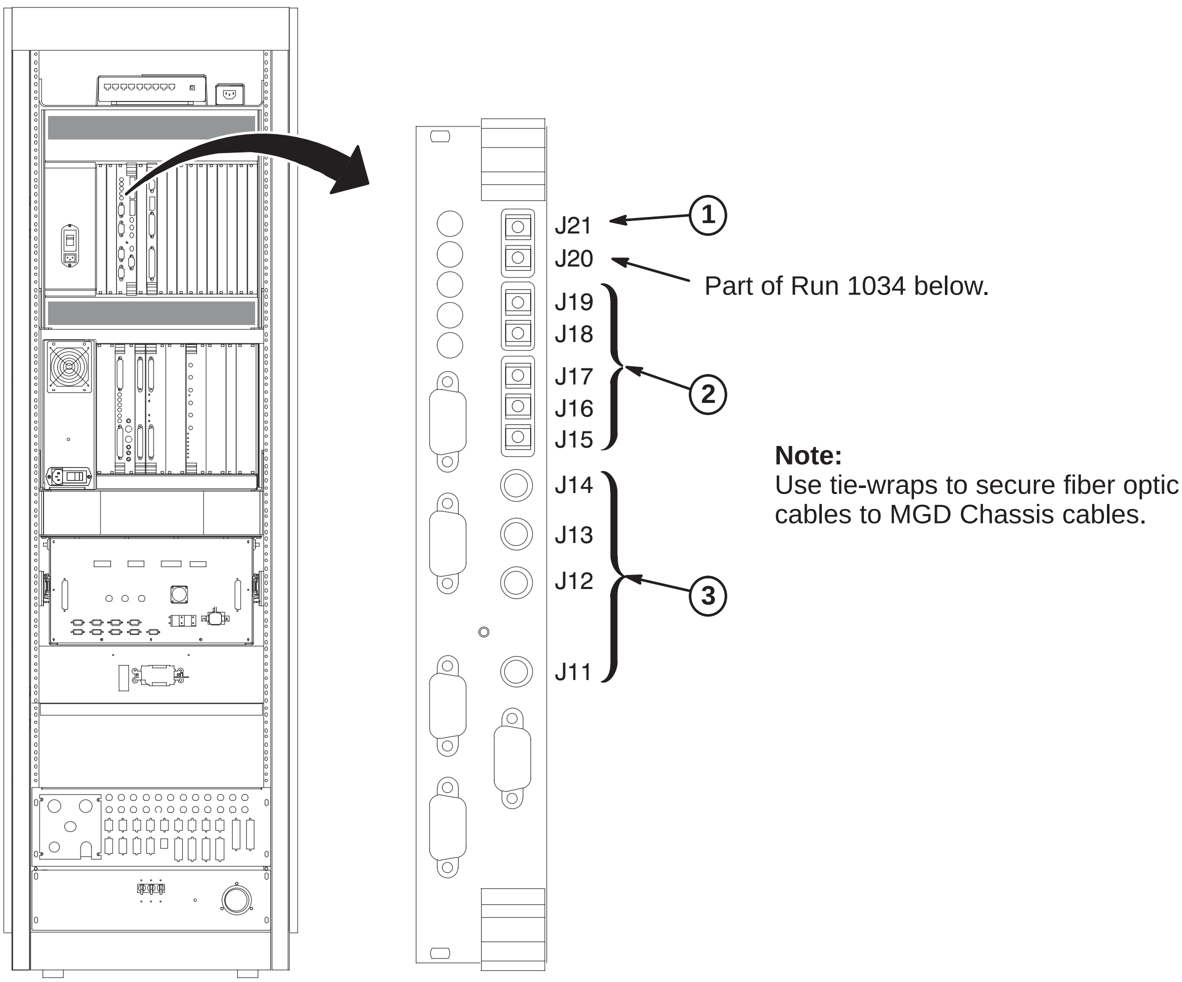
SYSTEM CABINET
(FRONT VIEW)

1–5 ROUTE/CONNECT FIBER OPTIC RUNS1033, 1034, AND 1045 TO STIF BOARD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

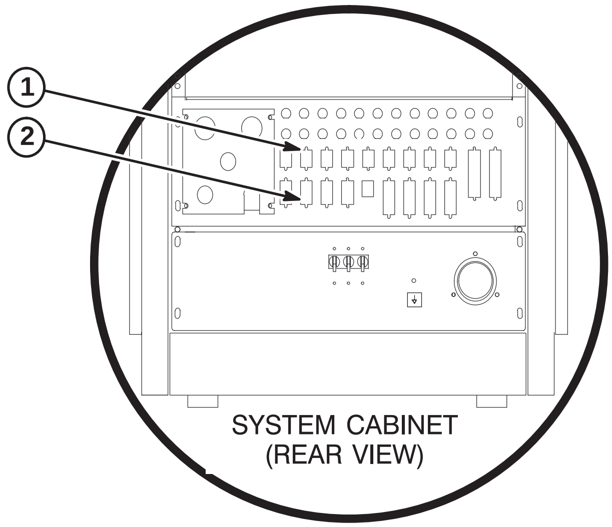
1. Route/Connect Run 1033 fiber optic cable to J21 on the STIF Board.
2. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
3. Route/Connect Run 1034 fiber optic cables to J11, J12, J13, J14, and J20 above on the STIF Board.



SYSTEM CABINET
(REAR VIEW)

1–6 CONNECT COMMUNICATION CABLES TO SYSTEM CABINET I/F

1. Connect RF Door Switch cable Run 701 to J15
2. Connect Magnet Monitor cable Run 823 to J24
3. Connect remaining cables as marked. Refer to cable map in Cable section of Overview tab.



ACCESSORIES AND OPTIONS INSTALLATION INSTRUCTIONS

1–1 Tab 11, Accessories & Options, contains installation instructions for the following:

LCC Wide-Open Magnet Enclosures

- *DIRECTION 2123158* – Patient Transport Installation Instructions
- *DIRECTION 2231762* – Signa WideOpen Pneumatic Patient Alert Installation

PATIENT TRANSPORT INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

- ❑ **A – Patient Transport Installation**
- ❑ **B – Head Coil Holder**
- ❑ **C – Head Coil Adapters**

❑ A – PATIENT TRANSPORT INSTALLATION

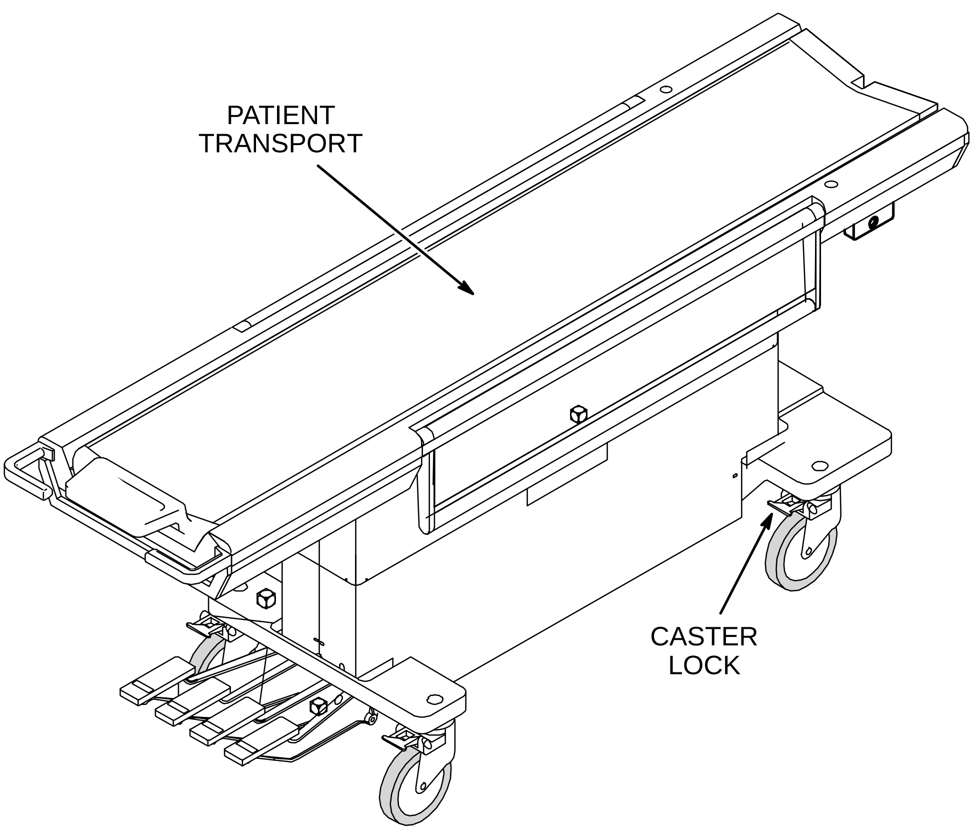
Patient Transport is shipped covered with protective cardboard which should remain in place until ready to be moved into the magnet room. Cover table with plastic previously removed when checks and adjustments are completed.

1. Unlock casters and move Patient Transport to magnet room for adjustments and function checks at the dock.
2. Remove protective plastic sheet from table.
3. Perform mechanial procedures in MR CD–ROM, Set up and Calibration: Patient Transport, LIGHTWEIGHT PATIENT TRANSPORT.

Note

If hardware fastening procedures for installation of the dock and support brackets were followed, the hardware used to fasten the dock and dock supports was left loose. The loose hardware permits a quick and easy dock centering adjustment with table docked.

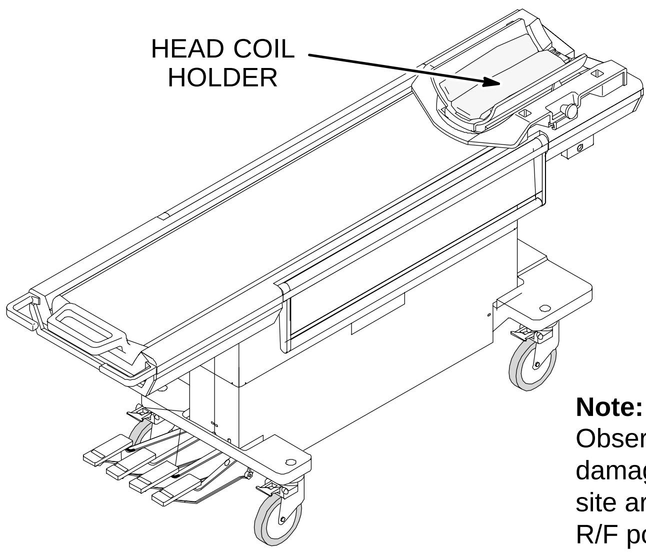
4. If additional Patient Transports are supplied, refer to *Direction 15477, Second Patient Transport Option Installation*, (delivered with M1020BD option).



PATIENT TRANSPORT INSTALLATION INSTRUCTIONS

❑ B – HEAD COIL HOLDER

1. Unpack Head Coil Holder and install Head Coil Holder on Patient Transport cradle.
2. With Patient Transport docked, check mechanical operation of Head Coil with holder.



Note:
Observe warning Label on head coil. To prevent head coil damage, be sure that all service personnel and users at the site are aware that the head coil must be removed before R/F power is applied to the body coil. This warning also applies to service procedures.

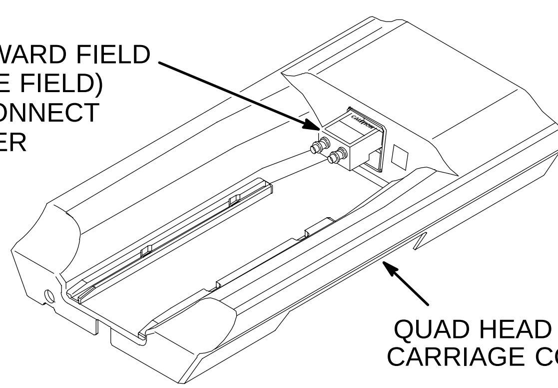
❑ C – HEAD COIL ADAPTERS

1. Unpack Head Coil, Coil Adapters (Head Coil Forward Field, Extremity Coil , Surface Coil), and adapter cables.
2. Install Head Coil Adapter (Head Coil Forward or Reverse Field) into upper coil adapter port on carriage cover.

Note

If magnet is configured with reverse polarity, “reverse field” Head and Surface Coil Adapters are required to be ordered. The “forward field” Coil Adapters must be returned. Since an unaware operator could use the wrong adapter, do not leave both adapters on site.

3. Remove head coil cables from bag attached to Head Coil. Place Head Coil on carriage cover and connect cables to verify connections. Connect blue cable to connector indicated with blue.
3. Set Head Coil with cables, holder, and adapters aside for safe storage (outside of magnet room) until needed during set–up calibrations, functional checks, and system tests.



INSTALLATION STEPS SUMMARY

- ❑ **A1 – Preinstallation Planning**
- ❑ **A2 – Squeeze Bulb Hanger Installation**
- ❑ **A3 – Squeeze Bulb Installation**
- ❑ **A4 – Magnet Room Routing of Tubing & Wave Guide Installation**
- ❑ **A5 – Determining Need of Extender Box**

❑ A1 – PREINSTALLATION PLANNING

The customer and users should be involved in the following decisions:

Routing of Pneumatic Tubing:

Location of Control Box

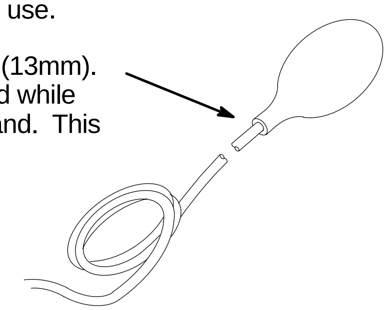
CAUTION

Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

❑ A2 – SQUEEZE BULB INSTALLATION

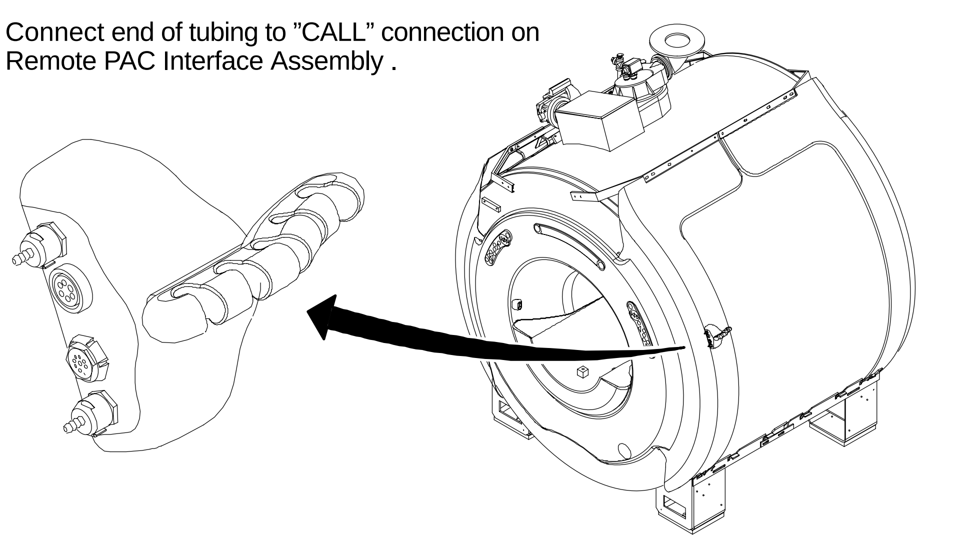
(2) Before cutting pneumatic tubing from roll, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the “CALL” connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.

(1) Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.



❑ A3 – CONNECT TUBE TO REMOTE PAC INTERFACE ASSEMBLY

(1) Connect end of tubing to ”CALL” connection on Remote PAC Interface Assembly .



❑ A4 – MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION

CAUTION

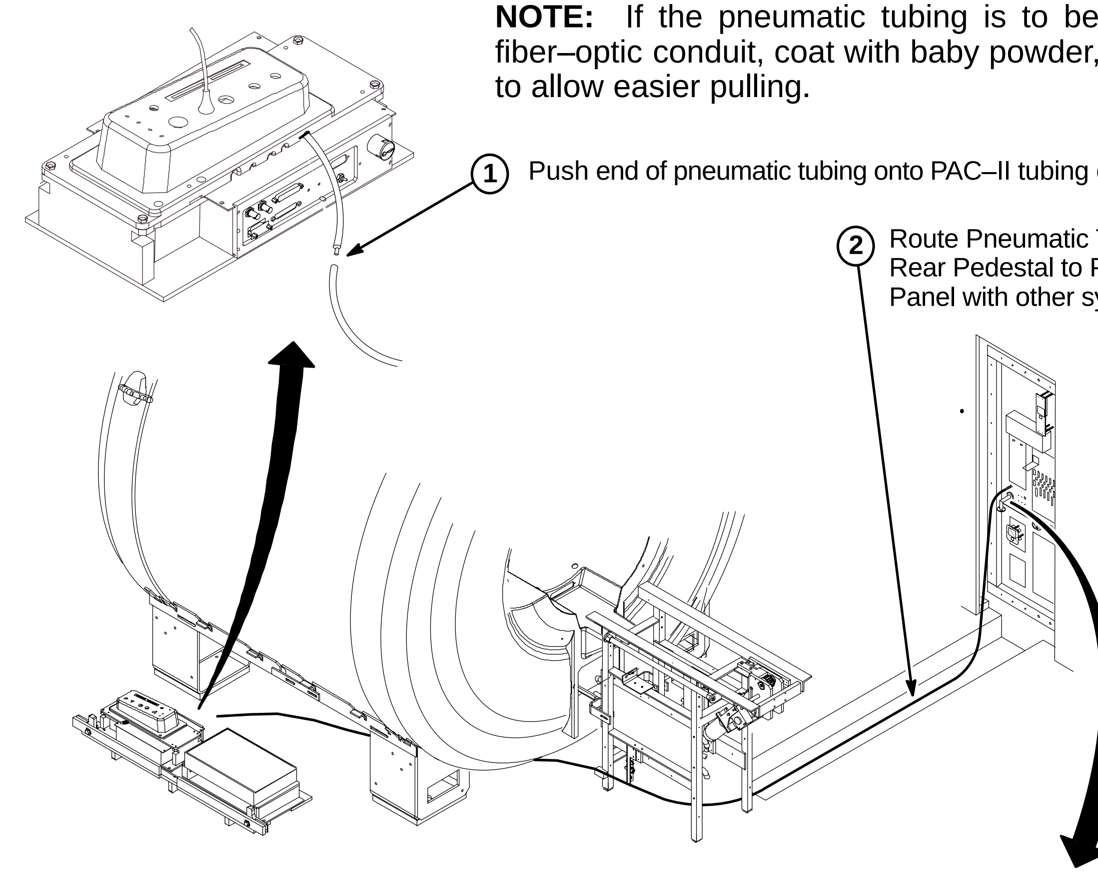
Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

NOTE: Also, make sure that the pneumatic tubing can be routed from the PAC connector through the Rear Pedestal, and to the Penetration panel without being pinched along the route by other cables.

NOTE: If the pneumatic tubing is to be pulled thru fiber–optic conduit, coat with baby powder, not grease, to allow easier pulling.

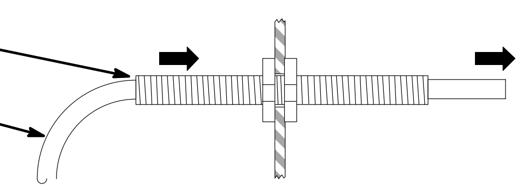
(1) Push end of pneumatic tubing onto PAC–II tubing connector.

(2) Route Pneumatic Tubing from Rear Pedestal to Penetration Panel with other system cables.



(3) Insert Wave Guide thru “ALERT AIR LINE” hole and fasten to Penetration Panel with nut on each side.

(4) Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide into the Equipment Room where the Control Box will be installed.



❑ A5 – DETERMINING NEED OF EXTENDER BOX

Note: If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator’s Console), Extender Kit, 46–317758P2, must be ordered.

If the pneumatic tubing is long enough to reach control box:
The extender is not needed. Proceed to Step **B3**, CONTROL BOX INSTALLATION.

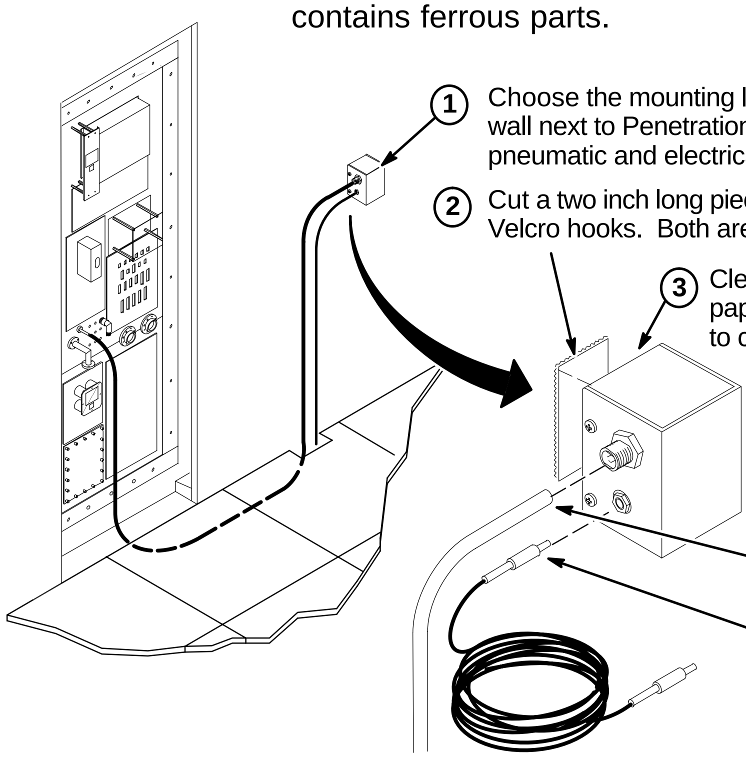
If the pneumatic tubing is not long enough to reach control box:
The extender is needed. Proceed to Step **B1**, INSTALLATION OF EXTENDER KIT.

INSTALLATION STEPS SUMMARY

- ❑ **B1 – Installation of Extender Kit (46–317758P2)**
- ❑ **B2 – Connection of Extender Wire to Control Box**
- ❑ **B3 – Control Box Installation**
- ❑ **B4 – Control Box Installation on Wall or Under Shelf**
- ❑ **B5 – Control Box Mounted with Velcro**

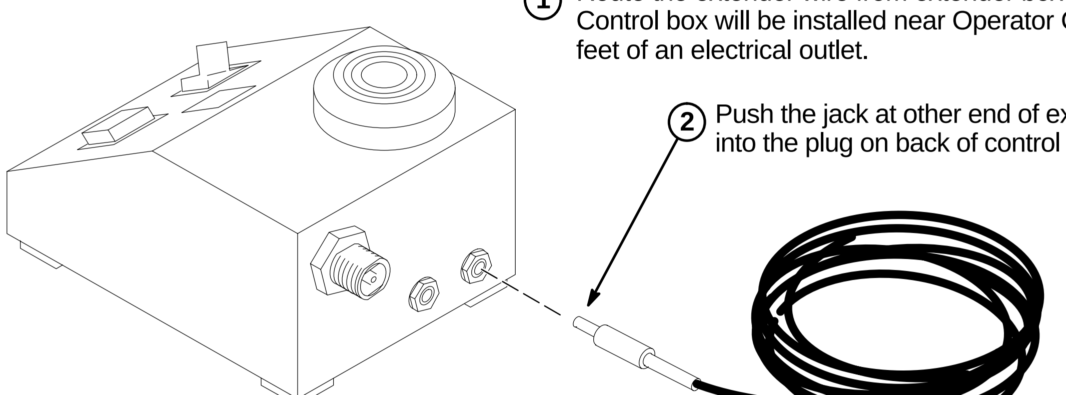
❑ B1 – INSTALLATION OF EXTENDER KIT (46–317758P2)

Note: Do not mount Extender Box in Magnet Room because the box contains ferrous parts.



1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed–thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.

❑ B2 – CONNECTION OF EXTENDER WIRE TO CONTROL BOX



1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.

❑ B3 – CONTROL BOX INSTALLATION

Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

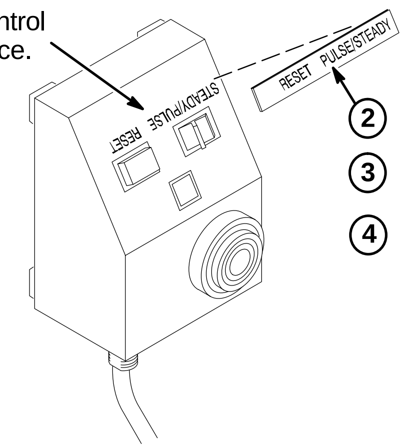
Mount control box on a wall or other vertical surface per Step **B4**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.

Mount control box under shelf per Step **B4**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.

Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed within sight of operator. Proceed to Step **C5**, FINAL CONNECTIONS TO CONTROL BOX.

Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

❑ B4 – CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF

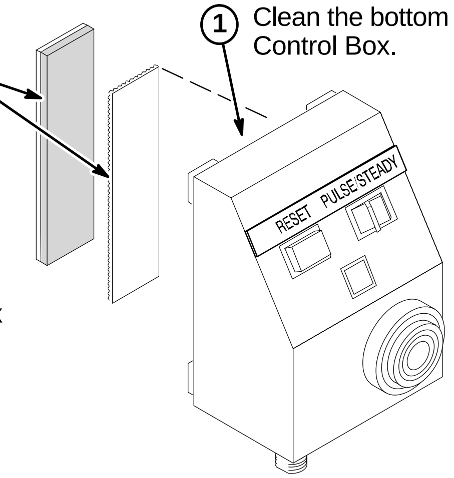


1. Clean Control Box surface.
2. Remove adhesive backing from metalized nameplate.
3. Apply nameplate over silk screened lettering as shown.
4. **If control box will be mounted to WALL or SHELF with Velcro:** Proceed to Step **B5**.

   If control box will be mounted on WALL with screws: Proceed to Step **C1**.

   If control box will be mounted under SHELF with screws: Proceed to Step **C3**.

❑ B5 – CONTROL BOX MOUNTED WITH VELCRO



1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted within sight of operator.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.
7. Mount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.

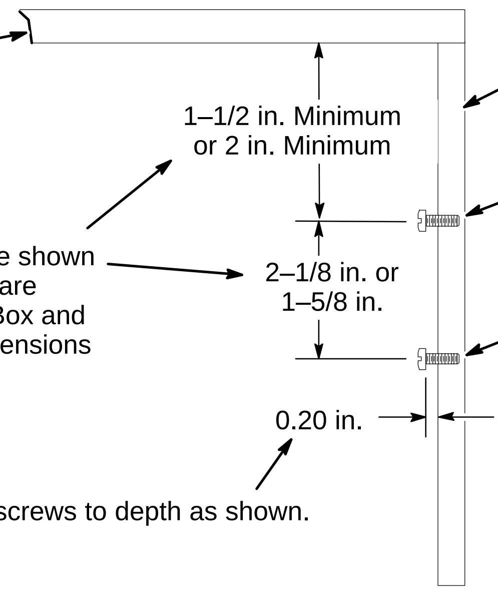
INSTALLATION STEPS SUMMARY

- ❑ **C1 – Installing Mounting Screws in Vertical Surface**
- ❑ **C2 – Installing Control Box on Vertical Surface**
- ❑ **C3 – Installing Mounting Screws Under Shelf**
- ❑ **C4 – Installing Control Box Under Shelf**
- ❑ **C5 – Final Connections to Control Box**

❑ C1 – INSTALLING MOUNTING SCREWS IN VERTICAL SURFACE

① Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted <u>within sight of operator</u>.

Ceiling or Shelf Bottom



② Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.

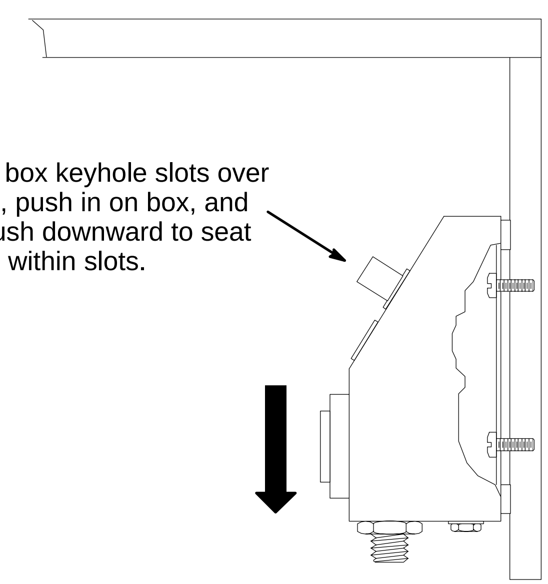
③ Mark top mounting hole position on wall.

④ Mark bottom mounting hole position on wall.

⑤ Drill a 1/16 in. pilot hole at locations if furnished screws are used.

⑥ Install screws to depth as shown.

❑ C2 – INSTALLING CONTROL BOX ON VERTICAL SURFACE



① Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.

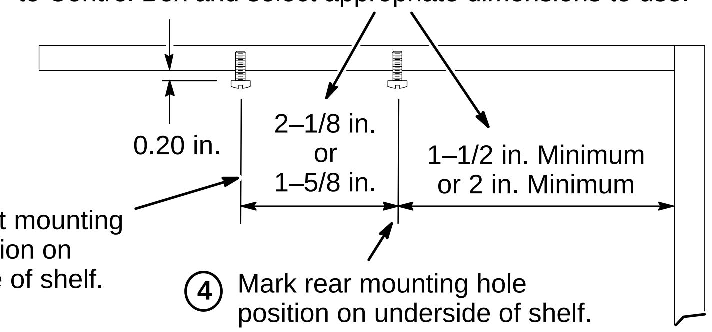
② Tighten or loosen screws as necessary to obtain a snug fit of box against wall.

③ Proceed to Step **C5**, FINAL CONNECTIONS TO CONTROL BOX.

❑ C3 – INSTALLING MOUNTING SCREWS UNDER SHELF

① Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted <u>within sight of operator</u>.

② Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.

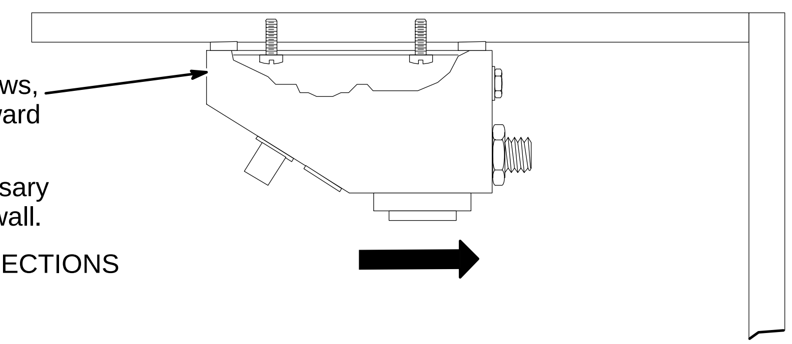


③ Mark front mounting hole position on underside of shelf.

④ Mark rear mounting hole position on underside of shelf.

⑤ Drill a 1/16 in. pilot hole at locations if furnished screws are used.

❑ C4 – INSTALLING CONTROL BOX UNDER SHELF



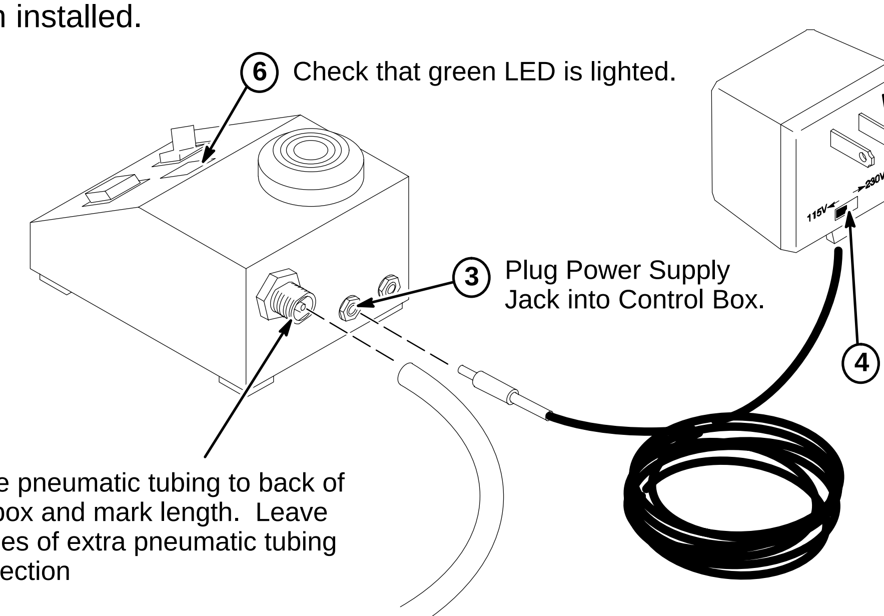
① Center box keyhole slots over screws, push up on box, and then push inward to seat screws within slots.

② Tighten or loosen screws as necessary to obtain a snug fit of box against wall.

③ Proceed to Step **C5**, FINAL CONNECTIONS TO CONTROL BOX.

❑ C5 – FINAL CONNECTIONS TO CONTROL BOX

Note: Steps 1 and 2 are not needed if Extender Box (46–317758P2) has been installed.



① Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection

② Cut tubing and push fully onto feed–thru connector on back of control box.

③ Plug Power Supply Jack into Control Box.

④ Set Power Supply to outlet voltage (115VAC or 230VAC).

⑤ Plug Power Supply into AC outlet.

⑥ Check that green LED is lighted.

OPERATOR WORKSPACE INSTALLATION INSTRUCTIONS

1–1 Tab 13, Operator Workspace, contains installation instructions for the following:

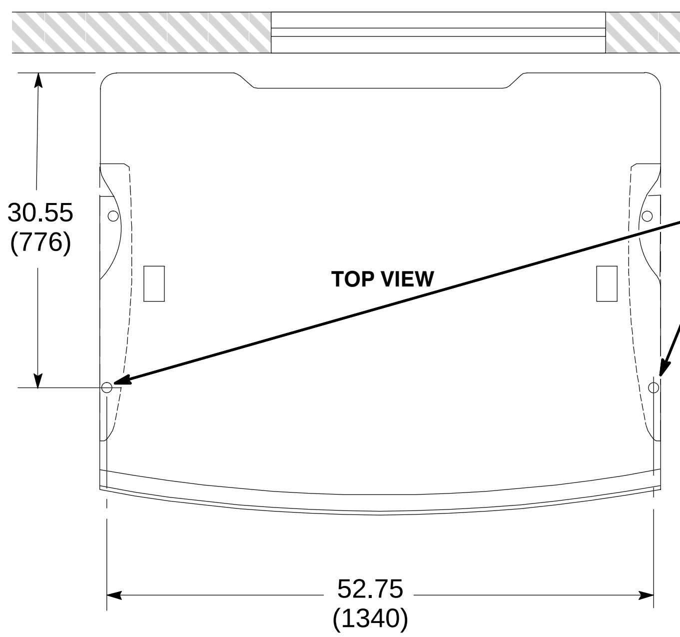
- *DIRECTION 2210982* – Signa Octane Operator Workspace Installation Instructions
- *DIRECTION 2386129* – Signa Linux Operator Workspace Installation Instructions
- *DIRECTION 5234966* – Signa EXCITE Operator Workspace Installation Instructions

INSTALLATION STEPS SUMMARY

- ❑ **A1 – Operator Workspace Table Anchor Installation**
- ❑ **A2 – Securing Operator Workspace Table to Floor**
- ❑ **A3 – Attach Workspace Interface Module (2141978 series) to Table**
- ❑ **A4 – Attach Power Distribution Box (46–307830G1) to Table**

Note: The CERD Spike Noise Test Cable (2168505) is part of the 8.x Octane Operator Workspace Cable collector (2154392–2). Remove this cable from the collector and place it in the TPS RF Service Interface Kit (46–301927G1).

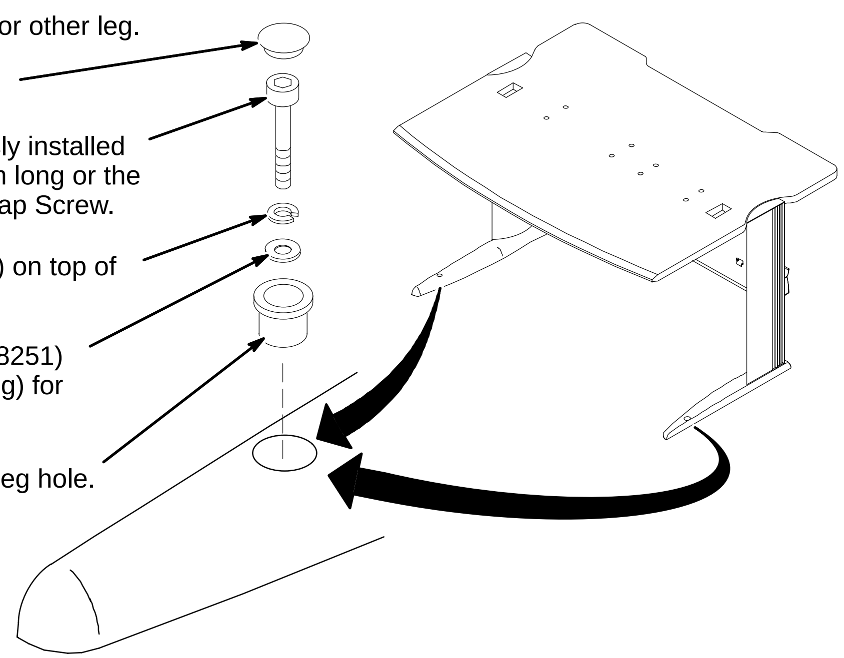
❑ A1 – OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION



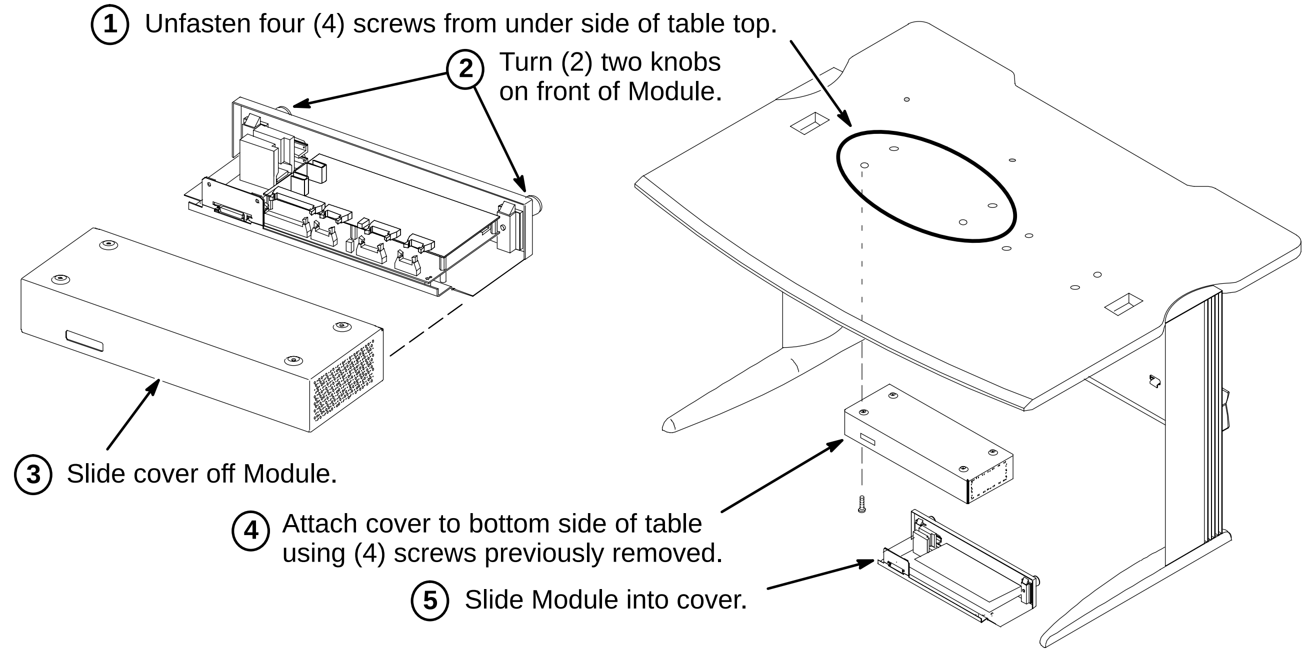
1. Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table (2135998–2).
2. Remove and discard the two (2) cover plugs. Locate and mark on floor the two hole locations at the front of the table legs.
3. Move table to prepare for installing the floor anchors in Steps 4 and 5.
4. At each hole marked in Step 2, drill a .38 inch hole and insert 1/4–20 x .375 diameter anchor (2168253).
5. Secure anchor in place with supplied Setting Tool (2168781).

❑ A2 – SECURING OPERATOR WORKSPACE TABLE TO FLOOR

6. Repeat anchoring process below for other leg.
5. Install black nylon Plug (2168254).
4. Depending upon depth of previously installed anchor, install either the 1–3/4 inch long or the 2 inch long 1/4 x 20 Hex Socket Cap Screw.
3. Insert one Lock Washer (2168252) on top of Flat Washer(s).
2. Insert as many Flat Washers (2168251) as required (3 provided for each leg) for securing table to floor.
1. Insert Bushing (2151454) in table leg hole.

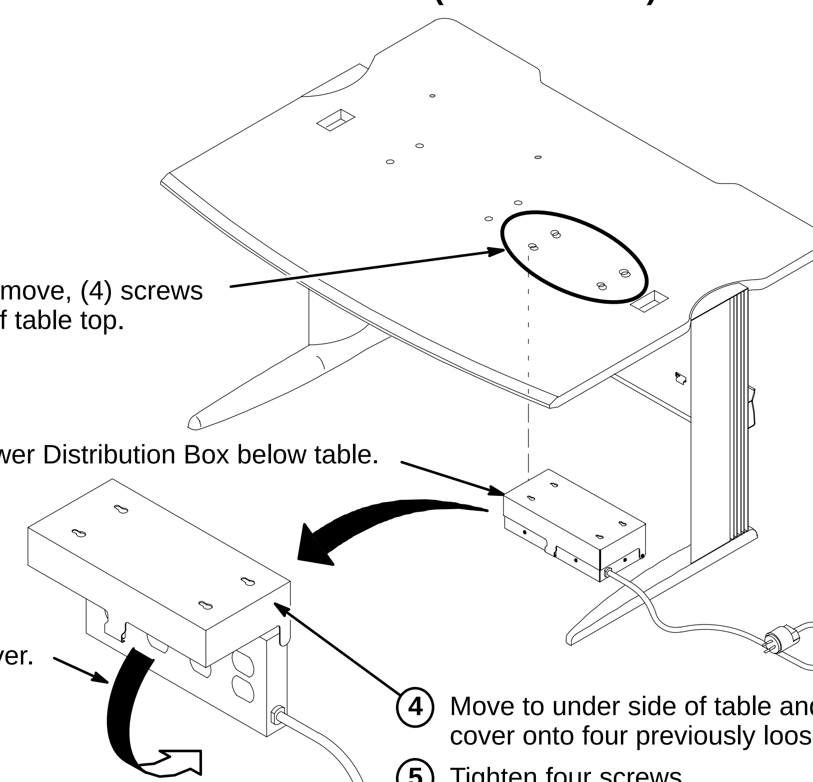


❑ A3 – ATTACH WORKSPACE INTERFACE MODULE (2141978 series) TO TABLE



1. Unfasten four (4) screws from under side of table top.
2. Turn (2) two knobs on front of Module.
3. Slide cover off Module.
4. Attach cover to bottom side of table using (4) screws previously removed.
5. Slide Module into cover.

❑ A4 – ATTACH POWER DISTRIBUTION BOX (46–307830G1) TO TABLE

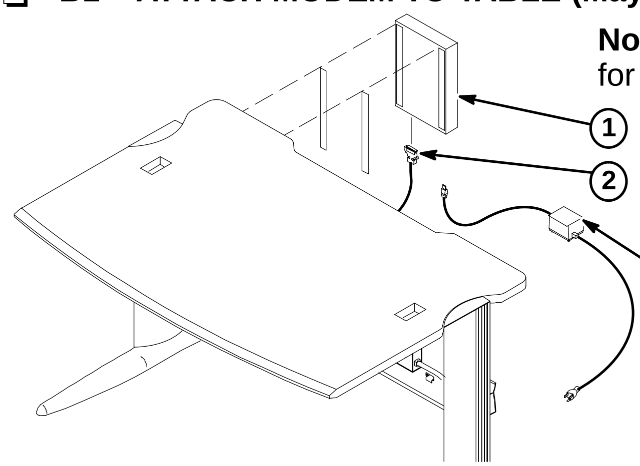


1. Loosen, do not remove, (4) screws from under side of table top.
2. Position Power Distribution Box below table.
3. Swing Box down from mounting cover.
4. Move to under side of table and slide mounting cover onto four previously loosened screws.
5. Tighten four screws.

INSTALLATION STEPS SUMMARY

- ❑ **B1 – Attach Modem to Table (May be replaced by ethernet connection)**
- ❑ **B2 – Attach DASM to Table**
- ❑ **B3 – Attach Serial Expansion Box Asm to Underside of Table Top**
- ❑ **B4 – Attach Serial Coverter to Underside of Table Top (MGD & Open*Speed* only)**

❑ B1 – ATTACH MODEM TO TABLE (May be replaced by ethernet connection)



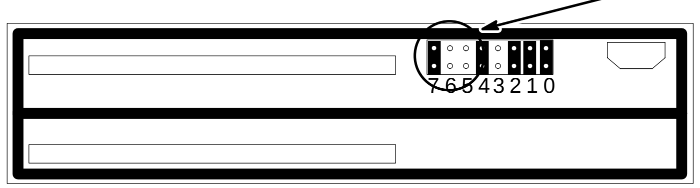
Note: Modem may be replaced by Ethernet cable for connection to InSite (See Procedure **E4**).

1. Attach to back plate of table.
2. Connect Run 812 (2188746) to Modem. Other end to be connected to Operator Workspace Cabinet SER 1 later.
3. Modem Power Adapter to be supplied by District office and connected to Power Distribution Box J7.

❑ B2 – ATTACH DASM TO TABLE

Note: The Analog and Digital DASMs are shipped with the jumper set to SCSI ID=0. This is the correct ID for an AW, but incorrect for Signa Horizon LX (SCSI ID=3).

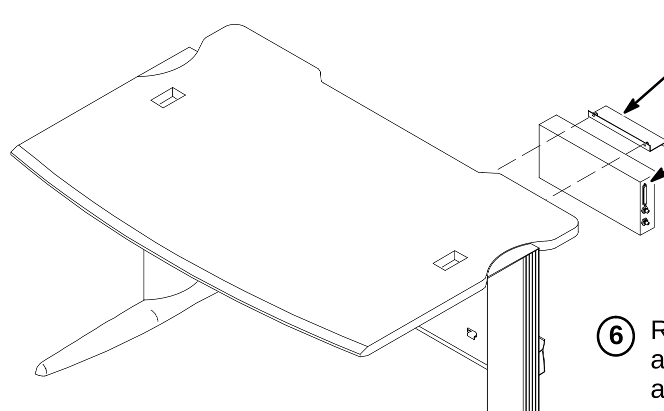
1. Unfasten DASM cover screw and remove cover.



2. For SCSI ID = 3, set Configuration Jumper J2 as follows:
   5=OFF, 6=OFF, 7=ON
   Note: A jumper in the <u>OFF</u> position <u>enables</u> the SCSI ID bit.
3. Re-attach DASM cover.

Note: Consult with Site customer for preference of DASM location. This procedure installs DASM in the cable trough. Customer may prefer DASM on top of table.

CAUTION: Longer SCSI cables cannot be substituted.

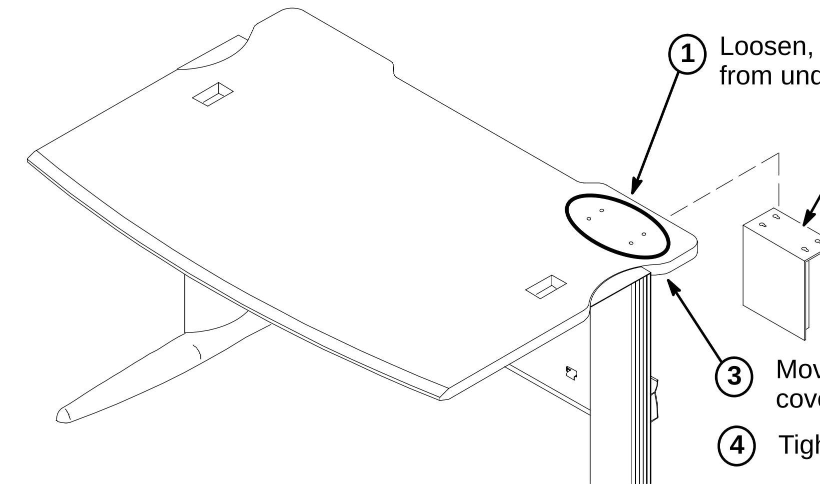


4. Unfasten DASM Bracket from back plate. Use only for 46–269566P3 or 2143388–2. Not used for 2251040 or 2251043.
5. Install DASM (Analog, 46–269566P3, or Digital, 2143388–2) in cable trough and fasten in place with DASM Bracket.

   OR

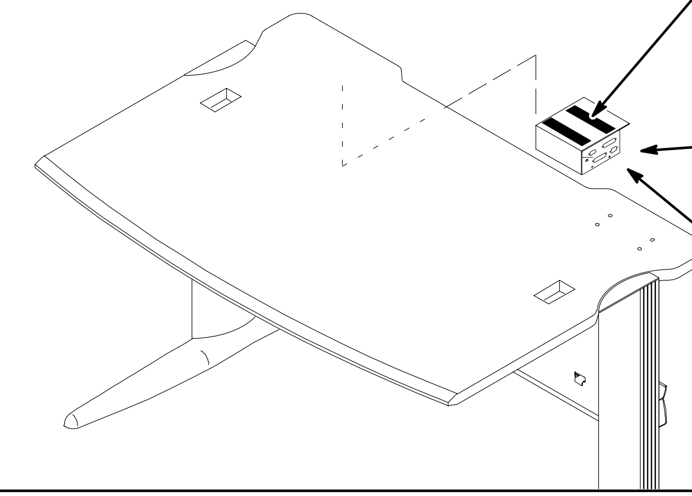
   Install DASM (Analog, 2251040, or Digital, 2251043) in cable trough. Refer to installation instructions that are shipped with DASM.
6. Route and connect DASM cables and accessories according to instructions provided with DASM Module and Cable Map in Overview Tab of this manual.
7. Connect power cable to Power Distribution Box J2.

❑ B3 – ATTACH SERIAL EXPANSION BOX ASM TO UNDERSIDE OF TABLE TOP



1. Loosen, do not remove, (4) screws from under side of table top.
2. Position Serial Expansion Box Asm (2204796) below table.
3. Move to under side of table and slide mounting cover onto four previously loosened screws.
4. Tighten four screws.

❑ B4 – ATTACH SERIAL CONVERTER TO BOTTOM OF TABLE (MGD & Open*Speed*)



1. Cut supplied velcro strip into two pieces and attach to module.
2. Attach Serial Converter Module to underside of table top.

3A. **FOR Open*Speed*:** Connect to module Runs 837 (J4), 838 (J3), 910 (J2), and 911 (J1).

OR

3B. **FOR MGD: If provided, c**onnect to module Runs 838 (J3) and 911 (J1). (future feature)

4. Connect Power cable to Power Distribution Box J8.

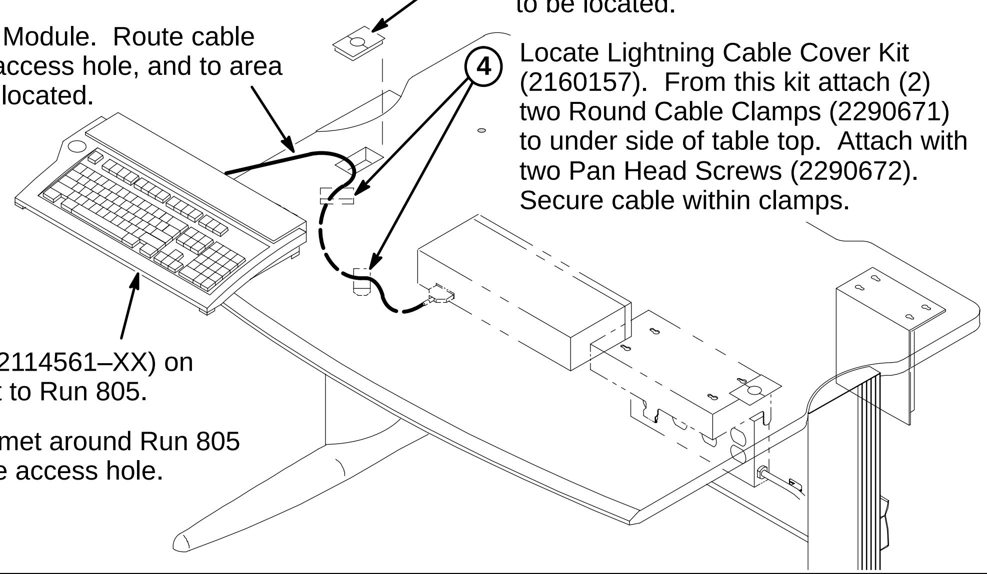
INSTALLATION STEPS SUMMARY

- ❑ **C1 – Install Keyboard on Table**
  OR
- ❑ **C2 – Install SCIM/Keyboard Data Cable on Table**
- ❑ **C3 – Install SCIM/Keyboard on Table**
- ❑ **C4 – Install Flat Screen Color Monitor on Table**

NOTE:
Consult with Site customer for exact location of Color Monitor on Operator Workspace Table.

❑ C1 – INSTALL KEYBOARD ON TABLE

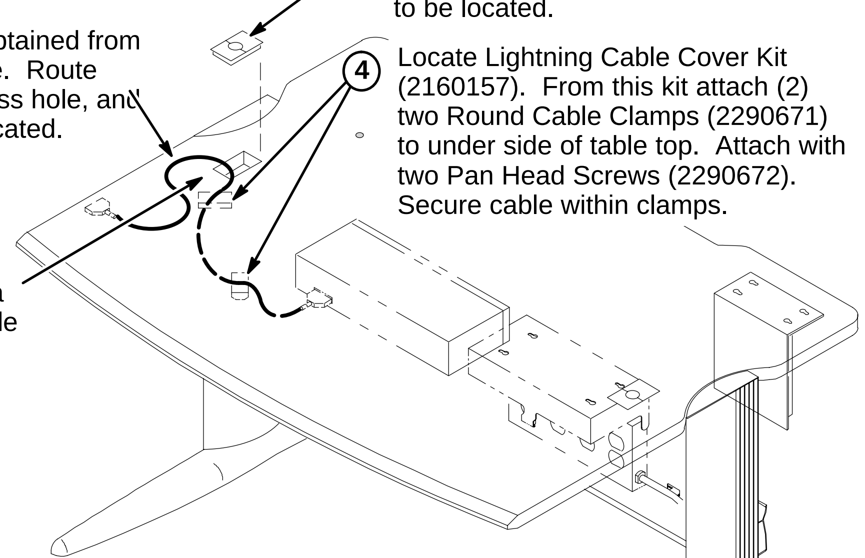
1. Consult with Site customer for exact location of Keyboard and Color Monitor on Operator Workspace (OW) Table.
2. Remove cable grommet from side of table where Monitor and Keyboard are to be located.
3. Connect Run 805 to I/F Module. Route cable under table, thru cable access hole, and to area where Keyboard will be located.
4. Locate Lightning Cable Cover Kit (2160157). From this kit attach (2) two Round Cable Clamps (2290671) to under side of table top. Attach with two Pan Head Screws (2290672). Secure cable within clamps.
5. Place Keyboard (2114561–XX) on table and connect to Run 805.
6. Place cable grommet around Run 805 and insert in cable access hole.



OR

❑ C2 – INSTALL SCIM/KEYBOARD DATA CABLE ON TABLE

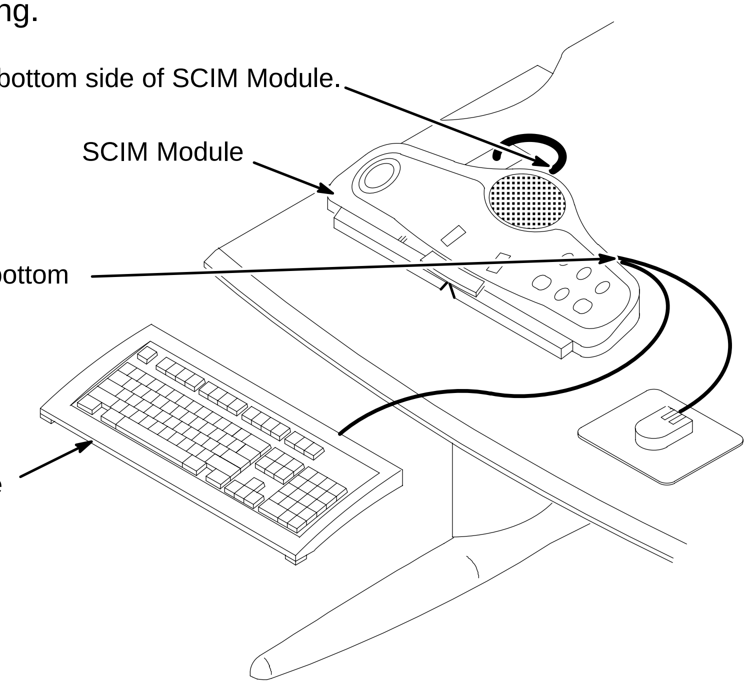
1. Consult with Site customer for exact location of Keyboard and Color Monitor on Operator Workspace (OW) Table.
2. Remove cable grommet from side of table where Monitor and Keyboard are to be located.
3. Connect Data Cable (Run 805), obtained from SCIM Kit (2307175), to I/F Module. Route cable under table, thru cable access hole, and to area where Keyboard will be located.
4. Locate Lightning Cable Cover Kit (2160157). From this kit attach (2) two Round Cable Clamps (2290671) to under side of table top. Attach with two Pan Head Screws (2290672). Secure cable within clamps.
5. Place cable grommet around Data Cable (Run 805) and insert in cable access hole.



❑ C3 – INSTALL SCIM/KEYBOARD ON TABLE

NOTE: Refer to Installation document OW1INA3.DOC on Service Methods CD–ROM or MR Service Engineering web site for installation information on language labels and keyboard mounting.

1. Connect Data Cable to bottom side of SCIM Module.
2. Connect Keyboard and Mouse cables to bottom side of SCIM Module as marked.
3. Place Keyboard on table in front of SCIM Module.

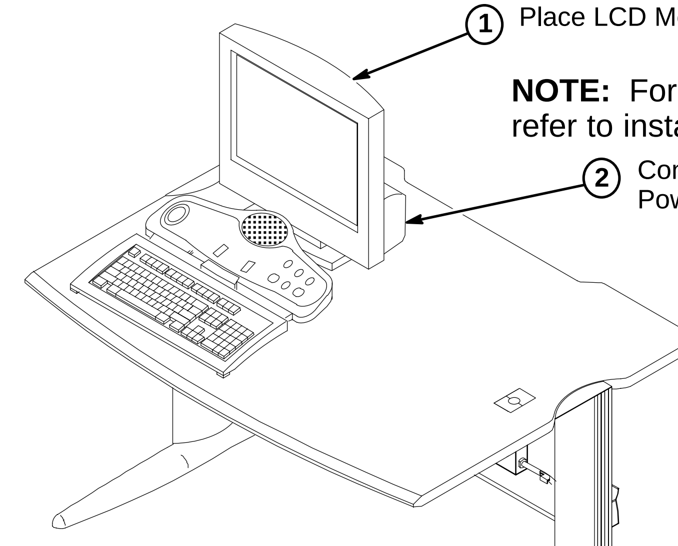


❑ C4 – INSTALL FLAT SCREEN COLOR MONITOR ON TABLE

1. Place LCD Monitor on table in back of SCIM.

NOTE: For fastening monitor to surface, refer to installation drawing 2204493DDW.

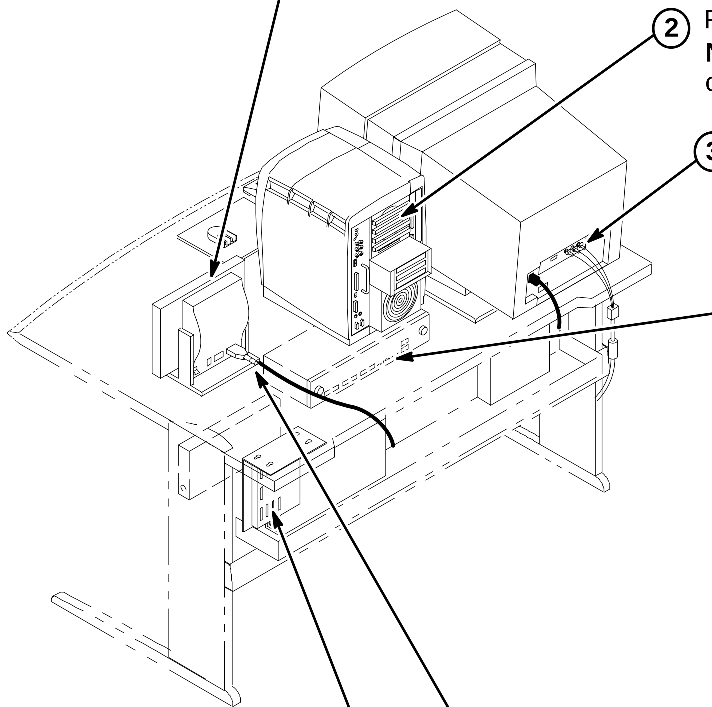
2. Connect power cable to Power Distribution Box J1.



INSTALLATION STEPS SUMMARY

- ❑ **D1 – Install LCD Display on Table and Position Octane Computer**
- ❑ **D2 – Connect Table Component Cables**
- ❑ **D3 – Serial Expansion Box Set Up**
- ❑ **D4 – SGI Octane Computer Cables Connections**

❑ D2 – CONNECT TABLE COMPONENT CABLES



1. Place LCD Display (2138599) on table. Connect power cable to Power Distribution Box J5.
2. Place Octane Computer on table.
   NOTE: Refer to Procedure **D4** for Run connections to SGI Octane Computer.
3. **For Color Monitor:** Connect Run 798/821 (2150533–2) to R IN, B IN, G IN on Monitor and route/connect to MONITOR on the Octane Computer module.
4. Connect the following cables from the OW I/F Module (OW A1 A4) for later connection to Operator Workspace Cabinet:

RUN #	CONNECTION	PART #
805	J6	2114561–38
810	J8	2141980–2
814	J9	2196990
817	J17	46–328000G958

Connect the following cables to OW I/F Module. These cables will be routed to the System Cabinet and Penetration Panel:

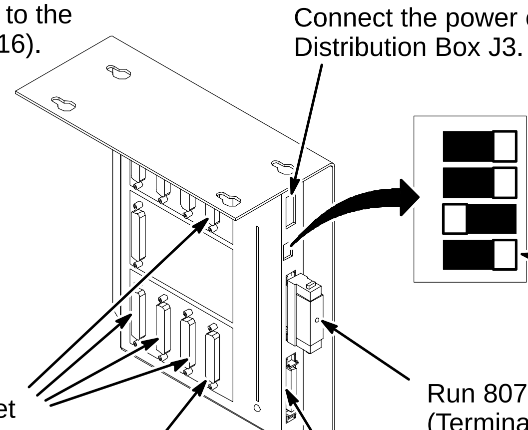
RUN #	CONNECTION	PART #
789	J18	46–317068G938
788	J7	46–328000G937

5. Connect Run 797 (2138599–2) to LCD Display. Other end is connected later to PC in the Operator Workspace Cabinet.

NOTE: Refer to Procedure **D3** for Run connections and DIP switch settings.

❑ D3 – SERIAL EXPANSION BOX SET UP FOR MGD (10.x)

1. Connect the following cables to the SCSI Expansion Box (OW A16).



Connect the power cord to Power Distribution Box J3.

2. Set SCSI ID switches on side of module as shown. (Addr = 4)

Termination Enabled

Run 1046 from System cabinet to Ports 1, 6, 7, and 8.

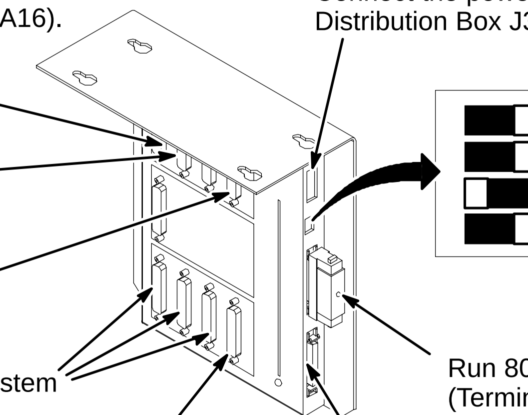
Run 807 (2163886) from DASM. (Terminator shown)

Run 911 from Serial Converter J1 to Port 5. (Future)

Run 815 (2198983) from Octane computer

❑ D3 – SERIAL EXPANSION BOX SET UP (9.x)

1. Connect the following cables to the SCSI Expansion Box (OW A16).



Connect the power cord to Power Distribution Box J3.

Run 1004 from TAC Cabinet J27 to Port 2. (Twin*Speed* option only)

Run 910 from Serial Converter J2 to Port 3. (Open*Speed* only)

Run 791 from System cabinet to Port 1.

Run 818 (2198993) from System cabinet to Ports 6, 7, and 8.

Run 911 from Serial Converter J1 to Port 5. (Open*Speed* only)

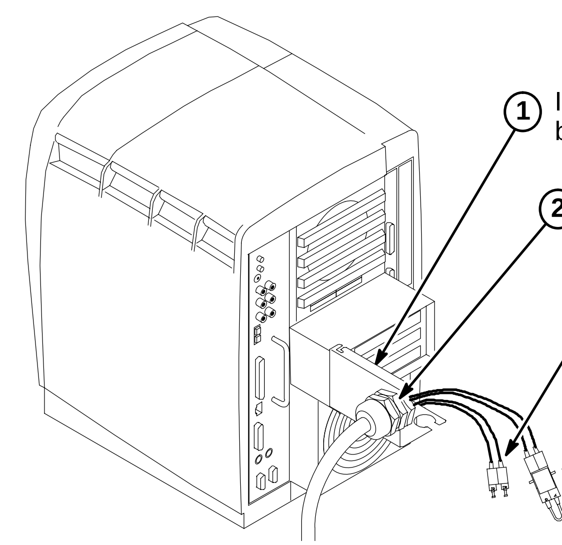
2. Set SCSI ID switches on side of module as shown. (Addr = 4)

Termination Enabled

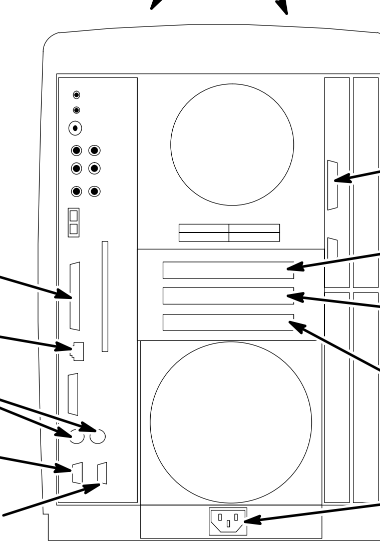
Run 807 (2163886) from DASM. (Terminator shown)

Run 815 (2198983) from Octane computer

❑ D4 – SGI OCTANE COMPUTER CABLES CONNECTIONS



1. Install fiber optic Strain Relief Bracket (2245994) to back of Octane Computer using existing screws.
2. Route and connect Bit3 Fiber Optic cable Run 836 (2241052) to bracket.
3. Connect Run 836 (or Run 1044, MGD) orange and blue fiber optic cables as indicated in Step 5 below.
4. Spare Fiber Optic cable and Loopback Assembly (2241052–6).
5. Connect the following cables to the SGI Octane Computer (OW A15) as marked.

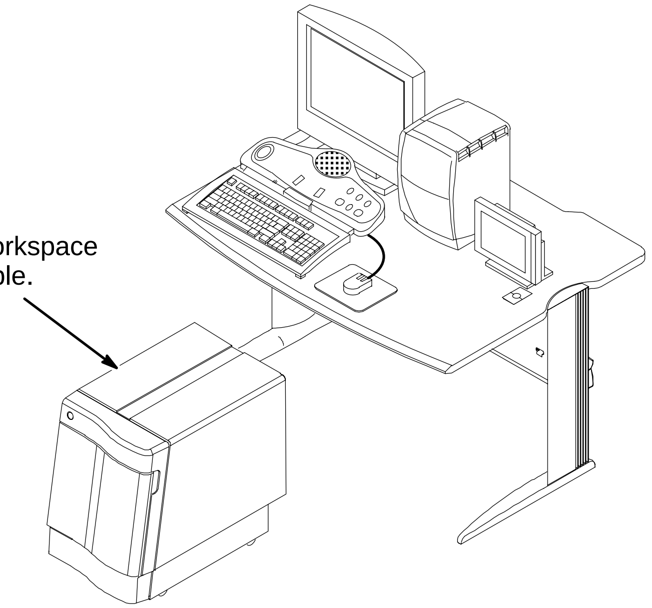


INSTALLATION STEPS SUMMARY

- ❑ **E1 – Locate Operator Workspace Cabinet in Front of Table**
- ❑ **E2 – Remove Operator Workspace Cabinet Rear Cover**
- ❑ **E3 – Remove Operator Workspace Cabinet Front And Side Covers**
- ❑ **E4 – Connect Ethernet Cables**
- ❑ **E5 – Route/Connect Runs 792, 797, 813, and 814 to PC Module**

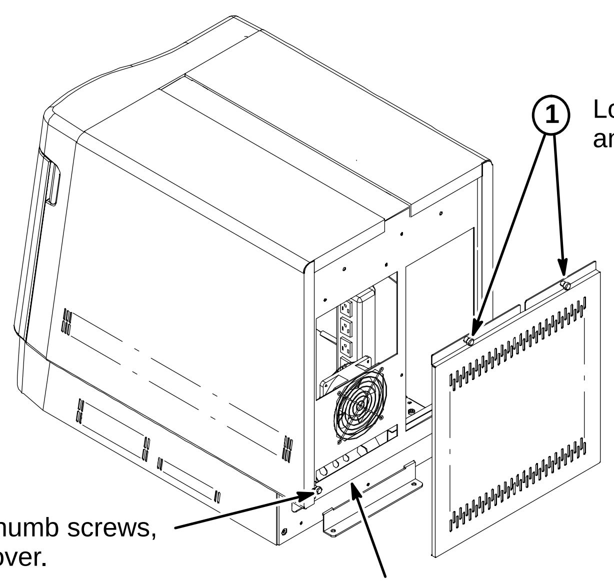
❑ E1 – LOCATE OPERATOR WORKSPACE CABINET IN FRONT OF TABLE

1. Position Operator Workspace Cabinet in front of table.



❑ E2 – REMOVE OPERATOR WORKSPACE CABINET REAR COVER

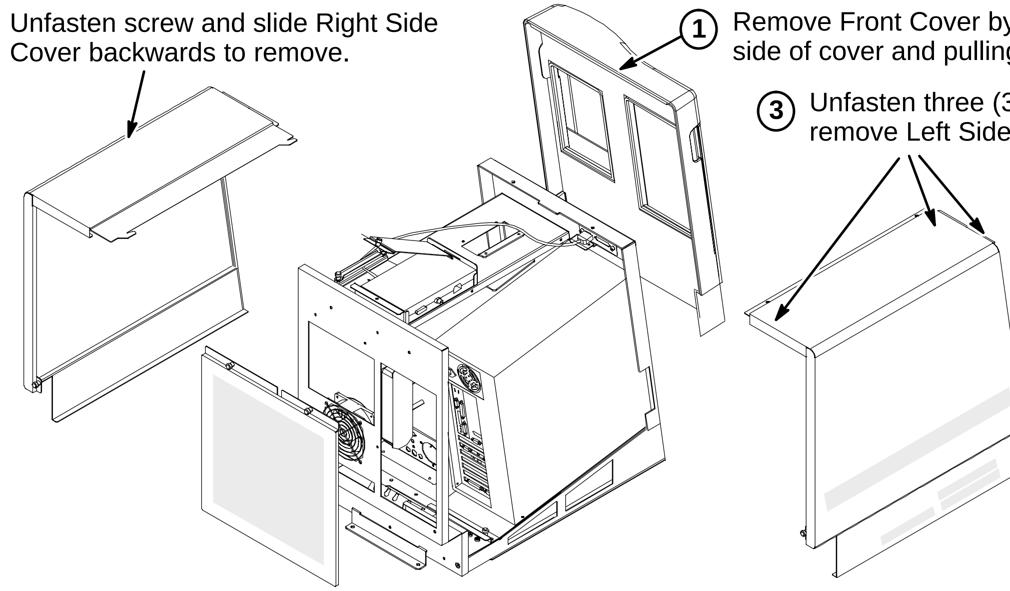
1. Loosen (2) two thumb screws and remove Rear Cover.
2. Loosen the (2) two thumb screws, one on each Side Cover.



Cable Access Hole

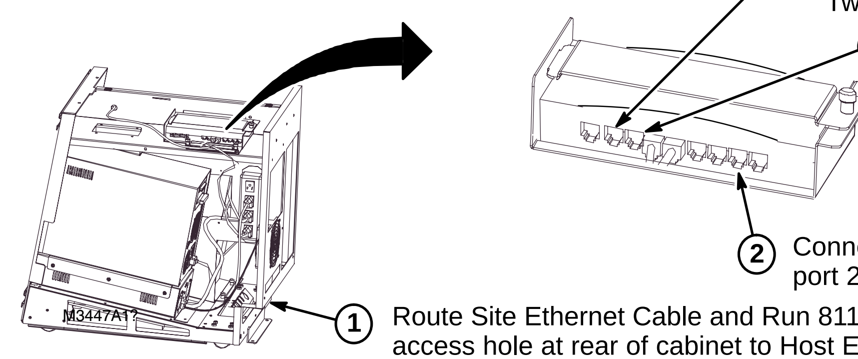
❑ E3 – REMOVE OPERATOR WORKSPACE CABINET FRONT AND SIDE COVERS

1. Remove Front Cover by grasping each side of cover and pulling forward.
2. Unfasten screw and slide Right Side Cover backwards to remove.
3. Unfasten three (3) screws and remove Left Side Cove.



❑ E4 – CONNECT ETHERNET CABLES

Note: Site Ethernet cable for InSite connection is supplied by customer. For Europe, connection to ISDN router is required. Refer to catalog required for country or region for specific equipment to be installed.

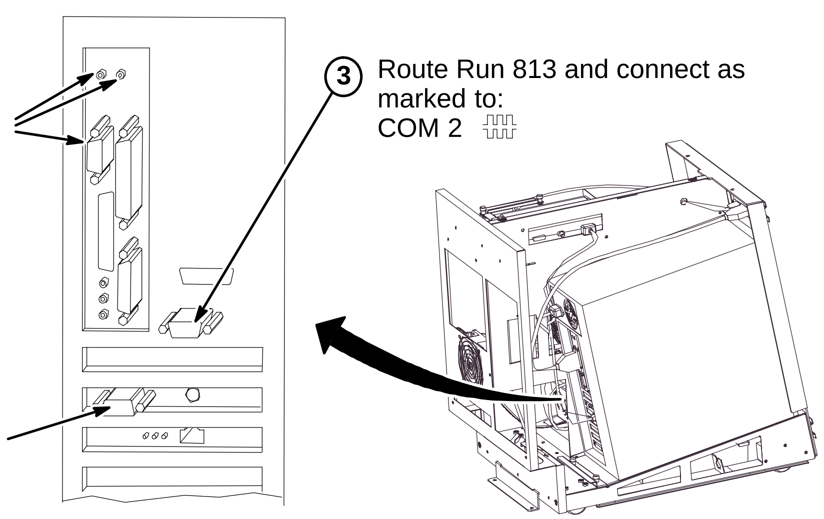


1. Route Site Ethernet Cable and Run 811 thru cable access hole at rear of cabinet to Host Ethernet Module.
2. Connect Run 811 from Octane Computer to port 2, FE laptop to port 3, and PC to port 4.
3. Connect Run 942 from Magnet Monitor J5 to port 7 (Open*Speed* and Twin*Speed*).
4. Site Ethernet cable is supplied by customer. If cable supplied is a Shielded Twisted Pair Cable, connect to port 8.

❑ E5 – ROUTE/CONNECT RUNS 792, 797, 813, AND 814 TO PC MODULE

Note: Route cables thru access hole at rear of cabinet.

1. Route Run 814 and connect as marked to:
   KEYBOARD
   MOUSE
   COM1
2. Route Run 797 and connect as marked to:
   VIDEO
3. Route Run 813 and connect as marked to:
   COM 2



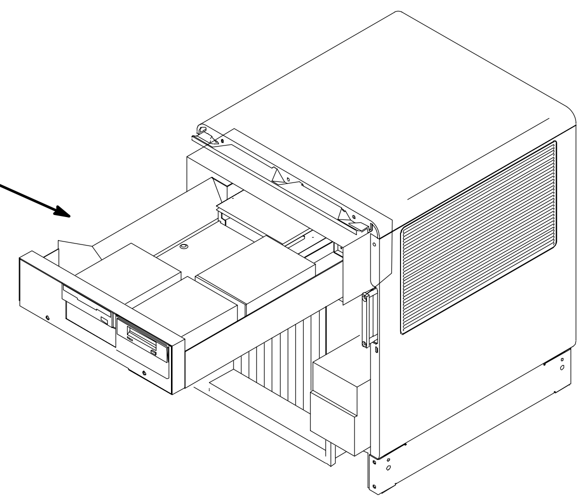
PROCEDURES F1 THRU F5 ARE USED ONLY WHEN UPGRADING FROM 5.x PLATFORMS, <u>DO NOT PERFORM PROCEDURES F1 THRU F5 FOR NEW OCTANE INSTALLATIONS</u>

<u>INSTALLATION STEPS SUMMARY</u>

- ❑ **F1 – Removal of DAT and MOD Drives from Genesis Computer**
- ❑ **F2 – Remove SCSI Expansion Box**
- ❑ **F3 – DAT Drive Settings**
- ❑ **F4 – MOD Drive Settings**
- ❑ **F5 – Install DAT and MOD Drives (Legacy Media) in SCSI Expansion Box**

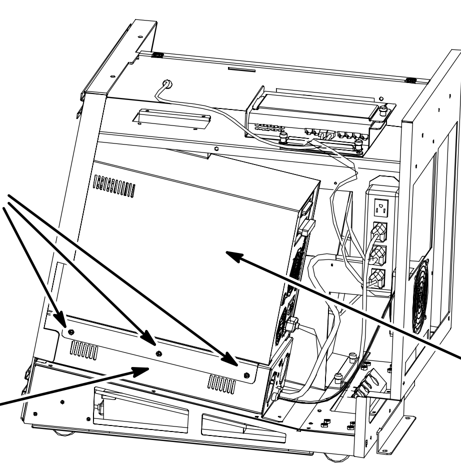
❑ F1 – REMOVAL OF DAT AND MOD DRIVES FROM GENESIS COMPUTER

UPGRADE FROM 5.x OPTIONS:
For transferring Legacy Media DAT and MOD Drives from Genesis Computer – refer to Tab 13, Directions 2174955 and 2174956, steps **A1** thru **A4** only for removal of existing drives.



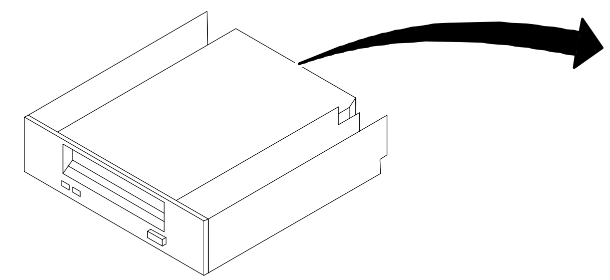
❑ F2 – REMOVE SCSI EXPANSION BOX

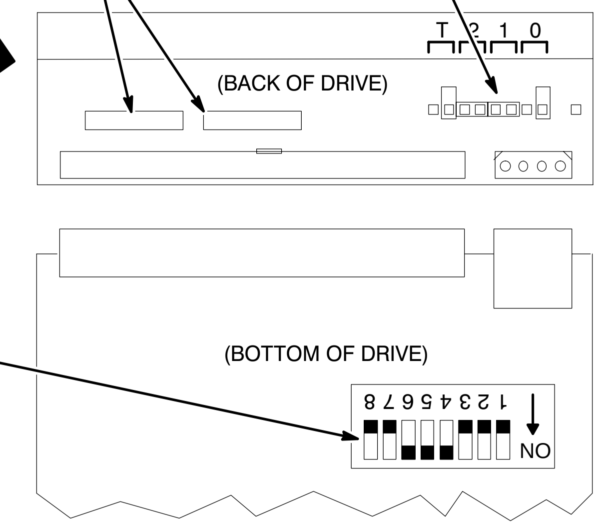
1. Unfasten three (3) screws from SCSI Expansion Box mounting bracket and remove Box and bracket from power module.
2. Unfasten six (6) screws that hold bracket to bottom of SCSI Expansion Box and remove bracket.
3. Remove SCSI Expansion Box cover.



❑ F3 – DAT DRIVE SETTINGS

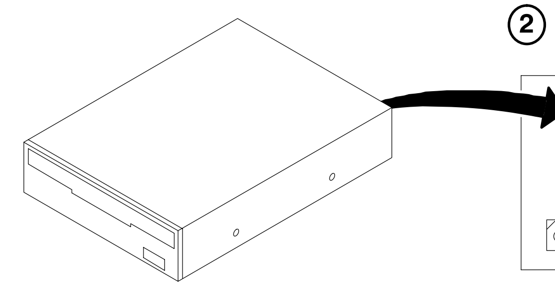
1. Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.
2. For Octane, set SCSI ID = 6
3. Set the swithches on the bottom of the drive as shown. This sets the drive to NOT perform “data disconnects” when there is no data available. This is required because of the odd byte count record packing done on GEMS tapes and SGI’s inability to handle odd byte “save pointer” SCSI requests.

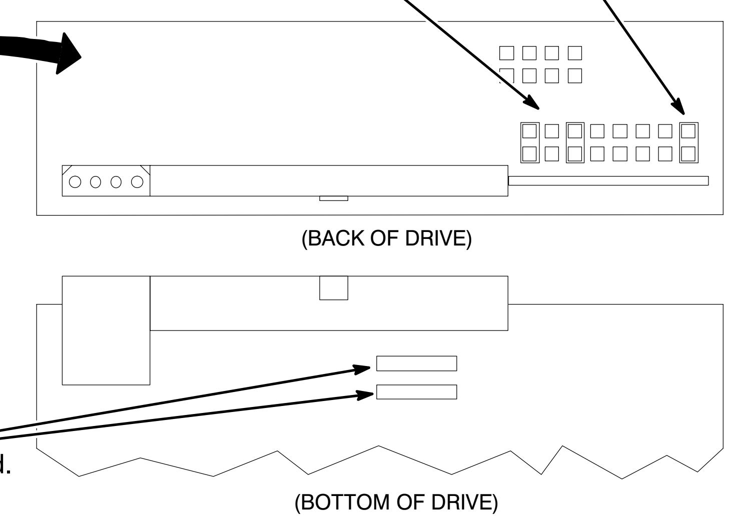




❑ F4 – MOD DRIVE SETTINGS

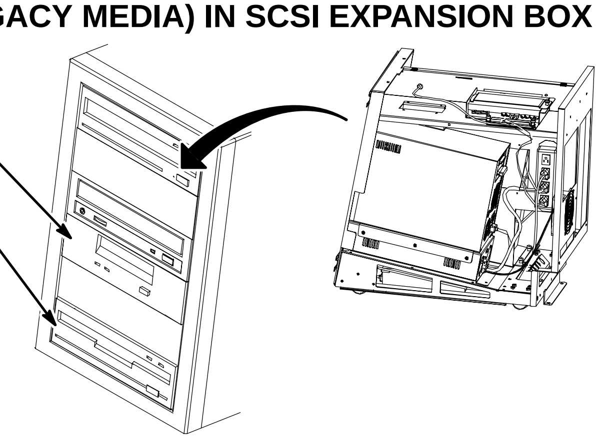
1. Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.
2. For Octane, SCSI ID = 5
3. Make sure jumper is installed on pins at position 8 for SCSI Optical Disk.





❑ F5 – INSTALL DAT AND MOD DRIVES (LEGACY MEDIA) IN SCSI EXPANSION BOX

1. Install DAT Drive in slot 4 with existing hardware.
2. Install MOD Drive in slot 6 with existing hardware.
3. Connect power cords to DAT and MOD Drives.
4. Connect ribbon cable to DAT and MOD Drives.
5. Install SCSI Expansion Box cover. Reattach bracket to bottom of Box. Reattach bracket and Box assembly to power module.

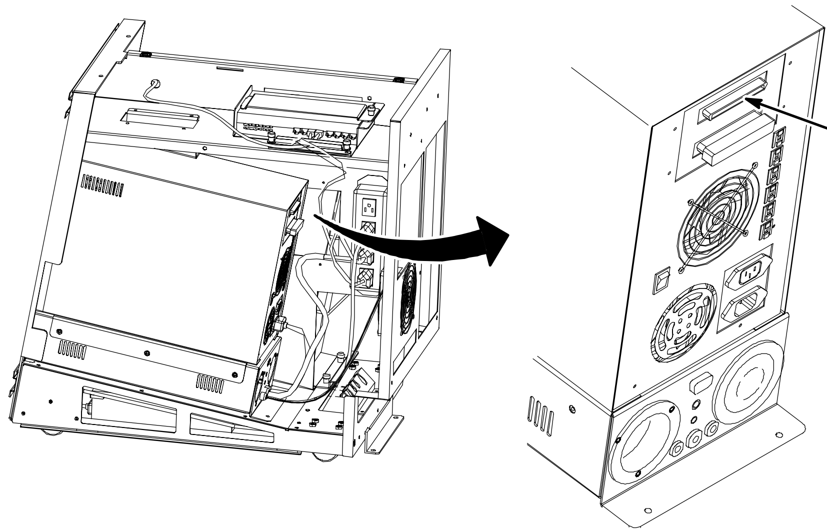


INSTALLATION STEPS SUMMARY

- ❑ **G1 – Route/Connect Run 819 to SCSI Expansion Box**
- ❑ **G2 – Route/Connect Runs 047, 048, 049/969, and 817 to Power Module**
- ❑ **G3 – Align Operator Workspace Cabinet Cables & Attach InSite "Computer" Magnet**
- ❑ **G4 – Bundle Operator Workspace Cabinet Cables**

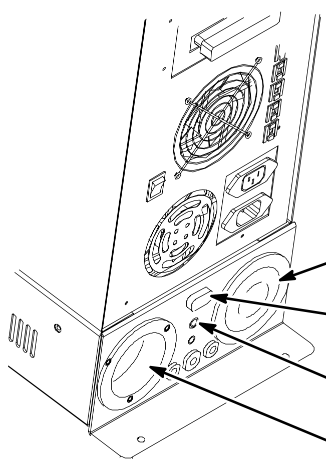
❑ G1 – ROUTE/CONNECT RUN 819 TO SCSI EXPANSION BOX

Note: Route all cables from SGI Octane Computer thru cable access hole at rear of cabinet.



1. Route/Connect Run 819 from SGI Octane Computer.

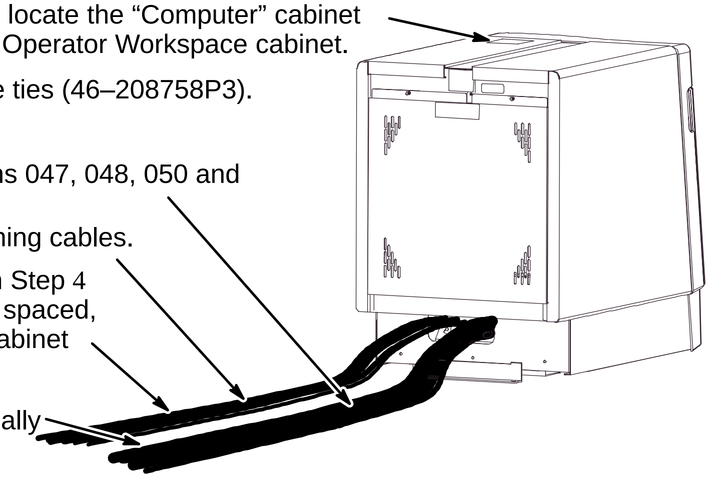
❑ G2 – ROUTE/CONNECT RUNS 047, 048, 049/969, AND 817 TO POWER MODULE



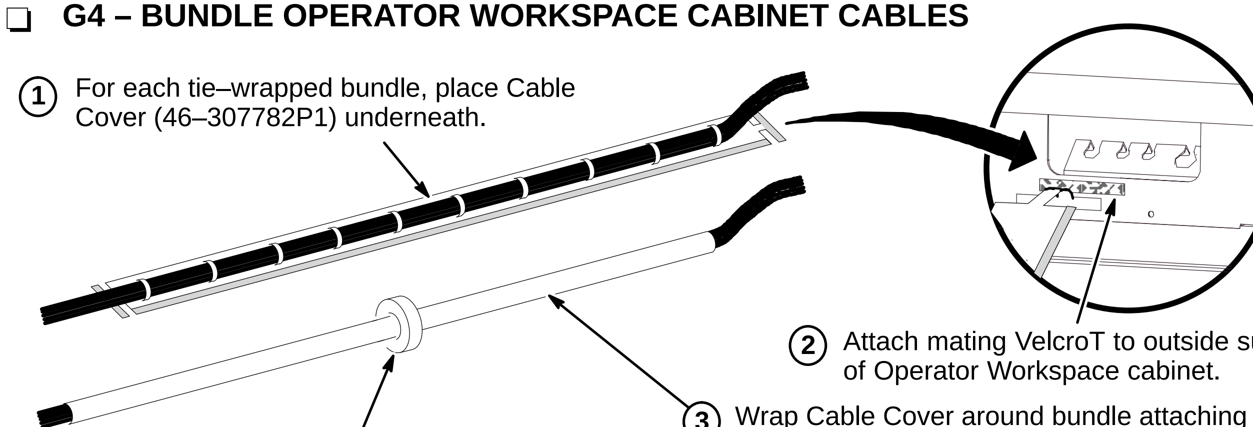
1. Route/Connect Run 049 or Run 969 Power Cable from Operator Workspace I/F to J2.
2. Route/Connect Run 817 Power Cable from Operator Workspace Power Module to J3.
3. Route/Connect Run 048 Ground Cable from PDU to ground screw.
4. Route/Connect Run 047 Power Cable from PDU to J1.

❑ G3 – ALIGN CABINET CABLES & ATTACH InSite "COMPUTER" MAGNET

1. Install Operator Workspace Cabinet right and left Side covers, Front cover and Rear cover.
2. From the 8.0 LX InSite Kit (46–301708G14), locate the "Computer" cabinet magnet (46–320095P1) and attach to top of Operator Workspace cabinet.
3. Locate the twenty (20) seven inch long cable ties (46–208758P3).
4. Seperate cables into two bundles;
   The first bundle will include power cable Runs 047, 048, 050 and fiber optic cable Run 790.
   The second bundle will include all the remaining cables.
5. Align the cables of the first bundle created in Step 4 and attach ten (10) of the cable ties, equally spaced, around the cable bundle between the OW Cabinet and the OW Table.
6. Attach the remaining ten (10) cable ties, equally spaced, around the second cable bundle.



❑ G4 – BUNDLE OPERATOR WORKSPACE CABINET CABLES



1. For each tie–wrapped bundle, place Cable Cover (46–307782P1) underneath.
2. Attach mating VelcroT to outside surface of Operator Workspace cabinet.
3. Wrap Cable Cover around bundle attaching the VelroT hooks and loops.
4. Attach cable bundle Clamp (46–307873P1) to center of Cable Cover.
5. Carefully roll OW Cabinet into final position. Make sure cable bundles are unobstructed and slide easily. Attach to mating Velcro on cabinet.
6. Repeat Step 5 as often as necessary.

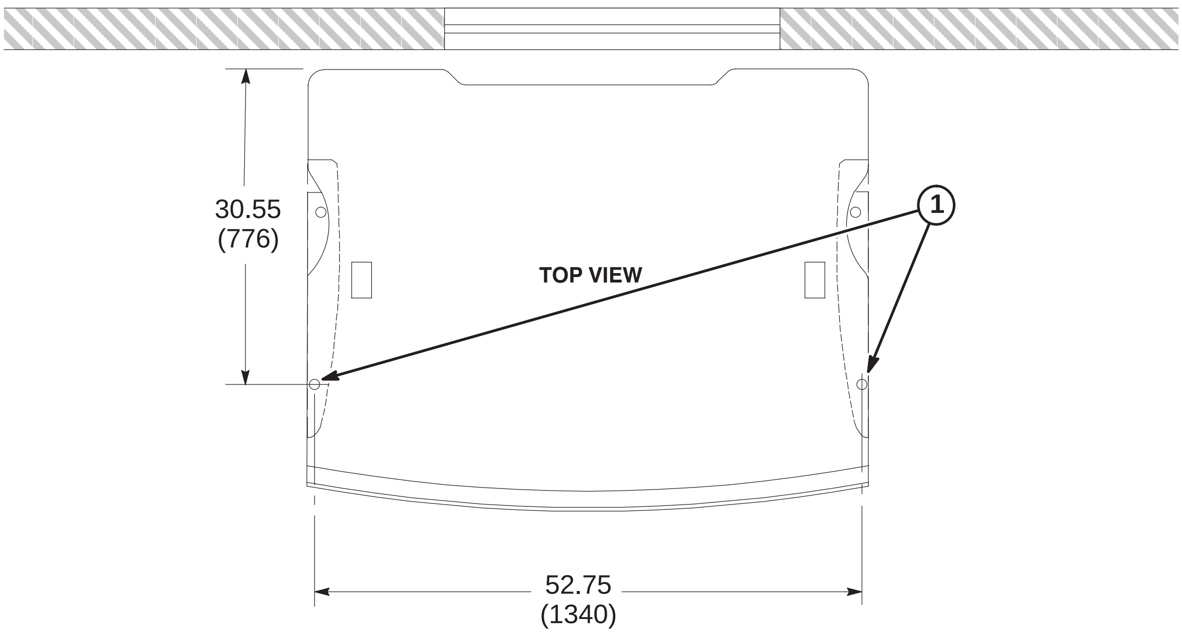
TABLE OF CONTENTS

SECTION **PAGE**

1–1 OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION

Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table. Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

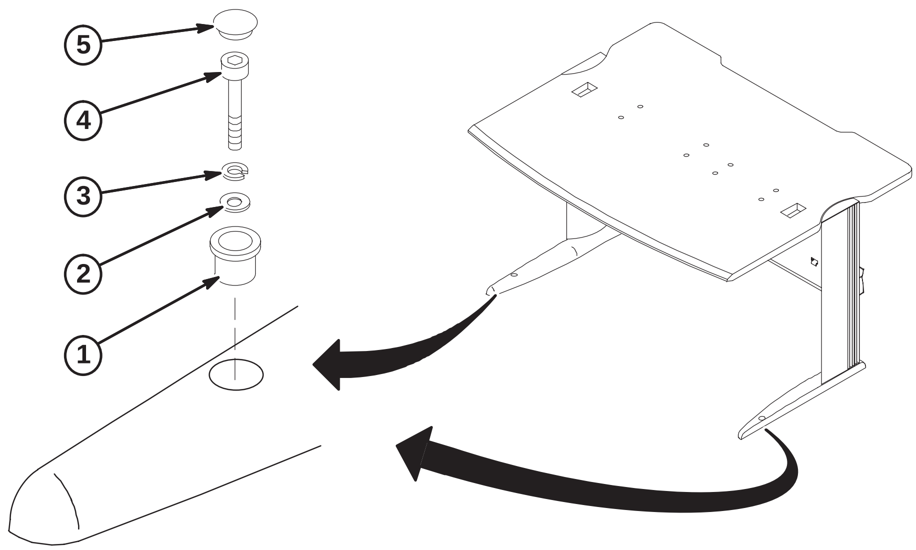
1. Remove and discard the two (2) cover plugs. Locate and mark on floor the two hole locations at the front of the table legs.
2. Move table to prepare for installing the floor anchors in Steps 3 and 4.
3. At each hole marked in Step 2, drill a .38 inch hole and insert 1/4–20 x .375 diameter anchor (2168253).
4. Secure anchor in place with supplied Setting Tool (2168781).



1–2 SECURE OPERATOR WORKSPACE TABLE TO FLOOR

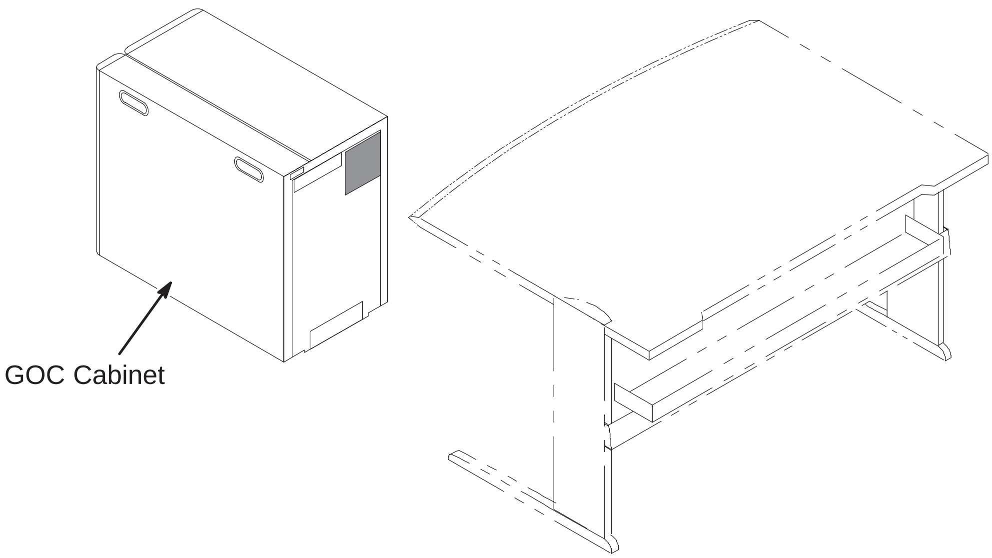
Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

1. Insert Bushing (2151454) in table leg hole.
2. Insert as many Flat Washers (2168251) as required (3 provided for each leg) for securing table to floor.
3. Insert one Lock Washer (2168252) on top of Flat Washer(s).
4. Depending upon depth of previously installed anchor, install either the 1–3/4 inch long or the 2 inch long 1/4 x 20 Hex Socket Cap Screw.
5. Install black nylon Plug (2168254).
6. Repeat anchoring process below for other leg.



1–3 POSITION COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.

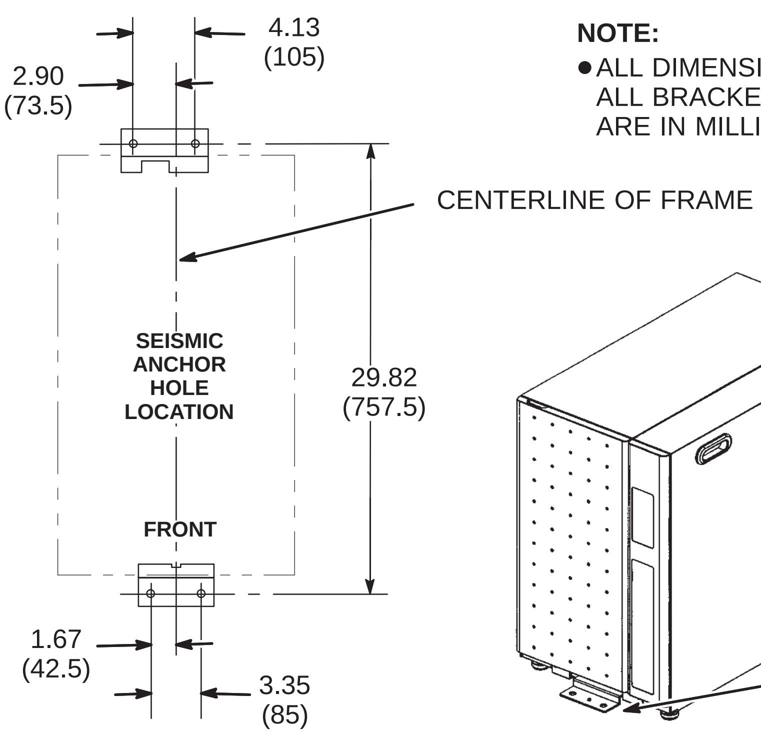


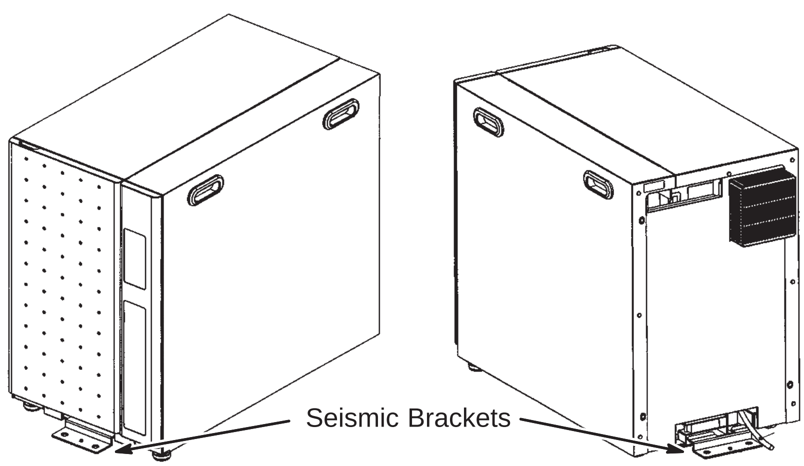
1–4 SEISMIC ANCHORING (IF REQUIRED)

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre–installation manual. Illustration below indicates hole location for installing brackets.

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.





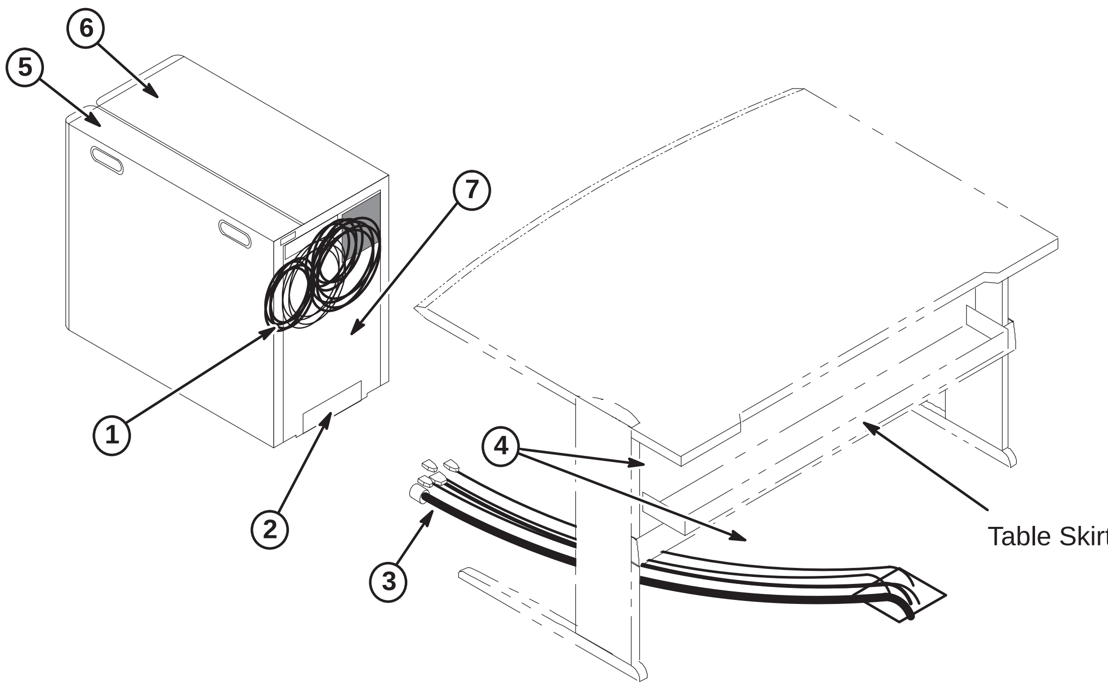
1–5 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Three power cords
- Two video ouput cables
- Keyboard output cable
- SCSI output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1062, 1063, 1064, 1065, 1068, 1069, and 1070 are pre–connected to computer and are routed thru top opening.
2. Runs 1077, 1081, 1082, 1083, 1084, 1085, power cord for the Patient Alert system, and any other option cables will be routed thru the lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the table skirt. The cables coming out of the lower opening will be routed below the tabel skirt.



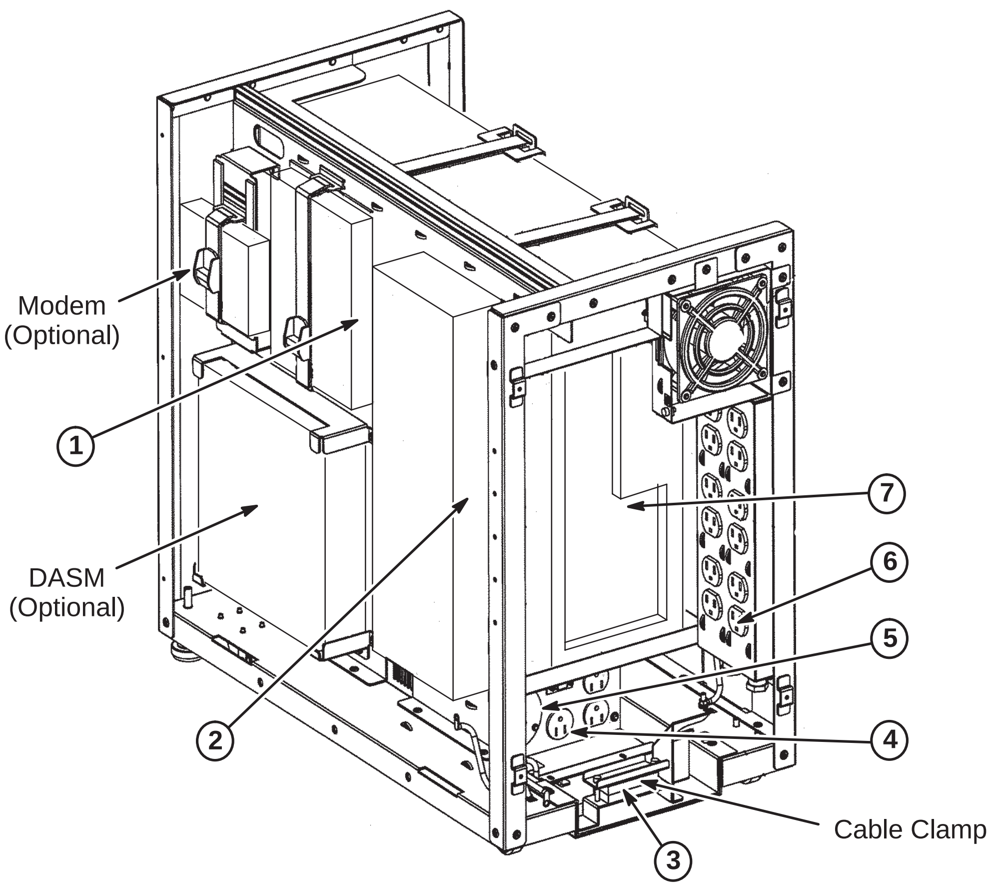
The Computer Cabinet covers need to be removed to access the cable connection points.

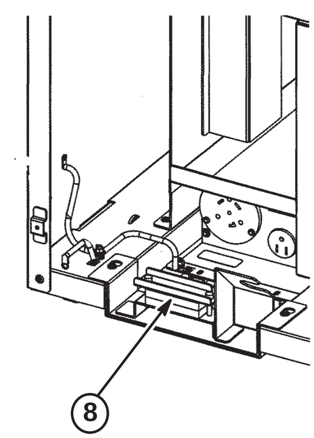
5. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
6. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
7. Loosen four screws holding the back cover. Lift and remove cover.

1–6 ROUTE CABLES FOR LOWER OPENING

Refer to cable map for information on cables to be connected. Route cables thru clamp at lower rear of cabinet and connect cable runs as shown in illustration below.

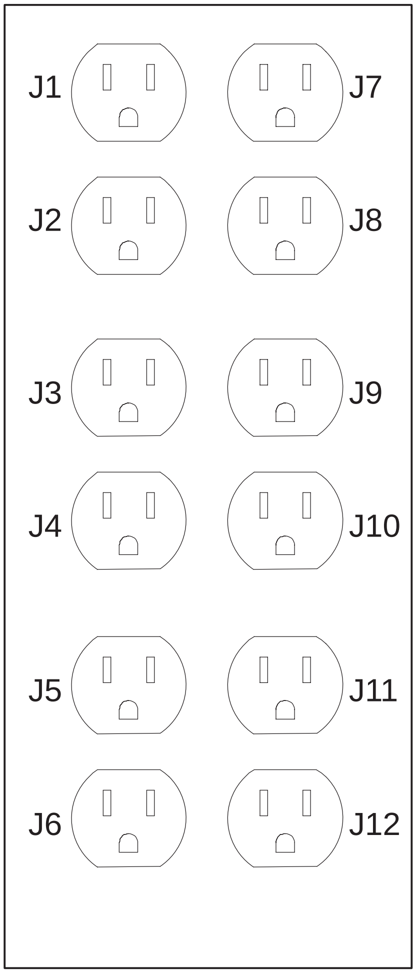
1. Run 1077 to Lan Hub.
2. Run 1085 to WIM.
3. Run 1084 routes from GOC Cabinet thru clamp to J26 on System Cabinet.
4. Make sure Modem power cable is connected to J5 on Computer Cabinet PDU.
5. Run 1081 to PDU J1.
6. Patient Alert Power cable to power strip at J12. Refer to Section 1–7 for other power cable connections to power strip.
7. Runs 1082 and 1083 to Linux PC computer. Refer to Section 1–8 for all cables connected to Linux PC computer.
8. Tighten clamp to secure cables in place.





1–7 POWER STRIP CONNECTIONS

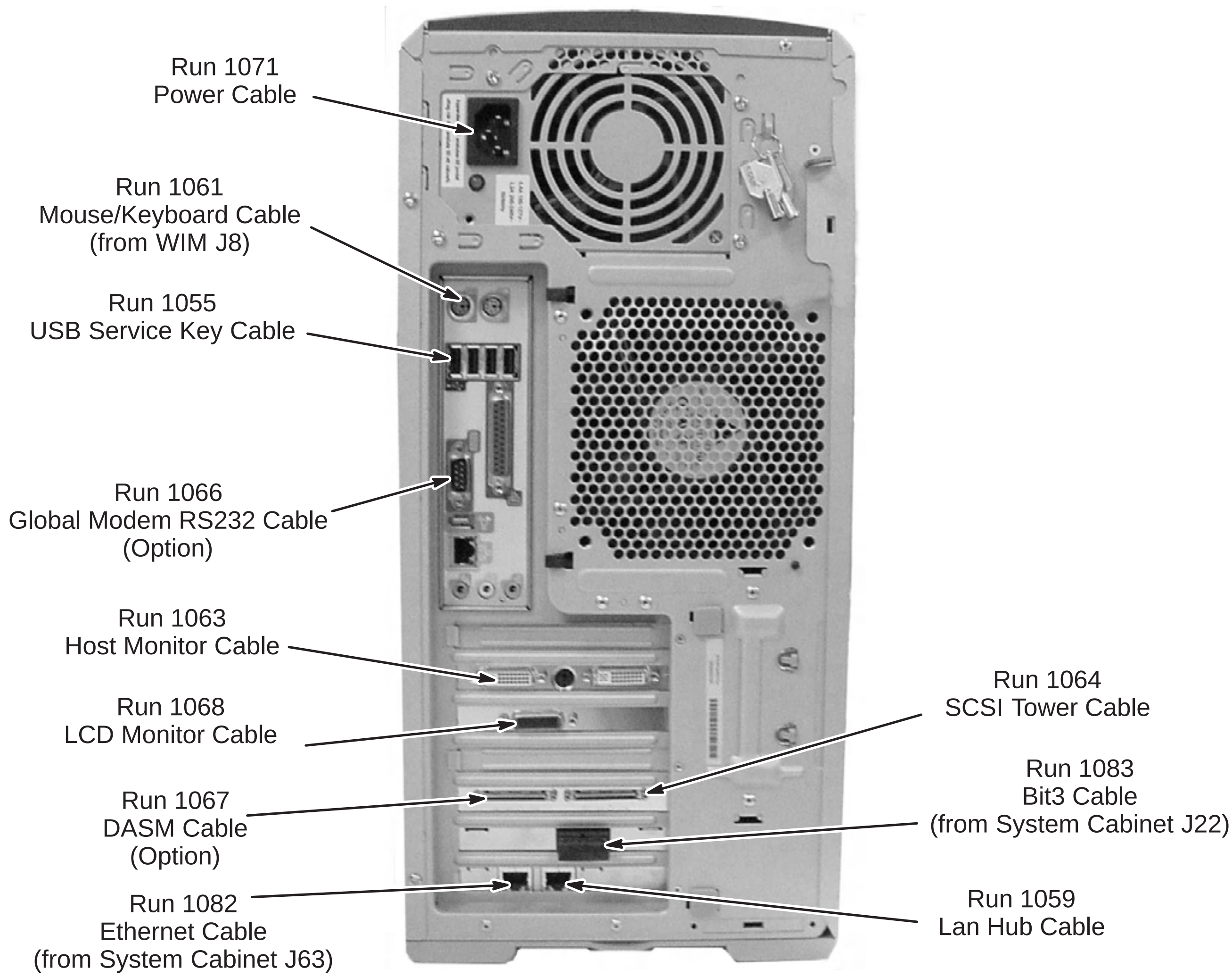
The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.



POSITION	NAME	MAX AMP CIRCUIT 1	MAX AMP CIRCUIT 2
J1	Host	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6			
J7	Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.4	8.8

1–8 LINUX PC CABLE CONNECTIONS

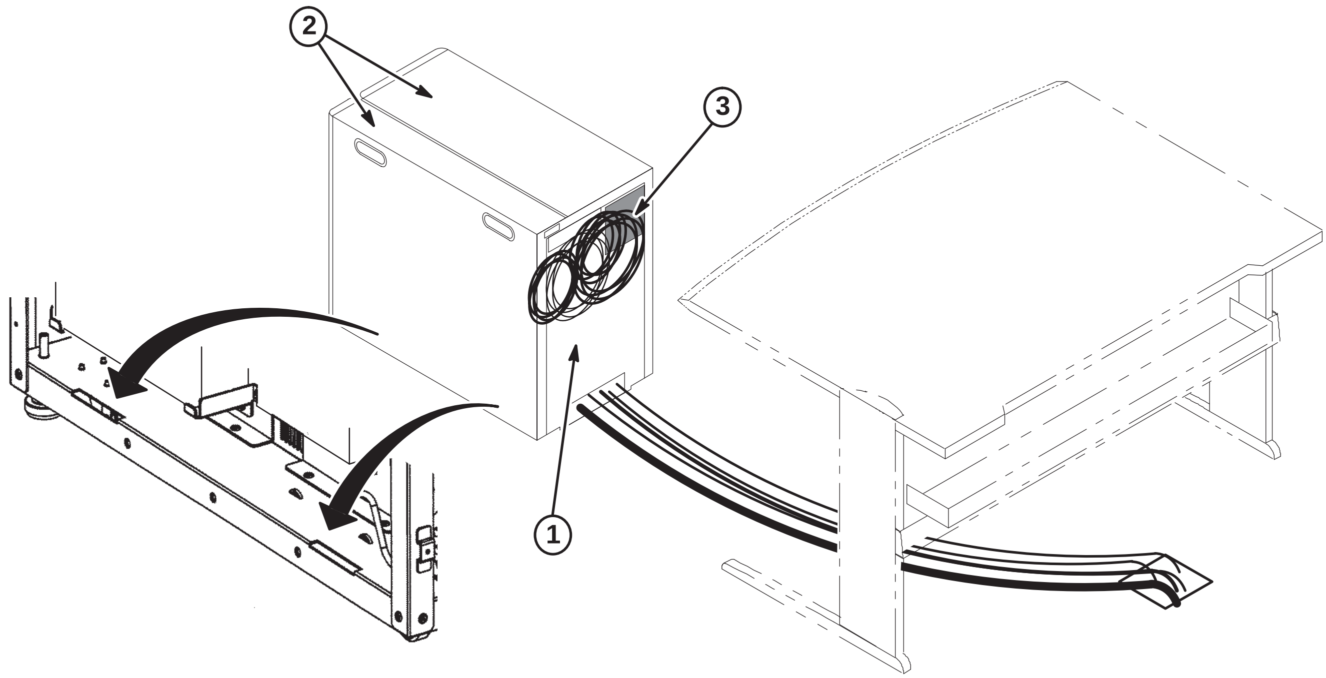
The illustration below shows the connection points of cables routed to the Linux PC. All cables shown are preconnected except for Runs 1082 and 1083, which have to be routed and connected at site.



1–9 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

1. Re–attach rear cover and tighten screws.
2. Install both side covers by inserting tabs at bottom of covers into slots (as indicated in illustration below) and slide forward. Install screws at rear of cover that were removed in earlier procedure.
3. Route remaining cables from top opening at rear of GOC Cabinet thru opening of table skirt.
4. Place cabinet in final position as determined by customer.



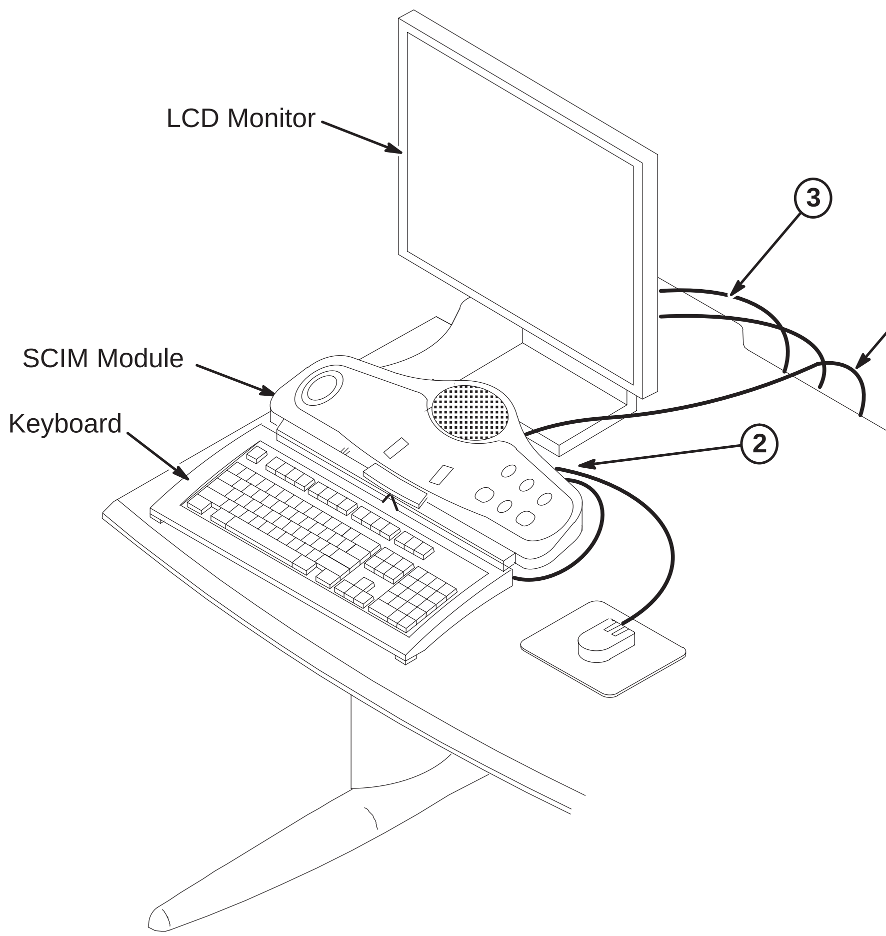
1–10 INSTALL SCIM/KEYBOARD AND MONITOR

Consult with Site customer for exact location of Keyboard and Monitor on Operator Workspace (OW) Table.

Note

Refer to Installation document OW1INA3.DOC on Service Methods CD–ROM or MR Service Engineering web site for installation information on language labels and keyboard mounting.

1. Route data cable Run 1062 from Computer Cabinet to bottom side of SCIM Module and connect as marked. (Run 805, supplied with SCIM Kit (2307175) is not used for EXCITE) Run 1062 is preconnected to the Computer Cabinet and is located on back side of cabinet.
2. Place Keyboard and Mouse on table and connect cables to bottom side of SCIM Module as marked.
3. Place LCD Host Monitor on table in back of SCIM. Route Runs 1063 and 1065 from Computer Cabinet and connect to monitor.



1–11 INSTALL SCSI EXPANSION BOX AND LCD DISPLAY

1. Place the SCSI Tower on the OW table and connect Runs 1064 and 1070 from GOC Cabinet. Install terminator.
2. Place LCD Display on OW table and connect Runs 1068 and 1069 from GOC Cabinet.

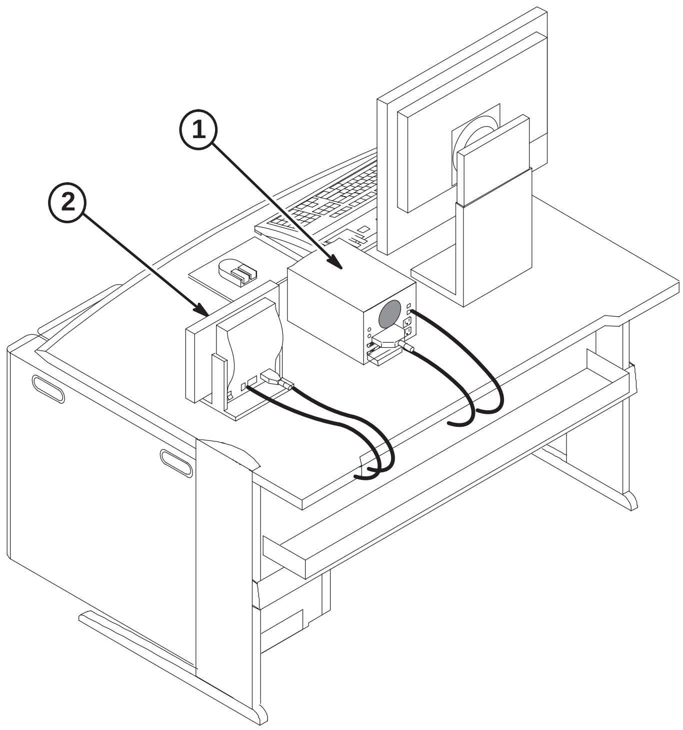


TABLE OF CONTENTS

1–1 OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION

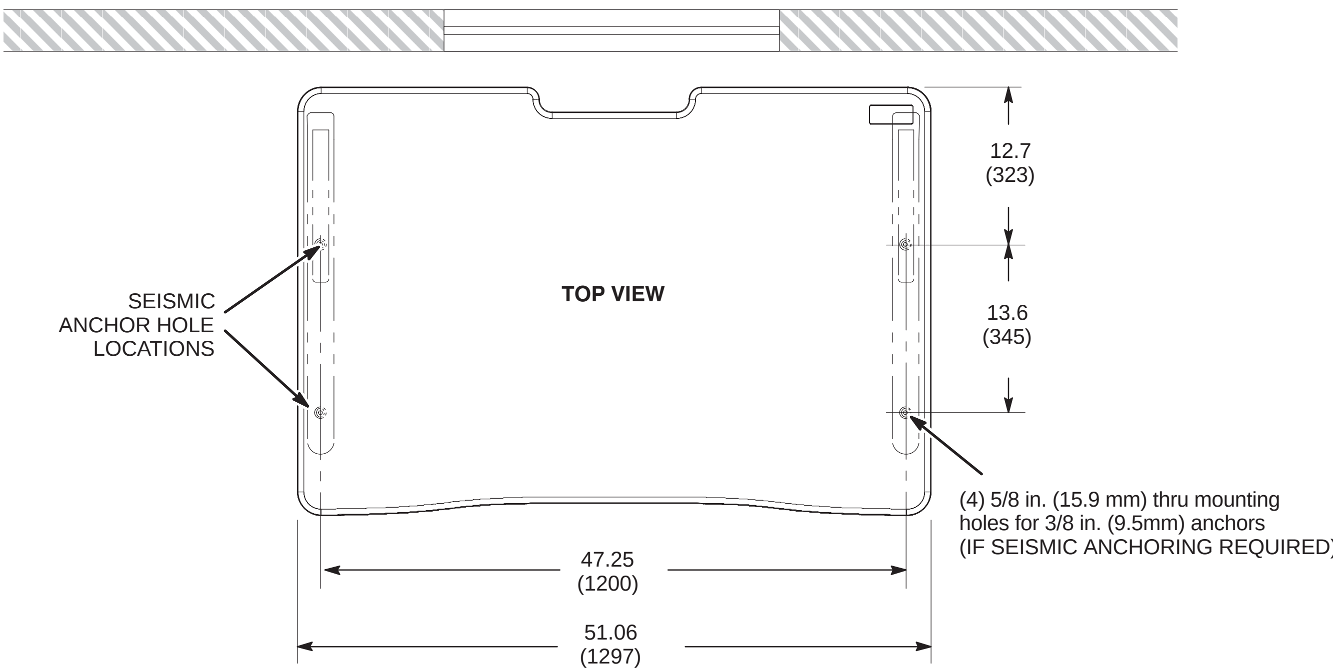
Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.

1. Open palletized M1000MW box that contains the Operator Workspace table. Use furnished *Operator's Workspace Table Assembly Instructions* (5189137–50) to assemble the model 5191031 table.
2. Move table into designated position. Refer to room layout plan and/or consult with Site customer for exact location of Operator Workspace Table. Typical location with table in front of Magnet Room viewing window is shown below.

Note

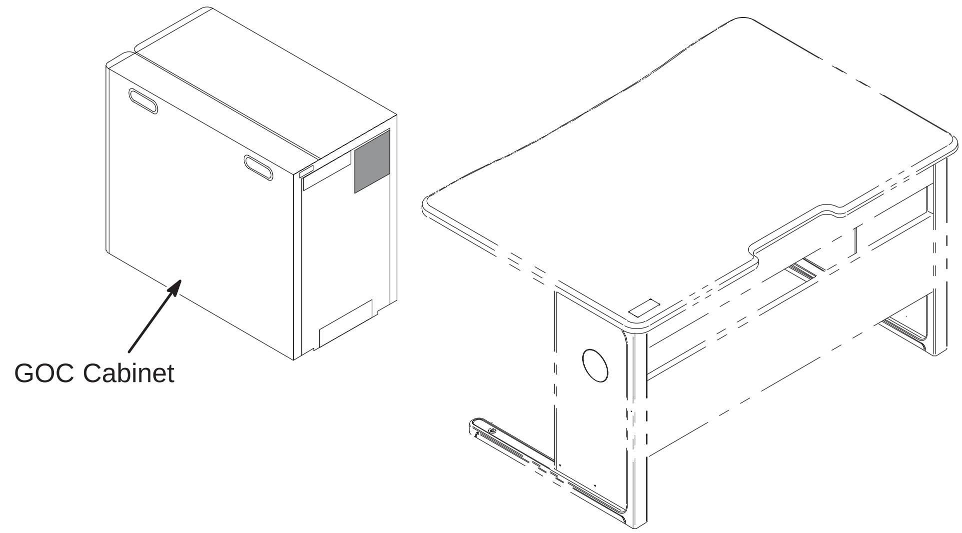
The design of the new model 5191031 table does not require anti–tiip anchors. This table meets the ANSA/BIFMA anti–tip standards.

3. If seismic anchoring is required, refer to site architectural drawings and procure and install specified anchors.



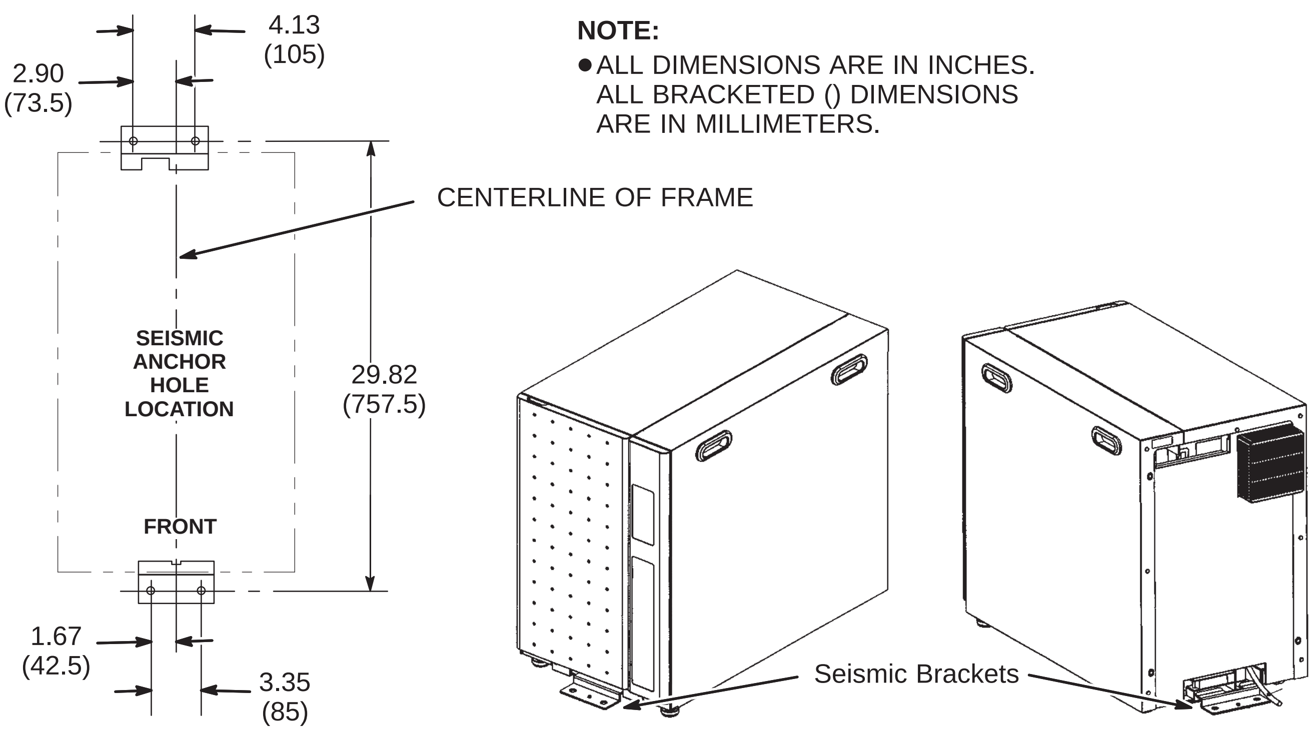
1–2 POSITION COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



1–3 SEISMIC ANCHORING (IF REQUIRED)

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre–installation manual. Illustration below indicates hole location for installing brackets.



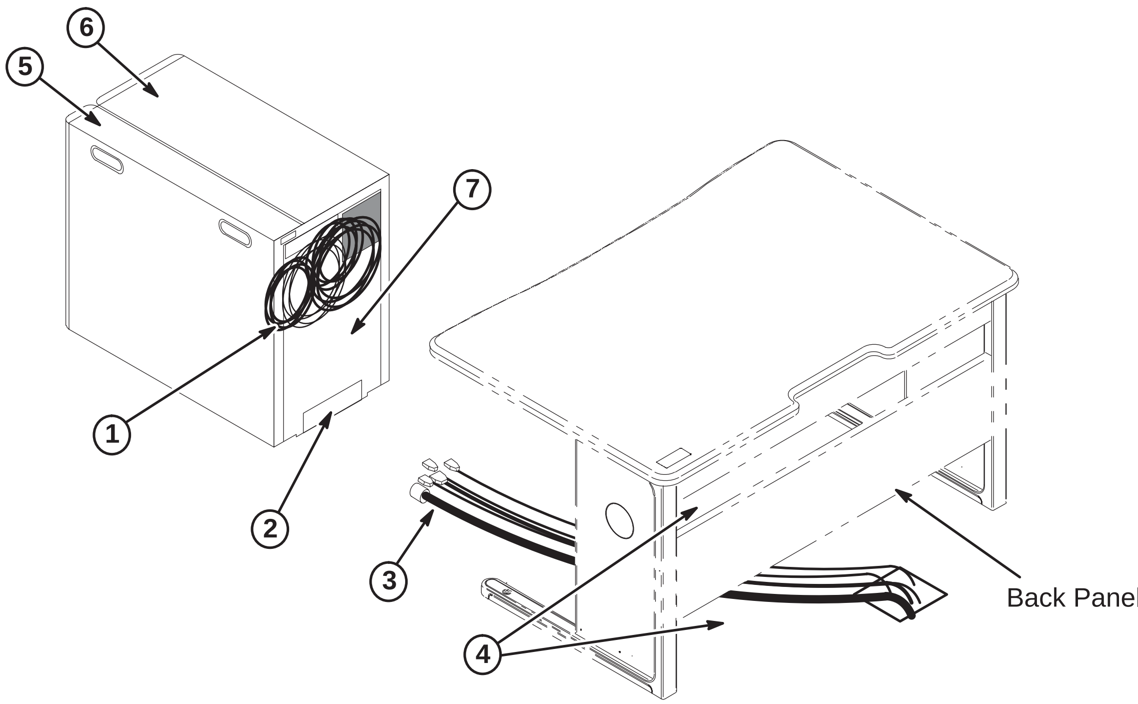
1–4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Three power cords
- Two video ouput cables
- Keyboard output cable
- SCSI output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1062, 1063, 1064, 1065, 1068, 1069, and 1070 are pre–connected to computer and are routed thru top opening.
2. Runs 1081, 1082, 1083, 1084, 1085, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening. Also, for 3.0T: Run 1077.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



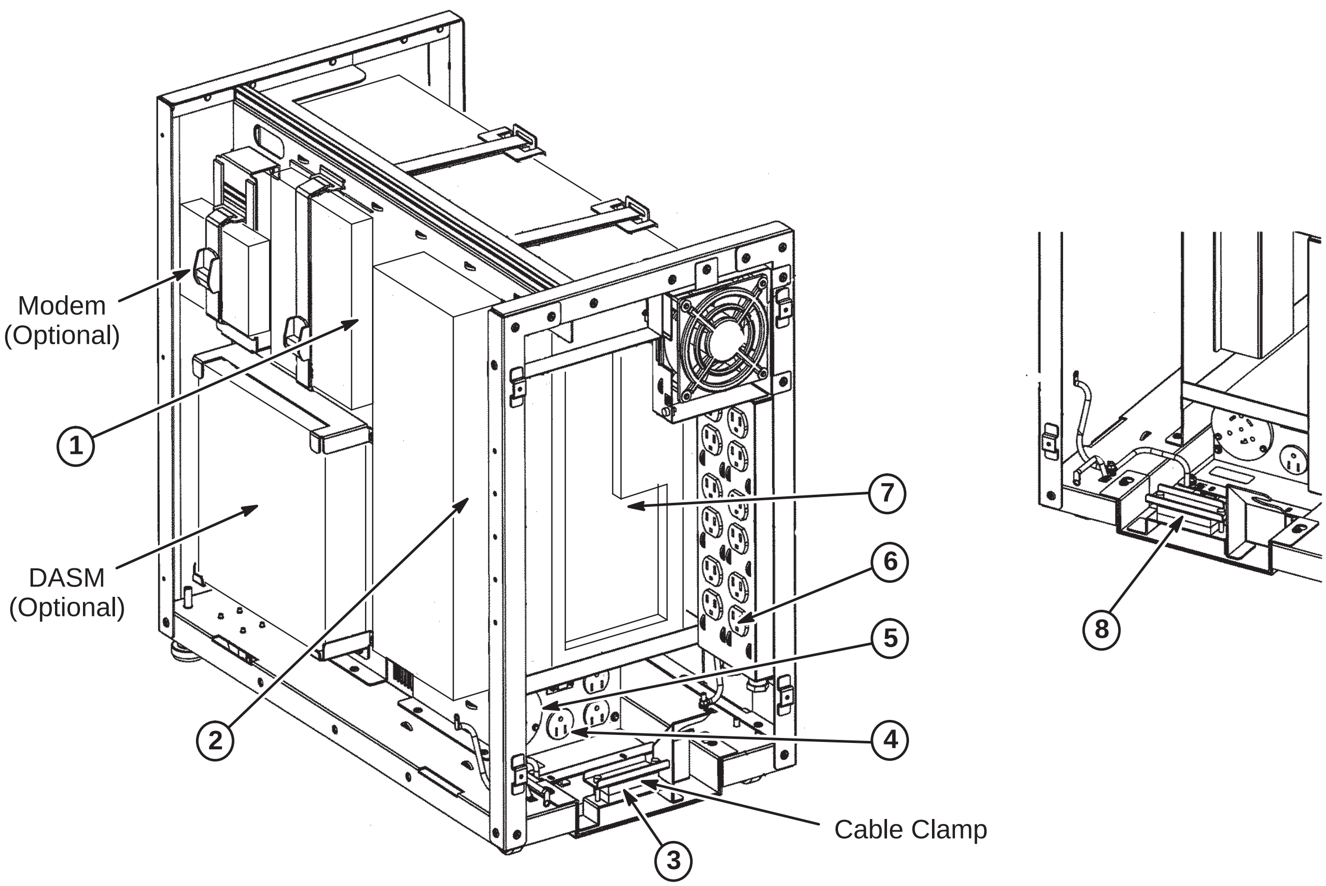
The Computer Cabinet covers need to be removed to access the cable connection points.

5. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
6. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
7. Loosen four screws holding the back cover. Lift and remove cover.

1–5 ROUTE CABLES FOR LOWER OPENING

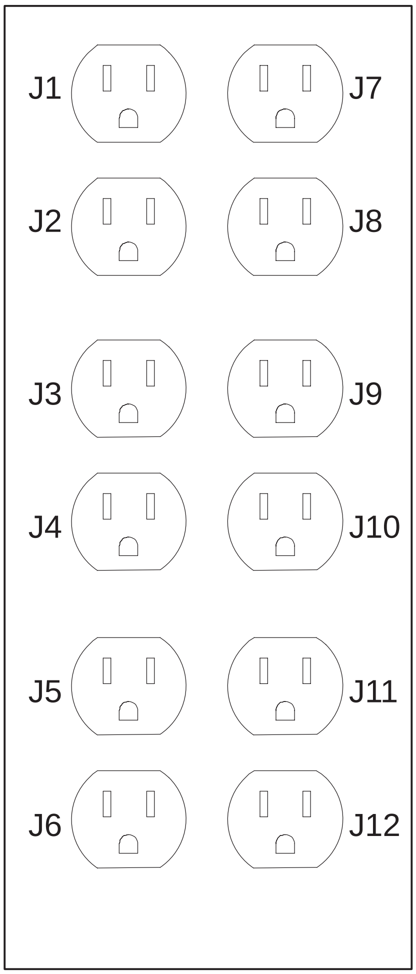
Refer to cable map for information on cables to be connected. Route cables thru clamp at lower rear of cabinet and connect cable runs as shown in illustration below.

1. If part of installation, route Run 1077 to Lan Hub (for 3.0T).
2. Run 1085 to WIM.
3. Run 1084 routes from GOC Cabinet thru clamp to J26 on System Cabinet.
4. Make sure Modem power cable is connected to J5 on Computer Cabinet PDU.
5. Run 1081 to PDU J1.
6. Patient Alert Power cable to power strip at J12. Refer to Section 1–6 for other power cable connections to power strip.
7. Runs 1082 and 1083 to Linux PC computer. Refer to Section 1–7 for all cables connected to Linux PC computer.
8. Tighten clamp to secure cables in place.



1–6 POWER STRIP CONNECTIONS

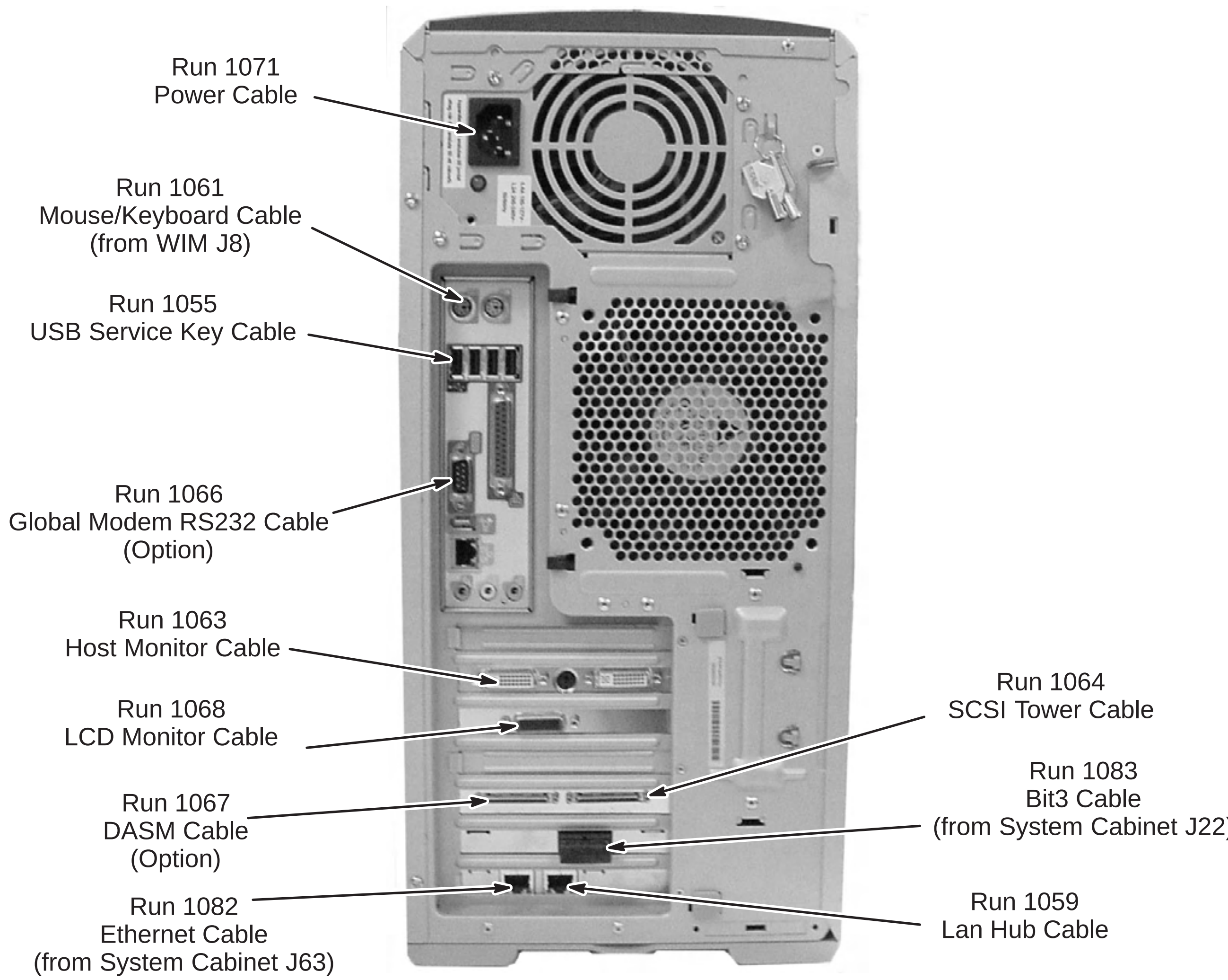
The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.



POSITION	NAME	MAX AMP CIRCUIT 1	MAX AMP CIRCUIT 2
J1	Host	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6			
J7	Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.4	8.8

1–7 LINUX PC CABLE CONNECTIONS

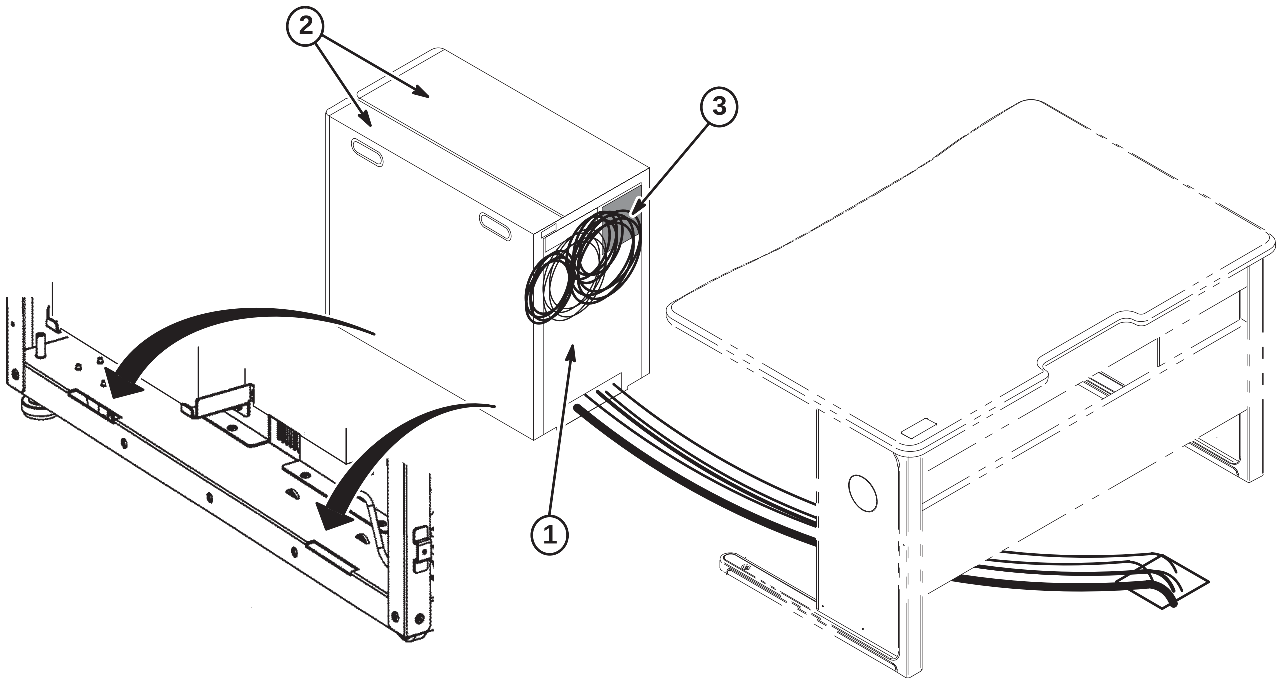
The illustration below shows the connection points of cables routed to the Linux PC. All cables shown are preconnected except for Runs 1082 and 1083, which have to be routed and connected at site.



1–8 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

1. Re–attach rear cover and tighten screws.
2. Install both side covers by inserting tabs at bottom of covers into slots (as indicated in illustration below) and slide forward. Install screws at rear of cover that were removed in earlier procedure.
3. Route remaining cables from top opening at rear of GOC Cabinet thru opening above Back Panel on OW table.
4. Place cabinet in final position as determined by customer.



1–9 INSTALL SCIM/KEYBOARD

The Scan–Control Intercom Module (SCIM) is the current keyboard on all Signa LX and OpenSpeed systems. Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

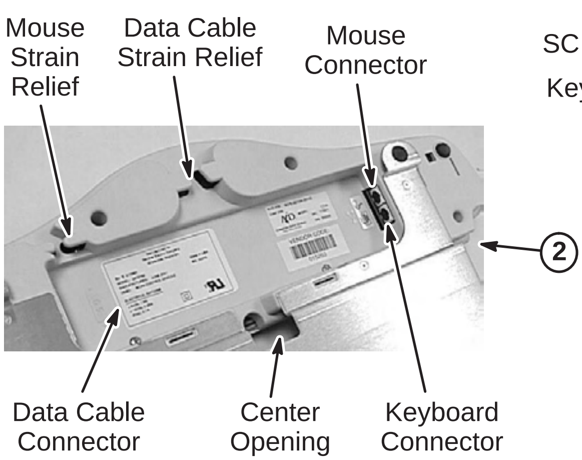
1. Remove the SCIM, Data Cable, and Keyboard From the box.

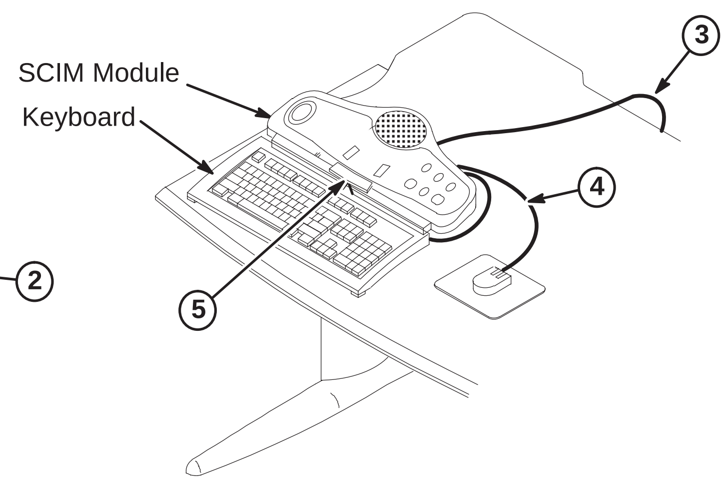

SCIM Module


Data Cable


Keyboard

2. Turn the SCIM unit up side down exposing the cable connections.
3. Route data cable Run 1062, preconnected to the GOC Cabinet and is located on back side of cabinet, from Computer Cabinet to bottom side of SCIM Module. Route thru data cable strain relief and connect as marked. (Run 805, supplied with SCIM Kit (2307175) is not used for EXCITE)
4. Attach the mouse to the jack labeled with a Mouse icon. Route the cable through the strain relief notches in at the rear of the SCIM.
5. Guide the keyboard cable through the center opening and attach the keyboard cable to the jack labeled with a Keyboard icon (Store excess cable in the base of the SCIM by tucking it in the depressed area).

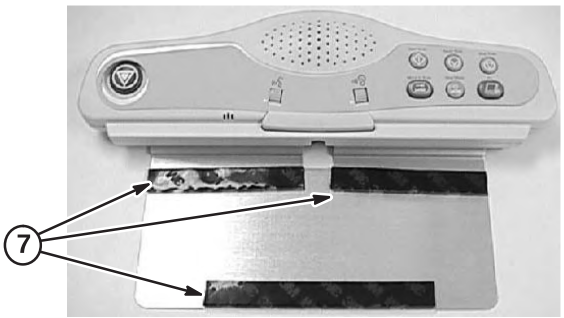


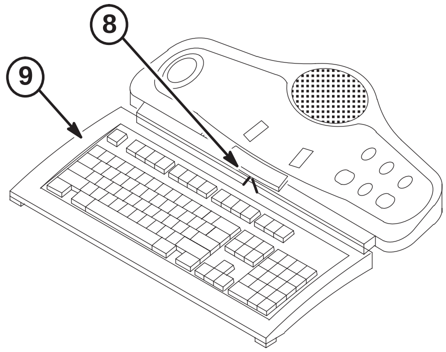


6. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
7. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface (see illustration on next page).

1–9 INSTALL SCIM/KEYBOARD (Continued)

8. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
9. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.



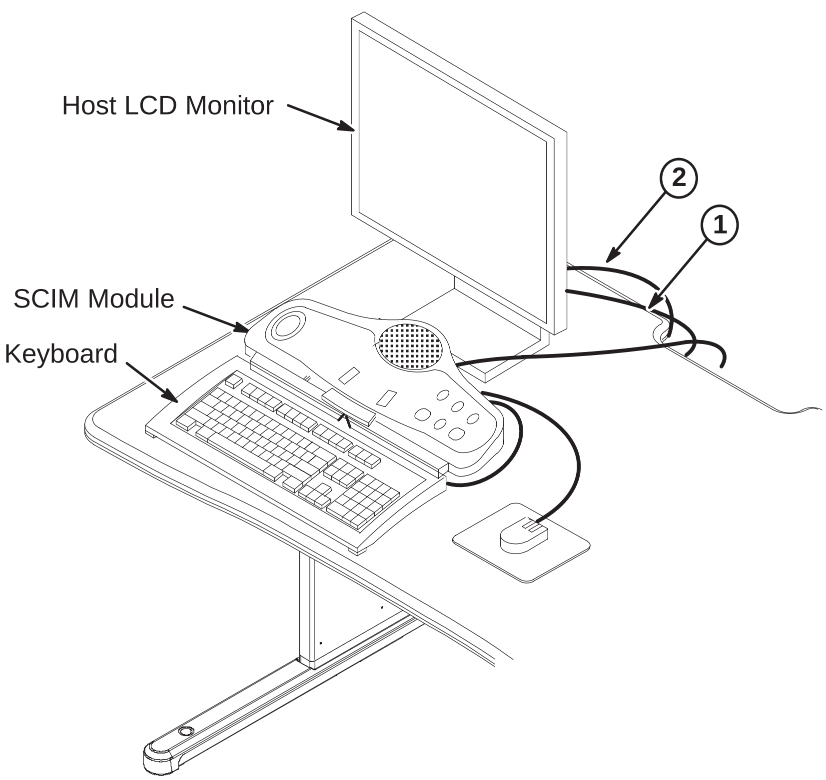


Note

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

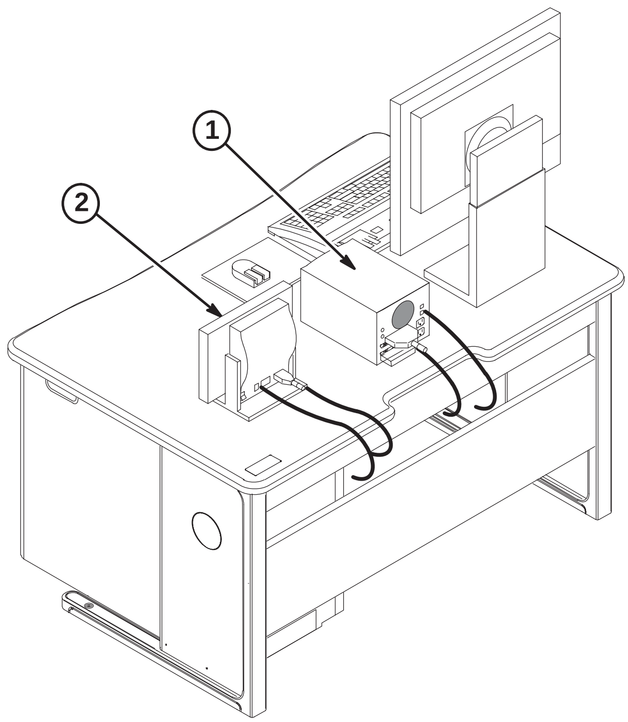
1–10 INSTALL MONITOR

1. Place Host LCD Monitor on table in back of SCIM. Route Run 1063 from GOC Cabinet and connect to monitor.
2. Route power cable Run1065 from Computer Cabinet and connect to monitor. Power cable must be routed and secured thru ty–wrap located on the under side of table top.



1–11 INSTALL SCSI TOWER AND LCD DISPLAY

1. Place the SCSI Tower on the OW table and connect Runs 1064 and 1070 from GOC Cabinet. Install terminator on back of SCSI Tower. Power cable Run 1070 must be routed and secured thru ty–wrap located on the under side of table top.
2. Place LCD Display (Waveform Monitor) on OW table and connect Runs 1068 and 1069 from GOC Cabinet. Power cable Run 1069 must be routed and secured thru ty–wrap located on the under side of table top.



1–12 POWER ON INFORMATION

IMPORTANT
When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note
When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the WIM and Modem power.

ACGD/PDU CABINET INSTALLATION INSTRUCTIONS

Tab 15, ACGD/PDU Cabinet, contains installation instructions for the following:

- *DIRECTION 2286811* – Signa ACGD/PDU Cabinet Installation Instructions
- *DIRECTION 2379890* – Signa SmartSpeed ACGD Lite/PDU Cabinet Installation Instructions

Use only the applicable ACGD/PDU Cabinet Installation Instruction Direction for the site being upgraded

INSTALLATION STEPS SUMMARY

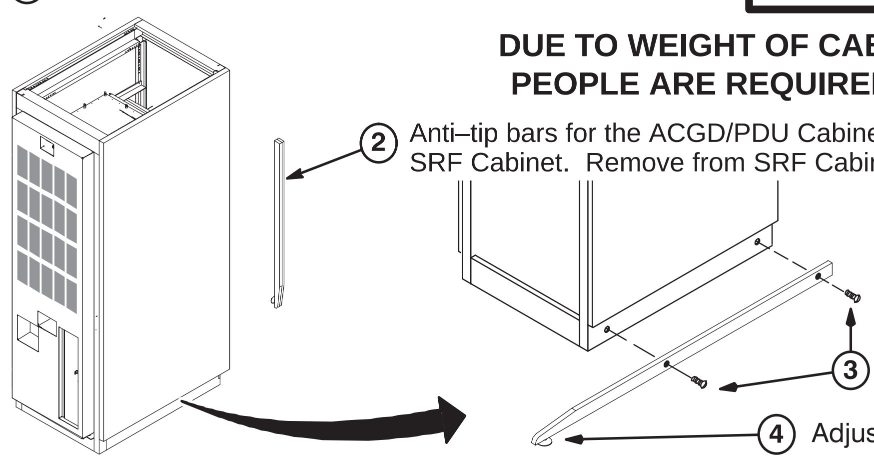
- ❑ **A1 – Position ACGD/PDU Cabinet**
- ❑ **A2 – Install Fan Asembly and Attach InSite Cabinet Magnet**
- ❑ **A3 – Input Voltage Selection**
- ❑ **A4 – Circuit Breaker DIP Switch Settings**

❑ A1 – POSITION ACGD/PDU CABINET

WARNING!

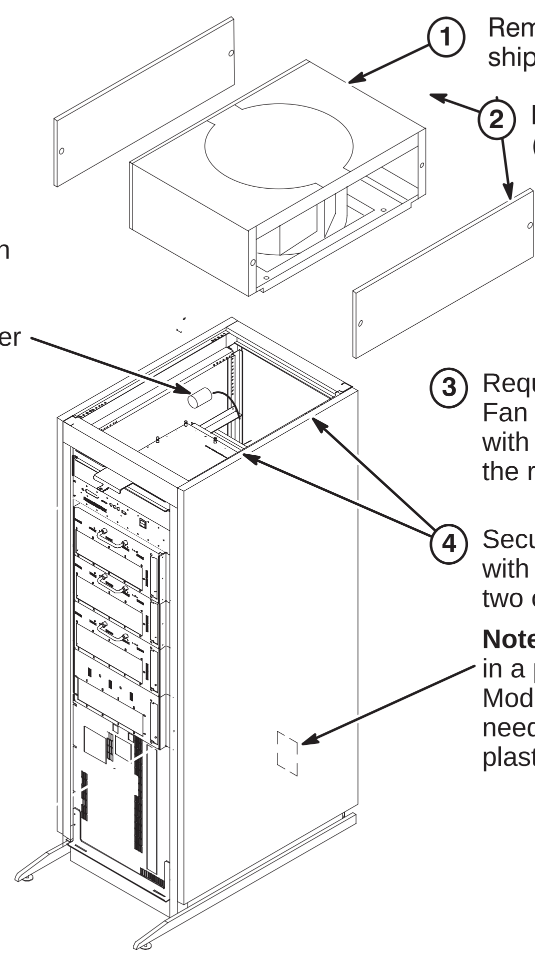
DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

1. Move cabinet to final position.



2. Anti–tip bars for the ACGD/PDU Cabinet are shipped in the SRF Cabinet. Remove from SRF Cabinet for this procedure.
3. Attach bars to cabinet.
4. Adjust leveling pads to level cabinet.

❑ A2 – INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET



1. Remove Fan Assembly from shipping container.
2. Remove two side trim panels (two screws from each).
3. Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
4. Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet.

   Note: The fasteners are provided in a plastic bag located in PDU Module. Rear PDU covers will need to be removed for access to plastic bag with fasteners.
5. Install two side trim panels with previously removed screws.
6. Connect cable from SGA Power Supply to the fan connector.

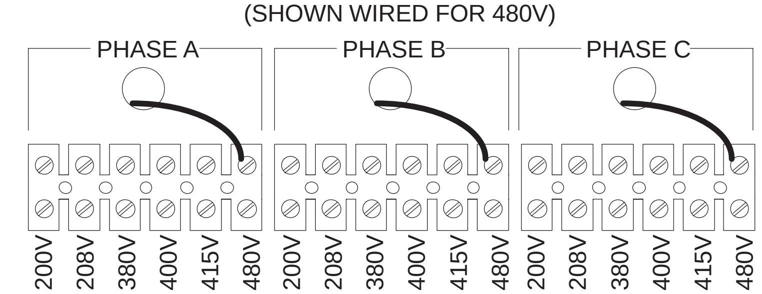
DANGER!!

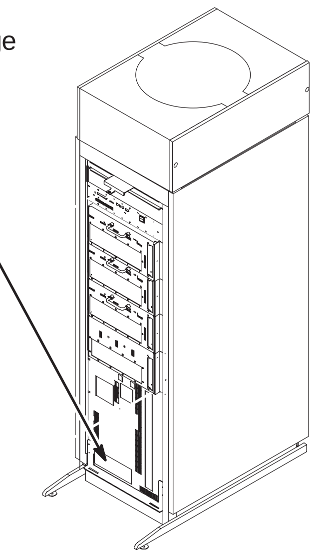
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

❑ A3 – INPUT VOLTAGE SELECTION

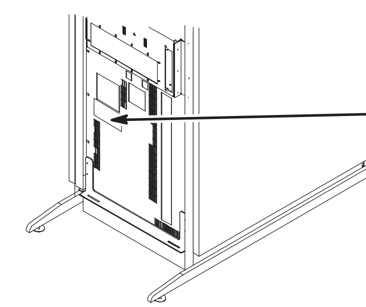
1. Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
2. Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
3. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.

(SHOWN WIRED FOR 480V)





❑ A4 – CIRCUIT BREAKER DIP SWITCH SETTINGS



1. Remove small cover plate under the main circuit breaker to access DIP switches controlling the circuit breaker operation.
2. Table below shows the position of the DIP switches for each input voltage selection. The switches immediately under the letter "L" and "I" are the only ones that should be changed.

INPUT VOLTAGE	DIP SWITCH "L" SETTING	L	I	DIP SWITCH "I" SETTING	INPUT VOLTAGE
200V		Set these switches according to the input voltage shown at the left.	Set these switches according to the input voltage shown at the right.		200V
208V					208V
380V		If these switches are present, set both of them in the **DOWN** position as shown.			380V
400V					400V
415V					415V
480V					480V

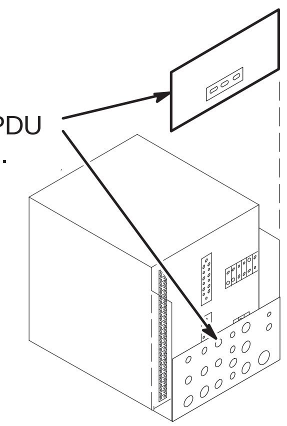
INSTALLATION STEPS SUMMARY

- ❑ **B1 – Facility Power and Ground Connention**
- ❑ **B2 – Route Power Cables Thru Interface Panel**
- ❑ **B3 – Run 967 Power Cable Alteration**
- ❑ **B4 – Connect Power Cables to PDU Module**

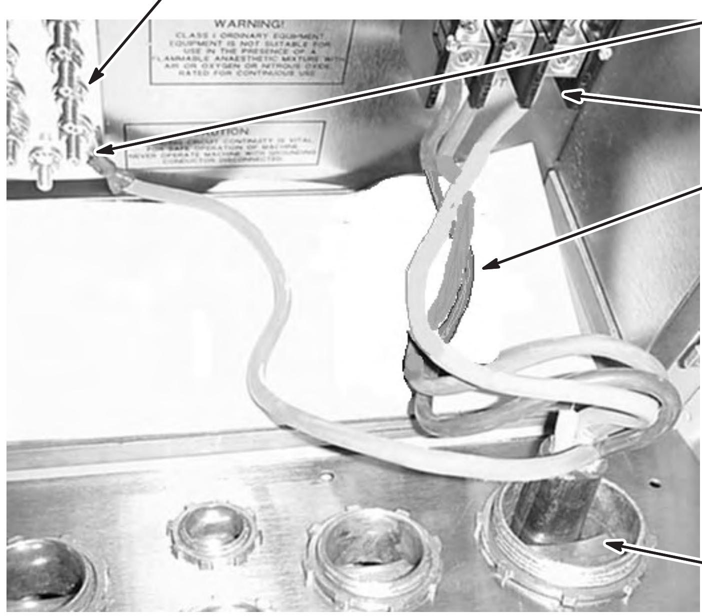
❑ B1 – FACILITY POWER AND GROUND CONNECTION

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE ACGD/PDU CABINET.



1. Remove rear PDU module covers.



6. Connect RF screen room ground conductor to Ground Bus Bar.
5. Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.
4. Connect L1–L3 conductors to terminal block.
3. **NOTE:** Air flow thru cabinet must exit thru top of cabinet. If not, fan at top of cabinet is rotating in wrong direction. To reverse rotation of fan, switch any two of the three incoming L1–L3 power wires.

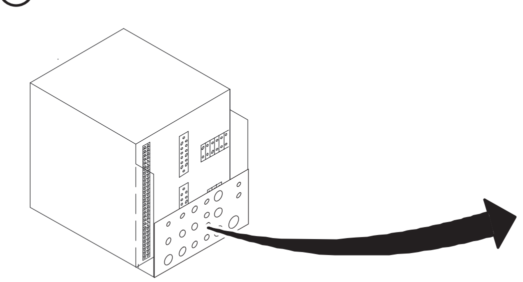
DANGER!!

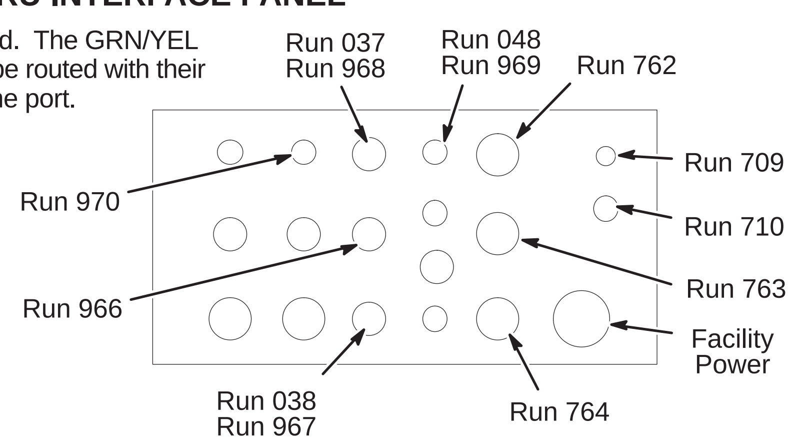
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE SWITCHING ANY ABOVE WIRES.

2. Route facility input power cable and ground wire thru interface panel.

❑ B2 – ROUTE POWER CABLES THRU INTERFACE PANEL

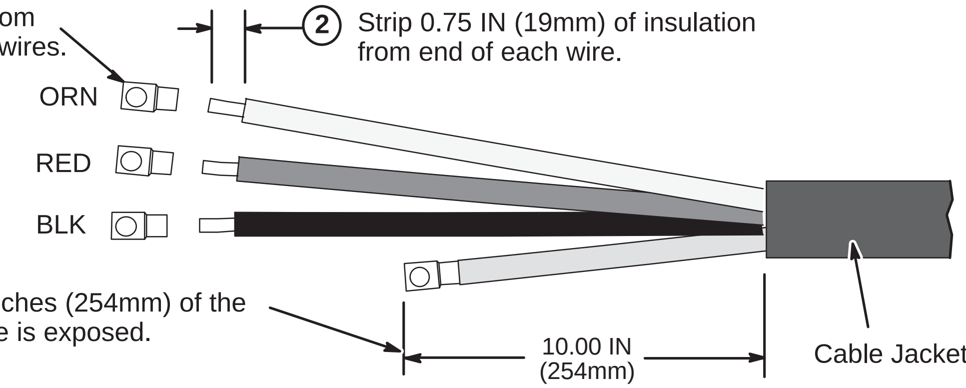
1. Route Runs thru interface panel as marked. The GRN/YEL ground wires (Runs 037 and 048) **MUST** be routed with their respective power cable Runs thru the same port.
2. Route any option Runs as marked.



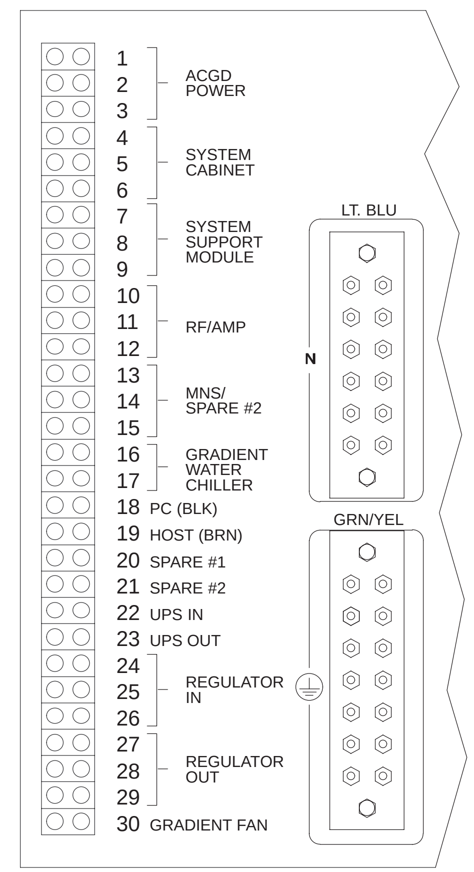


❑ B3 – RUN 967 POWER CABLE ALTERATION

1. Cut off ring terminal from ORN, RED, and BLK wires.
2. Strip 0.75 IN (19mm) of insulation from end of each wire.
3. Strip jacket until 10 Inches (254mm) of the GRN/YEL ground wire is exposed.

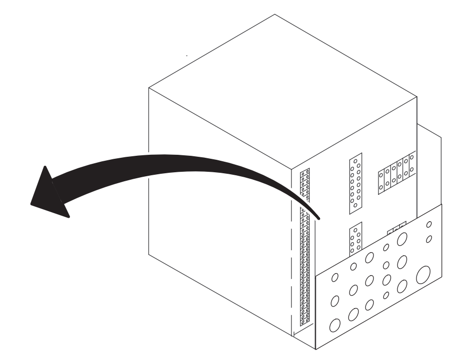


❑ B4 – CONNECT POWER CABLES TO PDU MODULE



WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.



1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
2. Trim and connect ground wire cables (Runs 037, 038, and 048) to ground bus connectors.
3. Tighten clamps of interface panel to secure cables in place.

CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	ϕ A, B, C	System Cabient 3ϕ, Neutral, Ground
7,8,9	966	ϕ A, B, C	SSM 3ϕ Neutral, Ground
10,11,12	967	ϕ A, B, C	RFI/AMP 3ϕ, Ground
16,17	970	ϕ A,B	Water Chiller Single ϕ, Ground
18,19	969	ϕ C,A	Operator Workspace Single ϕ, Neutral, GND
20	973	ϕ A	Heat Exchanger Single ϕ, Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	038		SRF Cabinet Ground Wire
GRN/YEL	048		Operator Workspace Ground Wire

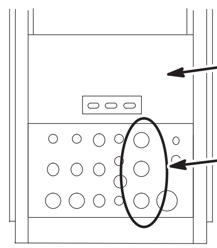
COLOR CODE: for remainder of cables:
3ϕ= Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel
Single ϕ = Brown, Black

INSTALLATION STEPS SUMMARY

- ❑ **C1 – Gradient Output Cable Interface Panel Access**
- ❑ **C2 – Gradient Output Cable Routing**
- ❑ **C3 – Route/Connect Fiber Optics Cables to GP Module**
- ❑ **C4 – Connect Runs 971 and 972 to PDU Module**
- ❑ **C5 – Atach InSite Cabinet Magnets**

❑ C1 – GRADIENT OUTPUT CABLE INTERFACE PANEL ACCESS

ACGD CABINET
(REAR VIEW)

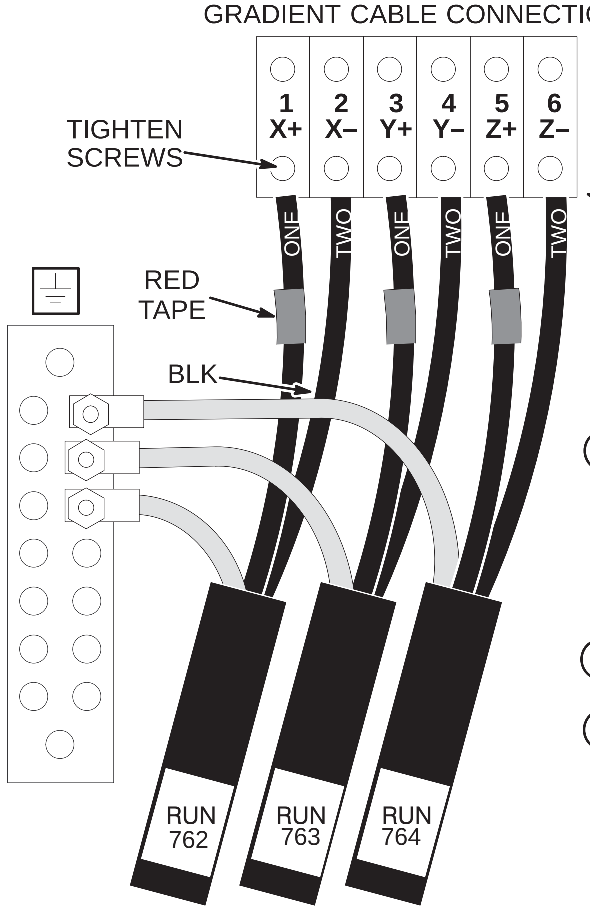


NOTE: Output cables (Runs 762, 763, and 764) were previously routed, cut to length, and terminated.

1. Remove Rear Cover Plate by removing seven screws that hold it in place.
2. Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

❑ C2 – GRADIENT OUTPUT CABLE ROUTING

GRADIENT CABLE CONNECTIONS

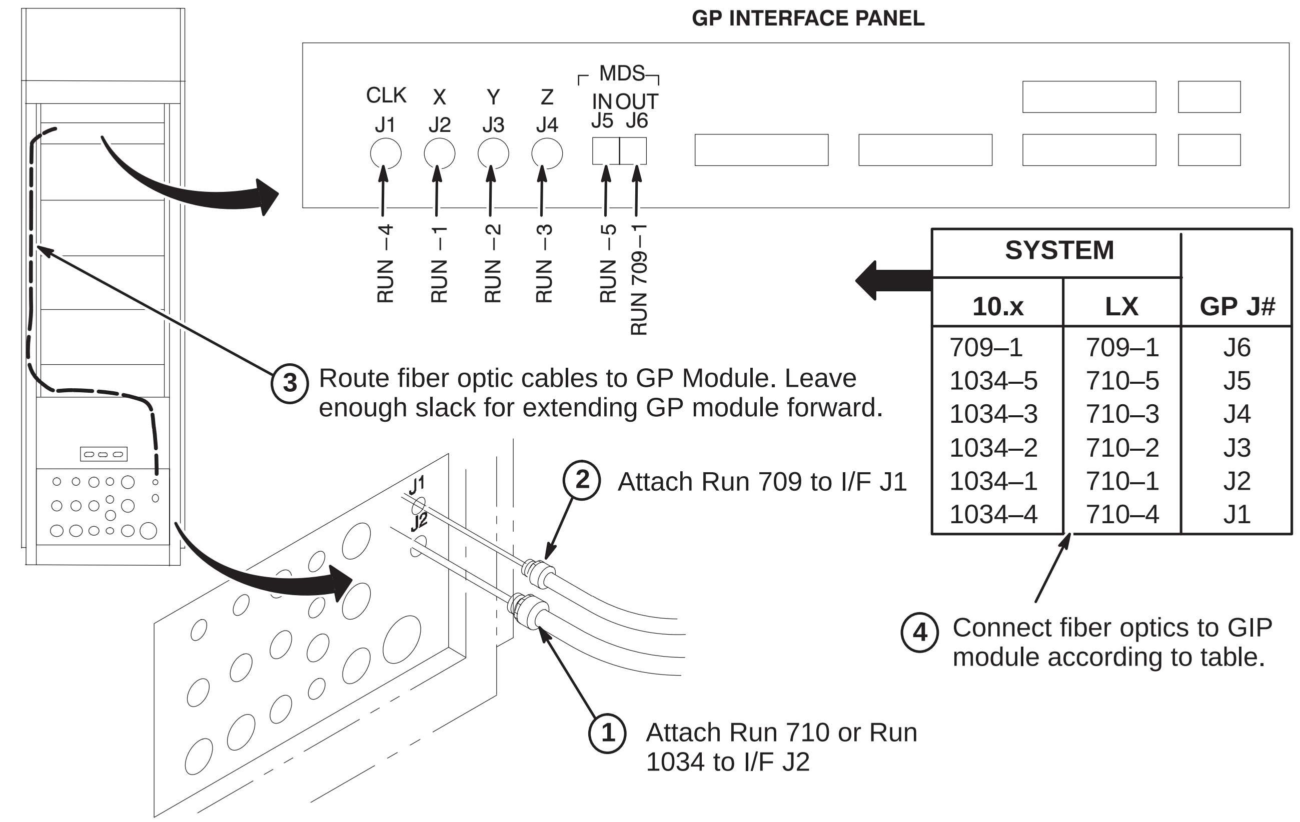


1. Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals.

NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.

2. Tighten screws to <u>52–65 inch–lbs (5,88–7,34 N–M).</u> This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
3. Connect the GND leads of each run to the Ground Bus Bar studs.
4. Tighten clamps on Interface Panel to secure gradient output cables.

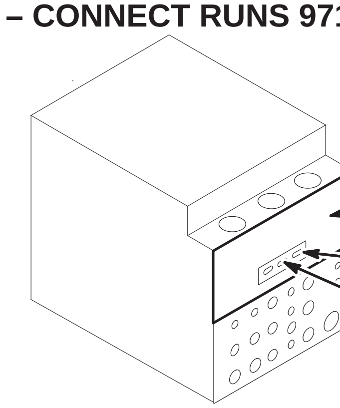
❑ C3 – ROUTE/CONNECT FIBER OPTICS TO GP MODULE



SYSTEM		
10.x	**LX**	**GP J#**
709–1	709–1	J6
1034–5	710–5	J5
1034–3	710–3	J4
1034–2	710–2	J3
1034–1	710–1	J2
1034–4	710–4	J1

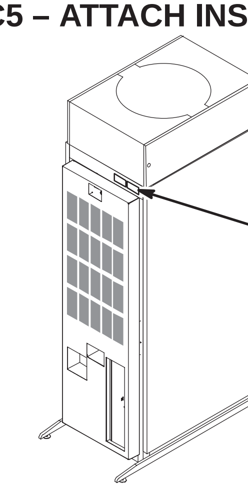
1. Attach Run 710 or Run 1034 to I/F J2
2. Attach Run 709 to I/F J1
3. Route fiber optic cables to GP Module. Leave enough slack for extending GP module forward.
4. Connect fiber optics to GIP module according to table.

❑ C4 – CONNECT RUNS 971 AND 972 TO PDU MODULE



1. Attach rear PDU module covers removed in Procedure **B1.**
2. Connect Run 972 to J3.
3. Connect Run 971 to J2.

❑ C5 – ATTACH INSITE CABINET MAGNETS



1. Attach front and rear covers.
2. From the 8.0 LX InSite Kit (46–301708G14), locate the “Gradient” cabinet magnet (46–320095P7) and the “PDU” cabinet magnet (46–320095P4). Attach to front of cabinet as shown.

INSTALLATION STEPS SUMMARY

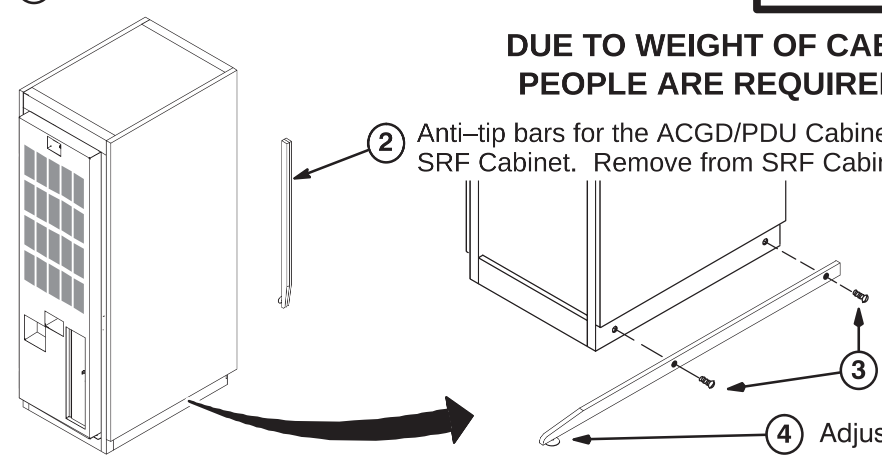
- ❑ A1 – Position ACGD/PDU Cabinet
- ❑ A2 – Input Voltage Selection
- ❑ A3 – Circuit Breaker DIP Switch Settings

❑ A1 – POSITION ACGD/PDU CABINET

1. Move cabinet to final position.

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET



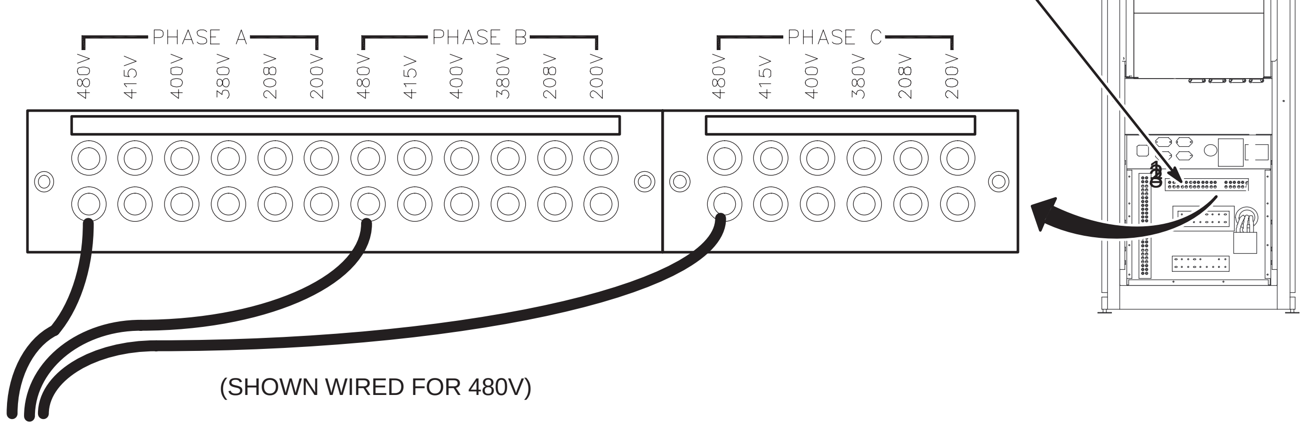
2. Anti–tip bars for the ACGD/PDU Cabinet are shipped in the SRF Cabinet. Remove from SRF Cabinet for this procedure.
3. Attach bars to cabinet.
4. Adjust leveling pads to level cabinet.

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

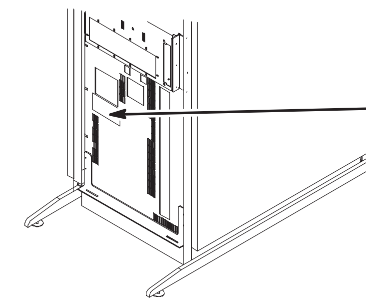
❑ A2 – INPUT VOLTAGE SELECTION

1. Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
2. Remove the cover from the rear of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
3. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



(SHOWN WIRED FOR 480V)

❑ A3 – CIRCUIT BREAKER DIP SWITCH SETTINGS



1. Remove small cover plate under the main circuit breaker to access DIP switches controlling the circuit breaker operation.
2. Table below shows the position of the DIP switches for each input voltage selection. The switches immediately under the letter “L” and “I” are the only ones that should be changed.

INPUT VOLTAGE	DIP SWITCH “L” SETTING	DIP SWITCH “I” SETTING	INPUT VOLTAGE
200V			200V
208V			208V
380V			380V
400V			400V
415V			415V
480V			480V

L	I
Set these switches according to the input voltage shown at the left.	Set these switches according to the input voltage shown at the right.
If these switches are present, make sure they are in the **DOWN** position as shown.	

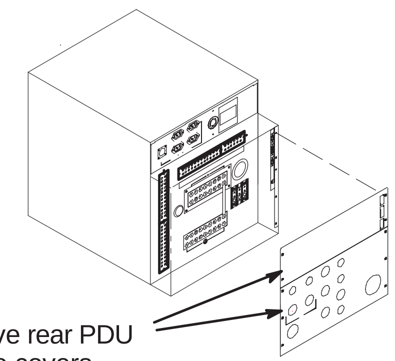
INSTALLATION STEPS SUMMARY

- ❑ **B1 – Facility Power and Ground Connention**
- ❑ **B2 – Route Power Cables Thru Interface Panel**
- ❑ **B3 – Run 967 Power Cable Alteration**
- ❑ **B4 – Connect Power Cables to PDU Module**

❑ B1 – FACILITY POWER AND GROUND CONNECTION

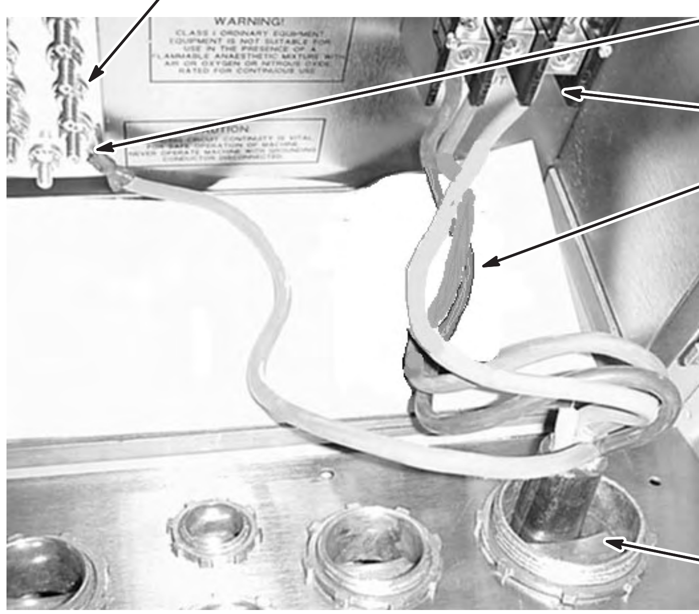
WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE ACGD/PDU CABINET.



1. Remove rear PDU module covers.

6. Connect RF screen room ground conductor to Ground Bus Bar.



5. Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.

4. Connect L1–L3 conductors to terminal block.

3. **NOTE:** Air flow thru cabinet must exit thru top of cabinet. If not, fan at top of cabinet is rotating in wrong direction. To reverse rotation of fan, switch any two of the three incoming L1–L3 power wires.

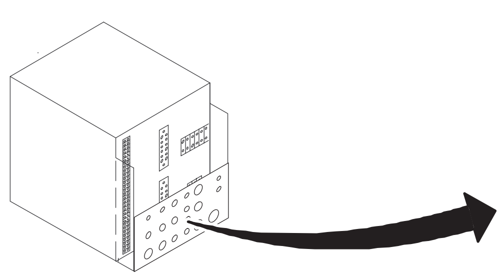
DANGER!!

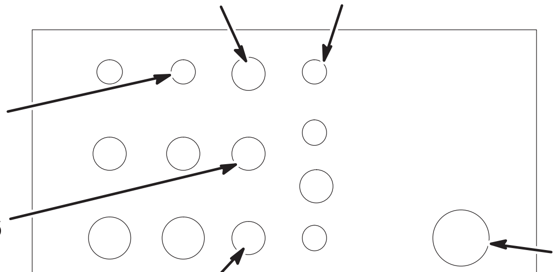
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE SWITCHING ANY ABOVE WIRES.

2. Route facility input power cable and ground wire thru interface panel.

❑ B2 – ROUTE POWER CABLES THRU INTERFACE PANEL

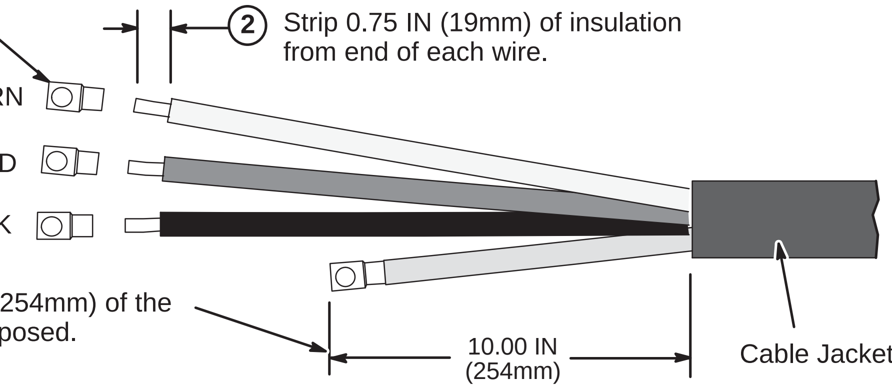
1. Route Runs thru interface panel as marked. The GRN/YEL ground wires (Runs 037 and 048) **MUST** be routed with their respective power cable Runs thru the same port.
2. Route any option Runs as marked.





❑ B3 – RUN 967 POWER CABLE ALTERATION

1. Cut off ring terminal from ORN, RED, and BLK wires.
2. Strip 0.75 IN (19mm) of insulation from end of each wire.
3. Strip jacket until 10 Inches (254mm) of the GRN/YEL ground wire is exposed.



❑ B4 – CONNECT POWER CABLES TO PDU MODULE

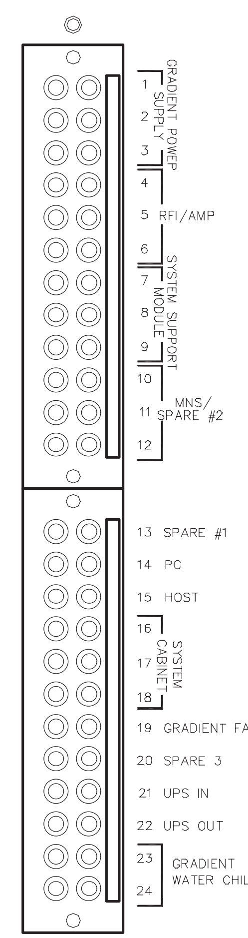
WARNING!

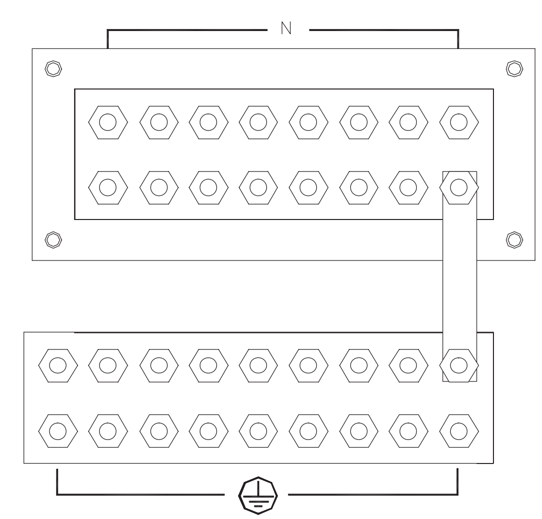
ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.

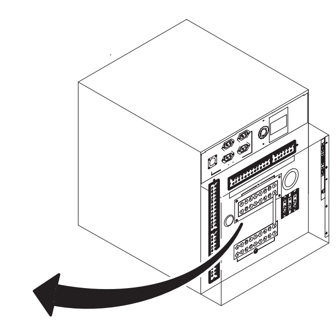
1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
2. Trim and connect ground wire cables (Runs 037, 038, and 048) to ground bus connectors.
3. Tighten clamps of interface panel to secure cables in place.

CONNECT TO	RUN	PHASE	DESTINATION
16, 17, 18	968	ϕ A, B, C	System Cabient 3ϕ, Neutral, Ground
7,8,9	966	ϕ A, B, C	SSM 3ϕ Neutral, Ground
4, 5, 6	967	ϕ A, B, C	RFI/AMP 3ϕ, Ground
23, 24	970	ϕ A,B	Water Chiller Single ϕ, Ground
14, 15	969	ϕ C,A	Operator Workspace Single ϕ, Neutral, GND
20	973	ϕ A	Heat Exchanger Single ϕ, Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	038		SRF Cabinet Ground Wire
GRN/YEL	048		Operator Workspace Ground Wire

COLOR CODE: for remainder of cables:
3ϕ= Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel
Single ϕ = Brown, Black





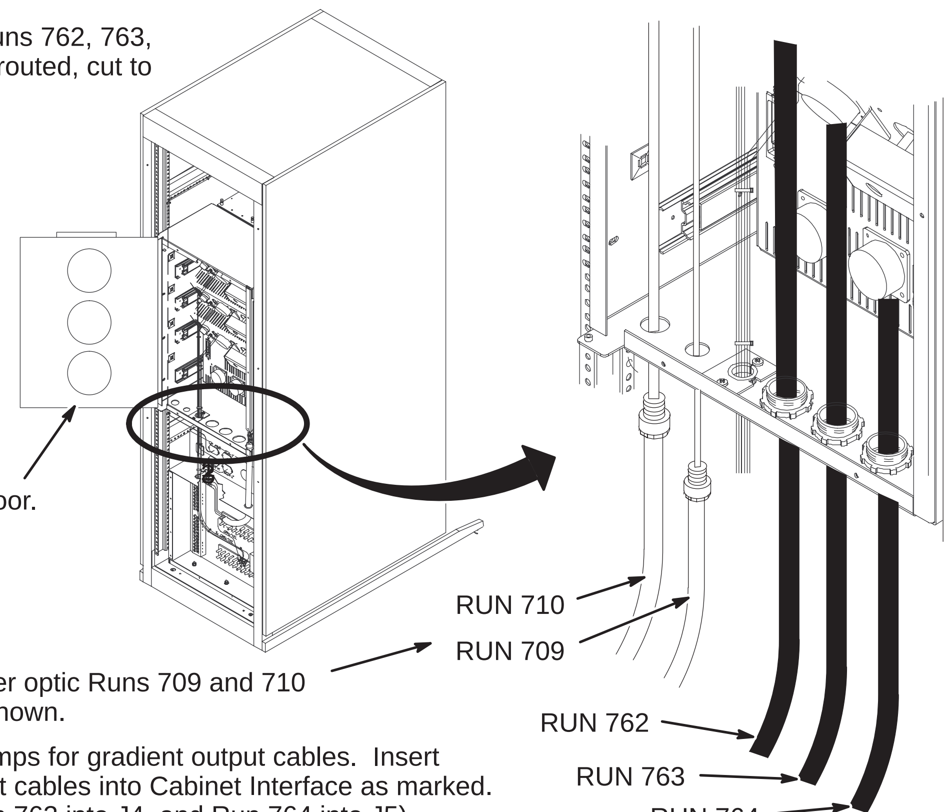


INSTALLATION STEPS SUMMARY

- ❑ C1 – Fiber Optic and Gradient Cable Interface Panel Access
- ❑ C2 – Route/Connect Fiber Optics Cables to GP Module
- ❑ C3 – Route/Connect Runs 762, 763, & 764 to Gradient Amplifiers
- ❑ C4 – Connect Runs 971 and 972 to PDU Module
- ❑ C5 – Atach InSite Cabinet Magnets

❑ C1 – FIBER OPTIC AND GRADIENT CABLE INTERFACE PANEL ACCESS

NOTE: Output cables (Runs 762, 763, and 764) were previously routed, cut to length, and terminated.

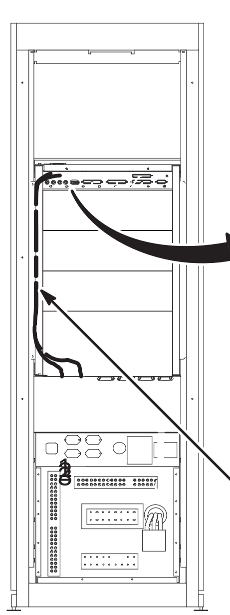


(1) Open rear fan panel door.

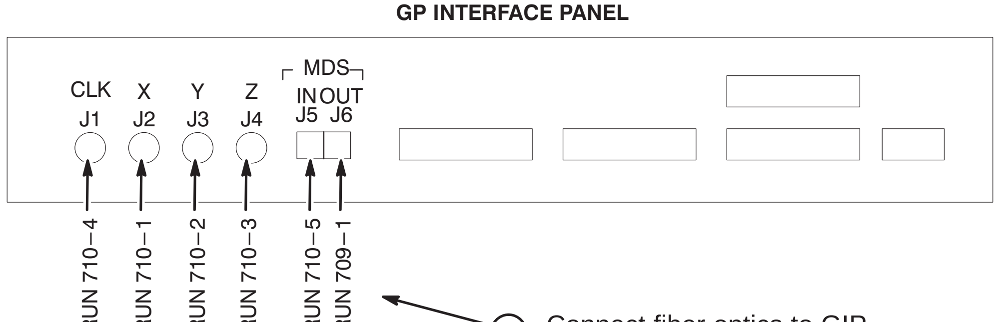
(2) Route and connect fiber optic Runs 709 and 710 to interface panel as shown.

(3) Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

❑ C3 – ROUTE/CONNECT FIBER OPTICS TO GP MODULE



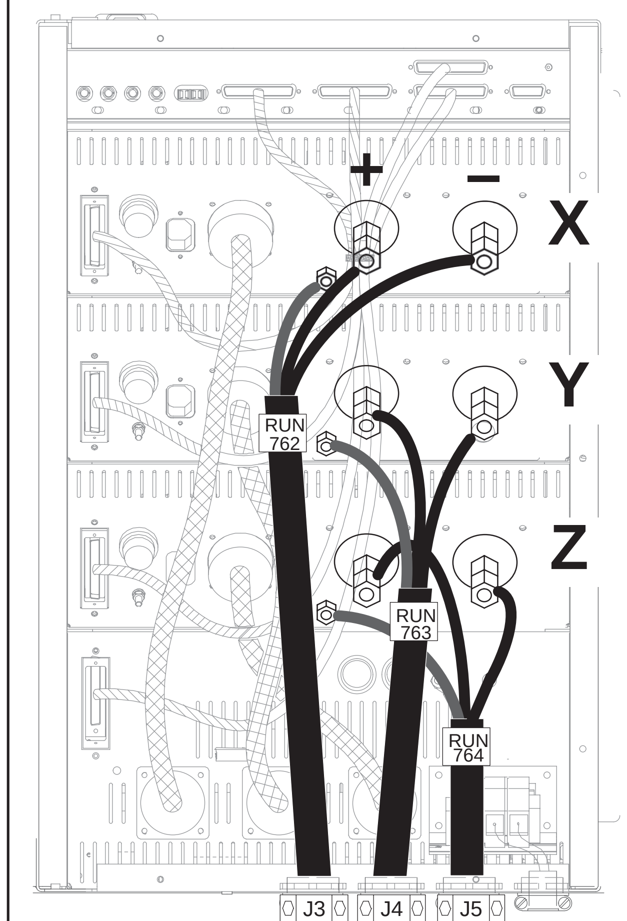
GP INTERFACE PANEL



(2) Connect fiber optics to GIP module according to table.

(1) Route fiber optic cables along this side of cabinet to GP Module. Leave enough slack for extending GP module forward.

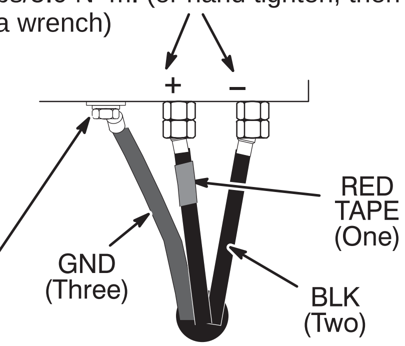
❑ C3 – ROUTE/CONNECT RUNS 762, 763, & 764 TO GRADIENT AMPLIFIERS



CAUTION

DO NOT OVER TIGHTEN nuts when connecting gradient output cables to gradient amp output lugs in step 2 below. STUDS CAN BREAK OFF.

(2) Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals of the Z Amplifier. Tighten to torque of 44 in–lbs/5.0 N–m. (or hand tighten, then one full turn with a wrench)



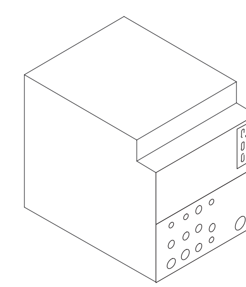
(3) Connect the GND leads of each run to the Ground stud located on the back of amplifier.

WARNING!

MAKE SURE GRADIENT CABLES ARE CONNECTED CORRECTLY. FAILURE TO DO SO WILL DESTROY GRADIENT AMPLIFIER.

(4) Tighten clamp blocks to secure output power cables to I/F panel.

❑ C4 – CONNECT RUNS 971 AND 972 TO PDU MODULE

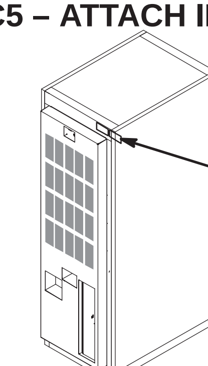


(1) Attach rear PDU module covers removed in Procedure **B1.**

(2) Connect Run 972 to J3.

(3) Connect Run 971 to J2.

❑ C5 – ATTACH INSITE CABINET MAGNETS



(1) Attach front and rear covers.

(2) From the 8.0 LX InSite Kit (46–301708G14), locate the “Gradient” cabinet magnet (46–320095P7) and the “PDU” cabinet magnet (46–320095P4). Attach to front of cabinet as shown.

SRF CABINET UPGRADE INSTRUCTIONS

Tab 16, SRF Cabinet, contains installation instructions for the following:

- *DIRECTION* 2286812 – Signa SRF Cabinet Installation Instructions
- *DIRECTION 2*346942 – Signa SRFD2 Cabinet Installation Instructions

Use only the applicable SRF Cabinet Installation Instruction Direction for the site being upgraded

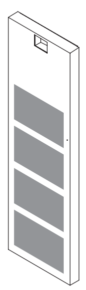
SIGNA SRF CABINET INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

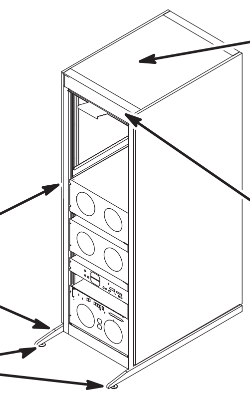
- ❑ **A – Install New SRF Cabinet**
- ❑ **B1 – Route/Connect Run 967 to Circuit Breaker**
  OR } **Use the applicable procedure.**
- ❑ **B2 – Route/Connect Run 967 to Terminal Strip**
- ❑ **C – Connect Runs 235/745 and 236/748 to RF Interface (RFI) Module**
- ❑ **D – Connect Fiber Optics and Run 229**
- ❑ **E – System Support Module Cable Connections**

❑ A – INSTALL NEW SRF CABINET

Note: The 1.5T SRF (Scaleable RF) cabinet is shown in these procedures. The 1.0T is the same except for the Heliax cables installed in Procedure **C** below.



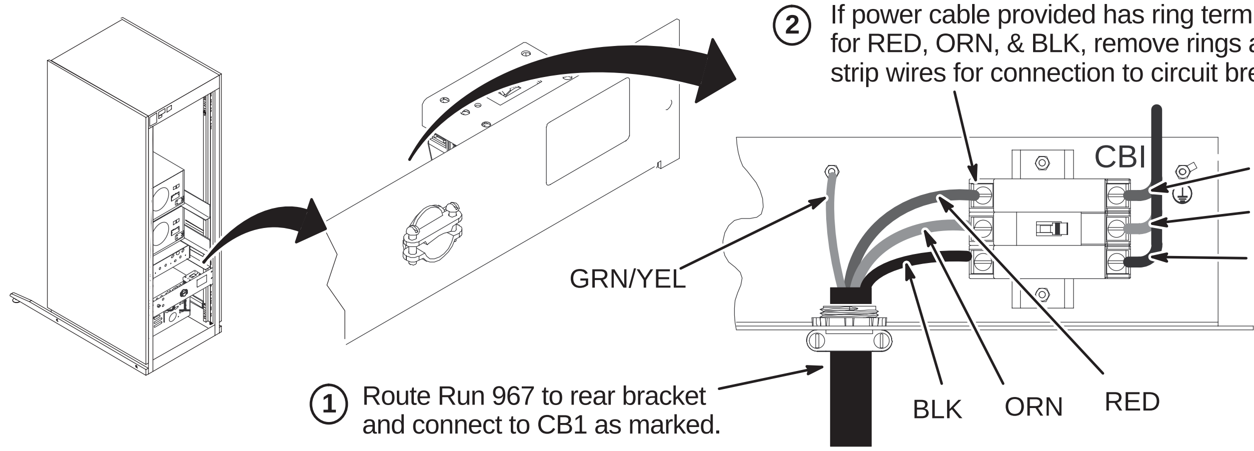
Note: <u>Comply with UL requirements</u> by either installing anti tip legs or securing the cabinet to the floor with local supplied brackets and floor anchors.



Note: UFI Label Kit (2146243) is attached to upper rail inside of cabinet.

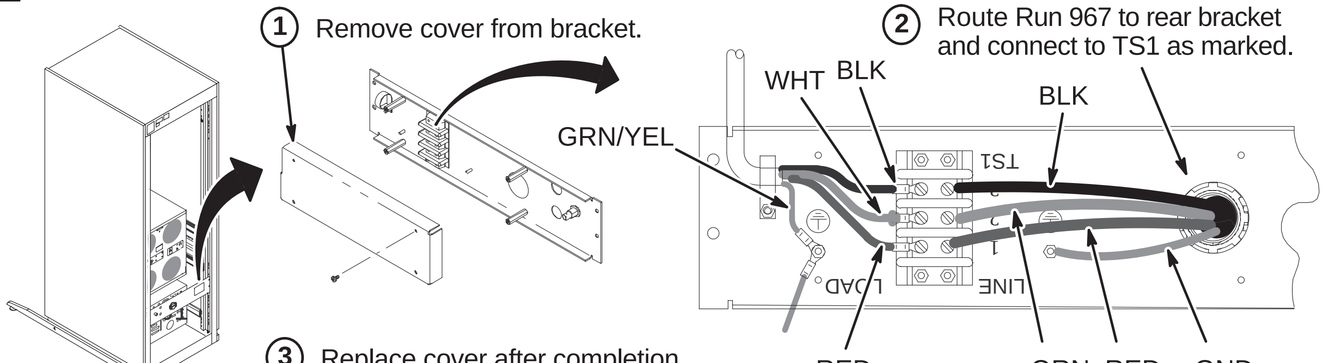
1. Move new SRF Cabinet to final position.
2. Remove anti–tip bars from inside of cabinet and attach to each side.
3. Adjust leveling pads to level cabinet.
4. From the 8.0 LX InSite Kit (46–301708G14), locate the "RF/Penetration" cabinet magnet (46–320095P13). Attach to front of cabinet as shown.

❑ B1 – ROUTE/CONNECT RUN 967 TO CIRCUIT BREAKER



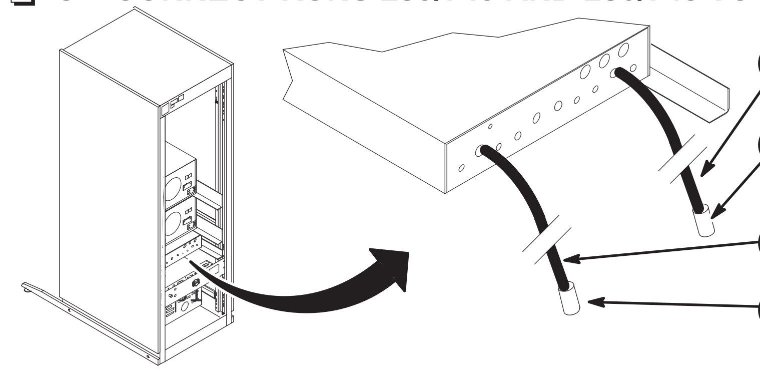
1. Route Run 967 to rear bracket and connect to CB1 as marked.
2. If power cable provided has ring terminals for RED, ORN, & BLK, remove rings and strip wires for connection to circuit breaker.

❑ B2 – ROUTE/CONNECT RUN 967 TO TERMINAL STRIP



1. Remove cover from bracket.
2. Route Run 967 to rear bracket and connect to TS1 as marked.
3. Replace cover after completion of power cable connection.

❑ C – CONNECT RUNS 235/745 AND 236/748 TO RF INTERFACE (RFI) MODULE

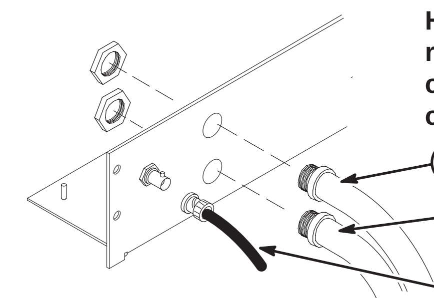


1. Connect Extension Cable with HN plugs (2105362) to J3 Head Output on RFI Module.
2. Connect Run 236 (1.5T) or Run 748 (1.0T) to above Extension cable using HN Bulkhead connector (46–221942P1).
3. Connect Extension Cable with N plugs (2105358) to J4 Body Output on RFI Module.
4. Connect Run 235 (1.5T) or Run 745 (1.0T) to above Extension cable using N Bulkhead connector (46–221865P1).

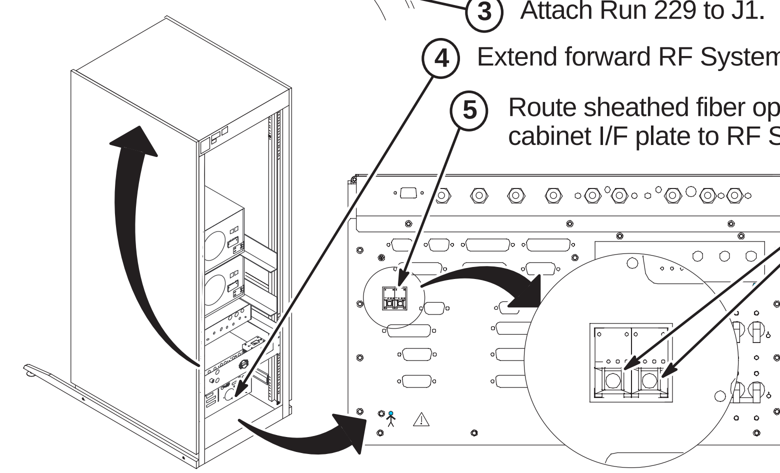
❑ D – ROUTE/CONNECT FIBER OPTICS AND RUN 229

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



1. Attach Run 709 to I/F RX*.
2. Attach Run 708 (or 1033) to I/F TX*.
   * Instructions provided with connectors.
3. Attach Run 229 to J1.
4. Extend forward RF System Support Module into servicing position.
5. Route sheathed fiber optic cables (Run 708 and Run 709) from cabinet I/F plate to RF System Support Module.

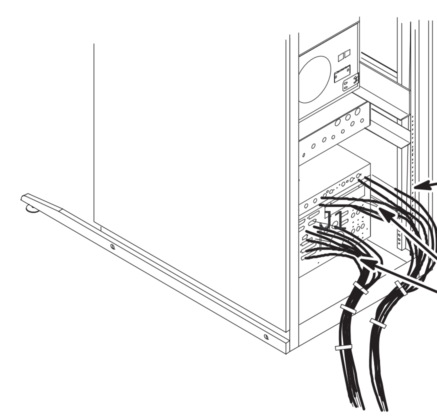


6. Connect fiber optic cables as marked to TR1 and TR2 at rear of Module:
   – Run 708 to Transmit (TX)
   – Run 709 to Receive (RX)
7. Dress and secure fiber optic cables along route from Module to I/F Panel. While checking Fiber Optic Cable routing, push RF System Support Module into cabinet.

❑ E – SYSTEM SUPPORT MODULE CABLE CONNECTIONS

CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



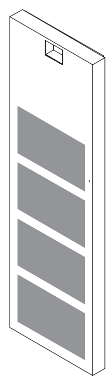
1. Connect remainder of SRF Cabinet cables to System Support Module.
   Note:
   If needed, refer to Overview: Section 2 – System Cables, for Run descriptions and designator information located at end of each cable.
2. Dress and secure cables connected to System Support Module. Verify sufficient slack by moving Module completely forward and back again. Adjust as necessary.
3. Attach all front and rear covers.

INSTALLATION STEPS SUMMARY

- ❑ A – Install New SRFD2 Cabinet
- ❑ B – Route/Connect Run 967
- ❑ C – Connect Runs 235 and 236 to RF Amp & Interface Module
- ❑ D – Connect Fiber Optics and Run 229
- ❑ E – System Support Module Cable Connections
- ❑ F – Attach InSite Cabinet Magnets

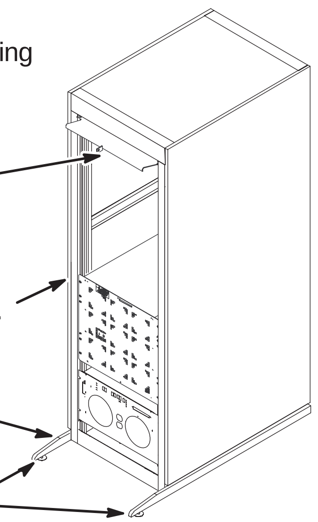
❑ A – INSTALL NEW SRFD2 CABINET

Note: Comply with UL requirements by either installing anti tip legs or securing the cabinet to the floor with local supplied brackets and floor anchors.



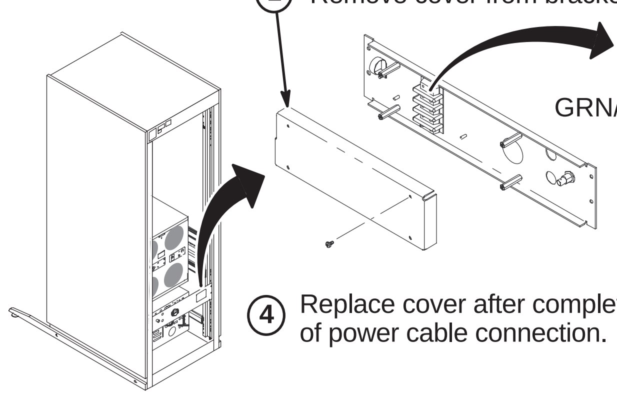
Note: UFI Label Kit (2146243) is attached to upper rail inside of cabinet.

1. Move new SRFD2 Cabinet to final position.
2. Remove anti–tip bars from inside of cabinet and attach to each side.
3. Adjust leveling pads to level cabinet.

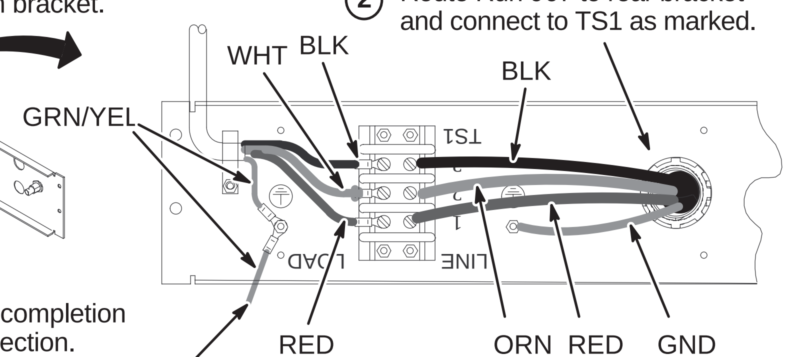


❑ B – ROUTE/CONNECT RUN 967

1. Remove cover from bracket.

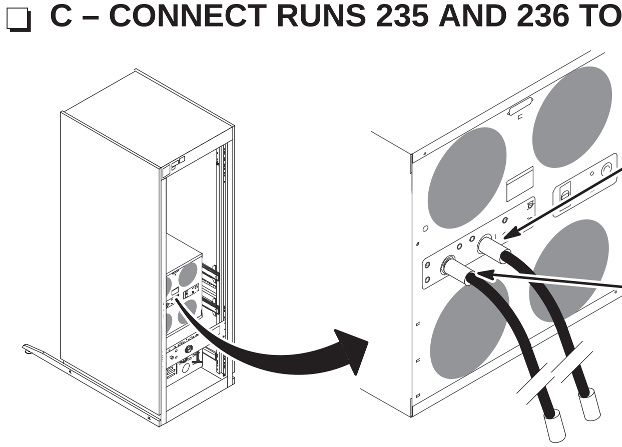


2. Route Run 967 to rear bracket and connect to TS1 as marked.



3. Connect Run 038 to stud GND stud on rear bracket.
4. Replace cover after completion of power cable connection.

❑ C – CONNECT RUNS 235 AND 236 TO RF AMP & INTERFACE MODULE

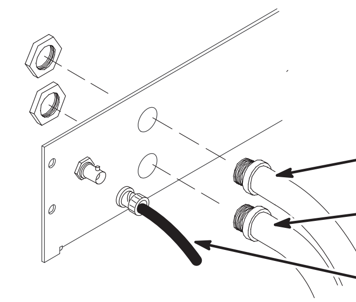


1. Connect Extension Cable with HN plugs (2105362) to J3 Head Output on RFI Module.
2. Connect Run 236 (1.5T) to above Extension cable using HN Bulkhead connector (46–221942P1).
3. Connect Extension Cable with N plugs (2105358) to J4 Body Output on RFI Module.
4. Connect Run 235 (1.5T) to above Extension cable using N Bulkhead connector (46–221865P1).

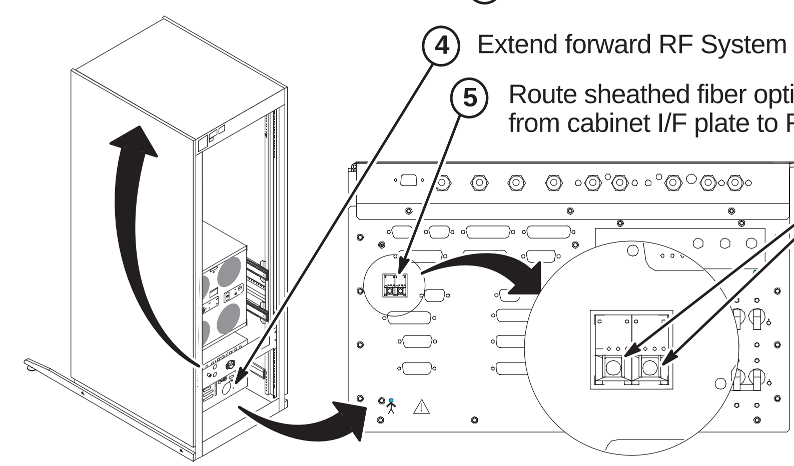
❑ D – ROUTE/CONNECT FIBER OPTICS AND RUN 229

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



1. Attach Run 709 to I/F RX*.
2. Attach Run 708 (or1033) to I/F TX*.
   * Instructions provided with connectors.
3. Attach Run 229 to J1.

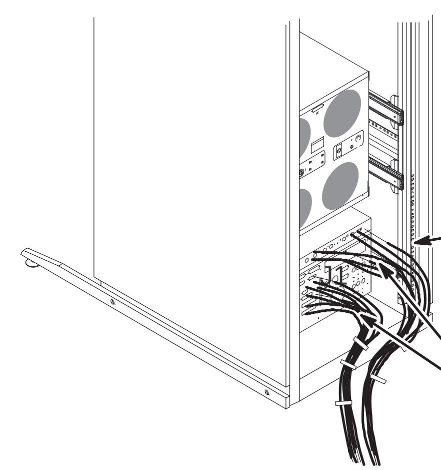


4. Extend forward RF System Support Module into servicing position.
5. Route sheathed fiber optic cables (Run 708/1033 and Run 709) from cabinet I/F plate to RF System Support Module.
6. Connect fiber optic cables as marked to TR1 and TR2 at rear of Module:
   – Run 708/1033 to Transmit (TX)
   – Run 709 to Receive (RX)
7. Dress and secure fiber optic cables along route from Module to I/F Panel. While checking Fiber Optic Cable routing, push RF System Support Module into cabinet.

❑ E – SYSTEM SUPPORT MODULE CABLE CONNECTIONS

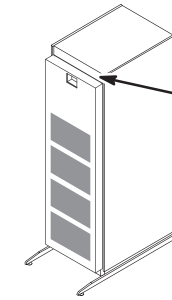
CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



1. Connect remainder of SRFD2 Cabinet cables to System Support Module.
   Note:
   If needed, refer to Overview: Section 2 – System Cables, for Run descriptions and designator information located at end of each cable.
2. Dress and secure cables connected to System Support Module. Verify sufficient slack for service loop by moving Module completely forward and back again. Adjust as necessary.

❑ F – ATTACH INSITE CABINET MAGNET



1. Attach all front and rear covers.
2. From the 8.0 LX InSite Kit (46–301708G14), locate the “RF/Penetration” cabinet magnet (46–320095P13). Attach to front of cabinet as shown.